

2024	LEVEL I VOLUME 6	Portfolio Management and Ethical	 CFA Institute® CFA Program
2024	LEVEL I VOLUME 5	Fixed Income, Derivatives, Alternative Investments, and Portfolio Management	 CFA Institute® CFA Program
2024	LEVEL I VOLUME 4	Corporate Issuers, Equity, Investments, and Fixed Income	 CFA Institute® CFA Program
2024	LEVEL I VOLUME 3	Financial Statement Analysis and Corporate Issuers	 CFA Institute® CFA Program
2024	LEVEL I VOLUME 2	Economics and Financial Statement Analysis	 CFA Institute® CFA Program
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2024 CFA® PROGRAM CURRICULUM LEVEL I VOLUMES 1-6

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FINANCIAL STATEMENT ANALYSIS, EQUITY INVESTMENTS



CFA® Program Curriculum
2024 • LEVEL 1 • VOLUME 3

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HOW TO USE THE CFA PROGRAM CURRICULUM

The CFA® Program exams measure your mastery of the core knowledge, skills, and abilities required to succeed as an investment professional. These core competencies are the basis for the Candidate Body of Knowledge (CBOK™). The CBOK consists of four components:

- A broad outline that lists the major CFA Program topic areas (www.cfainstitute.org/programs/cfa/curriculum/cbok)
- Topic area weights that indicate the relative exam weightings of the top-level topic areas (www.cfainstitute.org/programs/cfa/curriculum)
- Learning outcome statements (LOS) that advise candidates about the specific knowledge, skills, and abilities they should acquire from curriculum content covering a topic area: LOS are provided at the beginning of each block of related content and the specific lesson that covers them. We encourage you to review the information about the LOS on our website (www.cfainstitute.org/programs/cfa/curriculum/study-sessions), including the descriptions of LOS “command words” on the candidate resources page at www.cfainstitute.org.
- The CFA Program curriculum that candidates receive upon exam registration

Therefore, the key to your success on the CFA exams is studying and understanding the CBOK. You can learn more about the CBOK on our website: www.cfainstitute.org/programs/cfa/curriculum/cbok.

The entire curriculum, including the practice questions, is the basis for all exam questions and is selected or developed specifically to teach the knowledge, skills, and abilities reflected in the CBOK.

Errata

The curriculum development process is rigorous and includes multiple rounds of reviews by content experts. Despite our efforts to produce a curriculum that is free of errors, there are instances where we must make corrections. Curriculum errata are periodically updated and posted by exam level and test date online on the Curriculum Errata webpage (www.cfainstitute.org/en/programs/submit-errata). If you believe you have found an error in the curriculum, you can submit your concerns through our curriculum errata reporting process found at the bottom of the Curriculum Errata webpage.

Designing Your Personal Study Program

An orderly, systematic approach to exam preparation is critical. You should dedicate a consistent block of time every week to reading and studying. Review the LOS both before and after you study curriculum content to ensure that you have mastered the applicable content and can demonstrate the knowledge, skills, and abilities described by the LOS and the assigned reading. Use the LOS self-check to track your progress and highlight areas of weakness for later review.

Successful candidates report an average of more than 300 hours preparing for each exam. Your preparation time will vary based on your prior education and experience, and you will likely spend more time on some study sessions than on others.

CFA Institute Learning Ecosystem (LES)

Your exam registration fee includes access to the CFA Program Learning Ecosystem (LES). This digital learning platform provides access, even offline, to all of the curriculum content and practice questions and is organized as a series of short online lessons with associated practice questions. This tool is your one-stop location for all study materials, including practice questions and mock exams, and the primary method by which CFA Institute delivers your curriculum experience. The LES offers candidates additional

practice questions to test their knowledge, and some questions in the LES provide a unique interactive experience.

Prerequisite Knowledge

The CFA[®] Program assumes basic knowledge of Economics, Quantitative Methods, and Financial Statements as presented in introductory university-level courses in Statistics, Economics, and Accounting. CFA Level I candidates who do not have a basic understanding of these concepts or would like to review these concepts can study from any of the three pre-read volumes.

Feedback

Please send any comments or feedback to info@cfainstitute.org, and we will review your suggestions carefully.

Financial Statement Analysis

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Analysis of Income Taxes

by Elbie Louw PhD CFA CIPM and Michael A. Broihahn CPA CIA CFA

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Learning Outcomes

The candidate should be able to:

- ☐ contrast accounting profit, taxable income, taxes payable, and income tax expense and temporary versus permanent differences between accounting profit and taxable income
- ☐ explain how deferred tax liabilities and assets are created and the factors that determine how a company's deferred tax liabilities and assets should be treated for the purposes of financial analysis
- ☐ calculate, interpret, and contrast an issuer's effective tax rate, statutory tax rate, and cash tax rate
- ☐ analyze disclosures relating to deferred tax items and the effective tax rate reconciliation and explain how information included in these disclosures affects a company's financial statements and financial ratios

1. Introduction

Differences between tax laws and financial accounting standards result in differences between accounting profit (i.e., income before taxes on the income statement) and taxable income, or income computed under the prevailing tax laws in a given jurisdiction. These differences can be temporary or permanent. Temporary differences result in deferred tax assets and liabilities on the balance sheet and are important for capturing the income tax effects of all current period activities, even if tax consequences occur in the future. Current and deferred income tax expenses are used to calculate the effective tax rate, which is commonly used by analysts in estimating after-tax profitability measures like free cash flow. Given temporary and permanent differences, the effective tax rate typically differs from the statutory and cash tax rates for an issuer. Company disclosures of income tax related information in the notes to financial statements is typically one of the most extensive note disclosures.

Learning Module Overview

- Accounting profit is reported on a company's income statement in accordance with prevailing accounting standards and does not include a provision for income tax expense.
- A company's taxable income is its income subject to income taxes under the tax laws of the relevant jurisdiction and is the basis for its income tax payable (a liability), which appears on its balance sheet.
- Taxable and deductible temporary differences reverse in future periods while permanent differences do not.
- Deferred tax assets and liabilities arise from temporary differences in accounting profit and taxable income.
- Deferred tax assets represent taxes that have been paid but have not yet been recognized on the income statement, and deferred tax liabilities occur when financial accounting income tax expense is greater than regulatory income tax expense.
- The changes in deferred tax assets and liabilities are added to income tax payable to determine the company's income tax expense as it is reported on the income statement.
- Income taxes payable are primarily determined by the geographic composition of taxable income and the tax rates in each jurisdiction.
- Three types of tax rates are relevant to analysts: the statutory tax rate, the effective tax rate, and the cash tax rate.
- The notes in the financial statements disclose a reconciliation of the statutory tax rate to the effective rate and identify the items that significantly contribute to a temporarily high or low effective tax rate.
- Companies present and disclose income tax-related information through income statements, balance sheets, and income tax note disclosures.
- Companies will disclose how the income tax provision was derived from the US federal statutory rate.
- In the income tax note disclosure, companies will provide detailed information about the derivation of the deferred tax assets and deferred tax liabilities.

2. Differences between Accounting Profit and Taxable Income

LEARNING OUTCOME

The candidate should be able to:

- ❑ **contrast accounting profit, taxable income, taxes payable, and income tax expense and temporary versus permanent differences between accounting profit and taxable income**

A company's **accounting profit** is reported on its income statement in accordance with prevailing accounting standards. Accounting profit (also referred to as income before taxes or pretax income) does not include a provision for income tax expense.¹ A company's **taxable income** is its income subject to income taxes under the tax laws of the relevant jurisdiction. A company's taxable income is the basis for its **income tax payable** (a liability) or recoverable (an asset), which appears on its balance sheet. **Income tax paid** in a period is the cash amount paid for income and reduces the income tax payable.

The **tax base** of an asset or liability is the amount at which the asset or liability is valued for tax purposes, whereas the **carrying amount** is the amount at which the asset or liability is recorded in the financial statements.² The tax bases and carrying amounts of assets and liabilities can differ based on differences in accounting standards and the relevant tax laws. Common differences are as follows:

- Revenues and expenses may be recognized in one period for accounting purposes and a different period for tax purposes.
- Specific revenues and expenses may be either recognized for accounting purposes and not at all for tax purposes, or vice versa.
- The deductibility of gains and losses of assets and liabilities may vary for accounting and income tax purposes.
- Subject to tax rules, tax losses in prior years might be used to reduce taxable income in later years, resulting in differences in accounting and taxable income (tax loss carryforward).
- Adjustments of reported financial data from prior years might not be recognized equally for accounting and tax purposes or might be recognized in different periods.

A common example is accelerated depreciation of an asset for tax reporting (to increase expense and lower tax payments in the early years) while using the straight-line depreciation method on the financial statements. Although different on a year-to-year basis (e.g., depreciation of 10 percent on a straight-line basis may be used for accounting purposes, whereas 50 percent might be allowed for tax purposes in the first year) both approaches allow for the total cost of the asset to be depreciated over its useful life.

Differences between the tax base and carrying amount of liabilities (and, by extension, between taxable income and accounting profit) can either be temporary or permanent. Temporary differences, like the aforementioned accelerated versus straight-line depreciation example, reverse in future periods, whereas permanent differences do not.

2.1. Taxable Temporary Differences

Temporary differences are further divided into two categories, namely taxable temporary differences and deductible temporary differences. **Taxable temporary differences** result from the carrying amount of an asset exceeding its tax base (like the aforementioned accelerated depreciation example at the end of Year 1) or when the tax base of a liability exceeds its carrying amount. Taxable temporary differences result in the recognition of **deferred tax liabilities**.

2.2. Deductible Temporary Differences

Deductible temporary differences are temporary differences that result in a reduction or deduction of taxable income in a future period when the balance sheet item is recovered or settled. Deductible temporary differences result in a **deferred tax asset** when the tax base of an asset exceeds its carrying amount and, in the case of a liability, when the carrying amount of the liability exceeds its tax base. The recognition of a deferred tax asset is allowed only to the extent there is a reasonable expectation of future profits against which the asset or liability (that gave rise to the deferred tax asset) can be recovered or settled.

To determine the probability of sufficient future profits for utilization, one must consider the following: (1) sufficient taxable temporary differences must exist that are related to the same tax authority and the same taxable entity; and (2) the taxable temporary differences that are expected to reverse in the same periods as expected for the reversal of the deductible temporary differences.

2.3. Taxable and Deductible Temporary Differences

Exhibit 1 summarizes how differences between the tax bases and carrying amounts of assets and liabilities give rise to deferred tax assets or deferred tax liabilities.

Exhibit 1: Treatment of Temporary Differences

Balance Sheet Item	Carrying Amount vs. Tax Base	Results in Deferred Tax Asset/Liability
Asset	Carrying amount > tax base	Deferred tax liability
Asset	Carrying amount < tax base	Deferred tax asset
Liability	Carrying amount > tax base	Deferred tax asset
Liability	Carrying amount < tax base	Deferred tax liability

Example 1 and 2 illustrate the difference in the tax base and carrying amount of the assets and liabilities, whether they are temporary or permanent differences, and whether a deferred tax asset or liability will be recognized.

EXAMPLE 1

Differences in Tax Base and Carrying Amount of Assets and Liabilities

Exhibit 2: Tax Base and Carrying Amounts

	Carrying Amount (euros)	Tax Base (euros)	Temporary Difference (euros)	Will Result in Deferred Tax Asset/Liability
1. Dividends receivable	1,000,000	1,000,000	0	N/A
2. Development costs	2,500,000	2,250,000	250,000	Deferred tax liability
3. Research costs	0	375,000	(375,000)	Deferred tax asset
4. Accounts receivable	1,500,000	1,218,750	281,250	Deferred tax liability

1. *Dividends receivable*: As a result of non-taxability, the carrying amount equals the tax base of dividends receivable. This constitutes a permanent difference and will not result in the recognition of any deferred tax asset or liability. A temporary difference constitutes a difference that will, at some future date, be reversed. Although the timing of recognition is different for tax and accounting purposes, in the end the full carrying amount will be expensed/recognized as income. A permanent difference will never be reversed. Based on tax legislation, dividends from a subsidiary are not recognized as income. Therefore, no amount will be reflected as dividend income when calculating the taxable income, and the tax base of dividends receivable must be the total amount received, namely EUR1,000,000. The taxable income and accounting profit will permanently differ with the amount of dividends receivable, even on future financial statements as an effect on the retained earnings reflected on the balance sheet.
2. *Development costs*: The difference between the carrying amount and tax base is a temporary difference that, in the future, will reverse. In this fiscal year, it will result in a deferred tax liability.
3. *Research costs*: The difference between the carrying amount and tax base is a temporary difference that results in a deferred tax asset. Remember that a deferred tax asset arises because of an excess amount paid for taxes (when taxable income is greater than accounting profit), which is expected to be recovered from future operations. Based on accounting principles, the full amount was deducted resulting in a lower accounting profit, while the taxable income by implication, should be greater because of the lower amount expensed.
4. *Accounts receivable*: The difference between the carrying amount and tax base of the asset is a temporary difference that will result in a deferred tax liability.

EXAMPLE 2

Differences in Tax Base and Carrying Amount of Assets and Liabilities

Exhibit 3: Tax Base and Carrying Amounts

	Carrying Amount (euros)	Tax Base (euros)	Temporary Difference (euros)	Will Result in Deferred Tax Asset/Liability
1. Donations	0	0	0	N/A
2. Interest received in advance	300,000	0	(300,000)	Deferred tax asset
3. Rent received in advance	10,000,000	0	(10,000,000)	Deferred tax asset
4. Loan (capital)	550,000	550,000	0	N/A
4. Interest paid	0	0	0	N/A

1. *Donations*: It was assumed that tax legislation does not allow donations to be deducted for tax purposes. No temporary difference results from donations, and thus a deferred tax asset or liability will not be recognized. This constitutes a permanent difference.

2. *Interest received in advance*: Interest received in advance results in a temporary difference that gives rise to a deferred tax asset. A deferred tax asset arises because of an excess amount paid for taxes (when taxable income is greater than accounting profit), which is expected to be recovered from future operations.
3. *Rent received in advance*: The difference between the carrying amount and tax base is a temporary difference that leads to the recognition of a deferred tax asset.
4. *Loan*: There are no temporary differences, as a result of the loan or interest paid, and thus no deferred tax item is recognized.

2.4. Permanent Differences

Permanent differences are differences between tax laws and accounting standards that *will not* be reversed at some future date. Because they will not be reversed at a future date, these differences do not give rise to deferred tax. These items typically include the following:

- income or expense items not allowed by tax legislation, such as penalties and fines that are considered expenses for financial reporting purposes, but are not deductible for tax purposes; and
- tax credits for some expenditures that directly reduce taxes. An example is tax credits provided by tax authorities to encourage the purchase of solar power or an electric vehicle.

Because no deferred tax item is created for permanent differences, all permanent differences result in a difference between the company's tax rate and its statutory corporate income tax rate.

2.5. Tax Expense

A company's **tax expense** or its provision for income taxes, appears on its income statement and is an aggregate of its income tax payable (or recoverable in the case of a tax benefit) and any changes in deferred tax assets and liabilities. This approach, rather than simply reporting income taxes paid, follows the matching principle by reporting the tax consequences of all current period activities.

Question Set

1. When accounting standards require recognition of an expense that is not permitted under tax laws, the result is a:
 - A. deferred tax liability.
 - B. temporary difference.
 - C. permanent difference.

Solution:

C is correct. Accounting items that are not deductible for tax purposes will not be reversed and thus result in permanent differences.

2. When certain expenditures result in tax credits that directly reduce taxes, the company will *most likely* record:
 - A. a deferred tax asset.
 - B. a deferred tax liability.

C. no deferred tax asset or liability.

Solution:

C is correct. Tax credits that directly reduce taxes are a permanent difference, and permanent differences do not give rise to deferred tax.

3. In early 2018, Sanborn Company must pay the tax authority EUR37,000 on the income it earned in 2017. This amount was *most likely* recorded on the company's 31 December 2017 financial statements as:

A. taxes payable.

B. income tax expense.

C. a deferred tax liability.

Solution:

A is correct. The taxes a company must pay in the immediate future are taxes payable.

3. DEFERRED TAX ASSETS AND LIABILITIES

LEARNING OUTCOME

The candidate should be able to:

☐ **explain how deferred tax liabilities and assets are created and the factors that determine how a company's deferred tax liabilities and assets should be treated for the purposes of financial analysis**

Deferred tax assets and liabilities arise from temporary differences in accounting profit and taxable income. Deferred tax assets represent taxes that have been paid (or often the carrying forward of losses from previous periods) but have not yet been recognized on the income statement. Deferred tax liabilities occur when financial accounting income tax expense is greater than regulatory income tax expense. At the end of each reporting period, deferred tax assets and liabilities are recalculated by comparing the tax bases and carrying amounts of the balance sheet items. The changes in deferred tax assets and liabilities are added to income tax payable to determine the company's income tax expense (or credit) as it is reported on the income statement.

If statutory tax rates change, the recorded value of a deferred tax asset or deferred tax liability would also change. For example, assume a tax authority reduces the statutory corporate tax rate from 35 percent to 21 percent. Because the future tax benefit would be reduced, the recorded value of a deferred tax asset would decrease. Similarly, because the amount of a future tax obligation decreases, the value of a corresponding deferred tax liability would also decrease.

3.1. Realizability of Deferred Tax Assets

Assume Pinto Construction (a hypothetical company) depreciates equipment on a straight-line basis of 10 percent per year. The tax authorities allow depreciation of 15 percent per year. At the end of the fiscal year, the carrying amount of the equipment for accounting purposes would be greater than the tax base of the equipment thus resulting in a temporary difference. A deferred tax asset may be created only if the company expects to be able to realize the economic benefit of the deferred tax asset in the future. In this example, the equipment is used in the core business of Pinto Construction. If the company is a going

concern and has stable earnings, there should be no doubt that future economic benefits will result from the equipment, and it would be appropriate to create the deferred tax item.

If, however, it were doubtful that future economic benefits will be realized from a temporary difference (i.e., if Pinto Construction was being liquidated), the temporary difference will not lead to recognition of a deferred tax asset. If a deferred tax asset was recognized previously, but there was sufficient doubt about the economic benefits being realized, then, under IFRS, an existing deferred tax asset would be reversed. Under US GAAP, a valuation allowance would be established to reduce the amount of the deferred tax asset to the amount that is more likely than not to be realized. In assessing future economic benefits, much is left to the discretion of management in assessing the temporary differences and the issue of future economic benefits.

EXAMPLE 3

Reston Partners

The information in [Exhibit 4](#) pertains to a hypothetical company, Reston Partners.

Exhibit 4: Reston Partners Consolidated Income Statement

Period Ending 31 March	Year 3	Year 2	Year 1
Revenue	GBP40,000	GBP30,000	GBP25,000
Other net gains	2,000	0	0
Changes in inventories of finished goods and work in progress	400	180	200
Raw materials and consumables used	(5,700)	(4,000)	(8,000)
Depreciation expense	(2,000)	(2,000)	(2,000)
Other expenses	(6,000)	(5,900)	(4,500)
Interest expense	(2,000)	(3,000)	(6,000)
Profit before tax	GBP26,700	GBP15,280	GBP4,700

The financial performance and accounting profit of Reston Partners on this income statement is based on accounting principles appropriate for the jurisdiction in which Reston Partners operates. The principles used to calculate accounting profit (profit before tax) may differ from the principles applied for tax purposes (the calculation of taxable income). For illustrative purposes, however, assume that all income and expenses on the income statement are treated identically for tax and accounting purposes *except* depreciation.

The depreciation is related to equipment owned by Reston Partners. For simplicity, assume that the equipment was purchased at the beginning of Year 1. Depreciation should thus be calculated and expensed for the full year. Assume that accounting standards permit equipment to be depreciated on a straight-line basis over a 10-year period, whereas the tax standards in the jurisdiction specify that equipment should be depreciated on a straight-line basis over a seven-year period. For simplicity, assume a salvage value of GBP0 at the end of the equipment's useful life. Both methods will result in the full depreciation of the asset over the respective tax or accounting life.

The equipment was originally purchased for GBP20,000. In accordance with accounting standards, over the next 10 years the company will recognize annual depreciation of GBP2,000 ($\text{GBP}20,000 \div 10$) as an expense on its income statement and for the determination of accounting profit. For tax purposes, however, the company will recognize GBP2,857 ($\text{GBP}20,000 \div 7$) in depreciation each year. Each fiscal year the depreciation expense related to the use of the equipment will, therefore, differ for tax and accounting purposes (tax base vs. carrying amount), resulting in a difference between accounting profit and taxable income.

The previous income statement reflects accounting profit (depreciation at GBP2,000 per year). [Exhibit 5](#) shows the taxable income for each fiscal year.

Exhibit 5: Taxable Income (British pound millions)			
Taxable Income	Year 3	Year 2	Year 1
Revenue	GBP40,000	GBP30,000	GBP25,000
Other net gains	2,000	0	0
Changes in inventories of finished goods and work in progress	400	180	200
Raw materials and consumables used	(5,700)	(4,000)	(8,000)
Depreciation expense	(2,857)	(2,857)	(2,857)
Other expenses	(6,000)	(5,900)	(4,500)
Interest expense	(2,000)	(3,000)	(6,000)
Taxable income	GBP25,843	GBP14,423	GBP3,843

The carrying amount and tax base for the equipment is shown in [Exhibit 6](#):

Exhibit 6: Tax Base for Equipment (British pound millions)			
	Year 3	Year 2	Year 1
Equipment value for accounting purposes (<i>carrying amount</i>) (depreciation of GBP2,000/year)	GBP14,000	GBP16,000	GBP18,000
Equipment value for tax purposes (<i>tax base</i>) (depreciation of GBP2,857/year)	GBP11,429	GBP14,286	GBP17,143
Difference	GBP2,571	GBP1,714	GBP857

At each balance sheet date, the tax base and carrying amount of all assets and liabilities must be determined. The income tax payable by Reston Partners will be based on the taxable income of each fiscal year. If a tax rate of 30 percent is assumed, then the income taxes payable for years 1, 2, and 3 are GBP1,153 (30% × 3,843), GBP4,327 (30% × 14,423), and GBP7,753 (30% × 25,843), respectively.

Remember, though, that if the tax obligation is calculated based on accounting profits, it will differ because of the differences between the tax base and the carrying amount of equipment. The difference in each fiscal year is reflected in the table above. In each fiscal year the carrying amount of the equipment exceeds its tax base. For tax purposes, therefore, the asset tax base is less than its carrying value under financial accounting principles. The difference results in a deferred tax liability as shown in [Exhibit 7](#).

Exhibit 7: Deferred Tax Liability (British pound millions)			
	Year 3	Year 2	Year 1
Deferred tax liability	GBP771	GBP514	GBP257
(Difference between tax base and carrying amount) × tax rate			
Year 1: GBP(18,000 – 17,143) × 30 percent = 257			
Year 2: GBP(16,000 – 14,286) × 30 percent = 514			

	Year 3	Year 2	Year 1
Deferred tax liability	GBP771	GBP514	GBP257
Year 3: $\text{GBP}(14,000 - 11,429) \times 30 \text{ percent} = 771$			

The comparison of the tax base and carrying amount of equipment shows what the deferred tax liability should be on a particular balance sheet date. In each fiscal year, only the change in the deferred tax liability should be included in the calculation of the income tax expense reported on the income statement prepared for accounting purposes.

On the income statement, the company's income tax expense will be the sum of change in the deferred tax liability and the income tax payable.

Exhibit 8: Deferred Tax Liability (British pound millions)

	Year 3	Year 2	Year 1
Income tax payable (based on tax accounting)	GBP7,753	GBP4,327	GBP1,153
Change in deferred tax liability	257	257	257
Income tax (based on financial accounting)	GBP8,010	GBP4,584	GBP1,410

Note that because the different treatment of depreciation is a temporary difference, the income tax on the income statement is 30 percent of the accounting profit, although only a part is income tax payable and the rest is a deferred tax liability.

The consolidated income statement of Reston Partners including income tax is presented in [Exhibit 9](#):

Exhibit 9: Reston Partners Consolidated Income Statement (British pound millions)

Period Ending 31 March	Year 3	Year 2	Year 1
Revenue	GBP40,000	GBP30,000	GBP25,000
Other net gains	2,000	0	0
Changes in inventories of finished goods and work in progress	400	180	200
Raw materials and consumables used	(5,700)	(4,000)	(8,000)
Depreciation expense	(2,000)	(2,000)	(2,000)
Other expenses	(6,000)	(5,900)	(4,500)
Interest expense	(2,000)	(3,000)	(6,000)
Profit before tax	GBP26,700	GBP15,280	GBP4,700
Income tax	(8,010)	(4,584)	(1,410)
Profit after tax	GBP18,690	GBP10,696	GBP3,290

Any amount paid to the tax authorities will reduce the liability for income tax payable and be reflected on the statement of cash flows of the company.

Question Set

1. Using the straight-line method of depreciation for reporting purposes and accelerated depreciation for tax purposes would *most likely* result in a:
- A. deferred tax asset.
 - B. valuation allowance.
 - C. temporary difference.

Solution:

C is correct. Because the differences between tax and financial accounting will correct over time, the resulting deferred tax liability, for which the expense was charged to the income statement but the tax authority has not yet been paid, will be a temporary difference. A valuation allowance would only arise if there was doubt over the company's ability to earn sufficient income in the future to require paying the tax.

2. Income tax expense reported on a company's income statement equals taxes payable, plus the net increase in:
- A. deferred tax assets and deferred tax liabilities.
 - B. deferred tax assets, less the net increase in deferred tax liabilities.
 - C. deferred tax liabilities, less the net increase in deferred tax assets.

Solution:

C is correct. Higher reported tax expense relative to taxes paid will increase the deferred tax liability, whereas lower reported tax expense relative to taxes paid increases the deferred tax asset.

3. Analysts should treat deferred tax liabilities that are expected to reverse as:
- A. equity.
 - B. liabilities.
 - C. neither liabilities nor equity.

Solution:

B is correct. If the liability is expected to reverse (and thus require a cash tax payment) the deferred tax represents a future liability.

4. When accounting standards require an asset to be expensed immediately but tax rules require the item to be capitalized and amortized, the company will *most likely* record:
- A. a deferred tax asset.
 - B. a deferred tax liability.
 - C. no deferred tax asset or liability.

Solution:

A is correct. The capitalization will result in an asset with a positive tax base and zero carrying value. The amortization means the difference is temporary. Because there is a temporary difference on an asset resulting in a higher tax base than carrying value, a deferred tax asset is created.

5. A company incurs a capital expenditure that may be amortized over five years for accounting purposes, but over four years for tax purposes. The company will *most likely* record:
- A. a deferred tax asset.
 - B. a deferred tax liability.
 - C. no deferred tax asset or liability.

Solution:

B is correct. The difference is temporary, and the tax base will be lower (because of more rapid amortization) than the carrying value of the asset. The result will be a deferred tax liability.

6. A company receives advance payments from customers that are immediately taxable but will not be recognized for accounting purposes until the company fulfills its obligation. The company will *most likely* record:
- A. a deferred tax asset.
 - B. a deferred tax liability.
 - C. no deferred tax asset or liability.

Solution:

A is correct. The advances represent a liability for the company. The carrying value of the liability exceeds the tax base (which is now zero). A deferred tax asset arises when the carrying value of a liability exceeds its tax base.

The information in [Exhibit 10](#) pertains to questions 7–9.

The tax effects of temporary differences that give rise to deferred tax assets and liabilities are as follows (US dollar thousands):

Exhibit 10: Tax Assets and Liabilities

	Year 3	Year 2
Deferred tax assets:		
Accrued expenses	USD8,613	USD7,927
Tax credit and net operating loss carryforwards	2,288	2,554
LIFO and inventory reserves	5,286	4,327
Other	2,664	2,109
Deferred tax assets	18,851	16,917
Valuation allowance	(1,245)	(1,360)
Net deferred tax assets	USD17,606	USD15,557
Deferred tax liabilities:		
Depreciation and amortization	(USD27,338)	(USD29,313)
Compensation and retirement plans	(3,831)	(8,963)
Other	(1,470)	(764)
Deferred tax liabilities	(32,639)	(39,040)
Net deferred tax liability	(USD15,033)	(USD23,483)

7. A reduction in the statutory tax rate would *most likely* benefit the company's:

- A. income statement and balance sheet.
- B. income statement but not the balance sheet.
- C. balance sheet but not the income statement.

Solution:

A is correct. A lower tax rate would increase net income on the income statement, and because the company has a net deferred tax liability, the net liability position on the balance sheet would also improve (be smaller).

8. If the valuation allowance had been the same in Year 3 as it was in Year 2, the company would have reported USD115 *higher*:

- A. net income.
- B. deferred tax assets.
- C. income tax expense.

Solution:

C is correct. The reduction in the valuation allowance resulted in a corresponding reduction in the income tax provision.

9. Relative to the provision for income taxes in Year 3, the company's cash tax payments were:

- A. lower.
- B. higher.
- C. the same.

Solution:

B is correct. The net deferred tax liability was smaller in Year 3 than it was in Year 2, indicating that in addition to meeting the tax payments provided for in Year 3 the company also paid taxes that had been deferred in prior periods.

4. CORPORATE INCOME TAX RATES

LEARNING OUTCOME

The candidate should be able to:

- ☐ **calculate, interpret, and contrast an issuer's effective tax rate, statutory tax rate, and cash tax rate**

Income taxes payable are primarily determined by the geographic composition of taxable income and the tax rates in each jurisdiction but can also be influenced by the nature of a business. Some companies benefit from special tax treatment—for example, from R&D tax credits or accelerated depreciation of fixed assets. Analysts should also be aware of any governmental or business changes that can alter tax rates.

Differences in tax rates can be an important driver of value. Generally, three types of tax rates are relevant to analysts:

- The **statutory tax rate**, which is the corporate income tax rate in the country in which the company is domiciled.
- The **effective tax rate**, which is calculated as the reported income tax expense amount on the income statement divided by the pre-tax income.
- The **cash tax rate**, which is the tax paid in cash that period (cash tax) divided by pre-tax income.

As discussed previously, differences between cash taxes and reported taxes typically result from differences between financial accounting standards and tax laws and result from changes in deferred tax assets or deferred tax liabilities.

In forecasting tax expense and cash taxes, respectively, the effective tax rate and cash tax rate are key. A good understanding of their operational drivers and the financial structure of a company is useful in forecasting these tax rates.

Differences between the statutory tax rate and the effective tax rate can arise for many reasons. Tax credits, withholding tax on dividends, adjustments to previous years, and expenses not deductible for tax purposes are among the reasons for differences. Effective tax rates can also differ when companies are active outside the country in which they are domiciled. The effective tax rate becomes a blend of the different tax rates of the countries in which the activities take place in relation to the profit generated in each country. If a company reports a high profit in a country with a high tax rate and a low profit in a country with a low tax rate, the effective tax rate will be the weighted average of the rates and higher than the simple average tax rate of both countries.

In general, an effective tax rate that is consistently lower than statutory rates or the effective tax rates reported by competitors is not necessarily unusual but might warrant additional attention when forecasting future tax expenses. The notes in the financial statements should disclose a reconciliation of the statutory tax rate to the effective rate and identify the items that significantly contribute to a temporarily high or low effective tax rate. The cash tax rate is used for forecasting cash flows, and the effective tax rate is relevant for projecting earnings on the income statement.

In developing an estimated tax rate for forecasts, analysts should adjust for one-time events. If the income from equity-method investees is a substantial part of pre-tax income and, also a volatile component of it, the effective tax rate excluding this amount is likely to be a better estimate for the future tax costs for a company. The tax impact from income from participations is disclosed in the notes on the financial statements.

Often, a good starting point for estimating future tax expense is a tax rate based on normalized operating income, before the results from associates and special items. This normalized tax rate should be a good indication of the future tax expense, adjusted for special items, in an analyst's earnings model.

Building a model allows the effective tax amount to be found in the profit and loss projections and the cash tax amount on the cash flow statement (or given as supplemental information). The reconciliation between the profit and loss tax amount and the cash flow tax figures should be the change in the deferred tax asset or liability.

EXAMPLE 4

Tax Rate Estimates

ABC, a hypothetical company, operates in Countries A and B. The tax rate in Country A is 40 percent, and the tax rate in Country B is 10 percent. In the first year, the company generates an equal amount of profit before tax in each country, as shown in [Exhibit 11](#).

Exhibit 11: Tax Rates That Differ by Jurisdiction

	A	B	Total
Profit before tax	100	100	200
Effective tax rate	40%	10%	25%
Tax	40	10	50
Net profit	60	90	150

1. What will happen to the effective tax rate for the next three years if the profit before tax in Country A is stable but the profit before tax in Country B grows 15 percent annually?

Solution:

The effective tax rate will gradually decline because a higher proportion of profit will be generated in the country with the lower tax rate each year. In [Exhibit 12](#), the effective tax rate declines from 25 percent in the beginning to 22 percent in the third year.

Exhibit 12: Worksheet for Tax Rate Estimates Problem

	Year			
	0	1	2	3
Profit before tax, Country A	100	100	100	100
Growth rate		0%	0%	0%
Profit before tax, Country B	100	115	132	152
Growth rate		15%	15%	15%
Total profit before tax	200	215	232	252
Effective tax rate, Country A	40%	40%	40%	40%
Effective tax rate, Country B	10%	10%	10%	10%
Total tax	50	52	53	55
Total effective tax rate	25%	24%	23%	22%

2. Evaluate the cash tax and effective tax rates for the next three years if the tax authorities in Country A allow some costs (e.g., accelerated depreciation) to be taken sooner for tax purposes. Specifically, assume for Country A, the result is a 50 percent reduction in taxes paid in the current year (Year 0) but an increase in taxes paid by the same amount in the following year (Year 1) and in subsequent years. Assume stable profit before tax in Country A and 15 percent annual before-tax-profit growth in Country B.

Solution:

The combined cash tax rate (last line in [Exhibit 13](#)) will be 15 percent in the first year and then rebound in subsequent years. Only the rate for the first year will benefit from a tax deferral; in subsequent years, the deferral for a given year will be offset by the addition of the amount postponed from the previous year. The combined effective tax rate will be unaffected by the deferral. As shown in [Exhibit 13](#), beginning with the second year (Year 1), the combined cash tax and effective tax rates decline over time because the growth in taxable income occurs in Country B, which has the lower tax rate.

Exhibit 13: Worksheet for Tax Rate Estimates Problem

	Year			
	0	1	2	3
Profit before tax, Country A	100	100	100	100
Growth rate		0%	0%	0%
Profit before tax, Country B	100	115	132	152
Growth rate		15%	15%	15%
Total profit before tax	200	215	232	252
Effective tax rate, Country A	40%	40%	40%	40%
Effective tax rate, Country B	10%	10%	10%	10%
Total tax per income statement	50	52	53	55
Total effective tax rate	25%	24%	23%	22%
Cash taxes, Country A	20	40	40	40
Cash taxes, Country B	10	12	13	15
Total cash tax	30	52	53	55
Cash tax rate	15%	24%	23%	22%

3. Repeat the exercise of the Tax Rate Estimates Problem, but now assume that Country B, rather than Country A, allows some costs to be taken sooner for tax purposes and that the tax effect described applies to Country B. Continue to assume stable profit before tax in Country A and 15 percent annual profit growth in Country B.

Solution:

The combined effective tax rate is unchanged from [Exhibit 12](#) and [Exhibit 13](#). Because of the growth assumed for Country B, however, the annual tax postponement will result in a lower cash tax rate in Country B than the effective tax rate in Country B. Consequently, as shown in [Exhibit 14](#), the combined cash tax rate will be less than the effective tax rate in Year 0.

Exhibit 14: Worksheet for Tax Rate Estimates Problem

	Year			
	0	1	2	3
Profit before tax, Country A	100	100	100	100
Growth rate		0%	0%	0%
Profit before tax, Country B	100	115	132	152
Growth rate		15%	15%	15%
Total profit before tax	200	215	232	252
Effective tax rate, Country A	40%	40%	40%	40%
Effective tax rate, Country B	10%	10%	10%	10%
Total tax per income statement	50	52	53	55
Total effective tax rate	25%	24%	23%	22%
Cash taxes, Country A	40	40	40	40
Cash taxes, Country B	5	11	12	14

	Year			
	0	1	2	3
Total cash tax	45	51	52	54
Cash tax rate	23%	24%	23%	22%

EXAMPLE 5

Johnson & Johnson

The difference between the effective tax rate and the cash tax rate for Johnson & Johnson (JNJ) is shown in [Exhibit 15](#).

Exhibit 15: Johnson & Johnson Consolidated Statement of Earnings (US dollar millions)

	2021	2020	2019
Sales to customers	93,775	82,584	82,059
Cost of products sold	29,855	28,427	27,556
Gross profit	63,920	54,157	54,503
Selling, marketing, and administrative expense	24,659	22,084	22,178
Research and development expense	14,714	12,159	11,355
In-process research and development	900	181	890
Interest income	(53)	(111)	(357)
Interest expense	183	201	318
Other (income) expense, net	489	2,899	2,525
Restructuring	252	247	266
Income before income taxes	22,776	16,497	17,328
Provision for income taxes	1,898	1,783	2,209
Net income	20,878	14,714	15,119

JNJ's income tax expense for 2021 was USD1,898 million. Accordingly, JNJ's effective tax rate was 8.3 percent (USD1,898/USD22,776). This is substantially lower than JNJ's statutory tax rate of 21 percent as a US corporation. JNJ's reconciliation of its statutory tax rate to its effective tax rate reported in its notes to financial statements shows that the primary driver of this differences is a lower tax rate on its business outside the United States.

Tax rates:	2021	2020	2019
US statutory rate	21.0%	21.0%	21.0%
International operations	(16.4)	(9.9)	(5.9)
US taxes on international income	6.7	2.7	1.8
Tax benefits from loss on capital assets	(1.3)	(1.2)	(0.3)
Tax benefits on share-based compensation	(1.0)	(1.5)	(0.5)
Tax Cuts And Jobs Act related impacts	(0.5)	0.7	(3.9)
All Other	(0.2)	(1.0)	0.5
Effective tax rate	8.3%	10.8%	12.7%

Question Set

1. Which of the following *best* describes a statutory tax rate?

- A. Tax paid in cash that period divided by pre-tax income
- B. Corporate income tax rate in the country in which the company is domiciled
- C. Reported income tax expense amount on the income statement divided by the pre-tax income

Solution:

B is correct. The statutory tax rate is the corporate income tax rate in the country in which the company is domiciled. A is incorrect because it describes the cash tax rate. C is incorrect because it describes the effective tax rate.

Please use the following information and [Exhibit 16](#) and [Exhibit 17](#) from Walmart's 2021 Form 10-K to answer questions 2 and 3:

Walmart reported total revenues of USD572,762 million, income before taxes of USD18,696 million and consolidated net income of USD13,940 million. In addition, it disclosed the following information in its tax footnote:

Exhibit 16: Walmart's 2021 Form 10-K

Note 9: Taxes

The components of income before income taxes are as follows:

(in millions USD)	Fiscal Years Ended 31 January		
	2022	2021	2020
US	15,536	18,068	15,019
Non-US	3,160	2,496	5,097
Total income before income taxes	18,696	20,564	20,116

A summary of the provision for income taxes is as follows:

Exhibit 17: Walmart's 2021 Form 10-K

(in millions USD)	Fiscal Years Ended 31 January		
	2022	2021	2020
Current:			
US Federal	3,313	2,991	2,794
US State and Local	649	742	587
International	1,553	1,127	1,205
Total current tax provision	5,515	4,860	4,586

(in millions USD)	Fiscal Years Ended 31 January		
	2022	2021	2020
<i>Deferred:</i>			
US Federal	(671)	2,316	663
US State and Local	41	23	35
International	(129)	(341)	(369)
Total deferred tax expense (benefit)	(759)	1,998	329
Total provision for income taxes	4,756	6,858	4,915

2. Based on the information in [Exhibit 16](#) and [Exhibit 17](#), Walmart's effective tax rate is *closest* to:

- A. 25.4 percent.
- B. 29.5 percent.
- C. 34.1 percent.

Solution:

A is correct. The effective tax rate is calculated as the total provision for income taxes (USD4,756 million) divided by Total income before taxes (USD18,696 million).

3. Based on the information in [Exhibit 16](#) and [Exhibit 17](#), Walmart's cash tax rate is *closest* to:

- A. 25.4 percent.
- B. 29.5 percent.
- C. 34.1 percent.

Solution:

B is correct. The cash tax rate is calculated as the current tax expense (USD5,515 million) divided by Total income before taxes (USD18,696 million).

Please use the following information to answer questions 4 and 5 below:

Neutrino is a hypothetical company that is domiciled in the United States and that has significant operations in Ireland. The Statutory Tax rate in the United States is 21 percent, and the statutory tax rate in Ireland is 12 percent. Assume that Neutrino earns USD1,000 in profit before taxes in each country during year 20X1.

4. Assuming that there are no other differences between Neutrino's effective and statutory tax rates, Neutrino's combined effective tax rate is *closest* to:

- A. 12.0 percent.
- B. 16.5 percent.
- C. 21.0 percent.

Solution:

B is correct. Taxes are calculated as: $(\text{USD}1,000 \times 21\%) + (\text{USD}1,000 \times 12\%) = \text{USD}330$; Effective tax rate = $\text{USD}330 / \text{USD}2,000 = 16.5\%$.

5. Assume on January 1 the following year, 20X2, Neutrino acquires company EFG, which is domiciled in South Korea. The statutory tax rate in South Korea is 25 percent. EFG

earns USD500 in profits in 20X2. Assuming US and Ireland operations each increase pre-tax profits by 25 percent, the effective tax rate in 20X2 for the consolidated entity is *closest* to:

- A. 6.5 percent.
- B. 17.9 percent.
- C. 22.0 percent.

Solution:

B is correct. The effective tax rate in 20X2 for the consolidated entity is calculated as follows:

Country	Taxable Income	Statutory Rate	Taxes
United States	USD1,250	21%	USD262.50
Ireland	1,250	12	150.00
South Korea	500	25	125.00
Total	USD3,000		537.50
Effective tax rate:		537.50/3,000	17.9%

5. PRESENTATION AND DISCLOSURE

LEARNING OUTCOME

The candidate should be able to:

- ☐ **analyze disclosures relating to deferred tax items and the effective tax rate reconciliation and explain how information included in these disclosures affects a company's financial statements and financial ratios**

The Consolidated Statements of Operations (Income Statements) and Consolidated Balance Sheets for Micron Technology (MU), a global technology company based in the US, are provided in [Exhibit 18](#) and [Exhibit 19](#), respectively. [Exhibit 18](#) provides the income tax note disclosures for MU for the 2015, 2016, and 2017 fiscal years.

MU's income tax provision (i.e., income tax expense) for fiscal year 2017 is USD114 million (see [Exhibit 18](#)). The income tax note disclosure in [Exhibit 20](#) reconciles how the income tax provision was determined beginning with MU's reported income before taxes (shown in [Exhibit 20](#) as USD5,196 million for fiscal year 2017). The note disclosure then denotes the income tax provision for 2017 that is current (USD153 million), which is then offset by the deferred tax benefit for foreign taxes (USD39 million), for a net income tax provision of USD114 million. [Exhibit 20](#) further shows a reconciliation of how the income tax provision was derived from the US federal statutory rate. Many public companies comply with this required disclosure by displaying the information in percentage terms, but MU has elected to provide the disclosure in absolute dollar amounts. From this knowledge, for 2017, we can see that the dollar amount shown for US federal income tax provision at the statutory rate (USD1,819 million) was determined by multiplying MU's income before taxes by the 35 percent US federal statutory rate ($\text{USD5,196} \times 0.35 = \text{USD1,819}$).

In addition, the note disclosure in [Exhibit 20](#) provides detailed information about the derivation of the deferred tax assets (USD766 million for 2017) and deferred tax liabilities (USD17 million for 2017). These

deferred tax assets are shown separately on MU's consolidated balance sheet for fiscal year 2017 with noncurrent assets (see [Exhibit 19](#)), while the deferred tax liabilities are included in other noncurrent liabilities (also see [Exhibit 19](#)).

Exhibit 18: Micron Technology, Inc. Consolidated Statements of Operations (US dollar millions, except per share)

For the Year Ended	31 Aug. 2017	1 Sept. 2016	3 Sept. 2015
Net sales	20,322	USD12,399	USD16,192
Cost of goods sold	11,886	9,894	10,977
Gross margin	8,436	2,505	5,215
Selling, general and administrative	743	659	719
Research and development	1,824	1,617	1,540
Restructure and asset impairments	18	67	3
Other operating (income) expense, net	(17)	(6)	(45)
Operating income	5,868	168	2,998
Interest income (expense), net	(560)	(395)	(336)
Other non-operating income (expense), net	(112)	(54)	(53)
Income tax (provision) benefit	(114)	(19)	(157)
Equity in net income (loss) of equity method investees	8	25	447
Net income (loss) attributable to noncontrolling interests	(1)	(1)	—
Net income (loss) attributable to Micron	USD5,089	USD(276)	USD2,899
Earnings (loss) per share:			
Basic	USD4.67	USD(0.27)	USD2.71
Diluted	USD4.41	USD(0.27)	USD2.47
Number of shares used in per share calculations:			
Basic	1,089	1,036	1,070
Diluted	1,154	1,036	1,170

Exhibit 19: Micron Technology, Inc. Consolidated Balance Sheets (US dollar millions)

As of	31 Aug. 2017	1 Sept. 2016
Assets		
Cash and equivalents	USD5,109	USD4,140
Short-term investments	319	258
Receivables	3,759	2,068
Inventories	3,123	2,889
Other current assets	147	140
Total current assets	12,457	9,495
Long-term marketable investments	617	414
Property, plant and equipment, net	19,431	14,686
Equity method investments	16	1,364
Intangible assets, net	387	464

As of	31 Aug. 2017	1 Sept. 2016
Deferred tax assets	766	657
Other noncurrent assets	1,662	460
Total assets	USD35,336	USD27,540
Liabilities and shareholders' equity		
Accounts payable and accrued expenses	USD3,664	USD3,879
Deferred income	408	200
Current debt	1,262	756
Total current liabilities	5,334	4,835
Long-term debt	9,872	9,154
Other noncurrent liabilities	639	623
Total liabilities	15,845	14,612
Redeemable convertible notes	21	—
Micron shareholder's equity		
Common stock of USD0.10 par value, 3,000 shares authorized, 1,116 shares issued and 1,112 shares outstanding (1,094 issued and 1,040 outstanding as of September 1, 2016)	112	109
Additional capital	8,287	7,736
Retained earnings	10,260	5,299
Treasury stock, 4 shares held (54 as of 1 September 2016)	(67)	(1,029)
Accumulated other comprehensive income (loss)	29	(35)
Total Micron shareholders' equity	18,621	12,080
Noncontrolling interests in subsidiaries	849	848
Total equity	19,470	12,928
Total liabilities and shareholders' equity	35,336	USD27,540

Exhibit 20: Micron Technology, Inc. Income Taxes Note to the Consolidated Financial Statements

Income (loss) before taxes and the income tax (provision) benefit consisted of the following:

(USD millions)	2017	2016	2015
Income (loss) before income taxes, net income (loss) attributable to noncontrolling interests, and equity in net income (loss) of equity method investees			
Foreign	USD5,252	(USD353)	USD2,431
US	(56)	72	178
	USD5,196	(USD281)	USD2,609
Income tax (provision) benefit:			
Current:			
Foreign	(USD152)	(USD27)	(USD93)
State	(1)	(1)	(1)
US federal	—	—	6
	(153)	(28)	(88)
Deferred:			
US federal	—	39	15

(USD millions)	2017	2016	2015
State	—	2	1
Foreign	39	(32)	(85)
	39	9	(69)
Income tax (provision)	(USD114)	(USD19)	(USD157)

The company's income tax (provision) computed using the US federal statutory rate and the company's income tax (provision) benefit is reconciled as shown in [Exhibit 21](#):

Exhibit 21: Company Income Tax (US dollar millions)

	2017	2016	2015
US federal income tax (provision) benefit at statutory rate	(USD1,819)	USD98	(USD913)
Foreign tax rate differential	1,571	(300)	515
Change in valuation allowance	64	63	260
Change in unrecognized tax benefits	12	52	(118)
Tax credits	66	48	53
Noncontrolling investment transactions	—	—	57
Other	(8)	20	(11)
Income tax (provision) benefit	(114)	(USD19)	(USD157)

State taxes reflect investment tax credits of USD233 million at 31 August 2017. Deferred income taxes reflect the net tax effects of temporary differences between the bases of assets and liabilities for financial reporting and income tax purposes. The company's deferred tax assets and liabilities consist of the following as of the end of the periods shown in [Exhibit 22](#):

Exhibit 22: Deferred Tax Assets and Liabilities (US dollar millions)

	2017	2016
Deferred tax assets:		
Net operating loss and tax credit carryforwards	USD3,426	USD3,014
Accrued salaries, wages, and benefits	211	142
Other accrued liabilities	59	76
Other	86	65
Gross deferred assets	3,782	3,297
Less valuation allowance	(2,321)	(2,107)
Deferred tax assets, net of valuation allowance	1,461	1,190
Deferred tax liabilities:		
Debt discount	(145)	(170)
Property, plant, and equipment	(300)	(135)
Unremitted earnings on certain subsidiaries	(123)	(121)
Product and process technology	(85)	(81)
Other	(59)	(28)
Deferred tax liabilities	(712)	(535)
Net deferred tax assets	USD749	USD655

	2017	2016
Reported as:		
Current deferred tax assets (included in other current assets)	USD—	USD—
Deferred tax assets	766	657
Current deferred tax liabilities (included in accounts payable and accrued expenses)	—	—
Deferred tax liabilities (included in other noncurrent liabilities)	(17)	(2)
Net deferred tax assets	USD749	USD655

The company has a valuation allowance against substantially all of its US net deferred tax assets. As of 31 August 2017, the company had aggregate US tax net operating loss carryforwards of USD3.88 billion and unused US tax credit carryforwards of USD416 million. The company also has unused state tax net operating loss carryforwards of USD1.95 billion and unused state tax credits of USD233 million. The net operating loss carryforwards and the tax credit carryforwards expire between 2018 to 2037.

The changes in valuation allowance of USD64 million and USD63 million in 2017 and 2016, respectively, are primarily a result of uncertainties of realizing certain US and foreign net operating losses and certain tax credit carryforwards.

Provision has been made for deferred taxes on undistributed earnings of non-US subsidiaries to the extent that dividend payments from such companies are expected to result in additional tax liability. Remaining undistributed earnings of USD12.91 billion as of 31 August 2017 have been indefinitely reinvested. Determination of the amount of unrecognized deferred tax liability on these unremitted earnings is not practicable.

EXAMPLE 6

Financial Analysis

Use the financial statement information and disclosures provided by MU in [Exhibit 18–Exhibit 20](#) to answer the following questions:

1. MU discloses a valuation allowance of USD2,321 million (see [Exhibit 20](#)) against gross deferred assets of USD3,782 million in 2017. Does the existence of this valuation allowance have any implications concerning MU's future earnings prospects?

Solution:

According to [Exhibit 20](#), MU's deferred tax assets expire gradually until 2037 (2018 to 2037 for the net operating loss carryforwards and the tax credit carryforwards).

Because the company is still relatively young, it is likely that most of these expirations occur toward the end of that period. Because cumulative US tax net operating loss carryforwards total USD3.88 billion, the valuation allowance could imply that MU is not reasonably expected to earn USD3.88 billion over the next 20 years. However, as we can see in [Exhibit 18](#), MU earned a profit for 2017 and 2015, thereby showing that the allowance could be adjusted downward if the company continues to generate profits in the future and making it more likely than not that the deferred tax asset would be recognized.

2. How would MU's deferred tax assets and deferred tax liabilities be affected if the federal statutory tax rate was changed to 21 percent?

Solution:

MU's total deferred tax assets exceed total deferred tax liabilities by USD749 million. A change in the federal statutory tax rate to 21 percent from the current rate of 35 percent would make these net deferred assets less valuable. Also, because it is possible that the deferred tax asset

valuation allowance could be adjusted downward in the future (see discussion to solution 1), the impact could be far greater in magnitude.

3. How would reported earnings have been affected if MU were not using a valuation allowance?

Solution:

The disclosure in [Exhibit 20](#) shows that the increase in the valuation allowance increased the income tax provision as reported on the income statement by USD64 million in 2017.

Additional potential reductions in the valuation allowance could similarly reduce reported income taxes (actual income taxes would not be affected by a valuation allowance established for financial reporting) in future years (see discussion to solution 1).

4. How would MU's USD3.88 billion in net operating loss carryforwards in 2017 (see [Exhibit 20](#)) affect the valuation that an acquiring company would be willing to offer?

Solution:

If an acquiring company is profitable, it may be able to use MU's tax loss carryforwards to offset its own tax liabilities. The value to an acquirer would be the present value of the carryforwards, based on the acquirer's tax rate and expected timing of realization. The higher the acquiring company's tax rate, and the more profitable the acquirer, the sooner it would be able to benefit. Therefore, an acquirer with a high current tax rate would theoretically be willing to pay more than an acquirer with a lower tax rate.

5. Under what circumstances should the analyst consider MU's deferred tax liability as debt or as equity? Under what circumstances should the analyst exclude MU's deferred tax liability from both debt and equity when calculating the debt-to-equity ratio?

Solution:

The analyst should classify the deferred tax liability as debt if the liability is expected to reverse with subsequent tax payment. If the liability is not expected to reverse, there is no expectation of a cash outflow and the liability should be treated as equity. By way of example, future company losses may preclude the payment of any income taxes, or changes in tax laws could result in taxes that are never paid. The deferred tax liability should be excluded from both debt and equity when both the amounts and timing of tax payments resulting from the reversals of temporary differences are uncertain.

Question Set

1. Deferred tax liabilities should be treated as equity when:

- A. they are not expected to reverse.
- B. the timing of tax payments is uncertain.
- C. the amount of tax payments is uncertain.

Solution:

A is correct. If the liability will not reverse, there will be no required tax payment in the future and the "liability" should be treated as equity.

2. When both the timing and amount of tax payments are uncertain, analysts should treat deferred tax liabilities as:

- A. equity.

B. liabilities.

C. neither liabilities nor equity.

Solution:

C is correct. The deferred tax liability should be excluded from both debt and equity when both the amounts and timing of tax payments resulting from the reversals of temporary differences are uncertain.

Note I: Income Taxes

The components of earnings before income taxes are shown in [Exhibit 23](#):

Exhibit 23: Earnings before Income Taxes (US dollar thousands)

	Year 3	Year 2	Year 1
Earnings before income taxes:			
United States	USD88,157	USD75,658	USD59,973
Foreign	116,704	113,509	94,760
Total	USD204,861	USD189,167	USD154,733

The components of the provision for income taxes are shown in [Exhibit 24](#):

Exhibit 24: Provision for Income Taxes (US dollar thousands)

	Year 3	Year 2	Year 1
Income taxes			
Current:			
Federal	USD30,632	USD22,031	USD18,959
Foreign	28,140	27,961	22,263
	USD58,772	USD49,992	USD41,222
Deferred:			
Federal	(USD4,752)	USD5,138	USD2,336
Foreign	124	1,730	621
	(4,628)	6,868	2,957
Total	USD54,144	USD56,860	USD44,179

3. In Year 3, the company's US GAAP income statement recorded a provision for income taxes *closest* to:

A. USD30,632.

B. USD54,144.

C. USD58,772.

Solution:

B is correct. The income tax provision in Year 3 was USD54,144, consisting of USD58,772 in current income taxes, of which USD4,628 were deferred.

4. The company's effective tax rate was *highest* in:

- A. Year 1.
- B. Year 2.
- C. Year 3.

Solution:

B is correct. The effective tax rate of 30.1 percent (USD56,860/USD189,167) was higher than the effective rates in Year 1 and Year 3.

5. Relative to the company's effective tax rate on US income, the company's effective tax rate on foreign income was:

- A. lower in each year presented.
- B. higher in each year presented.
- C. higher in some periods and lower in others.

Solution:

A is correct. In Year 3 the effective tax rate on foreign operations was 24.2 percent $[(USD28,140 + USD124)/USD116,704]$, and the effective US tax rate was $[(USD30,632 - USD4,752)/USD88,157] = 29.4$ percent. In Year 2 the effective tax rate on foreign operations was 26.2 percent, and the US rate was 35.9 percent. In Year 1 the foreign rate was 24.1 percent, and the US rate was 35.5 percent.

A company's provision for income taxes resulted in effective tax rates attributable to loss from continuing operations before cumulative effect of change in accounting principles that varied from the statutory federal income tax rate of 34 percent, as summarized in [Exhibit 25](#).

Exhibit 25: Effective Tax Rates

Year Ended 30 June	Year 3	Year 2	Year 1
Expected federal income tax expense (benefit) from continuing operations at 34 percent	(USD112,000)	USD768,000	USD685,000
Expenses not deductible for income tax purposes	357,000	32,000	51,000
State income taxes, net of federal benefit	132,000	22,000	100,000
Change in valuation allowance for deferred tax assets	(150,000)	(766,000)	(754,000)
Income tax expense	USD227,000	USD56,000	USD82,000

6. In Year 3, the company's net income (loss) was *closest* to:

- A. (USD217,000).
- B. (USD329,000).
- C. (USD556,000).

Solution:

C is correct. The income tax provision at the statutory rate of 34 percent is a benefit of USD112,000, suggesting that the pre-tax income was a loss of $\text{USD}112,000 / 0.34 = (\text{USD}329,412)$. The income tax provision was USD227,000. $(\text{USD}329,412) - \text{USD}227,000 = (\text{USD}556,412)$.

7. The USD357,000 adjustment in Year 3 *most likely* resulted in:

- A. an increase in deferred tax assets.
- B. an increase in deferred tax liabilities.
- C. no change to deferred tax assets and liabilities.

Solution:

C is correct. Accounting expenses that are not deductible for tax purposes result in a permanent difference, and thus do not give rise to deferred taxes.

8. Over the three years presented, changes in the valuation allowance for deferred tax assets were *most likely* indicative of:

- A. decreased prospect for future profitability.
- B. increased prospects for future profitability.
- C. assets being carried at a higher value than their tax base.

Solution:

B is correct. Over the three-year period, changes in the valuation allowance reduced cumulative income taxes by USD1,670,000. The reductions to the valuation allowance were a result of the company being “more likely than not” to earn sufficient taxable income to offset the deferred tax assets.

PRACTICE PROBLEMS

1. In the current year, Michaels Company has a carrying amount of USD3,500,000 and tax base of USD5,000,000 for accounts receivable. Michaels will most likely recognize:
 - A. a deferred tax asset.
 - B. a deferred tax liability.
 - C. no deferred tax asset or liability.
2. James Company has received USD500,000 of tax credits from the recent installation of solar panels that will directly reduce their taxes. Which of the following *best* describes these tax credits?
 - A. Permanent difference
 - B. Taxable temporary difference
 - C. Deductible temporary difference
3. Please use the selected data in [Exhibit 1](#) for the Samuels Corporation.

Exhibit 1: Selected Data for Samuels Company (US dollar millions)

	Year 3	Year 2	Year 1
Equipment value for accounting purposes (<i>carrying amount</i>) (depreciation of USD1,000/year)	USD7,000	USD8,000	USD9,000
Equipment value for tax purposes (<i>tax base</i>) (depreciation of USD1,429/year)	USD5,714	USD7,143	USD8,571

Assuming a 35 percent tax rate and the selected data below for the Samuels Company, the company's deferred tax liability in Year 3 is *closest* to:

- A. USD450.
 - B. USD750.
 - C. USD900.
4. Which of the following is added to income tax payable to determine the company's income tax expense as reported on the income statement?
 - A. Deferred tax assets
 - B. Deferred tax liabilities
 - C. Changes in deferred tax assets and liabilities
 5. Jamison Corp. is domiciled in the United States and has significant operations in the United Kingdom and Australia. The statutory tax rates are 21 percent in the United States, 19 percent in the United Kingdom, and 30 percent in Australia. The company generates Profit before tax of USD2,000,000 in the United States, USD500,000 in the United Kingdom, and USD750,000 in

Australia. There are no other differences between Jamison's effective and statutory tax rates. Jamison's combined effective tax rate is closest to:

- A. 21.0 percent.
- B. 22.8 percent.
- C. 23.3 percent.

6. Which of the following statements about tax rates is correct?

- A. The effective tax rate is typically used for forecasting cash flows.
- B. The cash tax rate is relevant for projecting earnings on the income statement.
- C. A company's income tax expense equals the sum of current taxes plus the change in deferred tax assets and liabilities.

The following information relates to questions 7-8

Please use the selected disclosure data in [Exhibit 1](#) and [Exhibit 2](#) for the Marcy Corporation.

Note I: Income Taxes

The components of earnings before income taxes are as shown in [Exhibit 1](#):

Exhibit 1: Earnings before Income Taxes (US dollars thousands)

	Year 3	Year 2	Year 1
Earnings before income taxes:			
United States	USD117,758	USD107,053	USD97,321
Foreign	57,526	52,296	47,542
Total	USD175,284	USD159,349	USD144,863

The components of the provision for income taxes are as shown in [Exhibit 2](#):

Exhibit 2: Provision for Income Taxes (US dollars thousands)

	Year 3	Year 2	Year 1
Income taxes			
Current:			
Federal	USD34,739	USD34,257	USD31,143
Foreign	14,382	13,074	17,591
	USD49,121	USD47,331	USD48,734
Deferred:			
Federal	(USD6,524)	(USD6,002)	(USD5,325)
Foreign	389	305	262
	(6,135)	(5,697)	(5,063)
Total	USD42,986	USD41,634	USD43,671

7. Marcy's effective tax rate was lowest in:

A. Year 1.

B. Year 2.

C. Year 3.

8. Relative to Marcy's effective tax rate on foreign income, the company's effective tax rate on US income was:

A. lower in each year presented.

B. higher in each year presented.

C. higher in some periods and lower in others.

SOLUTIONS

1. A is correct. Because the carrying amount is less than the tax base for this asset, this difference is a temporary difference that will result in a deferred tax asset. B is incorrect because a deferred tax liability would apply if the carrying amount was greater than the asset base. C is incorrect because this is not a permanent difference thus there will be either a deferred tax asset or deferred tax liability.
2. A is correct. Permanent differences are differences between tax laws and accounting standards that will not be reversed at some future date. Because they will not be reversed at a future date, these differences do not give rise to deferred tax. These items include tax credits for expenditures that directly reduce taxes, such as tax credits related to the purchase of solar power. B is incorrect because taxable temporary differences result in the recognition of deferred tax liabilities. C is incorrect because deductible temporary differences result in a deferred tax asset.
3. A is correct. USD450 is calculated as: $(\text{USD}7,000 - \text{USD}5,714) \times 0.35 = \text{USD}450$. B is incorrect because it incorrectly sums the deferred tax liabilities from Years 2 and 3: $(\text{USD}7,000 - \text{USD}5,714) \times 0.35 + (\text{USD}8,000 - \text{USD}7,143) \times 0.35 = \text{USD}750$. C is incorrect because it incorrectly sums the deferred tax liabilities from Years 1, 2 and 3: $(\text{USD}7,000 - \text{USD}5,714) \times 0.35 + (\text{USD}8,000 - \text{USD}7,143) \times 0.35 + (\text{USD}9,000 - \text{USD}8,571) \times 0.35 = \text{USD}900$.
4. C is correct. The changes in deferred tax assets and liabilities are added to income tax payable to determine the company's income tax expense (or credit) as it is reported on the income statement. A and B are incorrect because it is the changes in deferred tax assets and liabilities that are added to income tax payable.
5. B is correct. The combined effective tax rate is calculated as:

Country	Taxable Income	Statutory Rate	Taxes
U.S.	USD2,000,000	21%	USD420,000
U.K.	500,000	19	95,000
Australia	750,000	30	225,000
Total	USD3,250,000		740,000
Effective tax rate:		$740,000/3,250,000$	22.8%

The effective tax rate is a blend of the different tax rates of the countries in which the activities take place in relation to the profit generated in each country.

A is incorrect because 21.0 percent is the statutory tax rate in the US and does not incorporate statutory tax rates in the United Kingdom and Australia. C is incorrect because 23.3 percent is the simple average of all three statutory tax rates.

6. C is correct. A company's income tax expense equals the sum of current taxes (i.e., the amount currently payable) plus the change in deferred tax assets and liabilities. A is incorrect because the cash tax rate is typically used for forecasting cash flows. B is incorrect because the effective tax rate is relevant for projecting earnings on the income statement.
7. C is correct. The effective tax rate of 24.5 percent ($\text{USD}42,986/\text{USD}175,284$) in Year 3 was lower than the effective tax rates in Year 1 and Year 2. A is incorrect because its effective tax rate of 30.1 percent is higher than that of Year 3. B is incorrect because its effective tax rate of 26.1 percent is higher than that of Year 3.
8. C is correct. In Year 1, the effective tax rate on foreign operations was 37.6 percent $[(\text{USD}17,591 + \text{USD}262)/\text{USD}47,542]$, and the effective US tax rate was $[(\text{USD}31,143 - \text{USD}5,325)/\text{USD}97,321] =$

26.5 percent. In Year 2, the effective tax rate on foreign operations was 25.6 percent, and the US rate was 26.4 percent. In Year 3, the foreign rate was 25.7 percent, and the US rate was 24.0 percent.

NOTES

¹As defined under International Accounting Standard 12 (IAS 12), paragraph 5.

²The terms “tax base” and “tax basis” are interchangeable. “Tax basis” is more commonly used in the United States. Similarly, “carrying amount” and “book value” refer to the same concept.

Financial Reporting Quality

by Jack T. Ciesielski CPA CFA, Elaine Henry PhD CFA and Thomas I. Selling PhD CPA

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LEARNING OUTCOMES

The candidate should be able to:

- ☐ compare financial reporting quality with the quality of reported results (including quality of earnings, cash flow, and balance sheet items)
- ☐ describe a spectrum for assessing financial reporting quality
- ☐ explain the difference between conservative and aggressive accounting
- ☐ describe motivations that might cause management to issue financial reports that are not high quality and conditions that are conducive to issuing low-quality, or even fraudulent, financial reports
- ☐ describe mechanisms that discipline financial reporting quality and the potential limitations of those mechanisms
- ☐ describe presentation choices, including non-GAAP measures, that could be used to influence an analyst's opinion
- ☐ describe accounting methods (choices and estimates) that could be used to manage earnings, cash flow, and balance sheet items
- ☐ describe accounting warning signs and methods for detecting manipulation of information in financial reports

1. INTRODUCTION

Financial reporting quality varies across companies. The ability to assess the quality of a company's financial reporting is an important skill for analysts. Indications of low-quality financial reporting can prompt an analyst to maintain heightened skepticism when reading a company's reports, to review disclosures critically when undertaking financial statement analysis, and to incorporate appropriate adjustments in assessments of past performance and forecasts of future performance.

Learning Module Overview

- Financial reporting quality can be thought of as spanning a continuum. Reporting of the highest quality contains information that is relevant, correct, complete, and unbiased, whereas the lowest quality reporting contains information that is not just biased or incomplete but possibly pure fabrication.
- Reporting quality, the focus of this module, pertains to the quality of the information disclosed. High-quality reporting represents the economic reality of the company's activities during the reporting period and the company's financial condition at the end of the period.
- Results quality (commonly referred to as earnings quality) pertains to the earnings and cash generated by the company's actual economic activities and the resulting financial condition, relative to expectations of current and future financial performance. Quality earnings can be regarded as more sustainable, providing a sound platform for forecasts.
- An aspect of financial reporting quality is the degree to which accounting choices are conservative or aggressive. "Aggressive" typically refers to choices that aim to enhance the company's reported performance and financial position by inflating the amount of revenues, earnings, and/or operating cash flow reported in the period; or by decreasing expenses for the period and/or the amount of debt reported on the balance sheet.
- Conservatism in financial reports can result from either (1) accounting standards that specifically require a conservative treatment of a transaction or an event or (2) judgments made by managers when applying accounting standards that result in conservative results.
- Managers may be motivated to issue less-than-high-quality financial reports to mask poor performance, boost the company's stock price, to increase personal compensation, and/or to avoid violation of debt covenants.
- Conditions that are conducive to the issuance of low-quality financial reports include a cultural environment that result in fewer or less transparent financial disclosures, book/tax conformity that shifts emphasis toward legal compliance and away from fair presentation, and limited capital markets regulation.
- Mechanisms that discipline financial reporting quality include open capital markets and incentives for companies to minimize cost of capital, independent auditors, contract provisions specifically tailored to penalize misreporting, and enforcement by regulatory entities.
- Pro forma earnings (also commonly referred to as non-GAAP or non-IFRS earnings) adjust earnings as reported on the income statement. Pro forma earnings that exclude negative items are a hallmark of aggressive presentation choices.
- Companies are required to make additional disclosures when presenting any non-GAAP or non-IFRS metric.
- Managers' considerable flexibility in choosing their companies' accounting policies and in formulating estimates provides opportunities for aggressive accounting.
- Examples of accounting choices that affect earnings and balance sheets include inventory cost flow assumptions, estimates of uncollectible accounts receivable, estimated realizability of deferred tax assets, depreciation method, estimated salvage value of depreciable assets, and estimated useful life of depreciable assets.
- Cash flow from operations is an important metric for investors that can be enhanced by management's operating choices, such as stretching accounts payable, and potentially by classification choices.

2. CONCEPTUAL OVERVIEW

LEARNING OUTCOME

The candidate should be able to:

- ☐ **compare financial reporting quality with the quality of reported results (including quality of earnings, cash flow, and balance sheet items)**

Ideally, analysts would always have access to financial reports that are based on sound financial reporting standards, such as those from the International Accounting Standards Board (IASB) and the Financial Accounting Standards Board (FASB), and that are free from manipulation. But, in practice, the quality of financial reports can vary greatly. High-quality financial reporting provides information that is useful to analysts in assessing a company’s performance and prospects. Low-quality financial reporting contains inaccurate, misleading, or incomplete information.

Extreme lapses in financial reporting quality have given rise to high-profile scandals that resulted not only in investor losses but also in reduced confidence in the financial system. Financial statement users who were able to accurately assess financial reporting quality were better positioned to avoid losses. These lapses illustrate the challenges analysts face as well as the potential costs of failing to recognize practices that result in misleading or inaccurate financial reports.¹ Examples of misreporting can provide an analyst with insight into various signals that may indicate poor-quality financial reports.

This module addresses *financial reporting quality*, which pertains to the quality of information in financial reports, including disclosures in notes. High-quality reporting provides decision-useful information, which is relevant and faithfully represents the economic reality of the company’s activities during the reporting period as well as the company’s financial condition at the end of the period. A separate but interrelated attribute of quality is *quality of reported results* or *earnings quality*, which pertains to the earnings and cash generated by the company’s actual economic activities and the resulting financial condition. The term “earnings quality” is commonly used in practice and will be used broadly to encompass the quality of earnings, cash flow, or balance sheet items. High-quality earnings result from activities that a company likely will be able to sustain in the future and provide a sufficient return on the company’s investment. The concepts of earnings quality and financial reporting quality are interrelated because a correct assessment of earnings quality is possible only when there is some basic level of financial reporting quality. Beyond this basic level, as the quality of reporting increases, the ability of financial statement users to correctly assess earnings quality and to develop expectations for future performance also increases.

After providing a conceptual overview of reporting quality, this module discusses motivations that might cause, and conditions that might enable, management to issue financial reports that are not high quality and mechanisms that aim to provide discipline to financial reporting quality. We also describe choices made by management that can affect financial reporting quality—presentation choices, accounting methods, and estimates—as well as warning signs of poor-quality financial reporting.

2.1. Conceptual Overview

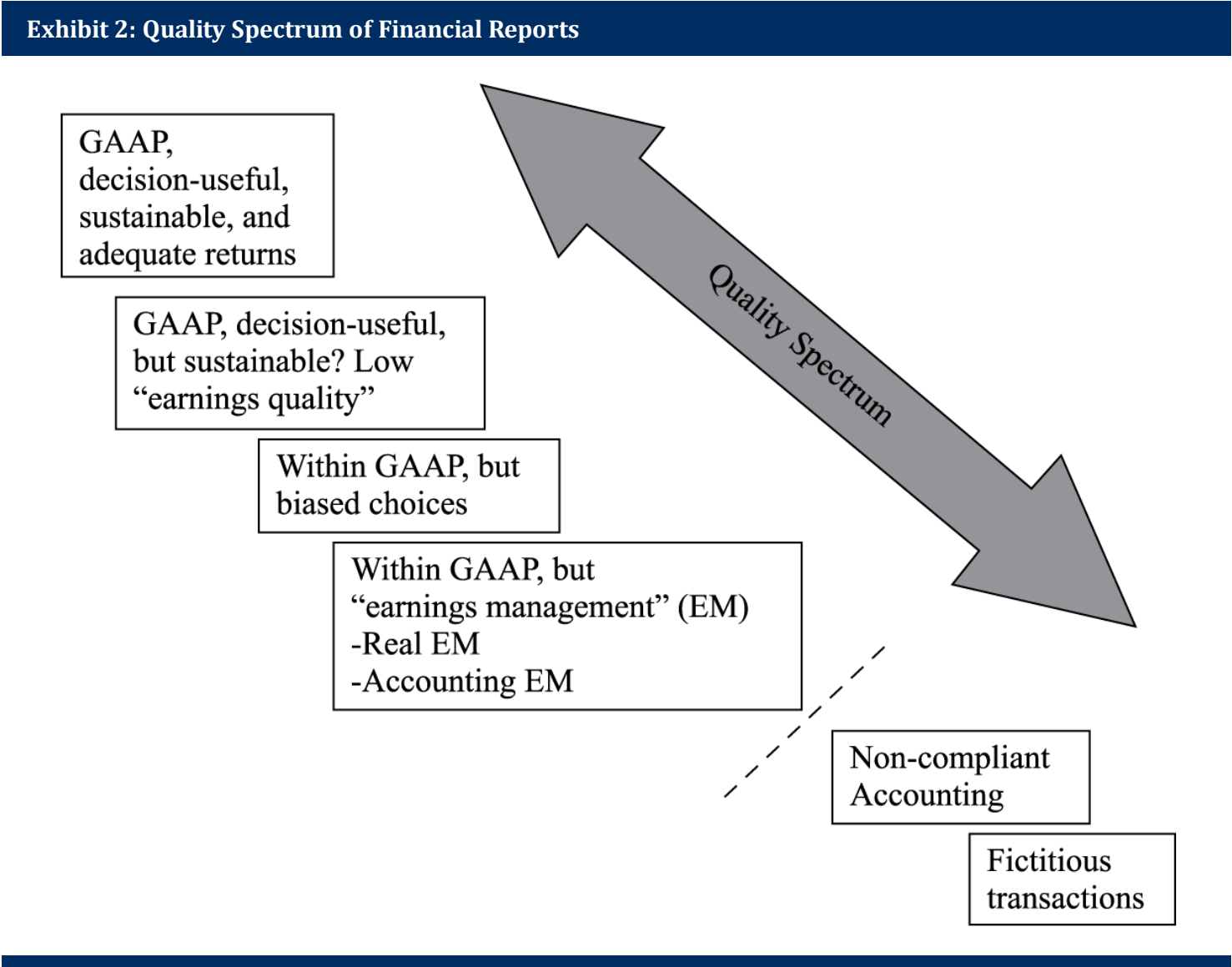
Financial reporting quality and results or earnings quality are interrelated attributes of quality. [Exhibit 1](#) illustrates this interrelationship and its implications.

Exhibit 1: Relationships between Financial Reporting Quality and Earnings Quality

		Financial Reporting Quality	
		Low	High
Earnings (Results) Quality	High	LOW financial reporting quality impedes assessment of earnings quality and impedes valuation.	HIGH financial <u>reporting</u> quality enables assessment. HIGH <u>earnings</u> quality increases company value.
	Low		HIGH financial <u>reporting</u> quality enables assessment. LOW <u>earnings</u> quality decreases company value.

As shown in [Exhibit 1](#), if financial reporting quality is low, the information provided is of little use in assessing the company’s performance, and thus in making investment and other decisions.

Financial reporting quality varies across companies. High-quality reports contain information that is relevant, complete, neutral, and free from error. The lowest-quality reports contain information that is pure fabrication. Earnings (results) quality can range from high and sustainable to low and unsustainable. Providers of resources prefer high and sustainable earnings. Combining the two measures of quality—financial reporting and earnings—the overall quality of financial reports from a user perspective can be thought of as spanning a continuum from the highest to the lowest. [Exhibit 2](#) presents a quality spectrum that provides a basis for evaluating better versus poorer quality reports. This spectrum ranges from reports that are of high financial reporting quality and reflect high and sustainable earnings quality to reports that are not useful because of poor financial reporting quality.



3. GAAP, DECISION USEFUL FINANCIAL REPORTING

LEARNING OUTCOME

The candidate should be able to:

- ☐ **describe a spectrum for assessing financial reporting quality**

At the top of the spectrum, labeled in Exhibit 2 as “GAAP, decision-useful, sustainable, and adequate returns,” are high-quality reports that provide useful information about high-quality earnings.

- High-quality financial reports conform to the generally accepted accounting principles (GAAP) of the jurisdiction, such as International Financial Reporting Standards (IFRS), US GAAP, or other home-country GAAP. The exhibit uses the term GAAP to refer generically to the accounting standards accepted in a company’s jurisdiction.
- In addition to conforming to GAAP, high-quality financial reports also embody the characteristics of decision-useful information, such as those defined in the *Conceptual Framework*.² Recall that the fundamental

characteristics of useful information are relevance and faithful representation. Relevant information is defined as information that can affect a decision and encompasses the notion of materiality. (Information is considered material if “omitting it or misstating it could influence decisions that users make on the basis of the financial information of a specific reporting entity.”³) Faithful representation of economic events is complete, neutral, and free from error.

The *Conceptual Framework* also enumerates enhancing characteristics of useful information: comparability, verifiability, timeliness, and understandability. Of course, the desirable characteristics for financial information require trade-offs. For example, financial reports must balance the aim of providing information that is produced quickly enough to be timely and thus relevant, and yet not so quickly that errors occur. Financial reports must balance the aim of providing information that is complete but not so exhaustive that immaterial information is included. High-quality information results when these and other trade-offs are made in an unbiased, skillful manner.

- High-quality earnings indicate an adequate level of return on investment and derive from activities that a company likely will be able to sustain in the future. An adequate level of return on investment exceeds the cost of the investment and also equals or exceeds the expected return. Sustainable activities and sustainable earnings are those expected to recur in the future. Sustainable earnings that provide a high return on investment contribute to higher valuation of a company and its securities.

3.1. GAAP, Decision-Useful, but Sustainable?

The next level down in Exhibit 2, “GAAP, decision-useful, but sustainable?” refers to circumstances in which high-quality reporting provides useful information, but that information reflects results or earnings that are not sustainable (lower earnings quality). The earnings may not be sustainable because the company cannot expect earnings that generate the same level of return on investment in the future or because the earnings, although replicable, will not generate sufficient return on investment to sustain the company. Earnings quality is low in both cases. Reporting can be high quality even when the economic reality being depicted is not of high quality. For example, consider a company that generates a loss, or earnings that do not provide an adequate return on investment, or earnings that resulted from non-recurring activities. The relatively undesirable economic reality could nonetheless be depicted in financial reporting that provides high-quality, decision-useful information.

Exhibit 3 presents an excerpt from the fiscal year 2014 first-quarter results of Toyota Motor Corporation, a Japanese automobile company. As highlighted by a *Wall Street Journal* article,⁴ the company sold fewer cars but reported an 88 percent increase in operating profits compared with the prior year, primarily because of the change in exchange rates. The weaker yen benefited Toyota both because the company manufactures more cars in Japan (compared with its competitors) and because the company sells a significant number of cars outside of Japan. Exchange rate weakening is a less sustainable source of profits than manufacturing and selling cars. In summary, this example is a case of high-quality financial reporting coupled with lower earnings quality.

Exhibit 3: Excerpt from Toyota Motor Corporation’s Consolidated Financial Results for FY2014 First Quarter Ending 30 June 2013

Consolidated vehicle unit sales in Japan and overseas decreased by 37 thousand units, or 1.6%, to 2,232 thousand units in FY2014 first quarter (the three months ended June 30, 2013) compared with FY2013 first quarter (the three months ended June 30, 2012). Vehicle unit sales in Japan decreased by 51 thousand units, or 8.8%, to 526 thousand units in FY2014 first quarter compared with FY2013 first quarter. Meanwhile, overseas vehicle unit sales increased by 14 thousand units, or 0.8%, to 1,706 thousand units in FY2014 first quarter compared with FY2013 first quarter.

As for the results of operations, net revenues increased by 753.7 billion yen, or 13.7%, to 6,255.3 billion yen in FY2014 first quarter compared with FY2013 first quarter, and operating income increased by 310.2 billion yen, or 87.9%, to 663.3 billion yen in FY2014 first quarter compared with FY2013 first quarter. The factors contributing to an increase in operating income were the effects of changes in exchange rates of 260.0 billion yen, cost reduction efforts of 70.0 billion yen, marketing efforts of 30.0 billion yen and other factors of 10.2 billion yen. On the other hand, the factors contributing to a decrease in operating income were the increase in expenses and others of 60.0 billion yen.

Source: Back Aaron 2013. “Toyota, What a Difference the Yen Makes.” *Wall Street Journal* (4 August 2013).

4. BIASED ACCOUNTING CHOICES

LEARNING OUTCOME

The candidate should be able to:

☐ **describe a spectrum for assessing financial reporting quality**

The next level down in the spectrum shown in Exhibit 2 is “Within GAAP, but biased choices.” Biased choices result in financial reports that do not faithfully represent the economic substance of what is being reported. The problem with bias in financial reporting, as with other deficiencies in reporting quality, is that it impedes an investor’s ability to correctly assess a company’s past performance, to accurately forecast future performance, and thus to appropriately value the company.

Choices are deemed to be “aggressive” if they increase a company’s reported performance and financial position in the period under review. The choice can increase the amount of revenues, earnings, or operating cash flow reported for the period, or decrease expenses, or reduce the level of debt reported on the balance sheet. Aggressive choices may lead to a reduction in the company’s reported performance and in its financial position in later periods. In contrast, choices are deemed “conservative” if they decrease a company’s performance and financial position in the reporting period. This can include lowering the reported revenues, earnings, or operating cash flow reported or increasing expenses, or recording a higher level of debt on the balance sheet. Conservative choices may lead to a rise in the company’s reported performance and financial position in later periods.

Another type of bias is understatement of earnings volatility, so-called earnings smoothing. Earnings smoothing can result from conservative choices to understate earnings in periods when a company’s operations are performing well, building up (often hidden) reserves that allow aggressive choices in periods when its operations are struggling.

Biased choices can be made not only in the context of reported amounts but also in the context of how information is presented. For example, companies can disclose information transparently, which facilitates analysis, or they can disclose it in a manner that aims to obscure unfavorable or emphasize favorable information.

EXAMPLE 1

Quality of Financial Reports

PACCAR Inc. designs, manufactures, and distributes trucks and related aftermarket parts that are sold worldwide under the Kenworth, Peterbilt, and DAF nameplates. In 2013, the US SEC charged PACCAR for various accounting deficiencies that “clouded their financial reporting to investors in the midst of the financial crisis.” The SEC complaint cites the company’s 2009 segment reporting. Exhibit 4 presents an excerpt from the notes to PACCAR’s financial statements and an excerpt from the management discussion and analysis (MD&A) of PACCAR’s annual report.

Exhibit 4: PACCAR’S 2009 Financial Statements and Annual Report

A. Excerpt from Notes

S. SEGMENT AND RELATED INFORMATION

PACCAR operates in two principal segments, Truck and Financial Services.

The Truck segment includes the manufacture of trucks and the distribution of related aftermarket parts, both of which are sold through a network of independent dealers. . . . The Financial Services segment is composed of finance and leasing products and services provided to truck customers and dealers ... Included in All Other is PACCAR’s industrial winch manufacturing business. Also within this category are other sales, income and expenses not attributable to a reportable segment, including a portion of corporate expense.

Pre-tax Income by Business Segment (USD millions)

	2009	2008	2007
Truck	USD25.9	USD1,156.5	USD1,352.8
All other	42.2	6.0	32.0
	68.1	1,162.5	1,384.8
Financial services	84.6	216.9	284.1

	2009	2008	2007
Investment income	22.3	84.6	95.4
	USD175.0	USD1,464.0	USD1,764.3

B. Excerpt from MD&A

Net sales and revenues and gross margins for truck units and aftermarket parts are provided below. The aftermarket parts gross margin includes direct revenues and costs, but excludes certain truck segment costs.

	2009	2008	Change (%)
Net Sales and Revenues			
Trucks	USD5,103.30	USD11,281.30	-55
Aftermarket parts	1,890.70	2,266.10	-17
	USD6,994.00	USD13,547.40	-48
Gross Margin			
Trucks	-USD46.6	USD1,141.70	-104
Aftermarket parts	625.7	795.20	-21
	USD579.1	USD1,936.90	-70

1. Based on the segment data excerpted from the notes to the financial statements, was PACCAR's truck segment profitable in 2009?

Solution:

Yes, the segment data presented in the note to the financial statements indicates that the truck segment earned USD25.9 million in 2009.

2. Based on the data about the truck's gross margin presented in the MD&A, was PACCAR's truck segment profitable in 2009?

Solution:

No, the segment data presented in the MD&A indicates that the truck segment had a negative gross margin.

3. What is the main difference between the note presentation and the MD&A presentation?

Solution:

The main difference between the note presentation and the MD&A presentation is that the aftermarket parts business is combined with the trucks business in the notes but is separated in the MD&A. Although the data are not exactly comparable in the two disclosures (because the note shows income before taxes and the MD&A shows gross profit), the two disclosures present a different picture of PACCAR's profits from truck sales.

4. The SEC complaint stated that "PACCAR failed to report the operating results of its aftermarket parts business separately from its truck sales business as required under segment reporting requirements, which are in place to ensure that investors gain the same insight into a company as its executives." Is the PACCAR situation an example of issues with financial reporting quality, earnings quality, or both?

Solution:

The PACCAR situation appears to be an example of issues with both financial reporting quality and earnings quality. The substantial decrease in truck sales and the negative gross margin reflect poor earnings quality. The failure to disclose clear segment information is an instance of poor financial reporting quality.

Although choices exist within GAAP for the presentation of a desired economic picture, non-GAAP reporting adds yet another dimension of management discretion. Non-GAAP reporting of financial metrics not in compliance with generally accepted accounting principles, such as US GAAP and IFRS, includes both financial metrics and operating metrics.⁵ Non-GAAP financial metrics relate directly to the financial statements. A common non-GAAP financial metric is "non-GAAP earnings," which are created by companies "that adjust standards-compliant earnings to *exclude items*

required by accounting standards or to include items not permitted by accounting standards” (Ciesielski and Henry, 2017). In contrast, non-GAAP operating metrics do not relate directly to the financial statements and include metrics that are typically industry-driven, such as subscriber numbers, active users, and occupancy rates.

Non-GAAP financial reporting has become increasingly common, presenting challenges to analysts. An important challenge is that non-GAAP financial reporting diminishes comparability across financial statements. The adjustments that companies make to create non-GAAP earnings, for example, are generally ad hoc and thus differ significantly. When evaluating non-GAAP metrics, investors must decide the extent to which specific adjustments should be incorporated into their analyses and forecasts.⁶

Another challenge arises from differences in terminology. Non-GAAP earnings are sometimes referred to as underlying earnings, adjusted earnings, recurring earnings, core earnings, or similar. Exhibit 5 provides an example from Jaguar Land Rover Automotive plc (JLR), a subsidiary of Tata Motors Ltd. The company prepares its financial reports under IFRS. The exhibit is an excerpt from JLR’s 2016/17 annual report and uses the term “alternative performance measures.” Exhibit 6 is from Tata Motors Ltd.’s Form 6-K filed with the US SEC, containing supplemental information regarding JLR and using the term “non-IFRS Financial Measures.” The information in the two exhibits is essentially identical, but the terminology and formatting differ.

Exhibit 5: Excerpt from JLR’s 2016/17 Annual Report: Footnote 3

3) ALTERNATIVE PERFORMANCE MEASURES

Many companies use alternative performance measures (APMs) to provide helpful additional information for users of their financial statements, telling a clearer story of how the business has performed over the period. . . . These measures exclude certain items that are included in comparable statutory measures....

Reconciliations between these alternative performance measures and statutory reported measures are shown below.

Fiscal year ended 31 March 2017	GBP millions
EBITDA	2,955
Depreciation and amortization	-1,656
Share of profit/(loss) of equity accounted investments	159
EBIT	1,458
Foreign exchange (loss)/gain on derivatives	-11
Unrealised gain/(loss) on commodities	148
Foreign exchange loss on loans	-101
Finance income	33
Finance expense (net)	-68
Exceptional item	151
Profit before tax	1,610

Exhibit 6: Excerpt from Tata Motors Ltd. SEC Form 6-K

Non-IFRS Financial Measures

This Report includes references to certain non-IFRS measures, including EBITDA, EBIT . . . [These measures] and related ratios should not be considered in isolation and are not measures of JLR’s financial performance or liquidity under IFRS and should not be considered as an alternative to profit or loss for the period or any other performance measures derived in accordance with IFRS or as an alternative to cash flow from operating, investing or financing activities or any other measure of JLR’s liquidity derived in accordance with IFRS. . . . In addition, EBITDA, EBIT... as defined, may not be comparable to other similarly titled measures used by other companies.

Form 6-K Supplemental Information Regarding the Jaguar and Land Rover Business of Tata Motors Limited

The reconciliation of JLR’s EBIT and EBITDA to profit for the period line item is:

Fiscal year ended 31 March 2017	GBP millions
Profit for the period	1,272
Add back taxation	338
Add/(less) back exceptional charge/(credit)	-151
Add back/(less) foreign exchange (gains)/loss – financing	101
Add back/(less) foreign exchange (gains)/loss – derivatives	11
Add back/(less) unrealized commodity losses/(gains) – unrealized derivatives	-148
Less finance income	-33
Add back finance expense (net)	68
EBIT	1,458
Add back depreciation and amortization	1,656
Add/(less) back share of loss/(profit) from equity accounted investees	-159
EBITDA	2,955

Management emphasis on non-GAAP financial measures to deflect attention from less-than-desirable GAAP financial results is an example of an aggressive presentation choice. Since 2003, if a company uses a non-GAAP financial measure⁷ in an SEC filing, it is required to display the most directly comparable GAAP measure with equal prominence and to provide a reconciliation between the non-GAAP measure and the equivalent GAAP measure. In other words, a company is not allowed to give more prominence to a non-GAAP financial measure in an SEC filing.

Similarly, the IFRS Practice Statement “Management Commentary,” issued December 2010, requires disclosures when non-IFRS measures are included in financial reports:

If information from the financial statements has been adjusted for inclusion in management commentary, that fact should be disclosed. If financial performance measures that are not required or defined by IFRSs are included within management commentary, those measures should be defined and explained, including an explanation of the relevance of the measure to users. When financial performance measures are derived or drawn from the financial statements, those measures should be reconciled to measures presented in the financial statements that have been prepared in accordance with IFRSs. (page 17)

The reconciliation between as-reported measures (GAAP financial measures presented in the financial statements) and as-adjusted measures (non-GAAP financial measures presented in places other than the financial statements) can provide important information.

The European Securities and Markets Authority (ESMA) published guidelines in October 2015 (*ESMA Guidelines on Alternative Performance Measures*) covering such points as the definition of APMs, reconciliation to GAAP, explanation of the metrics’ relevance, and consistency over time. We discuss ESMA in more detail later in this module.

EXAMPLE 2

Presentation of Non-GAAP Financial Measures

Convatec Group PLC (Convatec), a global medical products manufacturer, raised USD1.8 billion through an initial public offering (IPO) on the London Stock Exchange in 2016. The company had been purchased by private equity firms from Bristol-Myers Squibb in 2008 for USD4.1 billion. Exhibit 7 presents excerpts from the company’s regulatory filing at the London Stock Exchange announcing its full year 2016 results.

Exhibit 7: Excerpt from Convatec’s Press Release for Full Year 2016 Results

A. Strong results, delivering on strategy

CEO Review

At constant currency, revenue grew 4% to \$1,688 million and adjusted EBITDA was \$508 million, up 6.5% at constant currency. . .

[Footnote] Constant currency growth ‘CER’ is calculated by restating 2016 results using 2015 foreign exchange rates for the relevant period.

Consolidated Statement of Profit or Loss for the year ended 31 December 2016 (USD millions)

	2016	2015
Revenue	1,688.3	1,650.4
Cost of goods sold	-821.0	-799.9
Gross profit	867.3	850.5
Selling and distribution expenses	-357.0	-346.7
General and administrative expenses	-318.2	-233.1
Research and development expenses	-38.1	-40.3
Operating profit	154.0	230.4
Finance costs	-271.4	-303.6
Other expense, net	-8.4	-37.1
Loss before income taxes	-125.8	-110.3
Income tax (expense) benefit	-77.0	16.9
Net loss	-202.8	-93.4

B. Non-IFRS Financial Information

This release contains certain financial measures that are not defined or recognised under IFRS. These measures are referred to as "Adjusted" measures. . . . These measures are not measurements of financial performance or liquidity under IFRS and should not replace measures of liquidity or operating profit that are derived in accordance with IFRS.

C. Reconciliation to adjusted earnings

2016	Reported	(a)	(b)	(c)	(d)	(e)	(f)	(g)	Adjusted
Revenue	1,688.3	—	—	—	—	—	—	—	1,688.3
⋮									
Operating profit	154.0	155.1	30.9	11.7	0.8	—	90.2	29.5	472.2
⋮									
(Loss) profit before income taxes	-125.8	155.1	30.9	11.7	0.8	37.6	90.2	29.5	230.0
Income tax expense ^(h)	-77.0								-51.2
Net (loss) profit	-202.8								178.8

(a) Represents an adjustment to exclude (i) acquisition-related amortisation expense ... (ii) accelerated depreciation ... related to the closure of certain manufacturing facilities, and (iii) impairment charges and assets write offs related to property, plant and equipment and intangible assets

(b) Represents restructuring costs and other related costs ...

(c) Represents remediation costs which include regulatory compliance costs related to FDA activities, IT enhancement costs, and professional service fees associated with activities that were undertaken in respect of the Group's compliance function and to strengthen its control environment within finance.

(d) Represents costs primarily related to (i) corporate development activities and (ii) a settlement of ordinary course multi-year patent-related litigations in 2015

(e) Represents adjustments to exclude (i) loss on extinguishment of debt and write off of deferred financing fees ... and (ii) foreign exchange related transactions.

(f) Represents an adjustment to exclude (i) share-based compensation expense ... arising from pre-IPO employee equity grants and (ii) pre-IPO ownership structure related costs, including management fees to Nordic Capital and Avista (refer to Note 6 Related Party Transactions for further information).

(g) Represents IPO related costs, primary advisory fees.

(h) Adjusted income tax expense/benefit is income tax (expense) benefit net of tax adjustments.

D. Adjusted EBITDA

Adjusted EBITDA is defined as Adjusted EBIT . . . further adjusted to exclude (i) software and R&D amortisation, (ii) depreciation and (iii) post-IPO share-based compensation.

The following table reconciles the Group's Adjusted EBIT to Adjusted EBITDA.

2016 (USD millions)

	2016 (USD millions)
Adjusted EBIT	472.2
Software and R&D amortization	6.7
Depreciation	27.9
Post-IPO share-based compensation	0.8
Adjusted EBITDA	507.6

1. Based on the information provided in Exhibit 7, explain the differences between the following two disclosures contained in Convatec's press release:

A. CEO Review of 2016 results states that revenue grew 4 percent to USD1,688 million.

B. Convatec's Consolidated Statement of Profit or Loss shows 2016 revenues of USD1,688.3 million and 2015 revenues of USD1,650.4 million.

Solution:

The amount of revenue reported on the company's income statement conforms to International Financial Reporting Standards (IFRS). Using the amounts from the income statement, the company's total revenue increased by 2.3% ($= \text{USD}1,688.3 / \text{USD}1,650.4 - 1$). The revenue growth rate of 4 percent in the CEO review is a non-IFRS measure, calculated on a "constant currency" basis, which the footnote describes as a comparison using 2016 revenues restated at 2015 foreign exchange rates.

2. Based on the information provided, explain the differences between the following two disclosures contained in Convatec's earnings release:

A. The CEO Review of 2016 results states that adjusted EBITDA was USD508 million, up 6.5 percent at constant currency.

B. Convatec's Consolidated Statement of Profit or Loss shows 2016 net loss of USD202.8 million and 2015 net loss of USD93.4 million.

Solution:

The amounts reported on the company's income statement conform to IFRS. Using amounts from the income statement, the company reported a loss in 2016 of USD202.8 million, which was more than twice as large a loss as the USD93.4 million loss reported in 2015. Also referring to the income statement, the company reported 2016 operating profit (referred to elsewhere as EBIT) of USD154.0 million, a decline of 33.2 percent from the USD230.4 million operating profit reported in 2015.

In contrast, the adjusted EBITDA amount highlighted in the CEO Review is neither defined nor recognized under IFRS. It is a non-IFRS measure. To create the adjusted EBITDA, the company first begins with EBIT (called Operating profit in excerpts II and III) of USD154.0 and creates adjusted EBIT (USD472.2 million) by adding back eight different expenses that IFRS requires the company to recognize. These adjustments are listed beneath the first tabular reconciliation in items (a) through (g). After developing Adjusted EBIT, the company creates adjusted EBITDA (USD507.6 million) by adding back a further three different expenses that IFRS requires the company to recognize.

Overall, there are three key differences between Disclosures A and B: (1) Most important, disclosure A refers to a non-IFRS metric rather than an IFRS-compliant metric; (2) Disclosure A refers to operating profit, which was positive, rather than to net income, which was negative; and (3) Disclosure A highlights a positive economic outcome—that is, an increase, on a currency-adjusted basis. An analyst should be aware of the alternative means by which earnings announcements can paint a positive picture of companies' results.

Often, poor reporting quality occurs simultaneously with poor earnings quality; for example, aggressive accounting choices are made to obscure poor performance. It is also possible, of course, for poor reporting quality to occur with high-quality earnings. Although a company with good performance would not require aggressive accounting choices to obscure poor performance, it might nonetheless produce poor-quality reports for other reasons. A company with good performance might not be able to produce high-quality reports because of inadequate internal systems.

Another scenario in which poor reporting quality might occur simultaneously with high-quality earnings is that a company with good performance might deliberately produce reports based on "conservative" rather than aggressive accounting choices—that is, choices that make current performance look worse. One motivation might be to avoid unwanted political attention. Another motivation could arise in a period in which management had already exceeded targets before the end of the period and thus made conservative accounting choices that would delay reporting profits

until the following period (so-called hidden reserves). Similar motivations might also contribute to accounting choices that create the appearance that the trajectory of future results would appear more attractive. For example, a company might make choices to accelerate losses in the first year of an acquisition or the first year of a new CEO’s tenure so that the trajectory of future results would appear more attractive.

Overall, *unbiased* financial reporting is optimal. Some investors may prefer conservative choices rather than aggressive ones, however, because a positive surprise is easier to tolerate than a negative surprise. Biased reporting, whether conservative or aggressive, adversely affects a user’s ability to assess a company.

The quality spectrum considers the more intuitive situation in which less-than-desired underlying economics are the central motivation for poor reporting quality. In addition, it is necessary to have some degree of reporting quality in order to evaluate earnings quality. Proceeding down the spectrum, therefore, the concepts of reporting quality and earnings quality become progressively less distinguishable.

4.1. Within GAAP, but “Earnings Management”

The next level down on the spectrum shown in Exhibit 2 is labeled “Within GAAP, but ‘earnings management.’” The term “earnings management” is defined as making intentional choices that create biased financial reports.⁸ The distinction between earnings management and biased choices is subtle and, primarily, a matter of intent. Earnings management represents “deliberate actions to influence reported earnings and their interpretation” (Ronen and Yaari, 2008). Earnings can be “managed” upward (increased) by taking *real* actions, such as deferring research and development (R&D) expenses into the next reporting period. Alternatively, earnings can be increased by *accounting* choices, such as changing accounting estimates. For example, the amount of estimated product returns, bad debt expense, or asset impairment could be decreased to create higher earnings. Because it is difficult to determine intent, we include earnings management under the biased choices discussion.

5. DEPARTURES FROM GAAP

LEARNING OUTCOME

The candidate should be able to:

- ☐ describe a spectrum for assessing financial reporting quality

The next levels down on the spectrum shown in Exhibit 2 mark departures from GAAP. Financial reporting that departs from GAAP generally can be considered to be low quality. In such situations, earnings quality is likely difficult or impossible to assess because comparisons with earlier periods and/or other entities cannot be made. An example of improper accounting was Enron (accounting issues revealed in 2001), whose inappropriate use of off-balance-sheet structures and other complex transactions resulted in vastly understated indebtedness as well as overstated profits and operating cash flow. Another notorious example of improper accounting was WorldCom (accounting issues discovered in 2002), a company that by improperly capitalizing certain expenditures dramatically understated its expenses and thus overstated its profits. New Century Financial (whose accounting issues were revealed in 2007) issued billions of dollars of subprime mortgages and improperly reserved only minimal amounts for loan repurchase losses. Each of these companies subsequently filed for bankruptcy.

In the 1980s, Polly Peck International (PPI) reported currency losses, incurred in the normal course of operations, directly through equity rather than in its profit and loss statements. In the 1990s, Sunbeam improperly reported revenues from “bill-and-hold” sales and also manipulated the timing of expenses in an effort to falsely portray outstanding performance of its then-new chief executive.

At the bottom of the quality spectrum, fabricated reports portray fictitious events, either to deceive investors by misrepresenting the company’s performance or to obscure fraudulent misappropriation of the company’s assets. Examples of fraudulent reporting are unfortunately easy to find, although they were not necessarily easy to identify at the time. In the 1970s, Equity Funding Corp. created fictitious revenues and even fictitious policyholders. In the 1980s, Crazy Eddie’s reported fictitious inventory as well as fictitious revenues supported by fake invoices. In 2004, Parmalat reported fictitious bank balances.

EXAMPLE 3

Spectrum for Assessing Quality of Financial Reports

Jake Lake, a financial analyst, has identified several items in the financial reports of several (hypothetical) companies. Describe each of these items in the context of the financial reporting quality spectrum.

1. ABC Co.'s 2018 earnings totaled USD233 million, including a USD100 million gain from selling one of its less profitable divisions. ABC's earnings for the prior three years totaled USD120 million, USD107 million, and USD111 million. The company's financial reports are extremely clear and detailed, and the company's earnings announcement highlights the one-time nature of the USD100 million gain.

Solution:

ABC's 2018 total earnings quality can be viewed as low because nearly half of the earnings are derived from a non-sustainable activity, namely the sale of a division. ABC's quality of earnings in 2018 from continuing operations may be high because the amounts are fairly consistent from year to year, although an analyst would undertake further analysis to confirm earnings quality. In general, a user of financial reports should look beyond the bottom-line net income. The description provided suggests that the company's reporting quality is high; the reports are clear and detailed, and the one-time nature of the USD100 million gain is highlighted.

2. DEF Co. discloses that, in 2018, it changed the depreciable life of its equipment from 3 years to 15 years. Equipment represents a substantial component of the company's assets. The company's disclosures indicate that the change is permissible under the accounting standards of its jurisdiction, but it provides only limited explanation of the change.

Solution:

DEF's accounting choice appears to be within permissible accounting standards, but its effect is to substantially lower depreciation expense and thus to increase earnings for the year. The quality of reported earnings is questionable. Although the new level of earnings may be sustainable, similar increases in earnings for future periods might not be achievable, because increasing earnings solely by changing accounting estimates is likely not sustainable. In addition, the description provided suggests that the company's reporting quality is low because it offers only a limited explanation for the change.

3. GHI Co.'s R&D expenditures for the past five years have been approximately 3 percent of sales. In 2018, the company significantly reduced its R&D expenditures. Without the reduction in R&D expenditures, the company would have reported a loss. No explanation is disclosed.

Solution:

GHI's operational choice to reduce its R&D may reflect real earnings management because the change enabled the company to avoid reporting a loss. In addition, the description provided suggests that the company's reporting quality is low because it does not offer an explanation for the change.

6. DIFFERENTIATE BETWEEN CONSERVATIVE AND AGGRESSIVE ACCOUNTING

LEARNING OUTCOME

The candidate should be able to:

- ☐ explain the difference between conservative and aggressive accounting

This lesson returns to the implications of conservative and aggressive accounting choices. As mentioned earlier, *unbiased* financial reporting is ideal. But some investors may prefer or be perceived to prefer conservative rather than aggressive accounting choices, because a positive surprise is acceptable. In contrast, management may make, or be perceived to make, aggressive accounting choices because they increase the company's reported performance and financial position.

Aggressive accounting choices in the period under review may decrease the company's reported performance and financial position in later periods, which creates a sustainability issue. Conservative choices do not typically create a sustainability issue because they decrease the company's reported performance and financial position, and may increase them in later periods. In terms of establishing expectations for the future, however, financial reporting that is relevant and faithfully representative is the most useful.

A common presumption is that financial reports are typically biased upward, but that is not always the case. Although accounting standards ideally promote unbiased financial reporting, some accounting standards may specifically require a conservative treatment of a transaction or an event. Also, managers may choose to take a conservative approach when applying standards. It is important that an analyst consider the possibility of conservative choices and their effects.

At its most extreme, conservatism follows accounting practices that “anticipate no profit, but anticipate all losses” (Bliss, 1924). But in general, conservatism means that revenues may be recognized once a verifiable and legally enforceable receivable has been generated and that losses need not be recognized until it becomes “probable” that an actual loss will be incurred. Conservatism is not an absolute but is characterized by degrees, such as “the accountant’s tendency to require a higher degree of verification to recognize good news as gains than to recognize bad news as losses” (Basu, 1997). From this perspective, “verification” (e.g., physical existence of inventories, evidence of costs incurred or to be incurred, or establishment of rights and obligations on legal grounds) drives the degree of conservatism. For recognition of revenues, a higher degree of verification would be required than for expenses.

6.1. Conservatism in Accounting Standards

The *Conceptual Framework* supports neutrality of information: “A neutral depiction is without bias in the selection or presentation of financial information.”⁹ Neutrality—lack of upward or downward bias—is considered a desirable characteristic of financial reporting. Conservatism directly conflicts with the characteristic of neutrality because the asymmetric nature of conservatism leads to bias in measuring assets and liabilities—and, ultimately, earnings.

Despite efforts to support neutrality in financial reporting, many conservatively biased standards remain. Standards across jurisdictions may differ on the extent of conservatism embedded within them. An analyst should be aware of the implications of accounting standards for the financial reports.

An example is the different treatment by IFRS and US GAAP of the impairment of long-lived assets.¹⁰ Both IFRS and US GAAP specify an impairment analysis protocol that begins with an assessment of whether recent events indicate that the economic benefit from an individual or group of long-lived assets may be less than its carrying amount(s). From that point on, however, the two regimes diverge:

- Under IFRS, if the “recoverable amount” (the higher of fair value less costs to sell and value in use) is less than the carrying amount, then an impairment charge will be recorded.
- Under US GAAP, an impairment charge will be recorded only when the sum of the undiscounted future cash flows expected to be derived from the asset(s) is less than the carrying amount(s). If the undiscounted future cash flows are less than the carrying amount, the asset is written down to fair value.

To illustrate the difference in application, assume that a factory is the unit of account eligible for impairment testing. Its carrying amount is USD10,000,000; “fair value” and “recoverable amount” are both USD6,000,000; and the undiscounted future net cash flows associated with the factory total USD10,000,000. Under IFRS, an impairment charge of USD4,000,000 would be recorded; under US GAAP, however, no impairment charge would be recognized.

Thus, on its face, IFRS would be regarded as more conservative than US GAAP because impairment losses normally would be recognized earlier under IFRS than under US GAAP. But, taking the analysis one step further, such a broad generalization may not hold up. For example, if an asset is impaired under both IFRS and US GAAP, and the asset’s value in use exceeds its fair value, the impairment loss under US GAAP will be greater. Also, IFRS permits the recognition of recoveries of the recoverable amount in subsequent periods if evidence indicates that the recoverable amount has subsequently increased. In contrast, US GAAP prohibits the subsequent write-up of an asset after an impairment charge has been taken; it would recognize the asset’s increased value only when the asset is ultimately sold.

Common examples of conservatism in accounting standards include the following:

- *Research costs.* Because the future benefit of research costs is uncertain at the time the costs are incurred, both US GAAP and IFRS require immediate expensing instead of capitalization.
- *Litigation losses.* When it becomes “probable” that a cost will be incurred, both US GAAP and IFRS require expense recognition, even though a legal liability may not be incurred until a future date.
- *Insurance recoverables.* Generally, a company that receives payment on an insurance claim may not recognize a receivable until the insurance company acknowledges the validity of the claimed amount.

Watts (2003) reviews empirical studies of conservatism and identifies the following four potential benefits of conservatism:

- Given asymmetrical information, conservatism may protect the contracting parties with less information and greater risk. This protection is necessary because the contracting party may be at a disadvantage. For example, corporations that access debt markets have limited liability, and lenders thus have limited recourse to recover their losses from shareholders. As another example, executives who receive earnings-based bonuses might not be subject to having those bonuses “clawed back” if earnings are subsequently discovered to be overstated.

- Conservatism reduces the possibility of litigation and, by extension, litigation costs. Rarely, if ever, is a company sued because it understated good news or overstated bad news.
- Conservative rules may protect the interests of regulators and politicians by reducing the possibility that fault will be found with them if companies overstate earnings or assets.
- In many tax jurisdictions, financial and tax reporting rules are linked. For example, in Germany and Japan, only deductions taken against reported income can be deducted against taxable income. Hence, companies can reduce the present value of their tax payments by electing conservative accounting policies for certain types of events.

Analysts should consider possible conservative and aggressive biases and their consequences when examining financial reports. Current-period financial reports may be unbiased, upward biased through aggressive accounting choices, downward biased through conservative accounting choices, or biased through a combination of conservative and aggressive accounting choices.

6.2. Bias in the Application of Accounting Standards

Any application of accounting standards, whether or not the standard is neutral, often requires significant amounts of judgment. Characterizing the application of an accounting standard as conservative or aggressive is more a matter of intent than definition.

Careful analysis of disclosures, facts, and circumstances contributes to making an accurate inference of intent. Management seeking to manipulate earnings may take a longer view by sacrificing short-term profitability in order to ensure higher profits in later periods. One example of biased accounting in the guise of conservatism is the so-called big bath restructuring charge. Both US GAAP and IFRS provide for accrual of future costs associated with restructurings, and these costs are often associated with and presented along with asset impairments. But in some instances, companies use the accounting provisions to estimate “big” losses in the period under review so that performance in future periods will appear better. Having observed numerous instances of manipulative practices in the late 1990s, in which US companies set up opportunities to report higher profits in future periods that were not connected with performance in those periods, the SEC staff issued rules that narrowed the circumstances under which costs can be categorized as part of a “non-recurring” restructuring event and enhanced the transparency surrounding restructuring charges and asset impairments.¹¹

A similar manifestation of big bath accounting is often referred to as “cookie jar reserve accounting.” Both US GAAP and IFRS require accruals of estimates of future non-payments of loans. In his 1998 speech “The Numbers Game,” SEC chair Arthur Levitt expressed the general concern that corporations were overstating loans and other forms of loss allowances for the purpose of smoothing income over time.¹² In 2003, the SEC issued interpretive guidance that essentially requires a company to provide a separate section in management discussion and analysis (MD&A) titled “Critical Accounting Estimates.”¹³ If the effects of subjective estimates and judgments of highly uncertain matters are material to stakeholders (investors, customers, suppliers, and other users of the financial statements), disclosures of their nature and exposure to uncertainty should be made in the MD&A. This requirement is in addition to required disclosures in the notes to the financial statements.

7. CONTEXT FOR ASSESSING FINANCIAL REPORTING QUALITY

LEARNING OUTCOME

The candidate should be able to:

- ☐ **describe motivations that might cause management to issue financial reports that are not high quality and conditions that are conducive to issuing low-quality, or even fraudulent, financial reports**

In assessing financial reporting quality, it is useful to consider whether a company’s managers may be motivated to issue financial reports that are not high quality. If motivation exists, an analyst should consider whether the reporting environment is conducive to managers’ misreporting. It is important to consider mechanisms within the reporting environment that discipline financial reporting quality, such as the regulatory regime.

7.1. Motivations

Managers may be motivated to issue financial reports that are not high quality to mask poor performance, such as a loss of market share or lower profitability than competitors. As [Lewis \(2012\)](#) stated, “A firm experiencing performance problems, particularly those it considers transient, may induce a response that inflates current earnings numbers in exchange for lower future earnings.”

- Even when there is no need to mask poor performance, managers frequently have incentives to meet or beat market expectations as reflected in analysts’ forecasts or management’s own forecasts. Exceeding forecasts may increase the stock price, if only temporarily. Additionally, exceeding forecasts can increase management compensation that is linked to increases in stock price or to reported earnings. [Graham, Harvey, and Rajgopal \(2005\)](#) found that the chief financial officers (CFOs) they surveyed view earnings as the most important financial metric to financial markets. Achieving (or exceeding) particular benchmarks, including prior-year earnings and analysts’ forecasts, is very important. The authors examined a variety of motivations for why managers might “exercise accounting discretion to achieve some desirable earnings goal.” Motivations to meet earnings benchmarks include equity market effects (e.g., building credibility with market participants and positively affecting stock price) and trade effects (e.g., enhancing reputation with customers and suppliers). Equity market effects are the most powerful incentives, but trade effects are important, particularly for smaller companies.
- Career concerns and incentive compensation may motivate accounting choices. For example, managers might be concerned that working for a company that performs poorly will limit their future career opportunities or that they will not receive a bonus based on exceeding a particular earnings target. In both cases, management might be motivated to make accounting choices to increase earnings. In a period of marginally poor performance, a manager might accelerate or inflate revenues or delay or under-report expenses. Conversely, in a period of strong performance, a manager might delay revenue recognition or accelerate expense recognition to increase the probability of exceeding the next period’s targets (i.e., to “bank” some earnings for the next period.) The surveyed managers indicated a greater concern with career implications of reported results than with incentive compensation implications.

Avoiding debt covenant violations can motivate managers to inflate earnings. Graham, Harvey, and Rajgopal’s survey indicates that avoidance of bond covenant violation is important to highly leveraged and unprofitable companies but relatively unimportant overall.

7.2. Conditions Conducive to Issuing Low-Quality Financial Reports

As discussed, deviations from a neutral presentation of financial results could be driven by management choices or by a jurisdiction’s financial reporting standards. Ultimately, a decision to issue low-quality, or even fraudulent, financial reports is made by an individual or individuals. Why individuals make such choices is not always immediately apparent. For example, why would the newly appointed CEO of Sunbeam, who already had a net worth of more than USD100 million, commit accounting fraud by improperly reporting revenues from “bill-and-hold” sales and manipulating the timing of expenses, rather than admit to lower-than-expected financial results?

Typically, three conditions exist when low-quality financial reports are issued: opportunity, pressure or motivation, and rationalization—sometimes referred to as the **fraud triangle**. Opportunity can be the result of internal conditions, such as poor internal controls or an ineffective board of directors, or external conditions, such as accounting standards that provide scope for divergent choices or minimal consequences for an inappropriate choice. Motivation can result from pressure to meet some criteria for personal reasons, such as a bonus, or corporate reasons, such as concern about financing in the future. Rationalization is important because if a decision maker is concerned about a choice, that person needs to be able to justify it to him- or herself.

Former Enron CFO Andrew Fastow, speaking at the 2013 Association of Certified Fraud Examiners Annual Fraud Conference, indicated that he knew at the time he was doing something wrong but followed procedure to justify his decision ([Pavlo, 2013](#)). He made sure to get management and board approval, as well as legal and accounting opinions, and to include appropriate disclosures. The incentive and corporate culture was to create earnings rather than focus on long-term value. Clearly, as reflected in his prison sentence, he did something that was not only wrong but illegal.

8. MECHANISMS THAT DISCIPLINE FINANCIAL REPORTING QUALITY

LEARNING OUTCOME

The candidate should be able to:

□ describe mechanisms that discipline financial reporting quality and the potential limitations of those mechanisms

Markets may discipline poor financial reporting quality. Companies and nations compete for capital, and the cost of capital is a function of perceived risk—including the risk that a company's financial statements will skew investors' expectations. Thus, in the absence of other conflicting economic incentives, a company seeking to minimize its long-term cost of capital should aim to provide high-quality financial reports. In addition to markets, other mechanisms that discipline financial reporting quality include market regulatory authorities, auditors, and private contracts.

8.1. Market Regulatory Authorities

Companies seeking to minimize the cost of capital should maximize reporting quality, but as discussed earlier, conflicting incentives often exist. For this reason, national regulations, and the regulators that establish and enforce rules, can play a significant role in financial reporting quality. Many of the world's securities regulators are members of the International Organization of Securities Commissions (IOSCO). IOSCO is recognized as the "global standard setter for the securities sector" although it does not actually set standards but rather establishes objectives and principles to guide securities and capital market regulation. IOSCO's membership includes more than 120 securities regulators and 80 other securities market participants, such as stock exchanges.¹⁴

One member of IOSCO is The European Securities and Markets Authority (ESMA), an independent EU authority with a mission to "enhance the protection of investors and reinforce stable and well-functioning financial markets in the European Union."¹⁵ ESMA organizes financial reporting enforcement activities through a forum consisting of European enforcers from European Economic Area countries. Direct supervision and enforcement activities are performed at the national level. For example, the Financial Conduct Authority (FCA) is the IOSCO member with primary responsibility for securities regulation in the United Kingdom. ESMA reported that European enforcers examined the interim or annual financial statements of 1,141 issuers in 2017, which in turn led to enforcement actions for 328 issuers with the following outcomes: 12 required reissuances of financial statements, 71 required public corrective notes, and 245 required corrections to be made in future financial statements.¹⁶

Another member of IOSCO is the US regulatory authority, the Securities and Exchange Commission. The SEC is responsible for overseeing approximately 9,100 US public companies (along with investment advisers, broker/dealers, securities exchanges, and other entities) and reviews the disclosures of these companies at least once every three years with the aim of improving information available to investors and potentially uncovering possible violations of securities laws.¹⁷ In 2017, the SEC reported that it had filed 754 total and 446 standalone enforcement actions, about 20 percent of which concerned issuer reporting or accounting and auditing.¹⁸

Examples of regulatory bodies in Asia include the Financial Services Agency in Japan, the China Securities Regulatory Commission, and the Securities and Exchange Board of India. Examples of regulatory bodies in South America include the Comisión Nacional de Valores in Argentina, Comissão de Valores Mobiliários in Brazil, and Comisión para el Mercado Financiero in Chile. A full list of IOSCO members can be found on the organization's website.

Typical features of a regulatory regime that most directly affect financial reporting quality include the following:

- *Registration requirements.* Market regulators typically require publicly traded companies to register securities before offering the securities for sale to the public. A registration document typically contains current financial statements, other relevant information about the risks and prospects of the company issuing the securities, and information about the securities being offered.
- *Disclosure requirements.* Market regulators typically require publicly traded companies to make public periodic reports, including financial reports and management comments. Standard-setting bodies, such as the IASB and FASB, are typically private sector, self-regulated organizations with board members who are experienced accountants, auditors, users of financial statements, and academics. Regulatory authorities, such as the Accounting and Corporate Regulatory Authority in Singapore, the Securities and Exchange Commission in the United States, the Securities and Exchange Commission in Brazil, and the Financial Reporting Council in the United Kingdom, have the legal authority to enforce financial reporting requirements and exert other controls over entities that participate in the capital markets within their jurisdiction. In other words, *generally*, standard-setting bodies set the standards, and regulatory authorities recognize and enforce those standards. Without the recognition of standards by regulatory authorities, the private-sector standard-setting bodies would have no authority. Regulators often retain the legal authority to establish financial reporting standards in their jurisdiction and can overrule the private-sector standard-setting bodies.
- *Auditing requirements.* Market regulators typically require companies' financial statements to be accompanied by an audit opinion attesting that the financial statements conform to the relevant set of accounting standards. Some regulators, such as the SEC in the United States, require an additional audit opinion attesting to the effectiveness of the company's internal controls over financial reporting.
- *Management commentaries.* Regulations typically require publicly traded companies' financial reports to include statements by management. For example, the FCA in the United Kingdom requires a management report

containing “(1) a fair review of the issuer’s business; and (2) a description of the principal risks and uncertainties facing the issuer” (Disclosure Guidance and Transparency Rules sourcebook).

- *Responsibility statements.* Regulations typically require a statement from the person or persons responsible for the company’s filings. Such statements require the responsible individuals to explicitly acknowledge responsibility and to attest to the correctness of the financial reports. Some regulators, such as the SEC in the United States, require formal certifications that carry specific legal penalties for false certifications.
- *Regulatory review of filings.* Regulators typically undertake a review process to ensure that the rules have been followed. The review process typically covers all initial registrations and a sample of subsequent periodic financial reports.
- *Enforcement mechanisms.* Regulators are granted various powers to enforce the securities market rules. Such powers can include assessing fines, suspending or permanently barring market participants, and bringing criminal prosecutions. Public announcements of disciplinary actions are also a type of enforcement mechanism.

In summary, market regulatory authorities play a central role in encouraging high-quality financial reporting.

8.2. Auditors

As noted, regulatory authorities typically require that publicly traded companies’ financial statements be audited by an independent auditor. Private companies also obtain audit opinions for their financial statements, either voluntarily or because audit reports are required by an outside party, such as providers of debt or equity capital.

Audit opinions provide financial statement users with some assurance that the information complies with the relevant set of accounting standards and presents the company’s information fairly. [Exhibit 8](#), [Exhibit 9](#), [Exhibit 10](#), and [Exhibit 11](#) provide excerpts from the independent auditors’ reports for GlaxoSmithKline plc, Alibaba Group Holding Limited, Apple Inc., and Tata Motors Limited, respectively. For each company, the auditor issued an unqualified opinion on the financial statements, indicating that the financial statements present fairly the company’s performance in accordance with relevant standards. (Note: The term “unqualified opinion” means that the opinion did not include any qualifications or exceptions; the term is synonymous with the less formal term “clean opinion.” Unqualified opinions are the most common opinions issued.) Other items in the audit reports reflect the specific requirements of the company’s regulatory regime. For example, the audit report for GlaxoSmithKline spans nine pages and includes opinions on the company’s financial statements as well as the Strategic Report and the Directors’ Report. This audit report also includes disclosures about “Key audit matters,” in accordance with International Standards on Auditing (ISAs) issued by the International Auditing and Assurance Standards Board (IAASB) in 2015 and effective for periods ending on or after 15 December 2016.

The excerpts for Alibaba, Apple, and Tata Motors show the auditors’ opinions on the companies’ financial statements and additionally the SEC-required opinions on the effectiveness of the companies’ internal controls because these companies are listed in the United States. For Alibaba, a single report includes both unqualified opinions: (1) the financial statements present fairly the financial position, results of operations, and cash flows . . . in conformity with US GAAP; and (2) the company maintained effective control over financial reporting. For Apple, the first report includes the unqualified opinion on the financial statements, and the second report includes the unqualified opinion on the company’s effective internal controls. For Tata Motors, the first report includes the unqualified opinion that the financial statements present the company’s position and results fairly in accordance with IFRS. (The SEC permits non-US companies to report using US GAAP, IFRS as issued by the IASB, or home-country GAAP.) However, the second report includes an *adverse* opinion on the effectiveness of the company’s internal controls: “In our opinion, because of the effect of the material weakness . . . the company has not maintained effective internal control.” The report explains that the material weakness involved a third party’s inappropriate access to the company’s systems. The report further states that although the material weakness resulted in ineffective internal controls, it did not affect the audit opinion on the financial statements. Elsewhere in Tata Motors’ annual report (not shown in the excerpt), the company discloses that the weakness did not result in misstatement and that it has undertaken remedial measures.

Exhibit 8: Excerpt from Audit Opinion of PricewaterhouseCoopers LLP from the 2017 Annual Report (pages 149–157) of GlaxoSmithKline plc

In our opinion, GlaxoSmithKline e Group financial statements (the “financial statements”):

- give a true and fair view of the state of the Group’s affairs as at 31 December 2017 and of its profit and cash flows for the year then ended;
- have been properly prepared in accordance with International Financial Reporting Standards (“IFRSs”) as adopted by the European Union; and
- have been prepared in accordance with the requirements of the Companies Act 2006 and Article 4 of the IAS Regulation.

...

In our opinion, the Group financial statements have been properly prepared in accordance with IFRSs as issued by the IASB.

...

Key audit matters

Key audit matters are those matters that, in the auditors' professional judgement, were of most significance in the audit of the financial statements of the current period and include the most significant assessed risks of material misstatement (whether or not due to fraud) identified by the auditors, including those which had the greatest effect on: the overall audit strategy; the allocation of resources in the audit; and directing the efforts of the engagement team. These matters, and any comments we make on the results of our procedures thereon, were addressed in the context of our audit of the financial statements as a whole, and in forming our opinion thereon, and we do not provide a separate opinion on these matters. This is not a complete list of all risks identified by our audit.

...

In our opinion, based on the work undertaken in the course of the audit, the information given in the Strategic Report and Directors' Report for the year ended 31 December 2017 is consistent with the financial statements and has been prepared in accordance with applicable legal requirements.

Exhibit 9: Excerpt from Audit Opinion of PricewaterhouseCoopers Hong Kong, SAR from the Annual Report (SEC Form 20-F, pages F-2 and F-3) of Alibaba Group Holding Limited for the year ended 31 March 2018

In our opinion, the consolidated financial statements referred to above present fairly, in all material respects, the financial position of the Company as of March 31, 2017 and 2018, and the results of their operations and their cash flows for each of the three years in the period ended March 31, 2018 in conformity with accounting principles generally accepted in the United States of America. Also in our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of March 31, 2018, based on criteria established in Internal Control — Integrated Framework (2013) issued by the COSO.

Exhibit 10: Excerpt from Audit Opinion of Ernst & Young from the Annual Report (SEC Form 10-K, pages 70 and 71) of Apple Inc. for the year ended 30 September 30 2017

[From the Financial Statement Opinion]

We have audited the accompanying consolidated balance sheets of Apple Inc. as of September 30, 2017 and September 24, 2016, and the related consolidated statements of operations, comprehensive income, shareholders' equity and cash flows for each of the three years in the period ended September 30, 2017.

...

In our opinion, the financial statements referred to above present fairly, in all material respects, the consolidated financial position of Apple Inc. at September 30, 2017 and September 24, 2016, and the consolidated results of its operations and its cash flows for each of the three years in the period ended September 30, 2017, in conformity with U.S. generally accepted accounting principles.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States), Apple Inc.'s internal control over financial reporting as of September 30, 2017, based on criteria established in Internal Control – Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) and our report dated November 3, 2017 expressed an unqualified opinion thereon.

[From the Internal Controls Opinion]

We have audited Apple Inc.'s internal control over financial reporting as of September 30, 2017, based on criteria established in Internal Control – Integrated Framework issued by the Committee of Sponsoring Organizations of the Treadway Commission (2013 framework) ("the COSO criteria").

...

In our opinion, Apple Inc. maintained, in all material respects, effective internal control over financial reporting as of September 30, 2017, based on the COSO criteria.

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States), the 2017 consolidated financial statements of Apple Inc. and our report dated November 3, 2017 expressed an unqualified opinion thereon.

Exhibit 11: Excerpt from Audit Opinion of KPMG Mumbai, India from the Annual Report (SEC Form 20-F, pages F2 to F4) of Tata Motors Limited for the year ended 31 March 2018

Opinion on the Consolidated Financial Statements

We have audited the accompanying consolidated balance sheet of Tata Motors Limited and its subsidiaries (the “Company”) as of March 31, 2018, the related consolidated income statement, statement of comprehensive income, statement of cash flows, and statement of changes in equity for the year ended March 31, 2018, and the related notes and financial statement schedule 1 (collectively, the consolidated financial statements).

In our opinion, the consolidated financial statements present fairly, in all material respects, the financial position of the Company as of March 31, 2018, and the results of its operations and its cash flows for the year ended March 31, 2018, in conformity with the International Financial Reporting Standards as issued by the International Accounting Standards Board (“IFRS”).

We also have audited, in accordance with the standards of the Public Company Accounting Oversight Board (United States) (PCAOB), the Company’s internal control over financial reporting as of March 31, 2018, based on criteria established in Internal Control – Integrated Framework (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission, and our report dated July, 31, 2018 expressed an adverse opinion on the effectiveness of the Company’s internal control over financial reporting.

...

Opinion on Internal Control Over Financial Reporting

We have audited Tata Motors Limited’s and subsidiaries’ (the Company) internal control over financial reporting as of March 31, 2018, based on criteria established in Internal Control – Integrated Framework (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission. In our opinion, because of the effect of the material weakness described below, on the achievement of the objectives of the control criteria, the Company has not maintained effective internal control over financial reporting as of March 31, 2018, based on criteria established in *Internal Control – Integrated Framework (2013)* issued by the Committee of Sponsoring Organizations of the Treadway Commission.

...

A material weakness is a deficiency, or a combination of deficiencies, in internal control over financial reporting, such that there is a reasonable possibility that a material misstatement of the company’s annual or interim financial statements will not be prevented or detected on a timely basis. A material weakness related to inappropriate system access restrictions at a third party logistics provider has been identified and included in management’s assessment. The material weakness was considered in determining the nature, timing, and extent of audit tests applied in our audit of the 2018 consolidated financial statements, and this report does not affect our report on those consolidated financial statements.

Although audit opinions provide discipline for financial reporting quality, inherent limitations exist. First, an audit opinion is based on a review of information prepared by the company. If a company deliberately intends to deceive its auditor, a review of information might not uncover misstatements. Second, an audit is based on sampling, and the sample might not reveal misstatements. Third, an “expectations gap” may exist between the auditor’s role and the public’s expectation of auditors. An audit is not typically intended to detect fraud; it is intended to provide assurance that the financial reports are fairly presented. Finally, the company being audited pays the audit fees, often established through a competitive process. This situation could provide an auditor with an incentive to show leniency to the company being audited, particularly if the auditor’s firm provides additional services to the company.

8.3. Private Contracting

Aspects of private contracts, such as loan agreements or investment contracts, can also serve as mechanisms to discipline poor financial reporting quality. Many parties that have a contractual arrangement with a company have an incentive to monitor that company’s performance and to ensure that the company’s financial reports are high quality.

For example, loan agreements often contain loan covenants, which create specifically tailored financial reporting requirements that are legally binding for the issuer. As noted earlier, avoidance of debt covenant violation is a potential motivation for managers to inflate earnings. As another example, an investment contract could contain provisions giving investors the option to recover all or part of their investment if certain financial triggers occur. Such provisions could motivate the investee’s managers to manipulate reported results to avoid the financial triggers.

Because the financial reports prepared by the investees or borrowers directly affect the contractual outcomes—potentially creating a motivation for misreporting—investors and lenders are motivated to monitor financial reports and to ensure that they are high quality.

EXAMPLE 4

Financial Reporting Manipulation: Motivations and Disciplining Mechanisms

For each of the following two scenarios, identify (1) factors that might motivate the company’s managers to manipulate reported financial amounts and (2) applicable mechanisms that could discipline poor financial reporting quality.

1. ABC Co. is a private company. Bank NTBig has made a loan to ABC Co. ABC is required to maintain a minimum 2.0 interest coverage ratio. In its most recent financial reports, ABC reported earnings before interest and taxes of USD1,200 and interest expense of USD600. In the report’s notes, the company discloses that it changed the estimated useful life of its property, plant, and equipment during the year. Depreciation was approximately USD150 lower as a result of this change in estimate.

Solution:

The need to maintain a minimum interest coverage ratio of 2.0 might motivate ABC’s managers to manipulate reported financial amounts. The company’s coverage ratio based on the reported amounts is exactly equal to 2.0. If ABC’s managers had not changed the estimated useful life of the property, plant, and equipment, the coverage ratio would have fallen below the required level.

EBIT, as reported	USD1,200
Impact on depreciation expense of changed assumptions about useful life	150
EBIT, as adjusted	USD1,050
Interest expense	USD600
Coverage ratio, as reported	2.00
Coverage ratio, as adjusted	1.75

The potential disciplining mechanisms include the auditors, who will assess the reasonableness of the depreciable lives estimates. In addition, the lenders will carefully scrutinize the change in estimate because the company only barely achieved the minimum coverage ratio and would not have achieved the minimum without the change in accounting estimate.

2. DEF Co. is a publicly traded company. For the most recent quarter, the average of analysts’ forecasts for earnings per share was USD2.50. In its quarterly earnings announcement, DEF reported net income of USD3,458,780. The number of common shares outstanding was 1,378,000. DEF’s main product is a hardware device that includes a free two-year service contract in the selling price. Based on management estimates, the company allocates a portion of revenues to the hardware device, which it recognizes immediately, and a portion to the service contract, which it defers and recognizes over the two years of the contract. Based on the disclosures, a higher percentage of revenue was allocated to hardware than in the past, with an estimated after-tax impact on net income of USD27,000.

Solution:

The desire to meet or exceed the average of analysts’ forecasts for earnings per share might motivate DEF Co.’s managers to manipulate reported financial amounts. As illustrated in the following calculations, the impact of allocating a greater portion of revenue to hardware enabled the company to exceed analysts’ earnings per share forecasts by USD0.01.

Net income, as reported	USD3,458,780
Impact on gross profit of changed revenue recognition, net of tax	27,000
Net income, as adjusted	USD3,431,780
Weighted average number of shares	1,378,000

Earnings per share, as reported	USD2.51
Earnings per share, as adjusted	USD2.49

Potential disciplining mechanisms include the auditors, market regulators, financial analysts, and financial journalists.

9. DETECTION OF FINANCIAL REPORTING QUALITY ISSUES: INTRODUCTION AND PRESENTATION CHOICES

LEARNING OUTCOME

The candidate should be able to:

- ☐ **describe presentation choices, including non-GAAP measures, that could be used to influence an analyst's opinion**

Choices in the application of accounting standards abound, which is perhaps one reason why accounting literature and texts are so voluminous. Compounding the complexity, measurement often depends on estimates of economic phenomena. Two estimates might be justifiable, but they may have significantly different effects on the company's financial statements. As discussed earlier, the choice of a particular estimate may depend on the motivations of the reporting company's managers. With many choices available, and the inherent flexibility of estimates in the accounting process, managers have many tools for managing and meeting analysts' expectations through financial reporting.

An understanding of the choices that companies make in financial reporting is fundamental to evaluating the overall quality—both financial reporting and earnings quality—of the reports produced. Choices exist both in how information is presented (financial reporting quality) and in how financial results are calculated (earnings quality). Choices in presentation (financial reporting quality) may be transparent to investors. Choices in the calculation of financial results (earnings quality), however, are more difficult to discern because they can be deeply embedded in the construction of reported financial results.

The availability of accounting choices enables managers to affect the reporting of financial results. Some choices increase performance and financial position in the reporting period (aggressive choices), and others increase them in later periods (conservative choices). A manager who wants to increase performance and financial position in the reporting period could:

- recognize revenue prematurely;
- use non-recurring transactions to increase profits;
- defer expenses to later periods;
- measure and report assets at higher values; and/or
- measure and report liabilities at lower values.
- A manager who wants to increase performance and financial position in a later period could:
- defer current income to a later period (save income for a "rainy day"); and/or
- recognize future expenses in a current period, setting the table for improving future performance.

This lesson describes some of the potential choices for how information is presented and how accounting elements [assets, liabilities, owners' equity, revenue and gains (income), and expenses and losses] are recognized, measured, and reported. In addition to choices within GAAP, companies may prepare fraudulent reports. For example, these reports may include non-existent revenue or assets. Later lessons discuss warning signs that may indicate poor-quality financial reports.

9.1. Presentation Choices

The technology boom of the 1990s and the internet bubble of the early 2000s featured companies, popular with investors, that often shared the same characteristic: They could not generate enough current earnings to justify their stock prices using the traditional price-to-earnings ratio (P/E) approaches to valuation. Many investors chose to explain these apparent anomalies by rationalizing that the old focus on profits and traditional valuation approaches no longer applied to such companies. Strange new metrics for determining operating performance emerged. Website operators spoke of the “eyeballs” they had captured in a quarter, or the “stickiness” of their websites for web surfers’ visits. Various versions of “pro forma earnings”—that is, “non-GAAP earnings measures”—became a financial reporting staple of the era.

Many technology companies were accomplished practitioners of pro forma reporting, but they were not the first to use it. In the early 1990s, downsizing of large companies was a commonplace event, and massive restructuring charges obscured the operating performance at many established companies. For example, as it learned to cope in a world that embraced the personal computer rather than mainframe computing, International Business Machines (IBM) reported massive restructuring charges in 1991, 1992, and 1993: USD3.7 billion, USD11.6 billion, and USD8.9 billion, respectively. IBM was not alone. Sears incurred USD2.7 billion of restructuring charges in 1993, and AT&T reported restructuring charges of USD7.7 billion in 1995. These events were not isolated; restructuring charges were a standard quarterly reporting event. To counter perceptions that their operations were floundering, and supposedly to assist investors in evaluating operating performance, companies often sanitized earnings releases by excluding restructuring charges in pro forma measures of financial performance.

Accounting principles for reporting business combinations also played a role in boosting the popularity of pro forma earnings. Before 2001, acquisitions of one company by another often resulted in goodwill amortization charges that made subsequent earnings reports look weak. Complicating matters, there were two accounting methods for recording acquisitions: pooling-of-interests and purchase methods. The now-extinct pooling-of-interests treatment was difficult for companies to achieve because of the many restrictive criteria for its use, but it was greatly desired because it did not result in goodwill amortization charges. In the technology boom period, acquisitions were common and many were reported as purchases, with consequential goodwill amortization dragging down earnings for as long as 40 years under the then-existing rules. Acquisitive companies reporting under purchase accounting standards perceived themselves to be at a reporting disadvantage compared with companies able to apply pooling-of-interests. They responded by presenting earnings adjusted for the exclusion of amortization of intangible assets and goodwill.

Because investors try to make intercompany comparisons on a consistent basis, earnings before interest, taxes, depreciation, and amortization (EBITDA) has become an extremely popular performance measure. EBITDA is widely viewed as eliminating noisy reporting signals. That noise may be introduced by different accounting methods among companies for depreciation, amortization of intangible assets, and restructuring charges. Companies may construct and report their own version of EBITDA, sometimes referring to it as “adjusted EBITDA,” by adding to the list of items to exclude from net income. Items that analysts might encounter include the following:

- rental payments for operating leases, resulting in EBITDAR (earnings before interest, taxes, depreciation, amortization, and rentals);
- equity-based compensation, usually justified on the grounds that it is a non-cash expense;
- acquisition-related charges;
- impairment charges for goodwill or other intangible assets;
- impairment charges for long-lived assets;
- litigation costs; and
- loss/gain on debt extinguishments.

Among other incentives for the spread of non-GAAP earnings measures are loan covenants. Lenders may make demands on a borrowing company that require achieving and maintaining performance criteria that use GAAP net income as a starting point but arrive at a measure suitable to the lender. The company may use this measure as its preferred non-GAAP metric in earnings releases and also when describing its liquidity or solvency situation in the management commentary (called management discussion and analysis in the United States).

As mentioned earlier, if a company uses a non-GAAP financial measure in an SEC filing, it must display the most directly comparable GAAP measure with equal prominence and provide a reconciliation between the two. Management must explain why it believes that the non-GAAP financial measure provides useful information regarding the company’s financial condition and operations. Management must also disclose additional purposes, if material, for which it uses the non-GAAP financial measures.

Similarly, IFRS requires a definition and explanation of any non-IFRS measures included in financial reports, including why the measure is potentially relevant to users. Management must provide reconciliations of non-IFRS measures with IFRS measures. The concern is that management may use non-GAAP measures to distract attention from GAAP measures.

The SEC intended that the definition of non-GAAP financial measures would capture all measures with the effect of depicting either:

- a measure of performance that differs from that presented in the financial statements, such as income or loss before taxes or net income or loss, as calculated in accordance with GAAP; or
- a measure of liquidity that differs from cash flow or cash flow from operations computed in accordance with GAAP.¹⁹

The SEC prohibits the exclusion of charges or liabilities requiring cash settlement from any non-GAAP liquidity measures, other than EBIT and EBITDA. Also prohibited is the calculation of a non-GAAP performance measure intended to eliminate or smooth items tagged as non-recurring, infrequent, or unusual when such items are very likely to occur again. The SEC views the period within two years of either before or after the reporting date as the relevant time frame for considering whether a charge or gain is a recurring item. [Example 5](#) describes a case of misuse and misreporting of non-GAAP measures.

EXAMPLE 5

Misuse and Misreporting of Non-GAAP Measures

Groupon is an online discount merchant. In the company's initial S-1 registration statement in 2011, then-CEO Andrew Mason gave prospective investors an upfront warning in a section entitled "We don't measure ourselves in conventional ways", which described Groupon's adjusted consolidated segment operating income (adjusted CSOI) measure. [Exhibit 12](#) provides excerpts from a section entitled "Non-GAAP Financial Measures," which offered a more detailed explanation. [Exhibit 13](#), also from the initial registration statement, shows a reconciliation of CSOI to the most comparable US GAAP measure. In its review, the SEC took the position that online marketing expenses were a recurring cost of business. Groupon responded that the marketing costs were similar to acquisition costs, not recurring costs, and that "we'll ramp down marketing just as fast as we ramped it up, reducing the customer acquisition part of our marketing expenses" as time passes.²⁰

Eventually, and after much negative publicity, Groupon changed its non-GAAP measure. [Exhibit 14](#) shows an excerpt from the final prospectus filed in November, after the SEC's review. Use the three exhibits to answer the questions that follow.

Exhibit 12: Groupon's "Non-GAAP Financial Measures"

Disclosures from June S-1 Filing

Adjusted CSOI is operating income of our two segments, North America and International, adjusted for online marketing expense, acquisition-related costs and stock-based compensation expense. Online marketing expense primarily represents the cost to acquire new subscribers and is dictated by the amount of growth we wish to pursue. Acquisition-related costs are non-recurring non-cash items related to certain of our acquisitions. Stock-based compensation expense is a non-cash item. We consider Adjusted CSOI to be an important measure of the performance of our business as it excludes expenses that are non-cash or otherwise not indicative of future operating expenses. We believe it is important to view Adjusted CSOI as a complement to our entire consolidated statements of operations.

Our use of Adjusted CSOI has limitations as an analytical tool, and you should not consider this measure in isolation or as a substitute for analysis of our results as reported under GAAP. Some of these limitations are:

- Adjusted CSOI does not reflect the significant cash investments that we currently are making to acquire new subscribers;
- Adjusted CSOI does not reflect the potentially dilutive impact of issuing equity-based compensation to our management team and employees or in connection with acquisitions;
- Adjusted CSOI does not reflect any interest expense or the cash requirements necessary to service interest or principal payments on any indebtedness that we may incur;
- Adjusted CSOI does not reflect any foreign exchange gains and losses;
- Adjusted CSOI does not reflect any tax payments that we might make, which would represent a reduction in cash available to us;
- Adjusted CSOI does not reflect changes in, or cash requirements for, our working capital needs; and

- Other companies, including companies in our industry, may calculate Adjusted CSOI differently or may use other financial measures to evaluate their profitability, which reduces the usefulness of it as a comparative measure.

Because of these limitations, Adjusted CSOI should not be considered as a measure of discretionary cash available to us to invest in the growth of our business. When evaluating our performance, you should consider Adjusted CSOI alongside other financial performance measures, including various cash flow metrics, net loss and our other GAAP results.

Exhibit 13: Groupon's Adjusted CSOI

Excerpt from June S-1 Filing

The following is a reconciliation of CSOI to the most comparable US GAAP measure, "loss from operations," for the years ended December 31, 2008, 2009, and 2010 and the three months ended March 31, 2010 and 2011:

(in USD thousands)	Year Ended 31 December			Three Months Ended 31 March	
	2008	2009	2010	2010	2011
(Loss) Income from operations	(1,632)	(1,077)	(420,344)	8,571	(117,148)
Adjustments:					
Online marketing	162	4,446	241,546	3,904	179,903
Stock-based compensation	24	115	36,168	116	18,864
Acquisition- related	—	—	203,183	—	—
Total adjustments	186	4,561	480,897	4,020	198,767
Adjusted CSOI	(1,446)	3,484	60,553	12,591	81,619

Exhibit 14: Groupon's CSOI

Excerpt from Revised S-1 Filing

The following is a reconciliation of CSOI to the most comparable US GAAP measure, "loss from operations," for the years ended December 31, 2008, 2009, and 2010 and the nine months ended September 30, 2010 and 2011:

(in USD thousands)	Year Ended 31 December			Nine Months Ended 30 September	
	2008	2009	2010	2010	2011
Loss from operations	(1,632)	(1,077)	(420,344)	(84,215)	(218,414)
Adjustments:					
Stock-based compensation	24	115	36,168	8,739	60,922
Acquisition- related	—	—	203,183	37,844	(4,793)
Total adjustments	24	115	239,351	46,583	56,129
CSOI	(1,608)	(962)	(180,993)	(37,632)	(162,285)

1. What cautions did Groupon include along with its description of the adjusted CSOI metric?

Solution:

Groupon cautioned that the adjusted CSOI metric should not be considered in isolation, should not be considered as a substitute for analysis using GAAP results, and "should not be considered a measure of discretionary cash flow." The company lists numerous limitations, primarily citing items that adjusted CSOI did not reflect.

2. Groupon excludes “online marketing” from adjusted CSOI. How does the exclusion of this expense compare with the SEC’s limits on non-GAAP performance measures?

Solution:

The SEC specifies that non-GAAP measures should not eliminate items tagged as non-recurring, infrequent, or unusual when such items may be very likely to occur again. Because the online marketing expense occurred in every period reported and is likely to occur again, exclusion of this item appears contrary to SEC requirements.

3. In the first quarter of 2011, what was the effect of excluding online marketing expenses on the calculation of adjusted CSOI?

Solution:

As shown in [Exhibit 13](#), in the first quarter of 2011, the exclusion of the online marketing expense was enough to swing the company from a net loss under US GAAP reporting to a profit—at least, a profit as defined by adjusted CSOI. Using adjusted CSOI as a performance measure, the company showed results that were 35 percent higher for the first *quarter* of 2011 compared with the entire previous *year*.

4. For 2010, how did results under the revised non-GAAP metric compare with the originally reported metric?

Solution:

As shown in [Exhibit 14](#), the revised metric is now called CSOI and no longer refers to adjusted CSOI. For 2010, results under the revised non-GAAP metric, which includes online marketing costs, shows a loss of USD180,993,000 instead of a profit of USD60,553,000.

In [Example 5](#), Groupon changed its reporting and corrected the non-GAAP metric that the SEC had identified as misleading. In other cases, the SEC has pursued enforcement actions against companies for reporting misleading non-GAAP information. One such action was brought in 2009 against SafeNet Inc., in which the SEC charged the company with improperly classifying ordinary operating expenses as non-recurring. This related to the integration of an acquired company and exclusion of the expenses from non-GAAP earnings to exceed earnings targets. A second action was brought by the SEC in 2017 against MDC Partners Inc. (MDCA) for improper reconciliation of a non-GAAP measure and for improperly displaying the non-GAAP measure with greater prominence in its earnings releases. The case was brought after the company agreed to follow the rules but then failed to do so, as evidenced by the remark in the SEC’s action: “Despite agreeing to comply with non-GAAP financial measure disclosure rules in December 2012 correspondence with the [SEC’s] Division of Corporation Finance, MDCA continued to violate those rules for six quarters.” [Exhibit 15](#) presents the headline and subheadings for one of MDC Partners’ earnings announcements that was the subject of the enforcement action.

Exhibit 15: Excerpt from MDC Partners Inc. Press Release

This excerpt shows the headline, subheads, and lead sentence of the company’s press release announcing periodic earnings.

SEC Form 8-K filed 24 April 2014

MDC PARTNERS INC. REPORTS RECORD RESULTS FOR THE THREE MONTHS ENDED MARCH 31, 2014

ORGANIC REVENUE GROWTH OF 8.3%, EBITDA GROWTH OF 18.1% AND 90 BASIS POINTS OF MARGIN IMPROVEMENT

FREE CASH FLOW GROWTH OF 34.0%

INCREASED 2014 GUIDANCE IMPLIES YEAR-OVER-YEAR EBITDA GROWTH OF +13.5% TO +16.1%, MARGIN IMPROVEMENT OF 60 TO 70 BASIS POINTS, AND FREE CASH FLOW GROWTH OF +15.8% TO +20.2%

FIRST QUARTER HIGHLIGHTS

- Revenue increased to \$292.6 million from \$265.6 million, an increase of 10.1%
- Organic revenue increased 8.3%
- EBITDA increased to \$36.4 million from \$30.8 million, an increase of 18.1%

- EBITDA margin increased 90 basis points to 12.5% from 11.6%
- Free Cash Flow increased to \$20.6 million from \$15.4 million, an increase of 34.0%
- Net New Business wins totaled \$24.4 million

NEW YORK, NY (April 24, 2014) – MDC Partners Inc. (NASDAQ: MDCA; TSX: MDZ.A) today announced financial results for the three months ended March 31, 2014.

...

In general, management may choose to construct non-GAAP financial measures not only to help investors better understand the company's performance but also to paint a more flattering picture of its performance. In some cases, management may attempt to present non-GAAP measures in a way that diverts attention from the standards-compliant financial information that it is required to present.

10. ACCOUNTING CHOICES AND ESTIMATES

LEARNING OUTCOME

The candidate should be able to:

- ☐ **describe accounting methods (choices and estimates) that could be used to manage earnings, cash flow, and balance sheet items**

Management's accounting policies and choices do not necessarily involve complex accounting standards. Something as simple as the shipping terms for goods delivered to customers can have a profound effect on the timing of revenue. On the last day of the first quarter, suppose a company ships USD10,000 of goods to a customer on the terms "free on board (FOB) shipping point," arriving the next day. This shipping term means that the customer takes title to the goods, and bears the risk of loss, at the time the goods leave the seller's loading dock. Barring any issues with collectability of the receivable, or a likelihood of a return, the seller would be able to recognize revenue on the sale along with the associated profit. That revenue and profit would be recognized in the first quarter of the year. If the point at which the goods' title transfers to the customer is changed to "FOB destination," then the revenue pattern will be completely different. Under these terms, the title—and risk of loss—transfers to the customer when the goods arrive at their destination, which is the customer's address. The seller cannot recognize the sale and profit until the shipment arrives the following day, which is the start of a new accounting period.

A simple change in shipping terms can make the difference between revenue and profits in the reporting period or postponing them until the next period. Shipping terms can also influence management behavior. To "make the numbers," managers might push product out the door prematurely under FOB shipping point arrangements to reflect as much revenue as possible in the reporting period. Alternatively, in the case of an over-abundance of orders, the company could run the risk of exceeding analysts' consensus estimates by a large margin. Management might be uncomfortable with this situation because investors might extrapolate too much from one reporting period in which expectations were exceeded. Management might want to prevent investors from becoming too optimistic and, if possible, delay revenue recognition until the next quarter. This result could be accomplished by fulfilling customer orders by initiating delivery on the last day of the quarter, with shipping terms set as FOB destination. By doing so, title would transfer in the next accounting period. Another possibility in this scenario is that if the customers insisted on FOB shipping point terms, the selling company could simply delay shipment until after the close of the quarter.

This illustration also highlights a difficult distinction for investors to make. A company may use accounting as a tool to aggressively promote earnings growth—as in the example with the premature shipment of goods with FOB shipping point terms—but it may be aggressively managing the business flow by slacking off on shipping goods when business is "too good," as in the second example. In either case, a desired management outcome is obtained by a simple change in shipping terms. Yet, many investors might be inclined to say that the second example is a conservative kind of earnings management and accept it, even though it artificially masks the actual economic activity that occurred at the time.

10.1. How Accounting Choices and Estimates Affect Earnings and Balance Sheets

Assumptions about inventory cost flows provide another example of how accounting choices can affect financial reporting. Companies may assume that their purchases of inventory items are sold to customers on a first-in, first-out

(FIFO) basis, with the result that the remaining inventory reflects the most recent costs. Alternatively, they may assume that their purchases of inventory items are sold to customers on a weighted-average cost basis. [Example 6](#) makes the point that merely choosing a cost flow assumption can affect profitability.

EXAMPLE 6

Effect of Cost Flow Assumption

A company starts operations with no inventory at the beginning of a fiscal year and makes purchases of a good for resale five times during the period at increasing prices. Each purchase is for the same number of units of the good. The purchases, and the cost of goods available for sale, appear in the following table. Notice that the price per unit has increased by 140 percent by the end of the period.

	Units	Price	Cost
Purchase 1	5	USD100	USD500
Purchase 2	5	150	750
Purchase 3	5	180	900
Purchase 4	5	200	1,000
Purchase 5	5	240	1,200
Cost of goods available for sale			USD4,350

During the period, the company sells, at USD250 each, all of the goods purchased except for five of them. Although the ending inventory consists of five units, the cost attached to those units can vary greatly.

1. What are the ending inventory and cost of goods sold if the company uses the FIFO method of inventory costing?

Solution:

The ending inventory and cost of goods sold if the company uses the FIFO method of inventory costing are USD1,200 and USD3,150.
2. What are the ending inventory and cost of goods sold if the company uses the weighted-average method of inventory costing?

Solution:

The ending inventory and cost of goods sold if the company uses the weighted-average method of inventory costing are USD870 and USD3,480.
3. Compare cost of goods sold and gross profit calculated under the two methods.

Solution:

The following table shows how the choice of inventory costing methods—FIFO versus weighted average—affects the cost of goods sold and gross profit.

Cost Flow Assumption	FIFO	Weighted Average
Cost of goods available for sale	USD4,350	USD4,350
Ending inventory (5 units)	(1,200)	(870)
Cost of goods sold	USD3,150	USD3,480
Sales	USD5,000	USD5,000
Cost of goods sold	3,150	3,480
Gross profit	USD1,850	USD1,520
Gross profit margin	37.0%	30.4%

Note: Average inventory cost is calculated as Cost of goods available for sale/Units purchased = \$4,350/25 = \$174. There are five units in ending inventory, yielding an inventory value of \$870.

Depending on which cost flow assumption the company uses, the end-of-period inventory is either USD870 (under the weighted-average method) or USD1,200 (under FIFO). The choice of method results in a difference of USD330 in gross profit and 6.6 percent in gross profit margin.

The previous example is simplified and extreme for purposes of illustration clarity, but the point is important: Management's choice among acceptable inventory assumptions and methods affects profit. The selection of an inventory costing method is a policy decision, and companies cannot arbitrarily switch from one method to another. The selection does matter to profitability, however, and it also matters to the balance sheet.

In periods of changing prices, the FIFO cost assumption will provide a more current picture of ending inventory value, because the most recent purchases will remain in inventory. The balance sheet will be more relevant to investors. Under the weighted-average cost assumption, however, the balance sheet will display a blend of old and new costs. During inflationary periods, the value of the inventory will be understated: The company will not be able to replenish its inventory at the value shown. At the same time, the weighted-average inventory cost method ensures that the more current costs are shown in cost of sales, making the income statement more relevant than under the FIFO assumption. Trade-offs exist, and investors should be aware of how accounting choices affect financial reports. High-quality financial reporting provides users with sufficient information to assess the effects of accounting choices.

Estimates abound in financial reporting because of the use of accrual accounting, which attempts to show the effects of all economic events on a company during a particular period. Accrual accounting stands in contrast to cash basis accounting, which shows only the cash transactions conducted by a company. Although a high degree of certainty exists with reporting only cash transactions, much information is hidden. For instance, a company with growing revenues that makes the majority of its sales on credit would be understating its revenues for each period if it reported only cash transactions. On an accrual basis, revenues reflect all transactions that occurred, whether they transacted on a cash basis or credit-extended basis. Estimates enter the process because some facts related to events occurring in a particular period might not yet be known. Estimates can be well grounded in reality and applied to present a complete picture of the events affecting a company, or they can be management tools for achieving a desired financial picture.

To illustrate how estimates can affect financial reporting, consider sales made on credit. A company sells USD1,000,000 of merchandise on credit and records the sale just before year end. Under accrual accounting, that amount is included in revenues and accounts receivable. The company's managers know from experience that they will never collect every dollar of the accounts receivable. Past experience is that, on average, only 97 percent of accounts receivable is collected. The company would estimate an amount of the uncollectible accounts at the time the sales occur and record an uncollectible accounts expense of USD30,000, lowering earnings. The other side of the entry would be to establish an allowance for uncollectible accounts of USD30,000. This allowance would be a contra asset account, presented as an offset to accounts receivable. The accounts receivable, net of the allowance for uncollectible accounts, would be stated at USD970,000, which is the amount of cash the company ultimately expects to receive. If cash-basis accounting had been used, no revenues or accounts receivable would have been reported even though sales of merchandise had occurred. Accrual accounting, which contains estimates about future events, provides a much fuller picture of what transpired in the period than pure cash-basis accounting.

Yet, accrual accounting poses temptations to managers to manage the numbers, rather than to manage the business. Suppose a company's managers realize that the company will not meet analysts' consensus estimates in a particular quarter, and further, their bonus pay is dependent on reaching specified earnings targets. By offering special payment terms, or discounts, the managers may induce customers to take delivery of products that they normally would not order, so they could ship the products on FOB shipping point terms and recognize the revenues in the current quarter. They could even be so bold as to ship the goods under those terms even if the customer did not order them, in the hope that the customer would keep them or, at worst, return them in the next accounting period. Their aim would be to move the product off the company's property with FOB shipping point terms.

To further improve earnings in order to meet the consensus estimates, the company's managers might revise their estimate of the uncollectible accounts. The company's collection history shows a typical non-collection rate of 3 percent of sales, but the managers might rationalize the use of a 2 percent non-collection rate. This change will reduce the allowance for uncollectible accounts and the expense reported for the period. The managers might be able to justify the reduction on the grounds that the sales occurred in a part of the country that was experiencing an improved economic outlook, or that the company's collection history had been biased by the inclusion of a prolonged period of economic downturn. Whatever the justification, it would be hard to prove that the new estimate was completely right or wrong until time had passed. Because proof of the reliability of estimates is rarely available at the time the estimate is recorded, managers have a readily available means for manipulating earnings at their discretion.

ConAgra Foods Inc. provides an example of how the allowance for uncollectible accounts may be manipulated to manage earnings.²¹ A subsidiary, called United Agri-Products (UAP), engaged in several improper accounting practices, one of them being the understatement of uncollectible account expenses for several years. [Exhibit 16](#) presents an excerpt from the SEC's Accounting and Auditing Enforcement Release.

Exhibit 16: SEC's Accounting and Auditing Enforcement Release Regarding United Agri-Products

Generally, UAP's policy required that accounts which were past due between 90 days and one year should be reserved at 50%, and accounts over one year past due were to be reserved at 100%.

... In FY 1999 and continuing through FY 2000, UAP had substantial bad debt problems. In FY 2000, certain former UAP senior executives were informed that UAP needed to record an additional \$50 million of bad debt expense. Certain former UAP senior executives were aware that in FY 1999 the size of the bad debt at certain

IOCs had been substantial enough that it could have negatively impacted those IOC's ability to achieve PBT (profits before taxes) targets. In addition, just prior to the end of UAP's FY 2000, the former UAP COO (chief operating officer), in the presence of other UAP employees, ordered that UAP's bad debt reserve be reduced by \$7 million in order to assist the Company in meeting its PBT target for the fiscal year.

... At the end of FY 2000, former UAP senior executives reported financial results to ConAgra which they knew, or were reckless in not knowing, overstated UAP's income before income taxes because UAP had failed to record sufficient bad debt expense. The misconduct with respect to bad debt expense caused ConAgra to overstate its reported income before income taxes by \$7 million, or 1.13%, in FY 2000. At the Agricultural Products' segment level, the misconduct caused that segment's reported operating profit to be overstated by 5.05%.

Deferred-tax assets provide a similar example of choices in estimates affecting the earnings outcome. Deferred-tax assets may arise when a company reports a net operating loss under tax accounting rules. A company may record a deferred-tax asset based on the expectation that losses in the reporting period will offset expected future profits and reduce the company's future income tax liability. Accounting standards require that the deferred tax asset be reduced by a "valuation allowance" to account for the possibility that the company will not be able to generate enough profit to use all of the available tax benefits.²²

Assume a company loses EUR1 billion in 2012, generating a net operating loss of the same amount for tax purposes. The company's income tax rate is 25 percent, and it will be able to apply the net operating loss to its taxable income for the next 10 years. The net operating loss results in a deferred tax asset with a nominal value of EUR250 million ($25\% \times \text{EUR}1,000,000,000$). Initial recognition would result in a deferred tax asset of EUR250 million and a credit to deferred tax expense of EUR250 million. The company must address the question of whether or not the EUR250 million will ever be completely applied to future income. It may be experiencing increased competition and other circumstances that resulted in the EUR1 billion loss, and it may be unreasonable to assume it will have taxable income against which to apply the loss. In fact, the company's managers might believe it is reasonable to assume only that it will survive for five years, and with marginal profitability. The EUR250 million deferred tax asset is thus overstated if no valuation allowance is recorded to offset it.

The managers believe that only EUR100 million of the net operating losses actually will be applied to the company's taxable income. That belief implies that only EUR25 million of the tax benefits will ever be realized. The deferred tax assets reported on the balance sheet should not exceed this amount. The company should record a valuation allowance of EUR225 million, which would offset the deferred tax asset balance of EUR250 million, resulting in a net deferred tax asset balance of EUR25 million. There also would be a EUR225 million credit to the deferred tax provision. It is important to understand that the valuation allowance should be revised whenever facts and circumstances change.

The ultimate value of the deferred tax asset is driven by management's outlook for the future—and that outlook may be influenced by other factors. If the company needs to stay in compliance with debt covenants and needs every euro of value that can be justified by the outlook, its managers may take a more optimistic view of the future and keep the valuation allowance artificially low (in other words, the net deferred tax asset high).

PowerLinx Inc. provides an example of how over-optimism about the realizability of a deferred tax asset can lead to misstated financial reports. PowerLinx was a maker of security video cameras, underwater cameras, and accessories. Aside from fraudulently reporting 90 percent of its fiscal year 2000 revenue, PowerLinx had problems with valuation of its deferred tax assets. [Exhibit 17](#) provides an excerpt from the SEC's Accounting and Auditing Enforcement Release with emphasis added.²³

Exhibit 17: SEC's Accounting and Auditing Enforcement Release Regarding PowerLinx

PowerLinx improperly recorded on its fiscal year 2000 balance sheet a deferred tax asset of \$1,439,322 without any valuation allowance. The tax asset was material, representing almost forty percent of PowerLinx's total assets of \$3,841,944. PowerLinx also recorded deferred tax assets of \$180,613, \$72,907, and \$44,921, respectively, in its financial statements for the first three quarters of 2000.

PowerLinx did not have a proper basis for recording the deferred tax assets. The company had accumulated significant losses in 2000 and had no historical operating basis from which to conclude that it would be profitable in future years. Underwater camera sales had declined significantly and the company had devoted most of its resources to developing its SecureView product. The sole basis for PowerLinx's "expectation" of future profitability was the purported \$9 million backlog of SecureView orders, which management assumed would generate taxable income; however, this purported backlog, which predated Bauer's hiring, did not reflect actual demand for SecureView cameras and, consequently, was not a reasonable or reliable indicator of future profitability.

Another example of misstated financial results caused by improper reflection of the realizability of a deferred tax asset occurred with Hampton Roads Bankshares Inc. (HRBS), a commercial bank with deteriorating loan portfolio quality and commensurate losses in the years following the financial crisis. The company reported a deferred tax asset related to its loan losses; however, it did not establish a valuation allowance against its deferred tax asset. This decision was based on dubious projections indicating that the company would earn the necessary future taxable income “to fully utilize the [deferred tax asset] DTA over the applicable carry-forward period.”²⁴ Over time, it became clear that the earnings projections were not realistic, and ultimately the company restated its financial results to include a valuation allowance against almost the entire deferred tax asset. [Exhibit 18](#) presents an excerpt from the company’s amended Form 10-Q/A containing the restatement.

Exhibit 18: Excerpt from Hampton Roads Bankshares, Inc. Form 10-Q/A filed August 13, 2010

NOTE B – RESTATEMENT OF CONSOLIDATED FINANCIAL STATEMENTS

Subsequent to filing the Company’s annual report on Form 10-K for the year ended December 31, 2009 and its Form 10-Q for the three months ended March 31, 2010 the Company determined that a valuation allowance on its deferred tax assets should be recognized as of December 31, 2009. The Company decided to establish a valuation allowance against the deferred tax asset because it is uncertain when it will realize this asset.

Accordingly, the December 31, 2009 consolidated balance sheet and the March 31, 2010 consolidated financial statements have been restated to account for this determination. The effect of this change in the consolidated financial statements was as follows (in thousands, except per share amounts).

Consolidated Balance Sheet at 31 March 2010

	As Reported	Adjustment	As Restated
Deferred tax assets, net	USD70,323	USD(70,323)	—
Total assets	3,016,470	(70,323)	USD2,946,147
Retained earnings deficit	(158,621)	(70,323)	(228,944)
Total shareholder’s equity	156,509	(70,323)	86,186
Total liabilities and shareholders’ equity	3,016,470	(70,323)	2,946,147

Consolidated Balance Sheet at 31 December 2009

	As Reported	Adjustment	As Restated
Deferred tax assets, net	USD56,380	USD(55,983)	USD397
Total assets	2,975,559	(55,983)	2,919,576
Retained earnings deficit	(132,465)	(55,983)	(188,488)
Total shareholder’s equity	180,996	(55,983)	125,013
Total liabilities and shareholders’ equity	2,975,559	(55,983)	2,919,576

Another example of how choices and estimates can affect reported results lies in the selection of a depreciation method for allocating the cost of long-lived assets to accounting periods subsequent to their acquisition. A company’s managers may choose to depreciate long-lived assets (1) on a straight-line basis, with each year bearing the same amount of depreciation expense; (2) using an accelerated method, with greater depreciation expense recognition in the earlier part of an asset’s life; or (3) using an activity-based depreciation method, which allocates depreciation expense based on units of use or production. Depreciation expense is affected by another set of choices and estimates regarding the salvage value of the assets being depreciated. A salvage value of zero will always increase depreciation expense under any method compared with the choice of a non-zero salvage value.

Assume a company invests USD1,000,000 in manufacturing equipment and expects it to have a useful economic life of 10 years. During its expected life, the equipment will produce 400,000 units of product, or USD2.50 depreciation expense per unit produced. When it is disposed of at the end of its expected life, the company’s managers expect to realize no value for the equipment. [Exhibit 19](#) shows the differences in the three alternative methods of depreciation: straight-line, accelerated on a double-declining balance basis, and units-of-production method, with no salvage value assumed at the end of the equipment’s life.

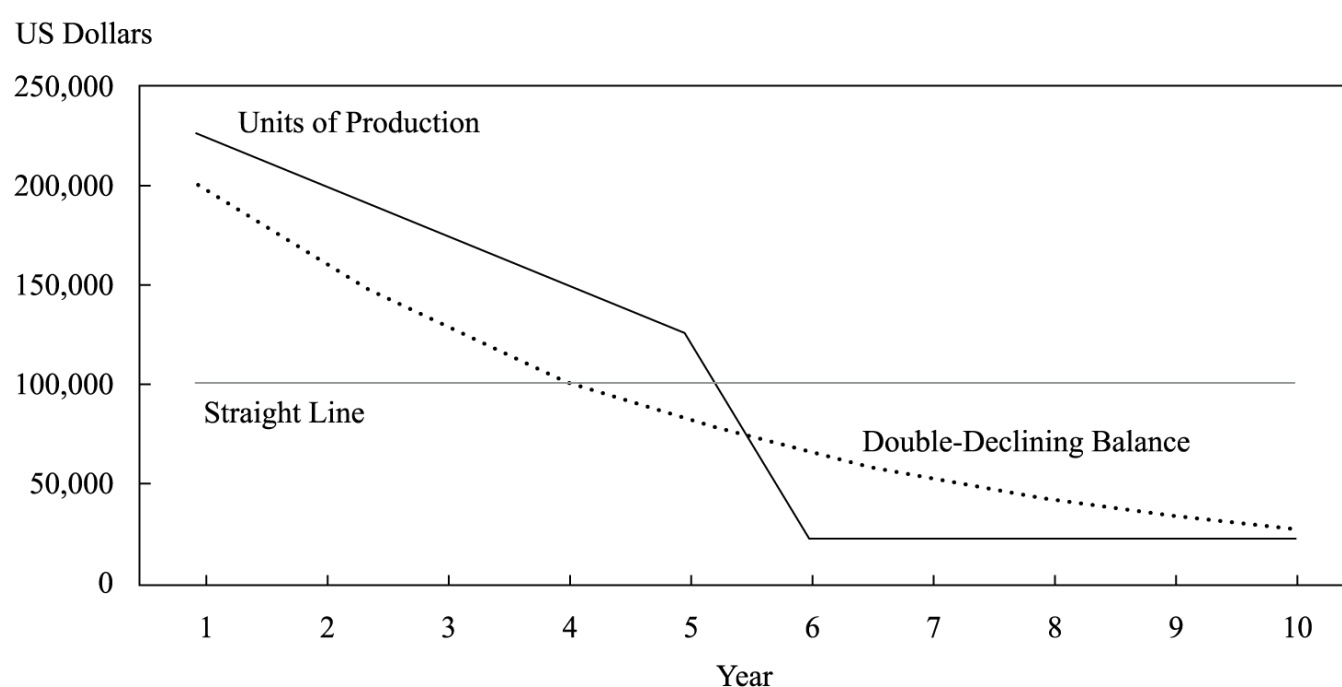
Exhibit 19: Alternative Methods of Depreciation

Straight-Line Method		Double-Declining Balance Method			Units-of-Production Method		
Year	Depreciation Expense	Balance	Declining Balance Rate	Depreciation Expense	Units Produced	Depreciation Rate/Unit	Depreciation Expense
1	USD100,000	USD1,000,000	20%	USD200,000	90,000	USD2.50	USD225,000
2	100,000	800,000	20%	160,000	80,000	USD2.50	200,000
3	100,000	640,000	20%	128,000	70,000	USD2.50	175,000
4	100,000	512,000	20%	102,400	60,000	USD2.50	150,000
5	100,000	409,600	20%	81,920	50,000	USD2.50	125,000
6	100,000	327,680	20%	65,536	10,000	USD2.50	25,000
7	100,000	262,144	20%	52,429	10,000	USD2.50	25,000
8	100,000	209,715	20%	41,943	10,000	USD2.50	25,000
9	100,000	167,772	20%	33,554	10,000	USD2.50	25,000
10	100,000	134,218	20%	26,844	10,000	USD2.50	25,000
Total	USD1,000,000			USD892,626	400,000		USD1,000,000

The straight-line method allocates the cost of the equipment evenly to all 10 years of the equipment's life. The double-declining balance method will have a higher allocation of cost to the earlier years of the equipment's life. As its name implies, the depreciation expense will decline in each succeeding year because it is based on a fixed rate applied to a declining balance. The rate used was double the straight-line rate, but it could have been any other rate that the company's managers believed was representative of the way the actual equipment depreciation occurred. Notice that the double-declining balance method also results in an incomplete depreciation of the machine at the end of 10 years; a balance of USD107,374 (= USD1,000,000 – USD892,626) remains at the end of the expected life, which will result in a loss upon the retirement of the equipment if the company's expectation of zero salvage value turns out to be correct. Some companies may choose to depreciate the equipment to its expected salvage value, zero in this case, in its final year of use. Some companies may use a policy of switching to straight-line depreciation after the midlife of its depreciable assets in order to fully depreciate them. That particular pattern is coincidentally displayed in the units-of-production example, in which the equipment is used most heavily in the earliest part of its useful life, and then levels off to much less utilization in the second half of the expected life.

Exhibit 20 shows the different expense allocation patterns of the methods over the same life. Each will affect earnings differently.

Exhibit 20: Expense Allocation Patterns of Different Depreciation Methods



The company's managers could justify any of these methods. Each might fairly represent the way the equipment will be consumed over its expected economic life, which is a subjective estimate. The choices of methods and lives can profoundly affect reported income. These choices are not proven right or wrong until far into the future—but managers must estimate their effects in the present.

[Exhibit 21](#) shows the effects of the three different methods on operating profit and operating profit margins, assuming that the production output of the equipment generates revenues of USD500,000 each year and USD200,000 of cash operating expenses are incurred, leaving USD300,000 of operating profit before depreciation expense.

Exhibit 21: Effects of Depreciation Methods on Operating Profit

Straight Line			
Year	Depreciation	Operating Profit	Operating Profit Margin
1	USD100,000	USD200,000	40.0%
2	100,000	200,000	40.0%
3	100,000	200,000	40.0%
4	100,000	200,000	40.0%
5	100,000	200,000	40.0%
6	100,000	200,000	40.0%
7	100,000	200,000	40.0%
8	100,000	200,000	40.0%
9	100,000	200,000	40.0%
10	100,000	200,000	40.0%

Double-Declining Balance			
Year	Depreciation	Operating Profit	Operating Profit Margin
1	USD200,000	USD100,000	20.0%
2	160,000	140,000	28.0%
3	128,000	172,000	34.4%
4	102,400	197,600	39.5%
5	81,920	218,080	43.6%
6	65,536	234,464	46.9%
7	52,429	247,571	49.5%
8	41,943	258,057	51.6%
9	33,554	266,446	53.3%
10	134,218*	165,782	33.2%

Units of Production			
Year	Depreciation	Operating Profit	Operating Profit Margin
1	USD225,000	USD75,000	15.0%
2	200,000	100,000	20.0%
3	175,000	125,000	25.0%
4	150,000	150,000	30.0%
5	125,000	175,000	35.0%
6	25,000	275,000	55.0%
7	25,000	275,000	55.0%
8	25,000	275,000	55.0%
9	25,000	275,000	55.0%
10	25,000	275,000	55.0%

* Includes \$107,374 of unde depreciated basis, treated as depreciation expense in final year of service.

The straight-line method shows consistent operating profit margins, and the other two methods show varying degrees of increasing operating profit margins as the depreciation expense decreases over time.

[Exhibit 21](#) shows the differences among alternative methods, but even more depreciation expense variation is possible by changing estimated lives and assumptions about salvage value. For instance, change the expected life assumption to 5 years from 10 and add an expectation that the equipment will have a 10 percent salvage value at the end of its expected life. [Exhibit 22](#) shows the revised depreciation calculations. Notice that under the double-declining balance method, the depreciation rate is applied to the gross cost, unlike the other two methods. The straight-line method and the units-of-production method subtract the salvage value from the cost before depreciation expense is calculated. Also note that

the assumption about the usage of the equipment is revised so that it is depreciated only to its salvage value of USD100,000 by the end of its estimated life. The total depreciation under each method is USD900,000.

Exhibit 22: Depreciation Calculations for Each Method in Changed Scenario

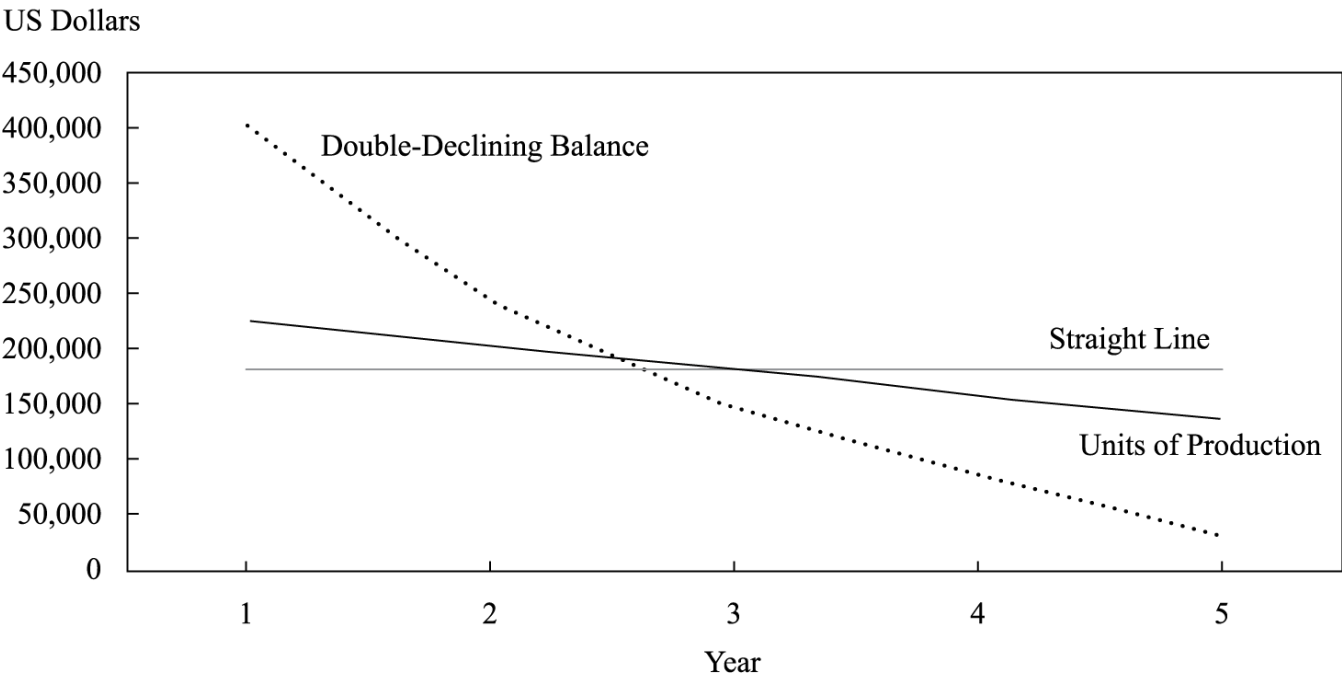
Year	Straight-Line Method	Double-Declining Balance Method			Units-of-Production Method		
	Depreciation Expense	Balance	Declining Balance Rate ¹	Depreciation Expense	Units Produced	Depreciation Rate/Unit	Depreciation Expense
1	USD180,000	USD1,000,000	40%	USD400,000	100,000	USD2.25	USD225,000
2	180,000	600,000	40%	240,000	90,000	USD2.25	202,500
3	180,000	360,000	40%	144,000	80,000	USD2.25	180,000
4	180,000	216,000	40%	86,400	70,000	USD2.25	157,500
5	180,000	129,600	40%	29,600 ²	60,000	USD2.25	135,000
Total	USD900,000			USD900,000	400,000		USD900,000

¹Declining balance rate of 20% calculated as 10-year life being equivalent to 10% annual depreciation rate, multiplied by 2 = 20%.

² Depreciation calculated as \$29,600 instead of 40% × \$129,600. Rote application of the declining-balance rate would have resulted in \$51,840 of expense, which would have depreciated the asset below salvage value.

Exhibit 23 shows the different expense allocation patterns of the methods over the five-year expected life, and assuming a 10 percent salvage value. Although each method is distinctly different in the timing of the cost allocation over time, the variation is less pronounced than over the longer life used in the previous example.

Exhibit 23: Expense Allocation Patterns of Depreciation Methods in Changed Scenario



One of the clearest examples of how choices affect both the balance sheet and income statement can be found in capitalization practices. In classifying a payment made, management must determine whether the payment will benefit only the current period—making it an expense—or whether it will benefit future periods, leading to classification as a cost to be capitalized as an asset. This management judgment embodies an implicit forecast of how the item acquired by the payment will be used, or not used, in the future.

That judgment can be biased by the powerful effect a capitalization policy can have on earnings. Every amount capitalized on the balance sheet as a building, an item of inventory, a deferred cost, or any “other asset” is an amount that has not been recognized as an expense in the reporting period.

A real-life example can be found in the case of WorldCom Inc., a telecom concern that grew rapidly in the late 1990s. Much of WorldCom’s financial reporting was eventually found to be fraudulent. An important part of the misreporting centered on its treatment of what is known in the telecom industry as “line costs.” These are the costs of carrying a voice call or data transmission from its starting point to its ending point, and they represented WorldCom’s largest expense. WorldCom’s CFO decided to capitalize such costs instead of treating them as an operating expense. As a consequence, from the second quarter of 1999 through the first quarter of 2002, WorldCom increased its operating income by USD7 billion. In three of the five quarters in which the improper line cost capitalization took place, WorldCom would have recognized pre-tax losses instead of profits.²⁵

Similarly, acquisitions are an area in which managers must exercise judgment. An allocation of the purchase price must be made to all of the different assets acquired based on their fair values, and those fair values are not always objectively verifiable. Management may have to make its own estimate of fair values for assets acquired, and it may be biased toward a low estimate for the values of depreciable assets in order to depress future depreciation expense. Another benefit to keeping depreciable asset values low is that the amount of the purchase price that cannot be allocated to specific assets is classified as goodwill, which is neither depreciated nor amortized in future reporting periods.

Goodwill reporting has choices of its own. Although goodwill has no effect on future earnings when unimpaired, annual testing of its fair value may reveal that the excess of price paid over the fair value of assets may not be recoverable, which should lead to a write-down of goodwill. The estimation process for the fair value of goodwill may depend heavily on projections of future performance. Those projections may be biased upward to avoid a goodwill write-down.

11. ACCOUNTING CHOICES THAT AFFECT THE CASH FLOW STATEMENT

LEARNING OUTCOME

The candidate should be able to:

- ☐ describe accounting methods (choices and estimates) that could be used to manage earnings, cash flow, and balance sheet items

The cash flow statement consists of three sections: the operating section, which shows the cash generated or used by operations; the investing section, which shows cash used for investments or provided by their disposal; and the financing section, which shows transactions attributable to financing activities.

The operating section is closely scrutinized by investors. Many of them consider it a reality check on reported earnings, on the grounds that earnings attributable to accrual accounting only and unsupported by actual cash flows may indicate earnings manipulation. Such investors believe that amounts shown for cash generated by operations is more insulated from managerial manipulation than the income statement. Cash generated by operations can be managed to an extent, however.

The operating section of the cash flow statement can be shown either under the direct method or the indirect method. Under the direct method, “entities are encouraged to report major classes of gross cash receipts and gross cash payments and their arithmetic sum—the net cash flow from operating activities.”²⁶ In practice, companies rarely use the direct method. Instead, they use the indirect method, which shows a reconciliation of net income to cash provided by operations. The reconciliation shows the non-cash items affecting net income along with changes in working capital accounts affecting cash from operations. [Exhibit 24](#) provides an example of the indirect presentation method.

Exhibit 24: Indirect Presentation Method

Cash Flows from Operating Activities (USD millions)	2018
Net income	USD3,000
<i>Adjustments to reconcile net income to net cash provided by operating activities:</i>	
Provision for doubtful receivables	10
Provision for depreciation and amortization	1,000
Goodwill impairment charges	35
Share-based compensation expense	100
Provision for deferred income taxes	200
<i>Changes in assets and liabilities:</i>	

Cash Flows from Operating Activities (USD millions)	2018
Trade, notes, and financing receivables related to sales	(2,000)
Inventories	(1,500)
Accounts payable	1,200
Accrued income taxes payable/receivable	(80)
Retirement benefits	90
Other	(250)
Net cash provided by operating activities	USD1,805

Whether the indirect method or direct method is used, simple choices exist for managers to improve the *appearance* of cash flow provided by operations without actually improving it. One such choice is in the area of accounts payable management, shaded in [Exhibit 24](#). Assume that the accounts payable balance is USD5,200 million at the end of the period, an increase of USD1,200 million from its previous year-end balance of USD4,000 million. The USD1,200 million increase in accounts payable matched increased expenses or assets but did not require cash. If the company's managers had further delayed paying creditors USD500 million until the day *after* the balance sheet date, they could have increased the cash provided by operating activities by USD500 million. If the managers believe that cash generated from operations is a metric of focus for investors, they can impress them with artificially strong cash flow by simply stretching the accounts payable credit period.

What might alert investors to such machinations? They need to examine the composition of the operations section of the cash flow statement—if they do not, then *nothing* will ever alert them. Studying changes in the working capital can reveal unusual patterns that may indicate manipulation of the cash provided by operations.

Another practice that might lead an investor to question the quality of cash provided by operations is to compare a company's cash generation with an industry-wide level or with the cash operating performance of one or more similar competitors. Cash generation performance can be measured several ways. One way is to compare the relationship between cash generated by operations and net income. Cash generated by operations in excess of net income signifies better quality of earnings, whereas a chronic excess of net income over cash generated by operations should be a cause for concern; it may signal the use of accounting methods to simply raise net income instead of depicting financial reality. Another way to measure cash generation performance is to compare cash generated by operations with debt service, capital expenditures, and dividends (if any). When there is a wide variance between the company's cash generation performance and that of its benchmarks, investors should seek an explanation and carefully examine the changes in working capital accounts.

Because investors may focus on cash from operations as an important metric, managers may resort to managing the working capital accounts as described in order to present the most favorable picture. But this can be done in other ways. A company may misclassify operating uses of cash into either the investing or financing sections of the cash flow statement, which enhances the appearance of cash generated by operating activities.

Dynegy Inc. provides an example of manipulation of cash from operations through clever construction of contracts and assistance from an unconsolidated special purpose entity named ABG Gas Supply LLC (ABG). In April 2001, Dynegy entered into a contract for the purchase of natural gas from ABG. According to the contract, Dynegy would purchase gas at *below-market* rates from ABG for nine months and sell it at the current market rate. The nine-month term coincided with Dynegy's 2001 year-end and would result in gains backed by cash flows. Dynegy also agreed to buy gas at *above-market* rates from ABG for the following 51 months and sell it at the current market rate. The contract was reported at its fair value at the end of fiscal year 2001. It had no effect on net income for the year. The earlier portion of the contract resulted in a gain, supported by USD300 million of cash flow, but the latter portion of the contract resulted in non-cash losses that offset the profit. The mark-to-market rules required the recognition of both gains and losses from all parts of the contract, and hence the net effect on earnings was zero.

In April 2002, a *Wall Street Journal* article exposed the chicanery, thanks to leaked documents. The SEC required Dynegy to restate the cash flow statement by reclassifying USD300 million from the operating section of the cash flow statement to the financing section, on the grounds that Dynegy had used ABG as a conduit to effectively borrow USD300 million from Citigroup. The bank had extended credit to ABG, which it used to finance its losses on the contract (Lee, 2012).

Another area of flexibility in cash flow reporting is found in the area of interest capitalization, which creates differences between total interest payments and total interest costs.²⁷ Assume a company incurs total interest cost of USD30,000, composed of USD3,000 of discount amortization and USD27,000 of interest payments. Of the USD30,000, two-thirds of it (USD20,000) is expensed; the remaining third (USD10,000) is capitalized as plant assets. If the company uses the same interest expense/capitalization proportions to allocate the interest payments between operating and investing activities, then it will report USD18,000 ($2/3 \times \text{USD27,000}$) as an operating outflow and USD9,000 ($1/3 \times \text{USD27,000}$) as an investing outflow. The company might also choose to offset the entire USD3,000 of non-cash discount amortization against the USD20,000 treated as expense, resulting in an operating outflow as low as USD17,000, or as much as USD20,000 if it allocated all of the non-cash discount amortization to interest capitalized as investing activities.

Similarly, the investing outflow could be as much as USD10,000 or as little as USD7,000, depending on the treatment of the non-cash discount amortization. There are choices within the choices, all in areas in which investors believe choices do not even exist. [Nurnberg and Largay \(1998\)](#) have noted that companies apparently favor the method that reports the lowest operating outflow, presumably to maximize reported cash from operations.

Investors and analysts need to be aware that presentation choices permitted in IAS 7, *Statement of Cash Flows*, offer flexibility in classification of certain items in the cash flow statement. This flexibility can drastically change the results in the operating section of the cash flow statement. An excerpt from IAS 7, Paragraphs 33 and 34, provides the background:

33. Interest paid and interest and dividends received are usually classified as operating cash flows for a financial institution. However, there is no consensus on the classification of these cash flows for other entities. Interest paid and interest and dividends received may be classified as operating cash flows because they enter into the determination of profit or loss. *Alternatively, interest paid and interest and dividends received may be classified as financing cash flows and investing cash flows respectively, because they are costs of obtaining financial resources or returns on investments.*

34. Dividends paid may be classified as a financing cash flow because they are a cost of obtaining financial resources. *Alternatively, dividends paid may be classified as a component of cash flows from operating activities in order to assist users to determine the ability of an entity to pay dividends out of operating cash flows.* [Emphasis added.]

By allowing a choice of operating or financing for the placement of interest and dividends received or paid, IAS 7 gives a company’s managers the opportunities to select the presentation that gives the best-looking picture of operating performance. An example is Norse Energy Corp. ASA, a Norwegian gas explorer and producer, which changed its classifications of interest paid and interest received in 2007 ([Gordon, Henry, Jorgensen, and Linthicum, 2017](#)). Interest paid was switched to financing instead of decreasing cash generated from operations. Norse Energy also switched its classification of interest received to investing from operating cash flow. The net effect of these changes was to report positive, rather than negative, operating cash flows in both 2007 and 2008. With these simple changes, the company could also change the perception of its operations. The cash flow statement formerly presented the appearance of a company with operations that used more cash than it generated, and it possibly raised questions about the sustainability of operations. After the revision, the operating section of the cash flow statement depicted a much more viable operation.

[Exhibit 25](#) shows the net effect of the reclassifications on Norse Energy’s cash flows.

Exhibit 25: Reclassification of Cash Flows (amounts in USD millions)						
	As Reported (following 2007 reclassification)		Adjustments (without reclassification*)		Pro forma (without reclassification)	
	2008	2007	2008	2007	2008	2007
Operating	USD5.30	USD2.80	(USD13.70)	(USD14.40)	(USD8.40)	(USD11.60)
Investing	USD0.90	(USD56.80)	(USD9.00)	(USD3.50)	(USD8.10)	(USD60.30)
Financing	(USD16.60)	USD34.50	USD22.70	USD17.90	USD6.10	USD52.40
Total	(USD10.40)	(USD19.50)	USD0	USD0	(USD10.40)	(USD19.50)
* The adjustments reverse the addition of interest received to investing and instead add it to operating. The adjustments also reverse the deduction of interest paid from financing and instead subtract it from operating.						

12. ACCOUNTING CHOICES THAT AFFECT FINANCIAL REPORTING

LEARNING OUTCOME

The candidate should be able to:

- ☐ describe accounting methods (choices and estimates) that could be used to manage earnings, cash flow, and balance sheet items

Exhibit 26 summarizes some of the areas in which choices can be made that affect financial reports.

Exhibit 26: Areas in Which Choices and Estimates Affect Financial Reporting

Area of Choice/Estimate	Analyst Concerns
<i>Revenue recognition</i>	- How is revenue recognized: upon shipment or upon delivery of goods? - Is the company engaging in “channel stuffing”—the practice of overloading a distribution channel with more product than it is normally capable of selling? This can be accomplished by inducing customers to buy more through unusual discounts, the threat of near-term price increases, or both—or simply by shipping goods that were not ordered. These transactions may be corrected in a subsequent period and may result in restated results. Are accounts receivable relative to revenues abnormally high for relative to the company’s history or to its peers? If so, channel stuffing may have occurred. - Is there unusual activity in the allowance for sales returns relative to past history? - Does the company’s days sales outstanding show any collection issues that might indicate shipment of unneeded or unwanted goods to customers? - Does the company engage in “bill-and-hold” transactions? This is when a customer purchases goods but requests that they remain with the seller until a later date. This kind of transaction makes it possible for a seller to manufacture fictitious sales by declaring end-of-period inventory as “sold but held,” with a minimum of effort and phony documentation. - Does the company use rebates as part of its marketing approach? If so, how significantly do the estimates of rebate fulfillment affect net revenues, and have any unusual breaks with history occurred? - Does the company separate its revenue arrangements into multiple deliverables of goods or services? This area is one of great revenue recognition flexibility and also is one that provides little visibility to investors. They simply cannot examine a company’s arrangements and decide for themselves whether or not revenue has been properly allocated to different components of a contract. If a company uses multiple deliverable arrangements with its customers as a routine matter, investors might be more sensitive to revenue reporting risks. In seeking a comfort level, they might ask the following questions: Does the company explain adequately how it determines the different allocations of deliverables and how revenue is recognized on each one? Do deferred revenues result? If not, does it seem reasonable that there are no deferred revenues for this kind of arrangement? Are there unusual trends in revenues and receivables, particularly with regard to cash conversion? If an investor is not satisfied with the answers to these questions, he or she might be more comfortable with other investment choices.
<i>Long-lived assets: Depreciation policies</i>	- Do the estimated life spans of the associated assets make sense, or are they unusually low compared with others in the same industry? - Have there been changes in depreciable lives that have a positive effect on current earnings? - Do recent asset write-downs indicate that company policy on asset lives might need to be reconsidered?

Area of Choice/Estimate	Analyst Concerns
<i>Intangibles: Capitalization policies</i>	- Does the company capitalize expenditures related to intangibles, such as software? Does its balance sheet show any R&D capitalized as a result of acquisitions? Or, if the company is an IFRS filer, has it capitalized any internally generated development costs? - How do the company's capitalization policies compare with the competition? - Are amortization policies reasonable?
<i>Allowance for doubtful accounts/loan loss reserves</i>	- Are additions to such allowances lower or higher than in the past? - Does the collection experience justify any difference from historical provisioning? - Is there a possibility that any lowering of the allowance may be the result of industry difficulties along with the difficulty of meeting earnings expectations?
<i>Inventory cost methods</i>	- Does the company use a costing method that produces fair reporting results in view of its environment? How do its inventory methods compare with others in its industry? Are there differences that will make comparisons uneven if there are unusual changes in inflation? - Does the company use reserves for obsolescence in its inventory valuation? If so, are they subject to unusual fluctuations that might indicate adjusting them to arrive at a specified earnings result? - If a company reports under US GAAP and uses last-in, first-out (LIFO) inventory accounting, does LIFO liquidation (the assumed sale of old, lower-cost layers of inventory) occur through inventory reduction programs? This inventory reduction may generate earnings without supporting cash flow, and management may intentionally reduce the layers to produce specific earnings benefits.
<i>Tax asset valuation accounts</i>	- Tax assets, if present, must be stated at the value at which management expects to realize them, and an allowance must be set up to restate tax assets to the level expected to eventually be converted into cash. Determining the allowance involves an estimate of future operations and tax payments. Does the amount of the valuation allowance seem reasonable, overly optimistic, or overly pessimistic? - Are there contradictions between the management commentary and the allowance level, or the tax note and the allowance level? There cannot be an optimistic management commentary and a fully reserved tax asset, or vice versa. One of them has to be wrong. - Look for changes in the tax asset valuation account. It may be 100 percent reserved at first, and then "optimism" increases whenever an earnings boost is needed. Lowering the reserve decreases tax expense and increases net income.
<i>Goodwill</i>	- Companies must annually assess goodwill balances for impairment on a qualitative basis. If further testing appears necessary, it is based on estimates of the fair value of the reporting units (US GAAP issuers), or cash-generating units (IFRS issuers), associated with goodwill balances. The tests are based on subjective estimates, including future cash flows and the employment of discount rates. - Do the disclosures on goodwill testing suggest that the exercise was skewed to avoid impairment charges?
<i>Warranty reserves</i>	Have additions to the reserves been reduced, perhaps to make earnings targets? Examine the trend in the charges of actual costs against the reserves: Do they support or contradict the warranty provisioning activity? Do the actual costs charged against the reserve give the analyst any indication about the quality of the products sold?

Area of Choice/Estimate	Analyst Concerns
<i>Related-party transactions</i>	- Is the company engaged in transactions that disproportionately benefit members of management? Does one company have control over another's destiny through supply contracts or other dealings? - Do extensive dealings take place with <i>non-public</i> companies that are under management control? If so, those companies could absorb losses (e.g., through supply arrangements that are unfavorable to them) to make the public company's performance look good. This scenario may provide opportunities for an owner to cash out.

The most important lesson is that choices exist among accounting methods and estimates, and an analyst needs a working knowledge of these options to understand whether management may have made choices to achieve a desired result.

13. WARNING SIGNS

LEARNING OUTCOME

The candidate should be able to:

- ☐ **describe accounting warning signs and methods for detecting manipulation of information in financial reports**

The choices management makes to achieve desired results leave a trail, like tracks in sand or snow. The evidence, or warning signs, of information manipulation in financial reports is directly linked to the basic means of manipulation: biased revenue recognition and biased expense recognition. The bias may relate to timing or location of recognition. An example of the timing issue is that a company may choose to defer expenses by capitalizing them. Regarding location, it may recognize a loss in other comprehensive income or directly through equity, rather than through the profit and loss statement. The alert investor or analyst should do the following to identify the warning signs.

13.1. Pay Attention to Revenue

The single largest number on the income statement is revenue, and revenue recognition is a recurring source of accounting manipulation and even outright fraud. Answering the question, "Is revenue higher or lower than the previous comparable period?" is not sufficient. Many analytical procedures can be routinely performed to identify warning signals of malfeasance:

- Examine the accounting policies note for a company's revenue recognition policies.
- Consider whether the policies make it easier to prematurely recognize revenue, such as recognizing revenue immediately upon shipment of goods, or if the company uses bill-and-hold arrangements whereby a sale is recognized before goods actually are shipped to the customer.
- Barter transactions may exist, which can be difficult to value properly.
- Rebate programs involve many estimates, including forecasts of the amount of rebates that ultimately will be incurred, which can have significant effects on revenue recognition.
- Multiple-deliverable arrangements of goods and services are common, but clarity about the timing of revenue recognition for each item or service delivered is necessary for the investor to be comfortable with the reporting of revenues.

Although none of these decisions necessarily violates accounting standards, each involves significant judgement and warrants close attention if other warning signs are present.

- Look at revenue relationships.* Compare a company's revenue growth with its primary competitors or its industry peer group.

- If a company's revenue growth is out of line with its competitors, its industry, or the economy, the investor or analyst needs to understand the reasons for the outperformance. It may be a result of superior management or products and services, but not all management is superior, nor are the products and services of their companies. Revenue quality might be suspect, and the investor should take additional analytical steps.
- Compare accounts receivable with revenues over several years.
 - Examine the trend to determine whether receivables are increasing as a percentage of total revenues. If so, a company might be engaging in channel-stuffing activities, or worse, recording fictitious sales transactions.
 - Calculate receivables turnover for several years:
- Examine the trend for unusual changes and seek an explanation if they exist.
- Compare a company's days sales outstanding (DSO) or receivables turnover with that of relevant competitors or an industry peer group and determine whether the company is an outlier.

An increase in DSO or decrease in receivables turnover could suggest that some revenues are recorded prematurely or are even fictitious, or that the allowance for doubtful accounts is insufficient.

- Examine asset turnover. If a company's managers make poor asset allocation choices, revenues may not be sufficient to justify the investment. Be particularly alert when asset allocation choices involve acquisitions of entire companies. If post-acquisition revenue generation is weak, managers might reach for revenue growth anywhere it can be found. That reach for growth might result in accounting abuses.

Revenues, divided by total assets, indicate the productivity of assets in generating revenues. If the company's asset turnover is continually declining, or lagging the asset turnover of competitors or the industry, it may portend future asset write-downs, particularly in the goodwill balances of acquisitive companies.

13.2. Pay Attention to Signals from Inventories

Although inventory is not a component of every company's asset base, its presence creates an opportunity for accounting manipulation.

- *Look at inventory relationships.* Because revenues involve items sold from inventory, the kind of examination an investor should perform on inventory is similar to that for revenues.
- Compare growth in inventories with competitors and industry benchmarks. If a company's inventory growth is out of line with its peers, without any concurrent sales growth, then it simply may be the result of poor inventory management—an operational inefficiency that might affect an investor's view of a company. It also may signal obsolescence problems in the company's inventory that have not yet been recognized through markdowns to the inventory's net realizable value. Reported gross and net profits could be overstated because of overstated inventory.
- Calculate the inventory turnover ratio. This ratio is the cost of sales divided by the average ending inventory. Declining inventory turnover could also suggest obsolescence problems that should be recognized.
- Companies reporting under US GAAP may use LIFO inventory cost flow assumptions. When this assumption is part of the accounting policies, and a company operates in an inflationary environment, investors should note whether old, low-cost inventory costs have been passed through current earnings and artificially improved gross, operating, and net profits.

13.3. Pay Attention to Capitalization Policies and Deferred Costs

In a study of enforcement actions over a five-year period, the SEC found that improper revenue recognition was the most prevalent accounting issue.²⁸ Suppression of expenses was the next most prevalent problem noted. As the earlier discussion of WorldCom showed, improper capitalization practices can result in a significant misstatement of financial results.

- *Examine the company's accounting policy note for its capitalization policy for long-term assets, including interest costs, and for its handling of other deferred costs.* Compare the company's policy with the industry practice. If the company is the only one capitalizing certain costs while other industry participants treat them as expenses, a red flag is raised. If an outlier company of this type is encountered, it would be useful to cross-check such a company's asset turnover and profitability margins with others in its industry. An investor might expect such a company to be more profitable than its competitors, but the investor might also have lower confidence in the quality of the reported numbers.

13.4. Pay Attention to the Relationship between Cash Flow and Net Income

Net income propels stock prices, but cash flow pays bills. Management can manipulate either one, but sooner or later, net income must be realized in cash if a company is to remain viable. When net income is higher than cash provided by operations, one possibility is that aggressive accrual accounting policies have shifted current expenses to later periods. Increasing earnings in the presence of declining cash generated by operations might signal accounting irregularities.

- *Construct a time series of cash generated by operations divided by net income.* If the ratio is consistently below 1.0 or has declined repeatedly, the company's accrual accounts may have problems.

13.5. Look for Other Potential Warnings Signs

Other areas that might suggest the need for further analysis include the following:

- *Depreciation methods and useful lives.* As discussed earlier, selection of depreciation methods and useful lives can greatly influence profitability. An investor should compare a company's policies with those of its peers to determine whether it is particularly lenient in its effects on earnings. Investors should likewise compare the length of depreciable lives used by a company with those used by its peers.
- *Fourth-quarter surprises.* An investor should be suspicious of possible earnings management if a company routinely disappoints investors with poor earnings or overachieves in the fourth quarter of the year when no seasonality exists in the business. The company may be over- or under-reporting profits in the first three quarters of the year.
- *Presence of related-party transactions.* Related-party transactions often arise when a company's founders are still very active in managing the company, with much of their wealth tied to the company's fortunes. They may be more biased in their view of a company's performance because it relates directly to their own wealth and reputations, and they may be able to transact business with the company in ways that may not be detected. For instance, they may purchase unsellable inventory from the company for disposal in another company of their own to avoid markdowns.
- *Non-operating income or one-time sales included in revenue.* To disguise weakening revenue growth, or just to enhance revenue growth, a company might classify non-operating income items into revenues or fail to clarify the nature of revenues. In the first quarter of 1997, Sunbeam Corporation included one-time disposal of product lines in sales without indicating that such non-recurring sales were included in revenues. This inclusion gave investors a false impression of the company's sustainable revenue-generating capability.
- *Classification of expenses as "non-recurring."* To make operating performance look more attractive, managers might carve out "special items" in the income statement. Particularly when these items appear period after period, equity investors might find their interests best served by not accepting the carve-out of serial "special items" and instead focusing on the net income line in evaluating performance over long periods.
- *Gross/operating margins out of line with competitors or industry.* This disparity is an ambivalent warning sign. It might signal superior management ability. But it might also signal the presence of accounting manipulations to add a veneer of superior management ability to the company's reputation. Only the compilation and examination of other warning signals will enable an investor or analyst to decide which signal is being given.

Warning signals are just that: signals, not indisputable declarations of accounting manipulation guilt. Investors and analysts need to evaluate them cohesively, not on an isolated basis. When an investor finds a number of these signals, the investee company should be viewed with caution or even discarded in favor of alternatives.

Furthermore, as discussed earlier, context is important in judging the value of warning signals. A few examples of facts and circumstances to be aware of are as follows.

- *Younger companies with an unblemished record of meeting growth projections.* It is plausible, especially for a younger company with new and popular product offerings, to generate above-average returns for a period of time. But, as demand dissipates, products mature, and competitors challenge for market share, management may seek to extend its recent record of rapid growth in sales and profitability by unconventional means. At this point, the "earnings games" begin, including aggressive estimates, drawing down "cookie jar" reserves, selling assets for accounting gains, taking on excess leverage, or entering into financial transactions with no apparent business purpose other than financial statement "window dressing."
- *Management has adopted a minimalist approach to disclosure.* Confidence in accounting quality depends on disclosure. If management does not seem to take seriously its obligation to provide information, one needs to be concerned. For example, for a large company, management might claim that it has only one reportable segment, or its commentary might be similar from period to period. A plausible explanation for minimalist disclosure policies could be that management is protecting investors' interests by withholding valuable information from competitors. But, this is not necessarily the case. For example, after Sony Corporation acquired CBS Records and Columbia Pictures, it incurred substantial losses for a number of years. Yet, Sony chose to hide its negative trends

and doubtful future prospects by aggregating the results within a much larger “Entertainment Division.” In 1998, after Sony ultimately wrote off much of the goodwill associated with these ill-fated acquisitions, the SEC sanctioned Sony and its CFO for failing to separately discuss them in the MD&A in a balanced manner.²⁹

- *Management fixation on earnings reports.* Beware of companies whose management appears to be fixated on reported earnings, sometimes to the detriment of attending to real drivers of value. Indicators of excessive earnings fixation include the aggressive use of non-GAAP measures of performance, special items, or non-recurring charges. Another indicator of earnings fixation is highly decentralized operations in which division managers’ compensation packages are heavily weighted toward the attainment of reported earnings or non-GAAP measures of performance.

13.5.1. *Company Culture*

A company’s culture is an intangible that investors should bear in mind when they are evaluating financial statements for the possibility of accounting manipulation. A management’s highly competitive mentality may serve investors well when the company conducts business (assuming that actions taken are not unethical, illegal, or harmfully myopic), but that kind of thinking should not extend to communications with the owners of the company: the shareholders. That mentality can lead to the kind of accounting gamesmanship seen in the early part of the century. In examining financial statements for warning signs of manipulation, the investor should consider whether that mindset exists in the preparation of the financial statements.

One notable example of this mindset comes from one of the most recognized corporate names in the world, General Electric. In the mid-1980s, GE acquired Kidder Peabody, and it was ultimately determined that much of the earnings that Kidder had reported were bogus. As a consequence, GE announced within two days of the acquisition that it would take a non-cash write-off of USD350 million. Here is how former CEO/Chair Jack Welch described the ensuing meeting with senior management in his memoir, *Straight from the Gut*:

The response of our business leaders to the crisis was typical of the GE *culture* [emphasis added]. Even though the books had closed on the quarter, many immediately offered to pitch in to cover the Kidder gap. Some said they could find an extra USD10 million, USD20 million, and even USD30 million from their businesses to offset the surprise. Though it was too late, their willingness to help was a dramatic contrast to the excuses I had been hearing from the Kidder people. (page 225)

It appears that the corporate governance apparatus fostered a GE culture that extended the concept of teamwork to the point of “sharing” profits to win one for the team as a whole, which is incompatible with the concept of neutral financial reporting. Although research is not conclusive on this question, it may be worth considering that predisposition to earnings manipulation is more likely to be present when the CEO and board chair positions are held by the same person, or when the audit committee of the board essentially serves at the pleasure of the CEO and lacks financial reporting sophistication. Finally, one could discuss whether the financial reporting environment today would reward or penalize a CEO who openly endorsed a view that he or she could legitimately exercise financial reporting discretion—albeit within limits—for the purpose of artificially smoothing earnings.

13.5.2. *Restructuring or Impairment Charges*

At times, a company’s stock price has been observed to rise after it recognized a big bath charge to reported earnings. The conventional wisdom explaining the stock price rise is that accounting recognition signals something positive: that management is now ready to part with the lagging portion of a company, so as to redirect its attention and talents to more profitable activities. Consequently, the earnings charge should be disregarded for being solely related to past events.

The analyst should also consider, however, that the events leading ultimately to the big bath on the financial statements did not happen overnight, even though the accounting for those events occurs at a subsequent point. Management may want to communicate that the accounting adjustments reflect the company’s new path, but the restructuring charge also indicates that the old path of reported earnings was not real. In particular, expenses reported in prior years were very likely understated—even assuming that no improper financial statement manipulation had occurred. To extrapolate historical earnings trends, an analyst should consider making pro forma analytical adjustments to prior years’ earnings to reflect a reasonable division of the latest period’s restructuring and impairment charges.

13.5.3. *Management Has a Merger and Acquisition Orientation*

Tyco International Ltd. acquired more than 700 companies from 1996 to 2002. Even assuming the best of intentions regarding financial reporting, a growth-at-any-cost corporate culture poses a severe challenge to operational and financial reporting controls. In Tyco’s case, the SEC found that it consistently and fraudulently understated assets acquired (lowering future depreciation and amortization charges) and overstated liabilities assumed (avoiding expense recognition and potentially increasing earnings in future periods).³⁰

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PRACTICE PROBLEMS

1. In contrast to earnings quality, financial reporting quality *most likely* pertains to:
 - A. sustainable earnings.
 - B. relevant information.
 - C. adequate return on investment.
2. The information provided by a low-quality financial report will *most likely*:
 - A. decrease company value.
 - B. indicate earnings are not sustainable.
 - C. impede the assessment of earnings quality.
3. To properly assess a company's past performance, an analyst requires:
 - A. high earnings quality.
 - B. high financial reporting quality.
 - C. both high earnings quality and high financial reporting quality.
4. Low quality earnings *most likely* reflect:
 - A. low-quality financial reporting.
 - B. company activities which are unsustainable.
 - C. information that does not faithfully represent company activities.
5. Earnings that result from non-recurring activities *most likely* indicate:
 - A. lower-quality earnings.
 - B. biased accounting choices.
 - C. lower-quality financial reporting.
6. Which attribute of financial reports would *most likely* be evaluated as optimal in the financial reporting spectrum?
 - A. Conservative accounting choices
 - B. Sustainable and adequate returns
 - C. Emphasized pro forma earnings measures
7. Financial reports of the lowest level of quality reflect:
 - A. fictitious events.
 - B. biased accounting choices.
 - C. accounting that is non-compliant with GAAP.

8. When earnings are increased by deferring research and development (R&D) investments until the next reporting period, this choice is considered:
- A. non-compliant accounting.
 - B. earnings management as a result of a real action.
 - C. earnings management as a result of an accounting choice.
9. A high-quality financial report may reflect:
- A. earnings smoothing.
 - B. low earnings quality.
 - C. understatement of asset impairment.
10. If a particular accounting choice is considered aggressive in nature, then the financial performance for the reporting period would *most likely*:
- A. be neutral.
 - B. exhibit an upward bias.
 - C. exhibit a downward bias.
11. Conservative accounting choices will most likely lead to:
- A. decreased reported earnings in later periods.
 - B. increased reported earnings in the period under review.
 - C. increased debt reported on the balance sheet at the end of the current period.
12. Which of the following is *most likely* to be considered a potential benefit of accounting conservatism?
- A. A reduction in litigation costs
 - B. Less biased financial reporting
 - C. An increase in current period reported performance
13. Which of the following statements *most likely* describes a situation that would motivate a manager to issue low-quality financial reports? The manager has:
- A. increased the market share of products significantly.
 - B. earned compensation that is linked to stock price performance.
 - C. brought the company's profitability to a level higher than competitors.
14. Which of the following concerns would *most likely* motivate a manager to make conservative accounting choices?
- A. Attention to future career opportunities
 - B. Debt covenant violation risk in the current period
 - C. Unexpected strength in the business environment
15. Which of the following conditions *best* explains why a company's manager would obtain legal, accounting, and board level approval prior to issuing low-quality financial reports?
- A. Motivation

B. Opportunity

C. Rationalization

16. A company is experiencing a period of strong financial performance. To increase the likelihood of exceeding analysts' earnings forecasts in the next reporting period, the company would *most likely* undertake accounting choices for the period under review that:
- A. inflate reported revenue.
 - B. delay expense recognition.
 - C. accelerate expense recognition.
17. Which of the following situations represents a motivation, rather than an opportunity, to issue low-quality financial reports?
- A. Poor internal controls
 - B. Search for a personal bonus
 - C. Inattentive board of directors
18. Which of the following situations will *most likely* motivate managers to inflate reported earnings?
- A. Possibility of bond covenant violation
 - B. Earnings that have exceeded analysts' forecasts
 - C. Earnings that have grown from the prior-year period
19. Which of the following *best* describes an opportunity for management to issue low-quality financial reports?
- A. Ineffective board of directors
 - B. Pressure to achieve some performance level
 - C. Corporate concerns about financing in the future
20. An audit opinion of a company's financial reports is *most likely* intended to:
- A. detect fraud.
 - B. reveal misstatements.
 - C. ensure that financial information is presented fairly.
21. If a company uses a non-GAAP financial measure in an SEC filing, then the company must:
- A. give more prominence to the non-GAAP measure if it is used in earnings releases.
 - B. provide a reconciliation of the non-GAAP measure and equivalent GAAP measure.
 - C. exclude charges requiring cash settlement from any non-GAAP liquidity measures.
22. A company wishing to increase earnings in the reporting period may choose to:
- A. decrease the useful life of depreciable assets.
 - B. lower estimates of uncollectible accounts receivables.
 - C. classify a purchase as an expense rather than a capital expenditure.
23. Which technique *most likely* increases the cash flow provided by operations?

- A. Stretching the accounts payable credit period
- B. Applying all non-cash discount amortization against interest capitalized
- C. Shifting classification of interest paid from financing to operating cash flows

24. Bias in revenue recognition would *least likely* be suspected if:

- A. the firm engages in barter transactions.
- B. reported revenue is higher than the previous quarter.
- C. revenue is recognized before goods are shipped to customers.

25. Which of the following is an indication that a company may be recognizing revenue prematurely? Relative to its competitors, the company's:

- A. asset turnover is decreasing.
- B. receivables turnover is increasing.
- C. days sales outstanding is increasing.

26. Which of the following would *most likely* signal that a company may be using aggressive accrual accounting policies to shift current expenses to later periods? Over the last five-year period, the ratio of cash flow to net income has:

- A. increased each year.
- B. decreased each year.
- C. fluctuated from year to year.

27. An analyst reviewing a firm with a large reported restructuring charge to earnings should:

- A. view expenses reported in prior years as overstated.
- B. disregard it because it is solely related to past events.
- C. consider making pro forma adjustments to prior years' earnings.

SOLUTIONS

1. B is correct. Financial reporting quality pertains to the quality of information in financial reports. High-quality financial reporting provides decision-useful information, which is relevant and faithfully represents the economic reality of the company's activities. Earnings of high quality are sustainable and provide an adequate level of return. Highest-quality financial reports reflect both high financial reporting quality and high earnings quality.
2. C is correct. Financial reporting quality pertains to the quality of the information contained in financial reports. High-quality financial reports provide decision-useful information that faithfully represents the economic reality of the company. Low-quality financial reports impede assessment of earnings quality. Financial reporting quality is distinguishable from earnings quality, which pertains to the earnings and cash generated by the company's actual economic activities and the resulting financial condition. Low-quality earnings are not sustainable and decrease company value.
3. B is correct. Financial reporting quality pertains to the quality of the information contained in financial reports. If financial reporting quality is low, the information provided is of little use in assessing the company's performance. Financial reporting quality is distinguishable from earnings quality, which pertains to the earnings and cash generated by the company's actual economic activities and the resulting financial condition.
4. B is correct. Earnings quality pertains to the earnings and cash generated by the company's actual economic activities and the resulting financial condition. Low-quality earnings are likely not sustainable over time because the company does not expect to generate the same level of earnings in the future or because earnings will not generate sufficient return on investment to sustain the company. Earnings that are not sustainable decrease company value. Earnings quality is distinguishable from financial reporting quality, which pertains to the quality of the information contained in financial reports.
5. A is correct. Earnings that result from non-recurring activities are unsustainable. Unsustainable earnings are an example of lower-quality earnings. Recognizing earnings that result from non-recurring activities is neither a biased accounting choice nor indicative of lower quality financial reporting because it faithfully represents economic events.
6. B is correct. At the top of the quality spectrum of financial reports are reports that conform to GAAP, are decision useful, and have earnings that are sustainable and offer adequate returns. In other words, these reports have both high financial reporting quality and high earnings quality.

7. Solution:

A is correct. Financial reports span a quality continuum from high to low based on decision-usefulness and earnings quality (see Exhibit 2). The lowest-quality reports portray fictitious events, which may misrepresent the company's performance or obscure fraudulent misappropriation of the company's assets.

8. Solution:

B is correct. Deferring R&D investments into the next reporting period is an example of earnings management by taking a *real* action.

9. B is correct. High-quality financial reports offer useful information, meaning information that is relevant and faithfully represents actual performance. Although low earnings quality may not be desirable, if the reported earnings are representative of actual performance, they are consistent with high-quality financial reporting. Highest-quality financial reports reflect both high financial reporting quality and high earnings quality.
10. B is correct. Aggressive accounting choices aim to enhance the company's reported performance by inflating the amount of revenues, earnings, or operating cash flow reported in the period.

Consequently, the financial performance for that period would most likely exhibit an upward bias.

11. C is correct. Accounting choices are considered conservative if they decrease the company's reported performance and financial position in the current period under review. Conservative choices may increase the amount of debt reported on the balance sheet. They may decrease the revenues, earnings, or operating cash flow reported for the period and increase those amounts in later periods.
12. A is correct. Conservatism reduces the possibility of litigation and, by extension, litigation costs. Rarely, if ever, is a company sued because it understated good news or overstated bad news. Accounting conservatism is a type of bias in financial reporting that decreases a company's reported performance. Conservatism directly conflicts with the characteristic of neutrality.
13. B is correct. Managers often have incentives to meet or beat market expectations, particularly if management compensation is linked to increases in share prices or to reported earnings.
14. C is correct. Managers may be motivated to understate earnings in a period with unexpected strong performance by delaying revenue recognition or accelerating expense recognition to increase the probability of exceeding expectations in a subsequent period (referred to as "banking" some earnings for the next period.)
15. C is correct. Typically, conditions of opportunity, motivation, and rationalization exist when individuals issue low-quality financial reports. Rationalization occurs when an individual is concerned about a choice and needs to be able to justify it to herself or himself. If the manager is concerned about a choice in a financial report, the manager may ask for other opinions to convince herself or himself that it is okay.
16. C is correct. In a period of strong financial performance, managers may pursue accounting choices that increase the probability of exceeding earnings forecasts for the next period. By accelerating expense recognition or delaying revenue recognition, managers may reduce financial performance in the current period in order to inflate earnings in the next period and increase the likelihood of exceeding targets.
17. B is correct. Motivation can result from pressure to meet some criteria for personal reasons, such as a bonus, or corporate reasons, such as concern about future financing. Poor internal controls and an inattentive board of directors offer opportunities to issue low-quality financial reports.
18. A is correct. The possibility of bond covenant violations may motivate managers to inflate earnings in the reporting period. In so doing, the company may be able to avoid the consequences associated with violating bond covenants.
19. A is correct. Opportunities to issue low-quality financial reports include internal conditions, such as an ineffective board of directors, and external conditions, such as accounting standards that provide scope for divergent choices. Pressure to achieve a certain level of performance and corporate concerns about future financing are examples of motivations to issue low-quality financial reports. Typically, three conditions exist when low-quality financial reports are issued: opportunity, motivation, and rationalization.
20. C is correct. An audit is intended to provide assurance that the company's financial reports are presented fairly, thus providing discipline regarding financial reporting quality. Regulatory agencies usually require that the financial statements of publicly traded companies be audited by an independent auditor to provide assurance that the financial statements conform to accounting standards. Privately held companies may also choose to obtain audit opinions either voluntarily or because an outside party requires it. An audit is not typically intended to detect fraud. An audit is based on sampling and it is possible that the sample might not reveal misstatements.
21. B is correct. If a company uses a non-GAAP financial measure in an SEC filing, it is required to provide the most directly comparable GAAP measure with equivalent prominence in the filing. In addition, the company is required to provide a reconciliation between the non-GAAP measure and the equivalent GAAP measure. Similarly, IFRS requires that any non-IFRS measures included in financial reports must be defined and their potential relevance explained. The non-IFRS measures must be reconciled with IFRS measures.

- 22.** B is correct. If a company wants to increase reported earnings, the company's managers may reduce the allowance for uncollected accounts and the related expense reported for the period. Decreasing the useful life of depreciable assets would increase depreciation expense and decrease earnings in the reporting period. Classifying a purchase as an expense, rather than capital expenditure, would decrease earnings in the reporting period. The use of accrual accounting may result in estimates in financial reports, because all facts associated with events may not be known at the time of recognition. These estimates can be grounded in reality or managed by the company to present a desired financial picture.
- 23.** A is correct. Managers can temporarily show a higher cash flow from operations by stretching the accounts payable credit period. In other words, the managers delay payments until the next accounting period. Applying all non-cash discount amortization against interest capitalized causes reported interest expenses and operating cash outflow to be higher, resulting in a lower cash flow provided by operations. Shifting the classification of interest paid from financing to operating cash flows lowers the cash flow provided by operations.
- 24.** B is correct. Bias in revenue recognition can lead to manipulation of information presented in financial reports. Addressing the question as to whether revenue is higher or lower than the previous period is insufficient to determine if there is bias in revenue recognition. Additional analytical procedures must be performed to identify warning signals of accounting malfeasance. Barter transactions are difficult to value properly and may result in bias in revenue recognition. Policies that make it easier to prematurely recognize revenue, such as before goods are shipped to customers, may be a warning sign of accounting malfeasance.
- 25.** C is correct. If a company's days sales outstanding (DSO) is increasing relative to competitors, this may be a signal that revenues are being recorded prematurely or are even fictitious. Numerous analytical procedures can be performed to provide evidence of manipulation of information in financial reporting. These warning signs are often linked to bias associated with revenue recognition and expense recognition policies.
- 26.** B is correct. If the ratio of cash flow to net income for a company is consistently below 1 or has declined repeatedly over time, this may be a signal of manipulation of information in financial reports through aggressive accrual accounting policies. When net income is consistently higher than cash provided by operations, one possible explanation is that the company may be using aggressive accrual accounting policies to shift current expenses to later periods.
- 27.** C is correct. To extrapolate historical earnings trends, an analyst should consider making pro forma analytical adjustments of prior years' earnings to reflect in those prior years a reasonable share of the current period's restructuring and impairment charges.

NOTES

¹In this module, the examples of misleading or inaccurate financial reports occurred in prior years—*not* because there are no current examples of questionable financial reporting, but rather because it has been conclusively resolved that misreporting occurred in the historical examples.

²The characteristics of decision-useful information are identical under IFRS and US GAAP. In September 2010, the IASB adopted the *Conceptual Framework for Financial Reporting* in place of the *Framework for the Preparation and Presentation of Financial Statements* (1989). The *Conceptual Framework* represents the partial completion of a joint convergence project between the IASB and FASB on an updated framework. The *Conceptual Framework* (2010) contains two updated chapters: “The Objective of Financial Reporting” and “Qualitative Characteristics of Useful Financial Information.” The remainder of the material in the *Conceptual Framework* is from the *Framework* (1989) and will be updated as the project is completed. Also in September 2010, the FASB issued Concepts Statement 8, “Conceptual Framework for Financial Reporting,” to replace Concepts Statements 1 and 2.

³Text from conceptual frameworks referenced in Note 4.

⁴Back Aaron 2013. “Toyota, What a Difference the Yen Makes.” *Wall Street Journal* (4 August 2013).

⁵The term “non-GAAP” refers generally to all metrics that are non-compliant with GAAP and thus includes “non-IFRS” metrics.

⁶A survey of non-GAAP earnings in the S&P 500 is presented in Ciesielski and Henry (2017). Some observers even recommend that investors shift their focus from a company’s earnings to a company’s “strategic assets” and the contribution of these assets to its competitive edge (Gu and Lev, 2017).

⁷Non-domestic private issuers can file financial statements prepared in accordance with IFRS without reconciliation to US GAAP. The SEC recognizes US GAAP and IFRS as GAAP.

⁸Various definitions have appeared in academic research. Closest to the discussion is Schipper (1989), which uses the term “earnings management” to mean “‘disclosure management’ in the sense of a purposeful intervention in the external financial reporting process, with the intent of obtaining some private gain (as opposed to, say, merely facilitating the neutral operation of the process).”

⁹International Accounting Standards Board (IASB) and Financial Accounting Standards Board (FASB), *The Conceptual Framework for Financial Reporting* (2010): QC 14.

¹⁰See International Accounting Standards (IAS) 36 and FASB, Accounting Standards Codification (ASC), Section 360-10-35.

¹¹SEC, “Restructuring and Impairment Charges,” Staff Accounting Bulletin (SAB) No. 100 (1999), www.sec.gov/interps/account/sab100.htm.

¹²Arthur Levitt, “The Numbers Game,” Remarks given at NYU Center for Law and Business (28 September 1998), www.sec.gov/news/speech/speecharchive/1998/spch220.txt.

¹³SEC, “Commission Guidance Regarding Management’s Discussion and Analysis of Financial Condition and Results of Operations,” Financial Reporting Release (FRR) No. 72 (2003), www.sec.gov/rules/interp/33-8350.htm.

¹⁴Visit www.iosco.org for more information.

¹⁵Text from ESMA’s mission statement on their website: www.esma.europa.eu.

¹⁶ESMA, “Enforcement and Regulatory Activities of Accounting Enforcers in 2017,” ESMA32-63-424, European Securities and Markets Authority (3 April 2018), www.esma.europa.eu.

¹⁷SEC, “FY2013 Congressional Justification,” Securities and Exchange Commission (February 2012), www.sec.gov/about/secfy13congbudgjust.pdf.

¹⁸SEC, Securities and Exchange Commission Division of Enforcement Annual Report, “A Look Back at Fiscal Year 2017,” www.sec.gov/report.

¹⁹SEC, “Final Rule: Conditions for Use of Non-GAAP Financial Measures,” Securities and Exchange Commission, www.sec.gov/rules/final/33-8176.htm.

²⁰Correspondence between Groupon and SEC, filed in EDGAR on 16 September 2011.

²¹Accounting and Auditing Enforcement Release No. 2542, “SEC v. James Charles Blue, Randy Cook, and Victor Campbell,” United States District Court for the District of Colorado, Civ. Action No. 07-CV-00095 REB-MEH (17 January 2007).

²²See Accounting Standards Codification (ASC) 740-10-30-16 to 25, *Establishment of a Valuation Allowance for Deferred Tax Assets*.

²³Accounting and Auditing Enforcement Release No. 2448, “In the Matter of Douglas R. Bauer, Respondent,” SEC (27 June 2006), www.sec.gov/litigation/admin/2006/34-54049.pdf.

²⁴Accounting and Auditing Enforcement Release No. 3600, “In the Matter of Hampton Roads Bankshares Inc., Respondent,” SEC (5 December 2014), www.sec.gov/litigation/admin/2014/34-73750.pdf.

²⁵See Report of Investigation by the Special Investigative Committee of the Board of Directors of WorldCom, Inc., by Dennis R. Beresford, Nicholas deB. Katzenbach, and C.B. Rogers, Jr., (31 March 2003), pages 9–11, www.sec.gov/Archives/edgar/data/723527/000093176303001862/dex991.htm.

²⁶Accounting Standards Codification (ASC), Section 230-10-45-25, *Reporting Operating, Investing, and Financing Activities*. The direct method and indirect method are similar in IFRS, as addressed in IAS 7, Paragraph 18.

²⁷See Nurnberg and Largay (1998) and Nurnberg (2006) .

²⁸SEC, “Report Pursuant to Section 704 of the Sarbanes–Oxley Act of 2002,” pages 5–6, www.sec.gov/news/studies/sox704report.pdf.

²⁹SEC, Accounting and Auditing Enforcement Release No. 1061, “In the Matter of Sony Corporation and Sumio Sano, Respondents” (5 August 1998).

³⁰SEC, Accounting and Auditing Enforcement Release No. 2414, “SEC Brings Settled Charges Against Tyco International Ltd. Alleging Billion Dollar Accounting Fraud” (17 April 2006), www.sec.gov/litigation/litreleases/2006/lr19657.htm.

Financial Analysis Techniques

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LEARNING OUTCOMES

The candidate should be able to:

- ☐ describe tools and techniques used in financial analysis, including their uses and limitations
- ☐ calculate and interpret activity, liquidity, solvency, and profitability ratios
- ☐ describe relationships among ratios and evaluate a company using ratio analysis
- ☐ demonstrate the application of DuPont analysis of return on equity and calculate and interpret effects of changes in its components
- ☐ describe the uses of industry-specific ratios used in financial analysis
- ☐ describe how ratio analysis and other techniques can be used to model and forecast earnings

1. INTRODUCTION

Analysts convert financial statement and other data into metrics that assist in decision making and help answer questions such as the following: How successfully has a target company performed, relative to its own past performance and relative to its competitors? How is the company likely to perform in the future? Based on expectations about future performance, what is the value of this company or the securities it issues? This module describes various techniques used to answer these and other questions. These financial analysis techniques are crucial to a wide range of analytical tasks, including valuing equity securities, assessing credit risk, conducting due diligence related to an acquisition, and evaluating business performance.

Learning Module Overview

- There is no single approach to structuring the financial analysis process, but a general framework entails the following phases: articulate the purpose of the analysis, collect input data, process the data, analyze and interpret the processed data, develop and communicate conclusions and recommendations, follow-up periodically to determine if any changes are necessary to recommendations or holdings.
- The purpose of analysis is not simply to compile information and do computations, but to integrate these into a cohesive result that addresses not just what happened, but why it happened and whether it created value. An analyst must be able to understand the “why” behind the numbers and ratios, not just what the numbers and ratios are.
- Evaluations require comparisons. It is difficult to say that a company’s financial performance was “good” or “bad” without clarifying the basis for comparison. Cross-sectional analysis compares multiple companies at the same point in time or over the same range of time, and trend or time-series analysis compares measures for a single company over a period of time.
- Ratios and common-size financial statements can remove size as a factor and enable more relevant comparisons. Financial statement ratios are helpful for valuing companies and securities, selecting investments, and predicting financial distress. The ratio is an indicator of some aspect of a company’s performance, telling what happened but not why it happened.
- Common-size analysis involves expressing financial data, including entire financial statements, in relation to a single financial statement item, or base. A vertical common-size balance sheet divides each balance sheet item by the same period’s total assets and expresses the results as percentages. A vertical common-size income statement divides each income statement item by revenue or by total assets. A horizontal common-size balance sheet divides the quantity of each item by a base year quantity of the same item to yield a percentage change in that item from the base year. Trend data generated by a horizontal common-size analysis can be compared across financial statements.
- Graphs facilitate comparison of performance and financial structure over time, provide a visual overview of changes and trends, and can be used to communicate the conclusions from financial analysis. Regression analysis can help identify relationships or correlation between variables.
- Activity ratios measure the efficiency of a company’s operations, such as a collection of receivables or management of inventory. Major activity ratios include inventory turnover, days of inventory on hand, receivables turnover, days of sales outstanding, payables turnover, number of days of payables, working capital turnover, fixed asset turnover, and total asset turnover.

- Liquidity ratios measure the ability of a company to meet short-term obligations. Major liquidity ratios include the current ratio, quick ratio, cash ratio, and defensive interval ratio. The cash conversion cycle is a measure of liquidity that is not a simple ratio.
- Solvency ratios measure the ability of a company to meet long-term obligations. Major solvency ratios include debt ratios (including the debt-to-assets ratio, debt-to-capital ratio, debt-to-equity ratio, and financial leverage ratio) and coverage ratios (including interest coverage and fixed charge coverage).
- Profitability ratios measure the ability of a company to generate profits from revenue and assets. Major profitability ratios include return on sales ratios (including gross profit margin, operating profit margin, pretax margin, and net profit margin) and return on investment ratios (including operating return on assets [ROA], ROA, return on total capital, return on equity [ROE], and return on common equity).
- It is important to examine a variety of financial ratios—not a single ratio or category of ratios in isolation—to ascertain the overall position and performance of a company.
- DuPont analysis breaks ROE into components that are indicators of different aspects of company performance. Many levels of decomposition are possible.
- The five-component DuPont decomposition expresses a company's ROE as a function of its tax rate, interest burden, operating profitability, efficiency, and leverage.
- Because aspects of performance that are considered important in one industry may be irrelevant in another, industry-specific ratios are used that reflect these differences.
- Techniques such as sensitivity analysis, scenario analysis, and simulation are used to forecast future financial performance.

2. THE FINANCIAL ANALYSIS PROCESS

LEARNING OUTCOME

The candidate should be able to:

- ☐ **describe tools and techniques used in financial analysis, including their uses and limitations**

In financial analysis, it is essential to clearly identify and understand the final objective and the steps required to reach that objective. In addition, the analyst needs to know where to find relevant data, how to process and analyze the data (in other words, know the typical questions to address when interpreting data), and how to communicate the analysis and conclusions.

2.1. The Objectives of the Financial Analysis Process

Because of the variety of reasons for performing financial analysis, the numerous available techniques, and the often substantial amount of data, it is important that the analytical approach be tailored to the specific situation. Prior to beginning any financial analysis, the analyst should clarify the purpose and context, and clearly understand the following:

- What is the purpose of the analysis? What questions will this analysis answer?

- What level of detail will be needed to accomplish this purpose?
- What data are available for the analysis?
- What are the factors or relationships that will influence the analysis?
- What are the analytical limitations, and will these limitations potentially impair the analysis?

Having clarified the purpose and context of the analysis, the analyst can select the set of techniques (e.g., ratios) that will best assist in making a decision. Although there is no single approach to structuring the analysis process, a general framework is set forth in [Exhibit 1](#). The steps in this process were discussed in more detail in an earlier module; the primary focus of this module is on Phases 3 and 4, processing and analyzing data.

Exhibit 1: A Financial Statement Analysis Framework

Phase	Sources of Information	Output
1. Articulate the purpose and context of the analysis.	■ The nature of the analyst's function, such as evaluating an equity or debt investment or issuing a credit rating. ■ Communication with client or supervisor on needs and concerns. ■ Institutional guidelines related to developing specific work product.	■ Statement of the purpose or objective of analysis. ■ A list (written or unwritten) of specific questions to be answered by the analysis. ■ Nature and content of report to be provided. ■ Timetable and budgeted resources for completion.
2. Collect input data.	■ Financial statements, other financial data, questionnaires, and industry/economic data. ■ Discussions with management, suppliers, customers, and competitors. ■ Company site visits (e.g., to production facilities or retail stores).	■ Organized financial statements. ■ Financial data tables. ■ Completed questionnaires, if applicable.
3. Process data.	■ Data from the previous phase.	■ Adjusted financial statements. ■ Common-size statements. ■ Ratios and graphs. ■ Forecasts.

Phase	Sources of Information	Output
4. Analyze/interpret the processed data.	Input data as well as processed data.	Analytical results.
5. Develop and communicate conclusions and recommendations (e.g., with an analysis report).	- Analytical results and previous reports. - Institutional guidelines for published reports.	- Analytical report answering questions posed in Phase 1. - Recommendation regarding the purpose of the analysis, such as whether to make an investment or grant credit.
6. Follow-up.	Information gathered by periodically repeating above steps as necessary to determine whether changes to holdings or recommendations are necessary.	Updated reports and recommendations.

2.2. Distinguishing between Computations and Analysis

An effective analysis encompasses both computations and interpretations. A well-reasoned analysis differs from a mere compilation of various pieces of information, computations, tables, and graphs by integrating the data collected into a cohesive whole. Analysis of past performance, for example, should address not only what happened but also why it happened and whether it created value. Some of the key questions to address include the following:

- What aspects of performance are critical for this company to successfully compete in this industry?
- How well did the company's performance meet these critical aspects? (Established through computation and comparison with appropriate benchmarks, such as the company's own historical performance or competitors' performance.)
- What were the key causes of this performance, and how does this performance reflect the company's strategy? (Established through analysis.)

If the analysis is forward looking, additional questions include the following:

- What is the likely impact of an event or trend? (Established through interpretation of analysis.)
- What is the likely response of management to this trend? (Established through evaluation of quality of management and corporate governance.)
- What is the likely impact of trends in the company, industry, and economy on future cash flows? (Established through assessment of corporate strategy and through forecasts.)
- What are the recommendations of the analyst? (Established through interpretation and forecasting of results of analysis.)
- What risks should be highlighted? (Established by an evaluation of major uncertainties in the forecast and in the environment within which the company operates.)

Example 1 demonstrates how a company's financial data can be analyzed in the context of its business strategy and changes in that strategy. An analyst must be able to understand the "why" behind the numbers and ratios, not just what the numbers and ratios are.

EXAMPLE 1

Strategy Reflected in Financial Performance

Apple Inc. engages in the design, manufacture, and sale of computer hardware, mobile devices, operating systems and related products, and services. It also operates retail and online stores. Microsoft develops, licenses, and supports software products, services, and technology devices through a variety of channels including retail stores in recent years. Selected financial data for 2015 through 2017 for these two companies are given in [Exhibit 2](#) and [Exhibit 3](#). Apple's fiscal year (FY) ends on the final Saturday in September (for example, FY2017 ended on 30 September 2017). Microsoft's fiscal year ends on 30 June (for example, FY2017 ended on 30 June 2017).

Exhibit 2: Selected Financial Data for Apple (US dollar millions)

Fiscal year	2017	2016	2015
Net sales (or Revenue)	229,234	215,639	233,715
Gross margin	88,186	84,263	93,626
Operating income	61,344	60,024	71,230

Exhibit 3: Selected Financial Data for Microsoft (US dollar millions)*

Fiscal year	2017	2016	2015
Net sales (or revenue)	89,950	85,320	93,580
Gross margin	55,689	52,540	60,542
Operating income	22,326	20,182	18,161

* Microsoft revenue for 2017 and 2016 were subsequently revised in the company's 2018 10-K report due to changes in revenue recognition and lease accounting standards.

Source: 10-K reports for Apple and Microsoft.

Apple reported a 7.7 percent decrease in net sales from FY2015 to FY2016 and an increase of 6.3 percent from FY2016 to FY2017 for an overall slight decline over the three-year period. Gross margin decreased 10.0 percent from FY2015 to FY2016 and increased 4.7 percent from FY2016 to FY2017. This also represented an overall decline in gross margin over the three-year period. The company's operating income exhibited similar trends.

Microsoft reported an 8.8 percent decrease in net sales from FY2015 to FY2016 and an increase of 5.4 percent from FY2016 to FY2017 for an overall slight decline over the three-year period. Gross margin decreased 13.2 percent from FY2015 to FY2016 and increased 6.0 percent from FY2016 to FY2017. Similar to Apple, this represented an overall decline in gross margin over the three-year period. Microsoft's operating income, in contrast, exhibited growth each year and for the three-year period. Overall growth in operating income was 23 percent.

What caused Microsoft's growth in operating income while Apple and Microsoft had similar negative trends in sales and gross margin? Apple's decline in sales, gross margin, and operating income from FY2015 to FY2016 was caused by declines in iPhone sales and weakness in foreign currencies relative to the US dollar. FY2017 saw a rebound in sales of iPhones, Mac computers, and services offset somewhat by continued weaknesses in foreign currencies. Microsoft similarly had

declines in revenue and gross margin from sales of its devices and Windows software in FY2016, as well as negative impacts from foreign currency weakness. Microsoft's increase in revenue and gross margin in FY2017 was driven by the acquisition of LinkedIn, higher sales of Microsoft Office software, and higher sales of cloud services. The driver in the continuous increase in operating income for Microsoft was a large decline over the three-year period in impairment, integration, and restructuring charges. Microsoft recorded a USD10 billion charge in FY2015 related to its phone business, and there were further charges of USD1.1 billion in FY2016 and USD306 million in FY2017. Absent these large write-offs, Microsoft would have had a trend similar to Apple's in operating income over the three-year period.

Analysts often need to communicate the findings of their analysis in a written report. Their reports should communicate how conclusions were reached and why recommendations were made. For example, a report might present the following:

- the purpose of the report, unless it is readily apparent;
- relevant aspects of the business context, including:
 - economic environment (country/region, macro economy, sector),
 - financial and other infrastructure (accounting, auditing, rating agencies), and
 - legal and regulatory environment (and any other material limitations on the company being analyzed);
- evaluation of corporate governance and assessment of management strategy, including the company's competitive advantage(s);
- assessment of financial and operational data, including key assumptions in the analysis; and
- conclusions and recommendations, including limitations of the analysis and risks.

An effective narrative and well supported conclusions and recommendations are normally enhanced by using 3–10 years of data as well as by analytic techniques appropriate to the purpose of the report.

3. ANALYTICAL TOOLS AND TECHNIQUES

LEARNING OUTCOME

The candidate should be able to:

- ☐ **describe tools and techniques used in financial analysis, including their uses and limitations**

The tools and techniques presented in this lesson facilitate evaluations of company data. Evaluations require comparisons. It is difficult to say that a company's financial performance was "good" or "bad" without clarifying the basis for comparison. In assessing a company's ability to generate and grow earnings and cash flow, and the risks related to those earnings and cash flows, the analyst draws comparisons to other companies at the same point in time or over the same range of time (cross-sectional analysis) and over time (trend or time-series analysis).

For example, an analyst may wish to compare the profitability of companies competing in a global industry. If the companies differ significantly in size or report their financial data in different currencies, comparing net income as reported is not useful. Ratios (which express one number in relation to another) and common-size financial statements can remove size as a factor and enable a more relevant comparison. To achieve comparability across companies reporting in different currencies, one approach

is to translate all reported numbers into a common currency using average or period-end exchange rates. Alternatively, if the focus is primarily on ratios, comparability can be achieved without translating the currencies.

The analyst may also want to examine comparable performance over time. Again, the nominal currency amounts of sales or net income may not highlight significant changes. To address this challenge, horizontal financial statements (whereby quantities are stated in terms of a selected base year value) can make such changes more apparent. Another obstacle to comparison is differences in fiscal year end. To achieve comparability, one approach is to develop trailing 12 months of data. Finally, it should be noted that differences in accounting standards can limit comparability.

EXAMPLE 2

Ratio Analysis

An analyst is examining the profitability of two international companies with large shares of the global personal computer market: Acer Inc. and Lenovo Group Limited. Acer has pursued a strategy of selling its products at affordable prices. In contrast, Lenovo aims to achieve higher selling prices by stressing the high engineering quality of its personal computers for business use. Acer reports in New Taiwan dollars (TWD) and Lenovo reports in US dollars (USD). For Acer, fiscal year end is 31 December. For Lenovo, fiscal year end is 31 March; thus, FY2017 ended 31 March 2018.

The analyst collects the data shown in [Exhibit 4](#). Use this information to answer the following questions:

Exhibit 4: Acer versus Lenovo Profitability

Acer					
TWD Millions	FY2013	FY2014	FY2015	FY2016	FY2017
Revenue	360,132	329,684	263,775	232,724	237,275
Gross profit	22,550	28,942	24,884	23,212	25,361
Net income	(20,519)	1,791	604	(4,901)	2,797

Lenovo					
USD Millions	FY2013	FY2014	FY2015	FY2016	FY2017
Revenue	38,707	46,296	44,912	43,035	45,350
Gross profit	5,064	6,682	6,624	6,105	6,272
Net income (Loss)	817	837	(145)	530	(127)

Note: Fiscal years for Lenovo end 31 March. Thus, FY2017 represents the fiscal year ended 31 March 2018; the same applies respectively for prior years.

1. Which company is larger based on the amount of revenue, in US dollars, reported in fiscal year 2017? For FY2017, assume the relevant, average exchange rate was 30.95 TWD/USD.

Solution:

Lenovo is much larger than Acer based on FY2017 revenues in US dollar terms. Lenovo's FY2017 revenues of USD45.35 billion are considerably higher than Acer's USD7.67 billion (= TWD237.275 million/30.95).

Acer: At the assumed average exchange rate of 30.95 TWD/USD, Acer's FY2017 revenues are equivalent to USD7.67 billion (= TWD237.275 million ÷ 30.95 TWD/USD).

Lenovo: Lenovo's FY2017 revenues totaled USD45.35 billion.

Note: Comparing the size of companies reporting in different currencies requires translating reported numbers into a common currency using exchange rates at some point in time. This solution converts the revenues of Acer to billions of US dollars using the average exchange rate of the fiscal period. It would be equally informative (and would yield the same conclusion) to convert the revenues of Lenovo to New Taiwan dollars.

2. Which company had the higher revenue growth from FY2016 to FY2017? FY2013 to FY2017?

Solution:

The growth in Lenovo's revenue was much higher than Acer's in the most recent fiscal year and for the five-year period.

	Change in Revenue FY2016 versus FY2017 (%)	Change in Revenue FY2013 to FY2017 (%)
Acer	1.96	(34.11)
Lenovo	5.38	17.16

The table shows two growth metrics. Calculations are illustrated using the revenue data for Acer:

The change in Acer's revenue for FY2016 versus FY2017 is 1.96 percent calculated as $(237,275 - 232,724) \div 232,724$ or equivalently $(237,275 \div 232,724) - 1$. The change in Acer's revenue from FY2013 to FY2017 is a decline of 34.11 percent.

3. How do the companies compare, based on profitability?

Solution:

Profitability can be assessed by comparing the amount of gross profit to revenue and the amount of net income to revenue. The following table presents these two profitability ratios—gross profit margin (gross profit divided by revenue) and net profit margin (net income divided by revenue)—for each year.

	FY2013 (%)	FY2014 (%)	FY2015 (%)	FY2016 (%)	FY2017 (%)
Acer					
Gross profit margin	6.26	8.78	9.43	9.97	10.69
Net profit margin	(5.70)	0.54	0.23	(2.11)	1.18
	FY2013 (%)	FY2014 (%)	FY2015 (%)	FY2016 (%)	FY2017 (%)
Lenovo					
Gross profit margin	13.08	14.43	14.75	14.19	13.83
Net profit margin	2.11	1.81	(0.32)	1.23	(0.28)

The net profit margins indicate that both companies' profitability is relatively low. Acer's net profit margin is lower than Lenovo's in three out of the five years. Acer's gross profit margin increased each year but remains significantly below that of Lenovo. Lenovo's gross profit margin grew from FY2013 to FY2015 and then declined in FY2016 and FY2017. Overall, Lenovo is the more profitable company, likely attributable to its larger size and commensurate economies of scale. (Lenovo has the largest share of the personal computer market relative to other personal computer companies.)

4. FINANCIAL RATIO ANALYSIS

LEARNING OUTCOME

The candidate should be able to:

- ☐ **describe tools and techniques used in financial analysis, including their uses and limitations**

There are many relationships among financial accounts and various expected relationships from one point in time to another. Ratios are a useful way of expressing these relationships. Ratios express one quantity in relation to another, usually as a quotient.

Extensive academic research has examined the importance of ratios in predicting stock returns (Ou and Penman, 1989; Abarbanell and Bushee, 1998) or credit failure (Altman, 1968; Ohlson, 1980; Hopwood et al., 1994). This research has found that financial statement ratios are effective in selecting investments and in predicting financial distress. Practitioners routinely use ratios to derive and communicate the value of companies and securities.

Several aspects of ratio analysis are important to understand. First, the computed ratio is not “the answer.” The ratio is an *indicator* of some aspect of a company’s performance, telling what happened but not why it happened. For example, an analyst might want to answer the question: Which of two companies was more profitable? As demonstrated in the previous example, the net profit margin, which expresses profit relative to revenue, can provide insight into this question. Net profit margin is calculated by dividing net income by revenue:

$$\frac{\text{Net income}}{\text{Revenue}} .$$

Assume Company A has EUR100,000 of net income and Company B has EUR200,000 of net income. Company B generated twice as much income as Company A, but was it more profitable? Assume further that Company A has EUR2,000,000 of revenue, and thus a net profit margin of 5 percent, and Company B has EUR6,000,000 of revenue, and thus a net profit margin of 3.33 percent. Expressing net income as a percentage of revenue clarifies the relationship: For each EUR100 of revenue, Company A earns EUR5 in net income, whereas Company B earns only EUR3.33 for each EUR100 of revenue. So, we can now answer the question of which company was more profitable in percentage terms: Company A was more profitable, as indicated by its higher net profit margin of 5 percent. Note that Company A was more *profitable* despite the fact that Company B reported higher absolute amounts of net income and revenue. However, this ratio by itself does not tell us *why* Company A has a higher profit margin. Further analysis is required to determine the reason (perhaps higher relative sales prices or better cost control or lower effective tax rates).

Company size sometimes confers economies of scale, so the absolute amounts of net income and revenue are useful in financial analysis. However, ratios control for the effect of size, which enhances comparisons between companies and over time.

A second important aspect of ratio analysis is that differences in accounting policies (across companies and across time) can distort ratios, and a meaningful comparison, therefore, may involve adjustments to the financial data. Third, not all ratios are necessarily relevant to a particular analysis. The ability to select a relevant ratio or ratios to answer the research question is an analytical skill. Finally, as with financial analysis in general, ratio analysis does not stop with computation; interpretation of the result is essential. In practice, differences in ratios across time and across companies can be subtle, and interpretation is situation specific.

4.1. The Universe of Ratios

No authoritative bodies specify the exact formulas for computing ratios or provide a standard, comprehensive list of ratios. Formulas and even names of ratios often differ from analyst to analyst or from database to database. The number of different ratios that can be created is practically limitless. Several widely accepted ratios, however, have been found to be useful, which are the focus of this module. The analyst should be aware that different ratios may be used in practice and that certain industries have

unique ratios tailored to the characteristics of that industry. When faced with an unfamiliar ratio, the analyst can examine the underlying formula to gain insight into what the ratio is measuring. For example, consider the following ratio formula:

$$\frac{\text{Operating income}}{\text{Average total assets}}$$

Never having seen this ratio, an analyst might question whether a result of 12 percent is better than 8 percent. The answer can be found in the ratio itself. The numerator is operating income and the denominator is average total assets, so the ratio can be interpreted as the amount of operating income generated per unit of assets. For every EUR100 of average total assets, generating EUR12 of operating income is better than generating EUR8 of operating income. Furthermore, it is apparent that this particular ratio is an indicator of profitability (as well as efficiency in use of assets in generating operating profits). When encountering a ratio for the first time, the analyst should evaluate the numerator and denominator to assess what the ratio is attempting to measure and how it should be interpreted. This is demonstrated in [Example 3](#).

EXAMPLE 3

Interpreting a Financial Ratio

A US insurance company reports that its “combined ratio” is determined by dividing losses and expenses incurred by net premiums earned. It reports the following combined ratios:

Exhibit 5: Combined Ratio					
Fiscal Year	5	4	3	2	1
Combined ratio	90.1%	104.0%	98.5%	104.1%	101.1%

1. Explain what this ratio is measuring and compare the results reported for each of the years shown in the chart. What other information might an analyst want to review before making any conclusions on this information?

Solution:

The combined ratio is a profitability measure. The ratio is explaining how much costs (losses and expenses) were incurred for every dollar of revenue (net premiums earned). The underlying formula indicates that a *lower* value for this ratio is better. The year 5 ratio of 90.1 percent means that for every dollar of net premiums earned, the costs were USD0.901, yielding a gross profit of \$0.099. Ratios greater than 100 percent indicate an overall loss. A review of the data indicates that there does not seem to be a consistent trend in this ratio. Profits were achieved in years 5 and 3. The results for years 4 and 2 show the most significant costs at approximately 104 percent.

The analyst would want to discuss this data further with management and understand the characteristics of the underlying business. He or she would want to understand why the results are so volatile. The analyst would also want to determine what should be used as a benchmark for this ratio.

The Operating income/Average total assets ratio is one of many versions of the **return on assets (ROA)** ratio. Note that there are other ways of specifying this formula based on how assets are defined. Some financial ratio databases compute ROA using the ending value of assets rather than average assets. In limited cases, one may also see beginning assets in the denominator. Which one is right? It depends on what you are trying to measure and the underlying company trends. If the company has a stable level of assets, the answer will not differ greatly under the three measures of assets (beginning, average, and ending). However, if the assets are growing (or shrinking), the results will differ among the three

measures. When assets are growing, operating income divided by ending assets may not make sense because some of the income would have been generated before some assets were purchased, and this would understate the company's performance. Similarly, if beginning assets are used, some of the operating income later in the year may have been generated only because of the addition of assets; therefore, the ratio would overstate the company's performance. Because operating income occurs throughout the period, it generally makes sense to use some average measure of assets. A good general rule is that when an income statement or cash flow statement number is in the numerator of a ratio and a balance sheet number is in the denominator, then an average should be used for the denominator. It is generally not necessary to use averages when only balance sheet numbers are used in both the numerator and denominator because both are determined as of the same date. However, in some instances, even ratios that only use balance sheet data may use averages. For example, **return on equity (ROE)**, which is defined as net income divided by average shareholders' equity, can be decomposed into other ratios, some of which only use balance sheet data. In decomposing ROE into component ratios, if an average is used in one of the component ratios, then it should be used in the other component ratios. The decomposition of ROE is discussed further in a later lesson.

If an average is used, judgment is also required about what average should be used. For simplicity, most ratio databases use a simple average of the beginning and end-of-year balance sheet amounts. If the company's business is seasonal so that levels of assets vary by interim period (semiannual or quarterly), then it may be beneficial to take an average over all interim periods, if available. (If the analyst is working within a company and has access to monthly data, this can also be used.)

4.2. Value, Purposes, and Limitations of Ratio Analysis

The value of ratio analysis is that it enables a financial analyst to evaluate past performance, assess the current financial position of the company, and gain insights useful for projecting future results. As noted previously, the ratio itself is not "the answer" but is an indicator of some aspect of a company's performance. Financial ratios provide insights into the following:

- economic relationships within a company that help analysts project earnings and free cash flow;
- a company's financial flexibility, or ability to obtain the cash required to grow and meet its obligations, even if unexpected circumstances develop;
- management's ability;
- changes in the company or industry over time; and
- comparability with peer companies or the relevant industry(ies).

Ratio analysis also has limitations. Factors to consider include the following:

- *The heterogeneity or homogeneity of a company's operating activities.* Companies may have divisions operating in many different industries. This can make it difficult to find comparable industry ratios to use for comparison purposes.
- *The need to determine whether the results of the ratio analysis are consistent.* One set of ratios may indicate a problem, whereas another set may indicate that the potential problem is only short term in nature.
- *The need to use judgment.* A key issue is whether a ratio for a company is within a reasonable range. Although financial ratios are used to help assess the growth potential and risk of a company, they cannot be used alone to directly value a company or its securities, or to determine its creditworthiness. The entire operation of the company must be examined, and the external economic and industry setting in which it is operating must be considered when interpreting financial ratios.
- *The use of alternative accounting methods.* Companies frequently have latitude when choosing certain accounting methods. Ratios taken from financial statements that employ different accounting choices may not be comparable unless adjustments are made. Some important accounting considerations include the following:

- FIFO (first in, first out), LIFO (last in, first out), or average cost inventory valuation methods (International Financial Reporting Standards [IFRS] does not allow LIFO);
- Cost or equity methods of accounting for unconsolidated affiliates;
- Straight-line or accelerated methods of depreciation; and
- Operating or finance lease treatment for lessors (under US GAAP, the type of lease affects classifications of expenses; under IFRS, operating lease treatment for lessors is not applicable).

Convergence efforts between IFRS and US GAAP make the financial statements of different companies more comparable and may overcome some of these difficulties. Nonetheless, there will remain accounting choices that the analyst must consider.

4.3. Sources of Ratios

Ratios may be computed using data obtained directly from companies' financial statements or from a database such as Bloomberg, Compustat, FactSet, or Thomson Reuters. The information provided by the database may include information as reported in companies' financial statements and ratios calculated based on the information. These databases are popular because they provide easy access to many years of historical data so that trends over time can be examined. They also allow for ratio calculations based on periods other than the company's fiscal year, such as for the trailing 12 months (TTM) or most recent quarter (MRQ).

EXAMPLE 4

Trailing 12 Months

On 15 July, an analyst is examining a company with a fiscal year ending on 31 December. Use the following data to calculate the company's TTM earnings (for the period ended 30 June 2018):

- Earnings for the year ended 31 December 2017: USD1,200;
- Earnings for the six months ended 30 June 2017: USD550; and
- Earnings for the six months ended 30 June 2018: USD750.

Solution:

The company's TTM earnings is USD1,400, calculated as $\text{USD1,200} - \text{USD550} + \text{USD750}$.

Analysts should be aware that the underlying formulas for ratios may differ by vendor. The formula used should be obtained from the vendor, and the analyst should determine whether any adjustments are necessary. Furthermore, database providers often exercise judgment when classifying items. For example, operating income may not appear directly on a company's income statement, and the vendor may use judgment to classify income statement items as "operating" or "non-operating." Variation in such judgments would affect any computation involving operating income. It is therefore a good practice to use the same source for data when comparing different companies or when evaluating the historical record of a single company. Analysts should verify the consistency of formulas and data classifications by the data source. Analysts should also be mindful of the judgments made by a vendor in data classifications and refer to the source financial statements until they are comfortable that the classifications are appropriate.

Collection of financial data from regulatory filings and calculation of ratios can be automated. The eXtensible Business Reporting Language (XBRL) is a mechanism that attaches "smart tags" to financial information (e.g., total assets), so that software can automatically collect the data and perform desired computations. The organization developing XBRL (www.xbrl.org) is an international nonprofit consortium of more than 600 members from companies, associations, and agencies, including the

International Accounting Standards Board (IASB). Many stock exchanges and regulatory agencies around the world now use XBRL for receiving and distributing public financial reports from listed companies.

Analysts can compare a subject company to similar (peer) companies in vendor databases or use aggregate industry data. For non-public companies, aggregate industry data can be obtained from such sources as Annual Statement Studies by the Risk Management Association or Dun & Bradstreet. These publications typically provide industry data with companies sorted into quartiles. By definition, 25 percent of companies' ratios fall within the lowest quartile, 25 percent have ratios between the lower quartile and median value, and so on. Analysts can then determine a company's relative standing in the industry.

5. COMMON SIZE BALANCE SHEETS AND INCOME STATEMENTS

LEARNING OUTCOME

The candidate should be able to:

- ☐ describe tools and techniques used in financial analysis, including their uses and limitations

Common-size analysis involves expressing financial data, including entire financial statements, in relation to a single financial statement item, or base. Items used most frequently as the bases are total assets or revenue. In essence, common-size analysis creates a ratio between every financial statement item and the base item. Common-size analysis was demonstrated in earlier modules for the income statement, balance sheet, and cash flow statement. In this lesson, we present common-size analysis of financial statements in greater detail and include further discussion of their interpretation.

Common-Size Analysis of the Balance Sheet A vertical common-size balance sheet, prepared by dividing each item on the balance sheet by the same period's total assets and expressing the results as percentages, highlights the composition of the balance sheet. What is the mix of assets being used? How is the company financing itself? How does one company's balance sheet composition compare with that of peer companies, and what are the reasons for any differences? A horizontal common-size balance sheet, prepared by computing the increase or decrease in percentage terms of each balance sheet item from the prior year or prepared by dividing the quantity of each item by a base year quantity of the item, highlights changes in items. These changes can be compared to expectations.

Exhibit 6 presents a vertical common-size (partial) balance sheet for a hypothetical company in two time periods. In this example, receivables have increased from 35 percent to 57 percent of total assets and the ratio has increased by 63 percent from Period 1 to Period 2. What are possible reasons for such an increase? The increase might indicate that the company is making more of its sales on a credit basis rather than a cash basis, perhaps in response to some action taken by a competitor. Alternatively, the increase in receivables as a percentage of assets may have occurred because of a change in another current asset category, for example, a decrease in the level of inventory; the analyst would then need to investigate why that asset category has changed. Another possible reason for the increase in receivables as a percentage of assets is that the company has lowered its credit standards, relaxed its collection procedures, or adopted more aggressive revenue recognition policies. The analyst can turn to other comparisons and ratios (e.g., comparing the rate of growth in accounts receivable with the rate of growth in sales) to help determine which explanation is most likely.

Exhibit 6: Vertical Common-Size (Partial) Balance Sheet for a Hypothetical Company

	Period 1	Period 2
	Percent of Total Assets	Percent of Total Assets

	Period 1 Percent of Total Assets	Period 2 Percent of Total Assets
Cash	25	15
Receivables	35	57
Inventory	35	20
Fixed assets, net of depreciation	5	8
Total assets	100	100

5.1. Common-Size Analysis of the Income Statement

A vertical common-size income statement divides each income statement item by revenue, or sometimes by total assets (especially in the case of financial institutions). If there are multiple revenue sources, a decomposition of revenue in percentage terms is useful. [Exhibit 7](#) presents a hypothetical company's vertical common-size income statement in two time periods. Revenue is separated into the company's four services, each shown as a percentage of total revenue.

In this example, revenues from Service A have become a far greater percentage of the company's total revenue (30 percent in Period 1 and 45 percent in Period 2). What are possible reasons for and implications of this change in business mix? Did the company make a strategic decision to sell more of Service A, perhaps because it is more profitable? Apparently not, because the company's earnings before interest, taxes, depreciation, and amortization (EBITDA) declined from 53 percent of sales to 45 percent, so other possible explanations should be examined. In addition, we note from the composition of operating expenses that the main reason for this decline in profitability is that salaries and employee benefits have increased from 15 percent to 25 percent of total revenue. Are more highly compensated employees required for Service A? Were higher training costs incurred to increase revenues from Service A? If the analyst wants to predict future performance, the causes of these changes must be understood.

In addition, [Exhibit 7](#) shows that the company's income tax as a percentage of sales has declined dramatically (from 15 percent to 8 percent). Furthermore, taxes as a percentage of earnings before tax (EBT) (the effective tax rate, which is usually the more relevant comparison), have decreased from 36 percent ($= 15/42$) to 24 percent ($= 8/34$). Is Service A, which in Period 2 is a greater percentage of total revenue, provided in a jurisdiction with lower tax rates? If not, what is the explanation for the change in effective tax rate?

The observations based on [Exhibit 7](#) summarize the issues that can be raised through analysis of the vertical common-size income statement.

Exhibit 7: Vertical Common-Size Income Statement for Hypothetical Company

	Period 1 Percent of Total Revenue	Period 2 Percent of Total Revenue
Revenue source: Service A	30	45
Revenue source: Service B	23	20
Revenue source: Service C	30	30
Revenue source: Service D	17	5
Total revenue	100	100
Operating expenses (excluding depreciation)		
Salaries and employee benefits	15	25

	Period 1Percent of Total Revenue	Period 2Percent of Total Revenue
Administrative expenses	22	20
Rent expense	10	10
EBITDA	53	45
Depreciation and amortization	4	4
EBIT	49	41
Interest paid	7	7
EBT	42	34
Income tax provision	15	8
Net income	27	26

EBIT = earnings before interest and tax.

6. CROSS-SECTIONAL, TREND ANALYSIS, AND RELATIONSHIPS IN FINANCIAL STATEMENTS

LEARNING OUTCOME

The candidate should be able to:

- ☐ describe tools and techniques used in financial analysis, including their uses and limitations

As noted previously, ratios and common-size statements derive their utility through comparison. **Cross-sectional analysis** (sometimes called “relative analysis”) compares a specific metric for one company with the same metric for another company or group of companies measured at the same point in time or over the same range of time, allowing comparisons even though the companies might be of significantly different sizes or operate in different currencies. This is illustrated in [Exhibit 8](#).

Exhibit 8: Vertical Common-Size (Partial) Balance Sheet for Two Hypothetical Companies

Assets	Company 1 Percent of Total Assets	Company 2 Percent of Total Assets
Cash	38	12
Receivables	33	55
Inventory	27	24
Fixed assets net of depreciation	1	2
Investments	1	7
Total Assets	100	100

[Exhibit 8](#) presents a vertical common-size (partial) balance sheet for two hypothetical companies at the same point in time. Company 1 is clearly more liquid (liquidity is a function of how quickly assets can be converted into cash) than Company 2, which has only 12 percent of assets available as cash, compared with the highly liquid Company 1, which has 38 percent of assets available as cash. Given that cash is generally a relatively low-yielding asset and thus not a particularly efficient use of excess funds, why does Company 1 hold such a large percentage of total assets in cash? Perhaps the company is preparing for an acquisition, or maintains a large cash position as insulation from a particularly volatile operating environment. Another issue highlighted by the comparison in this example is the relatively high percentage of receivables in Company 2's assets, which may indicate a greater proportion of credit sales, overall changes in asset composition, lower credit or collection standards, or aggressive accounting policies.

6.1. Trend Analysis

When looking at financial statements and ratios, trends in the data, whether they are improving or deteriorating, are as important as the current absolute or relative levels. Trend analysis provides important information regarding historical performance and growth and, given a sufficiently long history of accurate seasonal information, can be of great assistance as a planning and forecasting tool for management and analysts.

[Exhibit 9](#) presents a partial balance sheet for a hypothetical company over five periods. The last two columns of the table show the changes for Period 5 compared with Period 4, expressed both in absolute currency (in this case, dollars) and in percentages. A small percentage change could hide a significant currency change and vice versa, prompting the analyst to investigate the reasons despite one of the changes being relatively small. In this example, the largest percentage change was in investments, which decreased by 33.3 percent. However, an examination of the absolute currency amount of changes shows that investments changed by only USD2 million, and the more significant change was the USD12 million increase in receivables.

Another way to present data covering a period of time is to show each item in relation to the same item in a base year (i.e., a horizontal common-size balance sheet). [Exhibit 10](#) and [Exhibit 11](#) illustrate alternative presentations of horizontal common-size balance sheets. [Exhibit 10](#) presents the information from the same partial balance sheet as in [Exhibit 9](#), but indexes each item relative to the same item in Period 1. For example, in Period 2, the company had USD29 million cash, which is 74 percent or 0.74 of the amount of cash it had in Period 1. Expressed as an index relative to Period 1, where each item in Period 1 is given a value of 1.00, the value in Period 2 would be 0.74 ($\text{USD}29/\text{USD}39 = 0.74$). In Period 3, the company had USD27 million cash, which is 69 percent of the amount of cash it had in Period 1 ($\text{USD}27/\text{USD}39 = 0.69$).

[Exhibit 11](#) presents the percentage change in each item, relative to the previous year. For example, the change in cash from Period 1 to Period 2 was -25.6 percent ($\text{USD}29/\text{USD}39 - 1 = -0.256$), and the change in cash from Period 2 to Period 3 was -6.9 percent ($\text{USD}27/\text{USD}29 - 1 = -0.069$). An analyst will select the horizontal common-size balance that addresses the particular period of interest. [Exhibit 10](#) clearly highlights that in Period 5 compared to Period 1, the company has less than half the amount of cash, four times the amount of investments, and eight times the amount of property, plant, and equipment. [Exhibit 11](#) highlights year-to-year changes: For example, cash has declined in each period. Presenting data this way highlights significant changes. Again, note that a mathematically big change is not necessarily an important change. For example, fixed assets increased 100 percent (i.e., doubled between Period 1 and 2); however, as a proportion of total assets, fixed assets increased from 1 percent of total assets to 2 percent of total assets. The company's working capital assets (receivables and inventory) are a far higher proportion of total assets and would likely warrant more attention from an analyst.

An analysis of horizontal common-size balance sheets highlights structural changes that have occurred in a business. Past trends are obviously not necessarily an accurate predictor of the future, especially when the economic or competitive environment changes. An examination of past trends is more valuable when the macroeconomic and competitive environments are relatively stable and when the analyst is reviewing a stable or mature business. However, even in less stable contexts, historical analysis can serve as a basis for developing expectations. Understanding of past trends is helpful in assessing whether these trends are likely to continue or if the trend is likely to change direction.

Exhibit 9: Partial Balance Sheet for a Hypothetical Company over Five Periods

Assets (US dollar millions)	Period					Change 4 to 5 (US dollar millions)	Change 4 to 5 (%)
	1	2	3	4	5		
Cash	39	29	27	19	16	-3	-15.8
Investments	1	7	7	6	4	-2	-33.3
Receivables	44	41	37	67	79	12	17.9
Inventory	15	25	36	25	27	2	8.0
Fixed assets net of depreciation	1	2	6	9	8	-1	-11.1
Total assets	100	104	113	126	134	8	6.3

Exhibit 10: Horizontal Common-Size (Partial) Balance Sheet for a Hypothetical Company over Five Periods, with Each Item Expressed Relative to the Same Item in Period One

Assets	Period				
	1	2	3	4	5
Cash	1.00	0.74	0.69	0.49	0.41
Investments	1.00	7.00	7.00	6.00	4.00
Receivables	1.00	0.93	0.84	1.52	1.80
Inventory	1.00	1.67	2.40	1.67	1.80
Fixed assets net of depreciation	1.00	2.00	6.00	9.00	8.00
Total assets	1.00	1.04	1.13	1.26	1.34

Exhibit 11: Horizontal Common-Size (Partial) Balance Sheet for a Hypothetical Company over Five Periods, with Percent Change in Each Item Relative to the Prior Period

Assets	Period			
	2 (%)	3 (%)	4 (%)	5 (%)
Cash	-25.6	-6.9	-29.6	-15.8
Investments	600.0	0.0	-14.3	-33.3
Receivables	-6.8	-9.8	81.1	17.9
Inventory	66.7	44.0	-30.6	8.0
Fixed assets net of depreciation	100.0	200.0	50.0	-11.1
Total assets	4.0	8.7	11.5	6.3

One measure of success is for a company to grow at a rate greater than the rate of the overall market in which it operates. Companies that grow slowly may find themselves unable to attract equity capital. Conversely, companies that grow too quickly may find that their administrative and management information systems cannot keep up with the rate of expansion.

6.2. Relationships Among Financial Statements

Trend data generated by a horizontal common-size analysis can be compared across financial statements. For example, the growth rate of assets for the hypothetical company in [Exhibit 12](#) can be compared with the company’s growth in revenue over the same period of time. If revenue is growing more quickly than assets, the company may be increasing its efficiency (i.e., generating more revenue for every dollar invested in assets).

As another example, consider the following year-over-year percentage changes for a hypothetical company:

Exhibit 12: Year-over-Year Percentage Changes

Revenue	+20%
Net income	+25%
Operating cash flow	–10%
Total assets	+30%

Net income is growing faster than revenue, which indicates increasing profitability. However, the analyst would need to determine whether the faster growth in net income resulted from continuing operations or from non-operating, non-recurring items. In addition, the 10 percent decline in operating cash flow despite increasing revenue and net income clearly warrants further investigation because it could indicate a problem with earnings quality (perhaps aggressive reporting of revenue). Lastly, the fact that assets have grown faster than revenue indicates the company’s efficiency may be declining. The analyst should examine the composition of the increase in assets and the reasons for the changes. [Example 5](#) illustrates a historical example of a company for which comparisons of trend data from different financial statements were actually indicative of aggressive accounting policies.

EXAMPLE 5

Use of Comparative Growth Information¹

In July 1996, Sunbeam, a US company, brought in new management to turn the company around. In the following year, 1997, using 1996 as the base, the following was observed based on reported numbers:

Exhibit 13: Sunbeam Revenue

Revenue	+19 percent
Inventory	+58 percent
Receivables	+38 percent

It is generally more desirable to observe inventory and receivables growing at a slower (or similar) rate than revenue growth. Receivables growing faster than revenue can indicate operational issues, such as lower credit standards or aggressive accounting policies for revenue recognition. Similarly, inventory growing faster than revenue can indicate an operational problem with obsolescence or aggressive accounting policies, such as an improper overstatement of inventory to increase profits.

In this case, the explanation lay in aggressive accounting policies. Sunbeam was later charged by the US Securities and Exchange Commission with improperly accelerating the recognition of revenue and engaging in other practices, such as billing customers for inventory prior to shipment.

7. THE USE OF GRAPHS AND REGRESSION ANALYSIS

LEARNING OUTCOME

The candidate should be able to:

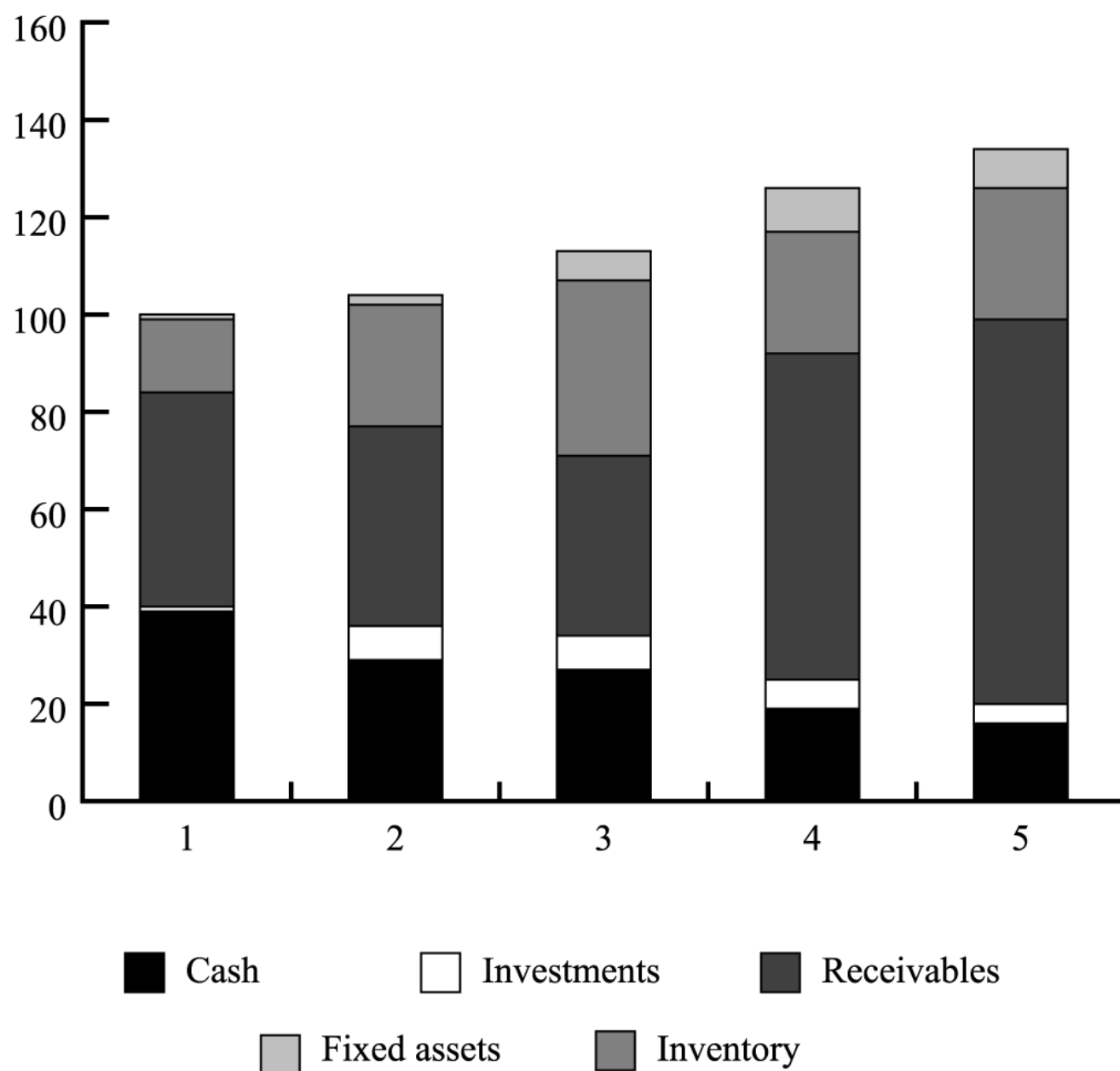
- ☐ **describe tools and techniques used in financial analysis, including their uses and limitations**

Graphs facilitate comparison of performance and financial structure over time, highlighting changes in significant aspects of business operations. In addition, graphs provide the analyst (and management) with a visual overview of risk trends in a business. Graphs may also be used effectively to communicate the analyst's conclusions regarding financial condition and risk management aspects.

[Exhibit 14](#) presents the information from Exhibit 9 in a stacked column format. The graph makes the significant decline in cash and growth in receivables (both in absolute terms and as a percentage of assets) readily apparent. In [Exhibit 14](#), the vertical axis shows US dollar millions and the horizontal axis denotes the period.

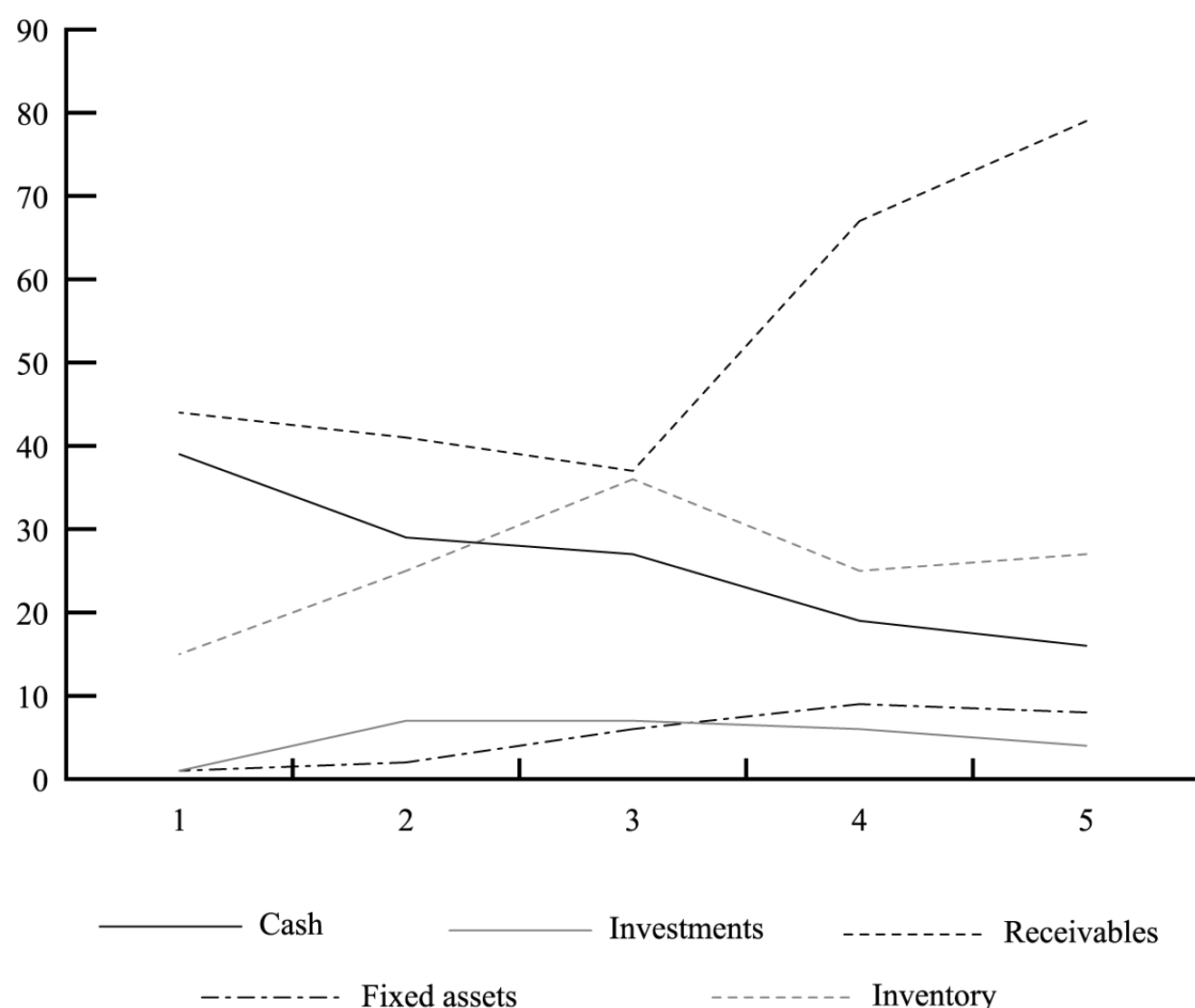
Choosing the appropriate graph to communicate the most significant conclusions of a financial analysis is a skill. In general, pie graphs are most useful to communicate the composition of a total value (e.g., assets over a limited amount of time, say one or two periods). Line graphs are useful when the focus is on the change in amount for a limited number of items over a relatively longer time period. When the composition and amounts, as well as their change over time, are all important, a stacked column graph can be useful.

Exhibit 14: Stacked Column Graph of Asset Composition of Hypothetical Company over Five Periods



When comparing Period 5 with Period 4, the growth in receivables appears to be within normal bounds; but when comparing Period 5 with earlier periods, the dramatic growth becomes apparent. In the same manner, a simple line graph will also illustrate the growth trends in key financial variables. [Exhibit 15](#) presents the information from Exhibit 9A as a line graph, illustrating the growth of assets of a hypothetical company over five periods. The steady decline in cash, volatile movements of inventory, and dramatic growth of receivables is clearly illustrated. Again, the vertical axis is shown in US dollar millions and the horizontal axis denotes periods.

Exhibit 15: Line Graph of Growth of Assets of Hypothetical Company over Five Periods



7.1. Regression Analysis

When analyzing the trend in a specific line item or ratio, frequently it is possible simply to visually evaluate the changes. For more complex situations, regression analysis can help identify relationships (or correlation) between variables. For example, a regression analysis could relate a company's sales to GDP over time, providing insight into whether the company is cyclical. In addition, the statistical relationship between sales and GDP could be used as a basis for forecasting sales.

Other examples in which regression analysis may be useful include the relationship between a company's sales and inventory over time, or the relationship between hotel occupancy and a company's hotel revenues. In addition to providing a basis for forecasting, regression analysis facilitates identification of items or ratios that are not behaving as expected, given historical statistical relationships.

8. COMMON RATIO CATEGORIES, INTERPRETATION, AND CONTEXT

LEARNING OUTCOME

The candidate should be able to:

- ☐ calculate and interpret activity, liquidity, solvency, and profitability ratios

In the previous lesson, we focused on ratios resulting from common-size analysis. In this lesson, we expand the discussion to include other commonly used financial ratios and the broad classes into which they are categorized. There is some overlap with common-size financial statement ratios. For example, a

common indicator of profitability is the net profit margin, which is calculated as net income divided by sales. This ratio appears on a vertical common-size income statement. Other ratios involve information from multiple financial statements or even data from outside the financial statements.

Because of the large number of ratios, it is helpful to think about ratios in terms of broad categories based on what aspects of performance a ratio is intended to detect. Financial analysts and data vendors use a variety of categories to classify ratios. The category names and the ratios included in each category can differ. Common ratio categories include activity, liquidity, solvency, and profitability, which were introduced in earlier modules in Corporate Issuers. These categories are summarized in [Exhibit 16](#). Each category measures a different aspect of the company’s business, but all are useful in evaluating a company’s overall ability to generate cash flows from operating its business and the associated risks.

Exhibit 16: Categories of Financial Ratios

Category	Description
Activity	Activity ratios measure the efficiency of a company’s operations, such as the collection of receivables and management of inventory.
Liquidity	Liquidity ratios measure the company’s ability to meet its short-term obligations.
Solvency	Solvency ratios measure a company’s ability to meet long-term obligations. Subsets of these ratios are also known as “leverage” and “long-term debt” ratios.
Profitability	Profitability ratios measure the company’s ability to generate profits from its resources (assets) or sales.

8.1. Interpretation and Context

Financial ratios can be interpreted only in the context of other information. In general, the financial ratios of a company are compared with those of its major competitors (cross-sectional and trend analysis) and to the company’s prior periods (trend analysis). The goal is to understand the underlying causes of divergence between a company’s ratios and those of the industry. Even ratios that remain consistent require understanding because consistency can sometimes indicate accounting policies selected to smooth earnings. An analyst should evaluate financial ratios in the context of the following:

- 1. Prior period results.** Trend analysis can reveal whether a company’s performance and position are weakening or strengthening.
- 2. Expectations.** These are point or range estimates for key values, such as sales growth, profit margins, and leverage ratios, that are specified by the analyst or external analysts before results are published. Differences from expectations should be scrutinized for setting expectations in subsequent periods.
- 3. Industry peers and competitors (cross-sectional analysis).** A company can be compared with others in its industry by relating its financial ratios to industry norms or to a subset of the companies in an industry. When industry norms are used to make judgments, care must be taken for the following reasons:
 - Companies may have several different lines of business. This will cause aggregate financial ratios to be distorted. It is better to examine industry-specific ratios by lines of business.
 - Differences in business model and corporate strategies can affect certain financial ratios.
 - Some ratios are industry specific, and not all ratios are important to all industries.
 - Differences in accounting methods used by companies can distort financial ratios.

- 4. *Company goals and strategy.* Actual ratios can be compared with company objectives to determine whether objectives are being attained and whether the results are consistent with the company’s strategy.
- 5. *Economic conditions.* For cyclical companies, financial ratios tend to improve when the economy is strong and weaken during recessions. Therefore, financial ratios should be examined in light of the current phase of the business cycle.

The following lessons discuss the calculation and interpretation of activity, liquidity, solvency, and profitability ratios using a company’s financial statements.

9. ACTIVITY RATIOS

LEARNING OUTCOME

The candidate should be able to:

- ☐ calculate and interpret activity, liquidity, solvency, and profitability ratios

Activity ratios, also known as **asset utilization ratios** or **operating efficiency ratios**, are measures of operational performance—how effectively the company is using working capital and longer term assets. Since working capital efficiency has a direct impact on liquidity, some activity ratios are also useful in assessing liquidity.

9.1. Calculation of Activity Ratios

Exhibit 17 presents commonly used activity ratios.

Exhibit 17: Definitions of Commonly Used Activity Ratios

Activity Ratios	Numerator	Denominator
Inventory turnover	Cost of sales or cost of goods sold	Average inventory
Days of inventory on hand (DOH)	Number of days in period	Inventory turnover
Receivables turnover	Revenue	Average receivables
Days of sales outstanding (DSO)	Number of days in period	Receivables turnover
Payables turnover	Cost of sales or cost of goods sold	Average trade payables
Number of days of payables	Number of days in period	Payables turnover
Working capital turnover	Revenue	Average working capital
Fixed asset turnover	Revenue	Average net fixed assets
Total asset turnover	Revenue	Average total assets

Activity ratios generally combine information from the income statement in the numerator with balance sheet items in the denominator. Because the income statement measures what happened *during* a period,

whereas the balance sheet shows the condition only at the end of the period, average balance sheet figures are normally used for consistency.

Activity ratios can be computed for any annual or interim period, but care must be taken in the interpretation and comparison across periods. For example, if the same company had cost of goods sold for the first quarter (90 days) of the following year of EUR35,000 and average inventory of EUR11,000, the inventory turnover would be 3.18 times. However, this turnover rate is 3.18 times per quarter, which is not directly comparable to the 12 times per year in the preceding year. In this case, we can annualize the quarterly inventory turnover rate by multiplying the quarterly turnover by 4 (12 months/3 months; or by 4.06, using 365 days/90 days) for comparison to the annual turnover rate. So, the quarterly inventory turnover is equivalent to a 12.72 annual inventory turnover (or 12.90 if we annualize the ratio using a 90-day quarter and a 365-day year). To compute the DOH using quarterly data, we can use the quarterly turnover rate and the number of days in the quarter for the numerator—or, we can use the annualized turnover rate and 365 days; either results in DOH of around 28.3, with slight differences due to rounding (90/3.18 = 28.30 and 365/12.90 = 28.29). Another time-related computational detail is that for companies using a 52/53-week annual period and for leap years, the actual days in the year should be used rather than 365.

In some cases, an analyst may want to know how many days of inventory are on hand at the end of the year rather than the average for the year. In this case, it would be appropriate to use the year-end inventory balance in the computation rather than the average. If the company is growing rapidly or if costs are increasing rapidly, analysts should consider using cost of goods sold just for the fourth quarter in this computation because the cost of goods sold of earlier quarters may not be relevant. [Example 6](#) further demonstrates computation of activity ratios using Hong Kong Stock Exchange (HKEX)–listed Lenovo Group Limited.

EXAMPLE 6

Computation of Activity Ratios

An analyst would like to evaluate Lenovo Group’s efficiency in collecting its trade accounts receivable during the fiscal year ended 31 March 2018 (FY2017). The analyst gathers the following information in [Exhibit 18](#) from Lenovo’s annual and interim reports:

Exhibit 18: Lenovo	
	US dollar thousands
Trade receivables as of 31 March 2017	4,468,392
Trade receivables as of 31 March 2018	4,972,722
Revenue for year ended 31 March 2018	45,349,943

Calculate Lenovo’s receivables turnover and number of days of sales outstanding (DSO) for the fiscal year ended 31 March 2018.

Solution:

Receivables turnover	=	Revenue/Average receivables
	=	45,349,943/[(4,468,392 + 4,972,722)/2]
	=	45,349,943/4,720,557
	=	9.6069 times, or 9.6 rounded
DSO	=	Number of days in period/Receivables turnover
	=	365/9.6
	=	38.0 days

On average, it took Lenovo 38 days to collect receivables during the fiscal year ended 31 March 2018.

9.2. Interpretation of Activity Ratios

9.2.1. Inventory Turnover and DOH

Inventory turnover indicates the resources tied up in inventory (i.e., the carrying costs) and, therefore, can be used to indicate inventory management effectiveness. In general, inventory turnover and DOH (days of inventory on hand) should be benchmarked against industry norms.

A high inventory turnover ratio relative to industry norms might indicate highly effective inventory management. Alternatively, a high inventory turnover ratio (and commensurately low DOH) could possibly indicate the company does not carry adequate inventory, so shortages could potentially hurt revenue. To assess which explanation is more likely, the analyst can compare the company's revenue growth with that of the industry. Slower growth combined with higher inventory turnover could indicate inadequate inventory levels. Revenue growth at or above the industry's growth supports the interpretation that the higher turnover reflects greater inventory management efficiency.

A low inventory turnover ratio (and commensurately high DOH) relative to the rest of the industry could be an indicator of slow-moving inventory, perhaps because of technological obsolescence or a change in fashion. Again, comparing the company's sales growth with the industry can offer insight.

9.2.2. Receivables Turnover and DSO

The number of DSO (days of sales outstanding) reflects how fast the company collects cash from customers to whom it offers credit. Although limiting the numerator to sales made on credit in the receivables turnover would be more appropriate, credit sales information is usually not available to analysts; therefore, revenue as reported in the income statement is generally used.

As with inventory management, comparison of the company's sales growth relative to the industry can help the analyst assess whether sales are being lost due to stringent credit policies. In addition, comparing the company's estimates of uncollectible accounts receivable and actual credit losses with past experience and with peer companies can help assess whether low turnover reflects credit management issues. Companies often provide details of receivables aging (how much receivables have been outstanding by age). This can be used along with DSO to understand trends in collection, as demonstrated in [Example 7](#).

EXAMPLE 7

Evaluation of an Activity Ratio

An analyst has computed the average DSO for Lenovo for fiscal years ended 31 March 2018 and 2017:

Exhibit 19: Average DSO

	FY2017	FY2016
Days of sales outstanding	38.0	37.6

Revenue increased from USD43.035 billion for fiscal year ended 31 March 2017 (FY2016) to USD45.350 billion for fiscal year ended 31 March 2018 (FY2017). The analyst would like to better understand the change in the company's DSO from FY2016 to FY2017 and whether the increase is indicative of any issues with the customers' credit quality. The analyst collects accounts receivable aging information from Lenovo's annual reports and computes the percentage of accounts receivable by days outstanding. This information is presented in [Exhibit 20](#):

Exhibit 20: Accounts Receivable

	FY2017		FY2016		FY2015	
	USD000	Percent	USD000	Percent	USD000	Percent
Accounts receivable						
0–30 days	3,046,240	59.95	2,923,083	63.92	3,246,600	71.99
31–60 days	1,169,286	23.01	985,251	21.55	617,199	13.69
61–90 days	320,183	6.30	283,050	6.19	240,470	5.33
Over 90 days	545,629	10.74	381,387	8.34	405,410	8.99
Total	5,081,338	100.00	4,572,771	100.00	4,509,679	100.00
Less: Provision for impairment	-108,616	-2.14	-104,379	-2.28	-106,172	-2.35
Trade receivables, net	4,972,722	97.86	4,468,392	97.72	4,403,507	97.65
<i>Total sales</i>	<i>45,349,943</i>		<i>43,034,731</i>		<i>44,912,097</i>	

Note: Lenovo's footnotes disclose that general trade customers are provided with credit terms ranging from 0 to 120 days.

These data indicate that total accounts receivable increased by 11 percent (net, after impairment) in FY2017 versus FY2016, while total sales increased by only 5.4 percent. Further, the percentage of receivables in all categories older than 30 days has increased over the three-year period, indicating that customers are indeed taking longer to pay. Conversely, the provision for impairment (estimate of uncollectible accounts) has declined as a percent of total receivables. Considering all this information, the company may be increasing customer financing purposely to drive its sales growth. They also may be underestimating the impairment. This should be investigated further by the analyst.

9.2.3. Payables Turnover and the Number of Days of Payables

The number of days of payables reflects the average number of days the company takes to pay its suppliers, and the payables turnover ratio measures how many times per year the company theoretically pays off all its creditors. A payables turnover ratio that is high (low days payable) relative to the industry could indicate that the company is not making full use of available credit facilities; alternatively, it could result from a company taking advantage of early payment discounts. An excessively low turnover ratio (high days payable) could indicate trouble making payments on time, or alternatively, exploitation of lenient supplier terms. This is another example in which it is useful to look simultaneously at other ratios. If liquidity ratios indicate that the company has sufficient cash and other short-term assets to pay obligations and yet the days payable ratio is relatively high, the analyst would favor the lenient supplier credit and collection policies as an explanation.

9.2.4. Working Capital Turnover

Working capital turnover indicates how efficiently the company generates revenue with its working capital. For example, a working capital turnover ratio of 4.0 indicates that the company generates EUR4 of revenue for every EUR1 of working capital. A high working capital turnover ratio indicates greater efficiency (i.e., the company is generating a high level of revenues relative to working capital). For some companies, working capital can be near zero or negative, rendering this ratio incapable of being interpreted. The following two ratios are more useful in those circumstances.

9.2.5. Fixed Asset Turnover

This ratio measures how efficiently the company generates revenues from its investments in fixed assets. Generally, a higher fixed asset turnover ratio indicates more efficient use of fixed assets in generating revenue. A low ratio can indicate inefficiency, a capital-intensive business environment, or a new business not yet operating at full capacity—in which case the analyst will not be able to link the ratio directly to efficiency. In addition, asset turnover can be affected by factors other than a company's efficiency. The fixed asset turnover ratio would be lower for a company whose assets are newer (and, therefore, less depreciated and so reflected in the financial statements at a higher carrying value) than the ratio for a company with older assets (that are thus more depreciated and so reflected at a lower carrying value). The fixed asset ratio can be erratic because, although revenue may have a steady growth rate, increases in fixed assets may not follow a smooth pattern; so, every year-to-year change in the ratio does not necessarily indicate important changes in the company's efficiency.

9.2.6. Total Asset Turnover

The total asset turnover ratio measures the company's overall ability to generate revenues with a given level of assets. A ratio of 1.20 would indicate that the company is generating EUR1.20 of revenues for every EUR1 of average assets. A higher ratio indicates greater efficiency. Because this ratio includes both fixed and current assets, inefficient working capital management can distort overall interpretations. It is therefore helpful to analyze working capital and fixed asset turnover ratios separately.

A low asset turnover ratio can be an indicator of inefficiency or of relative capital intensity of the business. The ratio also reflects strategic decisions by management—for example, the decision whether to use a more labor-intensive (and less capital-intensive) approach to its business or a more capital-intensive (and less labor-intensive) approach.

When interpreting activity ratios, the analysts should examine not only the individual ratios but also the collection of relevant ratios to determine the overall efficiency of a company. [Example 8](#) demonstrates the evaluation of activity ratios, both narrow (e.g., days of inventory on hand) and broad (e.g., total asset turnover) for a hypothetical manufacturer.

EXAMPLE 8

Evaluation of Activity Ratios

ZZZ Company is a hypothetical manufacturing company. As part of an analysis of management's operating efficiency, an analyst collects the following activity ratios from a data provider:

Exhibit 21: Operating Efficiency

Ratio	2018	2017	2016	2015
DOH	35.68	40.70	40.47	48.51
DSO	45.07	58.28	51.27	76.98
Total asset turnover	0.36	0.28	0.23	0.22

These ratios indicate that the company has improved on all three measures of activity over the four-year period. The company appears to be managing its inventory more efficiently, is collecting receivables faster, and is generating a higher level of revenues relative to total assets. The overall trend appears good, but thus far, the analyst has only determined *what* happened. A more important question is *why* the ratios improved, because understanding good changes as well as bad ones facilitates judgments about the company's future performance. To answer this question, the analyst examines company financial reports as well as external information about the industry and economy. In examining the annual report, the analyst notes that in the fourth quarter of 2018, the company experienced an "inventory correction" and that the company recorded an allowance for the decline in market value and obsolescence of inventory of about 15 percent of year-end inventory value (compared with about a 6 percent allowance in the prior year). This reduction in the value of inventory accounts for a large portion of the decline in DOH from 40.70 in 2017 to 35.68 in 2018. Management claims that this inventory obsolescence is a short-term issue; analysts can watch DOH in future interim periods to confirm this assertion. In any event, all else being equal, the analyst would likely expect DOH to return to a level closer to 40 days going forward.

More positive interpretations can be drawn from the total asset turnover. The analyst finds that the company's revenues increased more than 35 percent, whereas total assets increased only by about 6 percent. Based on external information about the industry and economy, the analyst attributes the increased revenues both to overall growth in the industry and to the company's increased market share. Management was able to achieve growth in revenues with a comparatively modest increase in assets, leading to an improvement in total asset turnover. Note further that part of the reason for the increase in asset turnover is lower DOH and DSO.

10. LIQUIDITY RATIOS

LEARNING OUTCOME

The candidate should be able to:

- ❑ **calculate and interpret activity, liquidity, solvency, and profitability ratios**

Liquidity analysis measures a company's ability to meet its short-term obligations. In the short run, a company's sources of liquidity typically include cash and marketable securities on hand and debt issuance. In the longer run, for non-financial companies, liquidity is addressed by cash flows from operations and managing the structure of liabilities, such as the timing of debt maturities (see the following discussion on the financial sector).

The level of liquidity needed differs from one industry to another. A particular company's liquidity position may vary according to the anticipated need for funds at any given time. Judging whether a company has adequate liquidity requires analysis of its historical funding requirements, current liquidity position, anticipated future funding needs, and options for reducing funding needs or attracting additional funds (including actual and potential sources of such funding).

Larger companies are usually better able to control the level and composition of their liabilities than smaller companies. Therefore, they may have more potential funding sources, including public capital and money markets. Greater discretionary access to capital markets also reduces the size of the liquidity buffer needed relative to companies without such access.

Contingent liabilities, such as letters of credit or financial guarantees, can also be relevant when assessing liquidity. The importance of contingent liabilities varies for the non-banking and banking sector. In the non-banking sector, contingent liabilities (usually disclosed in the footnotes to the company's financial statements) represent potential cash outflows, and when appropriate, should be included in an

assessment of a company’s liquidity. In the banking sector, contingent liabilities represent potentially significant cash outflows that are not dependent on the bank’s financial condition. Although outflows in normal market circumstances typically may be low, a general macroeconomic or market crisis can trigger a substantial increase in cash outflows related to contingent liabilities because of the increase in defaults and business bankruptcies that often accompany such events. In addition, such crises are usually characterized by diminished levels of overall liquidity, which can further exacerbate funding shortfalls. Therefore, for the banking sector, the effect of contingent liabilities on liquidity warrants particular attention.

10.1. Calculation of Liquidity Ratios

Common liquidity ratios, introduced in earlier modules in Corporate Issuers, are presented in [Exhibit 22](#). These liquidity ratios reflect a company’s position at a point in time and, therefore, typically use data from the ending balance sheet rather than averages.

Exhibit 22: Definitions of Commonly Used Liquidity Ratios

Liquidity Ratios	Numerator	Denominator
Current ratio	Current assets	Current liabilities
Quick ratio	Cash + Short-term marketable investments + Receivables	Current liabilities
Cash ratio	Cash + Short-term marketable investments	Current liabilities
Defensive interval ratio	Cash + Short-term marketable investments + Receivables	Daily cash expenditures
Additional Liquidity Measure		
Cash conversion cycle (net operating cycle)	DOH + DSO – Number of days of payables	

The **defensive interval ratio** measures how long a company can pay its daily cash expenditures using only its existing liquid assets, without additional cash flow coming in. This ratio is similar to the “burn rate” often computed for early-stage companies that are funded by venture capital funds and company insiders. The numerator of this ratio includes the same liquid assets used in the quick ratio, and the denominator is an estimate of daily cash expenditures. To obtain daily cash expenditures, the total of cash expenditures for the period is divided by the number of days in the period. Total cash expenditures for a period can be approximated by summing all expenses on the income statement—such as cost of goods sold; selling, general, and administrative expenses; and research and development expenses—and then subtracting any non-cash expenses, such as depreciation and amortization (typically, taxes are not included).

10.2. Interpretation of Liquidity Ratios

In the following, we discuss the interpretation of the five basic liquidity measures presented in [Exhibit 22](#).

10.2.1. Current Ratio

A higher current ratio indicates a higher level of liquidity (i.e., a greater ability to meet short-term obligations). A lower ratio indicates less liquidity, implying a greater reliance on operating cash flow and outside financing to meet short-term obligations. The current ratio implicitly assumes that inventories

and accounts receivable are indeed liquid (which is presumably not the case when related turnover ratios are low).

10.2.2. *Quick Ratio*

The quick ratio is more conservative than the current ratio because it includes only the more liquid current assets (sometimes referred to as “quick assets”) in relation to current liabilities. Like the current ratio, a higher quick ratio indicates greater liquidity.

The quick ratio reflects the fact that certain current assets—such as prepaid expenses, some taxes, and employee-related prepayments—represent costs of the current period that have been paid in advance and cannot usually be converted back into cash. This ratio also reflects the fact that inventory might not be easily and quickly converted into cash, and furthermore, that a company probably would not be able to sell all of its inventory for an amount equal to its carrying value, especially if it were required to sell the inventory quickly. In situations in which inventories are illiquid (as indicated, for example, by low inventory turnover ratios), the quick ratio may be a better indicator of liquidity than the current ratio.

10.2.3. *Cash Ratio*

The cash ratio normally represents a reliable measure of an entity’s liquidity in a crisis situation. Only highly marketable short-term investments and cash are included. In a general market crisis, the fair value of marketable securities could decrease significantly as a result of market factors, in which case even this ratio might not provide reliable information.

10.2.4. *Defensive Interval Ratio*

The defensive interval ratio measures how long the company can continue to pay its expenses from its existing liquid assets without receiving any additional cash inflow. A defensive interval ratio of 50 would indicate that the company can continue to pay its operating expenses for 50 days before running out of quick assets, assuming no additional cash inflows. A higher defensive interval ratio indicates greater liquidity. If a company’s defensive interval ratio is very low relative to peer companies or to the company’s own history, the analyst would want to ascertain whether there is sufficient cash inflow expected to mitigate the low defensive interval ratio.

10.2.5. *Cash Conversion Cycle (Net Operating Cycle)*

This cash conversion cycle metric indicates the amount of time that elapses from the point when a company invests in working capital until the point at which the company collects cash. A shorter cash conversion cycle indicates greater liquidity. A short cash conversion cycle implies that the company only needs to finance its inventory and accounts receivable for a short period of time. A longer cash conversion cycle indicates lower liquidity; it implies that the company must finance its inventory and accounts receivable for a longer period of time, possibly indicating a need for a higher level of capital to fund current assets. [Example 9](#) demonstrates the advantages of a short cash conversion cycle as well as how a company’s business strategies are reflected in financial ratios.

EXAMPLE 9

Evaluation of Liquidity Measures

An analyst is evaluating the liquidity of Apple and calculates the number of days of receivables, inventory, and accounts payable, as well as the overall cash conversion cycle, as shown in [Exhibit 23](#):

Exhibit 23: Liquidity of Apple

	FY2017	FY2016	FY2015
DSO	27	28	27
DOH	9	6	6
Less: Number of days of payables	112	101	86
Equals: Cash conversion cycle	(76)	(67)	(53)

The minimal DOH indicates that Apple maintains lean inventories, which is attributable to key aspects of the company's business model where manufacturing is outsourced. In isolation, the increase in number of days payable (from 86 days in FY2015 to 112 days in FY2017) might suggest an inability to pay suppliers; however, in Apple's case, the balance sheet (not shown here) indicates that the company has more than USD70 billion of cash and short-term investments, which would be more than enough to pay suppliers sooner if Apple chose to do so. Instead, Apple takes advantage of the favorable credit terms granted by its suppliers. The overall effect is a negative cash cycle, a somewhat unusual result. Instead of requiring additional capital to fund working capital as is the case for most companies, Apple has excess cash to invest for over 50 days during that three-year period (reflected on the balance sheet as short-term investments) on which it is earning, rather than paying, interest.

EXAMPLE 10

Bounds and Context of Financial Measures

The previous example focused on the cash conversion cycle, which many companies identify as a key performance metric. The less positive the number of days in the cash conversion cycle, typically, the better it is considered to be. However, is this always true?

This example considers the following question: If a larger negative number of days in a cash conversion cycle is considered to be a desirable performance metric, does identifying a company with a large negative cash conversion cycle necessarily imply good performance?

Using a historical example, National Datacomputer, a technology company, had large negative number of days in its cash conversion cycle during the 2005 to 2009 period. In 2008, its cash conversion cycle was -275.5 days.

Exhibit 24: National Datacomputer Inc. (US dollar millions)

Fiscal year	2004	2005	2006	2007	2008	2009
Sales	3.248	2.672	2.045	1.761	1.820	1.723
Cost of goods sold	1.919	1.491	0.898	1.201	1.316	1.228
Receivables, Total	0.281	0.139	0.099	0.076	0.115	0.045
Inventories, Total	0.194	0.176	0.010	0.002	0.000	0.000
Accounts payable	0.223	0.317	0.366	1.423	0.704	0.674
DSO		28.69	21.24	18.14	19.15	16.95
DOH		45.29	37.80	1.82	0.28	0.00
Less: Number of days of payables*		66.10	138.81	271.85	294.97	204.79
Equals: Cash conversion cycle		7.88	-79.77	-251.89	-275.54	-187.84

* Calculated using Cost of goods sold as an approximation of purchases. Ending inventories 2008 and 2009 are reported as \$0 million; therefore, inventory turnover for 2009 cannot be measured. However, given inventory and average sales per day, DOH in 2009 is 0.00.

Source: Raw data from Compustat. Ratios calculated."

National Datacomputer had a negative cash conversion cycle because the company's accounts payable increased substantially over the period. An increase from approximately 66 days in 2005 to 295 days in 2008 to pay trade creditors is clearly a negative signal. In addition, the company's inventories disappeared, most likely because the company did not have enough cash to purchase new inventory and was unable to get additional credit from its suppliers.

Of course, an analyst would have immediately noted the negative trends in these data, as well as additional data throughout the company's financial statements. In its management discussion and analysis (MD&A), the company clearly reports the risks as follows:

Because we have historically had losses and only a limited amount of cash has been generated from operations, we have funded our operating activities to date primarily from the sale of securities and from the sale of a product line in 2009. In order to continue to fund our operations, we may need to raise additional capital, through the sale of securities. We cannot be certain that any such financing will be available on acceptable terms, or at all. Moreover, additional equity financing, if available, would likely be dilutive to the holders of our common stock, and debt financing, if available, would likely involve restrictive covenants and a security interest in all or substantially all of our assets. If we fail to obtain acceptable financing when needed, we may not have sufficient resources to fund our normal operations which would have a material adverse effect on our business.

IF WE ARE UNABLE TO GENERATE ADEQUATE WORKING CAPITAL FROM OPERATIONS OR RAISE ADDITIONAL CAPITAL THERE IS SUBSTANTIAL DOUBT ABOUT THE COMPANY'S ABILITY TO CONTINUE AS A GOING CONCERN. (emphasis added by company)

Subsequently, the company's 2010 Form 10-K reported:

In January 2011, due to our inability to meet our financial obligations and the impending loss of a critical distribution agreement granting us the right to distribute certain products, our secured lenders ("Secured Parties") acting upon an event of default, sold certain of our assets (other than cash and accounts receivable) to Micronet, Ltd. ("Micronet"), an unaffiliated corporation pursuant to the terms of an asset purchase agreement between the Secured Parties and Micronet dated January 10, 2010 (the "Asset Purchase Agreement"). In order to induce Micronet to enter into the agreement, the Company also provided certain representations and warranties regarding certain business matters.

In summary, it is always necessary to consider ratios within bounds of reasonability and to understand the reasons underlying changes in ratios. Ratios must not only be calculated but also must be interpreted by an analyst.

Source: Form 10-K, National Datacomputer Inc., 2009, p. 7.

Source: Form 10-K, National Datacomputer Inc., 2010.

11. SOLVENCY RATIOS

LEARNING OUTCOME

The candidate should be able to:

□ calculate and interpret activity, liquidity, solvency, and profitability ratios

Solvency refers to a company's ability to fulfill its long-term debt obligations. Assessment of a company's ability to pay its long-term obligations (i.e., to make interest and principal payments) generally includes an in-depth analysis of the components of its financial structure. Solvency ratios, introduced in earlier modules in Corporate Issuers, provide information regarding the relative amount of debt in the company's capital structure and the adequacy of earnings and cash flow to cover interest expenses and other fixed charges (such as lease payments) as they come due.

By analyzing financial statements, an analyst aims to understand levels and trends in a company's use of financial leverage in relation to past practices and the practices of peer companies. Analysts also need to be aware that the greater a company's operating leverage, the greater the risk of the operating income stream available to cover debt payments; operating leverage can thus limit a company's capacity to use financial leverage.

A company's relative solvency is fundamental to valuation of its debt securities and its creditworthiness. Understanding a company's use of debt can provide analysts with insight into the company's future business prospects because management's decisions about financing may signal their beliefs about a company's future. For example, the issuance of long-term debt to repurchase common shares may indicate that management believes the market is underestimating the company's prospects and that the shares are undervalued.

11.1. Calculation of Solvency Ratios

The two primary types of solvency ratios are debt ratios and coverage ratios. Debt ratios focus on the balance sheet and measure the amount of debt capital relative to equity capital. Coverage ratios focus on the income statement and measure the ability of a company to cover its debt payments. These ratios are useful in assessing a company's solvency and, therefore, in evaluating the quality of a company's bonds and other debt obligations.

Exhibit 25 describes commonly used solvency ratios. The first three of the debt ratios presented use total debt in the numerator. The definition of total debt used in these ratios varies, with some market participants using the sum of interest-bearing short-term and long-term debt, excluding liabilities such as accrued expenses, accounts payable, and leases. (For calculations in this module, we use this definition.) Other market participants use definitions that are more inclusive (e.g., all liabilities) or restrictive (e.g., long-term debt only, in which case the ratio is sometimes qualified as "long-term," as in "long-term debt-to-equity ratio"). Finally, analysts also use solvency ratios that deduct cash, cash equivalents, and marketable securities from interest-bearing short-term and long-term debt to calculate **net debt** (or **net cash** if the former exceeds the latter). The assumption is that cash, cash equivalents, and marketable securities could be used to pay debt obligations, so it is only debt in excess of this amount that must be covered by future operating cash flows. Analysts should be transparent about their calculation methodologies and closely scrutinize ratios reported by issuers.

Exhibit 25: Definitions of Commonly Used Solvency Ratios

Solvency Ratios	Numerator	Denominator
Debt Ratios		
Debt-to-assets ratio ^a	Total debt ^b	Total assets
Debt-to-capital ratio	Total debt ^b	Total debt ^b + Total shareholders' equity
Debt-to-equity ratio	Total debt ^b	Total shareholders' equity
Financial leverage ratio ^c	Average total assets	Average total equity
Debt-to-EBITDA	Total or net debt	EBITDA
Coverage Ratios		
Interest coverage	EBIT	Interest payments

Solvency Ratios	Numerator	Denominator
Fixed charge coverage	EBIT + Lease payments	Interest payments + Lease payments

^a“Total debt ratio” is another name sometimes used for this ratio.

^bIn this reading, total debt is the sum of interest-bearing short-term and long-term debt.

^cAverage total assets divided by average total equity is used for the purposes of this reading (in particular, Dupont analysis covered later). In practice, period-end total assets divided by period-end total equity is often used."

11.2. Interpretation of Solvency Ratios

In the following, we discuss the interpretation of the basic solvency ratios presented in [Exhibit 25](#).

11.2.1. Debt-to-Assets Ratio

This ratio measures the percentage of total assets financed with debt. For example, a **debt-to-assets ratio** of 0.40 or 40 percent indicates that 40 percent of the company's assets are financed with debt. Generally, higher debt means higher financial risk and thus weaker solvency.

11.2.2. Debt-to-Capital Ratio

The **debt-to-capital ratio** measures the percentage of a company's capital (debt plus equity) represented by debt. As with the previous ratio, a higher ratio generally means higher financial risk and thus indicates weaker solvency.

11.2.3. Debt-to-Equity Ratio

The **debt-to-equity ratio** measures the amount of debt capital relative to equity capital. Interpretation is similar to the preceding two ratios (i.e., a higher ratio indicates weaker solvency). A ratio of 1.0 would indicate equal amounts of debt and equity, which is equivalent to a debt-to-capital ratio of 50 percent. Alternative definitions of this ratio use the market value of stockholders' equity rather than its book value (or use the market values of both stockholders' equity and debt).

11.2.4. Financial Leverage Ratio

The **financial leverage ratio** (often called simply the “leverage ratio”) measures the amount of total assets supported for each one money unit of equity. For example, a value of 3 for this ratio means that each EUR1 of equity supports EUR3 of total assets. The higher the financial leverage ratio, the more leveraged the company is in the sense of using debt and other liabilities to finance assets. This ratio is often defined in terms of average total assets and average total equity.

11.2.5. Debt-to-EBITDA Ratio

The debt-to-EBITDA ratio estimates how many years it would take to repay total debt based on earnings before income taxes, depreciation, and amortization (an approximation of operating cash flow). This ratio is commonly used in debt covenants between issuers and debt investors.

11.2.6. Interest Coverage

The **interest coverage ratio** measures the number of times a company's EBIT could cover its interest payments. Thus, it is sometimes referred to as "times interest earned." A higher **interest coverage** ratio indicates stronger solvency, offering greater assurance that the company can service its debt (i.e., bank debt, bonds, notes) from operating earnings. This ratio is commonly used in debt covenants between issuers and lenders or fixed income investors.

11.2.7. Fixed Charge Coverage

The **fixed charge coverage ratio** relates fixed charges, or obligations, to the cash flow generated by the company. It measures the number of times a company's earnings (before interest, taxes, and lease payments) can cover the company's interest and lease payments. Similar to the interest coverage ratio, a higher **fixed charge coverage** ratio implies stronger solvency, offering greater assurance that the company can service its debt (i.e., bank debt, bonds, notes, and leases) from normal earnings. The ratio is sometimes used as an indication of the quality of the preferred dividend, with a higher ratio indicating a more secure preferred dividend.

Example 11 demonstrates the use of solvency ratios in evaluating the creditworthiness of a company.

EXAMPLE 11

Evaluation of Solvency Ratios

A credit analyst is evaluating the solvency of Eskom, a South African public utility based on financial statements for the year ended 31 March 2017. The data in **Exhibit 26** are gathered from the company's 2017 annual report:

Exhibit 26: Eskom 2017 Annual Report

<i>South African rand, millions</i>	2017	2016	2015
Total Assets	710,009	663,170	559,688
Short-Term Debt	18,530	15,688	19,976
Long-Term Debt	336,770	306,970	277,458
Total Liabilities	534,067	480,818	441,269
Total Equity	175,942	182,352	118,419

1. Calculate the company's financial leverage ratios for 2016 and 2017.

Solution:

(Amounts are millions of Rand.)

For 2017, average total assets were $(710,009 + 663,170)/2 = 686,590$, and average total equity was $(175,942 + 182,352)/2 = 179,147$. Thus, financial leverage was $686,590/179,147 = 3.83$. For 2016, financial leverage was 4.07.

	2017	2016
Average Assets	686,590	611,429
Average Equity	179,147	150,386
Financial Leverage	3.83	4.07

2. Interpret the financial leverage ratio calculated in question 1.

Solution:

For 2017, every South African rand in total equity supported ZAR3.83 in total assets, on average. Financial leverage decreased from 2016 to 2017 on this measure.

3. What are the company's debt-to-assets, debt-to-capital, and debt-to-equity ratios for the three years?

Solution:

(Amounts are millions of South African rand other than ratios.)

	2017	2016	2015
Total Debt	355,300	322,658	297,434
Total Capital	531,242	505,010	415,853
Debt/Assets	50.0%	48.7%	53.1%
Debt/Capital	66.9%	63.9%	71.5%
Debt/Equity	2.02	1.77	2.51

4. What is the discernable trend over the three years?

Solution:

On all three metrics, the company's leverage decreased from 2015 to 2016 and increased from 2016 to 2017. For 2016 the decrease in leverage resulted from a conversion of subordinated debt into equity as well as additional issuance of equity. However, in 2017 debt levels increased again relative to assets, capital, and equity indicating that the company's solvency has weakened. From a creditor's perspective, lower solvency (higher debt) indicates higher risk of default on obligations.

As with all ratio analysis, it is important to consider leverage ratios in a broader context. In general, companies with lower business risk and operations that generate steady cash flows are better positioned to take on more leverage without a commensurate increase in the risk of insolvency. In other words, a higher proportion of debt financing poses less risk of non-payment of interest and debt principal to a company with steady cash flows than to a company with volatile cash flows.

12. PROFITABILITY RATIOS

LEARNING OUTCOME

The candidate should be able to:

☐ calculate and interpret activity, liquidity, solvency, and profitability ratios

The ability to generate profit on capital invested is a key determinant of a company's overall value and the value of the securities it issues. Consequently, many equity analysts consider profitability to be a key focus of their analytical efforts.

Profitability reflects a company's competitive position in the market, and by extension, the quality of its management. The income statement reveals the sources of earnings and the components of revenue and expenses. Earnings can be distributed to shareholders or reinvested in the company.

12.1. Calculation of Profitability Ratios

Profitability ratios measure the return earned by the company during a period. [Exhibit 27](#) provides the definitions of a selection of commonly used profitability ratios, some of which were introduced in earlier modules in Corporate Issuers. Return-on-sales profitability ratios express various subtotals on the income statement (e.g., gross profit, operating profit, net profit) as a percentage of revenue. Essentially, these ratios constitute part of a common-size income statement discussed earlier. Return on investment profitability ratios measure income relative to assets, equity, or total capital employed by the company. For operating ROA, returns are measured as operating income, that is, prior to deducting interest on debt capital. For ROA and ROE, returns are measured as net income, that is, after deducting interest paid on debt capital. For return on common equity, returns are measured as net income minus preferred dividends (because preferred dividends are a return to preferred equity).

Exhibit 27: Definitions of Commonly Used Profitability Ratios

Profitability Ratios	Numerator	Denominator
Return on Sales^a		
Gross profit margin	Gross profit	Revenue
Operating profit margin	Operating income ^b	Revenue
Pretax margin	EBT (earnings before tax but after interest)	Revenue
Net profit margin	Net income	Revenue
Return on Investment		
Operating ROA	Operating income	Average total assets
ROA	Net income	Average total assets
Return on invested capital	$EBIT \times (1 - \text{Effective Tax Rate})$	Average total short- and long-term debt and equity
ROE	Net income	Average total equity
Return on common equity	Net income – Preferred dividends	Average common equity

^a “Sales” is being used as a synonym for “revenue.”

^b Some analysts use EBIT as a shortcut representation of operating income. Note that EBIT, strictly speaking, includes non-operating items such as dividends received and gains and losses on investment securities. Of utmost importance is that the analyst compute ratios consistently whether comparing different companies or analyzing one company over time.”

12.2. Interpretation of Profitability Ratios

In the following, we discuss the interpretation of the profitability ratios presented in [Exhibit 27](#). For each of the profitability ratios, a higher ratio indicates greater profitability.

12.2.1. Gross Profit Margin

Gross profit margin indicates the percentage of revenue available to cover operating and other expenses and to generate profit. Higher gross profit margin indicates some combination of higher product pricing and lower product costs. The ability to charge a higher price is constrained by competition, so gross profits are affected by (and usually inversely related to) competition. If a product has a competitive advantage (e.g., superior branding, better quality, or exclusive technology), the company is better able to

charge more for it. On the cost side, higher gross profit margin can also indicate that a company has a competitive advantage in product costs.

12.2.2. *Operating Profit Margin*

Operating profit is calculated as gross profit minus operating costs. So, an **operating profit margin** increasing faster than the gross profit margin can indicate improvements in controlling operating costs, such as administrative overheads. In contrast, a declining operating profit margin could be an indicator of deteriorating control over operating costs.

12.2.3. *Pretax Margin*

Pretax income (also called “earnings before tax” or EBT) is calculated as operating profit minus interest, and the **pretax margin** is the ratio of pretax income to revenue. The pretax margin reflects the effects on profitability of leverage and other (non-operating) income and expenses. If a company’s pretax margin is increasing primarily as a result of increasing amounts of non-operating income, the analyst should evaluate whether this increase reflects a deliberate change in a company’s business focus and, therefore, the likelihood that the increase will continue.

12.2.4. *Net Profit Margin*

Net profit, or net income, is calculated as revenue minus all expenses. Net income includes both recurring and non-recurring components. Generally, the net income used in calculating the net profit margin is adjusted for non-recurring items to offer a better view of a company’s potential future profitability.

12.2.5. *ROA*

ROA measures the return earned by a company on its assets. The higher the ratio, the more income is generated by a given level of assets. Most databases compute this ratio as follows:

$$\frac{\text{Net income}}{\text{Average total assets}} .$$

An issue with this computation is that net income is the return to equity holders, whereas assets are financed by both equity holders and creditors. Interest expense (the return to creditors) has already been subtracted in the numerator. Some analysts, therefore, prefer to add back interest expense in the numerator. In such cases, interest must be adjusted for income taxes because net income is determined after taxes. With this adjustment, the ratio would be computed as follows:

$$\frac{\text{Net income} + \text{Interest expense}(1 - \text{Tax rate})}{\text{Average total assets}} .$$

Alternatively, some analysts elect to compute ROA on a pre-interest and pre-tax basis (operating ROA in [Exhibit 27](#)) as follows:

$$\frac{\text{Operating income or EBIT}}{\text{Average total assets}} .$$

In this ROA calculation, returns are measured prior to deducting interest on debt capital (i.e., as operating income or EBIT). This measure reflects the return on all assets invested in the company, whether financed with liabilities, debt, or equity. Whichever form of ROA is chosen, the analyst must use it consistently in comparisons to other companies or time periods.

12.2.6. *Return on Invested Capital*

Return on invested capital measures the after-tax profitability a company earns on all of the capital that it employs (short-term debt, long-term debt, and equity). As with operating ROA, returns are measured prior to deducting interest on debt capital (i.e., as operating income or EBIT).

12.2.7. ROE

ROE measures the return earned by a company on its equity capital, including minority equity, preferred equity, and common equity. As noted, return is measured as net income (i.e., interest on debt capital is not included in the return on equity capital). A variation of ROE is return on common equity, which measures the return earned by a company only on its common equity.

Both ROA and ROE are important measures of profitability and will be explored in more detail later. As with other ratios, profitability ratios should be evaluated individually and as a group to gain an understanding of what is driving profitability (operating versus non-operating activities). [Example 12](#) demonstrates the evaluation of profitability ratios and the use of the management report (sometimes called management discussion and analysis or management commentary) that accompanies financial statements to explain the trend in ratios.

EXAMPLE 12

Evaluation of Profitability Ratios

Recall from [Example 1](#) that an analysis found that Apple's gross margin declined over the three-year period FY2015 to FY2017. An analyst would like to further explore Apple's profitability using a five-year period. He gathers the following revenue data and calculates the following profitability ratios from information in Apple's annual reports, as shown in [Exhibit 28](#):

Exhibit 28: Profitability Ratios for Apple (US dollar millions)

	2017	2016	2015	2014	2013
Sales	229,234	215,639	233,715	182,795	170,910
Gross profit	88,186	84,263	93,626	70,537	64,304
Operating income	61,344	60,024	71,230	52,503	48,999
Pre-tax income	64,089	61,372	72,515	53,483	50,155
Net income	48,351	45,687	53,394	39,510	37,037
Gross profit margin	38.47%	39.08%	40.06%	38.59%	37.62%
Operating income margin	26.76%	27.84%	30.48%	28.72%	28.67%
Pre-tax margin	27.96%	28.46%	31.03%	29.26%	29.35%
Net profit margin	21.09%	21.19%	22.85%	21.61%	21.67%

1. Evaluate the overall trend in Apple's profitability ratios for the five-year period.

Solution:

Sales had increased steadily through 2015, dropped in 2016, and rebounded somewhat in 2017. As noted in [Example 1](#), the sales decline in 2016 was related to a decline in iPhone sales and weakness in foreign currencies. Margins also rose from 2013 to 2015 and declined in 2016. However, in spite of the increase in sales in 2017, all margins declined slightly indicating costs were rising faster than sales. In spite of the fluctuations, Apple's bottom line net profit margin was relatively stable over the five-year period.

Question Set

1. Which ratio would a company *most likely* use to measure its ability to meet short-term obligations?

- A. Current ratio
- B. Payables turnover
- C. Gross profit margin

Solution:

A is correct. The current ratio is a liquidity ratio. It compares the net amount of current assets expected to be converted into cash within the year with liabilities falling due in the same period. A current ratio of 1.0 would indicate that the company would have just enough current assets to pay current liabilities.

2. Which of the following ratios would be *most* useful in determining a company's ability to cover its lease and interest payments?

- A. ROA
- B. Total asset turnover
- C. Fixed charge coverage

Solution:

C is correct. The fixed charge coverage ratio is a coverage ratio that relates known fixed charges or obligations to a measure of operating profit or cash flow generated by the company. Coverage ratios, a category of solvency ratios, measure the ability of a company to cover its payments related to debt and leases.

3. Assuming no changes in other variables, which of the following would decrease ROA?

- A. An increase in average assets
- B. A decrease in interest expense
- C. A decrease in the effective tax rate

Solution:

A is correct. Assuming no changes in other variables, an increase in average assets (an increase in the denominator) would decrease ROA. A decrease in either the effective tax rate or interest expense, assuming no changes in other variables, would increase ROA.

13. INTEGRATED FINANCIAL RATIO ANALYSIS

LEARNING OUTCOME

The candidate should be able to:

- ☐ describe relationships among ratios and evaluate a company using ratio analysis

In prior lessons, the text presented separately activity, liquidity, solvency, and profitability ratios. Before discussing valuation ratios, this lesson demonstrates the importance of examining a variety of financial ratios—not a single ratio or category of ratios in isolation—to ascertain the overall position and performance of a company. Experience shows that the information from one ratio category can be helpful in answering questions raised by another category and that the most accurate overall picture comes from integrating information from all sources.

13.1. The Overall Ratio Picture: Examples

We present two simple examples to introduce the use of a variety of ratios to address an analytical task. [Example 13](#) shows how the analysis of a pair of activity ratios resolves an issue concerning a company’s liquidity. [Example 14](#) shows that examining the overall ratios of multiple companies can assist an analyst in drawing conclusions about their relative performances.

EXAMPLE 13

A Variety of Ratios

An analyst is evaluating the liquidity of a Canadian manufacturing company and obtains the liquidity ratios shown in [Exhibit 29](#):

Exhibit 29: Liquidity Ratios

Fiscal Year	10	9	8
Current ratio	2.1	1.9	1.6
Quick ratio	0.8	0.9	1.0

The ratios present a contradictory picture of the company’s liquidity. Based on the increase in its current ratio from 1.6 to 2.1, the company appears to have strong and improving liquidity; however, based on the decline of the quick ratio from 1.0 to 0.8, its liquidity appears to be deteriorating. Because both ratios have exactly the same denominator, current liabilities, the difference must be the result of changes in some asset that is included in the current ratio but not in the quick ratio (e.g., inventories). The analyst collects the activity ratios in [Exhibit 30](#):

Exhibit 30: Activity Ratios

DOH	55	45	30
DSO	24	28	30

The company’s DOH has deteriorated from 30 days to 55 days, meaning that the company is holding increasingly larger amounts of inventory relative to sales. The decrease in DSO implies that the company is collecting receivables faster. If the proceeds from these collections were held as cash, there would be no effect on either the current ratio or the quick ratio. However, if the proceeds from the collections were used to purchase inventory, there would be no effect on the current ratio and a decline in the quick ratio (i.e., the pattern shown in this example). Collectively, the ratios suggest that liquidity is declining and that the company may have an inventory problem that needs to be addressed.

EXAMPLE 14

A Comparison of Two Companies (1)

An analyst collects the information in [Exhibit 31](#) for two hypothetical companies.

Exhibit 31: Company Comparison

Anson Industries	Fiscal Year			
	5	4	3	2
Inventory turnover	76.69	89.09	147.82	187.64
DOH	4.76	4.10	2.47	1.95
Receivables turnover	10.75	9.33	11.14	7.56
DSO	33.95	39.13	32.77	48.29
Accounts payable turnover	4.62	4.36	4.84	4.22
Days payable	78.97	83.77	75.49	86.56
Cash from operations/Total liabilities	31.41%	11.15%	4.04%	8.81%
ROE	5.92%	1.66%	1.62%	-0.62%
ROA	3.70%	1.05%	1.05%	-0.39%
Net profit margin (Net income/Revenue)	3.33%	1.11%	1.13%	-0.47%
Total asset turnover (Revenue/Average assets)	1.11	0.95	0.93	0.84
Leverage (Average assets/Average equity)	1.60	1.58	1.54	1.60

Clarence Corporation	Fiscal Year			
	5	4	3	2
Inventory turnover	9.19	9.08	7.52	14.84
DOH	39.73	40.20	48.51	24.59
Receivables turnover	8.35	7.01	6.09	5.16
DSO	43.73	52.03	59.92	70.79
Accounts payable turnover	6.47	6.61	7.66	6.52
Days payable	56.44	55.22	47.64	56.00
Cash from operations/Total liabilities	13.19%	16.39%	15.80%	11.79%
ROE	9.28%	6.82%	-3.63%	-6.75%
ROA	4.64%	3.48%	-1.76%	-3.23%
Net profit margin (Net income/Revenue)	4.38%	3.48%	-1.60%	-2.34%
Total asset turnover (Revenue/Average assets)	1.06	1.00	1.10	1.38
Leverage (Average assets/Average equity)	2.00	1.96	2.06	2.09

Note: Ratios are expressed in terms of two decimal places and are rounded. Therefore, expected relationships may not hold perfectly.

1. Which of the following choices best describes reasonable conclusions an analyst might make about the companies' efficiency?

- A. In FY5, Anson's DOH of only 4.76 indicated that it was less efficient at inventory management than Clarence, which had DOH of 39.73.
- B. Over the past four years, Anson has shown greater improvement in efficiency than Clarence, as indicated by its total asset turnover ratio increasing from 0.84 to 1.11.

C. In FY5, Clarence's receivables turnover of 8.35 times indicated that it was more efficient at receivables management than Anson, which had receivables turnover of 10.75.

Solution:

B is correct. Over the past four years, Anson has shown greater improvement in efficiency than Clarence, as indicated by its total asset turnover ratio increasing from 0.84 to 1.11. Over the same period of time, Clarence's total asset turnover ratio has declined from 1.38 to 1.06. Choices B and C are incorrect because DOH and receivables turnover are misinterpreted.

14. DUPONT ANALYSIS—THE DECOMPOSITION OF ROE

LEARNING OUTCOME

The candidate should be able to:

- ☐ **demonstrate the application of DuPont analysis of return on equity and calculate and interpret effects of changes in its components**

As noted earlier, ROE measures the return a company generates on its equity capital. To understand what drives a company's ROE, a useful technique is to decompose ROE into its component parts. (Decomposition of ROE is sometimes referred to as **DuPont analysis** because it was developed originally at that company.) Decomposing ROE involves expressing the basic ratio (i.e., net income divided by average shareholders' equity) as the product of component ratios. Because each of these component ratios is an indicator of a distinct aspect of a company's performance that affects ROE, the decomposition allows us to evaluate how these different aspects of performance affected the company's profitability as measured by ROE.

Decomposing ROE is useful in determining the reasons for changes in ROE over time for a given company and for differences in ROE for different companies in a given time period. The information gained can also be used by management to determine which areas they should focus on to improve ROE. This decomposition will also show why a company's overall profitability, measured by ROE, is a function of its efficiency, operating profitability, taxes, and use of financial leverage. DuPont analysis shows the relationship between the various categories of ratios discussed in this module and how they all influence the return to the investment of the owners.

Analysts have developed several different methods of decomposing ROE. The decomposition presented here is one of the most commonly used and the one found in popular research databases, such as Bloomberg. Return on equity is calculated as follows:

$$\text{ROE} = \text{Net income} / \text{Average shareholders' equity}.$$

The decomposition of ROE makes use of simple algebra and illustrates the relationship between ROE and ROA. Expressing ROE as a product of only two of its components, we can write:

Equation 1

$$\begin{aligned} \text{ROE} &= \frac{\text{Net income}}{\text{Average shareholders' equity}} \\ &= \frac{\text{Net income}}{\text{Average total assets}} \times \frac{\text{Average total assets}}{\text{Average shareholders' equity}} \end{aligned}$$

which can be interpreted as follows:

$$\text{ROE} = \text{ROA} \times \text{Leverage}.$$

In other words, ROE is a function of a company's ROA and its use of financial leverage ("leverage" for short, in this discussion). A company can improve its ROE by improving ROA or by increasing leverage. Consistent with the definition given earlier, leverage is measured as average total assets divided by average shareholders' equity. If a company had no leverage (no liabilities), its leverage ratio would equal 1.0 and ROE would exactly equal ROA. As a company takes on liabilities, its leverage increases. As long as a company is able to borrow at a rate lower than the marginal rate it can earn investing the borrowed money, the company is making an effective use of leverage and ROE would increase as leverage increases. If a company's borrowing cost exceeds the marginal rate it can earn on investing in the business, ROE would decline as leverage increased because the effect of borrowing would be to depress ROA.

Using the data from [Example 14](#) for Anson Industries, an analyst can examine the trend in ROE and determine whether the increase from an ROE of -0.625 percent in FY2 to 5.925 percent in FY5 is a function of ROA or the use of leverage:

Exhibit 32: Anson Industries

	ROE	=	ROA	×	Leverage
FY5	5.92%		3.70%		1.60
FY4	1.66%		1.05%		1.58
FY3	1.62%		1.05%		1.54
FY2	-0.62%		-0.39%		1.60

Over the four-year period, the company's leverage factor was relatively stable. The primary reason for the increase in ROE is the increase in profitability measured by ROA.

Just as ROE can be decomposed, the individual components such as ROA can be decomposed. Further decomposing ROA, we can express ROE as a product of three component ratios:

Equation 2

$$\frac{\text{Net income}}{\text{Average shareholders' equity}} = \frac{\text{Net income}}{\text{Revenue}} \times \frac{\text{Revenue}}{\text{Average total assets}} \times \frac{\text{Average total assets}}{\text{Average shareholders' equity}}$$

which can be interpreted as follows:

$$\text{ROE} = \text{Net profit margin} \times \text{Total asset turnover} \times \text{Leverage}.$$

The first term on the right-hand side of this equation is the net profit margin, an indicator of profitability: how much income a company derives per one monetary unit (e.g., euro or US dollar) of sales. The second term on the right is the asset turnover ratio, an indicator of efficiency: how much revenue a company generates per one money unit of assets. Note that ROA is decomposed into these two components: net profit margin and total asset turnover. A company's ROA is a function of profitability (net profit margin) and efficiency (total asset turnover). The third term on the right-hand side of the equation is a measure of financial leverage, an indicator of solvency: the total amount of a company's assets relative to its equity capital. This decomposition illustrates that a company's ROE is a function of its net profit margin, its efficiency, and its leverage. Again, using the data from [Example 14](#) for Anson Industries, the analyst can evaluate in more detail the reasons behind the trend in ROE:

Exhibit 33: Anson Industries

	ROE	=	Net profit margin	×	Total asset turnover	×	Leverage
FY5	5.92%		3.33%		1.11		1.60
FY4	1.66%		1.11%		0.95		1.58
FY3	1.62%		1.13%		0.93		1.54

	ROE	=	Net profit margin	×	Total asset turnover	×	Leverage
FY2	-0.62%		-0.47%		0.84		1.60

This further decomposition confirms that increases in profitability (measured here as net profit margin) are indeed an important contributor to the increase in ROE over the four-year period. However, Anson's asset turnover has also increased steadily. The increase in ROE is, therefore, a function of improving profitability and improving efficiency. As noted earlier, ROE decomposition can also be used to compare the ROEs of peer companies, as demonstrated in [Example 15](#).

EXAMPLE 15

A Comparison of Two Companies (2)

Referring to the data for Anson Industries and Clarence Corporation in Example 14, which of the following choices best describes reasonable conclusions an analyst might make about the companies' ROE?

- A. Anson's inventory turnover of 76.69 indicates it is more profitable than Clarence.
- B. The main driver of Clarence's superior ROE in FY5 is its more efficient use of assets.
- C. The main drivers of Clarence's superior ROE in FY5 are its greater use of debt financing and higher net profit margin.

Solution:

C is correct. The main driver of Clarence's superior ROE (9.28 percent compared with only 5.92 percent for Anson) in FY5 is its greater use of debt financing (leverage of 2.00 compared with Anson's leverage of 1.60) and higher net profit margin (4.38 percent compared with only 3.33 percent for Anson).

A is incorrect because inventory turnover is not a direct indicator of profitability. An increase in inventory turnover may indicate more efficient use of inventory which in turn could affect profitability; however, an increase in inventory turnover would also be observed if a company was selling more goods even if it was not selling those goods at a profit. B is incorrect because Clarence has less efficient use of assets than Anson, indicated by turnover of 1.06 for Clarence compared with Anson's turnover of 1.11.

To separate the effects of taxes and interest, we can further decompose the net profit margin and write:

Equation 3

$$\frac{\text{Net income}}{\text{Average shareholders' equity}} = \frac{\text{Net income}}{\text{EBT}} \times \frac{\text{EBT}}{\text{EBIT}} \times \frac{\text{EBIT}}{\text{Revenue}} \\ \times \frac{\text{Revenue}}{\text{Average total assets}} \times \frac{\text{Average total assets}}{\text{Average shareholders' equity}}$$

which can be interpreted as follows:

$$\text{ROE} = \text{Tax burden} \times \text{Interest burden} \times \text{EBIT margin} \times \text{Total asset turnover} \times \text{Leverage}.$$

This five-way decomposition is the one found in financial databases such as Bloomberg. The first term on the right-hand side of this equation measures the effect of taxes on ROE. Essentially, it reflects one minus the average tax rate, or how much of a company's pretax profits it gets to keep. This can be expressed in decimal or percentage form. So, a 30 percent tax rate would yield a factor of 0.70 or 70 percent. A higher value for the tax burden implies that the company can keep a higher percentage of its pretax profits, indicating a lower tax rate. A decrease in the tax burden ratio implies the opposite (i.e., a higher tax rate leaving the company with less of its pretax profits).

The second term on the right-hand side captures the effect of interest on ROE. Higher borrowing costs reduce ROE. Some analysts prefer to use operating income instead of EBIT for this term and the following term. Either operating income or EBIT is acceptable as long as it is applied consistently. In such a case, the second term would measure both the effect of interest expense and non-operating income on ROE.

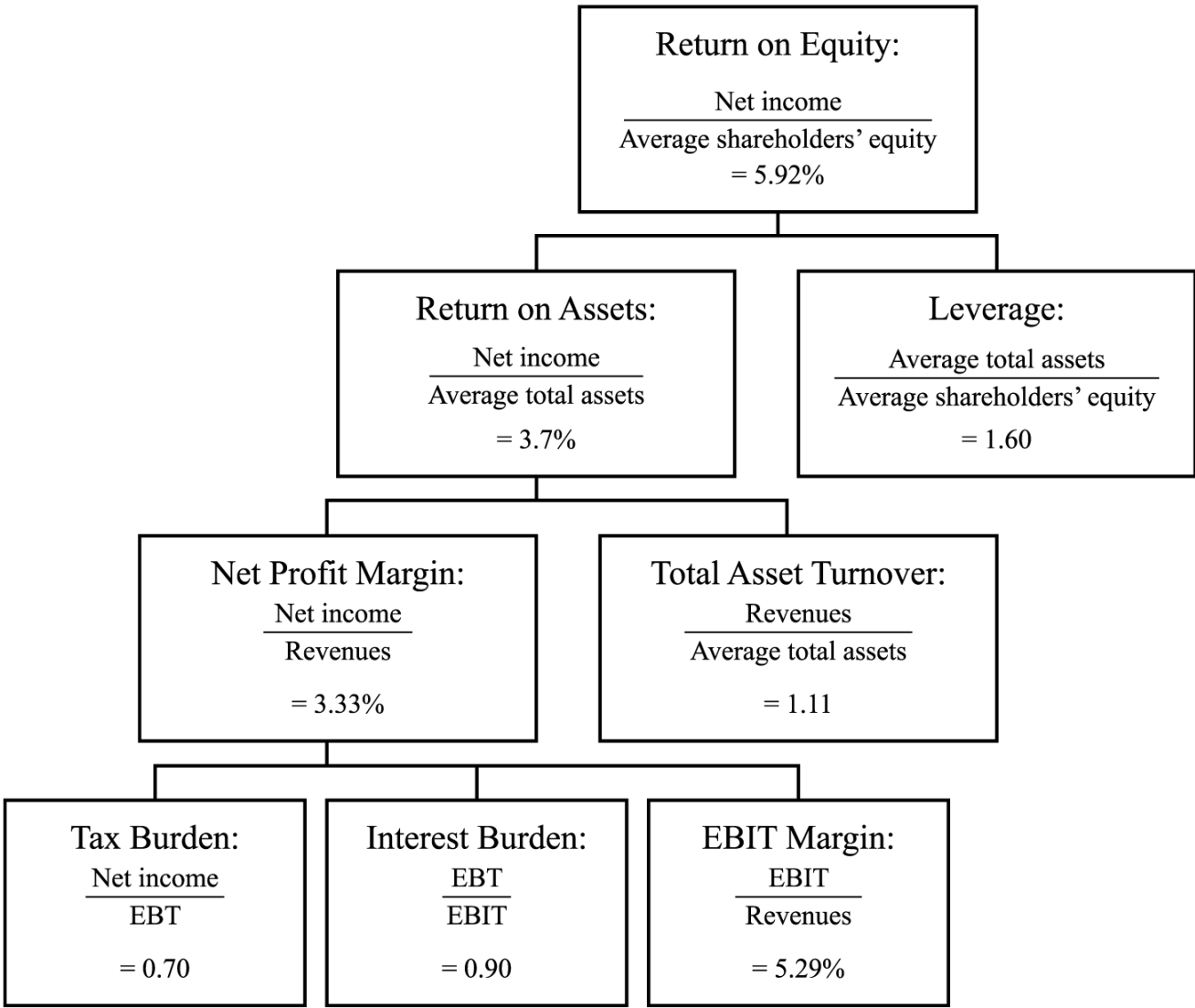
The third term on the right-hand side captures the effect of operating margin (if operating income is used in the numerator) or EBIT margin (if EBIT is used) on ROE. In either case, this term primarily measures the effect of operating profitability on ROE.

The fourth term on the right-hand side is again the total asset turnover ratio, an indicator of the overall efficiency of the company (i.e., how much revenue it generates per unit of total assets). The fifth term on the right-hand side is the financial leverage ratio described above—the total amount of a company’s assets relative to its equity capital.

This decomposition expresses a company’s ROE as a function of its tax rate, interest burden, operating profitability, efficiency, and leverage. An analyst can use this framework to determine what factors are driving a company’s ROE. The decomposition of ROE can also be useful in forecasting ROE based upon expected efficiency, profitability, financing activities, and tax rates. The relationship of the individual factors, such as ROA to the overall ROE, can also be expressed in the form of an ROE tree to study the contribution of each of the five factors, as shown in [Exhibit 34](#) for Anson Industries.

[Exhibit 34](#) shows that Anson’s ROE of 5.92 percent in FY5 can be decomposed into ROA of 3.70 percent and leverage of 1.60. ROA can further be decomposed into a net profit margin of 3.33 percent and total asset turnover of 1.11. Net profit margin can be decomposed into a tax burden of 0.70 (an average tax rate of 30 percent), an interest burden of 0.90, and an EBIT margin of 5.29 percent. Overall ROE is decomposed into five components.

Exhibit 34: DuPont Analysis of Anson Industries’ ROE: Fiscal Year 5



Example 16 demonstrates how the five-component decomposition can be used to determine reasons behind the trend in a company's ROE.

EXAMPLE 16

Five-Way Decomposition of ROE

An analyst examining Amsterdam PLC (a hypothetical company) wishes to understand the factors driving the trend in ROE over a four-year period. The analyst obtains and calculates the following data from Amsterdam's annual reports:

Exhibit 35: Amsterdam's Annual Reports

	2017	2016	2015	2014
ROE	9.53%	20.78%	26.50%	24.72%
Tax burden	60.50%	52.10%	63.12%	58.96%
Interest burden	97.49%	97.73%	97.86%	97.49%
EBIT margin	7.56%	11.04%	13.98%	13.98%
Asset turnover	0.99	1.71	1.47	1.44
Leverage	2.15	2.17	2.10	2.14

1. What might the analyst conclude?

Solution:

The tax burden measure has varied, with no obvious trend. In the most recent year, 2017, taxes declined as a percentage of pretax profit. (Because the tax burden reflects the relation of after-tax profits to pretax profits, the increase from 52.10 percent in 2016 to 60.50 percent in 2017 indicates that taxes declined as a percentage of pretax profits.) This decline in average tax rates could be a result of lower tax rates from new legislation or revenue in a lower tax jurisdiction. The interest burden has remained fairly constant over the four-year period indicating that the company maintains a fairly constant capital structure. Operating margin (EBIT margin) declined over the period, indicating the company's operations were less profitable. This decline is generally consistent with declines in oil prices in 2017 and declines in refining industry gross margins in 2016 and 2017. The company's efficiency (asset turnover) decreased in 2017. The company's leverage remained constant, consistent with the constant interest burden. Overall, the trend in ROE (declining substantially over the recent years) resulted from decreases in operating profits and a lower asset turnover. Additional research on the causes of these changes is required to develop expectations about the company's future performance.

The most detailed decomposition of ROE that we have presented is a five-way decomposition. Nevertheless, an analyst could further decompose individual components of a five-way analysis. For example, EBIT margin (EBIT/Revenue) could be further decomposed into a non-operating component (EBIT/Operating income) and an operating component (Operating income/Revenue). The analyst can also examine which other factors contributed to these five components. For example, an improvement in efficiency (total asset turnover) may have resulted from better management of inventory (DOH) or better collection of receivables (DSO).

15. INDUSTRY-SPECIFIC FINANCIAL RATIOS

LEARNING OUTCOME

The candidate should be able to:

❑ describe the uses of industry-specific ratios used in financial analysis

As stated earlier, a universally accepted definition and classification of ratios does not exist. The purpose of ratios is to serve as indicators of important aspects of a company’s performance and value. Aspects of performance that are considered important in one industry may be irrelevant in another, and industry-specific ratios reflect these differences. For example, companies in the retail industry may report same-store sales changes because, in the retail industry, it is important to distinguish between growth that results from opening new stores and growth that results from generating more sales at existing stores. Industry-specific metrics can be especially important to the value of equity in early-stage industries, where companies are not yet profitable.

In addition, regulated industries—especially in the financial sector—often are required to comply with specific regulatory ratios. For example, the banking sector’s liquidity and cash reserve ratios provide an indication of banking liquidity and reflect monetary and regulatory requirements. Banking capital adequacy requirements attempt to relate banks’ solvency requirements directly to their specific levels of risk exposure.

Exhibit 36 presents, for illustrative purposes only, some industry-specific ratios.²

Exhibit 36: Definitions of Some Common Industry-Specific Ratios

Ratio	Numerator	Denominator
Business Risk		
Coefficient of variation of operating income	Standard deviation of operating income	Average operating income
Coefficient of variation of net income	Standard deviation of net income	Average net income
Coefficient of variation of revenues	Standard deviation of revenue	Average revenue
Financial Sector Ratios		
Capital adequacy—banks	Various components of capital	Various measures such as risk-weighted assets, market risk exposure, or level of operational risk assumed
Monetary reserve requirement (Cash reserve ratio)	Reserves held at central bank	Specified deposit liabilities
Liquid asset requirement	Approved “readily marketable” securities	Specified deposit liabilities
Net interest margin	Net interest income	Total interest-earning assets
Retail Ratios		
Same (or comparable) store sales	Average revenue growth year over year for stores open in both periods	Not applicable

Retail Ratios	Numerator	Denominator
Sales per square meter (or square foot)	Revenue	Total retail space in square meters (or square feet)

Service Companies	Numerator	Denominator
Revenue per employee	Revenue	Total number of employees
Net income per employee	Net income	Total number of employees

Hotel	Numerator	Denominator
Average daily rate	Room revenue	Number of rooms sold
Occupancy rate	Number of rooms sold	Number of rooms available

Subscription or Relationship-Based Businesses	Numerator	Denominator
Average revenue per user (ARPU)	Revenue	Average number of subscribers or users

Note: Many other industry-specific ratios are outside the scope of this module. Resources such as Standard and Poor's Industry Surveys present useful ratios for each industry. Industry organizations may present useful ratios for the industry or a task specific to the industry.

16. MODEL BUILDING AND FORECASTING

LEARNING OUTCOME

The candidate should be able to:

- ☐ **describe how ratio analysis and other techniques can be used to model and forecast earnings**

Analysts often need to forecast future financial performance. For example, analysts' EPS forecasts and related equity valuations are widely followed by Wall Street. Analysts use data about the economy, industry, and company in arriving at a company's forecast. The results of an analyst's financial analysis, including common-size and ratio analyses, are integral to this process, along with the judgment of the analysts.

Based upon forecasts of growth and expected relationships among the financial statement data, the analyst can build a model to forecast future performance, which will be covered in later modules. In addition to budgets, pro forma financial statements are widely used in financial forecasting within companies, especially for use by senior executives and boards of directors. Last but not least, these budgets and forecasts are also used in presentations to credit analysts and others in obtaining external financing.

For example, based on a revenue forecast, an analyst may budget expenses based on expected common-size data. Forecasts of balance sheet and cash flow statements can be derived from expected ratio data, such as DSO. Forecasts are not limited to a single point estimate but should involve a range of possibilities. This can involve several techniques:

- **Sensitivity analysis:** Also known as "what if" analysis, sensitivity analysis shows the range of possible outcomes as specific assumptions are changed; this could, in turn, influence financing needs or investment in fixed assets.

- **Scenario analysis:** This type of analysis shows the changes in key financial quantities that result from given (economic) events, such as the loss of customers, the loss of a supply source, or a catastrophic event. If the list of events is mutually exclusive and exhaustive and the events can be assigned probabilities, the analyst can evaluate not only the range of outcomes but also standard statistical measures such as the mean and median value for various quantities of interest.
- **Simulation:** This is computer-generated sensitivity or scenario analysis based on probability models for the factors that drive outcomes. Each event or possible outcome is assigned a probability. Multiple scenarios are then run using the probability factors assigned to the possible values of a variable to determine an expected outcome for that variable.

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PRACTICE PROBLEMS

1. Comparison of a company's financial results to other peer companies for the same time period is called:
 - A. time-series analysis.
 - B. common-size analysis.
 - C. cross-sectional analysis.
2. An analyst observes a decrease in a company's inventory turnover. Which of the following would *most likely* explain this trend?
 - A. The company installed a new inventory management system, allowing more efficient inventory management.
 - B. Due to problems with obsolescent inventory last year, the company wrote off a large amount of its inventory at the beginning of the period.
 - C. The company installed a new inventory management system but experienced some operational difficulties resulting in duplicate orders being placed with suppliers.
3. Which of the following would *best* explain an increase in receivables turnover?
 - A. The company adopted new credit policies last year and began offering credit to customers with weak credit histories.
 - B. Due to problems with an error in its old credit scoring system, the company had accumulated a substantial amount of uncollectible accounts and wrote off a large amount of its receivables.
 - C. To match the terms offered by its closest competitor, the company adopted new payment terms now requiring net payment within 30 days rather than 15 days, which had been its previous requirement.
4. Brown Corporation had average days of sales outstanding of 19 days in the most recent fiscal year. Brown wants to improve its credit policies and collection practices and decrease its collection period in the next fiscal year to match the industry average of 15 days. Credit sales in the most recent fiscal year were \$300 million, and Brown expects credit sales to increase to \$390 million in the next fiscal year. To achieve Brown's goal of decreasing the collection period, the change in the average accounts receivable balance that must occur is *closest* to:
 - A. +USD0.41 million.
 - B. -USD0.41 million.
 - C. -USD1.22 million.

The following information relates to questions 5-5

An analyst is interested in assessing both the efficiency and liquidity of Spherion PLC. The analyst has collected the data in [Exhibit 1](#) for Spherion:

Exhibit 1: Spherion Data

	FY3	FY2	FY1
Days of inventory on hand	32	34	40
Days sales outstanding	28	25	23
Number of days of payables	40	35	35

5. Based on the data in [Exhibit 1](#), what is the analyst *least likely* to conclude?

- A. Inventory management has contributed to improved liquidity.
- B. Management of payables has contributed to improved liquidity.
- C. Management of receivables has contributed to improved liquidity.

6. To assess a company's ability to fulfill its long-term obligations, an analyst would *most likely* examine:

- A. activity ratios.
- B. liquidity ratios.
- C. solvency ratios.

The following information relates to questions 7-8

An analyst is evaluating the solvency and liquidity of Apex Manufacturing and has collected the data in [Exhibit 1](#):

Exhibit 1: Solvency and Liquidity of Apex Manufacturing (euro millions)

	FY5	FY4	FY3
Total debt	2,000	1,900	1,750
Total equity	4,000	4,500	5,000

7. Which of the following would be the analyst's *most likely* conclusion?

- A. The company is becoming less liquid, as evidenced by the increase in its debt-to-equity ratio from 0.35 to 0.50 from FY3 to FY5.
- B. The company is becoming increasingly more liquid, as evidenced by the increase in its debt-to-equity ratio from 0.35 to 0.50 from FY3 to FY5.
- C. The company is becoming increasingly less solvent, as evidenced by the increase in its debt-to-equity ratio from 0.35 to 0.50 from FY3 to FY5.

8. What would be the *most* reasonable explanation of the financial data?

- A. The decline in the company's equity results from a decline in the market value of this company's common shares.
- B. The EUR250 increase in the company's debt from FY3 to FY5 indicates that lenders are viewing the company as increasingly creditworthy.

C. The decline in the company's equity indicates that the company may be incurring losses, paying dividends greater than income, or repurchasing shares.

9. An analyst observes the data in [Exhibit 1](#) for two companies:

Exhibit 1: Data Comparison (US dollars)

	Company A	Company B
Revenue	4,500	6,000
Net income	50	1,000
Current assets	40,000	60,000
Total assets	100,000	700,000
Current liabilities	10,000	50,000
Total debt	60,000	150,000
Shareholders' equity	30,000	500,000

Which of the following choices *best* describes reasonable conclusions that the analyst might make about the two companies' ability to pay their current and long-term obligations?

- A. Company A's current ratio of 4.0 indicates it is more liquid than Company B, whose current ratio is only 1.2, but Company B is more solvent, as indicated by its lower debt-to-equity ratio.
- B. Company A's current ratio of 0.25 indicates it is less liquid than Company B, whose current ratio is 0.83, and Company A is also less solvent, as indicated by a debt-to-equity ratio of 200 percent compared with Company B's debt-to-equity ratio of only 30 percent.
- C. Company A's current ratio of 4.0 indicates it is more liquid than Company B, whose current ratio is only 1.2, and Company A is also more solvent, as indicated by a debt-to-equity ratio of 200 percent compared with Company B's debt-to-equity ratio of only 30 percent.

The following information relates to questions 10-13

The following data appear in the five-year summary of a major international company. A business combination with another major manufacturer took place in FY13.

Exhibit 1: Five-Year Summary of a Major International Company

	FY10	FY11	FY12	FY13	FY14
Financial statements	GBP millions	GBP millions	GBP millions	GBP millions	GBP millions
Income statements					
Revenue	4,390	3,624	3,717	8,167	11,366
Profit before interest and taxation (EBIT)	844	700	704	933	1,579
Net interest expense	-80	-54	-98	-163	-188
Taxation	-186	-195	-208	-349	-579
Minorities	-94	-99	-105	-125	-167
Profit for the year	484	352	293	296	645
Balance sheets					

	FY10	FY11	FY12	FY13	FY14
Fixed assets	3,510	3,667	4,758	10,431	11,483
Current asset investments, cash at bank and in hand	316	218	290	561	682
Other current assets	558	514	643	1,258	1,634
Total assets	4,384	4,399	5,691	12,250	13,799
Interest bearing debt (long term)	-602	-1,053	-1,535	-3,523	-3,707
Other creditors and provisions (current)	-1,223	-1,054	-1,102	-2,377	-3,108
Total liabilities	-1,825	-2,107	-2,637	-5,900	-6,815
Net assets	2,559	2,292	3,054	6,350	6,984
Shareholders' funds	2,161	2,006	2,309	5,572	6,165
Equity minority interests	398	286	745	778	819
Capital employed	2,559	2,292	3,054	6,350	6,984
Cash flow					
Working capital movements	-53	5	71	85	107
Net cash inflow from operating activities	864	859	975	1,568	2,292

- 10.** The company's total assets at year-end FY9 were GBP3,500 million. Which of the following choices *best* describes reasonable conclusions an analyst might make about the company's efficiency?
- A.** Comparing FY14 with FY10, the company's efficiency deteriorated, as indicated by its current ratio.
 - B.** Comparing FY14 with FY10, the company's efficiency deteriorated due to asset growth faster than turnover revenue growth.
 - C.** Comparing FY14 with FY10, the company's efficiency improved, as indicated by a total asset turnover ratio of 0.86 compared with 0.64.
- 11.** Which of the following choices *best* describes reasonable conclusions an analyst might make about the company's solvency?
- A.** Comparing FY14 with FY10, the company's solvency improved, as indicated by the growth in its profits to GBP 645 million.
 - B.** Comparing FY14 with FY10, the company's solvency deteriorated, as indicated by a decrease in interest coverage from 10.6 to 8.4.
 - C.** Comparing FY14 with FY10, the company's solvency improved, as indicated by an increase in its debt-to-assets ratio from 0.14 to 0.27.
- 12.** Which of the following choices *best* describes reasonable conclusions an analyst might make about the company's liquidity?
- A.** Comparing FY14 with FY10, the company's liquidity improved, as indicated by an increase in its current ratio from 0.71 to 0.75.
 - B.** Comparing FY14 with FY10, the company's liquidity deteriorated, as indicated by a decrease in interest coverage from 10.6 to 8.4.
 - C.** Comparing FY14 with FY10, the company's liquidity improved, as indicated by an increase in its debt-to-assets ratio from 0.14 to 0.27.
- 13.** Which of the following choices *best* describes reasonable conclusions an analyst might make about the company's profitability?

- A. Comparing FY14 with FY10, the company's profitability improved, as indicated by an increase in its debt-to-assets ratio from 0.14 to 0.27.
- B. Comparing FY14 with FY10, the company's profitability improved, as indicated by the growth in its shareholders' equity to GBP6,165 million.
- C. Comparing FY14 with FY10, the company's profitability deteriorated, as indicated by a decrease in its net profit margin from 11.0 percent to 5.7 percent.

14. An analyst compiles the data in [Exhibit 1](#) for a company:

Exhibit 1: Net Profit Margin

	FY13	FY14	FY15
ROE	19.8%	20.0%	22.0%
Return on total assets	8.1%	8.0%	7.9%
Total asset turnover	2.0	2.0	2.1

Based only on the information above, the *most* appropriate conclusion is that, over the period FY13 to FY15, the company's:

- A. net profit margin and financial leverage have decreased.
- B. net profit margin and financial leverage have increased.
- C. net profit margin has decreased but its financial leverage has increased.

15. A decomposition of ROE for Integra SA is as follows:

Exhibit 1: Integra SA ROE

	FY12	FY11
ROE	18.90%	18.90%
Tax burden	0.70	0.75
Interest burden	0.90	0.90
EBIT margin	10.00%	10.00%
Asset turnover	1.50	1.40
Leverage	2.00	2.00

Which of the following choices *best* describes reasonable conclusions an analyst might make based on this ROE decomposition?

- A. Profitability and the liquidity position both improved in FY12.
- B. The higher average tax rate in FY12 offset the improvement in profitability, leaving ROE unchanged.
- C. The higher average tax rate in FY12 offset the improvement in efficiency, leaving ROE unchanged.

16. A decomposition of ROE for Company A and Company B is as follows:

Exhibit 1: ROE for Company A and Company B

	Company A		Company B	
	FY15	FY14	FY15	FY14
ROE	26.46%	18.90%	26.33%	18.90%
Tax burden	0.7	0.75	0.75	0.75
Interest burden	0.9	0.9	0.9	0.9
EBIT margin	7.00%	10.00%	13.00%	10.00%
Asset turnover	1.5	1.4	1.5	1.4
Leverage	4	2	2	2

An analyst is *most likely* to conclude that:

- A. Company A's ROE is higher than Company B's in FY15, and one explanation consistent with the data is that Company A may have purchased new, more efficient equipment.
- B. the difference between the two companies' ROE in FY15 is very small and Company A's ROE remains similar to Company B's ROE mainly due to Company A increasing its financial leverage.
- C. Company A's ROE is higher than Company B's in FY15, and one explanation consistent with the data is that Company A has made a strategic shift to a product mix with higher profit margins.

17. When developing forecasts, analysts should *most likely*:

- A. develop possibilities relying exclusively on the results of financial analysis.
- B. aim to develop extremely precise forecasts using the results of financial analysis.
- C. use the results of financial analysis, analysis of other information, and judgment.

SOLUTIONS

1. C is correct. Cross-sectional analysis involves the comparison of companies with each other for the same time period. Time-series or trend analysis is the comparison of financial data across different time periods. Common-size analysis involves expressing financial data in relation to a single financial statement item, or base.
2. C is correct. The company's problems with its inventory management system causing duplicate orders would likely result in a higher amount of inventory and, therefore, would result in a decrease in inventory turnover. A more efficient inventory management system and a write-off of inventory at the beginning of the period would both likely decrease the average inventory for the period (the denominator of the inventory turnover ratio), thus increasing the ratio rather than decreasing it.
3. B is correct. A write-off of receivables would decrease the average amount of accounts receivable (the denominator of the receivables turnover ratio), thus increasing this ratio. Customers with weaker credit are more likely to make payments more slowly or to pose collection difficulties, which would likely increase the average amount of accounts receivable and thus decrease receivables turnover. Longer payment terms would likely increase the average amount of accounts receivable and thus decrease receivables turnover.
4. A is correct. The average accounts receivable balances (actual and desired) must be calculated to determine the desired change. The average accounts receivable balance can be calculated as an average day's credit sales times the DSO. For the most recent fiscal year, the average accounts receivable balance is USD15.62 million [= (USD300,000,000/365) × 19]. The desired average accounts receivable balance for the next fiscal year is USD16.03 million [= (USD390,000,000/365) × 15]. This is an increase of USD0.41 million [= 16.03 million – 15.62 million]. An alternative approach is to calculate the turnover and divide sales by turnover to determine the average accounts receivable balance. Turnover equals 365 divided by DSO. Turnover is 19.21 (= 365/19) for the most recent fiscal year and is targeted to be 24.33 (= 365/15) for the next fiscal year. The average accounts receivable balances are USD15.62 million (= USD300,000,000/19.21), and USD16.03 million [= USD390,000,000/24.33]. The change is an increase in receivables of USD0.41 million.
5. C is correct. The analyst is *unlikely* to reach the conclusion given in Statement C because days of sales outstanding increased from 23 days in FY1 to 25 days in FY2 to 28 days in FY3, indicating that the time required to collect receivables has increased over the period. This is a negative factor for Spherion's liquidity. By contrast, days of inventory on hand dropped over the period FY1 to FY3, a positive for liquidity. The company's increase in days payable, from 35 days to 40 days, shortened its cash conversion cycle, thus also contributing to improved liquidity.
6. C is correct. Solvency ratios are used to evaluate the ability of a company to meet its long-term obligations. An analyst is more likely to use activity ratios to evaluate how efficiently a company uses its assets. An analyst is more likely to use liquidity ratios to evaluate the ability of a company to meet its short-term obligations.
7. C is correct. The company is becoming increasingly less solvent, as evidenced by its debt-to-equity ratio increasing from 0.35 to 0.50 from FY3 to FY5. The amount of a company's debt and equity do not provide direct information about the company's liquidity position.

Debt to equity:

$$\text{FY5: } 2,000/4,000 = 0.5000$$

$$\text{FY4: } 1,900/4,500 = 0.4222$$

$$\text{FY3: } 1,750/5,000 = 0.3500$$

8. C is correct. The decline in the company's equity indicates that the company may be incurring losses, paying dividends greater than income, or repurchasing shares. Recall that Beginning equity –

Shares repurchased + Comprehensive income – Dividends = Ending equity. The book value of a company's equity is not affected by changes in the market value of its common stock. An increased amount of lending does not necessarily indicate that lenders view a company as increasingly creditworthy. Creditworthiness is not evaluated based on how much a company has increased its debt but rather on its willingness and ability to pay its obligations. (Its financial strength is indicated by its solvency, liquidity, profitability, efficiency, and other aspects of credit analysis.)

9. A is correct. Company A's current ratio of 4.0 ($= \text{USD}40,000/\text{USD}10,000$) indicates it is more liquid than Company B, whose current ratio is only 1.2 ($= \text{USD}60,000/\text{USD}50,000$). Company B is more solvent, as indicated by its lower debt-to-equity ratio of 30 percent ($= \text{USD}150,000/\text{USD}500,000$) compared with Company A's debt-to-equity ratio of 200 percent ($= \text{USD}60,000/\text{USD}30,000$).
10. B is correct. The company's efficiency deteriorated, as indicated by the decline in its total asset turnover ratio from 1.11 ($= 4,390/[(4,384 + 3,500)/2]$) for FY10 to 0.87 ($= 11,366/[(12,250 + 13,799)/2]$) for FY14. The decline in the total asset turnover ratio resulted from an increase in average total assets from GBP3,942 ($= (4,384 + 3,500)/2$) for FY10 to GBP13,024.5 for FY14, an increase of 230 percent, compared with an increase in revenue from GBP4,390 in FY10 to GBP11,366 in FY14, an increase of only 159 percent. The current ratio is not an indicator of efficiency.
11. B is correct. Comparing FY14 with FY10, the company's solvency deteriorated, as indicated by a decrease in interest coverage from 10.6 ($= 844/80$) in FY10 to 8.4 ($= 1,579/188$) in FY14. The debt-to-asset ratio increased from 0.14 ($= 602/4,384$) in FY10 to 0.27 ($= 3,707/13,799$) in FY14. This is also indicative of deteriorating solvency. In isolation, the amount of profits does not provide enough information to assess solvency.
12. A is correct. Comparing FY14 with FY10, the company's liquidity improved, as indicated by an increase in its current ratio from 0.71 ($= (316 + 558)/1,223$) in FY10 to 0.75 ($= (682 + 1,634)/3,108$) in FY14. Note, however, comparing only the cash ratio shows a decline in liquidity from 0.26 ($= 316/1,223$) in FY10 to 0.22 ($= 682/3,108$) in FY14. Debt-to-assets ratio and interest coverage are measures of solvency not liquidity.
13. C is correct. Comparing FY14 with FY10, the company's profitability deteriorated, as indicated by a decrease in its net profit margin from 11.0 percent ($= 484/4,390$) to 5.7 percent ($= 645/11,366$). Debt-to-assets ratio is a measure of solvency not an indicator of profitability. Growth in shareholders' equity, in isolation, does not provide enough information to assess profitability.
14. C is correct. The company's net profit margin has decreased and its financial leverage has increased. $\text{ROA} = \text{Net profit margin} \times \text{Total asset turnover}$. ROA decreased over the period despite the increase in total asset turnover; therefore, the net profit margin must have decreased.

 $\text{ROE} = \text{Return on assets} \times \text{Financial leverage}$. ROE increased over the period despite the drop in ROA; therefore, financial leverage must have increased.
15. C is correct. The increase in the average tax rate in FY12, as indicated by the decrease in the value of the tax burden (the tax burden equals one minus the average tax rate), offset the improvement in efficiency indicated by higher asset turnover) leaving ROE unchanged. The EBIT margin, measuring profitability, was unchanged in FY12 and no information is given on liquidity.
16. B is correct. The difference between the two companies' ROE in FY15 is very small and is mainly the result of Company A's increase in its financial leverage, indicated by the increase in its Assets/Equity ratio from 2 to 4. The impact of efficiency on ROE is identical for the two companies, as indicated by both companies' asset turnover ratios of 1.5. Furthermore, if Company A had purchased newer equipment to replace older, depreciated equipment, then the company's asset turnover ratio (computed as sales/assets) would have declined, assuming constant sales. Company A has experienced a significant decline in its operating margin, from 10 percent to 7 percent which, all else equal, would not suggest that it is selling more products with higher profit margins.
17. C is correct. The results of an analyst's financial analysis are integral to the process of developing forecasts, along with the analysis of other information and judgment of the analysts. Forecasts are not limited to a single point estimate but should involve a range of possibilities.

NOTES

¹Adapted from [Robinson and Munter \(2004, p. 2–15\)](#).

Introduction to Financial Statement Modeling

LEARNING OUTCOMES

The candidate should be able to:

- ☐ demonstrate the development of a sales-based pro forma company model
- ☐ explain how behavioral factors affect analyst forecasts and recommend remedial actions for analyst biases
- ☐ explain how the competitive position of a company based on a Porter's five forces analysis affects prices and costs
- ☐ explain how to forecast industry and company sales and costs when they are subject to price inflation or deflation
- ☐ explain considerations in the choice of an explicit forecast horizon and an analyst's choices in developing projections beyond the short-term forecast horizon

1. INTRODUCTION

Financial statement modeling is a key step in the process of valuing companies and the securities they have issued. We begin our discussion with an overview of developing a revenue forecast. We then describe the general approach to forecasting each of the financial statements and demonstrate the construction of a financial statement model, including forecasted income statements, balance sheets, and statements of cash flows. Then, we describe key behavioral biases that can influence the modeling process and strategies to mitigate them. We then turn to several important topics on the effects of micro- and macroeconomic conditions on financial statement models: the impact of competitive factors on prices and costs, the effects of inflation and deflation, technological developments, and long-term forecasting considerations.

Most of the examples and exhibits used throughout the reading can be downloaded as a Microsoft Excel workbook. Each worksheet in the workbook is labeled with the corresponding example or exhibit number in the text.

Learning Module Overview

- A financial statement model is the starting point for most valuation models, and valuation estimates can be made based on a variety of metrics, including free cash flow, EPS, EBITDA, and EBIT.
- Some balance sheet line items, such as retained earnings, flow directly from the income statement, whereas accounts receivable, accounts payable, and inventory are very closely linked to income statement projections.
- Working capital accounts are modeled by projecting working capital ratios (e.g., days of inventory, days sales outstanding, days payable outstanding) which are combined with the sales and cost of sales forecast to produce projected working capital accounts on the balance sheet.
- Five key behavioral biases that influence analyst forecasts are overconfidence, illusion of control, conservatism, representativeness, and confirmation bias.
- Illusion of control, a bias linked to overconfidence, is a tendency to overestimate the ability to control what cannot be controlled and to take ultimately fruitless actions in pursuit of control.
- A common manifestation of confirmation bias among investment analysts is to structure the research process in pursuit of only positive news or certain criteria, or with a narrow scope.
- Competitive factors affect a company's ability to negotiate lower input prices with suppliers and to raise prices for products and services. Porter's five forces framework can be used as a basis for identifying such factors.
- Porter's five forces are Threat of substitutes, Rivalry, Bargaining power of suppliers, Bargaining power of buyers, and Threat of new entrants.
- Return on invested capital, ROIC, defined as net operating profit less adjusted taxes divided by the difference between operating assets and operating liabilities, is an after-tax measure of profitability. High and persistent levels of ROIC are often associated with having a competitive advantage.
- Inflation and deflation can significantly affect the accuracy of forecasts for a company's future revenue, profit, and cash flow.
- Forecasting revenue for a company faced with inflation in input costs requires an understanding of the price elasticity of the products, the different rates of cost inflation in the countries where the company is active, and the likely inflation in costs relevant to a company's individual product categories.
- Faced with rising input prices, a company might decide to preserve its margins by passing on the costs to its customers, or it might decide to accept some margin reduction to increase its market share.
- The choice of the forecast time horizon can be influenced by certain factors, including the investment strategy for which the security is being considered, the cyclicity of the industry, company-specific factors, and the analyst's employer's preferences.
- Normalized earnings are the expected level of mid-cycle earnings for a company in the absence of any unusual or temporary factors that affect profitability.
- One of the greatest challenges facing the analyst is anticipating inflection points, such as from economic disruption, regulation, and technology, when the future will look significantly different from the recent past.

2. BUILDING A FINANCIAL STATEMENT MODEL

LEARNING OUTCOME

The candidate should be able to:

□ demonstrate the development of a sales-based pro forma company model

In this module, we apply the principles covered in earlier modules in Financial Statement Analysis and Corporate Issuers in a demonstration of building a financial statement model. The subject company is the Rémy Cointreau Group (Rémy), a French company that primarily sells spirits. After providing a brief overview of the company, we will focus primarily on the mechanics of constructing pro forma income statements, statements of cash flows, and balance sheets. Data sources for this example include the company's fiscal year ended 31 March 2021 and 2020 annual reports, the company's interim reports, and corresponding investor presentations for additional information on the underlying results of the respective divisions. While forecasts are described in some detail here, later modules in company analysis in the Equity topic area will discuss forecasting in much greater detail.

2.1. Company Overview

Rémy, whose reporting year ends 31 March, operates and reports three business segments:

1. **Cognac.** This division, composed primarily of Rémy Martin brand cognac, represented approximately 73 percent of FY2021 (year-end 31 March 2021) revenue and 94 percent of total current operating profit. Current operating profit is a non-IFRS measure reported by Rémy equal to IFRS operating profit excluding items related to discontinued brands or items deemed infrequent or immaterial, such as impairment or litigation provisions.
2. **Liqueurs & Spirits.** A diverse portfolio of spirits brands, the main brands in this segment are Cointreau, Metaxa, St-Rémy, Mount Gay, Bruichladdich, and The Botanist. The segment represented approximately 25 percent of FY2021 revenue and 14 percent of current operating profits.
3. **Partner Brands.** This segment includes other companies' brands that are marketed through Rémy's distribution network. They represented approximately 3 percent of FY2021 revenue and just under 0 percent of current operating profit, earning a slight operating loss in FY2021 of –EUR0.8 million. This division's importance has declined significantly over time as the company discontinued distribution ("partner brand") contracts.

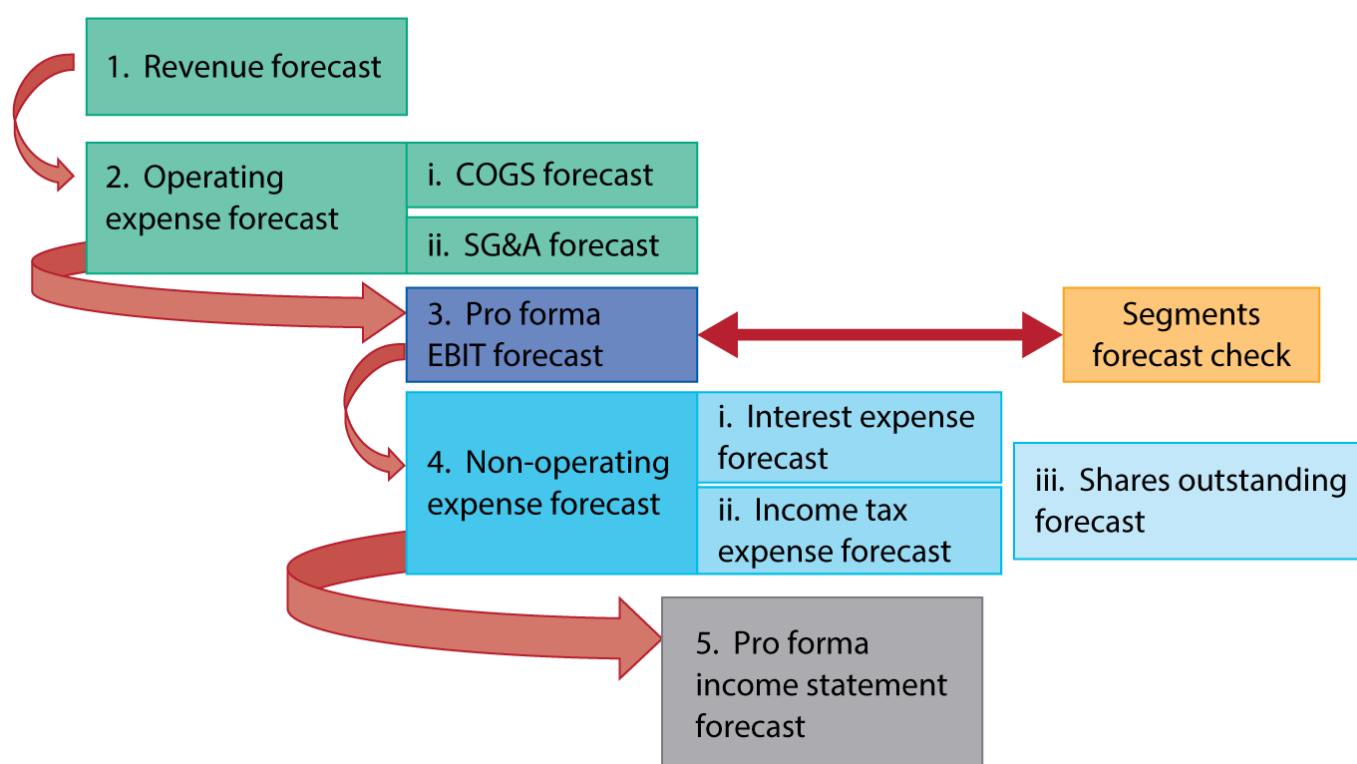
Segment financial information is summarized in [Exhibit 1](#). As shown, the company's largest business segment is also its most profitable: The Cognac segment earned a current operating profit margin of approximately 30 percent (= EUR221 million/EUR735 million) in fiscal year 2021. Exhibits are in the downloadable Microsoft Excel workbook in a single worksheet titled "Rémy." We strongly recommend following along with the Excel workbook and exploring the model construction in detail.

Exhibit 1: Analysis of Rémy's Turnover and Operating Profit

<i>Revenue (euro millions)</i>	FY2019	FY2020	FY2021
Cognac	774	736	735
Liqueurs and spirits	264	262	248
Partner brands	87	28	27
Total revenues	1,126	1,025	1,010
<i>Current Operating Profit (euro millions)</i>			
Cognac	236	200	221
Liqueurs and spirits	39	38	33
Partner brands	5	–2	–1
Holding/Corporate-level costs	–15	–20	–17
Total current operating profit	264	215	236
<i>Current Operating Profit Margins</i>			
Cognac	30.4%	27.1%	30.1%
Liqueurs and spirits	14.7%	14.3%	13.3%
Partner brands	5.6%	–6.2%	–3.0%
Holding/Corporate-level costs (percent of total revenue)	–1.3%	–2.0%	–1.7%
Total current operating margin	23.5%	21.0%	23.4%

Construction of pro forma income statements, as [Exhibit 2](#) illustrates, is composed of four forecasting steps: revenue, COGS, other operating expenses, and, finally, non-operating items.

Exhibit 2: Income Statement Forecast Process



2.2. Revenue Forecast

For each segment, the change in revenue is driven by volume, price, and foreign currency estimates that are based on historical trends as adjusted for expected deviations from trend. Price changes refer not only to price changes for a single product but also to changes in mix, which are defined as changes resulting from selling varying quantities of higher- and lower-priced products. Changes in revenue attributable to volume or price/mix are organic growth and are shown separately from the impact of acquisitions and divestitures (scope change) and foreign exchange (forex impact in the model).

In the Cognac segment, historical volume growth is usually in the 4 to –6 percent range. For future years, volume growth is expected to remain robust but slower than the 9.1 percent achieved in 2021 as the global recovery from the COVID-19 pandemic and associated recession fades (volumes were down 10.1 percent in FY2020). The growing number of affluent Asian consumers will likely keep demand high, while developed market consumption is likely to be rather flat. In the model, the assumption is for 7 percent volume growth in 2022, declining to 6 percent in 2023 and 2024.

Price/mix contributed approximately 6.0 percent, 2.6 percent, and –5.4 percent to the Cognac segment revenue growth in FY2019, FY2020, and FY2021, respectively. Although the impact of price/mix on revenue growth has fluctuated in recent years, price/mix will likely remain a relatively significant contributor to revenue growth in the future given the favorable structure of the industry and the company’s efforts to increase the share of revenues accounted for by what it calls “exceptional spirits” (those that cost more than USD50 per bottle and are seeing a 10 percent annual demand growth). A 4 percent price/mix contribution to revenue growth is assumed in 2022, with the trend maintained into 2023 and 2024. The combined projections for 2022 of 7 percent volume growth and 4 percent price/mix impact results in overall organic revenue growth of 11.3 percent, calculated as $[(1 + 0.07) \times (1 + 0.04)] - 1$.

In addition to the impact of volume and price/mix, Rémy’s revenues are affected by movements in exchange rates. Company disclosures indicate that more than 70 percent of revenues are realized outside the eurozone, whereas most of Rémy’s production occurs within the eurozone. The model forecasts no foreign currency impact on revenue in the 2022–24 forecast period.

Exhibit 3 summarizes historical and projected information for the Cognac segment’s revenue.

Exhibit 3: Historical and Projected Information for Cognac Segment Revenue

	FY2018	FY2019	FY2020	FY2021	FY2022E	FY2023E	FY2024E
Cognac Segment Revenues (euro millions)	760	774	736	735	818	902	994
YoY%	7.4%	1.9%	–5.0%	–0.1%	11.3%	10.2%	10.2%
Volume growth (%)	6.0%	5.9%	–10.1%	9.1%	7.0%	6.0%	6.0%
Price/mix (%)	7.2%	6.0%	2.6%	–5.4%	4.0%	4.0%	4.0%
Organic growth (%)	13.6%	12.3%	–7.8%	3.2%	11.3%	10.2%	10.2%
Forex impact and scope change (%)	–5.8%	–4.0%	2.5%	–3.8%	0.0%	0.0%	0.0%

	FY2018	FY2019	FY2020	FY2021	FY2022E	FY2023E	FY2024E
Effect of IFRS 15 adoption	0.0%	-6.0%	0.0%	0.0%	0.0%	0.0%	0.0%
YoY%	7.8%	2.3%	-5.3%	-0.6%	11.3%	10.2%	10.2%

A similar analysis can be performed to project revenue for the other segments. Then, the amounts can be summed to derive projected consolidated revenue.

2.3. COGS

Rémy's gross margin has remained roughly flat from FY2018 (67.5 percent) to FY2021 (67.3 percent) as total sales have decreased modestly. Going forward, we project gross margin to increase by 100 bps in each of the next three years based on increasing total revenues, particularly from price/mix, which is strongly accretive to gross margin (see the previous section on "Revenue Forecast"). Management has set a FY2030 objective of a 72.0 percent gross margin, largely in line with our forecasts. Should revenue growth prove more (less) robust than our forecast, we expect more (less) gross margin accretion.

2.4. SG&A Expenses and Other Operating Expenses

Distribution costs increased significantly over time, from 26.1 percent of revenue in FY2009 (not shown in the exhibits) to 38 percent in FY2018, and thereafter decreasing to 33.8 percent in FY2021. In particular, the setup of Rémy's distribution network in Asia increased the cost base. Rémy is very committed to its brand building and is also diversifying geographically. We estimate modest increases in distribution costs as a percentage of revenue, of 20 bps per year. Administrative costs as a percentage of revenue have increased from 8.1 percent to 10.1 percent as revenues have fallen, owing to the COVID-19 pandemic. However, the growth in absolute euro amounts has been modest, with costs of approximately EUR100 million in FY2019–FY2021. We expect 1 percent growth in administrative costs per year through FY2024E.

Other operating expenses (income), composed primarily of provisions for impairments of intangible assets, restructurings, and divestiture gains, has fluctuated from -EUR2 million to EUR20 million from FY2018 to FY2021. Because we do not anticipate any transactions that would result in other operating expenses or income, we forecast zero for this line in the model.

Exhibit 4 provides a consolidated income statement for Rémy through the EBIT and EBIT margin line.

Exhibit 4: Consolidated Historical and Projected Income Statement (Operating) for Rémy Cointreau Group (euro millions, unless noted)

	FY2018	FY2019	FY2020	FY2021	FY2022E	FY2023E	FY2024E
Sales	1,127	1,126	1,025	1,010	1,095	1,181	1,275
Cost of sales	366	415	348	330	347	362	379
Gross profit	761	711	677	680	748	819	897
Gross margin	67.5%	63.1%	66.1%	67.3%	68.3%	69.3%	70.3%
Change in gross margin	0.8%	-4.4%	2.9%	1.3%	1.0%	1.0%	1.0%
Distribution costs	433	346	355	342	373	404	439
Distribution costs as percent of sales	38.4%	30.7%	34.6%	33.8%	34.0%	34.2%	34.4%
Administrative expenses	92	101	107	103	104	105	106
Administrative expenses as % of sales	8.1%	8.9%	10.4%	10.1%	9.5%	8.9%	8.3%
Other operating expenses (income)	13	-2	20	0	0	0	0
EBIT	223	266	196	236	272	310	352
EBIT margin	19.8%	23.6%	19.1%	23.3%	24.8%	26.2%	27.6%

2.5. Operating Profit by Segment

In this section, we alternatively estimate operating profit and margin using a segment approach. Rémy discloses current operating profit for each of its segments as well as an operating cost at the corporate or holding company level. Recall that current operating profit is a non-IFRS measure that excludes certain items. These certain items are disclosed on Rémy's income statement as "Other operating expenses (income)." Therefore, the sum of the segment current operating profit equals consolidated EBIT before other operating expenses (income).

For the Cognac segment, the forecast of higher revenue growth, based partially on strong price/mix growth, assumes an improving product mix that will also result in a higher gross margin. But the benefit to gross margin will be somewhat mitigated by higher distribution costs. Thus, the expectation is that the Cognac segment's operating margin will increase to 33.4 percent by FY2024. As a benchmark, this forecast can be compared with the financial results reported by Hennessy (part of LVMH), another cognac brand. That company's operating margin in its Wine & Spirits segment in FY2017–2019 was 30–32 percent, though that business has a significantly higher mix of lower-priced products with lower gross margins.

For the other segments, there is not much upside. In the Liqueurs & Spirits division, we assume operating margin to increase modestly to 13.6 percent. In total, Rémy Cointreau Group's consolidated operating margin is forecast to improve from 23.4 percent in FY2021 to 27.6 percent in FY2024, largely because of growth and margin improvement in the Cognac segment, the most profitable division, and leverage from that sales growth on corporate-level costs.

While a segment approach like [Exhibit 5](#) can be used instead of a consolidated approach to forecasting revenue and operating profit, it is also commonly used as a "check" on the consolidated forecasts. This analysis revealed, for example, that the model relies significantly on margin improvement in the Cognac segment.

Exhibit 5: Historical and Projected Operating Profit by Segment for Rémy Cointreau Group

Revenue (euro millions)	FY2018	FY2019	FY2020	FY2021	FY2022E	FY2023E	FY2024E
Cognac	760	774	736	735	818	902	994
Liqueurs and spirits	267	264	262	248	251	253	256
Partner brands	100	87	28	27	26	26	26
Total revenues	1,127	1,126	1,025	1,010	1,095	1,181	1,275

Current Operating Profit (euro millions)	FY2018	FY2019	FY2020	FY2021	FY2022E	FY2023E	FY2024E
Cognac	204	236	200	221	255	291	332
Liqueurs and spirits	43	39	38	33	34	34	35
Partner brands	5	5	–2	–1	–1	–1	–1
Holding/Corporate-level costs	–16	–15	–20	–17	–16	–15	–14
Total current operating profit	237	264	215	236	271	309	352

Current Operating Profit Margins	FY2018	FY2019	FY2020	FY2021	FY2022E	FY2023E	FY2024E
Cognac	26.9%	30.4%	27.1%	30.1%	31.2%	32.3%	33.4%
Liqueurs and spirits	16.0%	14.7%	14.3%	13.3%	13.4%	13.5%	13.6%
Partner brands	5.3%	5.6%	–6.2%	–3.0%	–3.0%	–3.0%	–3.0%
Holding/Corporate-level costs	–1.4%	–1.3%	–2.0%	–1.7%	–1.5%	–1.3%	–1.1%
Total current operating profit	21.0%	23.5%	21.0%	23.4%	24.8%	26.2%	27.6%

2.6. Non-Operating Items

Three types of non-operating line items are included in the model: finance expenses (i.e., interest expenses), income taxes, and shares outstanding.

Net finance cost on Rémy's income statement is interest expense on debt less interest income earned on cash and investments. Forecasting net finance cost, therefore, requires estimating the debt and cash positions and interest rates paid and earned.

Companies pay a fixed or variable interest rate on debt. If the interest rate is variable, the rate is typically determined by a market reference rate plus a credit spread. As shown in [Exhibit 6](#), Rémy's interest expenses are fixed and calculated as 1.7 percent incurred on gross debt at the beginning of the period (EUR720 million at end of FY2020). Other financial expenses are assumed to be zero. Gross debt and the interest rate paid on it are estimated to remain flat from the year ended FY2021 level

Although interest income is typically forecasted after forecasting the cash position from the forecasted statement of cash flows, in this case we have simply estimated EUR0 in interest income through the model period; in each of FY2018–FY2021, annual interest income was EUR0, EUR0, EUR0.1, and EUR0.2 million, respectively, because Rémy maintains its liquidity in assets with zero or very low yields. For companies that own liquid assets with higher interest rates, or in higher interest rate environments, interest income should be forecast in the same manner as interest expense: forecasted cash and investments multiplied by a forecasted interest rate.

Exhibit 6: Debt Position and Financial Costs and Income for Rémy (EUR millions, unless noted)

	FY2018	FY2019	FY2020	FY2021	FY2022E	FY2023E	FY2024E
Long-term financial debt	397	424	452	424	424	424	424
Short-term financial debt and accrued interest	73	98	268	92	92	92	92
Gross debt	470	522	720	515	515	515	515
Interest expense	14.5	13.7	12.9	12.1	8.7	8.7	8.7
Interest rate (on beginning balance)		2.9%	2.5%	1.7%	1.7%	1.7%	1.7%
Interest income	0.0	0.0	0.1	0.2	0.0	0.0	0.0
Net finance cost	14.5	13.7	12.8	11.9	8.7	8.7	8.7

2.7. Corporate Income Tax Forecast

The French statutory corporate income tax rate at the time of analysis is 32 percent. Rémy Cointreau Group's effective tax rate has, over the longer run, been close to the statutory rate. Therefore, an estimated 32 percent effective tax rate is used in the forecast period. Rémy has no material minority interests in any of its subsidiaries.

2.8. Shares Outstanding

Shares outstanding to compute earnings per share (EPS) on the income statement are disclosed in two ways, both weighted averages throughout the fiscal year: basic and diluted. Basic shares outstanding includes common equity securities outstanding, while diluted shares outstanding is a type of what-if analysis; it is basic shares outstanding plus the number of shares from the exercise or conversion of in-the-money instruments, less an assumed repurchase of those if-issued shares.

Typically, the two major factors that affect shares outstanding over time are share issuance related to equity-based compensation of employees (increases shares outstanding) and share repurchases (decreases shares outstanding). Less common but sometimes significant transactions that also affect shares outstanding include acquisitions financed with stock, secondary issuance, and conversions of preferred stock or other instruments to common stock.

[Exhibit 7](#) shows beginning and ending basic shares outstanding for the past six fiscal years as well as the annual net amount of share repurchases and issuance, which were gathered from the statements of stockholders' equity and notes to financial statements. Additionally, the basic and diluted shares outstanding on the income statement used to calculate basic and diluted EPS (weighted averages) are shown and differed by approximately 2.6 million shares in each of the past five years.

Exhibit 7: Shares Outstanding for Rémy (euro millions, unless noted)

	FY2016	FY2017	FY2018	FY2019	FY2020	FY2021
Beginning basic shares outstanding	48.6	48.6	49.6	50.0	49.8	49.8
Share repurchases	-0.0	0.0	-0.3	-1.0	-0.0	0.0
Share issuance	0.0	1.0	0.7	0.8	0.1	0.4
Ending basic shares outstanding	48.6	49.6	50.0	49.8	49.8	50.3
Weighted average basic shares	48.6	49.1	49.8	50.1	49.8	50.1
Dilutive securities	0.1	2.7	2.6	2.6	2.6	2.6
Weighted average diluted shares	48.7	51.8	52.4	52.7	52.4	53.1

As evident in [Exhibit 7](#), shares outstanding for Rémy have not changed materially in six years because the company does not pay significant share-based compensation nor has it repurchased shares. Additionally, management has not disclosed an intention to repurchase shares in the near term. Therefore, the model assumes that weighted average basic and diluted shares outstanding on the income statement remain flat at the FY2021 level.

2.9. Pro Forma Income Statement

Now with the forecast components in place, a consolidated pro forma income statement can be constructed, as shown in [Exhibit 8](#). Although not presented on the face of the income statement as disclosed by the company, the calculation of EBITDA is shown after EBIT by adding depreciation and amortization expense from the statement of cash flows. It is not linked to other quantities on the income statement but merely shown as a useful profitability measure.

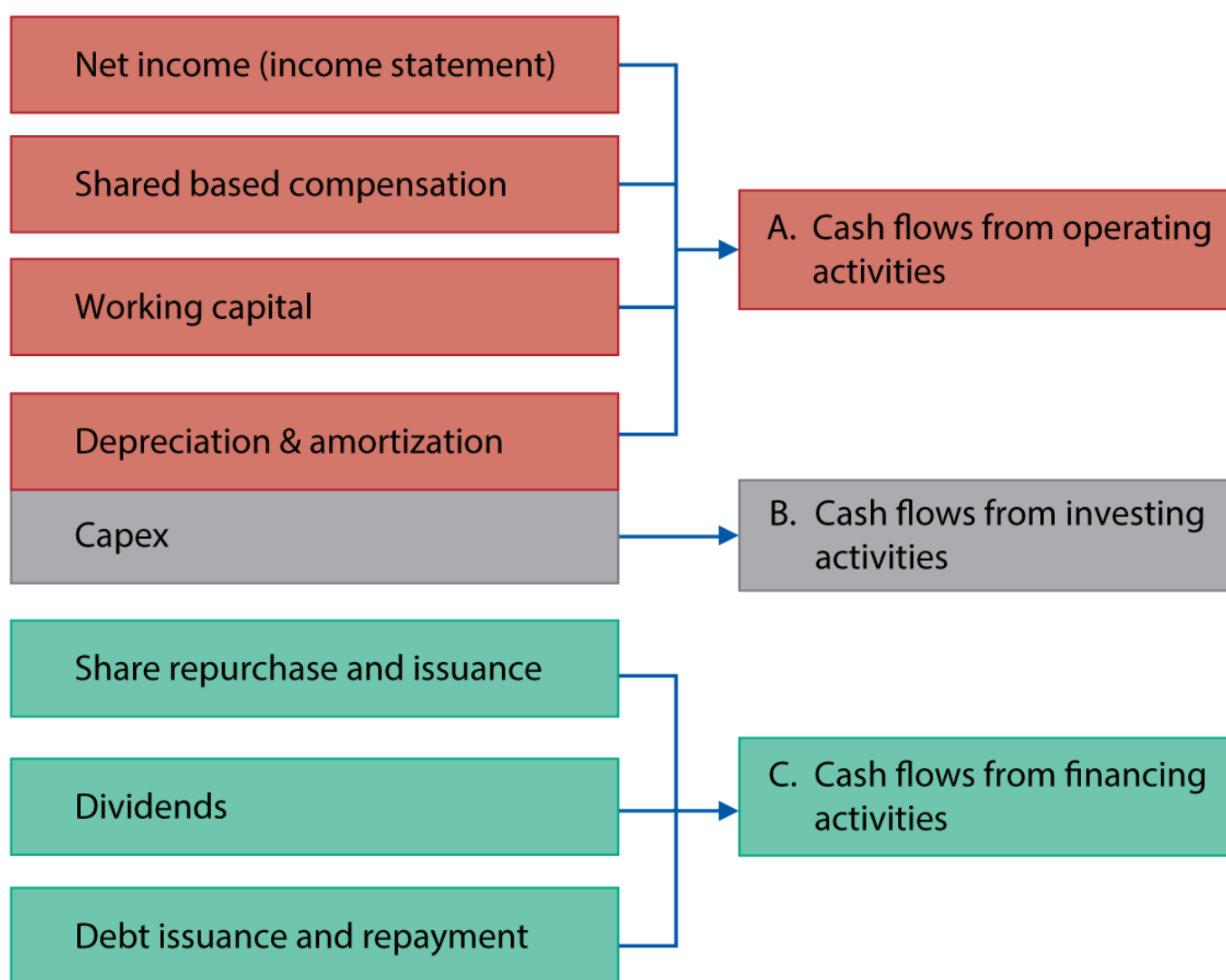
Exhibit 8: Consolidated Historical and Projected Income Statement for Rémy Cointreau Group (euro millions, unless noted)

	FY2018	FY2019	FY2020	FY2021	FY2022E	FY2023E	FY2024E
Sales	1,127	1,126	1,025	1,010	1,095	1,181	1,275
Cost of sales	366	415	348	330	347	362	379
Gross profit	761	711	677	680	748	819	897
Gross margin	67.5%	63.1%	66.1%	67.3%	68.3%	69.3%	70.3%
Change in gross margin	0.8%	-4.4%	2.9%	1.3%	1.0%	1.0%	1.0%
Distribution costs	433	346	355	342	373	404	439
Distribution costs as percent of sales	38.4%	30.7%	34.6%	33.8%	34.0%	34.2%	34.4%
Administrative expenses	92	101	107	103	104	105	106
Administrative expenses as percent of sales	8.1%	8.9%	10.4%	10.1%	9.5%	8.9%	8.3%
Other operating expenses (income)	13	-2	20	0	0	0	0
EBIT	223	266	196	236	272	310	352
EBIT margin	19.8%	23.6%	19.1%	23.3%	24.8%	26.2%	27.6%
Depreciation and amortization (add-back)	22	30	33	34			
Depreciation and amortization as percent of sales	1.9%	2.7%	3.3%	3.4%			
EBITDA	245	296	229	270			
EBITDA margin	21.7%	26.3%	22.3%	26.7%			
Net finance costs	15	14	13	12	9	9	9
Other financial expenses	8	19	15	3	0	0	0
Total financial expenses	22	33	28	15	9	9	9
Profit before tax	201	233	167	221	263	301	344
Income tax	54	68	61	78	84	96	110
Effective tax rate	26.6%	29.0%	36.4%	35.1%	32.0%	32.0%	32.0%
Income from associates	1	-7	0	1	0	0	0
Profit from continuing operations	148	159	107	144	179	205	234
Profit from discontinued operations	0	0	6	0	0	0	0
Net profit for the year	148	159	113	144	179	205	234
YoY%		8%	-29%	27%	24%	14%	14%
EPS basic continuing operations	2.97	3.18	2.14	2.88	3.58	4.09	4.67
EPS diluted continuing operations	2.82	3.02	2.04	2.74	3.40	3.89	4.44
EPS basic total	2.97	3.18	2.27	2.88	3.58	4.09	4.67
EPS diluted total	2.82	3.02	2.16	2.74	3.40	3.89	4.44
Average number of shares, basic (millions)	49.8	50.1	49.8	50.1	50.1	50.1	50.1
Average number of shares, diluted (millions)	52.4	52.7	52.4	52.6	52.6	52.6	52.6

2.10. Pro Forma Statement of Cash Flows

The forecast statements of cash flows begin with forecasted net income and other amounts from the forecast income statement, and then typically require estimates for capital expenditures, depreciation and amortization, working capital, share-based compensation, dividends, and share repurchases. Once the forecasted income statements and statements of cash flows are completed, forecasting the balance sheet is largely a matter of properly linking the spreadsheet, as illustrated in [Exhibit 9](#).

Exhibit 9: Statement of Cash Flows Projection Process



2.11. Capital Investments and Depreciation Forecasts

Capital investment, or capex, as a percentage of revenue was 5.3 percent in FY2021. Given the healthy volume growth prospects, we expect capex to remain at a modestly above historical average level of 5.0 percent of sales through FY2024. With Rémy's growing fixed asset base, it is logical that depreciation will increase. The model assumes that depreciation and amortization (D&A) is equal to 4.2 percent of prior year fixed assets, the average of the past three years. The breakdowns of capex and D&A are shown in [Exhibit 10](#).

Exhibit 10: Capex, D&A Breakdowns

	2018	2019	2020	2021	2022E	2023E	2024E
D&A (euro millions)	22	30	33	34	36	36	37
As percent of prior year fixed assets		4.0%	4.3%	4.2%	4.2%	4.2%	4.2%
Capex, PP&E, and intangibles (euro millions)	34	45	65	54	55	59	64
Capex as % of sales	3.0%	4.0%	6.3%	5.3%	5.0%	5.0%	5.0%
Capex/D&A ratio	1.6	1.5	1.9	1.6	1.5	1.6	1.7

2.12. Working Capital Forecasts

We have assumed that working capital ratios will remain similar to what the company experienced in the FY2018–21 period. In [Exhibit 11](#), we include only the relevant balance sheet items related to revenues and costs (i.e., inventories, accounts receivable, and accounts payable) and keep the other items constant. Rémy Cointreau Group had positive net working capital of 105 percent of its sales in fiscal year 2021. The largest working capital component is inventory because much of Rémy's cognac requires years of aging. Inventory days on hand in FY2021 was 1,493, which reflects an approximate 300-day increase owing to the volume slowdown during the COVID-19 pandemic. Inventory days are partially mitigated by extended payment terms to suppliers; days payable outstanding has averaged around 500 days since FY2018.

We model the working capital accounts by projecting working capital ratios (days of inventory, days sales outstanding, days payable outstanding) which are combined with the sales and cost of sales forecast to produce projected working capital accounts on the balance sheet. We expect inventory days to decline through FY2024 as the inventory increase that occurred during the COVID-19 pandemic is worked through, days sales outstanding to remain at FY2021 levels, and for model days

payable outstanding to decline back to an average level, again reflecting a normalization after the COVID-19 pandemic. As a result of the decrease in inventory days, the model projects a net positive contribution from working capital to the reconciliation of net income to cash flows from operations on the statement of cash flows, which is in stark contrast to prior years' negative contribution.

Exhibit 11: Working Capital Development for Rémy

	FY2018	FY2019	FY2020	FY2021	FY2022E	FY2023E	FY2024E
Inventories (euro millions)	1,170	1,246	1,364	1,493	1,426	1,340	1,245
Accounts receivable	210	271	199	158	171	185	200
Accounts payable	517	544	534	586	597	604	610
Working capital, net	863	973	1,029	1,065	1,000	922	835
Percent of sales	77%	86%	100%	105%	91%	78%	65%
Change in working capital		-110	-56	-36	64	79	87
Days inventories on hand	1,166	1,095	1,431	1,650	1,500	1,350	1,200
Days sales outstanding	68	88	71	57	57	57	57
Days payable outstanding	515	478	561	648	628	608	588

2.13. Forecasted Cash Flow Statement

With net income, D&A, change in working capital, capex, and debt estimates already in place, the cash flow statement shown in [Exhibit 12](#) is almost automatically generated by linking the relevant lines on a spreadsheet. The three significant items left to forecast are share-based compensation, share repurchases or issuance, and dividends. Going forward, the model assumes flat share-based compensation, no share repurchases or issuance, and dividends paid equal to the FY2021 level through FY2024. Lines labeled "other" are aggregated and zeroed out going forward because they are immaterial, difficult to forecast, or both.

Exhibit 12: Projected Statement of Cash Flows for Rémy (euro millions)

	FY2018	FY2019	FY2020	FY2021	FY2022E	FY2023E	FY2024E
Net income (loss)	148	159	113	144	179	205	234
D&A	22	30	33	34	36	36	37
Share-based compensation	3	3	4	2	2	2	2
Investment in working capital	-7	-162	-72	-13	64	79	87
Other non-cash amounts	20	22	3	10	0	0	0
Cash flows from operations	185	53	81	177	281	322	360
Capex (PP&E and intangibles)	-34	-45	-65	-54	-55	-59	-64
Other investing activities	2	92	12	62	0	0	0
Cash flows from investments	-32	47	-53	8	-55	-59	-64
Debt issuance (repayment)	0	11	196	-246	0	0	0
Share issuance (repurchases)	-27	-104	-2	2	0	0	0
Dividends paid	-25	-9	-132	-10	-10	-10	-10
Cash flows from financing	-52	-102	62	-253	-10	-10	-10
FX translation effects	8	-6	1	-1	0	0	0
Net change in cash	109	-8	91	-68	217	254	287
Cash and equivalents, beginning	78	187	179	269	201	418	671
Cash and equivalents, end	187	179	269	201	418	671	958

2.14. Forecasted Balance Sheet

The forecasted balance sheet is given in [Exhibit 13](#) and is based on the combination of the projected income statement ([Exhibit 8](#)), the projected statement of cash flows ([Exhibit 12](#)), and the historical starting balance sheet. The balance sheet items that were not specifically discussed are held constant, which preserves the accounting identity. For ease of presentation, the stockholders' equity lines (e.g., common stock, additional paid in capital, retained earnings, treasury shares, accumulated other comprehensive income) are aggregated. For each forecast period, common stockholders' equity is the prior year value plus net income and share-based compensation less dividends.

If each of the discussed lines is linked properly—and other lines are held constant from FY2021—the forecasted balance sheet should balance each year. Consult the Rémy worksheet in the downloadable Microsoft Excel workbook for greater detail.

Exhibit 13: Projected Balance Sheet for Rémy (euro millions)

	FY2018	FY2019	FY2020	FY2021	FY2022E	FY2023E	FY2024E
Cash and equivalents	186.8	178.6	269.4	201.0	418	671	958
Accounts receivable	210	271	199	158	171	185	200
Inventories	1,170	1,246	1,364	1,493	1,426	1,340	1,245
Other current assets	16	5	16	10	10	10	10
Total current assets	1,583	1,700	1,848	1,861	2,025	2,206	2,412
PP&E, intangibles, goodwill, net	752	785	808	845	864	887	913
Investment in associates	20	1	1	2	2	2	2
Other non-current assets	186	139	131	73	73	73	73
Total assets	2,542	2,625	2,789	2,781	2,964	3,168	3,400
Short-term/current debt	73	98	268	92	92	92	92
Accounts payable	517	544	534	586	597	604	610
Other current liabilities and accrued expenses	26	31	39	42	42	42	42
Total current liabilities	616	673	842	720	731	737	744
Long-term/non-current debt	397	424	452	424	424	424	424
Other non-current liabilities	121	102	92	88	88	88	88
Total common equity	1,407	1,425	1,403	1,548	1,720	1,918	2,144
NCI (Non-Controlling Interest)	1	1	1	1	1	1	1
Total equity and liabilities	2,542	2,625	2,789	2,781	2,964	3,168	3,400

2.15. Valuation Model Inputs

A financial statement model is the starting point for most valuation models. Valuation estimates can be made based on a variety of metrics, including free cash flow, EPS, EBITDA, and EBIT. The company-specific inputs needed to build a discounted cash flow (DCF) to the firm model (to estimate enterprise value) are shown in [Exhibit 14](#). All the variables are sourced from the forecasted income statements and statements of cash flows.

Exhibit 14: Calculating Free Cash Flow to the Firm as Basis for a DCF Valuation Model (euro millions)

	FY2021	FY2022E	FY2023E	FY2024E
EBIT	236	272	310	352
Taxes (32% tax rate)	-75	-87	-99	-113
After-tax EBIT	160	185	211	240
D&A	34	36	36	37
Change in working capital	-13	64	79	87
Capital expenditures	-54	-55	-59	-64
Free cash flow to the firm	127	230	267	300

3. BEHAVIORAL FINANCE AND ANALYST FORECASTS

LEARNING OUTCOME

The candidate should be able to:

- ☐ explain how behavioral factors affect analyst forecasts and recommend remedial actions for analyst biases

Studies have shown that experts in many fields persistently make forecasting errors arising from behavioral biases, and investment analysts’ models of financial statements are in no way immune. To improve forecasts and the investment decisions based on them, analysts must be aware of the impact of biases and potential remedies for them. Five key behavioral biases that influence analyst forecasts are overconfidence, illusion of control, conservatism, representativeness, and confirmation bias.

3.1. Overconfidence in Forecasting

Overconfidence bias occurs when people demonstrate unwarranted faith in their own abilities. Studies have identified that 90 percent *confidence intervals* for forecasts, which should leave only 10 percent error rates, turn out to be wrong as much as 40 percent of the time (Russo and Schoemaker 1992). Studies have also suggested that individuals are more confident when making contrarian predictions that counter the consensus. That is, overconfidence arises more frequently when forecasting what others do not expect (Dunning, Griffin, Milojkovic, and Ross 1990).

To mitigate overconfidence bias, analysts should record and share their forecasts and review them regularly, identifying *both* the correct and incorrect forecasts they have made. Given the wide range of outcomes for most financial variables, an analyst will likely find that they have been wrong as much or more often than they have been right. The goal is to recognize that forecast error rates are high, so mitigating actions that widen the confidence interval of forecasts should be taken. One such action is **scenario analysis**. By asking, “Where could I be wrong and by how much?,” an analyst can generate different forecast scenarios.

EXAMPLE 1

Mitigating Overconfidence: Scenario Analysis for Rémy

In the prior lesson, a financial statement model was constructed for Rémy Cointreau Group that includes only one set of forecasted numbers, or one scenario. Creating several more scenarios is an important modeling step because the range of outcomes for the most important variable is wider than a single point.

Three important variables in the forecast of free cash flow are organic sales growth in the Cognac segment, EBIT margin, and net working capital as a percentage of sales. A benefit of the spreadsheet-driven model is that the forecasts can be easily modified to calculate different free cash flow estimates. The base case inputs and forecast for 2024E free cash flow to the firm, as well as figures for two different scenarios, are shown in Exhibit 15.

Alternative Scenario 1 assumes that the Cognac segment’s organic growth remains the same as its FY2021 rate, an EBIT margin of 23.6 percent, where it was before the COVID-19 pandemic, and working capital of 86 percent of sales, also the pre-pandemic level from FY2019. Alternative Scenario 2 assumes the same Cognac segment organic growth rate as the base case but an EBIT margin of 25.0 percent and working capital of 90 percent of sales. This scenario reflects strong growth but a high level of reinvestment in sales and marketing costs and aged cognac inventory to support that growth.

As Exhibit 15 demonstrates, there is a wide range of free cash flow estimates for 2022E–2024E because of a wide range of reasonable inputs for key variables.

Exhibit 15: Calculating Free Cash Flow to the Firm as Basis for a DCF Valuation Model (euro millions)

Base Case	2022E	2023E	2024E
Cognac segment organic growth	11.3%	10.2%	10.2%
EBIT margin	24.8%	26.2%	27.6%
Working capital as percent of sales	91%	78%	65%
Free cash flow to the firm est.	230	267	300
Alternative Scenario 1	2022E	2023E	2024E
Cognac segment organic growth	4.0%	4.0%	4.0%
EBIT margin	23.6%	23.6%	23.6%
Working capital as percent of sales	86%	86%	86%
Free cash flow to the firm est.	318	133	129
Alternative Scenario 2	2022E	2023E	2024E
Cognac segment organic growth	11.3%	10.2%	10.2%
EBIT margin	25.0%	25.0%	25.0%
Working capital as percent of sales	90%	90%	90%
Free cash flow to the firm est.	240	109	105

3.2. Illusion of Control

A bias often linked to overconfidence, illusion of control is a tendency to overestimate the ability to control what cannot be controlled and to take ultimately fruitless actions in pursuit of control. This bias often manifests in analysts' beliefs that forecasts can be rendered more accurate in two ways: by acquiring more information and opinions from experts and by creating more granular and complex models. Although additional information and complexity in model specification can improve forecasting accuracy, there are diminishing marginal returns. The amount of material information available for an investment is finite and adding immaterial information will mislead. Complex models tend to be overfitted to historical data sets which do not prove robust in a range of environments that include never-before-seen outliers. Excessive breadth of data and model complexity can also conceal assumptions and make updating forecasts upon the receipt of new information difficult. Finally, analysts face significant opportunity costs; additional hours modeling one company could mean that the analyst will examine fewer opportunities in total.

Beyond awareness of the bias and the recognition that uncertainty is an inherent characteristic in investments, illusion of control can be mitigated by restricting modeling variables to those that are regularly disclosed by the company, focusing on the most important or impactful variables, and speaking only with those who are likely to have unique or significant perspectives.

EXAMPLE 2

Illusion of Control: How Much Model Complexity?

Rémy Cointreau Group regularly reports revenues by segment and by geographic region (Europe/Middle East/Africa, the Americas, and Asia Pacific). It does not disclose segment revenue by geographic region (e.g., the Cognac segment revenue in the Asia Pacific region), nor does it disclose revenue by sales channel, such as retailers versus bars and restaurants, travel retail, and so on. In its quarterly earnings calls, however, the company often makes numerous references to segment growth rates in specific regions and growth rates of specific channels, even though the actual numbers are not disclosed. Such a practice is common, especially during the COVID-19 pandemic because large sales channel shifts occurred: travel retail in most regions experienced declines >90 percent, sales shifted from bars and restaurants to retailers for at-home consumption, and different geographies were affected by the pandemic at different times.

An analyst might be tempted to collect all these growth rates and other anecdotal figures that management discloses on its earnings calls and, perhaps by combining them with third-party estimates of sales, to build an extensive revenue model for Rémy in which each segment is broken out into geographic regions and sales channels.

Although such an endeavor might be useful to set expectations and to monitor over time, building the revenue forecast in this way would introduce several problems and probably not materially improve accuracy. First, because the data used in the model are not regularly disclosed, there is no way to check actuals versus estimates. Second, model construction would take dozens of hours. Finally, and perhaps most importantly, whether the constituent small parts of such a model would be accurate is unclear, which would not make the consolidated revenue forecast any more accurate than a simpler model.

3.3. Conservatism Bias

Conservatism bias is a bias in which people maintain their prior views or forecasts by inadequately incorporating new information. This often happens in forecasting when an analyst does not update their forecasting after receiving conflicting information, such as disappointing earnings results or a competitor action. Although the most common form of conservatism is the reluctance to incorporate new negative information into a forecast, analysts could also fail to adequately incorporate positive information and thus have estimates that are too low. A different name for conservatism bias in this context is anchoring and adjustment, referring to an analyst using their prior estimates as an “anchor” that is subsequently adjusted. Although nothing is wrong with modifying a previous forecast, the previous forecast or anchor tends to exert significant influence; in other words, the adjustment is too small, and the updated forecast is too close to the previous forecast.

Conservatism bias can be mitigated by reviews of forecasts and models by an investment team at a regular interval, such as each quarter, and by creating flexible models with fewer variables, to make changing assumptions easier. Because conservatism bias is related to overconfidence and the illusion of control, mitigating those biases can also serve to mitigate conservatism.

EXAMPLE 3

Conservatism Bias: Rémy Management Guidance for FY2022

The base case forecasts in the Rémy Cointreau Group model call for organic revenue growth of 11.3 percent and net income growth of 24 percent in FY2022E over FY2021. However, during the earnings call for the fourth quarter of FY2021, Rémy management gave the following guidance for FY2022:

- “Fiscal year 2022 will be a strong year of growth and investment, and we are on track to achieve our 2030 [objectives of a 72% gross margin and 33% operating margin].”
- “Being ahead of [our] 2030 strategic plan and given the favorable environment, [we] have decided to revise up [our] strategic investments [in sales and marketing] to support brands through the recovery and boost their medium-

term growth potential by developing brand awareness and attractiveness.”

- Fiscal year 2022 will have “top-line and bottom-line growth in the mid-teens in organic terms.”

Based on these comments, your colleague suggests revising the Rémy model slightly by reducing the operating margin forecast to reduce net income growth from 24 percent to 20 percent.

What behavioral bias does your colleague’s suggestion exhibit, and what research or steps should be taken, if any, with respect to revising the Rémy model? Explain your answer.

Solution:

Your colleague is exhibiting conservatism bias or anchoring and adjustment; they are anchored to the prior forecast of 24 percent net income growth and not fully considering management’s guidance on profitability.

Changing the model to follow the guidance without further consideration is not necessarily appropriate because results can and often do under- or outperform guidance. However, in this case, management guidance differs quite significantly from the FY2022E forecast on both sales growth and net income growth. As a first step, management’s credibility should be assessed by examining the company’s performance against management guidance in the past. Second, the guidance should be considered as a scenario in the scenario analysis, and the investment implications of that scenario should be examined; for example, if the company will in fact increase sales and profits at a mid-teens rate in FY2022, does that result in an investment decision? Finally, the performance of, and guidance provided by, other alcohol and spirits companies should be compared to these figures as a check for reasonableness.

3.4. Representativeness Bias

Representativeness bias refers to the tendency to classify information based on past experiences and known classifications. New information might resemble or seem representative of familiar elements already classified but can in fact be very different and is better viewed from a different perspective. In these instances, the classification reflex can deceive, producing an incorrect understanding that biases all future thinking about the information. Base-rate neglect is a common form of representativeness bias in forecasting. In base-rate neglect, a phenomenon’s rate of incidence in a larger population, or characteristics of a larger class to which a specific member belongs—its base rate—is neglected in favor of situation- or member-specific information. Considering the base rate is sometimes known as the “outside view,” while the situation-specific is known as the “inside view.”

For example, an analyst is modeling operating costs and margins for a biopharmaceutical company. The “inside view” approach would consider company-specific factors such as the types of drugs the company sells, the number of salespeople needed in each geographic region for each drug, and so on. The “outside view” approach would view the company as a member of the “biopharmaceuticals” industry, of which there are many others, and use industry or sector averages for gross margin, R&D expense as percentage of sales, and so on in the model.

Neither the outside nor inside view is superior; what makes for a superior forecast is considering both. One way of doing so is by starting with the base rate but determining which factors make the target company different from the base rate or class average and what the implications of those differences are, if any. For example, the analyst modeling the biopharmaceuticals company might start with industry averages in the model but change some of the variables to account for factors such as royalties versus product sales revenues, geographic composition of revenues, and whether the company is likely to face patent expirations on its products over the forecast period.

EXAMPLE 4

Considering Base Rates for Rémy

While constructing the Rémy model in the prior lesson, little attention was given to comparable companies or to the broader industry to which Rémy belongs. In other words, the model was constructed primarily with the “inside view.” In this example, Rémy is put in the context of six other spirits-focused alcohol companies: Brown-Forman Corporation, Pernod Ricard SA, Davide Campari-Milano N.V., Diageo plc, Beclé S.A.B de C.V. (Cuervo), and the Wine & Spirits segment of LVMH (LVMH W&S) for the five most recently reported fiscal years at the time of analysis. The variable used for the industry comparison is the five-year average of EBIT margin because it is a key model input, and the profitability of an individual company is strongly influenced by industry profitability. Many of these peer companies are significantly larger by revenue than Rémy, which is useful because we have modeled Rémy becoming larger over time. The analysis for [Exhibit 16](#) is included in the [Exhibit 16](#) worksheet in the downloadable Microsoft Excel workbook.

Exhibit 16: EBIT Margin Comparison of Spirits Companies, Last Five Reported Fiscal Years (euro millions)

EBIT margin	MRY-4	MRY-3	MRY-2	MRY-1	Most Recent Year (MRY)
Rémy	20%	20%	24%	19%	23%
Brown Forman	34%	32%	34%	32%	34%

EBIT margin	MRY-4	MRY-3	MRY-2	MRY-1	Most Recent Year (MRY)
Pernod	24%	25%	26%	26%	12%
Campari	22%	26%	25%	25%	17%
Diageo	27%	30%	30%	31%	18%
Cuervo	23%	26%	20%	18%	20%
LVMH W&S	31%	31%	32%	31%	29%
Peer average (ex Rémy)	27%	28%	28%	27%	22%
Peer five-year average (ex Rémy)	26%				

Evaluate the base case forecasts in the Rémy model as well as Rémy’s management’s FY2030 objective of a 33 percent operating margin considering the analysis in [Exhibit 16](#).

Solution:

The base case forecasts in the Rémy model are for EBIT margins of 24.8 percent, 26.2 percent, and 27.6 percent in FY2022E, FY2023E, and FY2024E, respectively. The most recently reported fiscal year(s) for most of the peer companies include the effect of deleveraging from sales declines associated with the COVID-19 pandemic. Aside from that, the base case forecasts are close to the peer average and by that measure appear reasonable, though they are substantially higher than the past five years of profitability for Rémy itself.

Rémy management’s objective of 33 percent operating margin in 2030 appears high relative to those of its peers; only one company, Brown Forman, has achieved that level of profitability, on annual revenues ~3.0× that of Rémy. Industry-leading growth and profitability of Rémy’s Cognac segment will be required to meet this objective.

3.5. Confirmation Bias

Confirmation bias is the tendency to look for and notice what confirms prior beliefs and to ignore or undervalue whatever contradicts them. A common manifestation of this bias among investment analysts is to structure the research process in pursuit of only positive news or certain criteria, or with a narrow scope. For example, an analyst might research a particular company but conduct only cursory research on its competitors and companies that offer substitute products. An analyst who has a positive view on a company might speak only to other analysts who share that view and the company’s management, all of whom will likely tell the analyst what they want to hear and already know. Confirmation bias is closely related to overconfidence and representativeness biases.

The extent to which company management can be excessively optimistic is shown in [Exhibit 17](#), which analyzes the annual report of a major European bank for 2007, published mere months before it entered bankruptcy and was nationalized.

Speaking with management is valuable given their role and should not be excluded from the research process, but analysts must be aware of management’s inherent bias and seek differing perspectives, especially when examining a company with significant controversy. Two approaches to mitigating confirmation bias in the forecasting process are to speak to or read research from analysts with a negative opinion on the security under scrutiny and to seek perspectives from colleagues who are not economically or psychologically invested in the subject security.

EXAMPLE 5

Management Optimism

Consider this text analysis of the chairman’s statement and business review in the 2007 annual report of a major European bank published in 2008, a few months before the bank was rescued by the government.

Exhibit 17: Text Analysis

Occurrences of ...			
Negative words		Positive words	
Disappoint/disappointed	0	Good	55
Bad/badly	0	Excellent	12
Poor	0	Success/successful	35
Weaker/weakening	7	Improvement	23
Slowdown	6	Strong/stronger/strongly	78

4. THE IMPACT OF COMPETITIVE FACTORS IN PRICES AND COSTS

LEARNING OUTCOME

The candidate should be able to:

- explain how the competitive position of a company based on a Porter’s five forces analysis affects prices and costs

One of the tools that analysts can use to think about how competition will affect financial results is Michael Porter’s widely used “five forces” framework (see [Porter 1980](#)) introduced in earlier learning modules.

4.1. Cognac Industry Overview

The cognac segment is Rémy Cointreau Group’s most important business segment, accounting for over 90 percent of total operating profit. An important feature of the cognac market is that supply is limited and demand is growing. Supply is limited because the production of cognac, like that of champagne, is highly regulated, in this case through The Bureau National Interprofessionnel du Cognac. By regulation, cognac can be produced only in a limited geographic area, in and around the town of Cognac in southwest France. Furthermore, within the region, production volume is capped each year. Approximately 98 percent of production is exported. The cognac market is highly concentrated, with the top four players controlling 78 percent of world volume and 84 percent of global value. Rémy’s market share is approximately 16 percent and 18 percent of global volume and value, respectively (*The Spirits Business*, June 2018). Demand for cognac has been growing because of increasing demand from Asia, particularly China and Singapore, more than offsetting a weakening European market. The global spirits market has grown more than 5 percent annually during the 2000–17 period (*Source: IWSR drinks market analysis*). Simultaneously, Rémy has also seen a product mix improvement because consumers increasingly prefer superior quality and more expensive cognac. [Exhibit 18](#) summarizes Porter’s five forces analysis of the cognac industry.

Exhibit 18: Porter’s Five Forces Analysis of the Cognac Industry

Force	Degree	Factors to Consider
Threat of substitutes	Low	■ Cognac consumers show brand loyalty and do not easily shift to other beverages or high-end spirits.
Rivalry	Low	■ The market is consolidated, with four players controlling 78 percent of the world market in volume and 84 percent of global value. ■ Only the European market is fragmented, with less than half of the market controlled by the top four.
Bargaining power of suppliers	Low/medium	■ A large number of small independent vineyards supply inputs. ■ Most of the distillation is carried out by a large body of independent distillers that sell to the big houses.
Bargaining power of buyers	Low	■ Premium beverages are sold primarily to wine and spirits retail outlets that do not coordinate purchasing. ■ Premium beverages are consumed primarily in small and fragmented on-premises outlets (restaurants, etc.).
Threat of new entrants	Low	■ Producers have long-term contracts with suppliers in the Cognac area. ■ Barriers to entry are high. • Building brands is difficult because they must have heritage/pedigree. • A large capital investment is required to build an inventory with “aged” cognac and set up a distribution network.

In summary, the cognac market, Rémy's largest and most profitable operating segment, exhibits a favorable profitability profile. In addition to limited supply and growing demand, the industry faces a generally favorable situation with respect to substitutes, rivalry, suppliers, buyers, and potential new entrants.

EXAMPLE 6

Analysis of Anheuser-Busch InBev Using Porter's Five Forces

The competitive structure a company faces can vary among countries, with implications for modeling revenue growth, profit margins, capital expenditures, and return on investments. For example, Anheuser-Busch (AB) InBev, the largest global brewer, operates in many countries, two of which are the United Kingdom and Brazil, the world's third largest beer market. AB InBev's competitive position and prospects in the highly consolidated and growing Brazilian market are much more favorable than in the fragmented and declining UK market.

The Brazilian beer market is divided among four players. AmBev (AB InBev's subsidiary in Brazil, of which it owns a 61.9 percent stake) is the dominant brewer with an estimated 65 percent market share in 2018 versus 20 percent for Heineken and 12 percent for Petropolis, Brazil's largest privately owned brewing group. Helped by its dominant market position and strong distribution network, AmBev was able to report an EBITDA margin of nearly 50.4 percent in 2018 (ri.ambev.com.br), the highest in the global beer industry. The industry participants focus less on price competition and more on expanding distribution and "premiumization" (i.e., selling more expensive beers.) Although the 2015–18 time period saw challenging trading conditions due to subdued consumer demand, causing years of decline in the market by volume, Brazil is still considered a promising market. In this environment, an analyst would likely forecast solid revenue growth for AmBev. [Exhibit 19](#) presents an analysis of the Brazilian beer market using Porter's five forces framework. Most of the competitive forces represent a low threat to profitability (consistent with AmBev's historical profitability), implying that analysts would most likely forecast continued above-average profitability.

Exhibit 19: Analysis of the Brazilian Beer Market Using Porter's Five Forces

Force	Degree	Factors to Consider
Threat of substitutes	Medium	- Beer consumers do not easily shift to other beverages, but such alternatives as wine and spirits are available. - Unlike in many other countries, the range of beers is relatively limited.
Rivalry	Low	- AmBev dominates the market with a 65% market share. Its economies of scale in production and distribution yield significant cost advantages relative to competition. - Price competition is limited because of AmBev's cost advantages and because of typically increasing beer volumes.
Bargaining power of suppliers	Low	The primary inputs (water, hops, barley, and packaging) are basically commodities.
Bargaining power of buyers	Low	- Beer is mostly consumed in bars and restaurants. The owners of these outlets represent a large and highly fragmented group of beer buyers. - The supermarket industry in Brazil is relatively fragmented, and supermarkets are less likely to offer alternatives, such as private labels.
Threat of new entrants	Low	New entrants face relatively high barriers to entry because of the high costs of building a brewery, establishing a national distribution network, and establishing a nationally known brand name.

The UK beer market is also divided among four players, but the competitive structure is totally different than in Brazil. The market is more fragmented, with smaller market shares held by the largest players. Heineken, MolsonCoors, AB InBev, and Carlsberg had market shares of 24 percent (adbrands.net), 18 percent, 18 percent (www.ab-inbev.com), and 11 percent (carlsberggroup.com), respectively, in 2018. Consequently, the British market has no dominant brewer. Given the high fixed costs of a brewery, declining volumes of UK beer consumption, and the highly consolidated customer base, which provides the clients with substantial purchasing power (particularly in the retail channels), price competition is usually intense. A gradual switch from drinking beer in pubs and restaurants ("on-trade") to consumption at home ("off-trade") is making brewers even more exposed to the bargaining power of the dominant retail supermarket (grocers) chains. Increasing taxes on beer and rents faced by pub landlords add to the burden faced by the industry, leading to a steady decline of Britain's pub industry. Profitability has been lower than the beer industry's global average; operating margins are believed to be less than 10 percent. In this kind of environment, analysts would most likely forecast only very

cautious revenue growth, if any. [Exhibit 20](#) presents an analysis of the UK beer market using Porter’s five forces framework.

Exhibit 20: Analysis of the UK Beer Market Using Porter’s Five Forces

Force	Degree	Factors to Consider
Threat of substitutes	Medium	Beer consumers do not easily shift to other beverages, but such alternatives as wine, spirits, and cider are available.
Rivalry	High	The market is relatively fragmented with no dominant market leader and large numbers of small breweries. Declining beer volumes make price wars more likely. Brand loyalty is less developed because of the extensive range of alternative beers.
Bargaining power of suppliers	Low	The primary inputs (water, hops, barley, and packaging) are basically commodities.
Bargaining power of buyers	High	The large supermarket chains that dominate the grocery sector have significant bargaining power. Large pub chains in the “on-trade” business (where beer is sold in pubs and restaurants) also have strong bargaining power.
Threat of new entrants	Low	Barriers to entry are relatively high because of the high costs of building a brewery, establishing a national distribution network (particularly given the history of brewers owning pubs and bars), and establishing a nationally known brand. Because the United Kingdom consists of islands, companies with breweries in other countries face higher transportation costs than existing participants.

There is a distinction between Porter’s five forces and other factors that can affect profitability, such as government regulation and taxes:

Industry structure, as manifested in the strength of the five competitive forces, determines the industry’s long-run profit potential because it determines how the economic value created by the industry is divided. Government is not best understood as a sixth force because government involvement is neither inherently good nor bad for industry profitability. The best way to understand the influence of government on competition is to analyze how specific government policies affect the five competitive forces. ([Porter 2008](#), page 10)

EXAMPLE 7

EuroAlco Case

In 20X2, EuroAlco was the beer market leader in Eurolandia (a fictional country) with 35 percent market share. The other four large brewers held 15 percent, 15 percent, 10 percent, and 7 percent share, respectively. The Eurolandia market is considered a growth market. It historically had high overall alcohol consumption but a relatively low per capita consumption of beer, a product that is attracting interest from the growing, younger population and is further supported by increasing disposable incomes.

At the start of year 20X1, the Eurolandia government, in its fight to curb alcohol consumption, tripled the excise duty (a special tax) on beer from EUR0.3 per liter to EUR0.9 and announced that excise duty will further increase by EUR0.1 per liter.

In the following year, 20X2, EuroAlco made efforts to strengthen the position of the more expensive brands in its portfolio. These efforts led to a 20 percent increase in selling costs. Similar to most consumer staple companies, EuroAlco experienced higher production costs. Poor grain harvests put price pressure on buyers of almost all feedstocks, and rising oil prices resulted in higher packaging costs. In 20X2, competing companies were much more cautious with A&P spending than EuroAlco.

Two analysts research EuroAlco at the start of year 20X3. In making their EuroAlco forecasts, both analysts use market data and the published annual report from EuroAlco (see [Exhibit 21](#) and/or the [Example 7](#) worksheet in the

downloadable Microsoft Excel workbook). Based on the published data, they consider a number of scenarios and reach different conclusions.

Exhibit 21: EuroAlco Key Financial and Operational Data

€ millions	20X2	20X1	20X0	% change	
				20X2/20X1	20X1/20X0
Retailer gross sales	11,504	10,248	9,180	12%	12%
Excise duty	2,900	2,520	900	15%	180%
As % of retail revenues	25%	25%	10%		
Value-Added-Tax, VAT (20%)	1,434	1,288	1,380	11%	-7%
Retailer net sales	7,170	6,440	6,900	11%	-7%
Typical retailer profit	935	840	900	11%	-7%
As % of retailer net sales	13%	13%	13%		
Brewer net sales	6,235	5,600	6,000	11%	-7%

Key Financial Indicators	20X2	20X1	20X0	% change	
				20X2/20X1	20X1/20X0
Volume (mln hectoliters)	29	28	30	4%	-7%
Net sales	6,235	5,600	6,000	11%	-7%
Cost of sales	3,190	2,800	3,150	14%	-11%
Gross profit	3,045	2,800	2,850	9%	-2%
Selling expenses	2,088	1,680	1,650	24%	2%
Administrative expenses	145	140	150	4%	-7%
Operating profit	812	980	1,050	-17%	-7%
Average invested capital	3,000	3,000	3,100	0%	-3%
Gross margin	48.8%	50.0%	47.5%		
Selling expense %	33.5%	30.0%	27.5%		
Operating margin	13.0%	17.5%	17.5%		
Return on invested capital (pre-tax)	27%	33%	34%		

€ per hectoliter (hl)	20X2	20X1	20X0	% change	
				20X2/20X1	20X1/20X0
Retail price	397	366	306	8%	20%
Excise duty	100	90	30	11%	200%
VAT	49	46	46	7%	0%
Typical distributor profit	32	30	30	7%	0%
Brewer net sales	215	200	200	8%	0%
Cost of sales	110	100	105	10%	-5%
Gross profit	105	100	95	5%	5%
Selling expenses	72	60	55	20%	9%
Administrative expenses	5	5	5	0%	0%
Operating profit	28	35	35	-20%	0%

Both analysts assume that the government will impose a further increase in the excise duty (special tax on beer). They also assume that the excise duty increase will be borne by the consumers, who will face a 10 percent price increase that will allow the brewers to maintain their net (after-tax) revenues per hectoliter (hl). They assume that half the cost of sales is fixed per hectoliter and half is variable based on volume, that selling expenses will remain unchanged as a percentage of sales, and that administrative expenses are fixed.

1. Analyst A expects price elasticity of 0.8, indicating that volume will fall by 8 percent given the 10 percent retail price increase. Calculate the impact on operating profit and operating profit margin in 20X3 using [Exhibit 22](#), which is also in the [Example 7](#) sheet in the downloadable Microsoft Excel workbook.

Exhibit 22: EuroAlco's Costs Structure for 20X2-20X3E (euro millions, unless noted)

20X2	Analyst A	Analyst B
------	-----------	-----------

	20X2	Analyst A		Analyst B	
		20X3E	YoY%	20X3E	YoY%
Volume (millions of hl)	29	26.7	-8.0%	27.6	-5.0%
Brewer net sales (€ per hl)	215				
Net sales	6,235				
Cost of sales	3,190				
Gross profit	3,045				
Gross margin	48.8%				
Selling expenses	2,088				
Administrative expenses	145	145		145	
Operating profit	812				
Operating profit margin	13.0%				
Cost of sales (fixed)	1,595	1,595		1,595	
Cost of sales (variable)	1,595				
Cost of sales (variable) per hl	55	55		55	
Selling expenses as % of sales	33.5%	33.5%		33.5%	

Solution:

Exhibit 23 (see the **Example 7** worksheet in the downloadable Microsoft Excel workbook) shows the results for both analysts' projections. Analyst A predicts that operating profit will decrease by 25 percent to EUR608 in 20X3, resulting in an operating margin decline from 13.0 percent in 20X2 to 10.6 percent in 20X3. Analyst A calculates a revenue decline of 8 percent to EUR5,736 based on volume dropping by 8 percent and a constant price per hectoliter of EUR215. The decrease in volume reflects the price elasticity of 0.8 and the price increase of 10 percent as a result of the excise duty increase. COGS sold fell only 4 percent because part of the costs are fixed. COGS as the sum of fixed and variable costs is EUR1,595 + [26.68 (hl volume) × 55 (hl cost)] = EUR1,595 + 1,467 (ignoring rounding error) or EUR3,062. Analyst A predicts selling expenses will decline in line with sales by 8 percent and administrative costs will remain unchanged because of their fixed character in the short term.

- Analyst B expects price elasticity of 0.5, indicating that volume will fall by 5 percent given the 10 percent retail price increase. Calculate the impact on operating profit and operating profit margin in 20X3 using **Exhibit 22**, which is also in the **Example 7** sheet in the downloadable Microsoft Excel workbook.

Solution:

Analyst B forecasts that operating profit will decline by 16 percent to EUR684. Analyst B's calculations follow the same pattern as those of Analyst A, but Analyst B predicts a smaller, 5 percent, decline in volume. Analyst A's estimates are more pessimistic than those of Analyst B. Note that the net price per hectoliter for the brewer is held constant while the price for the consumer increased 10 percent as a result of the excise duty increase. Because of Analyst B's more optimistic volume forecast, fixed costs are spread over a higher level of sales than is the case for Analyst A. Consequently, Analyst B will have a higher operating margin estimate than Analyst A. However, both analysts are predicting a decline in operating margin in 20X3.

Exhibit 23: Analysts' Results for EuroAlco's Cost Structure and Projection (euro millions, unless noted)

		Analyst A		Analyst B	
	20X2	20X3E	YoY%	20X3E	YoY%
Volume (millions of hl)	29	26.7	-8%	27.6	-5%
Brewer net sales per hl	215	215	0%	215	0%
Net sales	6,235	5,736	-8%	5,923	-5%
Cost of sales	3,190	3,062	-4%	3,110	-3%
Gross profit	3,045	2,674	-12%	2,813	-8%
Gross margin	48.8%	46.6%	-5%	47.5%	-3%
Selling expenses	2,088	1,921	-8%	1,984	-5%
Administrative expenses	145	145	0%	145	0%
Operating profit	812	608	-25%	684	-16%
Operating profit margin	13.0%	10.6%	-19%	11.6%	-11%
Cost of sales (fixed)	1,595	1,595	0%	1,595	0%
Cost of sales (variable)	1,595	1,467	-8%	1,515	-5%

	20X2	Analyst A		Analyst B	
		20X3E	YoY%	20X3E	YoY%
Cost of sales (variable) per hl	55	55	0%	55	0%
Selling expenses as % of net sales	33.5%	33.5%	0%	33.5%	0%

3. Gross margin improved in 20X1 (50.0 percent) but fell in 20X2 (48.8 percent). Cost of sales was relatively high in 20X2 because of high barley costs, an important ingredient for brewing beer. Assume that in 20X2, half of the cost of sales is fixed and half is based on volume. Of the variable part of the cost of sales, assume that half the amount is related to the barley price in 20X2. Barley prices increased 25 percent in 20X2. Consider a scenario where no additional taxes are imposed in 20X3, revenues and volumes remain stable, and barley prices return to their 20X1 level. Calculate EuroAlco's estimated gross margin for 20X3.

Solution:

If barley prices return to their 20X1 level, they will decline 20 percent in 20X3. Because volumes are assumed to remain constant, other variable costs will not change. Gross profit in 20X2 was 48.8 percent of sales, which indicates the cost of sales was 51.2 percent ($100\% - 48.8\%$). Barley is 25 percent of the cost of sales (because barley represents half of variable costs, and variable cost of sales represents half of total cost of sales). Cost of sales is predicted to decline by $25\% \times 20\% = 5\%$. New cost of sales will be $51.2\% - (5\% \times 51.2\%)$ or 48.6 percent. Consequently, gross margin is predicted to be $100\% - 48.6\% = 51.4\%$ in 20X3. Compared with the gross margin of 48.8 percent in 20X2, gross margin is predicted to increase by 260 bps.

Exhibit 24: Gross Margin Analysis

	20X3	20X2	YoY%
Volume	29	29	0%
Revenue	6,235	6,235	0%
Cost of sales	3,031	3,190	-5%
Variable	1,436	1,595	-10%
Barley related	638	798	-20%
Not barley related	798	798	0%
Fixed	1,595	1,595	0%
Gross profit	3,205	3,045	5%
Gross margin	51.4%	48.8%	5%

4. EuroAlco's selling expenses increased from 30 percent of sales in 20X1 to 33.5 percent of sales in 20X2. Which competitive forces most likely influenced EuroAlco's significant increase in selling expenses?

Solution:

Intra-industry rivalry and threat of substitutes most likely influenced EuroAlco's significant increase in selling costs. By spending more on advertising, EuroAlco wanted to enhance the brand loyalty of its products, thus improving its competitive position versus its brewer rivals and makers of other alcoholic beverages. Furthermore, buyers' bargaining power probably also influenced EuroAlco's increased spending to the extent that advertising creates demand by the ultimate consumer. Strong demand at the ultimate consumer level for EuroAlco's specific brands could enhance the company's bargaining position with its direct customers, the distributors who serve as intermediaries.

5. Retailers are the direct customers of brewers. They buy directly from the brewer and sell to the ultimate consumer. Analyst A expects that the increase in mass retailers in Eurolandia will cause brewers' margins to decline. He expects EuroAlco's operating margin will decrease from 13 percent in 20X2 to 8 percent in 20X6, with stable sales (EUR6,235 million) and an unchanged amount of average invested capital (EUR3,000 million). Analyst B also sees the increasing importance of the larger food retailers but expects that EuroAlco can offset potential pricing pressure by offering more attractive trade credit (e.g., allowing the retailers longer payment terms). He thinks operating margin can remain stable at 13 percent with no sales growth. Average invested capital (EUR3,000 million), however, will double because of the extra investments in inventory and receivables. Describe the analysts' expectations about the impact of large retailers on brewers in terms of Porter's five forces and return on invested capital (ROIC; pre-tax). Which of the two scenarios would be better for EuroAlco?

Solution:

The increase in mass retailers in EuroAlco is expected to strengthen the bargaining power of buyers relative to brewers. According to Analyst A, this will lead to a lower operating margin of 8 percent, while Analyst B believes margins can be maintained if the company offers much more favorable credit terms reflected in doubling of invested

capital. Analyst A expects operating profit on invested capital to fall from 27.1 percent ($13 \text{ percent} \times \text{EUR}6,235/\text{EUR}3,000$) to 16.6 percent ($8 \text{ percent} \times \text{EUR}6,235/\text{EUR}3,000$). Analyst B's assumptions indicate that the ROIC (operating profit divided by invested capital) in 20X2 of 27 percent will fall by half to 13.5 percent as the operating profit is earned on double the amount of invested capital (i.e., $13 \text{ percent} \times \text{EUR}6,235/\text{EUR}6,000$). The scenario envisioned by Analyst A is better for EuroAlco. Full supporting calculations are in the [Example 7](#) worksheet in the downloadable Microsoft Excel workbook.

Porter's five forces framework and similar analytical tools can help analysts assess the relative profit potential of a company by helping them understand the company's industry and its position within that industry. Understanding the industry and competitive contexts of a company helps analysts estimate whether, for example, sales growth is likely to be relatively high or low (relative to history, relative to the overall growth in the economy or a sector, and/or relative to competing companies) and whether profit margins are likely to be relatively high or low (relative to historical profit margins and relative to competing companies). The process of incorporating an industry and competitive analysis into expectations for future financial performance requires judgment. Suppose analysts observe that a given company is the market leader in a moderately competitive industry with limited buyer and supplier power and relatively high barriers to entry. In broad terms, analysts might project that the company's future revenue growth will be in line with that of the overall industry and that its profit margins and ROIC might be somewhat higher than those of other companies in the industry. But there is no mechanical link between the analysts' observations and projecting the company's future sales growth and profit margin. Instead, the link is more subjective and probabilistic.

5. MODELING INFLATION AND DEFLATION

LEARNING OUTCOME

The candidate should be able to:

- ☐ **explain how to forecast industry and company sales and costs when they are subject to price inflation or deflation**

Inflation and deflation (i.e., general increase and decrease in the prices of goods and services) can significantly affect the accuracy of forecasts for a company's future revenue, profit, and cash flow. The impact of inflation or deflation on revenue and expenses differs from company to company. Even within a single company, the impact of inflation or deflation is generally different for revenue and expenses categories.

Some companies are better able to pass on higher input costs by raising the prices at which they sell their output. The ability to pass on price increases can be the result of, for example, strong branding (Coca-Cola) or proprietary technology (Apple). Companies that are well positioned to pass on price increases are, in turn, more likely to have higher and more stable profits and cash flow, relative to competitors.

We first consider the impact of inflation on sales and then on costs.

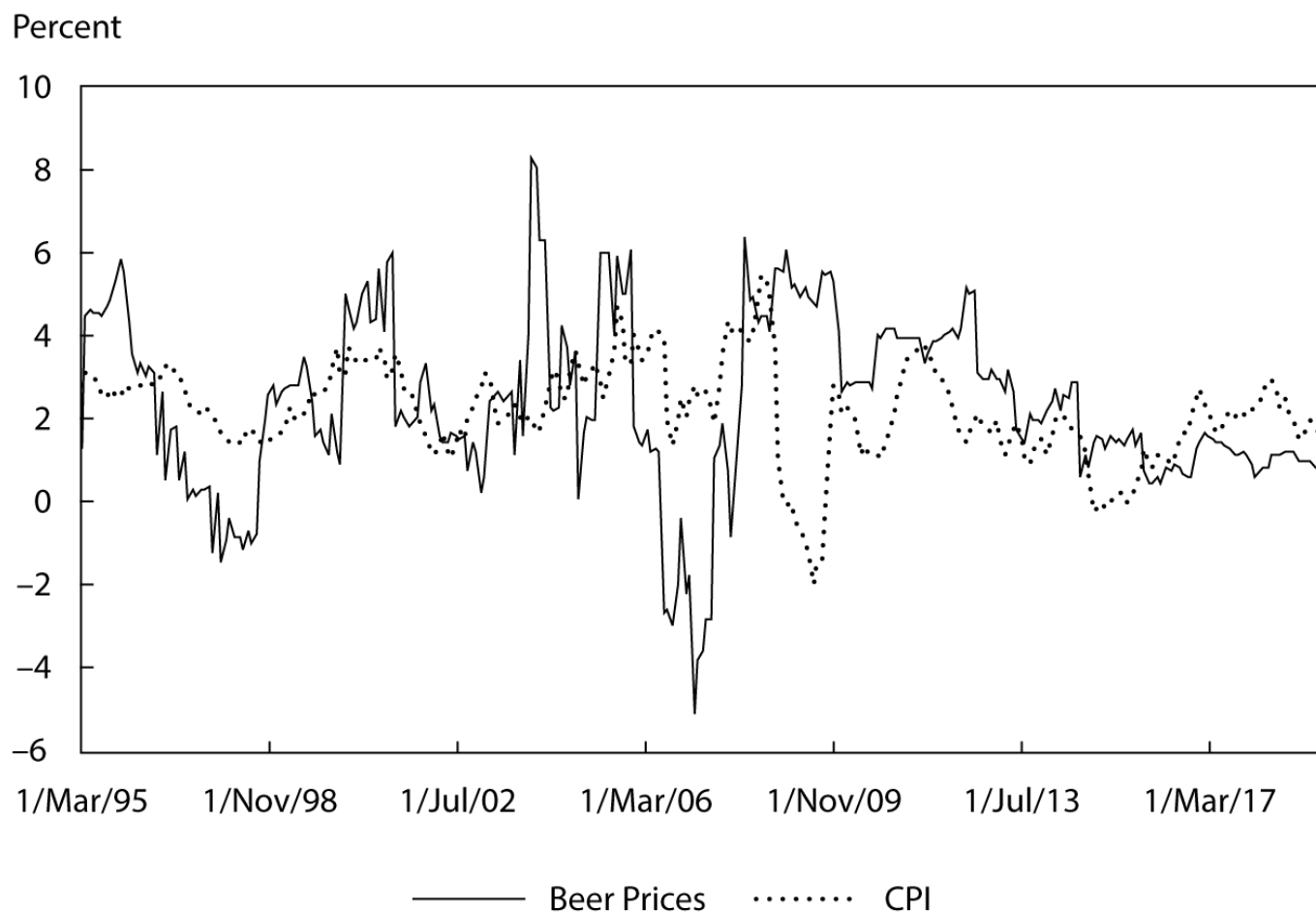
5.1. Sales Projections with Inflation and Deflation

The following analysis addresses the projection of industry sales and company sales in the presence of inflation.

5.1.1. *Industry Sales and Inflation or Deflation*

Most increases in the cost of inputs, such as commodities or labor, will eventually result in higher prices for end products. Industry structure can be an important factor in determining the relationship between increases in input costs and increases in the price of end products. For example, in the United States, the beer market is an oligopoly, with one player, AB InBev, controlling almost half of the market. Moreover, the three-tier structure of the US beer market, in which the producers (the brewers) must use a third party (the wholesalers) to get their products (beer) to the consumers (bars, restaurants, and retailers) results in a fragmented customer base because brewers are not allowed to deliver directly to the end consumer but rather must use wholesale distributors. These wholesalers often differ state by state. Large nationwide retailers, such as Walmart, still must negotiate with several different wholesalers instead of using their dominant national market position to negotiate directly with the brewers. The industry structure in the United States has likely contributed to increases in beer prices roughly in line with the US Consumer Price Index. In other words, beer prices have generally risen during years of inflation in input costs and decreased when costs have eased (though there have been brief exceptional periods where the opposite has occurred). If necessary, US brewers have been able to increase prices to compensate for costs of inflation. In contrast, European beer companies distribute through a more concentrated customer base—namely, such dominant retail outlets as Carrefour, Tesco, and Ahold—which results in a weaker pricing position for the brewers. Also, the European market lacks an overall dominant brewer. As a result of the industry structure and the lack of underlying volume growth, changes in beer prices in Europe have been on average 100 bps less than customer inflation.

Exhibit 25: US General Inflation and Inflation in Beer Prices

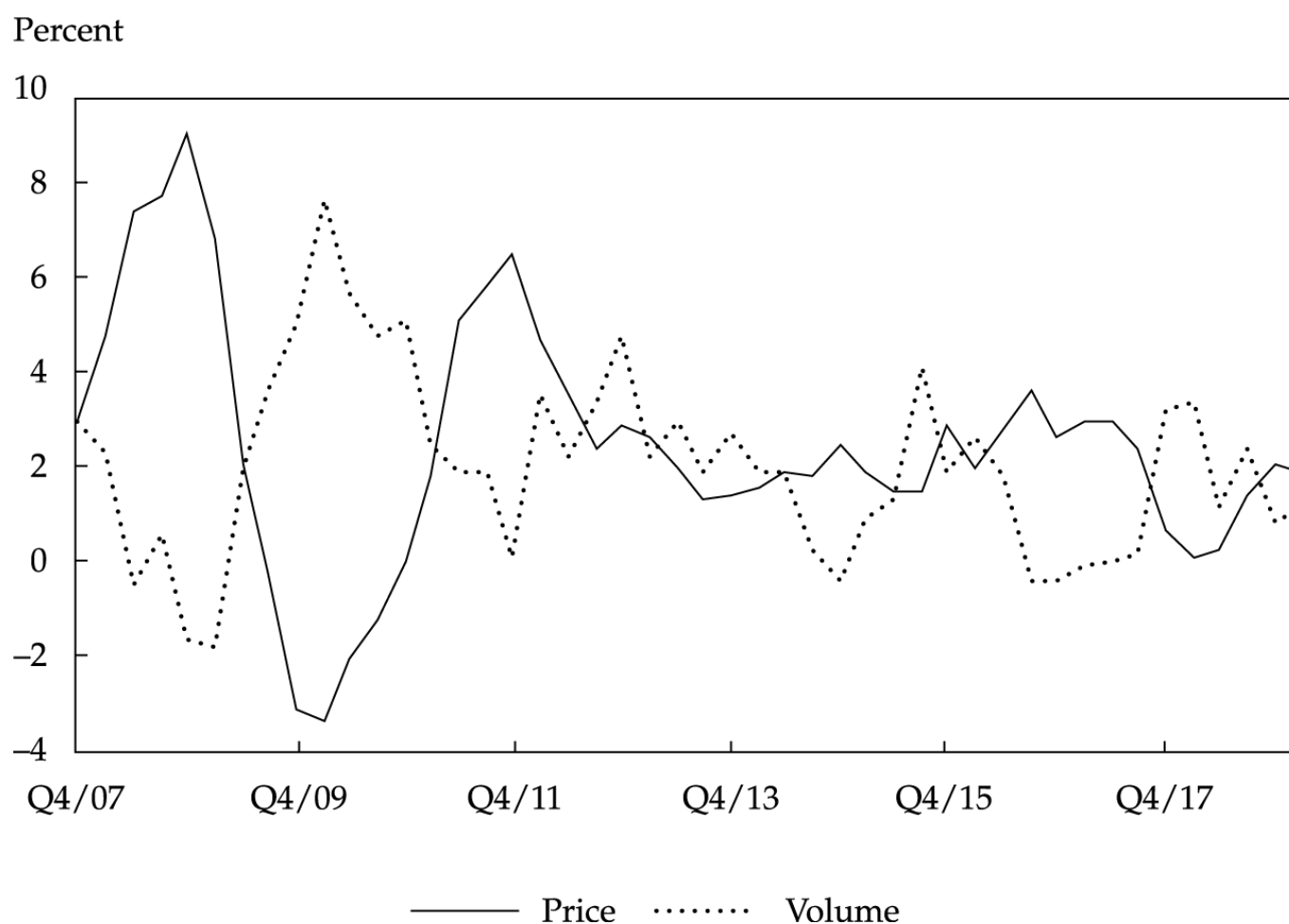


Source: US Bureau of Labor Statistics.

A company's efforts to pass on inflation through higher prices can have a negative impact on volume if the demand is price elastic, which is the case if cheaper substitutes are available. If selling prices could be increased 10 percent while maintaining unit sales volume to offset an increase of 10 percent in input costs, gross profit margin percentage would be the same but the absolute amount of gross profit would increase. In the short term, however, volumes will usually decline as result of a price increase. The decline would depend not only on the price elasticity of demand but also on the reaction of competitors and the availability of substitutes. Lower input costs also make lower consumer prices possible. The first competitor to lower prices will usually benefit with an uptick in volume. Competitors react quickly, however, resulting in a short-term benefit. The price-volume trade-off can make accurate revenue projections difficult. In an inflationary environment, raising prices too late will result in a profit margin squeeze but acting too soon could result in volume losses. In a deflationary environment, lowering prices too soon will result in a lower gross margin, but waiting too long will result in volume losses.

In the highly competitive consumer goods market, pricing is strongly influenced by movements in input prices, which can account for half of the COGS. In some time periods, customers' price sensitivity has resulted in a strong inverse relationship between volume and pricing. For example, [Exhibit 26](#) illustrates Unilever's annual underlying volume and price growth from 2001 to 2020. Increased input prices for packaging, wheat, and milk forced Anglo-Dutch consumer staple company Unilever to increase prices for its products significantly in 2008. Consequently, volumes deteriorated. But as raw material prices fell in 2009–2010, the company's prices were lowered and volumes recovered strongly. As the company started to increase prices in 2011, volume growth once again slowed. In 2016, the company faced challenging conditions in several emerging markets as currency-devaluation-led cost increases led to weaker volumes. Both volume and price growth have moderated to low-single digit growth rates, also exhibiting lower volatility.

Exhibit 26: Unilever Overall Revenue Growth by Percentage Change in Volume and Price



Sources: Unilever PLC filings.

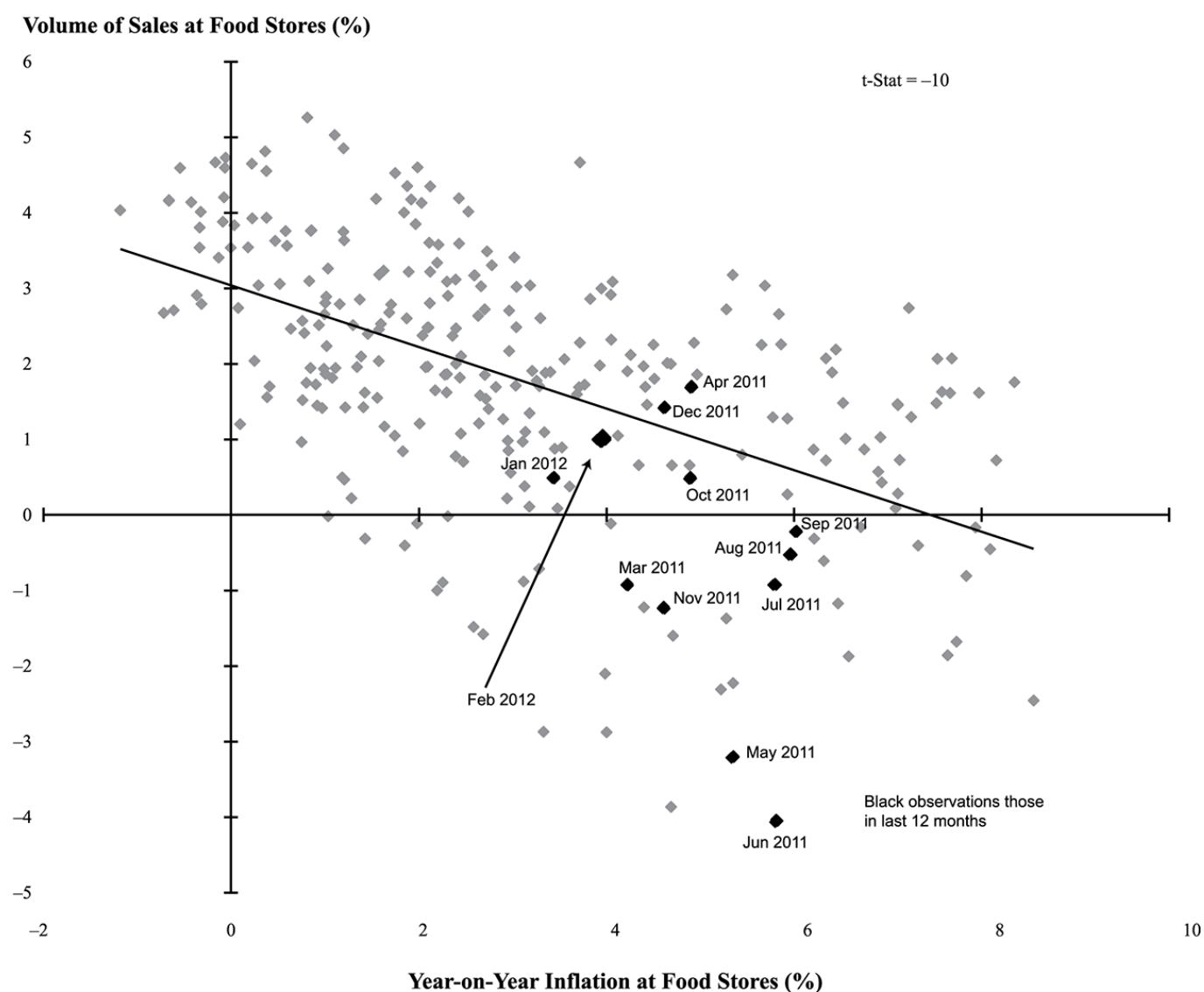
5.1.2. Company Sales and Inflation or Deflation

Revenue projections in a model are based on the expected volume and price development. Forecasting revenue for a company faced with inflation in input costs requires some understanding of the price elasticity of the products, the different rates of cost inflation in the countries where the company is active, and, if possible, the likely inflation in costs relevant to a company's individual product categories. Pricing strategy and market position are also important.

The impact of higher prices on volume depends on the price elasticity of demand (i.e., how the quantity demanded varies with price). If demand is relatively price inelastic, revenues will benefit from inflation. If demand is relatively price elastic (i.e., elasticity is greater than unit price elasticity), revenue can decline even if unit prices are raised. For example, a regression of volume on food inflation in UK food stores from 1989 to 2012 (shown in [Exhibit 27](#)) gives a regression slope coefficient of -0.398. (For every increase by 1 percentage point in year-on-year food prices, year-on-year sales decreased by approximately 0.4 percent.)

An analyst covering UK food retailers can use this information when building forecast profit models. By assuming an expected level of food inflation, volume growth can be estimated and revenue calculated.

Exhibit 27: UK Relationship between Food Inflation and Volume, January 1989–February 2012



Source: Based on data from Datastream. Analysis is the authors'.

The expected pricing component for an international company should consider the geographic mix of its revenues to reflect different rates of inflation among countries. Of course, strategy and competitive factors, in addition to inflation in input costs, play roles in price setting.

AB InBev's volume growth and pricing have been more robust in emerging markets, for example, thanks to strong demand for its new beer products. The impact of inflation is also an important factor. In its Latin America South division, which then mainly consisted of Argentina, the brewer reported strong 24.7 percent organic revenue growth in 2011, of which only 2.1 percent was driven by volume and the remainder by price. As costs increased in line with revenues, operating margin remained more or less stable, and organic operating profit growth was high at 27 percent. With only a limited negative currency impact, reported operating profit increased 24 percent in US dollars.

High inflation in a company's export market relative to a company's domestic inflation rate generally implies that the export country's currency will come under pressure and any pricing gain could be wiped out by the currency losses. The strong pricing increases AB InBev reported in its Latin America South division were clearly driven by input price inflation. The absence of a negative currency impact should be seen as a positive surprise but not as a typical outcome. A country's currency will usually come under pressure and depreciate if high rates of inflation persist for an extended period.

Most analysts adjust for recent high inflation in foreign countries by assuming a normalized growth rate for both revenues and costs after one or two years. This constant currency growth rate is based on an underlying growth rate assumption for the business. This approach can understate revenues in the short term. Other analysts reflect in their forecasts the high impact of inflation on revenues and expenses and adjust growth rates for the expected currency (interest rate parity) impact. This approach is also imperfect given the difficulty in projecting currency rates.

Identifying a company's major input costs provides an indication of likely pricing. For a specialist retail bakery chain, for example, the impact of increased grain prices will be more significant than for a diversified standard supermarket chain. Consequently, it seems logical that the bakery is likely to increase its prices by a higher percentage than the grocer in response to increased grain prices.

Company strategy is also an important factor. Faced with rising input prices, a company might decide to preserve its margins by passing on the costs to its customers, or it might decide to accept some margin reduction to increase its market share. In other words, the company could try to gain market share by not fully increasing prices to reflect increased costs. On the one hand, Sysco Company (the largest food distributor to restaurants and institutions in North America) has sometimes not passed on food price increases in recessionary conditions out of concern of not financially weakening already recession-affected customers (e.g., restaurants, private clubs, schools, nursing homes). On the other hand, in 2011 and 2012, the large French cognac houses substantially increased the prices of their products in China to reduce strong demand. Because older cognac generates a higher price, it can be more profitable to build an inventory of vintage cognac rather than maximizing short-term volumes.

EXAMPLE 8

Passing on Input Cost Increases or Not

Four food retail analysts are assessing the impact of a potential increase in input costs on the global supermarket chain Carrefour. In this hypothetical scenario, they believe that rising oil prices and packaging prices will affect many of the company's suppliers. They believe that Carrefour is likely to be confronted with 4 percent inflation in its COGS (with stable volume). The analysts have their own expectations about how the company will react. [Exhibit 28](#) shows Carrefour's 2020 results, and [Exhibit 29](#) shows the four analysts' estimates of input prices, volume growth, and pricing for the following year. Both exhibits are in the [Example 8](#) worksheet in the downloadable Microsoft Excel workbook.

Exhibit 28: Carrefour Data (euro millions, unless noted)

	2020
Total revenue	72,150
COGS	56,705
Gross profit	15,445
Gross margin	21.4%

Exhibit 29: Four Analysts' Estimates of Carrefour's Reaction to Inflation

	A	B	C	D
Price increase for revenues	0.00%	2.00%	3.00%	4.00%
Volume growth	5.00%	2.00%	1.00%	-4.00%
Total revenue growth	5.00%	4.04%	4.03%	-0.16%
Input costs increase	4.00%	4.00%	4.00%	4.00%

1. What are each analyst's predictions for gross profit and gross margin?

Solution:

The results for each analyst are shown in [Exhibit 30](#) and the [Example 8](#) worksheet in the downloadable Microsoft Excel workbook. For Analyst B, revenues increase 4 percent [= (1.02 × 1.02) – 1] and COGS 6.1 percent [= (1.02 × 1.04) – 1]. The difference between the calculated revenue and COGS is the new gross profit and gross margin is gross profit as a percentage of revenue.

Exhibit 30: Results for Analysts' Predictions (EUR millions, unless noted)

	2020	Analyst A 2021E	YoY%	Analyst B 2021E	YoY%	Analyst C 2021E	YoY%	Analyst D 2021E	YoY%
Total revenue	72,150	75,758	5.0%	75,065	4%	75,058	4.0%	72,035	-0.2%
COGS	56,705	61,922	9.2%	60,153	6%	59,563	5.0%	56,614	-0.2%
Gross profit	15,445	13,836	-10%	14,912	-3%	15,495	0%	15,420	-0.2%
Gross margin	21.4%	18.3%		19.9%		20.6%		21.4%	

2. Which analyst has the highest forecast for gross margin?

Solution:

The highest gross margin is projected by Analyst D, who assumes that selling prices would increase by 4 percent to offset rising input costs and keep gross margin stable from the 2020 level.

3. Which analyst has the highest forecast for gross profit?

Solution:

The highest gross profit is projected by Analyst D.

5.2. Cost Projections with Inflation and Deflation

The following analysis addresses the forecasting of industry and company costs in the presence of inflation and deflation.

5.2.1. Industry Costs and Inflation or Deflation

Familiarity with the specific purchasing characteristics of an industry can also be useful in forecasting costs. For example, long-term price-fixed forward contracts and hedges can delay the impact of price increases. Thus, an analyst forecasting costs for an industry in which companies customarily use such purchasing practices would incorporate any expected input price fluctuations more slowly than they would for an industry in which the participants do not use long-term contracts or hedges.

Monitoring the underlying drivers of input prices can also be useful in forecasting costs. For example, weather conditions can have a dramatic impact on the price of agricultural products and consequently on the cost base of industries that rely on them. An analyst observing a particular weather pattern might thus be able to incorporate this information into forecasts of costs.

How inflation or deflation affects an industry's cost structure depends on its competitive environment. For example, if the participants within the industry have access to alternative inputs or are vertically integrated, the impact of volatility in input costs can be mitigated. Jacobs Douwe Egberts (JDE) is a coffee company that has been facing high and volatile coffee prices. However, its coffee is a blend of different kinds of beans. By shifting the mix slightly, JDE can keep both taste and costs constant by reducing the amount of the more expensive types of coffee beans in the blend. But if all supplier countries significantly increase the price of coffee simultaneously, JDE cannot use blending as an offset and will be confronted with overall higher input costs. To sustain its profitability, JDE will have to increase its prices to its clients. But if competition from other companies, such as Nestlé (Nespresso, Dolce Gusto, Nescafé) makes it difficult to increase prices, JDE will have to look for alternatives if it wants to keep its profit margins stable. An easy solution for the short term could be reducing advertising and promotional (A&P) spending, which usually improves profit. For the longer term, however, it could be harmful for revenues because the company's brand position could be weakened.

For example, in 2010, Russia experienced a heat wave that destroyed large parts of its grain harvest, causing prices for malting barley, a major input for beer, to increase significantly. Carlsberg, as the largest Russian brewer at that time, was particularly hard hit because it had to pay more for its Russian barley and also needed to import grain into the country, incurring additional transportation costs. By increasing imports from Western Europe, Carlsberg also pushed up barley prices in this region, affecting the cost base of other Western European brewers.

5.2.2. Company Costs and Inflation or Deflation

In forecasting a company's costs, it is often helpful to segment the cost structure by category and geography. For each item of cost, an assessment should be made about the impact of potential inflation and deflation on input prices. This assessment should take into account the company's ability to substitute cheaper alternatives for expensive inputs or to increase efficiency to offset the impact of increases in input prices. For example, although a jump in raw material prices in 2011 caused Unilever's and Nestlé's gross margins to fall sharply (by 110–170 bps), increases in operational efficiencies, such as reducing advertising spending, enabled both companies to achieve slightly higher overall operating profit margins that year. [Example 9](#) shows the use of common size (percent-of-sales) analysis of inflation in input costs.

EXAMPLE 9

Inflation in Input Costs

Two fictional consumer staple companies—chocolate and sweets specialist “Choco A” and a food producer “Sweet B”—have costs that are constantly affected by inflation and deflation. [Exhibit 31](#) (see the [Example 9](#) worksheet in the downloadable Microsoft Excel workbook) presents a common size analysis.

Exhibit 31: Common Size Analysis for Sweet B and Choco A

	Sweet B	Choco A
Net sales	100%	100%
COGS	50%	36%
Gross margin	50%	64%
SG&A	31%	47%
Depreciation	3%	4%

	Sweet B	Choco A
EBIT	16%	13%
Raw materials	22%	22%
Packaging	12%	10%
Other COGS	16%	4%
Total COGS	50%	36%

Assume inflation of 10 percent for all costs (except depreciation) and that the companies are not able to pass on this increase through higher prices (total revenues will remain constant).

1. Calculate the gross profit margin for each company. Which company will experience the greater reduction in gross profit margin?

Solution:

The company with the higher COGS as a percent of net sales—equivalently, the lower gross margin—will experience the greater negative impact. Sweet B has a lower gross margin than Choco A: 50 percent compared with 64 percent, as shown in [Exhibit 31](#). After the 10 percent increase in COGS to $1.10 \times 50\% = 55\%$, Sweet B's gross margin will fall to 45 percent, as shown in [Exhibit 32](#). Sweet B's resulting gross margin of 45 percent represents a proportional decline of 10 percent from the initial value of 50 percent. In contrast, the proportional decline in Choco A's gross margin is approximately $4\%/64\% = 6\%$.

Exhibit 32: Effect of Cost Inflation

	All Costs (Except Depreciation) + 10%		Raw Materials + 10%	
	Sweet B	Choco A	Sweet B	Choco A
Net sales	100%	100%	100%	100%
COGS	55%	40%	52%	38%
Gross margin	45%	60%	48%	62%
SG&A	34%	52%	31%	47%
Depreciation	3%	4%	3%	4%
EBIT	8%	5%	14%	11%

2. Calculate the operating profit margin for each company. Which company will experience the greater reduction in operating profit (EBIT) margin?

Solution:

Choco A has higher overall costs than Sweet B, primarily as a consequence of its high SG&A expenses. Choco A's operating profit margin will drop to approximately 5 percent, as shown in [Exhibit 32](#), representing a proportional decline of approximately 62 percent compared with a proportional decline of approximately $8\%/16\% = 50\%$ for Sweet B.

3. Assume inflation of 10 percent only for the raw material costs (reflected in COGS) and that the companies are not able to pass on this increase through higher prices. Which company will be more affected negatively in terms of gross profit margin and operating profit margin?

Solution:

The company with the higher raw material expense component will experience the more negative effect. In this case, raw materials represent 22 percent of net sales for both Sweet B and Choco A. Gross margin and operating margin will decline by 220 bps for both. This impact is more severe on gross margin on a relative basis for Sweet B ($2.2\%/50\% = 4.4\%$ decline) than for Choco A ($2.2\%/64\% = 3.4\%$ decline). But the relative effect on operating margin will be more severe for Choco A ($2.2\%/13\% = 16.9\%$ decline) than for Sweet B ($2.2\%/16\% = 13.8\%$).

6. THE FORECAST HORIZON AND LONG-TERM FORECASTING

LEARNING OUTCOME

The candidate should be able to:

- explain considerations in the choice of an explicit forecast horizon and an analyst’s choices in developing projections beyond the short-term forecast horizon

The choice of the forecast time horizon can be influenced by certain factors, including the investment strategy for which the security is being considered, the cyclical­ity of the industry, company-specific factors, and the analyst’s employer’s preferences. Most professionally managed investment strategies describe the investment time frame, or average holding period, in the stated investment objectives of the strategy; the time frame should ideally correspond with average annual turnover of the portfolio. For example, a stated investment time horizon of three to five years would imply average annual portfolio turnover between 20 percent and 33 percent (average holding period is calculated as one/portfolio turnover). The cyclical­ity of the industry could also influence the analyst’s choice of time frame because the forecast period should be long enough to allow the business to reach an expected mid-cycle level of sales and profitability. Similar to cyclical­ity, various company-specific factors, including recent acquisition or restructuring activity, can influence the selection of the forecast period to allow enough time for the realization of the expected benefits from such activity to be reflected in the financial statements. In other cases, there might be no individual analyst choice in the sense that the analyst’s employer has specified more or less fixed parameters. Much of the discussion so far has focused on various methods of forecasting a company’s income statement, balance sheet, and cash flow for an explicit short-term forecast period. Although the underlying principles remain the same if one extends the time horizon, certain considerations and choices are available to the analyst when developing longer-term projections.

Longer-term projections often provide a better representation of the normalized earnings potential of a company than a short-term forecast, especially when certain temporary factors are present. **Normalized earnings** are the expected level of mid-cycle earnings for a company in the absence of any unusual or temporary factors that affect profitability (either positively or negatively). For example, at any given point in time, a company’s profitability can be influenced by a number of temporary factors, including the stage in the business cycle, recent merger and acquisition activity, and restructuring activity. Similarly, normalized free cash flow can be defined as the expected level of mid-cycle cash flow from operations adjusted for unusual items just described less recurring capital expenditures. By extending the forecast period, an analyst is able to adjust for these unusual or temporary factors and derive an estimate of earnings that the company is likely to earn in a normal year. We will consider various alternatives for two aspects of long-term forecasting: revenue forecasts and terminal value.

As with most income statement projections, a long-term forecast begins with a revenue projection, with most of the remaining income statement items subsequently derived from the level or change in revenue. Revenue projection methods were covered earlier.

6.1. Case Study: Estimating Normalized Revenue

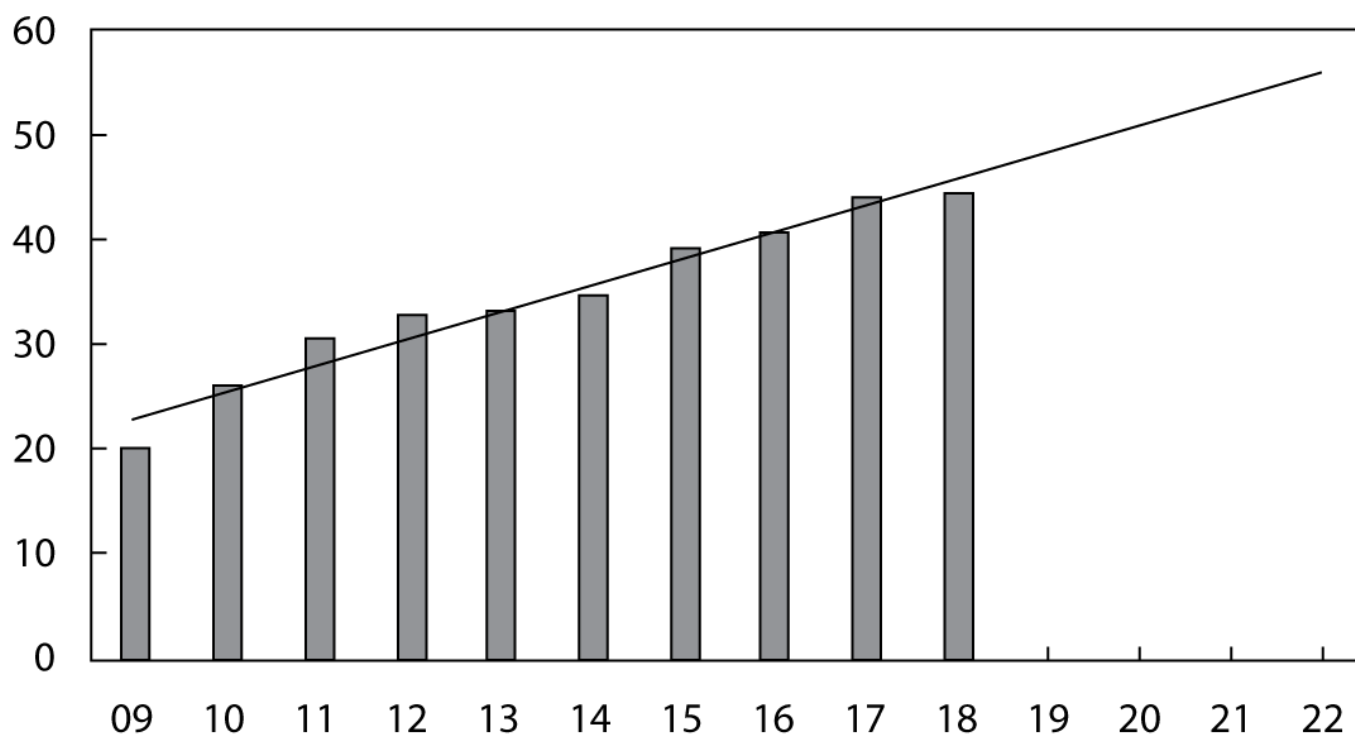
Exhibit 33 contains 10 years of historical revenue data and four years of estimated normalized data for Continental AG, a global automotive supplier. The accompanying bar chart in **Exhibit 34** graphically depicts the data and includes a trend line based on a linear regression of the data. The numerical values for each point along the trend line can be found by using the TREND formula in Microsoft Excel. The TREND formula uses observations on the dependent variable (in this case revenue) and observations on the explanatory (time) variable to perform a linear regression by using least squares criterion to find the best fit. After computing the best fit regression model, the TREND formula returns predicted values associated with new points in time. The worksheet for **Exhibit 33** and **Exhibit 34** in the downloadable Microsoft Excel workbook demonstrates the calculations used in the exhibits.

Exhibit 33: Historical and Estimated Revenue Data for Continental AG, 2011–2024E (euro billions)

	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024
Revenue	30.5	32.7	33.3	34.5	39.2	40.6	44.0	44.4	44.5	37.7				
Normalized revenue	31.8	33.2	34.6	36.0	37.4	38.9	40.3	41.7	43.1	44.5	45.9	47.3	48.7	50.1
Percent above/below trend	– 4.1%	– 1.4%	– 3.7%	– 4.2%	4.8%	4.4%	9.3%	– 6.6%	3.3%	– 15.2%				

Exhibit 34: Historical and Estimated Revenue for Continental AG, 2011–2024E

Revenues (€bn)



The “growth relative to GDP growth” and “market growth and market share” methods discussed earlier can also be applied to developing longer-term projections. Once a revenue projection has been established, previously described methods of forecasting costs can be used to complete the income statement, balance sheet, and cash flow statement.

If an analyst is creating a valuation model such as a DCF model, estimating a terminal value is required to capture the going-concern value of the company after the explicit forecast period. Certain considerations should be kept in mind when deriving the terminal value based on long-term projections.

First, an analyst should consider whether the terminal year free cash flow projection should be normalized before that cash flow is incorporated into a long-term projection. For example, if the explicitly forecasted terminal year free cash flow is “low” (e.g., because of business cycle reasons or capital investment projects), an adjustment to normalize the amount might be warranted. Second, an analyst should consider whether and how the future long-term growth rate will differ from the historical growth rate. For example, even some mature companies might be able to accelerate their long-term growth rate through product innovation and/or market expansion (e.g., Apple), whereas other seemingly well-protected “growers” could experience an unanticipated decline in their business as a result of technological change (e.g., Eastman Kodak Company, a global commercial printing and imaging company).

One of the greatest challenges facing the analyst is anticipating inflection points, when the future will look significantly different from the recent past. Most DCF models rely on a perpetuity calculation, which assumes that the cash flows from the last year of an explicit forecast grow at a constant rate forever. Because the perpetuity can account for a relatively large portion of the overall valuation of the company, it is critical that the cash flow used is representative of a “normalized” or “mid-cycle” result. If the analyst is examining a cyclical company, using a boom year as the starting point for the perpetuity could result in a grossly overstated valuation. Similarly, using a trough year could result in a valuation that is much too low.

Another important consideration is economic disruption. The economy can occasionally experience sudden, unprecedented changes that affect a wide variety of companies, such as the 2008 global financial crisis or the COVID-19 pandemic. Even a company with a sound strategy and solid operations can be thrown far off course by a sudden economic disruption, particularly if the company has a high degree of financial leverage.

Regulation and technology are also potential drivers of inflection points, and it is important for the analyst to keep a close eye on both. Government actions can have extreme, sudden, and unpredictable impacts on some businesses. Technological advances can turn fast-growing innovators obsolete in a matter of months. Both regulation and technology affect some industries more than others. Utilities experience intense regulation but might not see a significant technological change for decades. Semiconductor manufacturers must constantly keep up with new technology but experience relatively light regulation. Pharmaceutical manufacturers are heavily exposed to both regulation and technological advances.

Finally, long-term growth is a key input in the perpetuity calculation. Some companies and industries can grow faster than the overall economy for long periods of time, causing them to account for an increasing share of overall output. Examples include some technology companies, such as Tencent, Amazon, and Google. Other companies, such as those in the print media sector, are likely to grow slower than the overall economy or even shrink over time. Using an unrealistic long-term growth rate can put the analyst’s valuation far off the mark.

EXAMPLE 10

Important Considerations When Making Assumptions

1. Turkish Airlines (THYAO.IS) operates in the highly cyclical global airline industry. Operating margins for 2011–2019 are shown in the following table and in the [Example 10](#) worksheet in the downloadable Microsoft Excel workbook.

	2011	2012	2013	2014	2015	2016	2017	2018	2019
Operating margin	1.0%	10.8%	6.5%	5.6%	8.6%	–2.9%	9.0%	9.9%	7.9%

On the basis of only the information in the table, which of the following operating margins would *most likely* be appropriate to use in a perpetuity calculation for Turkish Airlines to arrive at a reasonable intrinsic value estimate?

- A. 6.0 percent
- B. 9.0 percent
- C. 9.9 percent

Solution:

A is correct. Because the airline industry is cyclical, an estimate of “mid-cycle” or “normalized” operating margin is necessary to estimate a perpetuity value. The nine-year average operating margin was 6.3 percent.

For each of the companies in the following problems, indicate which of the choices is *least likely* to cause a change in the company’s outlook.

2. ABC Diesel (hypothetical company), a manufacturer of diesel-power trucks

- A. Consumers have started switching to trucks with electric engines, threatening ABC’s historic strength in diesel engine trucks.
- B. ABC Diesel has formed a partnership with Electrico (hypothetical), a company involved in research and innovation in electric engines.
- C. Environmental regulations have been getting tighter in most regions, and consistent with prior experience, this need to make the engines less polluting is expected to continue over the next several years.

Solution:

C is correct. Although it is important that environmental regulations have been getting stricter, this is consistent with past experience and so does not represent a turning point.

3. Abbott Laboratories, a diversified manufacturer of health care products, including medical devices

- A. Management reiterates its long-standing approach to capital deployment.
- B. A competitor has demonstrated favorable efficacy data on a medical device candidate that will compete with an important Abbott product.
- C. It has become more difficult for medical device manufacturers to receive regulatory approval for new products because of heightened safety concerns.

Solution:

A is correct. Management is sticking with its historical approach to capital deployment, so this does not represent a turning point.

4. Grupo Aeroportuario del Sureste, operator of nine airports in Mexico, especially in the tourist-heavy southeast

- A. Global economic disruption has caused a sharp decline in international travel.
- B. Regulators will allow the construction of a new airport by a competitor in Grupo Aeroportuario del Sureste’s service territory.
- C. A technological advance will allow airlines to save 5 percent on fuel costs, but it is not expected to meaningfully alter passenger volumes. Similar developments in the past have benefited airlines but not airports, whose price per passenger is regulated.

Solution:

C is correct. Although the technological advance is good for the airlines, it will not have a meaningful effect on passenger volumes, which will likely prevent the airports from sharing in that benefit. In contrast, both A and B could have a significant impact on the long-run earnings power of Mexican airports.

5. LinkedIn, operator of an online social network for professionals and part of Microsoft Corporation, with limited investment needs and no debt

- A. Facebook, another online social network, announces a plan to enhance its offerings in the professional category.

B. Regulators announce an investigation of LinkedIn's privacy practices, which could result in significant changes to the service.

C. The US Federal Reserve has just increased interest rates. Although this will raise borrowing costs, the rate increase is not expected to have a negative impact on the economy.

Solution:

C is correct. Because LinkedIn carries no debt, it is unlikely that higher interest rates will cause a change in the company's outlook.

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PRACTICE PROBLEMS

The following information relates to questions 1-7

Nigel French, an analyst at Taurus Investment Management, is analyzing Archway Technologies, a manufacturer of luxury electronic auto equipment, at the request of his supervisor, Lukas Wright. French is asked to evaluate Archway's profitability over the past five years relative to its two main competitors, which are located in different countries with significantly different tax structures.

French begins by assessing Archway's competitive position within the luxury electronic auto equipment industry using Porter's five forces framework. A summary of French's industry analysis is presented in Exhibit 1.

Exhibit 1: Analysis of Luxury Electronic Auto Equipment Industry Using Porter's Five Forces Framework

Force	Factors to Consider
Threat of substitutes	Customer switching costs are high
Rivalry	Archway holds 60 percent of world market share; each of its two main competitors holds 15 percent
Bargaining power of suppliers	Primary inputs are considered basic commodities, and there are a large number of suppliers
Bargaining power of buyers	Luxury electronic auto equipment is very specialized (non-standardized)
Threat of new entrants	High fixed costs to enter industry

French notes that for the year just ended (2019), Archway's COGS was 30 percent of sales. To forecast Archway's income statement for 2020, French assumes that all companies in the industry will experience an inflation rate of 8 percent on the COGS. Exhibit 2 shows French's forecasts relating to Archway's price and volume changes.

Exhibit 2: Archway's 2020 Forecasted Price and Volume Changes

Average price increase per unit	5.00%
Volume growth	-3.00%

After putting together income statement projections for Archway, French forecasts Archway's balance sheet items. He uses Archway's historical efficiency ratios to forecast the company's working capital accounts.

Based on his financial forecast for Archway, French estimates a terminal value using a valuation multiple based on the company's average price-to-earnings multiple (P/E) over the past five years. Wright

discusses with French how the terminal value estimate is sensitive to key assumptions about the company's future prospects. Wright asks French:

"What change in the calculation of the terminal value would you make if a technological development that would adversely affect Archway was forecast to occur sometime beyond your financial forecast horizon?"

1. Which profitability metric should French use to assess Archway's five-year historic performance relative to its competitors?
 - A. Current ratio
 - B. Operating margin
 - C. Return on invested capital
2. Based on the current competitive landscape presented in Exhibit 1, French should conclude that Archway's ability to:
 - A. pass along price increases is high.
 - B. demand lower input prices from suppliers is low.
 - C. generate above-average returns on invested capital is low.
3. Based on the current competitive landscape presented in Exhibit 1, Archway's operating profit margins over the forecast horizon are *least likely* to:
 - A. decrease.
 - B. remain constant.
 - C. increase.
4. Based on Exhibit 2, Archway's forecasted gross profit margin for 2020 is *closest* to:
 - A. 62.7 percent.
 - B. 67.0 percent.
 - C. 69.1 percent.
5. French's approach to forecasting Archway's working capital accounts would be *most likely* classified as a:
 - A. hybrid approach.
 - B. top-down approach.
 - C. bottom-up approach.
6. The *most appropriate* response to Wright's question about the technological development is to:
 - A. increase the required return.
 - B. decrease the perpetual growth rate.
 - C. decrease the price-to-earnings multiple.
7. If the luxury electronic auto equipment industry is subject to rapid technological changes and market share shifts, how should French *best* adapt his approach to modeling?
 - A. Examine base rates
 - B. Forecast multiple scenarios

The following information relates to questions 8-14

Gertrude Fromm is a transportation sector analyst at Tucana Investments. She is conducting an analysis of Omikroon, N.V., a hypothetical European engineering company that manufactures and sells scooters and commercial trucks.

Omikroon's petrol scooter division is the market leader in its sector and has two competitors. Omikroon's petrol scooters have a strong brand name and a well-established distribution network. Given the strong branding established by the market leaders, the cost of entering the industry is high. But Fromm anticipates that small, inexpensive, imported petrol-fueled motorcycles could become substitutes for Omikroon's petrol scooters.

Fromm uses ROIC as the metric to assess Omikroon's performance.

Omikroon has just introduced the first electric scooter to the market at year-end 2019. The company's expectations are as follows:

- Competing electric scooters will reach the market in 2021.
- Electric scooters will not be a substitute for petrol scooters.
- The important research costs in 2020 and 2021 will lead to more efficient electric scooters.

Fromm decides to use a five-year forecast horizon for Omikroon after considering the following three factors:

Factor 1

The annual portfolio turnover at Tucana Investments is 30 percent.

Factor 2

The electronic scooter industry is expected to grow rapidly over the next 10 years.

Factor 3

Omikroon has announced it would acquire a light truck manufacturer that will be fully integrated into its truck division by 2021 and will add 2 percent to the company's total revenues.

Fromm uses the base case forecast for 2020 shown in Exhibit 1 to perform the following sensitivity analysis:

- The price of an imported specialty metal used for engine parts increases by 20 percent.
- This metal constitutes 4 percent of Omikroon's cost of sales.
- Omikroon will not be able to pass on the higher metal expense to its customers.

Exhibit 1: Omikroon's Selected Financial Forecasts for 2020 Base Case (euro millions)

	Petrol Scooter Division	Commercial Truck Division	Electric Scooter Division	Total
Sales	99.05	45.71	7.62	152.38
Cost of sales				105.38
Gross profit				47.00

	Petrol Scooter Division	Commercial Truck Division	Electric Scooter Division	Total
Operating profit				9.20

Omikroon will initially outsource its electric scooter parts. But manufacturing these parts in-house beginning in 2021 will imply changes to an existing factory. This factory cost EUR7 million three years ago and had an estimated useful life of 10 years. Fromm is evaluating two scenarios:

Scenario 1

Refit the existing factory for EUR27 million.

Scenario 2

Sell the existing factory for EUR5 million. Build a new factory costing EUR30 million with a useful life of 10 years.

8. Using Porter's five forces analysis, which of the following competitive factors is *most likely* to have the greatest impact on Omikroon's petrol scooter pricing power?
 - A. Rivalry
 - B. Threat of substitutes
 - C. Threat of new entrants
9. The metric used by Fromm to assess Omikroon's performance incorporates:
 - A. the degree of financial leverage.
 - B. operating liabilities relative to operating assets.
 - C. the firm's competitiveness relative to companies in other tax regimes.
10. Based on Omikroon's expectations, the gross profit margin of Omikroon's electric scooter division in 2021 is *most likely* to be affected by:
 - A. competition.
 - B. research costs.
 - C. cannibalization by petrol scooters.
11. Which factor *best* justifies the five-year forecast horizon for Omikroon selected by Fromm?
 - A. Factor 1
 - B. Factor 2
 - C. Factor 3
12. Fromm's sensitivity analysis will result in a decrease in the 2020 base case gross profit margin *closest to*:
 - A. 0.55 percent.
 - B. 0.80 percent.
 - C. 3.32 percent.

13. Fromm's estimate of growth capital expenditures included in Omikroon's PP&E under Scenario 1 should be:
- A. lower than under Scenario 2.
 - B. the same as under Scenario 2.
 - C. higher than under Scenario 2.
14. To validate the forecast for rapid growth in the electronic scooter market over the next 10 years, Fromm speaks to the management of Omikroon and investor relations of ZeroWheel, a competitor. Which behavioral bias is Fromm *most likely* subject to?
- A. Confirmation
 - B. Conservatism
 - C. Overconfidence
-

The following information relates to questions 15-21

Angela Green, an investment manager at Horizon Investments, intends to hire a new investment analyst. After conducting initial interviews, Green has narrowed the pool to three candidates. She plans to conduct second interviews to further assess the candidates' knowledge of industry and company analysis.

Prior to the second interviews, Green asks the candidates to analyze Chrome Network Systems, a company that manufactures internet networking products. Each candidate is provided Chrome's financial information presented in Exhibit 1.

Exhibit 1: Chrome Network Systems Selected Financial Information (US dollar millions)

	Year-End		
	2017	2018	2019
Net sales	46.8	50.5	53.9
Cost of sales	18.2	18.4	18.8
Gross profit	28.6	32.1	35.1
SG&A expenses	19.3	22.5	25.1
Operating income	9.3	9.6	10.0
Interest expense	0.5	0.7	0.6
Income before provision for income tax	8.8	8.9	9.4
Provision for income taxes	2.8	2.8	3.1
Net income	6.0	6.1	6.3

Green asks each candidate to forecast the 2020 income statement for Chrome and to outline the key assumptions used in their analysis. The job candidates are told to include Horizon's economic outlook for 2020 in their analysis, which assumes nominal GDP growth of 3.6 percent, based on expectations of real GDP growth of 1.6 percent and inflation of 2.0 percent.

Green receives the models from each of the candidates and schedules second interviews. To prepare for the interviews, Green compiles a summary of the candidates' key assumptions in Exhibit 2.

Exhibit 2: Summary of Key Assumptions Used in Candidates' Models

Metric	Candidate A	Candidate B	Candidate C
Net sales	Net sales will grow at the average annual growth rate in net sales over the 2017–19 time period.	Industry sales will grow at the same rate as nominal GDP, but Chrome will have a two-percentage-point decline in market share.	Net sales will grow 50 bps slower than nominal GDP.
Cost of sales	The 2020 gross margin will be the same as the average annual gross margin over the 2017–19 time period.	The 2020 gross margin will decline as costs increase by expected inflation.	The 2020 gross margin will increase by 20 bps from 2019.
SG&A expenses	The 2020 SG&A/net sales ratio will be the same as the average ratio over the 2017–19 time period.	The 2020 SG&A will grow at the rate of inflation.	The 2020 SG&A/net sales ratio will be the same as the 2019 ratio.
Interest expense	The 2020 interest expense assumes the effective interest rate will be the same as the 2019 rate.	The 2020 interest expense will be the same as the 2019 interest expense.	The 2020 interest expense will be the same as the average expense over the 2017–19 time period.
Income taxes	The 2020 effective tax rate will be the same as the 2019 rate.	The 2020 effective tax rate will equal the blended statutory rate of 30%.	The 2020 effective tax rate will be the same as the average effective tax rate over the 2017–19 time period.

15. Based on Exhibit 1, which of the following provides the strongest evidence that Chrome displays economies of scale?

- A. Increasing net sales
- B. Profit margins that are increasing with net sales
- C. Gross profit margins that are increasing with net sales

16. Based on Exhibit 2, the job candidate *most likely* using a bottom-up approach to model net sales is:

- A. Candidate A.
- B. Candidate B.
- C. Candidate C.

17. Based on Exhibit 2, the modeling approach used by Candidate B to project future net sales is *most accurately* classified as a:

- A. hybrid approach.
- B. top-down approach.
- C. bottom-up approach.

18. Based on Exhibits 1 and 2, Candidate C's forecast for cost of sales in 2020 is *closest* to:

- A. USD18.3 million.
- B. USD18.9 million.

C. USD19.3 million.

19. Based on Exhibits 1 and 2, Candidate A's forecast for SG&A expenses in 2020 is *closest* to:

A. USD23.8 million.

B. USD25.5 million.

C. USD27.4 million.

20. Based on Exhibit 2, forecasted interest expense will reflect changes in Chrome's debt level under the forecast assumptions used by:

A. Candidate A.

B. Candidate B.

C. Candidate C.

21. Candidate B asks Green if she had additional information on Horizon's industry peers and competitors, to put the profitability estimates in a richer context. By asking for this additional information for their analysis, Candidate B is *most likely* seeking to mitigate which behavioral bias?

A. Conservatism

B. Base rate neglect

C. illusion of control

SOLUTIONS

1. B is correct. Operating (EBIT) margin is a pre-tax profitability measure that can be useful in the peer comparison of companies in countries with different tax structures. Archway's two main competitors are located in different countries with significantly different tax structures; therefore, a pre-tax measure is better than an after-tax measure, such as ROIC. The current ratio is a liquidity measure, not a profitability measure.
2. A is correct. Porter's five forces framework in Exhibit 1 describes an industry with high barriers to entry, high customer switching costs (suggesting a low threat of substitutes), and a specialized product (suggesting low bargaining power of buyers). Furthermore, the primary production inputs from the large group of suppliers are considered basic commodities (suggesting low bargaining power of suppliers). These favorable industry characteristics will likely enable Archway to pass along price increases and generate above-average returns on invested capital.
3. A is correct. The current favorable characteristics of the industry (high barriers to entry, low bargaining power of suppliers and buyers, low threat of substitutes), coupled with Archway's dominant market share position, will likely lead to Archway's profit margins being at least equal to or greater than current levels over the forecast horizon.
4. C is correct. The calculation of Archway's gross profit margin for 2020, which reflects the industry-wide 8% inflation on COGS, is calculated as follows:

Revenue growth	1.85%
COGS increase	4.76%
Forecasted revenue (Base revenue = 100)	101.85
Forecasted COGS (Base COGS = 30)	31.43
Forecasted gross profit	70.42
Forecasted gross profit margin	69.14%

$$\begin{aligned}\text{Revenue growth} &= (1 + \text{Price increase for revenue}) \times (1 + \text{Volume growth}) - 1 \\ &= (1.05) \times (0.97) - 1 \\ &= 1.85\%.\end{aligned}$$

$$\begin{aligned}\text{COGS increase} &= (1 + \text{Price increase for COGS}) \times (1 + \text{Volume growth}) - 1 \\ &= (1.08) \times (0.97) - 1 \\ &= 4.76\%.\end{aligned}$$

$$\begin{aligned}\text{Forecasted revenue} &= \text{Base revenue} \times \text{Revenue growth increase} \\ &= 100 \times 1.0185 \\ &= 101.85.\end{aligned}$$

$$\begin{aligned}\text{Forecasted COGS} &= \text{Base COGS} \times \text{COGS increase} \\ &= 30 \times 1.0476 \\ &= 31.43.\end{aligned}$$

$$\begin{aligned}\text{Forecasted gross profit} &= \text{Forecasted revenue} - \text{Forecasted COGS} \\ &= 101.85 - 31.43\end{aligned}$$

= 70.42.

Forecasted gross profit margin = Forecasted gross profit/Forecasted revenue

= 70.42/101.85

= 69.14%.

5. C is correct. French is using a bottom-up approach to forecast Archway's working capital accounts by using the company's historical efficiency ratios to project future performance.
 6. C is correct. If the future growth or profitability of a company is likely to be lower than the historical average (in this case, because of a potential technological development), then the target multiple should reflect a discount to the historical multiple to reflect this difference in growth and/or profitability. If a multiple is used to derive the terminal value of a company, the choice of the multiple should be consistent with the long-run expectations for growth and required return. French tells Wright he believes that such a technological development could have an adverse impact on Archway beyond the forecast horizon.
 7. B is correct. Forecasting a single scenario would not be appropriate given the high degree of uncertainty and range of potential outcomes for companies in this industry.
 8. B is correct. Small, inexpensive, imported petrol-fueled motorcycles are substitutes for petrol scooters and could increasingly have an impact on Omikroon's petrol scooter pricing power.
 9. B is correct. Return on invested capital is net operating profit minus adjusted taxes divided by invested capital, where invested capital is defined as operating assets minus operating liabilities.
 10. A is correct. Competition from other electric scooter manufacturers is expected to begin in one year. After this time, competing electric scooters could lead to lower demand for Omikroon's electric scooters and affect Omikroon's gross profit margin.
 11. B is correct. The electric scooter market is expected to grow rapidly, so the contribution of Omikroon's new electric scooter division is forecast to expand significantly over the next 10 years. A is incorrect because the investment company's portfolio turnover is not relevant for forecasting Omikroon's future results. C is incorrect because the light truck division is expected to add only 2% to total revenues in the future.
 12. A is correct. The sensitivity analysis consists of an increase of 20 percent in the price of an input that constitutes 4 percent of cost of sales. Change in gross profit margin because of that increase is calculated as the change in cost of sales because of price increase divided by sales:
$$= (\text{Cost of sales} \times 0.04 \times 0.2) / \text{Sales}$$
$$= (105.38 \times 0.04 \times 0.2) / 152.38$$
$$= 0.0055 \text{ or } 0.55\%$$
 13. C is correct. In Scenario 1, growth capital expenditures of EUR27 million for the refit of the existing idle factory is higher than the growth capital expenditures in Scenario 2 of EUR25 million. The EUR25 million is the cost of building a new factory for EUR30 million less the proceeds from the sale of the existing idle factory of EUR5 million.
 14. A is correct. The management of Omikroon and investor relations of ZeroWheel are almost certainly biased in favor of expecting strong growth for the markets they participate in. To evaluate the forecast, Fromm should seek more independent sources and balance the biased sources with sources biased in the opposite direction or an analyst who is more skeptical.
 15. C is correct. Economies of scale are a situation in which average costs decrease with increasing sales volume. Chrome's gross margins have been increasing with net sales. Gross margins that increase with sales levels provide evidence of economies of scale, assuming that higher levels of sales reflect increased unit sales. Gross margin more directly reflects the cost of sales than does profit margin.
-

Metric	2017	2018	2019
Net sales	\$46.8	\$50.5	\$53.9
Gross profit	28.6	32.1	35.1
Gross margin (gross profit/net sales)	61.11%	63.56%	65.12%

16. A is correct. A bottom-up approach for developing inputs to equity valuation models begins at the level of the individual company or a unit within the company. By modeling net sales using the average annual growth rate, Candidate A is using a bottom-up approach. B and C are incorrect because both Candidate B and Candidate C are using a top-down approach, which begins at the level of the overall economy.
17. B is correct. A top-down approach usually begins at the level of the overall economy. Candidate B assumes industry sales will grow at the same rate as nominal GDP but that Chrome will have a 2-percentage-point decline in market share. A and C are incorrect because Candidate B is not using any elements of a bottom-up approach; therefore, a hybrid approach is not being employed.
18. C is correct. Candidate C assumes that the 2020 gross margin will increase by 20 bps from 2019 and that net sales will grow at 50 bps slower than nominal GDP (nominal GDP = Real GDP + Inflation = 1.6% + 2.0% = 3.6%). Accordingly, the 2020 forecasted cost of sales is USD19.27 million, rounded to USD19.3 million.

Metric	Calculation	Result
2020 gross margin = 2019 gm + 20 bps	USD35.1/USD53.9 = 65.12% + 0.20% =	65.32%
2020 CoS/net sales = 100% – gross margin	100% – 65.32% =	34.68%
2020 net sales = 2019 net sales × (1 + Nominal GDP – 0.50%)	USD53.9 million × (1 + 0.036 – 0.005) = USD53.9 million × 1.031 =	USD55.57 million
2020 cost of sales = 2020 net sales × CoS/net sales	USD55.57 × 34.68% =	USD19.27 million

19. B is correct. Candidate A assumes that the 2020 SG&A/net sales will be the same as the average SG&A/net sales over the 2017–19 time period and that net sales will grow at the annual average growth rate in net sales over the 2017–19 time period. Accordingly, the 2020 forecasted SG&A expenses are USD25.5 million.

Metric	Calculation	Result
Average SG&A/net sales, 2017–2019*	(41.24% + 44.55% + 46.57%)/3 =	44.12%
Average annual growth sales in net sales, 2017–2019**	(7.91% + 6.73%)/2 =	7.32%
2020 net sales = 2019 net sales × (1 + Average annual growth rate in net sales)	USD53.9 million × 1.0732 =	\$57.85 million
2020 SG&A = 2020 net sales × Average SG&A/net sales	USD57.85 million × 44.12% =	\$25.52 million

	2017	2018	2019
Net Sales	USD46.8	USD50.5	USD53.9
SG&A expenses	USD19.3	USD22.5	USD25.1
SG&A-to-sales ratio	41.24%	44.55%	46.57%

Year	Calculation
2018	(USD50.5/USD46.8) – 1 = 7.91%
2019	(USD53.9/USD50.5) – 1 = 6.73%

- 20.** A is correct. In forecasting financing costs, such as interest expense, the debt/equity structure of a company is a key determinant. Accordingly, a method that recognizes the relationship between the income statement account (interest expense) and the balance sheet account (debt) would be a preferable method for forecasting interest expense when compared with methods that forecast based solely on the income statement account. By using the effective interest rate (interest expense divided by average gross debt), Candidate A is taking the debt/equity structure into account. B and C are incorrect because Candidate B (who forecasts 2020 interest expense to be the same as 2019 interest expense) and Candidate C (who forecasts 2020 interest expense to be the same as the 2017–19 average interest expense) are not taking the balance sheet into consideration.
- 21.** B is correct. Base rates refer to attributes of a reference class and base rate neglect is ignoring such class information in favor of specific information. By incorporating industry data, Candidate B is seeking to mitigate this behavioral bias.

Equity Investments

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Market Organization and Structure

by Larry Harris PhD, CFA

Larry Harris, PhD, CFA, is at the USC Marshall School of Business (USA).

LEARNING OUTCOMES

The candidate should be able to:

- ☐ explain the main functions of the financial system
- ☐ describe classifications of assets and markets
- ☐ describe the major types of securities, currencies, contracts, commodities, and real assets that trade in organized markets, including their distinguishing characteristics and major subtypes
- ☐ describe types of financial intermediaries and services that they provide
- ☐ compare positions an investor can take in an asset
- ☐ calculate and interpret the leverage ratio, the rate of return on a margin transaction, and the security price at which the investor would receive a margin call
- ☐ compare execution, validity, and clearing instructions
- ☐ compare market orders with limit orders
- ☐ define primary and secondary markets and explain how secondary markets support primary markets
- ☐ describe how securities, contracts, and currencies are traded in quote-driven, order-driven, and brokered markets
- ☐ describe characteristics of a well-functioning financial system
- ☐ describe objectives of market regulation

1. INTRODUCTION

Financial analysts gather and process information to make investment decisions, including those related to buying and selling assets. Generally, the decisions involve trading securities, currencies, contracts, commodities, and real assets such as real estate. Consider several examples:

- Fixed income analysts evaluate issuer credit-worthiness and macroeconomic prospects to determine which bonds and notes to buy or sell to preserve capital while obtaining a fair rate of return.
- Stock analysts study corporate values to determine which stocks to buy or sell to maximize the value of their stock portfolios.
- Corporate treasurers analyze exchange rates, interest rates, and credit conditions to determine which currencies to trade and which notes to buy or sell to have funds available in a needed currency.
- Risk managers work for producers or users of commodities to calculate how many commodity futures contracts to buy or sell to manage inventory risks.

Financial analysts must understand the characteristics of the markets in which their decisions will be executed. This reading, by examining those markets from the analyst's perspective, provides that understanding.

This reading is organized as follows. Section 2 examines the functions of the financial system. Section 3 introduces assets that investors, information-motivated traders, and risk managers use to advance their financial objectives and presents ways practitioners classify these assets into markets. These assets include such financial instruments as securities, currencies, and some contracts; certain commodities; and real assets. Financial analysts must know the distinctive characteristics of these trading assets.

Section 4 is an overview of financial intermediaries (entities that facilitate the functioning of the financial system). Section 5 discusses the positions that can be obtained while trading assets. You will learn about the benefits and risks of long and short positions, how these positions can be financed, and how the financing affects their risks. Section 6 discusses how market participants order trades and how markets process those orders. These processes must be understood to achieve trading objectives while controlling transaction costs.

Section 7 focuses on describing primary markets. Section 8 describes the structures of secondary markets in securities. Sections 9 and 10 close the reading with discussions of the characteristics of a well-functioning financial system and of how regulation helps make financial markets function better. A summary reviews the reading's major ideas and points, and practice problems conclude.

2. THE FUNCTIONS OF THE FINANCIAL SYSTEM

LEARNING OUTCOME

The candidate should be able to:

- ☐ **explain the main functions of the financial system**

The financial system includes markets and various financial intermediaries that help transfer financial assets, real assets, and financial risks in various forms from one entity to another, from one place to another, and from one point in time to another. These transfers take place whenever someone exchanges one asset or financial contract for another. The assets and contracts that people (people act on behalf of themselves, companies, charities, governments, etc., so the term "people" has a broad definition in this

reading) trade include notes, bonds, stocks, exchange-traded funds, currencies, forward contracts, futures contracts, option contracts, swap contracts, and certain commodities. When the buyer and seller voluntarily arrange their trades, as is usually the case, the buyer and the seller both expect to be better off.

People use the financial system for six main purposes:

1. to save money for the future;
2. to borrow money for current use;
3. to raise equity capital;
4. to manage risks;
5. to exchange assets for immediate and future deliveries; and
6. to trade on information.

The main functions of the financial system are to facilitate:

1. the achievement of the purposes for which people use the financial system;
2. the discovery of the rates of return that equate aggregate savings with aggregate borrowings; and
3. the allocation of capital to the best uses.

These functions are extremely important to economic welfare. In a well-functioning financial system, transaction costs are low, analysts can value savings and investments, and scarce capital resources are used well.

Sections 2.1 through 2.3 expand on these three functions. The six subsections of Section 2.1 cover the six main purposes for which people use the financial system and how the financial system facilitates the achievement of those purposes. Sections 2.2 and 2.3 discuss determining rates of return and capital allocation efficiency, respectively.

2.1. Helping People Achieve Their Purposes in Using the Financial System

People often arrange transactions to achieve more than one purpose when using the financial system. For example, an investor who buys the stock of an oil producer may do so to move her wealth from the present to the future, to hedge the risk that she will have to pay more for energy in the future, and to exploit insightful research that she conducted that suggests the company's stock is undervalued in the marketplace. If the investment proves to be successful, she will have saved money for the future, managed her energy risk exposure, and obtained a return on her research.

The separate discussions of each of the six main uses of the financial system by people will help you better identify the reasons why people trade. Your ability to identify the various uses of the financial system will help you avoid confusion that often leads to poor financial decisions. The financial intermediaries that are mentioned in these discussions are explained further in Section 4.

2.1.1. *Saving*

People often have money that they choose not to spend now and that they want available in the future. For example, workers who save for their retirements need to move some of their current earnings into the future. When they retire, they will use their savings to replace the wages that they will no longer be earning. Similarly, companies save money from their sales revenue so that they can pay vendors when their bills come due, repay debt, or acquire assets (for example, other companies or machinery) in the future.

To move money from the present to the future, savers buy notes, certificates of deposit, bonds, stocks, mutual funds, or real assets such as real estate. These alternatives generally provide a better expected rate of return than simply storing money. Savers then sell these assets in the future to fund their future expenditures. When savers commit money to earn a financial return, they commonly are called investors. They invest when they purchase assets, and they divest when they sell them.

Investors require a fair rate of return while their money is invested. The required fair rate of return compensates them for the use of their money and for the risk that they may lose money if the investment fails or if inflation reduces the real value of their investments.

The financial system facilitates savings when institutions create investment vehicles, such as bank deposits, notes, stocks, and mutual funds, that investors can acquire and sell without paying substantial transaction costs. When these instruments are fairly priced and easy to trade, investors will use them to save more.

2.1.2. *Borrowing*

People, companies, and governments often want to spend money now that they do not have. They can obtain money to fund projects that they wish to undertake now by borrowing it. Companies can also obtain funds by selling ownership or equity interests (covered in Section 2.1.3). Banks and other investors provide those requiring funds with money because they expect to be repaid with interest or because they expect to be compensated with future disbursements, such as dividends and capital gains, as the ownership interest appreciates in value.

People may borrow to pay for such items as vacations, homes, cars, or education. They generally borrow through mortgages and personal loans, or by using credit cards. People typically repay these loans with money they earn later.

Companies often require money to fund current operations or to engage in new capital projects. They may borrow the needed funds in a variety of ways, such as arranging a loan or a line of credit with a bank, or selling fixed income securities to investors. Companies typically repay their borrowing with income generated in the future. In addition to borrowing, companies may raise funds by selling ownership interests.

Governments may borrow money to pay salaries and other expenses, to fund projects, to provide welfare benefits to their citizens and residents, and to subsidize various activities. Governments borrow by selling bills, notes, or bonds. Governments repay their debt using future revenues from taxes and in some instances from the projects funded by these debts.

Borrowers can borrow from lenders only if the lenders believe that they will be repaid. If the lenders believe, however, that repayment in full with interest may not occur, they will demand higher rates of interest to cover their expected losses and to compensate them for the discomfit they experience wondering whether they will lose their money. To lower the costs of borrowing, borrowers often pledge assets as collateral for their loans. The assets pledged as collateral often include those that will be purchased by the proceeds of the loan. If the borrowers do not repay their loans, the lenders can sell the collateral and use the proceeds to settle the loans.

Lenders often will not loan to borrowers who intend to invest in risky projects, especially if the borrowers cannot pledge other collateral. Investors may still be willing to supply capital for these risky projects if they believe that the projects will likely produce valuable future cash flows. Rather than lending money, however, they will contribute capital in exchange for equity in the projects.

The financial system facilitates borrowing. Lenders aggregate from savers the funds that borrowers require. Borrowers must convince lenders that they can repay their loans, and that, in the event they cannot, lenders can recover most of the funds lent. Credit bureaus, credit rating agencies, and governments promote borrowing; credit bureaus and credit rating agencies do so by collecting and disseminating information that lenders need to analyze credit prospects and governments do so by establishing bankruptcy codes and courts that define and enforce the rights of borrowers and lenders. When the transaction costs of loans (i.e., the costs of arranging, monitoring, and collecting them) are low, borrowers can borrow more to fund current expenditures with credible promises to return the money in the future.

2.1.3. *Raising Equity Capital*

Companies often raise money for projects by selling (issuing) ownership interests (e.g., corporate common stock or partnership interests). Although these equity instruments legally represent ownership in companies rather than loans to the companies, selling equity to raise capital is simply another mechanism for moving money from the future to the present. When shareholders or partners contribute capital to a company, the company obtains money in the present in exchange for equity instruments that will be entitled to distributions in the future. Although the repayment of the money is not scheduled as it would be for loans, equity instruments also represent potential claims on money in the future.

The financial system facilitates raising equity capital. Investment banks help companies issue equities, analysts value the securities that companies sell, and regulatory reporting requirements and accounting standards attempt to ensure the production of meaningful financial disclosures. The financial system helps promote capital formation by producing the financial information needed to determine fair prices for equity. Liquid markets help companies raise capital. In these markets, shareholders can easily divest their equities as desired. When investors can easily value and trade equities, they are more willing to fund reasonable projects that companies wish to undertake.

EXAMPLE 1

Financing Capital Projects

As a chief financial officer (CFO) of a large industrial firm, you need to raise cash within a few months to pay for a project to expand existing and acquire new manufacturing facilities. What are the primary options available to you?

Solution:

Your primary options are to borrow the funds or to raise the funds by selling ownership interests. If the company borrows the funds, you may have the company pledge some or all of the project as collateral to reduce the cost of borrowing.

2.1.4. *Managing Risks*

Many people, companies, and governments face financial risks that concern them. These risks include default risk and the risk of changes in interest rates, exchange rates, raw material prices, and sale prices, among many other risks. These risks are often managed by trading contracts that serve as hedges for the risks.

For example, a farmer and a food processor both face risks related to the price of grain. The farmer fears that prices will be lower than expected when his grain is ready for sale whereas the food processor fears that prices will be higher than expected when she has to buy grain in the future. They both can eliminate their exposures to these risks if they enter into a binding forward contract for the farmer to sell a specified quantity of grain to the food processor at a future date at a mutually agreed upon price. By entering into a forward contract that sets the future trade price, they both eliminate their exposure to changing grain prices.

In general, hedgers trade to offset or insure against risks that concern them. In addition to forward contracts, they may use futures contracts, option contracts, or insurance contracts to transfer risk to other entities more willing to bear the risks (these contracts will be covered in Section 3.4). Often the hedger and the other entity face exactly the opposite risks, so the transfer makes both more secure, as in the grain example.

The financial system facilitates risk management when liquid markets exist in which risk managers can trade instruments that are correlated (or inversely correlated) with the risks that concern them without incurring substantial transaction costs. Investment banks, exchanges, and insurance companies devote substantial resources to designing such contracts and to ensuring that they will trade in liquid markets.

When such markets exist, people are better able to manage the risks that they face and often are more willing to undertake risky activities that they expect will be profitable.

2.1.5. Exchanging Assets for Immediate Delivery (Spot Market Trading)

People and companies often trade one asset for another that they rate more highly or, equivalently, that is more useful to them. They may trade one currency for another currency, or money for a needed commodity or right. Following are some examples that illustrate these trades:

- Volkswagen pays its German workers in euros, but the company receives dollars when it sells cars in the United States. To convert money from dollars to euros, Volkswagen trades in the foreign exchange markets.
- A Mexican investor who is worried about the prospects for peso inflation or a potential devaluation of the peso may buy gold in the spot gold market. (This transaction may hedge against the risk of devaluation of the peso because the value of gold may increase with inflation.)
- A plastic producer must buy carbon credits to emit carbon dioxide when burning fuel to comply with environmental regulations. The carbon credit is a legal right that the producer must have to engage in activities that emit carbon dioxide.

In each of these cases, the trades are considered spot market trades because the instruments trade for immediate delivery. The financial system facilitates these exchanges when liquid spot markets exist in which people can arrange and settle trades without substantial transaction costs.

2.1.6. Information-Motivated Trading

Information-motivated traders trade to profit from information that they believe allows them to predict future prices. Like all other traders, they hope to buy at low prices and sell at higher prices. Unlike pure investors, however, they expect to earn a return on their information in addition to the normal return expected for bearing risk through time.

Active investment managers are information-motivated traders who collect and analyze information to identify securities, contracts, and other assets that their analyses indicate are under- or overvalued. They then buy those that they consider undervalued and sell those that they consider overvalued. If successful, they obtain a greater return than the unconditional return that would be expected for bearing the risk in their positions. The return that they expect to obtain is a conditional return earned on the basis of the information in their analyses. Practitioners often call this process active portfolio management.

Note that the distinction between pure investors and information-motivated traders depends on their motives for trading and not on the risks that they take or their expected holding periods. Investors trade to move wealth from the present to the future whereas information-motivated traders trade to profit from superior information about future values. When trading to move wealth forward, the time period may be short or long. For example, a bank treasurer may only need to move money overnight and might use money market instruments trading in an interbank funds market to accomplish that. A pension fund, however, may need to move money 30 years forward and might do that by using shares trading in a stock market. Both are investors although their expected holding periods and the risks in the instruments that they trade are vastly different.

In contrast, information-motivated traders trade because their information-based analyses suggest to them that prices of various instruments will increase or decrease in the future at a rate faster than others without their information or analytical models would expect. After establishing their positions, they hope that prices will change quickly in their favor so that they can close their positions, realize their profits, and redeploy their capital. These price changes may occur almost instantaneously, or they may take years to occur if information about the mispricing is difficult to obtain or understand.

The two categories of traders are not mutually exclusive. Investors also are often information-motivated traders. Many investors who want to move wealth forward through time collect and analyze information

to select securities that will allow them to obtain conditional returns that are greater than the unconditional returns expected for securities in their asset classes. If they have rational reasons to expect that their efforts will indeed produce superior returns, they are information-motivated traders. If they consistently fail to produce such returns, their efforts will be futile, and they would have been better off simply buying and holding well-diversified portfolios.

EXAMPLE 2

Investing versus Information-Motivated Trading

The head of a large labor union with a pension fund asks you, a pension consultant, to distinguish between investing and information-motivated trading. You are expected to provide an explanation that addresses the financial problems that she faces. How would you respond?

Solution:

The object of investing for the pension fund is to move the union's pension assets from the present to the future when they will be needed to pay the union's retired pensioners. The pension fund managers will typically do this by buying stocks, bonds, and perhaps other assets. The pension fund managers expect to receive a fair rate of return on the pension fund's assets without paying excessive transaction costs and management fees. The return should compensate the fund for the risks that it bears and for the time that other people are using the fund's money.

The object of information-motivated trading is to earn a return in excess of the fair rate of return. Information-motivated traders analyze information that they collect with the hope that their analyses will allow them to predict better than others where prices will be in the future. They then buy assets that they think will produce excess returns and sell those that they think will underperform. Active investment managers are information-motivated traders.

The characteristic that most distinguishes investors from information-motivated traders is the return that they expect. Although both types of traders hope to obtain extraordinary returns, investors rationally expect to receive only fair returns during the periods of their investments. In contrast, information-motivated traders expect to make returns in excess of required fair rates of return. Of course, not all investing or information-motivated trading is successful (in other words, the actual returns may not equal or exceed the expected returns).

The financial system facilitates information-motivated trading when liquid markets allow active managers to trade without significant transaction costs. Accounting standards and reporting requirements that produce meaningful financial disclosures reduce the costs of being well informed, but do not necessarily help informed traders profit because they often compete with each other. The most profitable well-informed traders are often those that have the most unique insights into future values.

2.1.7. Summary

People use the financial system for many purposes, the most important of which are saving, borrowing, raising equity capital, managing risk, exchanging assets in spot markets, and information-motivated trading. The financial system best facilitates these uses when people can trade instruments that interest them in liquid markets, when institutions provide financial services at low cost, when information about assets and about credit risks is readily available, and when regulation helps ensure that everyone faithfully honors their contracts.

2.2. Determining Rates of Return

Saving, borrowing, and selling equity are all means of moving money through time. Savers move money from the present to the future whereas borrowers and equity issuers move money from the future to the

present.

Because time machines do not exist, money can travel forward in time only if an equal amount of money is travelling in the other direction. This equality always occurs because borrowers and equity sellers create the securities in which savers invest. For example, the bond sold by a company that needs to move money from the future to the present is the same bond bought by a saver who needs to move money from the present to the future.

The aggregate amount of money that savers will move from the present to the future is related to the expected rate of return on their investments. If the expected return is high, they will forgo current consumption and move more money to the future. Similarly, the aggregate amount of money that borrowers and equity sellers will move from the future to the present depends on the costs of borrowing funds or of giving up ownership. These costs can be expressed as the rate of return that borrowers and equity sellers are expected to deliver in exchange for obtaining current funds. It is the same rate that savers expect to receive when delivering current funds. If this rate is low, borrowers and equity sellers will want to move more money to the present from the future. In other words, they will want to raise more funds.

Because the total money saved must equal the total money borrowed and received in exchange for equity, the expected rate of return depends on the aggregate supply of funds through savings and the aggregate demand for funds. If the rate is too high, savers will want to move more money to the future than borrowers and equity issuers will want to move to the present. The expected rate will have to be lower to discourage the savers and to encourage the borrowers and equity issuers. Conversely, if the rate is too low, savers will want to move less money forward than borrowers and equity issuers will want to move to the present. The expected rate will have to be higher to encourage the savers and to discourage the borrowers and equity issuers. Between rates too high and too low, an expected rate of return exists, in theory, in which the aggregate supply of funds for investing (supply of funds saved) and the aggregate demand for funds through borrowing and equity issuing are equal.

Economists call this rate the equilibrium interest rate. It is the price for moving money through time. Determining this rate is one of the most important functions of the financial system. The equilibrium interest rate is the only interest rate that would exist if all securities were equally risky, had equal terms, and were equally liquid. In fact, the required rates of return for securities vary by their risk characteristics, terms, and liquidity. For a given issuer, investors generally require higher rates of return for equity than for debt, for long-term securities than for short-term securities, and for illiquid securities than for liquid ones. Financial analysts recognize that all required rates of return depend on a common equilibrium interest rate plus adjustments for risk.

EXAMPLE 3

Interest Rates

For a presentation to private wealth clients by your firm's chief economist, you are asked to prepare the audience by explaining the most fundamental facts concerning the role of interest rates in the economy. You agree. What main points should you try to convey?

Solution:

Savers have money now that they will want to use in the future. Borrowers want to use money now that they do not have, but they expect that they will have money in the future. Borrowers are loaned money by savers and promise to repay it in the future.

The interest rate is the return that lenders, the savers, expect to receive from borrowers for allowing borrowers to use the savers' money. The interest rate is the price of using money.

Interest rates depend on the total amount of money that people want to borrow and the total amount of money that people are willing to lend. Interest rates are high when, in aggregate, people value having money now substantially more than they value having money in the future. In contrast, if many people with money want to use it in the future and few people presently need more money than they have, interest rates will be low.

2.3. Capital Allocation Efficiency

Primary capital markets (primary markets) are the markets in which companies and governments raise capital (funds). Companies may raise funds by borrowing money or by issuing equity. Governments may raise funds by borrowing money.

Economies are said to be **allocationally efficient** when their financial systems allocate capital (funds) to those uses that are most productive. Although companies may be interested in getting funding for many potential projects, not all projects are worth funding. One of the most important functions of the financial system is to ensure that only the best projects obtain scarce capital funds; the funds available from savers should be allocated to the most productive uses.

In market-based economies, savers determine, directly or indirectly, which projects obtain capital. Savers determine capital allocations directly by choosing which securities they will invest in. Savers determine capital allocations indirectly by giving funds to financial intermediaries that then invest the funds. Because investors fear the loss of their money, they will lend at lower interest rates to borrowers with the best credit prospects or the best collateral, and they will lend at higher rates to other borrowers with less secure prospects. Similarly, they will buy only those equities that they believe have the best prospects relative to their prices and risks.

To avoid losses, investors carefully study the prospects of the various investment opportunities available to them. The decisions that they make tend to be well informed, which helps ensure that capital is allocated efficiently. The fear of losses by investors and by those raising funds to invest in projects ensures that only the best projects tend to be funded. The process works best when investors are well informed about the prospects of the various projects.

In general, investors will fund an equity project if they expect that the value of the project is greater than its cost, and they will not fund projects otherwise. If the investor expectations are accurate, only projects that should be undertaken will be funded and all such projects will be funded. Accurate market information thus leads to efficient capital allocation.

EXAMPLE 4

Primary Market Capital Allocation

How can poor information about the value of a project result in poor capital allocation decisions?

Solution:

Projects should be undertaken only if their value is greater than their cost. If investors have poor information and overestimate the value of a project in which its true value is less than its cost, a wealth-diminishing project may be undertaken. Alternatively, if investors have poor information and underestimate the value of a project in which its true value is greater than its cost, a wealth-enhancing project may not be undertaken.

3. ASSETS AND CONTRACTS

LEARNING OUTCOME

The candidate should be able to:

□ describe classifications of assets and markets

People, companies, and governments use many different assets and contracts to further their financial goals and to manage their risks. The most common assets include financial assets (such as bank deposits, certificates of deposit, loans, mortgages, corporate and government bonds and notes, common and preferred stocks, real estate investment trusts, master limited partnership interests, pooled investment products, and exchange-traded funds), currencies, certain commodities (such as gold and oil), and real assets (such as real estate). The most common contracts are option, futures, forward, swap, and insurance contracts. People, companies, and governments use these assets and contracts to raise funds, to invest, to profit from information-motivated trading, to hedge risks, and/or to transfer money from one form to another.

3.1. Classifications of Assets and Markets

Practitioners often classify assets and the markets in which they trade by various common characteristics to facilitate communications with their clients, with each other, and with regulators.

The most actively traded assets are securities, currencies, contracts, and commodities. In addition, real assets are traded. Securities generally include debt instruments, equities, and shares in pooled investment vehicles. Currencies are monies issued by national monetary authorities. Contracts are agreements to exchange securities, currencies, commodities or other contracts in the future. Commodities include precious metals, energy products, industrial metals, and agricultural products. Real assets are tangible properties such as real estate, airplanes, or machinery. Securities, currencies, and contracts are classified as financial assets whereas commodities and real assets are classified as physical assets.

Securities are further classified as debt or equity. Debt instruments (also called fixed-income instruments) are promises to repay borrowed money. Equities represent ownership in companies. Pooled investment vehicle shares represent ownership of an undivided interest in an investment portfolio. The portfolio may include securities, currencies, contracts, commodities, or real assets. Pooled investment vehicles, such as exchange-traded funds, which exclusively own shares in other companies, generally are also considered equities.

Securities are also classified by whether they are public or private securities. Public securities are those registered to trade in public markets, such as on exchanges or through dealers. In most jurisdictions, issuers must meet stringent minimum regulatory standards, including reporting and corporate governance standards, to issue publicly traded securities.

Private securities are all other securities. Often, only specially qualified investors can purchase private equities and private debt instruments. Investors may purchase them directly from the issuer or indirectly through an investment vehicle specifically formed to hold such securities. Issuers often issue private securities when they find public reporting standards too burdensome or when they do not want to conform to the regulatory standards associated with public equity. Venture capital is private equity that investors supply to companies when or shortly after they are founded. Private securities generally are illiquid. In contrast, many public securities trade in liquid markets in which sellers can easily find buyers for their securities.

Contracts are derivative contracts if their values depend on the prices of other underlying assets. Derivative contracts may be classified as physical or financial depending on whether the underlying instruments are physical products or financial securities. Equity derivatives are contracts whose values depend on equities or indexes of equities. Fixed-income derivatives are contracts whose values depend on debt securities or indexes of debt securities.

Practitioners classify markets by whether the markets trade instruments for immediate delivery or for future delivery. Markets that trade contracts that call for delivery in the future are forward or futures markets. Those that trade for immediate delivery are called **spot markets** to distinguish them from forward markets that trade contracts on the same underlying instruments. Options markets trade contracts that deliver in the future, but delivery takes place only if the holders of the options choose to exercise them.

When issuers sell securities to investors, practitioners say that they trade in the **primary market**. When investors sell those securities to others, they trade in the **secondary market**. In the primary

market, funds flow to the issuer of the security from the purchaser. In the secondary market, funds flow between traders.

Practitioners classify financial markets as money markets or capital markets. **Money markets** trade debt instruments maturing in one year or less. The most common such instruments are repurchase agreements (defined in Section 3.2.1), negotiable certificates of deposit, government bills, and commercial paper. In contrast, **capital markets** trade instruments of longer duration, such as bonds and equities, whose values depend on the credit-worthiness of the issuers and on payments of interest or dividends that will be made in the future and may be uncertain. Corporations generally finance their operations in the capital markets, but some also finance a portion of their operations by issuing short-term securities, such as commercial paper.

Finally, practitioners distinguish between **traditional investment markets** and **alternative investment markets**. Traditional investments include all publicly traded debts and equities and shares in pooled investment vehicles that hold publicly traded debts and/or equities. Alternative investments include **hedge funds**, private equities (including venture capital), commodities, real estate securities and real estate properties, securitized debts, operating leases, machinery, collectibles, and precious gems. Because these investments are often hard to trade and hard to value, they may sometimes trade at substantial deviations from their intrinsic values. The discounts compensate investors for the research that they must do to value these assets and for their inability to easily sell the assets if they need to liquidate a portion of their portfolios.

The remainder of this section describes the most common assets and contracts that people, companies, and governments trade.

EXAMPLE 5

Asset and Market Classification

The investment policy of a mutual fund only permits the fund to invest in public equities traded in secondary markets. Would the fund be able to purchase:

1. Common stock of a company that trades on a large stock exchange?

Solution to 1:

Yes. Common stock is equity. Those common stocks that trade on large exchanges invariably are public equities that trade in secondary markets.

2. Common stock of a public company that trades only through dealers?

Solution to 2:

Yes. Dealer markets are secondary markets and the security is a public equity.

3. A government bond?

Solution to 3:

No. Although government bonds are public securities, they are not equities. They are debt securities.

4. A single stock futures contract?

Solution to 4:

No. Although the underlying instruments for single stock futures are invariably public equities, single stock futures are derivative contracts, not equities.

5. Common stock sold for the first time by a properly registered public company?

Solution to 5:

No. The fund would not be able to buy these shares because a purchase from the issuer would be in the primary market. The fund would have to wait until it could buy the shares from someone other than the issuer.

6. Shares in a privately held bank with €10 billion of capital?

Solution to 6:

No. These shares are private equities, not public equities. The public prominence of the company does not make its securities public securities unless they have been properly registered as public securities.

4. SECURITIES

LEARNING OUTCOME

The candidate should be able to:

☐ describe the major types of securities, currencies, contracts, commodities, and real assets that trade in organized markets, including their distinguishing characteristics and major subtypes

People, companies, and governments sell securities to raise money. Securities include bonds, notes, commercial paper, mortgages, common stocks, preferred stocks, warrants, mutual fund shares, unit trusts, and depository receipts. These can be classified broadly as fixed-income instruments, equities, and shares in pooled investment vehicles. Note that the legal definition of a security varies by country and may or may not coincide with the usage here. Securities that are sold to the public or that can be resold to the public are called issues. Companies and governments are the most common issuers.

4.1. Fixed Income

Fixed-income instruments contractually include predetermined payment schedules that usually include interest and principal payments. Fixed-income instruments generally are promises to repay borrowed money but may include other instruments with payment schedules, such as settlements of legal cases or prizes from lotteries. The payment amounts may be pre-specified or they may vary according to a fixed formula that depends on the future values of an interest rate or a commodity price. Bonds, notes, bills, certificates of deposit, commercial paper, repurchase agreements, loan agreements, and mortgages are examples of promises to repay money in the future. People, companies, and governments create fixed-income instruments when they borrow money.

Corporations and governments issue bonds and notes. Fixed-income securities with shorter maturities are called “notes,” those with longer maturities are called “bonds.” The cutoff is usually at 10 years. In practice, however, the terms are generally used interchangeably. Both become short-term instruments when the remaining time until maturity is short, usually taken to be one year or less.

Some corporations issue convertible bonds, which are typically convertible into stock, usually at the option of the holder after some period. If stock prices are high so that conversion is likely, convertibles are valued like stock. Conversely, if stock prices are low so that conversion is unlikely, convertibles are valued like bonds.

Bills, certificates of deposit, and commercial paper are respectively issued by governments, banks, and corporations. They usually mature within a year of being issued; certificates of deposit sometimes have longer initial maturities.

Repurchase agreements (repos) are short-term lending instruments. The term can be as short as overnight. A borrower seeking funds will sell an instrument—typically a high-quality bond—to a lender with an agreement to repurchase it later at a slightly higher price based on an agreed upon interest rate.

Practitioners distinguish between short-term, intermediate-term, and long-term fixed-income securities. No general consensus exists about the definition of short-term, intermediate-term, and long-term. Instruments that mature in less than one to two years are considered short-term instruments whereas those that mature in more than five to ten years are considered long-term instruments. In the middle are intermediate-term instruments.

Instruments trading in money markets are called money market instruments. Such instruments are traded debt instruments maturing in one year or less. Money market funds and corporations seeking a return on their short-term cash balances typically hold money market instruments.

4.2. Equities

Equities represent ownership rights in companies. These include common and preferred shares. Common shareholders own residual rights to the assets of the company. They have the right to receive any dividends declared by the boards of directors, and in the event of liquidation, any assets remaining after all other claims are paid. Acting through the boards of directors that they elect, common shareholders usually can select the managers who run the corporations.

Preferred shares are equities that have preferred rights (relative to common shares) to the cash flows and assets of the company. Preferred shareholders generally have the right to receive a specific dividend on a regular basis. If the preferred share is a cumulative preferred equity, the company must pay the preferred shareholders any previously omitted dividends before it can pay dividends to the common shareholders. Preferred shareholders also have higher claims to assets relative to common shareholders in the event of corporate liquidation. For valuation purposes, financial analysts generally treat preferred stocks as fixed-income securities when the issuers will clearly be able to pay their promised dividends in the foreseeable future.

Warrants are securities issued by a corporation that allow the warrant holders to buy a security issued by that corporation, if they so desire, usually at any time before the warrants expire or, if not, upon expiration. The security that warrant holders can buy usually is the issuer's common stock, in which case the warrants are considered equities because the warrant holders can obtain equity in the company by exercising their warrants. The warrant **exercise price** is the price that the warrant holder must pay to buy the security.

EXAMPLE 6

Securities

What factors distinguish fixed-income securities from equities?

Solution:

Fixed-income securities generate income on a regular schedule. They derive their value from the promise to pay a scheduled cash flow. The most common fixed-income securities are promises made by people, companies, and governments to repay loans.

Equities represent residual ownership in companies after all other claims—including any fixed-income liabilities of the company—have been satisfied. For corporations, the claims of preferred equities typically have priority over the claims of common equities. Common equities have the residual ownership in corporations.

4.3. Pooled Investments

Pooled investment vehicles are mutual funds, trusts, depositories, and hedge funds, that issue securities that represent shared ownership in the assets that these entities hold. The securities created by mutual funds, trusts, depositories, and hedge fund are respectively called *shares*, *units*, *depository receipts*, and *limited partnership interests* but practitioners often use these terms interchangeably. People invest in pooled investment vehicles to benefit from the investment management services of their managers and from diversification opportunities that are not readily available to them on an individual basis.

Mutual funds are investment vehicles that pool money from many investors for investment in a portfolio of securities. They are often legally organized as investment trusts or as corporate investment companies. Pooled investment vehicles may be open-ended or closed-ended. Open-ended funds issue new shares and redeem existing shares on demand, usually on a daily basis. The price at which a fund redeems and sells the fund's shares is based on the net asset value of the fund's portfolio, which is the difference between the fund's assets and liabilities, expressed on a per share basis. Investors generally buy and sell open-ended mutual funds by trading with the mutual fund.

In contrast, closed-end funds issue shares in primary market offerings that the fund or its investment bankers arrange. Once issued, investors cannot sell their shares of the fund back to the fund by demanding redemption. Instead, investors in closed-end funds must sell their shares to other investors in the secondary market. The secondary market prices of closed-end funds may differ—sometimes quite significantly—from their net asset values. Closed-end funds generally trade at a discount to their net asset values. The discount reflects the expenses of running the fund and sometimes investor concerns about the quality of the management. Closed-end funds may also trade at a discount or a premium to net asset value when investors believe that the portfolio securities are overvalued or undervalued. Many financial analysts thus believe that discounts and premiums on closed-end funds measure market sentiment.

Exchange-traded funds (ETFs) and exchange-traded notes (ETNs) are open-ended funds that investors can trade among themselves in secondary markets. The prices at which ETFs trade rarely differ much from net asset values because a class of investors, known as authorized participants (APs), has the option of trading directly with the ETF. If the market price of an equity ETF is sufficiently below its net asset value, APs will buy shares in the secondary market at market price and redeem shares at net asset value with the fund. Conversely, if the price of an ETF is sufficiently above its net asset value, APs will buy shares from the fund at net asset value and sell shares in the secondary market at market price. As a result, the market price and net asset values of ETFs tend to converge.

Many ETFs permit only in-kind deposits and redemptions. Buyers who buy directly from such a fund pay for their shares with a portfolio of securities rather than with cash. Similarly, sellers receive a portfolio of securities. The transaction portfolio generally is very similar—often essentially identical—to the portfolio held by the fund. Practitioners sometimes call such funds “depositories” because they issue depository receipts for the portfolios that traders deposit with them. The traders then trade the receipts in the secondary market. Some warehouses holding industrial materials and precious metals also issue tradable warehouse receipts.

Asset-backed securities are securities whose values and income payments are derived from a pool of assets, such as mortgage bonds, credit card debt, or car loans. These securities typically pass interest and principal payments received from the pool of assets through to their holders on a monthly basis. These payments may depend on formulas that give some classes of securities—called tranches—backed by the pool more value than other classes.

Hedge funds are investment funds that generally organize as limited partnerships. The hedge fund managers are the general partners. The limited partners are qualified investors who are wealthy enough and well informed enough to tolerate and accept substantial losses, should they occur. The regulatory requirements to participate in a hedge fund and the regulatory restrictions on hedge funds vary by jurisdiction. Most hedge funds follow only one investment strategy, but no single investment strategy characterizes hedge funds as a group. Hedge funds exist that follow almost every imaginable strategy ranging from long-short arbitrage in the stock markets to direct investments in exotic alternative assets.

The primary distinguishing characteristic of hedge funds is their management compensation scheme. Almost all funds pay their managers with an annual fee that is proportional to their assets and with an additional performance fee that depends on the wealth that the funds generate for their shareholders. A

secondary distinguishing characteristic of many hedge funds is the use of leverage to increase risk exposure and to hopefully increase returns.

5. CURRENCIES, COMMODITIES, AND REAL ASSETS

LEARNING OUTCOME

The candidate should be able to:

- ☐ **describe the major types of securities, currencies, contracts, commodities, and real assets that trade in organized markets, including their distinguishing characteristics and major subtypes**

Currencies are monies issued by national monetary authorities. Approximately 180 currencies are currently in use throughout the world. Some of these currencies are regarded as reserve currencies. Reserve currencies are currencies that national central banks and other monetary authorities hold in significant quantities. The primary reserve currencies are the US dollar and the euro. Secondary reserve currencies include the British pound, the Japanese yen, and the Swiss franc.

Currencies trade in foreign exchange markets. In spot currency transactions, one currency is immediately or almost immediately exchanged for another. The rate of exchange is called the spot exchange rate. Traders typically negotiate institutional trades in multiples of large quantities, such as US\$1 million or ¥100 million. Institutional trades generally settle in two business days.

Retail currency trades most commonly take place through commercial banks when their customers exchange currencies at a location of the bank, use ATM machines when travelling to withdraw a different currency than the currency in which their bank accounts are denominated, or use credit cards to buy items priced in different currencies. Retail currency trades also take place at airport kiosks, at store front currency exchanges, or on the street.

5.1. Commodities

Commodities include precious metals, energy products, industrial metals, agricultural products, and carbon credits. Spot commodity markets trade commodities for immediate delivery whereas the forward and futures markets trade commodities for future delivery. Managers seeking positions in commodities can acquire them directly by trading in the spot markets or indirectly by trading forward and futures contracts.

The producers and processors of industrial metals and agricultural products are the primary users of the spot commodity markets because they generally are best able to take and make delivery and to store physical products. They undertake these activities in the normal course of operating their businesses. Their ability to handle physical products and the information that they gather operating businesses also gives them substantial advantages as information-motivated traders in these markets. Many producers employ financial analysts to help them analyze commodity market conditions so that they can best manage their inventories to hedge their operational risks and to speculate on future price changes.

Commodities also interest information-motivated traders and investment managers because they can use them as hedges against risks that they hold in their portfolios or as vehicles to speculate on future price changes. Most such traders take positions in the futures markets because they usually do not have facilities to handle most physical products nor can they easily obtain them. They also cannot easily cope with the normal variation in qualities that characterizes many commodities. Information-motivated traders and investment managers also prefer to trade in futures markets because most futures markets are more liquid than their associated spot markets and forward markets. The liquidity allows them to easily close their positions before delivery so that they can avoid handling physical products.

Some information-motivated traders and investment managers, however, trade in the spot commodity markets, especially when they can easily contract for low-cost storage. Commodities for which delivery and storage costs are lowest are nonperishable products for which the ratio of value to weight is high and variation in quality is low. These generally include precious metals, industrial diamonds, such high-value industrial metals as copper, aluminum, and mercury, and carbon credits.

5.2. Real Assets

Real assets include such tangible properties as real estate, airplanes, machinery, or lumber stands. These assets normally are held by operating companies, such as real estate developers, airplane leasing companies, manufacturers, or loggers. Many institutional investment managers, however, have been adding real assets to their portfolios as direct investments (involving direct ownership of the real assets) and indirect investments (involving indirect ownership, for example, purchase of securities of companies that invest in real assets or real estate investment trusts). Investments in real assets are attractive to them because of the income and tax benefits that they often generate, and because changes in their values may have a low correlation with other investments that the managers hold.

Direct investments in real assets generally require substantial management to ensure that the assets are maintained and used efficiently. Investment managers investing in such assets must either hire personnel to manage them or hire outside management companies. Either way, management of real assets is quite costly.

Real assets are unique properties in the sense that no two assets are alike. An example of a unique property is a real estate parcel. No two parcels are the same because, if nothing else, they are located in different places. Real assets generally differ in their conditions, remaining useful lives, locations, and suitability for various purposes. These differences are very important to the people who use them, so the market for a given real asset may be very limited. Thus, real assets tend to trade in very illiquid markets.

The heterogeneity of real assets, their illiquidity, and the substantial costs of managing them are all factors that complicate the valuation of real assets and generally make them unsuitable for most investment portfolios. These same problems, however, often cause real assets to be misvalued in the market, so astute information-motivated traders may occasionally identify significantly undervalued assets. The benefits from purchasing such assets, however, are often offset by the substantial costs of searching for them and by the substantial costs of managing them.

Many financial intermediaries create entities, such as real estate investment trusts (REITs) and master limited partnerships (MLPs), to securitize real assets and to facilitate indirect investment in real assets. The financial intermediaries manage the assets and pass through the net benefits after management costs to the investors who hold these securities. Because these securities are much more homogenous and divisible than the real assets that they represent, they tend to trade in much more liquid markets. Thus, they are much more suitable as investments than the real assets themselves.

Of course, investors seeking exposure to real assets can also buy shares in corporations that hold and operate real assets. Although almost all corporations hold and operate real assets, many specialize in assets that particularly interest investors seeking exposure to specific real asset classes. For example, investors interested in owning aircraft can buy an aircraft leasing company such as Waha Capital (Abu Dhabi Securities Exchange) and Aircastle Limited (NYSE).

EXAMPLE 7

Assets and Contracts

Consider the following assets and contracts:

Bank deposits	Hedge funds
Certificates of deposit	Master limited partnership interests
Common stocks	Mortgages
Corporate bonds	Mutual funds

Currencies	Stock option contracts
Exchange-traded funds	Preferred stocks
Lumber forward contracts	Real estate parcels
Crude oil futures contracts	Interest rate swaps
Gold	Treasury notes

1. Which of these represent ownership in corporations?

Solution to 1:

Common and preferred stocks represent ownership in corporations.

2. Which of these are debt instruments?

Solution to 2:

Bank deposits, certificates of deposit, corporate bonds, mortgages, and Treasury notes are all debt instruments. They respectively represent loans made to banks, corporations, mortgagees (typically real estate owners), and the Treasury.

3. Which of these are created by traders rather than by issuers?

Solution to 3:

Lumber forward contracts, crude oil futures contracts, stock option contracts, and interest rate swaps are created when the seller sells them to a buyer.

4. Which of these are pooled investment vehicles?

Solution to 4:

Exchange-traded funds, hedge funds, and mutual funds are pooled investment vehicles. They represent shared ownership in a portfolio of other assets.

5. Which of these are real assets?

Solution to 5:

Real estate parcels are real assets.

6. Which of these would a home builder most likely use to hedge construction costs?

Solution to 6:

A builder would buy lumber forward contracts to lock in the price of lumber needed to build homes.

7. Which of these would a corporation trade when moving cash balances among various countries?

Solution to 7:

Corporations often trade currencies when moving cash from one country to another.

6. CONTRACTS

LEARNING OUTCOME

The candidate should be able to:

- ☐ **describe the major types of securities, currencies, contracts, commodities, and real assets that trade in organized markets, including their distinguishing characteristics and major subtypes**

A contract is an agreement among traders to do something in the future. Contracts include forward, futures, swap, option, and insurance contracts. The values of most contracts depend on the value of an **underlying** asset. The underlying asset may be a commodity, a security, an index representing the values of other instruments, a currency pair or basket, or other contracts.

Contracts provide for some physical or cash settlement in the future. In a physically settled contract, settlement occurs when the parties to the contract physically exchange some item, such as avocados, pork bellies, or gold bars. Physical settlement also includes the delivery of such financial instruments as bonds, equities, or futures contracts even though the delivery is electronic. In contrast, cash settled contracts settle through cash payments. The amount of the payment depends on formulas specified in the contracts.

Financial analysts classify contracts by whether they are physical or financial based on the nature of the underlying asset. If the underlying asset is a physical product, the contract is a physical; otherwise, the contract is a financial. Examples of assets classified as physical include contracts for the delivery of petroleum, lumber, and gold. Examples of assets classified as financial include option contracts, and contracts on interest rates, stock indexes, currencies, and credit default swaps.

Contracts that call for immediate delivery are called spot contracts, and they trade in spot markets. Immediate delivery generally is three days or less, but depends on each market. All other contracts involve what practitioners call futurity. They derive their values from events that will take place in the future.

EXAMPLE 8

Contracts for Difference

Contracts for difference (CFD) allow people to speculate on price changes for an underlying asset, such as a common stock or an index. Dealers generally sell CFDs to their clients. When the clients sell the CFDs back to their dealer, they receive any appreciation in the underlying asset's price between the time of purchase and sale (open and close) of the contract. If the underlying asset's price drops over this interval, the client pays the dealer the difference.

1. Are contracts for difference derivative contracts?

Solution to 1:

Contracts for difference are derivative contracts because their values are derived from changes in the prices of the underlying asset on which they are based.

2. Are contracts for difference based on copper prices cash settled or physically settled?

Solution to 2:

All contracts for difference are cash settled contracts regardless of the underlying asset on which they are based because they settle in cash and not in the underlying asset.

6.1. Forward Contracts

A **forward contract** is an agreement to trade the underlying asset in the future at a price agreed upon today. For example, a contract for the sale of wheat after the harvest is a forward contract. People often use forward contracts to reduce risk. Before planting wheat, farmers like to know the price at which they will sell their crop. Similarly, before committing to sell flour to bakers in the future, millers like to know the prices that they will pay for wheat. The farmer and the miller both reduce their operating risks by agreeing to trade wheat forward.

Practitioners call such traders hedgers because they use their contractual commitments to hedge their risks. If the price of wheat falls, the wheat farmer's crop will drop in value on the spot market but he has a contract to sell wheat in the future at a higher fixed price. The forward contract has become more valuable to the farmer. Conversely, if the price of wheat rises, the miller's future obligation to sell flour will become more burdensome because of the high price he would have to pay for wheat on the spot market, but the miller has a contract to buy wheat at a lower fixed price. The forward contract has become more valuable to the miller. In both cases, fluctuations in the spot price are hedged by the forward contract. The forward contract offsets the operating risks that the hedgers face.

Consider a simple example of hedging. An avocado farmer in Mexico expects to harvest 15,000 kilograms of avocados and that the price at harvest will be 60 pesos per kilogram. That price, however, could fluctuate significantly before the harvest. If the price of avocados drops to 50 pesos, the farmer would lose 10 pesos per kilogram (60 pesos – 50 pesos) relative to his expectations, or a total of 150,000 pesos. Now, suppose that the farmer can sell avocados forward to Del Rey Avocado at 58 pesos for delivery at the harvest. If the farmer sells 15,000 kilograms forward, and the price of avocados drops to 50 pesos, the farmer would still be able to sell his avocados for 58 pesos, and thus would not suffer from the drop in the price of avocados below this level.

EXAMPLE 9

Hedging Gold Production

A Zimbabwean gold producer invests in a mine expansion project on the expectation that gold prices will remain at or above \$1,200 USD per ounce when the new project starts producing ore.

1. What risks does the gold producer face with respect to the price of gold?

Solution to 1:

The gold producer faces the risk that the price of gold could fall below \$1,200 USD before it can sell its new production. If so, the investment in the expansion project will be less profitable than expected, and may even generate losses for the mine.

2. How might the gold producer hedge its gold price risk?

Solution to 2:

The gold producer could hedge the gold price risk by selling gold forward, hopefully at a price near \$1,200 USD. Even if the price of gold falls, the gold producer would get paid the contract price.

Forward contracts are very common, but two problems limit their usefulness for many market participants. The first problem is counterparty risk. **Counterparty risk** is the risk that the other party to a contract will fail to honor the terms of the contract. Concerns about counterparty risk ensure that generally only parties who have long-standing relationships with each other execute forward contracts. Trustworthiness is critical when prices are volatile because, after a large price change, one side or the other may prefer not to settle the contract.

The second problem is liquidity. Trading out of a forward contract is very difficult because it can only be done with the consent of the other party. The liquidity problem ensures that forward contracts tend to be executed only among participants for whom delivery is economically efficient and quite certain at the time of contracting so that both parties will want to arrange for delivery.

The counterparty risk problem and the liquidity problem often make it difficult for market participants to obtain the hedging benefits associated with forward contracting. Fortunately, futures contracts have been developed to mitigate these problems.

6.2. Futures Contracts

A **futures contract** is a standardized forward contract for which a clearinghouse guarantees the performance of all traders. The buyer of a futures contract is the side that will take physical delivery or its cash equivalent. The seller of a futures contract is the side that is liable for the delivery or its cash equivalent. A **clearinghouse** is an organization that ensures that no trader is harmed if another trader fails to honor the contract. In effect, the clearinghouse acts as the buyer for every seller and as the seller for every buyer. Buyers and sellers, therefore, can trade futures without worrying whether their counterparties are creditworthy. Because futures contracts are standardized, a buyer can eliminate his obligation to buy by selling his contract to anyone. A seller similarly can eliminate her obligation to deliver by buying a contract from anyone. In either case, the clearinghouse will release the trader from all future obligations if his or her long and short positions exactly offset each other.

To protect against defaults, futures clearinghouses require that all participants post with the clearinghouse an amount of money known as **initial margin** when they enter a contract. The clearinghouse then settles the margin accounts on a daily basis. All participants who have lost on their contracts that day will have the amount of their losses deducted from their margin by the clearinghouse. The clearinghouse similarly increases margins for all participants who gained on that day. Participants whose margins drop below the required **maintenance margin** must replenish their accounts. If a participant does not provide sufficient additional margin when required, the participant's broker will immediately trade to offset the participant's position. These **variation margin** payments ensure that the liabilities associated with futures contracts do not grow large.

EXAMPLE 10

Futures Margin

NYMEX's Light Sweet Crude Oil futures contract specifies the delivery of 1,000 barrels of West Texas Intermediate Crude Oil when the contract finally settles. A broker requires that its clients post an initial overnight margin of \$7,763 per contract and an overnight maintenance margin of \$5,750 per contract. A client buys ten contracts at \$75 per barrel through this broker. On the next day, the contract settles for \$72 per barrel. How much additional margin will the client have to provide to his broker?

Solution:

The client lost three dollars per barrel (he is the side committed to take delivery or its cash equivalent at \$75 per barrel). This results in a \$3,000 loss on each of his 10 contracts, and a total loss of \$30,000. His initial margin of \$77,630 is reduced by \$30,000 leaving \$47,630 in his margin account. Because his account has dropped below the maintenance margin requirement of \$57,500, the client will get a margin call. The client must provide an additional $\$30,000 = \$77,630 - \$47,630$ to replenish his margin account; the account is replenished to the amount of the initial margin. The client will only receive another margin call if his account drops to below \$57,500 again.

Futures contracts have vastly improved the efficiency of forward contracting markets. Traders can trade standardized futures contracts with anyone without worrying about counterparty risk, and they can close their positions by arranging offsetting trades. Hedgers for whom the terms of the standard contract are not ideal generally still use the futures markets because the contracts embody most of the price risk

that concerns them. They simply offset (close out) their futures positions, at the same time they enter spot contracts on which they make or take ultimate delivery.

EXAMPLE 11

Forward and Futures Contracts

What feature most distinguishes futures contracts from forward contracts?

Solution:

A futures contract is a standardized forward contract for which a clearinghouse guarantees the performance of all buyers and sellers. The clearinghouse reduces the counterparty risk problem. The clearinghouse allows a buyer who has bought a contract from one person and sold the same contract to another person to net out the two obligations so that she is no longer liable for either side of the contract; the positions are closed. The ability to trade futures contracts provides liquidity in futures contracts compared with forward contracts.

6.3. Swap Contracts

A **swap contract** is an agreement to exchange payments of periodic cash flows that depend on future asset prices or interest rates. For example, in a typical **interest rate swap**, at periodic intervals, one party makes fixed cash payments to the counterparty in exchange for variable cash payments from the counterparty. The variable payments are based on a pre-specified variable interest rate such as the London Interbank Offered Rate (Libor). This swap effectively exchanges fixed interest payments for variable interest payments. Because the variable rate is set in the future, the cash flows for this contract are uncertain when the parties enter the contract.

Investment managers often enter interest rate swaps when they own a fixed long-term income stream that they want to convert to a cash flow that varies with current short-term interest rates, or vice versa. The conversion may allow them to substantially reduce the total interest rate risk to which they are exposed. Hedgers often use swap contracts to manage risks.

In a **commodity swap**, one party typically makes fixed payments in exchange for payments that depend on future prices of a commodity such as oil. In a **currency swap**, the parties exchange payments denominated in different currencies. The payments may be fixed, or they may vary depending on future interest rates in the two countries. In an **equity swap**, the parties exchange fixed cash payments for payments that depend on the returns to a stock or a stock index.

EXAMPLE 12

Swap and Forward Contracts

What feature most distinguishes a swap contract from a cash-settled forward contract?

Solution:

Both contracts provide for the exchange of cash payments in the future. A forward contract only has a single cash payment at the end that depends on an underlying price or index at the end. In contrast, a swap contract has several scheduled periodic payments, each of which depends on an underlying price or index at the time of the payment.

6.4. Option Contracts

An **option contract** allows the holder (the purchaser) of the **option** to buy or sell, depending on the type of option, an underlying instrument at a specified price at or before a specified date in the future. Those that do buy or sell are said to **exercise** their contracts. An option to buy is a **call option**, and an option to sell is a **put** option. The specified price is called the strike price (exercise price). If the holders can exercise their contracts only when they mature, they are **European-style** contracts. If they can exercise the contracts earlier, they are **American-style** contracts. Many exchanges list standardized option contracts on individual stocks, stock indexes, futures contracts, currencies, swaps, and precious metals. Institutions also trade many customized option contracts with dealers in the over-the-counter derivative market.

Option holders generally will exercise call options if the strike price is below the market price of the underlying instrument, in which case, they will be able to buy at a lower price than the market price. Similarly, they will exercise put options if the strike price is above the underlying instrument price so that they sell at a higher price than the market price. Otherwise, option holders allow their options to expire as worthless.

The price that traders pay for an option is the option premium. Options can be quite expensive because, unlike forward and futures contracts, they do not impose any liability on the holder. The premium compensates the sellers of options—called option writers—for giving the call option holders the right to potentially buy below market prices and put option holders the right to potentially sell above market prices. Because the writers must trade if the holders exercise their options, option contracts may impose substantial liabilities on the writers.

EXAMPLE 13

Option and Forward Contracts

What feature most distinguishes option contracts from forward contracts?

Solution:

The holder of an option contract has the right, but not the obligation, to buy (for a call option) or sell (for a put option) the underlying instrument at some time in the future. The writer of an option contract must trade the underlying instrument if the holder exercises the option.

In contrast, the two parties to a forward contract must trade the underlying instrument (or its equivalent value for a cash-settled contract) at some time in the future if either party wants to settle the contract.

6.5. Other Contracts

Insurance contracts pay their beneficiaries a cash benefit if some event occurs. Life, liability, and automobile insurance are examples of insurance contracts sold to retail clients. People generally use insurance contracts to compensate for losses that they will experience if bad things happen unexpectedly. Insurance contracts allow them to hedge risks that they face.

Credit default swaps (CDS) are insurance contracts that promise payment of principal in the event that a company defaults on its bonds. Bondholders use credit default swaps to convert risky bonds into more secure investments. Other creditors of the company may also buy them to hedge against the risk they will not be paid if the company goes bankrupt.

Well-informed traders who believe that a corporation will default on its bonds may buy credit default swaps written on the corporation's bonds if the swap prices are sufficiently low. If they are correct, the traders will profit if the payoff to the swap is more than the cost of buying and maintaining the swap position.

People sometimes also buy insurance contracts as investments, especially in jurisdictions where payouts from insurance contracts are not subject to as much taxation as are payouts to other investment vehicles. They may buy these contracts directly from insurance companies, or they may buy already issued contracts from their owners. For example, the life settlements market trades life insurance contracts that people sell to investors when they need cash.

7. FINANCIAL INTERMEDIARIES

LEARNING OUTCOME

The candidate should be able to:

☐ **describe types of financial intermediaries and services that they provide**

Financial intermediaries help entities achieve their financial goals. These intermediaries include commercial, mortgage, and investment banks; credit unions, credit card companies, and various other finance corporations; brokers and exchanges; dealers and arbitrageurs; clearinghouses and depositories; mutual funds and hedge funds; and insurance companies. The services and products that financial intermediaries provide allow their clients to solve the financial problems that they face more efficiently than they could do so by themselves. Financial intermediaries are essential to well-functioning financial systems.

Financial intermediaries are called intermediaries because the services and products that they provide help connect buyers to sellers in various ways. Whether the connections are easy to identify or involve complex financial structures, financial intermediaries stand between one or more buyers and one or more sellers and help them transfer capital and risk between them. Financial intermediaries' activities allow buyers and sellers to benefit from trading, often without any knowledge of the other.

This section introduces the main financial intermediaries that provide services and products in well-developed financial markets. The discussion starts with those intermediaries whose services most obviously connect buyers to sellers and then proceeds to those intermediaries whose services create more subtle connections. Because many financial intermediaries provide many different types of services, some are mentioned more than once. The section concludes with a general characterization of the various ways in which financial intermediaries add value to the financial system.

7.1. Brokers, Exchanges, and Alternative Trading Systems

Brokers are agents who fill orders for their clients. They do not trade with their clients. Instead, they search for traders who are willing to take the other side of their clients' orders. Individual brokers may work for large brokerage firms, the brokerage arm of banks, or at exchanges. Some brokers match clients to clients personally. Others use specialized computer systems to identify potential trades and help their clients fill their orders. Brokers help their clients trade by reducing the costs of finding counterparties for their trades.

Block broker provide brokerage service to large traders. Large orders are hard to fill because finding a counterparty willing to do a large trade is often quite difficult. A large buy order generally will trade at a premium to the current market price, and a large sell order generally will trade at a discount to the current market price. These price concessions encourage other traders to trade with the large traders. They also make large traders reluctant, however, to expose their orders to the public before their trades are arranged because they do not want to move the market. Block brokers, therefore, carefully manage the exposure of the orders entrusted to them, which makes filling them difficult.

Investment banks provide advice to their mostly corporate clients and help them arrange transactions such as initial and seasoned securities offerings. Their corporate finance divisions help corporations finance their business by issuing securities, such as common and preferred shares, notes,

and bonds. Another function of corporate finance divisions is to help companies identify and acquire other companies (i.e., in mergers and acquisitions).

Exchanges provide places where traders can meet to arrange their trades. Historically, brokers and dealers met on an exchange floor to negotiate trades. Increasingly, exchanges arrange trades for traders based on orders that brokers and dealers submit to them. Such exchanges essentially act as brokers. The distinction between exchanges and brokers has become quite blurred. Exchanges and brokers that use electronic order matching systems to arrange trades among their clients are functionally indistinguishable in this respect. Examples of exchanges include the NYSE, Eurex, Frankfurt Stock Exchange, the Chicago Mercantile Exchange, the Tokyo Stock Exchange, and the Singapore Exchange.

Exchanges are easily distinguished from brokers by their regulatory operations. Most exchanges regulate their members' behavior when trading on the exchange, and sometimes away from the exchange.

Many securities exchanges regulate the issuers that list their securities on the exchange. These regulations generally require timely financial disclosure. Financial analysts use this information to value the securities traded at the exchange. Without such disclosure, valuing securities could be very difficult and market prices might not reflect the fundamental values of the securities. In such situations, well-informed participants may profit from less-informed participants. To avoid such losses, the less-informed participants may withdraw from the market, which can greatly increase corporate costs of capital.

Some exchanges also prohibit issuers from creating capital structures that would concentrate voting rights in the hands of a few owners who do not own a commensurate share of the equity. These regulations attempt to ensure that corporations are run for the benefit of all shareholders and not to promote the interests of controlling shareholders who do not have significant economic stakes in the company.

Exchanges derive their regulatory authority from their national or regional governments, or through the voluntary agreements of their members and issuers to subject themselves to the exchange regulations. In most countries, government regulators oversee the exchange rules and the regulatory operations. Most countries also impose financial disclosure standards on public issuers. Examples of government regulatory bodies include the Japanese Financial Services Agency, the British Financial Conduct Authority, the German Federal Financial Supervisory Authority (BaFin), the US Securities and Exchange Commission, the Ontario Securities Commission, and the Argentine National Securities Commission (CNV).

Alternative trading systems (ATSs), also known as **electronic communications networks** (ECNs) or **multilateral trading facilities** (MTFs) are trading venues that function like exchanges but that do not exercise regulatory authority over their subscribers except with respect to the conduct of their trading in their trading systems. Some ATSs operate electronic trading systems that are otherwise indistinguishable from the trading systems operated by exchanges. Others operate innovative trading systems that suggest trades to their customers based on information that their customers share with them or that they obtain through research into their customers' preferences. Many ATSs are known as **dark pools** because they do not display the orders that their clients send to them. Large investment managers especially like these systems because market prices often move to their disadvantage when other traders know about their large orders. ATSs may be owned and operated by broker-dealers, exchanges, banks, or by companies organized solely for this purpose, many of which may be owned by a consortia of brokers-dealers and banks. Examples of ATSs include MATCHNow (Canada), BATS (United States), POSIT (United States), Liquidnet (United States), Baxter-FX (Ireland), and Turquoise (Europe). Many of these ATSs provide services in many markets besides the ones in which they are domiciled.

7.2. Dealers

Dealers fill their clients' orders by trading with them. When their clients want to sell securities or contracts, dealers buy the instruments for their own accounts. If their clients want to buy securities, dealers sell securities that they own or have borrowed. After completing a transaction, dealers hope to reverse the transaction by trading with another client on the other side of the market. When they are successful, they effectively connect a buyer who arrived at one point in time with a seller who arrived at another point in time.

The service that dealers provide is liquidity. **Liquidity** is the ability to buy or sell with low transactions costs when you want to trade. By allowing their clients to trade when they want to trade, dealers provide liquidity to them. In over-the-counter markets, dealers offer liquidity when their clients ask them to trade with them. In exchange markets, dealers offer liquidity to anyone who is willing to trade at the prices that the dealers offer at the exchange. Dealers profit when they can buy at prices that on average are lower than the prices at which they sell.

Dealers may organize their operations within proprietary trading houses, investment banks, and hedge funds, or as sole proprietorships. Some dealers are traditional dealers in the sense that individuals make trading decisions. Others use computerized trading to make all trading decisions. Examples of companies with large dealing operations include Deutsche Bank (Germany), RBC Capital Markets (Canada), Nomura Securities (Japan), Timber Hill (United States), Goldman Sachs (United States), and IG Group (United Kingdom). Almost all investment banks have large dealing operations.

Most dealers also broker orders, and many brokers deal to their customers. Accordingly, practitioners often use the term **broker-dealer** to refer to dealers and brokers. Broker-dealers have a conflict of interest with respect to how they fill their customers' orders. When acting as a broker, they must seek the best price for their customers' orders. When acting as dealers, however, they profit most when they sell to their customers at high prices or buy from their customers at low prices. The problem is most serious when the customer allows the broker-dealer to decide whether to trade the order with another trader or to fill it as a dealer. Consequently, when trading with a broker-dealer, some customers specify how they want their orders filled. They may also trade only with pure agency brokers who do not also deal.

Primary dealers are dealers with whom central banks trade when conducting monetary policy. They buy bills, notes, and bonds when the central banks sell them to decrease the money supply. The dealers then sell these instruments to their clients. Similarly, when the central banks want to increase the money supply, the primary dealers buy these instruments from their clients and sell them to the central banks.

EXAMPLE 14

Brokers and Dealers

What characteristic *most likely* distinguishes brokers from dealers?

Solution:

Brokers are agents that arrange trades on behalf of their clients. They do not trade with their clients. In contrast, dealers are proprietary traders who trade with their clients.

7.3. Arbitrageurs

Arbitrageurs trade when they can identify opportunities to buy and sell identical or essentially similar instruments at different prices in different markets. They profit when they can buy in one market for less than they sell in another market. Arbitrageurs are financial intermediaries because they connect buyers in one market to sellers in another market.

The purest form of arbitrage involves buying and selling the same instrument in two different markets. Arbitrageurs who do such trades sell to buyers in one market and buy from sellers in the other market. They provide liquidity to the markets because they make it easier for buyers and sellers to trade when and where they want to trade.

Because dealers and arbitrageurs both provide liquidity to other traders, they compete with each other. The dealers connect buyers and sellers who arrive in the same market at different times whereas the arbitrageurs connect buyers and sellers who arrive at the same time in different markets. In practice, traders who profit from offering liquidity rarely are purely dealers or purely arbitrageurs. Instead, most traders attempt to identify and exploit every opportunity they can to manage their inventories profitably.

If information about prices is readily available to market participants, pure arbitrages involving the same instrument will be quite rare. Traders who are well informed about market conditions usually route their orders to the market offering the best price so that arbitrageurs will have few opportunities to match traders across markets when they want to trade the exact same instrument.

Arbitrageurs often trade securities or contracts whose values depend on the same underlying factors. For example, dealers in equity option contracts often sell call options in the contract market and buy the underlying shares in the stock market. Because the values of the call options and of the underlying shares are closely correlated (the value of the call increases with the value of the shares), the long stock position hedges the risk in the short call position so that the dealer's net position is not too risky.

Similar to the pure arbitrage that involves the same instrument in different markets, these arbitrage trades connect buyers in one market to sellers in another market. In this case, however, the buyers and sellers are interested in different instruments whose values are closely related. In the example, the buyer is interested in buying a call options contract, the value of which is a nonlinear function of the value of the underlying stock; the seller is interested in selling the underlying stock.

Options dealers buy stock and sell calls when calls are overpriced relative to the underlying stocks. They use complicated financial models to value options in relation to underlying stock values, and they use financial engineering techniques to control the risk of their portfolios. Successful arbitrageurs must know valuation relations well and they must manage the risk in their portfolios well to trade profitably. They profit by buying the relatively undervalued instrument and selling the relatively overvalued instrument.

Buying a risk in one form and selling it another form involves a process called replication. Arbitrageurs use various trading strategies to replicate the returns to securities and contracts. If they can substantially replicate those returns, they can use the replication trading strategy to offset the risk of buying or selling the actual securities and contracts. The combined effect of their trading is to transform risk from one form to another. This process allows them to create or eliminate contracts in response to the excess demand for, and supply of, contracts.

For example, when traders want to buy more call contracts than are presently available, they push the call contract prices up so that calls become overvalued relative to the underlying stock. The arbitrageurs replicate calls by using a particular financial engineering strategy to buy the underlying stock, and then create the desired call option contracts by selling them short. In contrast, if more calls have been created than traders want to hold, call prices will fall so that calls become undervalued relative to the underlying stock. The arbitrageurs will trade stocks and contracts to absorb the excess contracts. Arbitrageurs who use these strategies are financial intermediaries because they connect buyers and sellers who want to trade the same underlying risks but in different forms.

EXAMPLE 15

Dealers and Arbitrageurs

With respect to providing liquidity to market participants, what characteristics most clearly distinguish dealers from arbitrageurs?

Solution:

Dealers provide liquidity to buyers and sellers who arrive at the same market at different times. They move liquidity through time. Arbitrageurs provide liquidity to buyers and sellers who arrive at different markets at the same time. They move liquidity across markets.

8. SECURITIZERS, DEPOSITORY INSTITUTIONS AND INSURANCE COMPANIES

LEARNING OUTCOME

The candidate should be able to:

□ **describe types of financial intermediaries and services that they provide**

Banks and investment companies create new financial products when they buy and repackage securities or other assets. For example, mortgage banks commonly originate hundreds or thousands of residential mortgages by lending money to homeowners. They then place the mortgages in a pool and sell shares of the pool to investors as mortgage pass-through securities, which are also known as mortgage-backed securities. All payments of principal and interest are passed through to the investors each month, after deducting the costs of servicing the mortgages. Investors who purchase these pass-through securities obtain securities that in aggregate have the same net cash flows and associated risks as the pool of mortgages.

The process of buying assets, placing them in a pool, and then selling securities that represent ownership of the pool is called securitization.

Mortgage-backed securities have the advantage that default losses and early repayments are much more predictable for a diversified portfolio of mortgages than they are for individual mortgages. They are also attractive to investors who cannot efficiently service mortgages but wish to invest in mortgages. By securitizing mortgage pools, the mortgage banks allow investors who are not large enough to buy hundreds of mortgages to obtain the benefits of diversification and economies of scale in loan servicing.

Securitization greatly improves liquidity in the mortgage markets because it allows investors in the pass-through securities to buy mortgages indirectly that they otherwise would not buy. Because the financial risks associated with mortgage-backed securities (debt securities with specified claims on the cash flows of a portfolio of mortgages) are much more predictable than those of individual mortgages, mortgage-backed securities are easier to price and thus easier to sell when investors need to raise cash. These characteristics make the market for mortgage-backed securities much more liquid than the market for individual mortgages. Because investors value liquidity—the ability to sell when they want to—they will pay more for securitized mortgages than for individual mortgages. The homeowners benefit because higher mortgage prices imply lower interest rates.

The mortgage bank is a financial intermediary because it connects investors who want to buy mortgages to homeowners who want to borrow money. The homeowners sell mortgages to the bank when the bank lends them money.

Some mortgage banks form mortgage pools from mortgages that they buy from other banks that originate the loans. These mortgage banks are also financial intermediaries because they connect sellers of mortgages to buyers of mortgage-backed securities. Although the sellers of the mortgages are the originating lenders and not the borrowers, the benefits of creating liquid mortgage-backed securities ultimately flow back to the borrowers.

The creation of the pass-through securities generally takes place on the accounts of the mortgage bank. The bank buys mortgages and sells pass-through securities whose values depend on the mortgage pool. The mortgages appear on the bank's accounts as assets and the mortgage-backed securities appear as liabilities.

In many securitizations, the financial intermediary avoids placing the assets and liabilities on its balance sheet by setting up a special corporation or trust that buys the assets and issues the securities. That corporation or trust is called a **special purpose vehicle** (SPV) or alternatively a **special purpose entity** (SPE). Conducting a securitization through a special purpose vehicle is advantageous to investors because their interests in the asset pool are better protected in an SPV than they would be on the balance sheet of the financial intermediary if the financial intermediary were to go bankrupt.

Financial intermediaries securitize many assets. Besides mortgages, banks securitize car loans, credit card receivables, bank loans, and airplane leases, to name just a few assets. As a class, these securities are called asset-backed securities.

When financial intermediaries securitize assets, they often create several classes of securities, called tranches, that have different rights to the cash flows from the asset pool. The tranches are structured so that some produce more predictable cash flows than do others. The senior tranches have first rights to the cash flow from the asset pool. Because the overall risk of a given asset pool cannot be changed, the more junior tranches bear a disproportionate share of the risk of the pool. Practitioners often call the most junior tranche toxic waste because it is so risky. The complexity associated with slicing asset pools into tranches can make the resulting securities difficult to value. Mistakes in valuing these securities contributed to the financial crisis that started in 2007.

Investment companies also create pass-through securities based on investment pools. For example, an exchange-traded fund is an asset-backed security that represents ownership in the securities and contracts held by the fund. The shareholders benefit from the securitization because they can buy or sell an entire portfolio in a single transaction. Because the transaction cost savings are quite substantial, exchange-traded funds often trade in very liquid markets. The investment companies (and sometimes the arbitrageurs) that create exchange-traded funds are financial intermediaries because they connect the buyers of the funds to the sellers of the assets that make up the fund portfolios.

More generally, the creators of all pooled investment vehicles are financial intermediaries that transform portfolios of securities and contracts into securities that represent undivided ownership of the portfolios. The investors in these funds thus indirectly invest in the securities held by the fund. They benefit from the expertise of the investment manager and from obtaining a portfolio that may be more diversified than one they might otherwise be able to hold.

8.1. Depository Institutions and Other Financial Corporations

Depository institutions include commercial banks, savings and loan banks, credit unions, and similar institutions that raise funds from depositors and other investors and lend it to borrowers. The banks give their depositors interest and transaction services, such as check writing and check cashing, in exchange for using their money. They may also raise funds by selling bonds or equity in the bank.

These banks are financial intermediaries because they transfer funds from their depositors and investors to their borrowers. The depositors and investors benefit because they obtain a return (in interest, transaction services, dividends, or capital appreciation) on their funds without having to contract with the borrowers and manage their loans. The borrowers benefit because they obtain the funds that they need without having to search for investors who will trust them to repay their loans.

Many other financial corporations provide credit services. For example, acceptance corporations, discount corporations, payday advance corporations, and factors provide credit to borrowers by lending them money secured by such assets as consumer loans, machinery, future paychecks, or accounts receivables. They finance these loans by selling commercial paper, bonds, and shares to investors. These corporations are intermediaries because they connect investors to borrowers. The investors obtain investments secured by a diversified portfolio of loans while the borrowers obtain funds without having to search for investors.

Brokers also act as financial intermediaries when they lend funds to clients who want to buy securities on margin. They generally obtain the funds from other clients who deposit them in their accounts. Brokers who provide these services to hedge funds and other similar institutions are called prime brokers.

Banks, financial corporations, and brokers can only raise money from depositors and other lenders because their equity owners retain residual interests in the performance of the loans that they make. If the borrowers default, the depositors and other lenders have priority claims over the equity owners. If insufficient money is collected from the borrowers, shareholders' equity is used to pay their depositors and other lenders. The risk of losing capital focuses the equity owners' and management's attention so that credit is not offered foolishly.

Because the ability of these companies to cover their credit losses is limited by the capital that their owners invest in them, the depositors and other investors who lend them money pay close attention to how much money the owners have at risk. For example, if a finance corporation is poorly capitalized, its shareholders will lose little if its clients default on the loans that the finance corporation makes to them. In that case, the finance corporation will have little incentive to lend only to creditworthy borrowers and

to effectively manage collection on those loans once they have been made. Worse, it may even choose to lend to borrowers with poor credit because the interest rates that they can charge such borrowers are higher. Until those loans default, the higher income will make the corporation appear to be more profitable than it actually is. Depositors and other investors are aware of these problems and generally pay close attention to them. Accordingly, poorly capitalized financial institutions cannot easily borrow money to finance their operations at favorable rates.

Depository banks and financial corporations are similar to securitized asset pools that issue pass-through securities. Their depositors and investors own securities that ultimately are backed by an asset pool consisting of their loan portfolios. The depositors generally hold the most senior tranche, followed by the other creditors. The shareholders hold the most junior tranche. In the event of bankruptcy, they are paid only if everyone else is paid.

EXAMPLE 16

Commercial Banks

What services do commercial banks provide that make them financial intermediaries?

Solution:

Commercial banks collect deposits from investors and lend them to borrowers. They are intermediaries because they connect lenders to borrowers. Commercial banks also provide transaction services that make it easier for the banks' depository customers to pay bills and collect funds from their own customers.

8.2. Insurance Companies

Insurance companies help people and companies offset risks that concern them. They do this by creating insurance contracts (policies) that provide a payment in the event that some loss occurs. The insured buy these contracts to hedge against potential losses. Common examples of insurance contracts include auto, fire, life, liability, medical, theft, and disaster insurance contracts.

Credit default swaps are also insurance contracts, but historically they have not been subject to the same reserve requirements that most governments apply to more traditional insurance contracts. They may be sold by insurance companies or by other financial entities, such as investment banks or hedge funds.

Insurance contracts transfer risk from those who buy the contracts to those who sell them. Although insurance companies occasionally broker trades between the insured and the insurer, they more commonly provide the insurance themselves. In that case, the insurance company's owners and creditors become the indirect insurers of the risks that the insurance company assumes. Insurance companies also often transfer risks that they do not wish to bear by buying reinsurance policies from reinsurers.

Insurers are financial intermediaries because they connect the buyers of their insurance contracts with investors, creditors, and reinsurers who are willing to bear the insured risks. The buyers benefit because they can easily obtain the risk transfers that they seek without searching for entities that would be willing to assume those risks.

The owners, creditors, and reinsurers of the insurance company benefit because the company allows them to sell their tolerance for risk easily without having to manage the insurance contracts. Instead, the company manages the relationships with the insured—primarily collections and claims—and hopefully controls the various problems—fraud, moral hazard, and adverse selection—that often plague insurance markets. Fraud occurs when people deliberately cause or falsely report losses to collect on insurance. Moral hazard occurs when people are less careful about avoiding insured losses than they would be if they were not insured so that losses occur more often than they would otherwise. Adverse selection occurs when only those who are most at risk buy insurance so that insured losses tend to be greater than average.

Everyone benefits because insurance companies hold large diversified portfolios of policies. Loss rates for well-diversified portfolios of insurance contracts are much more predictable than for single contracts. For such contracts as auto insurance in which losses are almost uncorrelated across policies, diversification ensures that the financial performance of a large portfolio of contracts will be quite predictable and so holding the portfolio will not be very risky. The insured benefit because they do not have to pay the insurers much to compensate them for bearing risk (the expected loss is quite predictable so the risk is relatively low). Instead, their insurance premiums primarily reflect the expected loss rate in the portfolio plus the costs of running and financing the company.

9. SETTLEMENT AND CUSTODIAL SERVICES AND SUMMARY

LEARNING OUTCOME

The candidate should be able to:

- ❑ **describe types of financial intermediaries and services that they provide**

In addition to connecting buyers to sellers through a variety of direct and indirect means, financial intermediaries also help their customers settle their trades and ensure that the resulting positions are not stolen or pledged more than once as collateral.

Clearinghouses arrange for final settlement of trades. In futures markets, they guarantee contract performance. In other markets, they may act only as escrow agents, transferring money from the buyer to the seller while transferring securities from the seller to the buyer.

The members of a clearinghouse are the only traders for whom the clearinghouse will settle trades. To ensure that their members settle the trades that they present to the clearinghouse, clearinghouses require that their members have adequate capital and post-performance bonds (margins). Clearinghouses also limit the aggregate net (buy minus sell) quantities that their members can settle.

Brokers and dealers who are not members of the clearinghouse must arrange to have a clearinghouse member settle their trades. To ensure that the non-member brokers and dealers can settle their trades, clearinghouse members require that their customers (the non-member brokers and dealers) have adequate capital and post-margins. They also limit the aggregate net quantities that their customers can settle and they monitor their customers' trading to ensure that they do not arrange trades that they cannot settle.

Brokers and dealers similarly monitor the trades made by their retail and institutional customers, and regulate their customers to ensure that they do not arrange trades that they cannot settle.

This hierarchical system of responsibility generally ensures that traders settle their trades. The brokers and dealers guarantee settlement of the trades they arrange for their retail and institutional customers. The clearinghouse members guarantee settlement of the trades that their customers present to them, and clearinghouses guarantee settlement of all trades presented to them by their members. If a clearinghouse member fails to settle a trade, the clearinghouse settles the trade using its own capital or capital drafted from the other members.

Reliable settlement of all trades is extremely important to a well-functioning financial system because it allows strangers to confidently contract with each other without worrying too much about **counterparty risk**, the risk that their counterparties will not settle their trades. A secure clearinghouse system thus greatly increases liquidity because it greatly increases the number of counterparties with whom a trader can safely arrange a trade.

In many national markets, clearinghouses clear all securities trades so that traders can trade securities through any exchange, broker, alternative trading system, or dealer. These clearinghouse systems promote competition among these exchange service providers.

In contrast, most futures exchanges have their own clearinghouses. These clearinghouses usually will not accept trades arranged away from their exchanges so that a competing exchange cannot trade another exchange's contracts. Competing exchanges may create similar contracts, but moving traders from one established market to a new market is extraordinarily difficult because traders prefer to trade where other traders trade.

Depositories or custodians hold securities on behalf of their clients. These services, which are often offered by banks, help prevent the loss of securities through fraud, oversight, or natural disaster. Broker-dealers also often hold securities on behalf of their customers so that the customers do not have to hold the securities in certificate form. To avoid problems with lost certificates, securities increasingly are issued only in electronic form.

EXAMPLE 17

Financial Intermediaries

As a relatively new member of the business community, you decide it would be advantageous to join the local lunch club to network with businessmen. Upon learning that you are a financial analyst, club members soon enlist you to give a lunch speech. During the question and answer session afterwards, a member of the audience asks, "I keep reading in the newspaper about the need to regulate 'financial intermediaries', but really don't understand exactly what they are. Can you tell me?" How do you answer?

Solution:

Financial intermediaries are companies that help their clients achieve their financial goals. They are called intermediaries because, in some way or another, they stand between two or more people who would like to trade with each other, but for various reasons find it difficult to do so directly. The intermediary arranges the trade for them, or more often, trades with both sides.

For example, a commercial bank is an intermediary that connects investors with money to borrowers who need money. The investors buy certificates of deposit from the bank, buy bonds or stock issued by the bank, or simply are depositors in the bank. The borrowers borrow this money from the bank when they arrange loans. Without the bank's intermediation, the investors would have to find trustworthy borrowers themselves, which would be difficult, and the borrowers would have to find trusting lenders, which would also be difficult.

Similarly, an insurance company is an intermediary because it connects customers who want to insure risks with investors who are willing to bear those risks. The investors own shares or bonds issued by the insurance company, or they have sold reinsurance contracts to the insurance company. The insured benefit because they can more easily buy a policy from an insurance company than they can find counterparties who would be willing to bear their risks. The investors benefit because the insurance company creates a diversified portfolio of risks by selling insurance to thousands or millions of customers. Diversification ensures that the net risk borne by the insurance company and its investors will be predictable and thus financially manageable.

In both cases, the financial intermediary also manages the relationships with its customers and investors so that neither side has to worry about the credit-worthiness or trust-worthiness of its counterparties. For example, the bank manages credit quality and collections on its loans and the insurance company manages risk exposure and collections on its policies. These services benefit both sides by reducing the costs of connecting investors to borrowers or of insured to insurers.

These are only two examples of financial intermediation. Many others involve firms engaged in brokerage, dealing, arbitrage, securitization, investment management, and the clearing and settlement of trades. In all cases, the financial intermediary stands between a buyer and a seller, offering them services that allow them to better achieve their financial goals in a cost effective and efficient manner.

9.1. Summary

By facilitating transactions among buyers and sellers, financial intermediaries provide services essential to a well-functioning financial system. They facilitate transactions the following ways:

1. Brokers, exchanges, and various alternative trading systems match buyers and sellers interested in trading the same instrument at the same place and time. These financial intermediaries specialize in discovering and organizing information about who wants to trade.
2. Dealers and arbitrageurs connect buyers to sellers interested in trading the same instrument but who are not present at the same place and time. Dealers connect buyers to sellers who are present at the same place but at different times whereas arbitrageurs connect buyers to sellers who are present at the same time but in different places. These financial intermediaries trade for their own accounts when providing these services. Dealers buy or sell with one client and hope to do the offsetting transaction later with another client. Arbitrageurs buy from a seller in one market while simultaneously selling to a buyer in another market.
3. Many financial intermediaries create new instruments that depend on the cash flows and associated financial risks of other instruments. The intermediaries provide these services when they securitize assets, manage investment funds, operate banks and other finance corporations that offer investments to investors and loans to borrowers, and operate insurance companies that pool risks. The instruments that they create generally are more attractive to their clients than the instruments on which they are based. The new instruments also may be differentiated to appeal to diverse clienteles. Their efforts connect buyers of one or more instruments to sellers of other instruments, all of which in aggregate provide the same cash flows and risk exposures. Financial intermediaries thus effectively arrange trades among traders who otherwise would not trade with each other.
4. Arbitrageurs who conduct arbitrage among securities and contracts whose values depend on common factors convert risk from one form to another. Their trading connects buyers and sellers who want to trade similar risks expressed in different forms.
5. Banks, clearinghouses, and depositories provide services that ensure traders settle their trades and that the resulting positions are not stolen or pledged more than once as collateral.

10. POSITIONS AND SHORT POSITIONS

LEARNING OUTCOME

The candidate should be able to:

- ☐ **compare positions an investor can take in an asset**

People generally solve their financial and risk management problems by taking positions in various assets or contracts. A **position** in an asset is the quantity of the instrument that an entity owns or owes. A portfolio consists of a set of positions.

People have **long positions** when they own assets or contracts. Examples of long positions include ownership of stocks, bonds, currencies, contracts, commodities, or real assets. Long positions benefit from an appreciation in the prices of the assets or contracts owned.

People have **short positions** when they have sold assets that they do not own, or when they write and sell contracts. Short positions benefit from a decrease in the prices of the assets or contracts sold. Short sellers profit by selling at high prices and repurchasing at lower prices. Information-motivated traders sell assets and contracts short positions when they believe that prices will fall.

Hedgers also often sell instruments short. They short securities and contracts when the financial risks inherent in the instruments are positively correlated with the risks to which they are exposed. For

example, to hedge the risk associated with holding copper inventories, a wire manufacturer would sell short copper futures. If the price of copper falls, the manufacturer will lose on his copper inventories but gain on his short futures position. (If the risk in an instrument is inversely correlated with a risk to which hedgers are exposed, the hedgers will hedge with long positions.)

Contracts have long sides and short sides. The long side of a forward or futures contract is the side that will take physical delivery or its cash equivalent. The short side of such contracts is the side that is liable for the delivery. The long side of a futures contract increases in value when the value of the underlying asset increases in value.

The identification of the two sides can be confusing for option contracts. The long side of an option contract is the side that holds the right to exercise the option. The short side is the side that must satisfy the obligation. Practitioners say that the long side *holds* the option and the short side *writes* the option, so the long side is the holder and the short side is the writer. The put contracts are the source of the potential confusion. The put contract holder has the right to sell the underlying to the writer. The holder will benefit if the price of the underlying falls, in which case the price of the put contract will rise. The holder is long the put contract and has an indirect short position in the underlying instrument. Analysts call the indirect short position short exposure to the underlying. The put contract holders have long exposure to their option contract and short exposure to the underlying instrument.

Exhibit 1: Option Positions and Their Associated Underlying Risk Exposures

Type of Option	Option Position	Exposure to Underlying Risk
Call	Long	Long
Call	Short	Short
Put	Long	Short
Put	Short	Long

The identification of the long side in a swap contract is often arbitrary because swap contracts call for the exchange of contractually determined cash flows rather than for the purchase (or the cash equivalent) of some underlying instrument. In general, the side that benefits from an increase in the quoted price is the long side.

The identification of the long side in currency contracts also may be confusing. In this case, the confusion stems from symmetry in the contracts. The buyer of one currency is the seller of the other currency, and vice versa for the seller. Thus, a long forward position in one currency is a short forward position in the other currency. When practitioners describe a position, they generally will say, "I'm long the dollar against the yen," which means they have bought dollars and sold yen.

10.1. Short Positions

Short sellers create short positions in contracts by selling contracts that they do not own. In a sense, they become the issuers of the contract when they create the liabilities associated with their contracts. This analogy will also help you better understand risk when you study corporate finance: Corporations create short positions in their bonds when they issue bonds in exchange for cash. Although bonds are generally considered to be securities, they are also contracts between the issuer and the bondholder.

Short sellers create short positions in securities by borrowing securities from security lenders who are long holders. The short sellers then sell the borrowed securities to other traders. Short sellers close their positions by repurchasing the securities and returning them to the security lenders. If the securities drop in value, the short sellers profit because they repurchase the securities at lower prices than the prices at which they sold the securities. If the securities rise in value, they will lose. Short sellers who buy to close their positions are said to cover their positions.

The potential gains in a long position generally are unbounded. For example, the stock prices of such highly successful companies as Yahoo! have increased more than 50-fold since they were first publicly

traded. The potential losses on long positions, however, are limited to no more than 100 percent—a complete loss—for long positions without any associated liabilities.

In contrast, the potential gains on a short position are limited to no more than 100 percent whereas the potential losses are unbounded. The unbounded potential losses on short positions make short positions very risky in volatile instruments. As an extreme example of this, if you had shorted 100 shares of Yahoo! in July 1996 at \$20 and kept the position open for four years, you would have lost \$148,000 on your \$2,000 initial short position. During this period, Yahoo! rose 75-fold to \$1,500 on a split-adjusted equivalent basis.

Although security lenders generally believe that they are long the securities that they lend, in fact, they do not actually own the securities during the periods of their loans. Instead, they own promises made by the short sellers to return the securities. These promises are memorialized in security lending agreements. These agreements specify that the short sellers will pay the long sellers all dividends or interest that they otherwise would have received had they not lent their securities. These payments are called payments-in-lieu of dividends (or of interest), and they may have different tax treatments than actual dividends and interest. The security lending agreements also protect the lenders in the event of a stock split.

To secure the security loans, lenders require that the short seller leave the proceeds of the short sale on deposit with them as collateral for the stock loan. They invest the collateral in short-term securities, and they rebate the interest to the short sellers at rates called short rebate rates. The short rebate rates are determined in the market and generally are available only to institutional short-sellers and some large retail traders. If a security is hard to borrow, the rebate rate may be very small or even negative. Such securities are said to be “on special”. Most security lending agreements require various margin payments to keep the credit risk among the parties from growing when prices change.

Securities lenders lend their securities because the short rebate rates they pay on the collateral are lower than the interest rates they receive from investing the collateral. The difference is because of the implicit loan fees that they receive from the borrowers for borrowing the stock. The difference also compensates lenders for risks that the lenders take when investing the collateral and for the risk that the borrowers will default if prices rise significantly.

EXAMPLE 18

Short Positions in Securities and Contracts

How is the process of short selling shares of Siemens different from that of short selling a Siemens equity call option contract?

Solution:

To short sell shares of Siemens, the seller (or his broker) must borrow the shares from a long holder so that he can deliver them to the buyer. To short sell a Siemens equity call option contract, the seller simply creates the contract when he sells it to the buyer.

11. LEVERAGED POSITIONS

LEARNING OUTCOME

The candidate should be able to:

□ **calculate and interpret the leverage ratio, the rate of return on a margin transaction, and the security price at which the investor would receive a margin call**

In many markets, traders can buy securities by borrowing some of the purchase price. They usually borrow the money from their brokers. The borrowed money is called the **margin loan**, and they are said to buy on margin. The interest rate that the buyers pay for their margin loan is called the **call money rate**. The call money rate is above the government bill rate and is negotiable. Large buyers generally obtain more favorable rates than do retail buyers. For institutional-size buyers, the call money rate is quite low because the loans are generally well secured by securities held as collateral by the lender.

Trader's equity is that portion of the security price that the buyer must supply. Traders who buy securities on margin are subject to minimum margin requirements. The **initial margin requirement** is the minimum fraction of the purchase price that must be trader's equity. This requirement may be set by the government, the exchange, or the exchange clearinghouse. For example, in the United States, the Federal Reserve Board sets the initial margin requirement through Regulation T. In Hong Kong SAR, the Securities and Futures Commission sets the margin requirements. In all markets, brokers often require more equity than the government-required minimum from their clients when lending to them.

Many markets allow brokers to lend their clients more money if the brokers use risk models to measure and control the overall risk of their clients' portfolios. This system is called portfolio margining.

Buying securities on margin can greatly increase the potential gains or losses for a given amount of equity in a position because the trader can buy more securities on margin than he could otherwise. The buyer thus earns greater profits when prices rise and suffers greater losses when prices fall. The relation between risk and borrowing is called **financial leverage** (often simply called leverage). Traders leverage their positions when they borrow to buy more securities. A highly leveraged position is large relative to the equity that supports it.

The leverage ratio is the ratio of the value of the position to the value of the equity investment in it. The leverage ratio indicates how many times larger a position is than the equity that supports it. The maximum leverage ratio associated with a position financed by the minimum margin requirement is one divided by the minimum margin requirement. If the requirement is 40 percent, then the maximum leverage ratio is $2.5 = 100\% \text{ position} \div 40\% \text{ equity}$.

The leverage ratio indicates how much more risky a leveraged position is relative to an unleveraged position. For example, if a stock bought on 40 percent margin rises 10 percent, the buyer will experience a 25 percent ($2.5 \times 10\%$) return on the equity investment in her leveraged position. But if the stock falls by 10 percent, the return on the equity investment will be -25 percent (before the interest on the margin loan and before payment of commissions).

Financial analysts must be able to compute the total return to the equity investment in a leveraged position. The total return depends on the price change of the purchased security, the dividends or interest paid by the security, the interest paid on the margin loan, and the commissions paid to buy and sell the security. The following example illustrates the computation of the total return to a leveraged purchase of stock that pays a dividend.

EXAMPLE 19

Computing Total Return to a Leveraged Stock Purchase

A buyer buys stock on margin and holds the position for exactly one year, during which time the stock pays a dividend. For simplicity, assume that the interest on the loan and the dividend are both paid at the end of the year.

Purchase price	\$20/share
Sale price	\$15/share
Shares purchased	1,000
Leverage ratio	2.5
Call money rate	5%

Dividend	\$0.10/share
Commission	\$0.01/share

1. What is the total return on this investment?

Solution to 1:

To find the return on this investment, first determine the initial equity and then determine the equity remaining after the sale. The total purchase price is \$20,000. The leverage ratio of 2.5 indicates that the buyer's equity financed 40 percent = $(1 \div 2.5)$ of the purchase price. Thus, the equity investment is \$8,000 = 40% of \$20,000. The \$12,000 remainder is borrowed. The actual investment is slightly higher because the buyer must pay a commission of \$10 = $\$0.01/\text{share} \times 1,000 \text{ shares}$ to buy the stock. The total initial investment is \$8,010.

At the end of the year, the stock price has declined by \$5/share. The buyer lost \$5,000 = $\$5/\text{share} \times 1,000 \text{ shares}$ as a result of the price change. In addition, the buyer has to pay interest at 5 percent on the \$12,000 loan, or \$600. The buyer also receives a dividend of \$0.10/share, or \$100. The trader's equity remaining after the sale is computed from the initial equity investment as follows:

Initial investment	\$8,010
Purchase commission	-10
Trading gains/losses	-5,000
Margin interest paid	-600
Dividends received	100
Sales commission paid	-10
Remaining equity	\$2,490

or

Proceeds on sale	\$15,000
Payoff loan	-12,000
Margin interest paid	-600
Dividends received	100
Sales commission paid	-10
Remaining equity	\$2,490

so that the return on the initial investment of \$8,010 is $(2,490 - 8,010)/8,010 = -68.9\%$.

2. Why is the loss greater than the 25 percent decrease in the market price?

Solution to 2:

The realized loss is substantially greater than the stock price return of $(\$15 - \$20)/\$20 = -25\%$. Most of the difference is because of the leverage with the remainder primarily the result of the interest paid on the loan. Based on the leverage alone and ignoring the other cash flows, we would expect that the return on the equity would be $-62.5\% = 2.5 \text{ leverage times the } -25\% \text{ stock price return}$.

In the above example, if the stock dropped more than the buyer's original 40 percent margin (ignoring commissions, interest, and dividends), the trader's equity would have become negative. In that case, the investor would owe his broker more than the stock is worth. Brokers often lose money in such situations if the buyer does not repay the loan out of other funds.

To prevent such losses, brokers require that margin buyers always have a minimum amount of equity in their positions. This minimum is called the **maintenance margin requirement**. It is usually 25

percent of the current value of the position, but it may be higher or lower depending on the volatility of the instrument and the policies of the broker.

If the value of the equity falls below the maintenance margin requirement, the buyer will receive a **margin call**, or request for additional equity. If the buyer does not deposit additional equity with the broker in a timely manner, the broker will close the position to prevent further losses and thereby secure repayment of the margin loan.

When you buy securities on margin, you must know the price at which you will receive a margin call if prices drop. The answer to this question depends on your initial equity and on the maintenance margin requirement.

EXAMPLE 20

Margin Call Price

A trader buys stock on margin posting 40 percent of the initial stock price of \$20 as equity. The maintenance margin requirement for the position is 25 percent. Below what price will a margin call occur?

Solution:

The trader's initial equity is 40 percent of the initial stock price of \$20, or \$8 per share. Subsequent changes in equity per share are equal to the share price change so that equity per share is equal to $\$8 + (P - 20)$ where P is the current share price. The margin call takes place when equity drops below the 25 percent maintenance margin requirement. The price below which a margin call will take place is the solution to the following equation:

$$\frac{\text{Equity/share}}{\text{Price/share}} = \frac{\$8 + P - 20}{P} = 25\%$$

which occurs at $P = 16$. When the price drops below \$16, the equity will be under \$4/share, which is less than 25 percent of the price.

Traders who sell securities short are also subject to margin requirements because they have borrowed securities. Initially, the trader's equity supporting the short position must be at least equal to the margin requirement times the initial value of the short position. If prices rise, equity will be lost. At some point, the short seller will have to contribute additional equity to meet the maintenance margin requirement. Otherwise, the broker will buy the security back to cover the short position to prevent further losses and thereby secure repayment of the stock loan.

12. ORDERS AND EXECUTION INSTRUCTIONS

LEARNING OUTCOMES

The candidate should be able to:

- ☐ compare execution, validity, and clearing instructions
- ☐ compare market orders with limit orders

Buyers and sellers communicate with the brokers, exchanges, and dealers that arrange their trades by issuing **orders**. All orders specify what instrument to trade, how much to trade, and whether to buy or sell. Most orders also have other instructions attached to them. These additional instructions may include execution instructions, validity instructions, and clearing instructions. **Execution**

instructions indicate how to fill the order, **validity instructions** indicate when the order may be filled, and **clearing instructions** indicate how to arrange the final settlement of the trade.

In this section, we introduce various order instructions and explain how traders use them to achieve their objectives. We discuss execution mechanisms—how exchanges, brokers and dealers fill orders—in the next section. To understand the concepts in this section, however, you need to know a little about order execution mechanisms.

In most markets, dealers and various other proprietary traders often are willing to buy from, or sell to, other traders seeking to sell or buy. The prices at which they are willing to buy are called **bid** prices and those at which they are willing to sell are called **ask** prices, or sometimes **offer** prices. The ask prices are invariably higher than the bid prices.

The traders who are willing to trade at various prices may also indicate the quantities that they will trade at those prices. These quantities are called **bid sizes** and **ask sizes** depending on whether they are attached to bids or offers.

Practitioners say that the traders who offer to trade make a market. Those who trade with them take the market.

The highest bid in the market is the **best bid**, and the lowest ask in the market is the **best offer**. The difference between the best bid and the best offer is the **market bid–ask spread**. When traders ask, “What’s the market?” they want to know the best bid and ask prices and their associated sizes. Bid–ask spreads are an implicit cost of trading. Markets with small bid–ask spreads are markets in which the costs of trading are small, at least for the sizes quoted. Dealers often quote both bid and ask prices, and in that case, practitioners say that they quote a two-sided market. The market spread is never more than any dealer spread.

12.1. Execution Instructions

Market and limit orders convey the most common execution instructions. A **market order** instructs the broker or exchange to obtain the best price immediately available when filling the order. A **limit order** conveys almost the same instruction: Obtain the best price immediately available, but in no event accept a price higher than a specified limit price when buying or accept a price lower than a specified limit price when selling.

Many people mistakenly believe that limit orders specify the prices at which the orders will trade. Although limit orders do often trade at their limit prices, remember that the first instruction is to obtain the best price available. If better prices are available than the limit price, brokers and exchanges should obtain those prices for their clients.

Market orders generally execute immediately if other traders are willing to take the other side of the trade. The main drawback with market orders is that they can be expensive to execute, especially when the order is placed in a market for a thinly traded security, or more generally, when the order is large relative to the normal trading activity in the market. In that case, a market buy order may fill at a high price, or a market sell order may fill at a low price if no traders are willing to trade at better prices. High purchase prices and low sale prices represent price concessions given to other traders to encourage them to take the other side of the trade. Because the sizes of price concessions can be difficult to predict, and because prices often change between when a trader submits an order and when the order finally fills, the execution prices for market orders are often uncertain.

Buyers and sellers who are concerned about the possibility of trading at unacceptable prices add limit price instructions to their orders. The main problem with limit orders is that they may not execute. Limit orders do not execute if the limit price on a buy order is too low, or if the limit price on a sell order is too high. For example, if an investment manager submits a limit order to buy at the limit price of 20 (buy limit 20) and nobody is willing to sell at or below 20, the order will not trade. If prices never drop to 20, the manager will never buy. If the price subsequently rises, the manager will have lost the opportunity to profit from the price rise.

Whether traders use market orders or limit orders when trying to arrange trades depends on their concerns about price, trading quickly, and failing to trade. On average, limit orders trade at better prices

than do market orders, but they often do not trade. Traders generally regret when their limit orders fail to trade because they usually would have profited if they had traded. Limit buy orders do not fill when prices are rising, and limit sell orders do not fill when prices are falling. In both cases, traders would be better off if their orders had filled.

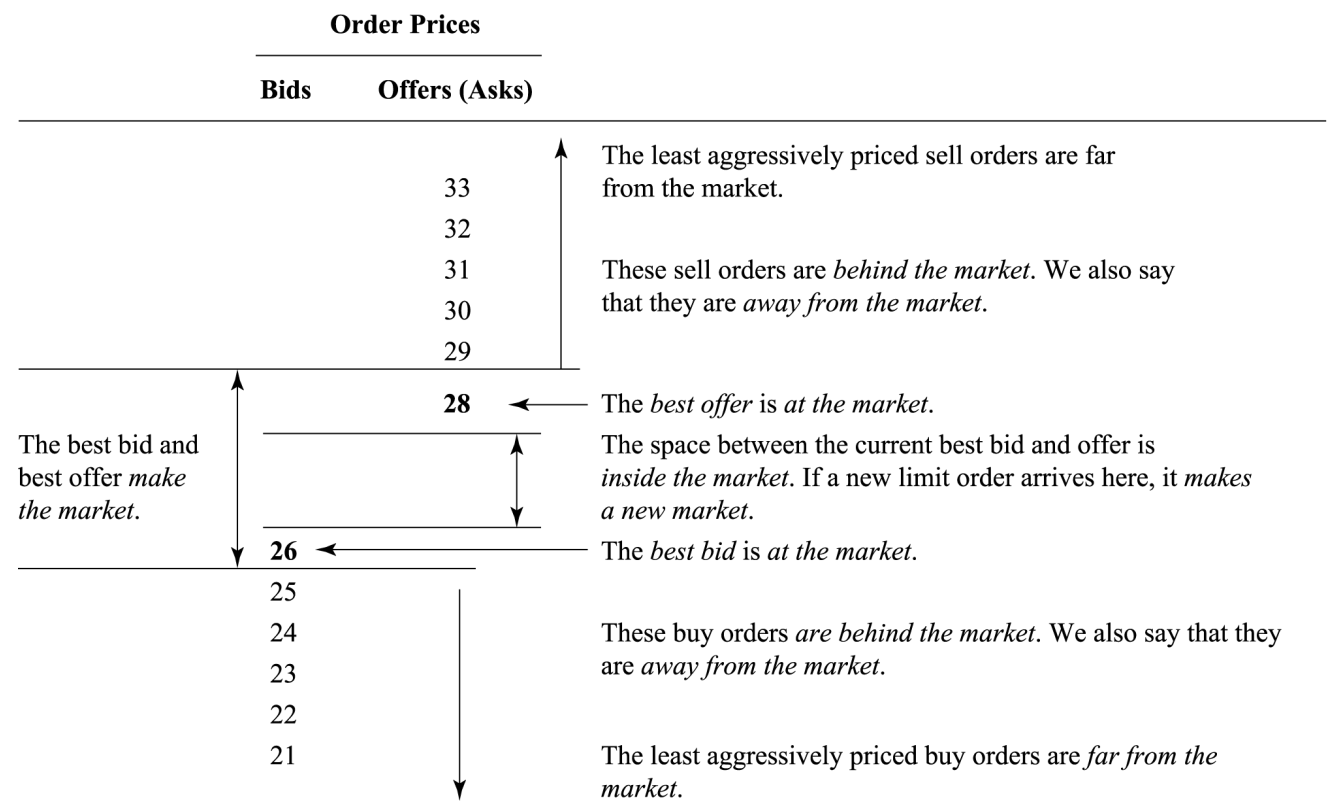
The probability that a limit order will execute depends on where the order is placed relative to market prices. An aggressively priced order is more likely to trade than is a less aggressively priced order. A limit buy order is aggressively priced when the limit price is high relative to the market bid and ask prices. If the limit price is placed above the best offer, the buy order generally will partially or completely fill at the best offer price, depending on the size available at the best offer. Such limit orders are called **marketable limit orders** because at least part of the order can trade immediately. A limit buy order with a very high price relative to the market is essentially a market order.

If the buy order is placed above the best bid but below the best offer, traders say the order makes a new market because it becomes the new best bid. Such orders generally will not immediately trade, but they may attract sellers who are interested in trading. A buy order placed at the best bid is said to make market. It may have to wait until all other buy orders at that price trade first. Finally, a buy order placed below the best bid is **behind the market**. It will not execute unless market prices drop. Traders call limit orders that are waiting to trade **standing limit orders**.

Sell limit orders are aggressively priced if the limit price is low relative to market prices. The limit price of a marketable sell limit order is below the best bid. A limit sell order placed between the best bid and the best offer makes a new market on the sell side, one placed at the best offer makes market, and one placed above the best offer is behind the market.

Exhibit 2 presents a simplified **limit order book** in which orders are presented ranked by their limit prices for a hypothetical market. The market is “26 bid, offered at 28” because the best bid is 26 and the best offer (ask) is 28.

Exhibit 2: Terms Traders Use to Describe Standing Limit Orders



Source: *Trading and Exchanges*.¹

EXAMPLE 21

Making and Taking

1. What is the difference between making a market and taking a market?

Solution to 1:

A trader makes a market when the trader offers to trade. A trader takes a market when the trader accepts an offer to trade.

2. What order types are most likely associated with making a market and taking a market?

Solution to 2:

Traders place standing limit orders to give other traders opportunities to trade. Standing limit orders thus make markets. In contrast, traders use market orders or marketable limit orders to take offers to trade. These marketable orders take the market.

A trade-off exists between how aggressively priced an order is and the ultimate trade price. Although aggressively priced orders fill faster and with more certainty than do less aggressively priced limit orders, the prices at which they execute are inferior. Buyers seeking to trade quickly must pay higher prices to increase the probability of trading quickly. Similarly, sellers seeking to trade quickly must accept lower prices to increase the probability of trading quickly.

Some order execution instructions specify conditions on size. For example, **all-or-nothing (AON) orders** can only trade if their entire sizes can be traded. Traders can similarly specify minimum fill sizes. This specification is common when settlement costs depend on the number of trades made to fill an order and not on the aggregate size of the order.

Exposure instructions indicate whether, how, and perhaps to whom orders should be exposed. **Hidden orders** are exposed only to the brokers or exchanges that receive them. These agencies cannot disclose hidden orders to other traders until they can fill them. Traders use hidden orders when they are afraid that other traders might behave strategically if they knew that a large order was in the market. Traders can discover hidden size only by submitting orders that will trade with that size. Thus, traders can only learn about hidden size after they have committed to trading with it.

Traders also often indicate a specific **display size** for their orders. Brokers and exchanges then expose only the display size for these orders. Any additional size is hidden from the public but can be filled if a suitably large order arrives. Traders sometimes call such orders **iceberg orders** because most of the order is hidden. Traders specify display sizes when they do not want to display their full sizes, but still want other traders to know that someone is willing to trade at the displayed price. Traders on the opposite side who wish to trade additional size at that price can discover the hidden size only if they trade the displayed size, at which point the broker or exchange will display any remaining size up to the display size. They also can discover the hidden size by submitting large orders that will trade with that size.

EXAMPLE 22

Market versus Limit and Hidden versus Displayed Orders

You are the buy-side trader for a very clever investment manager. The manager has hired a commercial satellite firm to take regular pictures of the parking lots in which new car dealers store their inventories. It has also hired some part-time workers to count the cars on the lots. With this information and some econometric analyses, the manager can predict weekly new car sale announcements more accurately than can most analysts. The manager typically makes a quarter percent each week on this strategy. Once a week, a day before the announcements are made, the manager gives you large orders to buy or sell car manufacturers based on his insights into their dealers' sales. What primary issues should you consider when deciding whether to:

1. use market or limit orders to fill his orders?

Solution to 1:

The manager's information is quite perishable. If his orders are not filled before the weekly sales are reported to the public, the manager will lose the opportunity to profit from the information as prices immediately adjust to the news. The manager, therefore, needs to get the orders filled quickly. This consideration suggests that the orders should be submitted as market orders. If submitted as limit orders, the orders might not execute and the firm would lose the opportunity to profit.

Large market orders, however, can be very expensive to execute, especially if few people are willing to trade significant size on the other side of the market. Because transaction costs can easily exceed the expected quarter percent return, you should submit limit orders to limit the execution prices that you are willing to accept. It is better to fail to trade than to trade at losing prices.

2. display the orders or hide them?

Solution to 2:

Your large orders could easily move the market if many people were aware of them, and even more so if others were aware that you are trading on behalf of a successful information-motivated trader. You thus should consider submitting hidden orders. The disadvantage of hidden orders is that they do not let people know that they can trade the other side if they want to.

13. VALIDITY INSTRUCTIONS AND CLEARING INSTRUCTIONS

LEARNING OUTCOME

The candidate should be able to:

☐ **compare execution, validity, and clearing instructions**

Validity instructions indicate when an order may be filled. The most common validity instruction is the **day order**. A day order is good for the day on which it is submitted. If it has not been filled by the close of business, the order expires unfilled.

Good-till-cancelled orders (GTC) are just that. In practice, most brokers limit how long they will manage an order to ensure that they do not fill orders that their clients have forgotten. Such brokers may limit their GTC orders to a few months.

Immediate or cancel orders (IOC) are good only upon receipt by the broker or exchange. If they cannot be filled in part or in whole, they cancel immediately. In some markets these orders are also known as **fill or kill** orders. When searching for hidden liquidity, electronic algorithmic trading systems often submit thousands of these IOC orders for every order that they fill.

Good-on-close orders can only be filled at the close of trading. These orders often are market orders, so traders call them **market-on-close** orders. Traders often use on-close orders when they want to trade at the same prices that will be published as the closing prices of the day. Mutual funds often like to trade at such prices because they value their portfolios at closing prices. Many traders also use **good-on-open** orders.

13.1. Stop Orders

A **stop order** is an order in which a trader has specified a stop price condition. The stop order may not be filled until the stop price condition has been satisfied. For a sell order, the stop price condition suspends execution of the order until a trade occurs at or below the stop price. After that trade, the stop condition is satisfied and the order becomes valid for execution, subject to all other execution instructions attached to it. If the market price subsequently rises above the sell order's stop price before the order trades, the order remains valid. Similarly, a buy order with a stop condition becomes valid only after a price rises above the specified stop price.

Traders often call stop orders **stop-loss orders** because many traders use them with the hope of stopping losses on positions that they have established. For example, a trader who has bought stock at 40 may want to sell the stock if the price falls below 30. In that case, the trader might submit a "GTC, stop 30, market sell" order. If the price falls to or below 30, the market order becomes valid and it should immediately execute at the best price then available in the market. That price may be substantially lower than 30 if the market is falling quickly. The stop-loss order thus does not guarantee a stop to losses at the stop price. If potential sellers are worried about trading at too low of a price, they can attach stop instructions to limit orders instead of market orders. In this example, if the trader is unwilling to sell below 25, the trader would submit a "GTC, stop 30, limit 25 sell" order.

If a trader wants to guarantee that he can sell at 30, the trader would buy a put option contract struck at 30. The purchase price of the option would include a premium for the insurance that the trader is buying. Option contracts can be viewed as limit orders for which execution is guaranteed at the strike price. A trader similarly might use a stop-buy order or a call option to limit losses on a short position.

A portfolio manager might use a stop-buy order when the manager believes that a security is undervalued but is unwilling to trade without market confirmation. For example, suppose that a stock currently trades for 50 RMB and a manager believes that it should be worth 100 RMB. Further, the manager believes that the stock will much more likely be worth 100 RMB if other traders are willing to buy it above 65 RMB. To best take advantage of this information, the manager would consider issuing a "GTC, stop 65 RMB, limit 100 RMB buy" order. Note that if the manager relies too much on the market when making this trading decision, however, he may violate CFA Standard of Professional Conduct V.A.2, which requires that all investment actions have a reasonable and adequate basis supported by appropriate research and investigation.

Because stop-sell orders become valid when prices are falling and stop-buy orders become valid when prices are rising, traders using stop orders contribute to market momentum as their sell orders push prices down further and their buy orders push prices up. Execution prices for stop orders thus are often quite poor.

EXAMPLE 23

Limit and Stop Instructions

In what ways do limit and stop instructions differ?

Solution:

Although both limit and stop instructions specify prices, the role that these prices play in the arrangement of a trade are completely different. A limit price places a limit on what trade prices will be acceptable to the trader. A buyer will accept prices only at or lower than the limit price whereas a seller will accept prices only at or above the limit price.

In contrast, a stop price indicates when an order can be filled. A buy order can only be filled once the market has traded at a price at or above the stop price. A sell order can only be filled once the market has traded at a price at or below the stop price.

Both order instructions may delay or prevent the execution of an order. A buy limit order will not execute until someone is willing to sell at or below the limit price. Similarly, a sell limit order will not execute until someone is willing to buy at or above the limit sell price. In contrast, a stop-buy

order will not execute if the market price never rises to the stop price. Similarly, a stop-sell order will not execute if the market price never falls to the stop price.

13.2. Clearing Instructions

Clearing instructions tell brokers and exchanges how to arrange final settlement of trades. Traders generally do not attach these instructions to each order—instead they provide them as standing instructions. These instructions indicate what entity is responsible for clearing and settling the trade. For retail trades, that entity is the customer's broker. For institutional trades, that entity may be a custodian or another broker. When a client uses one broker to arrange trades and another broker to settle trades, traders say that the first broker gives up the trade to the other broker, who is often known as the prime broker. Institutional traders provide these instructions so they can obtain specialized execution services from different brokers while maintaining a single account for custodial services and other prime brokerage services, such as margin loans.

An important clearing instruction that must appear on security sale orders is an indication of whether the sale is a long sale or a short sale. In either case, the broker representing the sell order must ensure that the trader can deliver securities for settlement. For a long sale, the broker must confirm that the securities held are available for delivery. For a short sale, the broker must either borrow the security on behalf of the client or confirm that the client can borrow the security.

14. PRIMARY SECURITY MARKETS

LEARNING OUTCOME

The candidate should be able to:

- ☐ **define primary and secondary markets and explain how secondary markets support primary markets**

When issuers first sell their securities to investors, practitioners say that the trades take place in the **primary markets**. An issuer makes an **initial public offering** (IPO)—sometimes called a placing—of a security issue when it sells the security to the public for the first time. A seasoned security is a security that an issuer has already issued. If the issuer wants to sell additional units of a previously issued security, it makes a **seasoned offering** (sometimes called a secondary offering). Both types of offerings occur in the **primary market** where issuers sell their securities to investors. Later, if investors trade these securities among themselves, they trade in **secondary markets**. This section discusses primary markets and the procedures that issuers use to offer their securities to the public.

14.1. Public Offerings

Corporations generally contract with an investment bank to help them sell their securities to the public. The investment bank then lines up subscribers who will buy the security. Investment bankers call this process **book building**. In London, the book builder is called the book runner. The bank tries to build a book of orders to which they can sell the offering. Investment banks often support their book building by providing investment information and opinion about the issuer to their clients and to the public. Before the offering, the issuer generally makes a very detailed disclosure of its business, of the risks inherent in it, and of the uses to which the new funds will be placed.

When time is of the essence, issuers in Europe may issue securities through an **accelerated book build**, in which the investment bank arranges the offering in only one or two days. Such sales often occur at discounted prices.

The first public offering of common stock in a company consists of newly issued shares to be sold by the company. It may also include shares that the founders and other early investors in the company seek to sell. The initial public offering provides these investors with a means of liquidating their investments.

In an **underwritten offering**—the most common type of offering—the investment bank guarantees the sale of the issue at an offering price that it negotiates with the issuer. If the issue is undersubscribed, the bank will buy whatever securities it cannot sell at the offering price. In the case of an IPO, the underwriter usually also promises to make a market in the security for about a month to ensure that the secondary market will be liquid and to provide price support, if necessary. For large issues, a syndicate of investment banks and broker-dealers helps the **lead underwriter** build the book. The issuer usually pays an underwriting fee of about 7 percent for these various services. The underwriting fee is a placement cost of the offering.

In a **best effort offering**, the investment bank acts only as broker. If the offering is undersubscribed, the issuer will not sell as much as it hoped to sell.

For both types of offerings, the issuer and the bank usually jointly set the offering price following a negotiation. If they set a price that buyers consider too high, the offering will be undersubscribed, and they will fail to sell the entire issue. If they set the price too low, the offering will be oversubscribed, in which case the securities are often allocated to preferred clients or on a pro-rata basis.

(Note that CFA Standard of Professional Conduct III.B—fair dealing—requires that the allocation be based on a written policy disclosed to clients and suggests that the securities be offered on a pro-rata basis among all clients who have comparable relationships with their broker-dealers.)

Investment banks have a conflict of interest with respect to the offering price in underwritten offerings. As agents for the issuers, they generally are supposed to select the offering price that will raise the most money. But as underwriters, they have strong incentives to choose a low price. If the price is low, the banks can allocate valuable shares to benefit their clients and thereby indirectly benefit the banks. If the price is too high, the underwriters will have to buy overvalued shares in the offering and perhaps also during the following month if they must support the price in the secondary market, which directly costs the banks. These considerations tend to lower initial offering prices so that prices in the secondary market often rise immediately following an IPO. They are less important in a seasoned offering because trading in the secondary market helps identify the proper price for the offering.

First time issuers generally accept lower offering prices because they and many others believe that an undersubscribed IPO conveys very unfavorable information to the market about the company's prospects at a time when it is most vulnerable to public opinion about its prospects. They fear that an undersubscribed initial public offering will make it substantially harder to raise additional capital in subsequent seasoned offerings.

EXAMPLE 24

The Healthybots Initial Public Offering

Healthybots is a health care company that treats diseases using artificial intelligence-based solutions. Healthybots raised approximately £265 million gross through an initial public offering of 103,142,466 ordinary shares at £2.57 per ordinary share. After the initial public offering, Healthybots had 213,333,333 ordinary shares issued and outstanding.

Healthybots received gross proceeds of approximately £34.3 million and net proceeds of £31.8 million. The ordinary shares that were sold to the public represented approximately 48 percent of Healthybots' total issued ordinary shares.

The shares commenced trading at 8:00 a.m. on the AIM market of the London Stock Exchange, where Healthybots opened at £2.74, traded 37 million shares between £2.68 and £2.74, and closed at £2.73.

1. Approximately how many new shares were issued by the company and how many shares were sold by the company's founders? What fraction of their holdings in the company did the founders sell?

Solution to 1:

Healthybots received gross proceeds of £34.3 million at £2.57 per share, so the company issued and sold 13,346,304 shares ($= £34.3 \text{ million} / £2.57 \text{ per share}$). The total placement was for 103,142,466 shares, so the founders sold 89,796,162 shares ($= 103,142,466 \text{ shares} - 13,346,304 \text{ shares}$). Because approximately 200 million shares ($= 213.3 \text{ million shares} - 13.3 \text{ million shares}$) were outstanding before the placement, the founders sold approximately 45 percent ($= 90 \text{ million shares} / 200 \text{ million shares}$) of the company.

2. Approximately what return did the subscribers who participated in the IPO make on the first day it traded?

Solution to 2:

The subscribers bought the stock for £2.57 per share, and it closed at £2.73. The first day return thus was $6.2\% = \frac{2.73 - 2.57}{2.57} \times 100$.

3. Approximately how much did Healthybots pay in placement costs as a percentage of the new funds raised?

Solution to 3:

Healthybots obtained gross proceeds of £34.3 million but only raised net proceeds of £31.8 million. The £2.5 million difference was the total cost of the placement to the firm, which is 7.9 percent of net proceeds, or new funds raised ($£2.5 \text{ million} / £31.8 \text{ million}$).

14.2. Private Placements and Other Primary Market Transactions

Corporations sometimes issue their securities in private placements. In a **private placement**, corporations sell securities directly to a small group of qualified investors, usually with the assistance of an investment bank. Qualified investors have sufficient knowledge and experience to recognize the risks that they assume, and sufficient wealth to assume those risks responsibly. Most countries allow corporations to do private placements without nearly as much public disclosure as is required for public offerings. Private placements, therefore, may be cheaper than public offerings, but the buyers generally require higher returns (lower purchase prices) because they cannot subsequently trade the securities in an organized secondary market.

Corporations sometimes sell new issues of seasoned securities directly to the public on a piecemeal basis via a shelf registration. In a **shelf registration**, the corporation makes all public disclosures that it would for a regular offering, but it does not sell the shares in a single transaction. Instead, it sells the shares directly into the secondary market over time, generally when it needs additional capital. Shelf registrations provide corporations with flexibility in the timing of their capital transactions, and they can alleviate the downward price pressures often associated with large secondary offerings.

Many corporations may also issue shares via dividend reinvestment plans (DRPs or DRIPs, for short) that allow their shareholders to reinvest their dividends in newly issued shares of the corporation (in particular, DRPs specify that the corporation issue new shares for the plan rather than purchase them on the open market). These plans sometimes also allow existing shareholders and other investors to buy additional stock at a slight discount to current prices.

Finally, corporations can issue new stock via a rights offering. In a rights offering, the corporation distributes rights to buy stock at a fixed price to existing shareholders in proportion to their holdings. Because the rights need not be exercised, they are options. The exercise price, however, is set below the current market price of the stock so that buying stock with the rights is immediately profitable. Consequently, shareholders will experience dilution in the value of their existing shares. They can offset the dilution loss by exercising their rights or by selling the rights to others who will exercise them. Shareholders generally do not like rights offerings because they must provide additional capital (or sell

their rights) to avoid losses through dilution. Financial analysts recognize that these securities, although called rights, are actually short-term stock warrants and value them accordingly.

The national governments of financially strong countries generally issue their bonds, notes, and bills in public auctions organized by a government agency (usually associated with the finance ministry). They may also sell them directly to dealers.

Smaller and less financially secure national governments and most regional governments often contract with investment banks to help them sell and distribute their securities. The laws of many governments, however, require that they auction their securities.

EXAMPLE 25

Private and Public Placements

In what ways do private placements differ from public placements?

Solution:

Issuers make private placements to a limited number of investors that generally are financially sophisticated and well informed about risk. The investors generally have some relationship to the issuer. Issuers make public placements when they sell securities to the general public. Public placements generally require substantially more financial disclosure than do private placements.

14.3. Importance of Secondary Markets to Primary Markets

Corporations and governments can raise money in the primary markets at lower cost when their securities will trade in liquid secondary markets. In a **liquid market**, traders can buy or sell with low transaction costs and small price concessions when they want to trade. Buyers value liquidity because they may need to sell their securities to meet liquidity needs. Investors thus will pay more for securities that they can easily sell than for those that they cannot easily sell. Higher prices translate into lower costs of capital for the issuers.

15. SECONDARY SECURITY MARKET AND CONTRACT MARKET STRUCTURES

LEARNING OUTCOMES

The candidate should be able to:

- ☐ **define primary and secondary markets and explain how secondary markets support primary markets**
- ☐ **describe how securities, contracts, and currencies are traded in quote-driven, order-driven, and brokered markets**

Trading is the successful outcome to a bilateral search in which buyers look for sellers and sellers look for buyers. Many market structures have developed to reduce the costs of this search. Markets are liquid when the costs of finding a suitable counterparty to a trade are low.

Trading in securities and contracts takes place in a variety of market structures. The structures differ by when trades can be arranged, who arranges the trades, how they do so, and how traders learn about

possible trading opportunities and executed trades. This section introduces the various market structures used to trade securities and contracts. We first consider trading sessions, then execution mechanisms, and finally market information systems.

15.1. Trading Sessions

Markets are organized as call markets or as continuous trading markets. In a **call market**, trades can be arranged only when the market is called at a particular time and place. In contrast in a **continuous trading market**, trades can be arranged and executed anytime the market is open.

Buyers can easily find sellers and vice versa in call markets because all traders interested in trading (or orders representing their interests) are present at the same time and place. Call markets thus have the potential to be very liquid when they are called. But they are completely illiquid between trading sessions. In contrast, traders can arrange and execute their trades at any time in continuous trading markets, but doing so can be difficult if the buyers and sellers (or their orders) are not both present at the same time.

Most call markets use single price auctions to match buyers to sellers. In these auctions, the market constructs order books representing all buy orders and all seller orders. The market then chooses a single trade price that will maximize the total volume of trade. The order books are supply and demand schedules, and the point at which they cross determines the trade price.

Call markets usually are organized just once a day, but some markets organize calls at more frequent intervals.

Many continuous trading markets start their trading with a call market auction. During a pre-opening period, traders submit their orders for the market call. At the opening, any possible trades are arranged and then trading continues in the continuous trading session. Some continuous trading markets also close their trading with a call. In these markets, traders who are only interested in trading in the closing call submit market- or limit-on-close orders.

EXAMPLE 26

Call Markets and Continuous Trading Markets

1. What is the main advantage of a call market compared with a continuous trading market?

Solution to 1:

By gathering all traders to the same place at the same time, a call market makes it easier for buyers to find sellers and vice versa. In contrast, if buyers and sellers (or their orders) are not present at the same time in a continuous market, they cannot trade.

2. What is the main advantage of a continuous trading market compared with a call market?

Solution to 2:

In a continuous trading market, a willing buyer and seller can trade at any time the market is open. In contrast, in a call market trading can take place only when the market is called.

15.2. Execution Mechanisms

The three main types of market structures are quote-driven markets (sometimes called price-driven or dealer markets), order-driven markets, and brokered markets. In **quote-driven markets**, customers trade with dealers. In **order-driven markets**, an order matching system run by an exchange, a broker,

or an alternative trading system uses rules to arrange trades based on the orders that traders submit. Most exchanges and ECNs organize order-driven markets. In **brokered markets**, brokers arrange trades between their customers. Brokered markets are common for transactions of unique instruments, such as real estate properties, intellectual properties, or large blocks of securities. Many trading systems use more than one type of market structure.

15.2.1. Quote-Driven Markets

Worldwide, most trading, other than in stocks, takes place in quote-driven markets. Almost all bonds and currencies and most spot commodities trade in quote-driven markets. Traders call them quote-driven (or price-driven or dealer) because customers trade at the prices quoted by dealers. Depending on the instrument traded, the dealers work for commercial banks, for investment banks, for broker-dealers, or for proprietary trading houses.

Quote-driven markets also often are called over-the-counter (OTC) markets because securities used to be literally traded over the dealer's counter in the dealer's office. Now, most trades in OTC markets are conducted over proprietary computer communications networks, by telephone, or sometimes over instant messaging systems.

15.2.2. Order-Driven Markets

Order-driven markets arrange trades using rules to match buy orders to sell orders. The orders may be submitted by customers or by dealers. Almost all exchanges use order-driven trading systems, and every automated trading system is an order-driven system.

Because rules match buyers to sellers, traders often trade with complete strangers. Order-driven markets thus must have procedures to ensure that buyers and sellers perform on their trade contracts. Otherwise, dishonest traders would enter contracts that they would not settle if a change in market conditions made settlement unprofitable.

Two sets of rules characterize order-driven market mechanisms: Order matching rules and trade pricing rules. The order matching rules match buy orders to sell orders. The trade pricing rules determine the prices at which the matched trades take place.

15.2.2.1. Order Matching Rules

Order-driven trading systems match buyers to sellers using rules that rank the buy orders and the sell orders based on price, and often along with other secondary criteria. The systems then match the highest-ranking buy order with the highest-ranking sell order. If the buyer is willing to pay at least as much as the seller is willing to receive, the system will arrange a trade for the minimum of the buy and sell quantities. The remaining size, if any, is then matched with the next order on the other side and the process continues until no further trades can be arranged.

The **order precedence hierarchy** determines which orders go first. The first rule is **price priority**: The highest priced buy orders and the lowest priced sell orders go first. They are the most aggressively priced orders. **Secondary precedence rules** determine how to rank orders at the same price. Most trading systems use time precedence to rank orders at the same price. The first order to arrive has precedence over other orders. In trading systems that permit hidden and partially hidden orders, displayed quantities at a given price generally have precedence over the undisplayed quantities. So the complete precedence hierarchy is given by price priority, display precedence at a given price, and finally time precedence among all orders with the same display status at a given price. These rules give traders incentives to improve price, display their orders, and arrive early if they want to trade quickly. These incentives increase market liquidity.

15.2.2.2. Trade Pricing Rules

After the orders are matched, the trading system then uses its trade pricing rule to determine the trade price. The three rules that various order-driven markets use to price their trades are the uniform pricing

rule, the discriminatory pricing rule, and the derivative pricing rule.

Call markets commonly use the uniform pricing rule. Under this rule, all trades execute at the same price. The market chooses the price that maximizes the total quantity traded.

Continuous trading markets use the **discriminatory pricing rule**. Under this rule, the limit price of the order or quote that first arrived—the standing order—determines the trade price. This rule allows a large arriving trader to discriminate among standing limit orders by filling the most aggressively priced orders first at their limit prices and then filling less aggressively priced orders at their less favorable (from the point of view of the arriving trader) limit prices. If trading systems did not use this pricing rule, large traders would break their orders into pieces to price discriminate on their own.

EXAMPLE 27

Filling a Large Order in a Continuous Trading Market

Before the arrival of a large order, the Tokyo Stock Exchange has the following limit orders standing on its book:

Buyer	Bid Size	Limit Price(¥)	Offer Size	Seller
Takumi	15	100.1		
Hiroto	8	100.2		
Shou	10	100.3		
		100.4	4	Hina
		100.5	6	Sakura
		100.6	12	Miku

Tsubasa submits a day order to buy 15 contracts, limit ¥100.5. With whom does he trade, what is his average trade price, and what does the limit order book look like afterward?

Solution:

Tsubasa's buy order first fills with the most aggressively priced sell order, which is Hina's order for four contracts. A trade takes place at ¥100.4 for four contracts, Hina's order fills completely, and Tsubasa still has 11 more contracts remaining.

The next most aggressively priced sell order is Sakura's order for six contracts. A second trade takes place at ¥100.5 for six contracts, Sakura's order fills completely, and Tsubasa still has five more contracts remaining.

The next most aggressively priced sell order is Miku's order at ¥100.6. No further trade is possible, however, because her limit sell price is above Tsubasa's limit buy price. Tsubasa's average trade price is $\text{¥}100.46 = \frac{4 \times \text{¥}100.4 + 6 \times \text{¥}100.5}{4 + 6}$.

Because Tsubasa issued a day order, the remainder of his order is placed on the book on the buy side at ¥100.5. The following orders are then on the book:

Buyer	Bid Size	Limit Price (¥)	Offer Size	Seller
Takumi	15	100.1		
Hiroto	8	100.2		
Shou	10	100.3		
		100.4		
Tsubasa	5	100.5		
		100.6	12	Miku

If Tsubasa had issued an immediate-or-cancel order, the remaining five contracts would have been cancelled.

Crossing networks use the derivative pricing rule. **Crossing networks** are trading systems that match buyers and sellers who are willing to trade at prices obtained from other markets. Most systems cross their trades at the midpoint of the best bid and ask quotes published by the exchange at which the security primarily trades. This pricing rule is called a **derivative pricing rule** because the price is derived from another market. In particular, the price does not depend on the orders submitted to the crossing network. Some crossing networks are organized as call markets and others as continuously trading markets. The most important crossing market is the equity trading system POSIT.

15.2.3. Brokered Markets

The third execution mechanism is the **brokered market**, in which brokers arrange trades among their clients. Brokers organize markets for instruments for which finding a buyer or a seller willing to trade is difficult because the instruments are unique and thus of interest only to a limited number of people or institutions. These instruments generally are also infrequently traded and expensive to carry in inventory. Examples of such instruments include very large blocks of stock, real estate properties, fine art masterpieces, intellectual properties, operating companies, liquor licenses, and taxi medallions. Because dealers generally are unable or unwilling to hold these assets in their inventories, they will not make markets in them. Organizing order-driven markets for these instruments is not sensible because too few traders would submit orders to them.

Successful brokers in these markets try to know everyone who might now or in the future be willing to trade. They spend most of their time on the telephone and in meetings building their networks.

EXAMPLE 28

Quote-Driven, Order-Driven, and Brokered Markets

What are the primary advantages of quote-driven, order-driven, and brokered markets?

Solution:

In a quote-driven market, dealers generally are available to supply liquidity. In an order-driven market, traders can supply liquidity to each other. In a brokered market, brokers help find traders who are willing to trade when dealers would not be willing to make markets and when traders would not be willing to post orders.

15.3. Market Information Systems

Markets vary in the type and quantity of data that they disseminate to the public. Traders say that a market is pre-trade transparent if the market publishes real-time data about quotes and orders. Markets are post-trade transparent if the market publishes trade prices and sizes soon after trades occur.

Buy-side traders value transparency because it allows them to better manage their trading, understand market values, and estimate their prospective and actual transaction costs. In contrast, dealers prefer to trade in opaque markets because, as frequent traders, they have an information advantage over those who know less than they do. Bid-ask spreads tend to be wider and transaction costs tend to be higher in opaque markets because finding the best available price is harder for traders in such markets.

16. WELL-FUNCTIONING FINANCIAL SYSTEMS

LEARNING OUTCOME

The candidate should be able to:

□ **describe characteristics of a well-functioning financial system**

The financial system allows traders to solve financing and risk management problems. In a well-functioning financial system:

- investors can easily move money from the present to the future while obtaining a fair rate of return for the risks that they bear;
- borrowers can easily obtain funds that they need to undertake current projects if they can credibly promise to repay the funds in the future;
- hedgers can easily trade away or offset the risks that concern them; and
- traders can easily trade currencies for other currencies or commodities that they need.

If the assets or contracts needed to solve these problems are available to trade, the financial system has **complete markets**. If the costs of arranging these trades are low, the financial system is **operationally efficient**. If the prices of the assets and contracts reflect all available information related to fundamental values, the financial system is informationally efficient.

Well-functioning financial systems are characterized by:

- the existence of well-developed markets that trade instruments that help people solve their financial problems (complete markets);
- liquid markets in which the costs of trading—commissions, bid–ask spreads, and order price impacts—are low (operationally efficient markets);
- timely financial disclosures by corporations and governments that allow market participants to estimate the fundamental values of securities (support **informationally efficient markets**); and
- prices that reflect fundamental values so that prices vary primarily in response to changes in fundamental values and not to demands for liquidity made by uninformed traders (informationally efficient markets).

Such complete and operationally efficient markets are produced by financial intermediaries who:

- organize exchanges, brokerages, and alternative trading systems that match buyers to sellers;
- provide liquidity on demand to traders;
- securitize assets to produce investment instruments that are attractive to investors and thereby lower the costs of funds for borrowers;
- run banks that match investors to borrowers by taking deposits and making loans;
- run insurance companies that pool uncorrelated risks;
- provide investment advisory services that help investors manage and grow their assets at low cost;
- organize clearinghouses that ensure everyone settles their trades and contracts; and
- organize depositories that ensure nobody loses their assets.

The benefits of a well-functioning financial system are huge. In such systems, investors who need to move money to the future can easily connect with entrepreneurs who need money now to develop new products and services. Similarly, producers who would otherwise avoid valuable projects because they

are too risky can easily transfer those risks to others who can better bear them. Most importantly, these transactions generally can take place among strangers so that the benefits from trading can be derived from an enormous number of potential matches.

In contrast, economies that have poorly functioning financial systems have great difficulties allocating capital among the many companies who could use it. Financial transactions tend to be limited to arrangements within families when people cannot easily find trustworthy counterparties who will honor their contracts. In such economies, capital is allocated inefficiently, risks are not easily shared, and production is inefficient.

An extraordinarily important byproduct of an operationally efficient financial system is the production of informationally efficient prices. Prices are informationally efficient when they reflect all available information about fundamental values. Informative prices are crucially important to the welfare of an economy because they help ensure that resources go where they are most valuable. Economies that use resources where they are most valuable are allocationally efficient. Economies that do not use resources where they are most valuable waste their resources and consequently often are quite poor.

Well-informed traders make prices informationally efficient. When they buy assets and contracts that they think are undervalued, they tend to push the assets' prices up. Similarly, when they sell assets and contracts that they think are overvalued, they tend to push the assets' prices down. The effect of their trading thus causes prices to reflect their information about values.

How accurately prices reflect fundamental information depends on the costs of obtaining fundamental information and on the liquidity available to well-informed traders. Accounting standards and reporting requirements that produce meaningful and timely financial disclosures reduce the costs of obtaining fundamental information and thereby allow analysts to form more accurate estimates of fundamental values. Liquid markets allow well-informed traders to fill their orders at low cost. If filling orders is very costly, informed trading may not be profitable. In that case, information-motivated traders will not commit resources to collect and analyze data and they will not trade. Without their research and their associated trading, prices would be less informative.

EXAMPLE 29

Well-Functioning Financial Systems

As a financial analyst specializing in emerging market equities, you understand that a well-functioning financial system contributes to the economic prosperity of a country. You are asked to start covering a new small market country. What factors will you consider when characterizing the quality of its financial markets?

Solution:

In general, you will consider whether:

- the country has markets that allow its companies and residents to finance projects, save for the future, and exchange risk;
- the costs of trading in those markets is low; and
- prices reflect fundamental values.

You may specifically check to see whether:

- fixed income and stock markets allow borrowers to easily obtain capital from investors;
- corporations disclose financial and operating data on a timely basis in conformity to widely respected reporting standards, such as IFRS;
- forward, futures, and options markets trade instruments that companies need to hedge their risks;
- dealers and arbitrageurs allow traders to trade when they want to;

- bid–ask spreads are small;
- trades and contracts invariably settle as expected;
- investment managers provide high-quality management services for reasonable fees;
- banks and other financing companies are well capitalized and thus able to help investors provide capital to borrowers;
- securitized assets are available and represent reasonable credit risks;
- insurance companies are well capitalized and thus able to help those exposed to risks insure against them; and
- price volatility appears consistent with changes in fundamental values.

17. MARKET REGULATION

LEARNING OUTCOME

The candidate should be able to:

☐ describe objectives of market regulation

Government agencies and practitioner organizations regulate many markets and the financial intermediaries that participate in them. The regulators generally seek to promote fair and orderly markets in which traders can trade at prices that accurately reflect fundamental values without incurring excessive transaction costs. This section identifies the problems that financial regulators hope to solve and the objectives of their regulations.

Regrettably, some people will steal from each other if given a chance, especially if the probability of detection is low or if the penalty for being caught is low. The number of ways that people can steal or misappropriate wealth generally increases with the complexity of their relationships and with asymmetries in their knowledge. Because financial markets tend to be complex, and because customers are often much less sophisticated than the professionals that serve them, the potential for losses through various frauds can be unacceptably high in unregulated markets.

Regulators thus ensure that systems are in place to protect customers from fraud. In principle, the customers themselves would demand such systems as a condition of doing business. When customers are unsophisticated or poorly informed, however, they may not know how to protect themselves. When the costs of learning are large—as they often are in complex financial markets—having regulators look out for the public interest can be economically efficient.

More customer money is probably lost in financial markets through negligence than through outright fraud. Most customers in financial markets use various agents to help them solve problems that they do not understand well. These agents include securities brokers, financial advisers, investment managers, and insurance agents. Because customers generally do not have much information about market conditions, they find it extremely difficult to measure the added value they obtain from their agents. This problem is especially challenging when performance has a strong random component. In that case, determining whether agents are skilled or lucky is very difficult. Moreover, if the agent is a good salesman, the customer may not critically evaluate their agent's performance. These conditions, which characterize most financial markets, ensure that customers cannot easily determine whether their agents are working faithfully for them. They tend to lose if their agents are unqualified or lazy, or if they unconsciously favor themselves and their friends over their clients, as is natural for even the most honest people.

Regulators help solve these agency problems by setting minimum standards of competence for agents and by defining and enforcing minimum standards of practice. CFA Institute provides significant standard setting leadership in the areas of investment management and investment performance reporting through its Chartered Financial Analyst Program, in which you are studying, and its Global Investment Performance Standards. In principle, regulation would not be necessary if customers could identify competent agents and effectively measure their performance. In the financial markets, doing so is very difficult.

Regulators often act to level the playing field for market participants. For example, in many jurisdictions, insider trading in securities is illegal. The rule prevents corporate insiders and others with access to corporate information from trading on material information that has not been released to the public. The purpose of the rule is to reduce the profits that insiders could extract from the markets. These profits would come from other traders who would lose when they trade with well-informed insiders. Because traders tend to withdraw from markets when they lose, rules against insider trading help keep markets liquid. They also keep corporate insiders from hoarding information.

Many situations arise in financial markets in which common standards benefit everyone involved. For example, having all companies report financial results on a common basis allows financial analysts to easily compare companies. Accordingly, the International Accounting Standards Board (IASB) and the US-based Financial Accounting Standards Board (FASB), among many others, promulgate common financial standards to which all companies must report. The benefits of having common reporting standards has led to a very successful and continuing effort to converge all accounting standards to a single worldwide standard. Without such regulations, investors might eventually refuse to invest in companies that do not report to a common standard, but such market-based discipline is a very slow regulator of behavior, and it would have little effect on companies that do not need to raise new capital.

Regulators generally require that financial firms maintain minimum levels of capital. These capital requirements serve two purposes. First, they ensure that the companies will be able to honor their contractual commitments when unexpected market movements or poor decisions cause them to lose money. Second, they ensure that the owners of financial firms have substantial interest in the decisions that they make. Without a substantial financial interest in the decisions that they make, companies often take too many risks and exercise poor judgment about extending credit to others. When such companies fail, they impose significant costs on others. Minimum capital requirements reduce the probability that financial firms will fail and they reduce the disruptions associated with those failures that do occur. In principle, a firm's customers and counterparties could require minimum capital levels as a condition of doing business with the firm, but they have more difficulty enforcing their contracts than do governments who can imprison people.

Regulators similarly regulate insurance companies and pension funds that make long-term promises to their clients. Such entities need to maintain adequate reserves to ensure that they can fund their liabilities. Unfortunately, their managers have a tendency to underestimate these reserves if they will not be around when the liabilities come due. Again, in principle, policyholders and employees could regulate the behavior of their insurance funds and their employers by refusing to contract with them if they do not promise to adequately fund their liabilities. In practice, however, the sophistication, information, and time necessary to write and enforce contracts that control these problems are beyond the reach of most people. The government thus is a sensible regulator of such problems.

Many regulators are self-regulating organizations (SROs) that regulate their members. Exchanges, clearinghouses, and dealer trade organizations are examples of self-regulating organizations. In some cases, the members of these organizations voluntarily subject themselves to the SRO's regulations to promote the common good. In other cases, governments delegate regulatory and enforcement authorities to SROs, usually subject to the supervision of a government agency, such as a national securities and exchange authority. Exchanges, dealer associations, and clearing agencies often regulate their members with these delegated powers.

By setting high standards of behavior, SROs help their members obtain the confidence of their customers. They also reduce the chance that members of the SRO will incur losses when dealing with other members of the SRO.

When regulators fail to solve the problems discussed here, the financial system does not function well. People who lose money stop saving and borrowers with good ideas cannot fund their projects. Similarly, hedgers withdraw from markets when the costs of hedging are high. Without the ability to hedge, producers become reluctant to specialize because specialization generally increases risk. Because

specialization also decreases costs, however, production becomes less efficient as producers chose safer technologies. Economies that cannot solve the regulatory problems described in this section tend to operate less efficiently than do better regulated economies, and they tend to be less wealthy.

To summarize, the objectives of market regulation are to:

1. control fraud;
2. control agency problems;
3. promote fairness;
4. set mutually beneficial standards;
5. prevent undercapitalized financial firms from exploiting their investors by making excessively risky investments; and
6. ensure that long-term liabilities are funded.

Regulation is necessary because regulating certain behaviors through market-based mechanisms is too costly for people who are unsophisticated and uninformed. Effectively regulated markets allow people to better achieve their financial goals.

EXAMPLE 30

Bankrupt Traders

You are the chief executive officer of a brokerage that is a member of a clearinghouse. A trader who clears through your firm is bankrupt at midday, but you do not yet know it even though your clearing agreement with him explicitly requires that he immediately report significant losses. The trader knows that if he takes a large position, prices might move in his favor so that he will no longer be bankrupt. The trader attempts to do so and succeeds. You find out about this later in the evening.

1. Why does the clearinghouse regulate its members?

Solution to 1:

The clearinghouse regulates its members to ensure that no member imposes costs on another member by failing to settle a trade.

2. What should you do about the trader?

Solution to 2:

You should immediately end your clearing relationship with the trader and confiscate his trading profits. The trader was trading with your firm's capital after he became bankrupt. Had he lost, your firm would have borne the loss.

3. Why would the clearinghouse allow you to keep his trading profits?

Solution to 3:

If the clearinghouse did not permit you to keep his trading profits, other traders similarly situated might attempt the same strategy.

SUMMARY

This reading introduces how the financial system operates and explains how well-functioning financial systems lead to wealthy economies. Financial analysts need to understand how the financial system works because their analyses often lead to trading decisions.

The financial system consists of markets and the financial intermediaries that operate in them. These institutions allow buyers to connect with sellers. They may trade directly with each other when they trade the same instrument or they only may trade indirectly when a financial intermediary connects the buyer to the seller through transactions with each that appear on the intermediary's balance sheet. The buyer and seller may exchange instruments, cash flows, or risks.

The following points, among others, were made in this reading:

- The financial system consists of mechanisms that allow strangers to contract with each other to move money through time, to hedge risks, and to exchange assets that they value less for those that they value more.
- Investors move money from the present to the future when they save. They expect a normal rate of return for bearing risk through time. Borrowers move money from the future to the present to fund current projects and expenditures. Hedgers trade to reduce their exposure to risks they prefer not to take. Information-motivated traders are active investment managers who try to identify under- and overvalued instruments.
- Securities are first sold in primary markets by their issuers. They then trade in secondary markets.
- People invest in pooled investment vehicles to benefit from the investment management services of their managers.
- Forward contracts allow buyers and sellers to arrange for future sales at predetermined prices. Futures contracts are forward contracts guaranteed by clearinghouses. The guarantee ensures that strangers are willing to trade with each other and that traders can offset their positions by trading with anybody. These features of futures contract markets make them highly attractive to hedgers and information-motivated traders.
- Many financial intermediaries connect buyers to sellers in a given instrument, acting directly as brokers and exchanges or indirectly as dealers and arbitrageurs.
- Financial intermediaries create instruments when they conduct arbitrage, securitize assets, borrow to lend, manage investment funds, or pool insurance contracts. These activities all transform cash flows and risks from one form to another. Their services allow buyers and sellers to connect with each other through instruments that meet their specific needs.
- Financial markets work best when strangers can contract with each other without worrying about whether their counterparts are able and willing to honor their contract. Clearinghouses, variation margins, maintenance margins, and settlement guarantees made by creditworthy brokers on behalf of their clients help manage credit risk and ultimately allow strangers to contract with each other.
- Information-motivated traders short sell when they expect that prices will fall. Hedgers short sell to reduce the risks of a long position in a related contract or commodity.
- Margin loans allow people to buy more securities than their equity would otherwise permit them to buy. The larger positions expose them to more risk so that gains and losses for a given amount of equity will be larger. The leverage ratio is the value of a position divided by the value of the equity supporting it. The returns to the equity in a position are equal to the leverage ratio times the returns to the unleveraged position.
- To protect against credit losses, brokers demand maintenance margin payments from their customers who have borrowed cash or securities when adverse price changes cause their customer's equity to drop below the maintenance margin ratio. Brokers close positions for customers who do not satisfy these margin calls.
- Orders are instructions to trade. They always specify instrument, side (buy or sell), and quantity. They usually also provide several other instructions.

- Market orders tend to fill quickly but often at inferior prices. Limit orders generally fill at better prices if they fill, but they may not fill. Traders choose order submission strategies on the basis of how quickly they want to trade, the prices they are willing to accept, and the consequences of failing to trade.
- Stop instructions are attached to other orders to delay efforts to fill them until the stop condition is satisfied. Although stop orders are often used to stop losses, they are not always effective.
- Issuers sell their securities using underwritten public offerings, best efforts public offerings, private placements, shelf registrations, dividend reinvestment programs, and rights offerings. Investment banks have a conflict of interests when setting the initial offering price in an IPO.
- Well-functioning secondary markets are essential to raising capital in the primary markets because investors value the ability to sell their securities if they no longer want to hold them or if they need to disinvest to raise cash. If they cannot trade their securities in a liquid market, they will not pay as much for them.
- Matching buyers and sellers in call markets is easy because the traders (or their orders) come together at the same time and place.
- Dealers provide liquidity in quote-driven markets. Public traders as well as dealers provide liquidity in order-driven markets.
- Order-driven markets arrange trades by ranking orders using precedence rules. The rules generally ensure that traders who provide the best prices, display the most size, and arrive early trade first. Continuous order-driven markets price orders using the discriminatory pricing rule. Under this rule, standing limit orders determine trade prices.
- Brokers help people trade unique instruments or positions for which finding a buyer or a seller is difficult.
- Transaction costs are lower in transparent markets than in opaque markets because traders can more easily determine market value and more easily manage their trading in transparent markets.
- A well-functioning financial system allows people to trade instruments that best solve their wealth and risk management problems with low transaction costs. Complete and liquid markets characterize a well-functioning financial system. Complete markets are markets in which the instruments needed to solve investment and risk management problems are available to trade. Liquid markets are markets in which traders can trade when they want to trade at low cost.
- The financial system is operationally efficient when its markets are liquid. Liquid markets lower the costs of raising capital.
- A well-functioning financial system promotes wealth by ensuring that capital allocation decisions are well made. A well-functioning financial system also promotes wealth by allowing people to share the risks associated with valuable products that would otherwise not be undertaken.
- Prices are informationally efficient when they reflect all available information about fundamental values. Information-motivated traders make prices informationally efficient. Prices will be most informative in liquid markets because information-motivated traders will not invest in information and research if establishing positions based on their analyses is too costly.
- Regulators generally seek to promote fair and orderly markets in which traders can trade at prices that accurately reflect fundamental values without incurring excessive transaction costs. Governmental agencies and self-regulating organizations of practitioners provide regulatory services that attempt to make markets safer and more efficient.
- Mandated financial disclosure programs for the issuers of publicly traded securities ensure that information necessary to estimate security values is available to financial analysts on a consistent basis.

PRACTICE PROBLEMS

1. Akihiko Takabe has designed a sophisticated forecasting model, which predicts the movements in the overall stock market, in the hope of earning a return in excess of a fair return for the risk involved. He uses the predictions of the model to decide whether to buy, hold, or sell the shares of an index fund that aims to replicate the movements of the stock market. Takabe would *best* be characterized as a(n):
 - A. hedger.
 - B. investor.
 - C. information-motivated trader.
2. James Beach is young and has substantial wealth. A significant proportion of his stock portfolio consists of emerging market stocks that offer relatively high expected returns at the cost of relatively high risk. Beach believes that investment in emerging market stocks is appropriate for him given his ability and willingness to take risk. Which of the following labels *most appropriately* describes Beach?
 - A. Hedger.
 - B. Investor.
 - C. Information-motivated trader.
3. Lisa Smith owns a manufacturing company in the United States. Her company has sold goods to a customer in Brazil and will be paid in Brazilian real (BRL) in three months. Smith is concerned about the possibility of the BRL depreciating more than expected against the US dollar (USD). Therefore, she is planning to sell three-month futures contracts on the BRL. The seller of such contracts generally gains when the BRL depreciates against the USD. If Smith were to sell these future contracts, she would *most appropriately* be described as a(n):
 - A. hedger.
 - B. investor.
 - C. information-motivated trader.
4. Which of the following is *not* a function of the financial system?
 - A. To regulate arbitrageurs' profits (excess returns).
 - B. To help the economy achieve allocational efficiency.
 - C. To facilitate borrowing by businesses to fund current operations.
5. An investor primarily invests in stocks of publicly traded companies. The investor wants to increase the diversification of his portfolio. A friend has recommended investing in real estate properties. The purchase of real estate would *best* be characterized as a transaction in the:
 - A. derivative investment market.
 - B. traditional investment market.
 - C. alternative investment market.
6. A hedge fund holds its excess cash in 90-day commercial paper and negotiable certificates of deposit. The cash management policy of the hedge fund is *best described* as using:

- A. capital market instruments.
 - B. money market instruments.
 - C. intermediate-term debt instruments.
7. An oil and gas exploration and production company announces that it is offering 30 million shares to the public at \$45.50 each. This transaction is *most likely* a sale in the:
- A. futures market.
 - B. primary market.
 - C. secondary market.
8. Consider a mutual fund that invests primarily in fixed-income securities that have been determined to be appropriate given the fund's investment goal. Which of the following is *least likely* to be a part of this fund?
- A. Warrants.
 - B. Commercial paper.
 - C. Repurchase agreements.
9. A friend has asked you to explain the differences between open-end and closed-end funds. Which of the following will you *most likely* include in your explanation?
- A. Closed-end funds are unavailable to new investors.
 - B. When investors sell the shares of an open-end fund, they can receive a discount or a premium to the fund's net asset value.
 - C. When selling shares, investors in an open-end fund sell the shares back to the fund whereas investors in a closed-end fund sell the shares to others in the secondary market.
10. The Standard & Poor's Depositary Receipts (SPDRs) is an investment that tracks the S&P 500 stock market index. Purchases and sales of SPDRs during an average trading day are *best* described as:
- A. primary market transactions in a pooled investment.
 - B. secondary market transactions in a pooled investment.
 - C. secondary market transactions in an actively managed investment.
11. Which of the following statements about exchange-traded funds is *most correct*?
- A. Exchange-traded funds are not backed by any assets.
 - B. The investment companies that create exchange-traded funds are financial intermediaries.
 - C. The transaction costs of trading shares of exchange-traded funds are substantially greater than the combined costs of trading the underlying assets of the fund.
12. The usefulness of a forward contract is limited by some problems. Which of the following is *most likely* one of those problems?
- A. Once you have entered into a forward contract, it is difficult to exit from the contract.
 - B. Entering into a forward contract requires the long party to deposit an initial amount with the short party.
 - C. If the price of the underlying asset moves adversely from the perspective of the long party, periodic payments must be made to the short party.

13. Tony Harris is planning to start trading in commodities. He has heard about the use of futures contracts on commodities and is learning more about them. Which of the following is Harris *least likely* to find associated with a futures contract?
- A. Existence of counterparty risk.
 - B. Standardized contractual terms.
 - C. Payment of an initial margin to enter into a contract.
14. A German company that exports machinery is expecting to receive \$10 million in three months. The firm converts all its foreign currency receipts into euros. The chief financial officer of the company wishes to lock in a minimum fixed rate for converting the \$10 million to euro but also wants to keep the flexibility to use the future spot rate if it is favorable. What hedging transaction is *most likely* to achieve this objective?
- A. Selling dollars forward.
 - B. Buying put options on the dollar.
 - C. Selling futures contracts on dollars.
15. A book publisher requires substantial quantities of paper. The publisher and a paper producer have entered into an agreement for the publisher to buy and the producer to supply a given quantity of paper four months later at a price agreed upon today. This agreement is a:
- A. futures contract.
 - B. forward contract.
 - C. commodity swap.
16. The Standard & Poor's Depository Receipts (SPDRs) is an exchange-traded fund in the United States that is designed to track the S&P 500 stock market index. The latest price of a share of SPDRs is \$290. A trader has just bought call options on shares of SPDRs for a premium of \$3 per share. The call options expire in six months and have an exercise price of \$305 per share. On the expiration date, the trader will exercise the call options (ignore any transaction costs) if and only if the shares of SPDRs are trading:
- A. below \$305 per share.
 - B. above \$305 per share.
 - C. above \$308 per share.
17. Jason Schmidt works for a hedge fund and he specializes in finding profit opportunities that are the result of inefficiencies in the market for convertible bonds—bonds that can be converted into a predetermined amount of a company's common stock. Schmidt tries to find convertibles that are priced inefficiently relative to the underlying stock. The trading strategy involves the simultaneous purchase of the convertible bond and the short sale of the underlying common stock. The above process could best be described as:
- A. hedging.
 - B. arbitrage.
 - C. securitization.
18. Pierre-Louis Robert just purchased a call option on shares of the Michelin Group. A few days ago he wrote a put option on Michelin shares. The call and put options have the same exercise price, expiration date, and number of shares underlying. Considering both positions, Robert's exposure to the risk of the stock of the Michelin Group is:
- A. long.

- B. short.
- C. neutral.
19. An online brokerage firm has set the minimum margin requirement at 55 percent. What is the maximum leverage ratio associated with a position financed by this minimum margin requirement?
- A. 1.55.
- B. 1.82.
- C. 2.22.
20. A trader has purchased 200 shares of a non-dividend-paying firm on margin at a price of \$50 per share. The leverage ratio is 2.5. Six months later, the trader sells these shares at \$60 per share. Ignoring the interest paid on the borrowed amount and the transaction costs, what was the return to the trader during the six-month period?
- A. 20 percent.
- B. 33.33 percent.
- C. 50 percent.
21. Jason Williams purchased 500 shares of a company at \$32 per share. The stock was bought on 75 percent margin. One month later, Williams had to pay interest on the amount borrowed at a rate of 2 percent per month. At that time, Williams received a dividend of \$0.50 per share. Immediately after that he sold the shares at \$28 per share. He paid commissions of \$10 on the purchase and \$10 on the sale of the stock. What was the rate of return on this investment for the one-month period?
- A. -12.5 percent.
- B. -15.4 percent.
- C. -50.1 percent.
22. Caroline Rogers believes the price of Gamma Corp. stock will go down in the near future. She has decided to sell short 200 shares of Gamma Corp. at the current market price of €47. The initial margin requirement is 40 percent. Which of the following is an appropriate statement regarding the margin requirement that Rogers is subject to on this short sale?
- A. She will need to contribute €3,760 as margin.
- B. She will need to contribute €5,640 as margin.
- C. She will only need to leave the proceeds from the short sale as deposit and does not need to contribute any additional funds.
23. The current price of a stock is \$25 per share. You have \$10,000 to invest. You borrow an additional \$10,000 from your broker and invest \$20,000 in the stock. If the maintenance margin is 30 percent, at what price will a margin call first occur?
- A. \$9.62.
- B. \$17.86.
- C. \$19.71.
24. A market has the following limit orders standing on its book for a particular stock. The bid and ask sizes are number of shares in hundreds.

Bid Size	Limit Price (€)	Offer Size
5	9.73	
12	9.81	

Bid Size	Limit Price (€)	Offer Size
4	9.84	
6	9.95	
	10.02	5
	10.10	12
	10.14	8

What is the market?

- A. 9.73 bid, offered at 10.14.
- B. 9.81 bid, offered at 10.10.
- C. 9.95 bid, offered at 10.02.

25. Consider the following limit order book for a stock. The bid and ask sizes are number of shares in hundreds.

Bid Size	Limit Price (¥)	Offer Size
3	122.80	
8	123.00	
4	123.35	
	123.80	7
	124.10	6
	124.50	7

A new buy limit order is placed for 300 shares at ¥123.40. This limit order is said to:

- A. take the market.
- B. make the market.
- C. make a new market.

26. Currently, the market in a stock is “\$54.62 bid, offered at \$54.71.” A new sell limit order is placed at \$54.62. This limit order is said to:

- A. take the market.
- B. make the market.
- C. make a new market.

27. You have placed a sell market-on-open order—a market order that would automatically be submitted at the market’s open tomorrow and would fill at the market price. Your instruction, to sell the shares at the market open, is a(n):

- A. execution instruction.
- B. validity instruction.
- C. clearing instruction.

28. Jim White has sold short 100 shares of Super Stores at a price of \$42 per share. He has also simultaneously placed a “good-till-cancelled, stop 50, limit 55 buy” order. Assume that if the stop condition specified by White is satisfied and the order becomes valid, it will get executed. Excluding transaction costs, what is the maximum possible loss that White can have?

- A. \$800.
- B. \$1,300.

C. Unlimited.

29. You own shares of a company that are currently trading at \$30 a share. Your technical analysis of the shares indicates a support level of \$27.50. That is, if the price of the shares is going down, it is more likely to stay above this level rather than fall below it. If the price does fall below this level, however, you believe that the price may continue to decline. You have no immediate intent to sell the shares but are concerned about the possibility of a huge loss if the share price declines below the support level. Which of the following types of orders could you place to most appropriately address your concern?

A. Short sell order.

B. Good-till-cancelled stop sell order.

C. Good-till-cancelled stop buy order.

30. In an underwritten offering, the risk that the entire issue may not be sold to the public at the stipulated offering price is borne by the:

A. issuer.

B. investment bank.

C. buyers of the part of the issue that is sold.

31. A British company listed on AIM (formerly the Alternative Investment Market) of the London Stock Exchange announced the sale of 6,686,665 shares to a small group of qualified investors at £0.025 per share. Which of the following *best describes* this sale?

A. Shelf registration.

B. Private placement.

C. Initial public offering.

32. A German publicly traded company, to raise new capital, gave its existing shareholders the opportunity to subscribe for new shares. The existing shareholders could purchase two new shares at a subscription price of €4.58 per share for every 15 shares held. This is an example of a(n):

A. rights offering.

B. private placement.

C. initial public offering.

33. Consider an order-driven system that allows hidden orders. The following four sell orders on a particular stock are currently in the system's limit order book. Based on the commonly used order precedence hierarchy, which of these orders will have precedence over others?

Order	Time of Arrival (HH:MM:SS)	Limit Price (€)	Special Instruction (If any)
I	9:52:01	20.33	
II	9:52:08	20.29	Hidden order
III	9:53:04	20.29	
IV	9:53:49	20.29	

A. Order I (time of arrival of 9:52:01).

B. Order II (time of arrival of 9:52:08).

C. Order III (time of arrival of 9:53:04).

34. Zhenhu Li has submitted an immediate-or-cancel buy order for 500 shares of a company at a limit price of CNY 74.25. There are two sell limit orders standing in that stock's order book at that time. One is for 300 shares at a limit price of CNY 74.30 and the other is for 400 shares at a limit price of CNY 74.35. How many shares in Li's order would get cancelled?
- A. None (the order would remain open but unfilled).
 - B. 200 (300 shares would get filled).
 - C. 500 (there would be no fill).

35. A market has the following limit orders standing on its book for a particular stock:

Buyer	Bid Size (Number of Shares)	Limit Price (£)	Offer Size (Number of Shares)	Seller
Keith	1,000	19.70		
Paul	200	19.84		
Ann	400	19.89		
Mary	300	20.02		
		20.03	800	Jack
		20.11	1,100	Margaret
		20.16	400	Jeff

Ian submits a day order to sell 1,000 shares, limit £19.83. Assuming that no more buy orders are submitted on that day after Ian submits his order, what would be Ian's average trade price?

- A. £19.70.
 - B. £19.92.
 - C. £20.05.
36. A financial analyst is examining whether a country's financial market is well functioning. She finds that the transaction costs in this market are low and trading volumes are high. She concludes that the market is quite liquid. In such a market:
- A. traders will find it hard to make use of their information.
 - B. traders will find it easy to trade and their trading will make the market less informationally efficient.
 - C. traders will find it easy to trade and their trading will make the market more informationally efficient.
37. The government of a country whose financial markets are in an early stage of development has hired you as a consultant on financial market regulation. Your first task is to prepare a list of the objectives of market regulation. Which of the following is *least likely* to be included in this list of objectives?
- A. Minimize agency problems in the financial markets.
 - B. Ensure that financial markets are fair and orderly.
 - C. Ensure that investors in the stock market achieve a rate of return that is at least equal to the risk-free rate of return.

SOLUTIONS

1. C is correct. Takabe is best characterized as an information-motivated trader. Takabe believes that his model provides him superior information about the movements in the stock market and his motive for trading is to profit from this information.
2. B is correct. Beach is an investor. He is simply investing in risky assets consistent with his level of risk aversion. Beach is not hedging any existing risk or using information to identify and trade mispriced securities. Therefore, he is not a hedger or an information-motivated trader.
3. A is correct. Smith is a hedger. The short position on the BRL futures contract offsets the BRL long position in three months. She is hedging the risk of the BRL depreciating against the USD. If the BRL depreciates, the value of the cash inflow goes down in USD terms but there is a gain on the futures contracts.
4. A is correct. Regulation of arbitrageurs' profits is not a function of the financial system. The financial system facilitates the allocation of capital to the best uses and the purposes for which people use the financial system, including borrowing money.
5. C is correct. The purchase of real estate properties is a transaction in the alternative investment market.
6. B is correct. The 90-day commercial paper and negotiable certificates of deposit are money market instruments.
7. B is correct. This transaction is a sale in the primary market. It is a sale of shares from the issuer to the investor and funds flow to the issuer of the security from the purchaser.
8. A is correct. Warrants are least likely to be part of the fund. Warrant holders have the right to buy the issuer's common stock. Thus, warrants are typically classified as equity and are least likely to be a part of a fixed-income mutual fund. Commercial paper and repurchase agreements are short-term fixed-income securities.
9. C is correct. When investors want to sell their shares, investors of an open-end fund sell the shares back to the fund whereas investors of a closed-end fund sell the shares to others in the secondary market. Closed-end funds are available to new investors but they must purchase shares in the fund in the secondary market. The shares of a closed-end fund trade at a premium or discount to net asset value.
10. B is correct. SPDRs trade in the secondary market and are a pooled investment vehicle.
11. B is correct. The investment companies that create exchange-traded funds (ETFs) are financial intermediaries. ETFs are securities that represent ownership in the assets held by the fund. The transaction costs of trading shares of ETFs are substantially lower than the combined costs of trading the underlying assets of the ETF.
12. A is correct. Once you have entered into a forward contract, it is difficult to exit from the contract. As opposed to a futures contract, trading out of a forward contract is quite difficult. There is no exchange of cash at the origination of a forward contract. There is no exchange on a forward contract until the maturity of the contract.
13. A is correct. Harris is least likely to find counterparty risk associated with a futures contract. There is limited counterparty risk in a futures contract because the clearinghouse is on the other side of every contract.
14. B is correct. Buying a put option on the dollar will ensure a minimum exchange rate but does not have to be exercised if the exchange rate moves in a favorable direction. Forward and futures contracts would lock in a fixed rate but would not allow for the possibility to profit in case the

value of the dollar three months later in the spot market turns out to be greater than the value in the forward or futures contract.

- 15.** B is correct. The agreement between the publisher and the paper supplier to respectively buy and supply paper in the future at a price agreed upon today is a forward contract.
- 16.** B is correct. The holder of the call option will exercise the call options if the price is above the exercise price of \$305 per share. Note that if the stock price is above \$305 but less than \$308, the option would be exercised even though the net result for the option buyer after considering the premium is a loss. For example, if the stock price is \$307, the option buyer would exercise the option to make $\$2 = \$307 - \$305$ per share, resulting in a loss of $\$1 = \$3 - \$2$ after considering the premium. It is better to exercise and have a loss of only \$1, however, rather than not exercise and lose the entire \$3 premium.
- 17.** B is correct. The process can best be described as arbitrage because it involves buying and selling instruments, whose values are closely related, at different prices in different markets.
- 18.** A is correct. Robert's exposure to the risk of the stock of the Michelin Group is long. The exposure as a result of the long call position is long. The exposure as a result of the short put position is also long. Therefore, the combined exposure is long.
- 19.** B is correct. The maximum leverage ratio is $1.82 = 100\% \text{ position} \div 55\% \text{ equity}$. The maximum leverage ratio associated with a position financed by the minimum margin requirement is one divided by the minimum margin requirement.
- 20.** C is correct. The return is 50 percent. If the position had been unleveraged, the return would be $20\% = (60 - 50)/50$. Because of leverage, the return is $50\% = 2.5 \times 20\%$.

Another way to look at this problem is that the equity contributed by the trader (the minimum margin requirement) is $40\% = 100\% \div 2.5$. The trader contributed $\$20 = 40\%$ of \$50 per share. The gain is \$10 per share, resulting in a return of $50\% = 10/20$.

- 21.** B is correct. The return is -15.4 percent.

$$\text{Total cost of the purchase} = \$16,000 = 500 \times \$32$$

$$\text{Equity invested} = \$12,000 = 0.75 \times \$16,000$$

$$\text{Amount borrowed} = \$4,000 = 16,000 - 12,000$$

$$\text{Interest paid at month end} = \$80 = 0.02 \times \$4,000$$

$$\text{Dividend received at month end} = \$250 = 500 \times \$0.50$$

$$\text{Proceeds on stock sale} = \$14,000 = 500 \times \$28$$

$$\text{Total commissions paid} = \$20 = \$10 + \$10$$

$$\text{Net gain/loss} = -\$1,850 = -16,000 - 80 + 250 + 14,000 - 20$$

$$\text{Initial investment including commission on purchase} = \$12,010$$

$$\text{Return} = -15.4\% = -\$1,850/\$12,010$$

- 22.** A is correct. She will need to contribute €3,760 as margin. In view of the possibility of a loss, if the stock price goes up, she will need to contribute $\text{€}3,760 = 40\%$ of €9,400 as the initial margin. Rogers will need to leave the proceeds from the short sale ($\text{€}9,400 = 200 \times \text{€}47$) on deposit.
- 23.** B is correct. A margin call will first occur at a price of \$17.86. Because you have contributed half and borrowed the remaining half, your initial equity is 50 percent of the initial stock price, or $\$12.50 = 0.50 \times \25 . If P is the subsequent price, your equity would change by an amount equal to the change in price. So, your equity at price P would be $12.50 + (P - 25)$. A margin call will occur when the percentage margin drops to 30 percent. So, the price at which a margin call will occur is the solution to the following equation.

$$\frac{\text{Equity/Share}}{\text{Price/Share}} = \frac{12.50+P-25}{P} = 30\%$$

The solution is $P = \$17.86$.

24. C is correct. The market is 9.95 bid, offered at 10.02. The best bid is at €9.95 and the best offer is €10.02.
25. C is correct. This order is said to make a new market. The new buy order is at ¥123.40, which is better than the current best bid of ¥123.35. Therefore, the buy order is making a new market. Had the new order been at ¥123.35, it would be said to make the market. Because the new buy limit order is at a price less than the best offer of ¥123.80, it will not immediately execute and is not taking the market.
26. A is correct. This order is said to take the market. The new sell order is at \$54.62, which is at the current best bid. Therefore, the new sell order will immediately trade with the current best bid and is taking the market.
27. B is correct. An instruction regarding when to fill an order is considered a validity instruction.
28. B is correct. The maximum possible loss is \$1,300. If the stock price crosses \$50, the stop buy order will become valid and will get executed at a maximum limit price of \$55. The maximum loss per share is $\$13 = \$55 - \$42$, or \$1,300 for 100 shares.
29. B is correct. The most appropriate order is a good-till-cancelled stop sell order. This order will be acted on if the stock price declines below a specified price (in this case, \$27.50). This order is sometimes referred to as a good-till-cancelled stop loss sell order. You are generally bullish about the stock, as indicated by no immediate intent to sell, and would expect a loss on short selling the stock. A stop buy order is placed to buy a stock when the stock is going up.
30. B is correct. The investment bank bears the risk that the issue may be undersubscribed at the offering price. If the entire issue is not sold, the investment bank underwriting the issue will buy the unsold securities at the offering price.
31. B is correct. This sale is a private placement. As the company is already publicly traded, the share sale is clearly not an initial public offering. The sale also does not involve a shelf registration because the company is not selling shares to the public on a piecemeal basis.
32. A is correct. This offering is a rights offering. The company is distributing rights to buy stock at a fixed price to existing shareholders in proportion to their holdings.
33. C is correct. Order III (time of arrival of 9:53:04) has precedence. In the order precedence hierarchy, the first rule is price priority. Based on this rule, sell orders II, III, and IV get precedence over order I. The next rule is display precedence at a given price. Because order II is a hidden order, orders III and IV get precedence. Finally, order III gets precedence over order IV based on time priority at same price and same display status.
34. C is correct. The order for 500 shares would get cancelled; there would be no fill. Li is willing to buy at CNY 74.25 or less but the minimum offer price in the book is CNY 74.30; therefore, no part of the order would be filled. Because Li's order is immediate-or-cancel, it would be cancelled.
35. B is correct. Ian's average trade price is:

$$£19.92 = \frac{300 \times £20.02 + 400 \times £19.89 + 200 \times £19.84}{300 + 400 + 200}$$

Ian's sell order first fills with the most aggressively priced buy order, which is Mary's order for 300 shares at £20.02. Ian still has 700 shares for sale. The next most aggressively priced buy order is Ann's order for 400 shares at £19.89. This order is filled. Ian still has 300 shares for sale. The next most aggressively priced buy order is Paul's order for 200 shares at £19.84. A third trade takes place. Ian still has 100 shares for sale.

The next buy order is Keith's order for 1,000 shares at £19.70. However, this price is below Ian's limit price of £19.83. Therefore, no more trade is possible.

- 36.** C is correct. In such a market, well-informed traders will find it easy to trade and their trading will make the market more informationally efficient. In a liquid market, it is easier for informed traders to fill their orders. Their trading will cause prices to incorporate their information and the prices will be more in line with the fundamental values.
- 37.** C is correct. Ensure that investors in the stock market achieve a rate of return that is at least equal to the risk-free rate of return is least likely to be included as an objective of market regulation. Stocks are risky investments and there would be occasions when a stock market investment would not only have a return less than the risk-free rate but also a negative return. Minimizing agency costs and ensuring that financial markets are fair and orderly are objectives of market regulation.

NOTES

¹Harris, Larry. 2003. *Trading and Exchanges: Market Microstructure for Practitioners*. New York: Oxford University Press.

Security Market Indexes

by Paul D. Kaplan PhD, CFA and Dorothy C. Kelly CFA

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LEARNING OUTCOMES

The candidate should be able to:

- ☐ describe a security market index
- ☐ calculate and interpret the value, price return, and total return of an index
- ☐ describe the choices and issues in index construction and management
- ☐ compare the different weighting methods used in index construction
- ☐ calculate and analyze the value and return of an index given its weighting method
- ☐ describe rebalancing and reconstitution of an index
- ☐ describe uses of security market indexes
- ☐ describe types of equity indexes
- ☐ compare types of security market indexes
- ☐ describe types of fixed-income indexes
- ☐ describe indexes representing alternative investments

1. INTRODUCTION

Investors gather and analyze vast amounts of information about security markets on a continual basis. Because this work can be both time consuming and data intensive, investors often use a single measure that consolidates this information and reflects the performance of an entire security market.

Security market indexes were first introduced as a simple measure to reflect the performance of the US stock market. Since then, security market indexes have evolved into important multi-purpose tools that help investors track the performance of various security markets, estimate risk, and evaluate the performance of investment managers. They also form the basis for new investment products.

in·dex, *noun* (pl. **in·dex·es** or **in·di·ces**) Latin *indic-*, *index*, from *indicare* to indicate: an indicator, sign, or measure of something.

ORIGIN OF MARKET INDEXES

Investors had access to regularly published data on individual security prices in London as early as 1698, but nearly 200 years passed before they had access to a simple indicator to reflect security market information. To give readers a sense of how the US stock market in general performed on a given day, publishers Charles H. Dow and Edward D. Jones introduced the Dow Jones Average, the world’s first security market index, in 1884. The index, which appeared in *The Customers’ Afternoon Letter*, consisted of the stocks of nine railroads and two industrial companies. It eventually became the Dow Jones Transportation Average. Convinced that industrial companies, rather than railroads, would be “the great speculative market” of the future, Dow and Jones introduced a second index in May 1896—the Dow Jones Industrial Average (DJIA). It had an initial value of 40.94 and consisted of 12 stocks from major US industries. Today, investors can choose from among thousands of indexes to measure and monitor different security markets and asset classes.

This reading is organized as follows. Section 2 defines a security market index and explains how to calculate the price return and total return of an index for a single period and over multiple periods. Section 3 describes how indexes are constructed and managed. Section 4 discusses the use of market indexes. Sections 5, 6, and 7 discuss various types of indexes, and the final section summarizes the reading. Practice problems follow the conclusions and summary.

2. INDEX DEFINITION AND CALCULATIONS OF VALUE AND RETURNS

LEARNING OUTCOMES

- The candidate should be able to:
- ☐ describe a security market index
 - ☐ calculate and interpret the value, price return, and total return of an index

A **security market index** represents a given security market, market segment, or asset class. Most indexes are constructed as portfolios of marketable securities.

The value of an index is calculated on a regular basis using either the actual or estimated market prices of the individual securities, known as **constituent securities**, within the index. For each security market index, investors may encounter two versions of the same index (i.e., an index with identical constituent securities and weights): one version based on price return and one version based on total return. As the name suggests, a **price return index**, also known as a **price index**, reflects *only* the prices of the constituent securities within the index. A **total return index**, in contrast, reflects not only the prices of the constituent securities but also the reinvestment of all income received since inception.

At inception, the values of the price and total return versions of an index are equal. As time passes, however, the value of the total return index, which includes the reinvestment of all dividends and/or interest received, will exceed the value of the price return index by an increasing amount. A look at how the values of each version are calculated over multiple periods illustrates why.

The value of a price return index is calculated as:

Equation 1

$$V_{PRI} = \frac{\sum_{i=1}^N n_i P_i}{D}$$

where

- V_{PRI} = the value of the price return index
- n_i = the number of units of constituent security i held in the index portfolio
- N = the number of constituent securities in the index
- P_i = the unit price of constituent security i
- D = the value of the divisor

The **divisor** is a number initially chosen at inception. It is frequently chosen so that the price index has a convenient initial value, such as 1,000. The index provider then adjusts the value of the divisor as necessary to avoid changes in the index value that are unrelated to changes in the prices of its constituent securities. For example, when changing index constituents, the index provider may adjust the divisor so that the value of the index with the new constituents equals the value of the index prior to the changes.

Index return calculations, like calculations of investment portfolio returns, may measure price return or total return. **Price return** measures only price appreciation or percentage change in price. **Total return** measures price appreciation plus interest, dividends, and other distributions.

2.1. Calculation of Single-Period Returns

For a security market index, price return can be calculated in two ways: either the percentage change in value of the price return index, or the weighted average of price returns of the constituent securities. The price return of an index can be expressed as:

Equation 2

$$PR_I = \frac{V_{PRI1} - V_{PRI0}}{V_{PRI0}}$$

where

PR_I = the price return of the index portfolio (as a decimal number, i.e., 12 percent is 0.12)

V_{PRI1} = the value of the price return index at the end of the period

V_{PRI0} = the value of the price return index at the beginning of the period

Similarly, the price return of each constituent security can be expressed as:

Equation 3

$$PR_i = \frac{P_{i1} - P_{i0}}{P_{i0}}$$

where

PR_i = the price return of constituent security i (as a decimal number)

P_{i1} = the price of constituent security i at the end of the period

P_{i0} = the price of constituent security i at the beginning of the period

Because the price return of the index equals the weighted average of price returns of the individual securities, we can write:

Equation 4

$$PR_I = \sum_{i=1}^N w_i PR_i = \sum_{i=1}^N w_i \left(\frac{P_{i1} - P_{i0}}{P_{i0}} \right)$$

where:

PR_I = the price return of index portfolio (as a decimal number)

PR_i = the price return of constituent security i (as a decimal number)

N = the number of individual securities in the index

w_i = the weight of security i (the fraction of the index portfolio allocated to security i)

P_{i1} = the price of constituent security i at the end of the period

P_{i0} = the price of constituent security i at the beginning of the period

Equation 4 can be rewritten simply as:

Equation 5

$$PR_I = w_1 PR_1 + w_2 PR_2 + \dots + w_N PR_N$$

where

PR_I = the price return of index portfolio (as a decimal number)

PR_i = the price return of constituent security i (as a decimal number)

w_i = the weight of security i (the fraction of the index portfolio allocated to security i)

N = the number of securities in the index

Total return measures price appreciation plus interest, dividends, and other distributions. Thus, the **total return** of an index is the price appreciation, or change in the value of the price return index, plus income (dividends and/or interest) over the period, expressed as a percentage of the beginning value of the price return index. The total return of an index can be expressed as:

Equation 6

$$TR_I = \frac{V_{PRI1} - V_{PRI0} + Inc_I}{V_{PRI0}}$$

where

TR_I = the total return of the index portfolio (as a decimal number)

V_{PRI1} = the value of the price return index at the end of the period

V_{PRI0} = the value of the price return index at the beginning of the period

Inc_I = the total income (dividends and/or interest) from all securities in the index held over the period

The total return of an index can also be calculated as the weighted average of total returns of the constituent securities. The total return of each constituent security in the index is calculated as:

Equation 7

$$TR_i = \frac{P_{1i} - P_{0i} + Inc_i}{P_{0i}}$$

where

TR_i = the total return of constituent security i (as a decimal number)

P_{1i} = the price of constituent security i at the end of the period

P_{0i} = the price of constituent security i at the beginning of the period

Inc_i = the total income (dividends and/or interest) from security i over the period

Because the total return of an index can be calculated as the weighted average of total returns of the constituent securities, we can express total return as:

Equation 8

$$TR_I = \sum_{i=1}^N w_i TR_i = \sum_{i=1}^N w_i \left(\frac{P_{1i} - P_{0i} + Inc_i}{P_{0i}} \right)$$

Equation 8 can be rewritten simply as

Equation 9

$$TR_I = w_1 TR_1 + w_2 TR_2 + \dots + w_N TR_N$$

where

TR_I = the total return of the index portfolio (as a decimal number)

TR_i = the total return of constituent security i (as a decimal number)

w_i = the weight of security i (the fraction of the index portfolio allocated to security i)

N = the number of securities in the index

2.2. Calculation of Index Values over Multiple Time Periods

The calculation of index values over multiple time periods requires geometrically linking the series of index returns. With a series of price returns for an index, we can calculate the value of the price return index with the following equation:

Equation 10

$$V_{PRI\ t} = V_{PRI\ 0} (1 + PR_{I\ 1}) (1 + PR_{I\ 2}) \dots (1 + PR_{I\ t})$$

where

$V_{PRI\ 0}$ = the value of the price return index at inception

$V_{PRI\ t}$ = the value of the price return index at time t

$PR_{I\ t}$ = the price return (as a decimal number) on the index over period t , $t = 1, 2, \dots, T$

For an index with an inception value set to 1,000 and price returns of 5 percent and 3 percent for Periods 1 and 2 respectively, the values of the price return index would be calculated as follows:

Period	Return (%)	Calculation	Ending Value
0		1,000(1.00)	1,000.00
1	5.00	1,000(1.05)	1,050.00
2	3.00	1,000(1.05)(1.03)	1,081.50

Similarly, the series of total returns for an index is used to calculate the value of the total return index with the following equation:

Equation 11

$$V_{TRIT\ t} = V_{TRI\ 0} (1 + TR_{I\ 1}) (1 + TR_{I\ 2}) \dots (1 + TR_{I\ t})$$

where

$V_{TRI\ 0}$ = the value of the index at inception

$V_{TRIT\ t}$ = the value of the total return index at time t

$TR_{I\ t}$ = the total return (as a decimal number) on the index over period t , $t = 1, 2, \dots, T$

Suppose that the same index yields an additional 1.5 percent return from income in Period 1 and an additional 2.0 percent return from income in Period 2, bringing the total returns for Periods 1 and 2, respectively, to 6.5 percent and 5 percent. The values of the total return index would be calculated as follows:

Period	Return (%)	Calculation	Ending Value
0		1,000(1.00)	1,000.00
1	6.50	1,000(1.065)	1,065.00
2	5.00	1,000(1.065)(1.05)	1,118.25

As illustrated above, as time passes, the value of the total return index, which includes the reinvestment of all dividends and/or interest received, exceeds the value of the price return index by an increasing amount.

3. INDEX CONSTRUCTION

LEARNING OUTCOMES

The candidate should be able to:

- describe the choices and issues in index construction and management

- compare the different weighting methods used in index construction
- calculate and analyze the value and return of an index given its weighting method

Constructing and managing a security market index is similar to constructing and managing a portfolio of securities. Index providers must decide the following:

1. Which target market should the index represent?
2. Which securities should be selected from that target market?
3. How much weight should be allocated to each security in the index?
4. When should the index be rebalanced?
5. When should the security selection and weighting decision be re-examined?

3.1. Target Market and Security Selection

The first decision in index construction is identifying the target market, market segment, or asset class that the index is intended to represent. The target market may be defined very broadly or narrowly. It may be based on asset class (e.g., equities, fixed income, real estate, commodities, hedge funds); geographic region (e.g., Japan, South Africa, Latin America, Europe); the exchange on which the securities are traded (e.g., Shanghai, Toronto, Tokyo), and/or other characteristics (e.g., economic sector, company size, investment style, duration, or credit quality).

The target market determines the investment universe and the securities available for inclusion in the index. Once the investment universe is identified, the number of securities and the specific securities to include in the index must be determined. The constituent securities could be nearly all those in the target market or a representative sample of the target market. Some equity indexes, such as the S&P 500 Index and the FTSE 100, fix the number of securities included in the index and indicate this number in the name of the index. Other indexes allow the number of securities to vary to reflect changes in the target market or to maintain a certain percentage of the target market. For example, the Tokyo Stock Price Index (TOPIX) represents and includes all of the largest stocks, known as the First Section, listed on the Tokyo Stock Exchange. To be included in the First Section—and thus the TOPIX—stocks must meet certain criteria, such as the number of shares outstanding, the number of shareholders, and market capitalization. Stocks that no longer meet the criteria are removed from the First Section and also the TOPIX. Objective or mechanical rules determine the constituent securities of most, but not all, indexes. The S&P Bombay Stock Exchange Sensitive Index, also called the S&P BSE SENSEX and the S&P 500, for example, use a selection committee and more subjective decision-making rules to determine constituent securities.

3.2. Index Weighting

The weighting decision determines how much of each security to include in the index and has a substantial impact on an index’s value. Index providers use a number of methods to weight the constituent securities in an index. Indexes can be price weighted, equal weighted, market-capitalization weighted, or fundamentally weighted. Each weighting method has its advantages and disadvantages.

3.2.1. Price Weighting

The simplest method to weight an index and the one used by Charles Dow to construct the Dow Jones Industrial Average is **price weighting**. In price weighting, the weight on each constituent security is determined by dividing its price by the sum of all the prices of the constituent securities. The weight is calculated using the following formula:

Equation 12

$$w_i^P = \frac{P_i}{\sum_{i=1}^N P_i}$$

Exhibit 1 illustrates the values, weights, and single-period returns following inception of a price-weighted equity index with five constituent securities. The value of the price-weighted index is determined by dividing the sum of the security values (101.50) by the divisor, which is typically set at inception to equal the initial number of securities in the index. Thus, in our example, the divisor is 5 and the initial value of the index is calculated as 101.50 ÷ 5 = 20.30.

Exhibit 1: Example of a Price-Weighted Equity Index

Security	Shares in Index	BOP Price	Value (Shares × BOP Price)	BOP Weight (%)	EOP Price	Dividends Per Share	Value (Shares × EOP Price)	Total Dividends	Price Return (%)	Total Return (%)	BOP Weight × Price Return (%)	BOP Weight × Total Return (%)	EOP Weight (%)
A	1	50.00	50.00	49.26	55.00	0.75	55.00	0.75	10.00	11.50	4.93	5.66	52.38
B	1	25.00	25.00	24.63	22.00	0.10	22.00	0.10	−12.00	−11.60	−2.96	−2.86	20.95
C	1	12.50	12.50	12.32	8.00	0.00	8.00	0.00	−36.00	−36.00	−4.43	−4.43	7.62
D	1	10.00	10.00	9.85	14.00	0.05	14.00	0.05	40.00	40.50	3.94	3.99	13.33
E	1	4.00	4.00	3.94	6.00	0.00	6.00	0.00	50.00	50.00	1.97	1.97	5.72
Total			101.50	100.00			105.00	0.90			3.45	4.33	100.00
Index Value			20.30				21.00	0.18	3.45	4.33			
Divisor = 5													
BOP = Beginning of period													
EOP = End of period													

Type of Index	BOP Value	Return (%)	EOP Value
Price Return	20.30	3.45	21.00
Total Return	20.30	4.33	21.18

As illustrated in this exhibit, Security A, which has the highest price, also has the highest weighting and thus will have the greatest impact on the return of the index. Note how both the price return and the total return of the index are calculated on the basis of the corresponding returns on the constituent securities.

A property unique to price-weighted indexes is that a stock split on one constituent security changes the weights on all the securities in the index.¹ To prevent the stock split and the resulting new weights from changing the value of the index, the index provider must adjust the value of the divisor as illustrated in Exhibit 2.

Given a 2-for-1 split in Security A, the divisor is adjusted by dividing the sum of the constituent prices *after* the split (77.50) by the value of the index *before* the split (21.00). This adjustment results in changing the divisor from 5 to 3.69 so that the index value is maintained at 21.00.

The primary advantage of price weighting is its simplicity. The main disadvantage of price weighting is that it results in arbitrary weights for each security. In particular, a stock split in any one security causes arbitrary changes in the weights of all the constituents' securities.

Exhibit 2: Impact of 2-for-1 Split in Security A

Security	Price before Split	Weight before Split (%)	Price after Split	Weight after Split (%)
A	55.00	52.38	27.50	35.48
B	22.00	20.95	22.00	28.39
C	8.00	7.62	8.00	10.32
D	14.00	13.33	14.00	18.07
E	6.00	5.72	6.00	7.74
Total	105.00	100.00	77.50	100.00
Divisor	5.00		3.69	
Index Value	21.00		21.00	

3.2.2. Equal Weighting

Another simple index weighting method is **equal weighting**. This method assigns an equal weight to each constituent security at inception. The weights are calculated as:

Equation 13

$$w_i^E = \frac{1}{N}$$

where

w_i = fraction of the portfolio that is allocated to security i or weight of security i

N = number of securities in the index

To construct an equal-weighted index from the five securities in [Exhibit 1](#), the index provider allocates one-fifth (20 percent) of the value of the index (at the beginning of the period) to each security. Dividing the value allocated to each security by each security's individual share price determines the number of shares of each security to include in the index. Unlike a price-weighted index, where the weights are arbitrarily determined by the market prices, the weights in an equal-weighted index are arbitrarily assigned by the index provider.

[Exhibit 3](#) illustrates the values, weights, and single-period returns following inception of an equal-weighted index with the same constituent securities as those in [Exhibit 1](#). This example assumes a beginning index portfolio value of 10,000 (i.e., an investment of 2,000 in each security). To set the initial value of the index to 1,000, the divisor is set to 10 (10,000 ÷ 10 = 1,000).

Exhibit 3: Example of an Equal-Weighted Equity Index

Security	Shares in Index	BOP Price	Value (Shares × BOP Price)	Weight (%)	EOP Price	Dividends Per Share	Value (Shares × EOP Price)	Total Dividends	Price Return (%)	Total Return (%)	Weight × Price Return (%)	Weight × Total Return (%)	EOP Weight (%)
A	40	50.00	2,000	20.00	55.00	0.75	2,200	30	10.00	11.50	2.00	2.30	19.93
B	80	25.00	2,000	20.00	22.00	0.10	1,760	8	-12.00	-11.60	-2.40	-2.32	15.94
C	160	12.50	2,000	20.00	8.00	0.00	1,280	0	-36.00	-36.00	-7.20	-7.20	11.60
D	200	10.00	2,000	20.00	14.00	0.05	2,800	10	40.00	40.50	8.00	8.10	25.36
E	500	4.00	2,000	20.00	6.00	0.00	3,000	0	50.00	50.00	10.00	10.00	27.17
Total			10,000	100.00			11,040	48			10.40	10.88	100.00
Index Value			1,000				1,104	4.80	10.40	10.88			
Divisor = 10													
BOP = Beginning of period													
EOP = End of period													

Type of Index	BOP Value	Return (%)	EOP Value
Price Return	1,000.00	10.40	1,104.00
Total Return	1,000.00	10.88	1,108.80

[Exhibit 1](#) and [Exhibit 3](#) demonstrate how different weighting methods result in different returns. The 10.4 percent price return of the equal-weighted index shown in [Exhibit 3](#) differs significantly from the 3.45 percent price return of the price-weighted index in [Exhibit 1](#).

Like price weighting, the primary advantage of equal weighting is its simplicity. Equal weighting, however, has a number of disadvantages. First, securities that constitute the largest fraction of the target market value are underrepresented, and securities that constitute a small fraction of the target market value are overrepresented. Second, after the index is constructed and the prices of constituent securities change, the index is no longer equally weighted. Therefore, maintaining equal weights requires frequent adjustments (rebalancing) to the index.

3.2.3. Market-Capitalization Weighting

In **market-capitalization weighting**, or value weighting, the weight on each constituent security is determined by dividing its market capitalization by the total market capitalization (the sum of the market capitalization) of all the securities in the index. Market capitalization or value is calculated by multiplying the number of shares outstanding by the market price per share.

The market-capitalization weight of security *i* is:

Equation 14

$$w_i^M = \frac{Q_i P_i}{\sum_{j=1}^N Q_j P_j}$$

where

w_i = fraction of the portfolio that is allocated to security *i* or weight of security *i*

Q_i = number of shares outstanding of security *i*

P_i = share price of security *i*

N = number of securities in the index

Exhibit 4 illustrates the values, weights, and single-period returns following inception of a market-capitalization-weighted index for the same five-security market. Security A, with 3,000 shares outstanding and a price of 50 per share, has a market capitalization of 150,000 or 26.29 percent (150,000/570,500) of the entire index portfolio. The resulting index weights in the exhibit reflect the relative value of each security as measured by its market capitalization.

Exhibit 4: Example of a Market-Capitalization-Weighted Equity Index

Stock	Shares Out- standing	BOP Price	BOP Market Cap	BOP Weight (%)	EOP Price	Dividends Per Share	EOP Market Cap	Total Dividends	Price Return (%)	Total Return (%)	BOP Weight × Price Return (%)	BOP Weight × Total Return (%)	EOP Weight (%)
A	3,000	50.00	150,000	26.29	55.00	0.75	165,000	2,250	10.00	11.50	2.63	3.02	28.50
B	10,000	25.00	250,000	43.82	22.00	0.10	220,000	1,000	-12.00	-11.60	-5.26	-5.08	38.00
C	5,000	12.50	62,500	10.96	8.00	0.00	40,000	0	-36.00	-36.00	-3.95	-3.95	6.91
D	8,000	10.00	80,000	14.02	14.00	0.05	112,000	400	40.00	40.50	5.61	5.68	19.34
E	7,000	4.00	28,000	4.91	6.00	0.00	42,000	0	50.00	50.00	2.46	2.46	7.25
Total			570,500	100.00			579,000	3,650			1.49	2.13	100.00
Index Value			1,000				1,014.90	6.40	1.49	2.13			

Divisor = 570.50

BOP = Beginning of period

EOP = End of period

Type of Index	BOP Value	Return (%)	EOP Value
Price Return	1,000.00	1.49	1,014.90
Total Return	1,000.00	2.13	1,021.30

As shown in Exhibit 1, Exhibit 3, and Exhibit 4, the weighting method affects the index’s returns. The price and total returns of the market-capitalization index in Exhibit 4 (1.49 percent and 2.13 percent, respectively) differ significantly from those of the price-weighted (3.45 percent and 4.33 percent, respectively) and equal-weighted (10.40 percent and 10.88 percent respectively) indexes. To understand the source and magnitude of the difference, compare the weights and returns of each security under each of the weighting methods. The weight of Security A, for example, ranges from 49.26 percent in the price-weighted index to 20 percent in the equal-weighted index. With a price return of 10 percent, Security A contributes 4.93 percent to the price return of the price-weighted index, 2.00 percent to the price return of the equal-weighted index, and 2.63 percent to the price return of the market-capitalization-weighted index. With a total return of 11.50 percent, Security A contributes 5.66 percent to the total return of the price-weighted index, 2.30 percent to the total return of the equal-weighted index, and 3.02 percent to the total return of the market-capitalization-weighted index.

3.2.3.1. Float-Adjusted Market-Capitalization Weighting

In float-adjusted market-capitalization weighting, the weight on each constituent security is determined by adjusting its market capitalization for its market float. Typically, market float is the number of shares of the constituent security that are available to the investing public. For companies that are closely held, only a portion of the shares outstanding are available to the investing public (the rest are held by a small group of controlling investors). In addition to excluding shares held by controlling shareholders, most float-adjusted market-capitalization-weighted indexes also exclude shares held by other corporations and governments. Some providers of indexes that are designed to represent the investment opportunities of global investors further reduce the number of shares included in the index by excluding shares that are not available to foreigner investors. The index providers may refer to these indexes as “free-float-adjusted market-capitalization-weighted indexes.”

Float-adjusted market-capitalization-weighted indexes reflect the shares available for public trading by multiplying the market price per share by the number of shares available to the investing public (i.e., the float-adjusted market capitalization) rather than the total number of shares outstanding (total market capitalization). Currently, most market-capitalization-weighted indexes are float adjusted. Therefore, unless otherwise indicated, for the remainder of this reading, “market-capitalization” weighting refers to float-adjusted market-capitalization weighting.

The float-adjusted market-capitalization weight of security *i* is calculated as:

Equation 15

$$w_i^M = \frac{f_i Q_i P_i}{\sum_{j=1}^N f_j Q_j P_j}$$

where

f_i = fraction of shares outstanding in the market float

w_i = fraction of the portfolio that is allocated to security *i* or weight of security *i*

Q_i = number of shares outstanding of security *i*

P_i = share price of security *i*

N = number of securities in the index

Exhibit 5 illustrates the values, weights, and single-period returns following inception of a float-adjusted market-capitalization-weighted equity index using the same five securities as before. The low percentage of shares of Security D in the market float compared with the number of shares outstanding indicates that the security is closely held.

Exhibit 5: Example of Float-Adjusted Market-Capitalization-Weighted Equity Index

Stock	Shares Out- standing	% Shares in Market Float	Shares in Index	BOP Price	BOP Float- Adjusted Market Cap	BOP Weight (%)	EOP Price	Dividends Per Share	Ending Float- Adjusted Market Cap	Total Dividends	Price Return (%)	Total Return (%)	BOP Weight × Price Return (%)	BOP Weight × Total Return (%)	EOP Weight (%)
A	3,000	100	3,000	50.00	150,000	35.40	55.00	0.75	165,000	2,250	10.00	11.50	3.54	4.07	39.61
B	10,000	70	7,000	25.00	175,000	41.31	22.00	0.10	154,000	700	−12.00	−11.60	−4.96	−4.79	36.97
C	5,000	90	4,500	12.50	56,250	13.28	8.00	0.00	36,000	0	−36.00	−36.00	−4.78	−4.78	8.64
D	8,000	25	2,000	10.00	20,000	4.72	14.00	0.05	28,000	100	40.00	40.50	1.89	1.91	6.72
E	7,000	80	5,600	4.00	22,400	5.29	6.00	0.00	33,600	0	50.00	50.00	2.65	2.65	8.06
Total					423,650	100.00			416,600	3,050			−1.66	−0.94	100.00
Index Value					1,000				983.36	7.20	−1.66	−0.94			

Divisor = 423.65

BOP = Beginning of period

EOP = End of period

Type of Index	Initial Value	Return (%)	Ending Value
Price Return	1,000.00	−1.66	983.36
Total Return	1,000.00	−0.94	990.56

The primary advantage of market-capitalization weighting (including float adjusted) is that constituent securities are held in proportion to their value in the target market. The primary disadvantage is that constituent securities whose prices have risen the most (or fallen the most) have a greater (or lower) weight in the index (i.e., as a security's price rises relative to other securities in the index, its weight increases; and as its price decreases in value relative to other securities in the index, its weight decreases). This weighting method leads to overweighting stocks that have risen in price (and may be overvalued) and underweighting stocks that have declined in price (and may be undervalued). The effect of this weighting method is similar to a momentum investment strategy in that over time, the securities that have risen in price the most will have the largest weights in the index.

3.2.4. Fundamental Weighting

Fundamental weighting attempts to address the disadvantages of market-capitalization weighting by using measures of a company's size that are independent of its security price to determine the weight on each constituent security. These measures include book value, cash flow, revenues, earnings, dividends, and number of employees.

Some fundamental indexes use a single measure, such as total dividends, to weight the constituent securities, whereas others combine the weights from several measures to form a composite value that is used for weighting.

Letting F_i denote a given fundamental size measure of company i , the fundamental weight on security i is:

Equation 16

$$w_i^F = \frac{F_i}{\sum_{j=1}^N F_j}$$

Relative to a market-capitalization-weighted index, a fundamental index with weights based on such an item as earnings will result in greater weights on constituent securities with earnings yields (earnings divided by price) that are higher than the earnings yield of the overall market-weighted portfolio. Similarly, stocks with earnings yields less than the yield on the overall market-weighted portfolio will have lower weights. For example, suppose there are two stocks in an index. Stock A has a market capitalization of €200 million, Stock B has a market capitalization of €800 million, and their aggregate market capitalization is €1 billion (€1,000 million). Both companies have earnings of €20 million and aggregate earnings of €40 million. Thus, Stock A has an earnings yield of 10 percent (20/200) and Stock B has an earnings yield of 2.5 percent (20/800). The earnings weight of Stock A is 50 percent (20/40), which is higher than its market-capitalization weight of 20 percent (200/1,000). The earnings weight of Stock B is 50 percent (20/40), which is less than its market-capitalization weight of 80 percent (800/1,000). Relative to the market-cap-weighted index, the earnings-weighted index over-weights the high-yield Stock A and under-weights the low-yield Stock B.

The most important property of fundamental weighting is that it leads to indexes that have a "value" tilt. That is, a fundamentally weighted index has ratios of book value, earnings, dividends, etc. to market value that are higher than its market-capitalization-weighted counterpart. Also, in contrast to the momentum "effect" of market-capitalization-weighted indexes, fundamentally weighted indexes generally will have a contrarian "effect" in that the portfolio weights will shift away from securities that have increased in relative value and toward securities that have fallen in relative value whenever the portfolio is rebalanced.

4. INDEX MANAGEMENT: REBALANCING AND RECONSTITUTION

LEARNING OUTCOME

The candidate should be able to:

- describe rebalancing and reconstitution of an index

So far, we have discussed index construction. Index management entails the two remaining questions:

- When should the index be rebalanced?

- When should the security selection and weighting decisions be re-examined?

4.1. Rebalancing

Rebalancing refers to adjusting the weights of the constituent securities in the index. To maintain the weight of each security consistent with the index’s weighting method, the index provider rebalances the index by adjusting the weights of the constituent securities on a regularly scheduled basis (rebalancing dates)—usually quarterly. Rebalancing is necessary because the weights of the constituent securities change as their market prices change. Note, for example, that the weights of the securities in the equal-weighted index ([Exhibit 3](#)) at the end of the period are no longer equal (i.e., 20 percent):

Security A	19.93%
Security B	15.94
Security C	11.60
Security D	25.36
Security E	27.17

In rebalancing the index, the weights of Securities D and E (which had the highest returns) would be decreased and the weights of Securities A, B, and C (which had the lowest returns) would be increased. Thus, rebalancing creates turnover within an index.

Price-weighted indexes are not rebalanced because the weight of each constituent security is determined by its price. For market-capitalization-weighted indexes, rebalancing is less of a concern because the indexes largely rebalance themselves. In our market-capitalization index, for example, the weight of Security C automatically declined from 10.96 percent to 6.91 percent, reflecting the 36 percent decline in its market price. Market-capitalization weights are only adjusted to reflect mergers, acquisitions, liquidations, and other corporate actions between rebalancing dates.

4.2. Reconstitution

Reconstitution is the process of changing the constituent securities in an index. It is similar to a portfolio manager deciding to change the securities in his or her portfolio. Reconstitution is part of the rebalancing cycle. The reconstitution date is the date on which index providers review the constituent securities, re-apply the initial criteria for inclusion in the index, and select which securities to retain, remove, or add. Constituent securities that no longer meet the criteria are replaced with securities that do meet the criteria. Once the revised list of constituent securities is determined, the weighting method is re-applied. Indexes are reconstituted to reflect changes in the target market (bankruptcies, de-listings, mergers, acquisitions, etc.) and/or to reflect the judgment of the selection committee.

Reconstitution creates turnover in a number of different ways, particularly for market-capitalization-weighted indexes. When one security is removed and another is added, the index provider has to change the weights of the other securities in order to maintain the market-capitalization weighting of the index.

The frequency of reconstitution is a major issue for widely used indexes and their constituent securities. The Russell 2000 Index, for example, reconstitutes annually. It is used as a benchmark by numerous investment funds, and each year, prior to the index’s reconstitution, the managers of these funds buy stocks they think will be added to the index—driving those stocks’ prices up—and sell stocks they think will be deleted from the index—driving those stocks’ prices down. [Exhibit 6](#) illustrates a historical example of the potential impact of these decisions. Beginning in late April 2009, some managers began acquiring and bidding up the price of Uranium Energy Corporation (UEC) because they believed that it would be included in the reconstituted Russell 2000 Index. On 12 June, Russell listed UEC as a preliminary addition to the Russell 2000 Index and the Russell 3000 Index.² By that time, the stock value had increased by more than 300 percent. Investors continued to bid up the stock price in the weeks following the announcement, and the stock closed on the reconstitution date of 30 June at USD2.90, up nearly 400 percent for the quarter.

Exhibit 6: Three-Month Performance of Uranium Energy Corporation and NASDAQ April through June 2009



Source: Yahoo! Finance and Capital IQ.

5. USES OF MARKET INDEXES

LEARNING OUTCOME

The candidate should be able to:

☐ describe uses of security market indexes

Indexes were initially created to give a sense of how a particular security market performed on a given day. With the development of modern financial theory, their uses in investment management have expanded significantly. Some of the major uses of indexes include:

- gauges of market sentiment;
- proxies for measuring and modeling returns, systematic risk, and risk-adjusted performance;
- proxies for asset classes in asset allocation models;
- benchmarks for actively managed portfolios; and
- model portfolios for such investment products as index funds and exchange-traded funds (ETFs).

Investors using security market indexes must be familiar with how various indexes are constructed in order to select the index or indexes most appropriate for their needs.

5.1. Gauges of Market Sentiment

The original purpose of stock market indexes was to provide a gauge of investor confidence or market sentiment. As indicators of the collective opinion of market participants, indexes reflect investor attitudes and behavior. The Dow Jones Industrial Average has a long history, is frequently quoted in the media, and remains a popular gauge of market sentiment. It may not accurately reflect the overall attitude of investors or the “market,” however, because the index consists of only 30 of the thousands of US stocks traded each day.

5.2. Proxies for Measuring and Modeling Returns, Systematic Risk, and Risk-Adjusted Performance

The capital asset pricing model (CAPM) defines beta as the systematic risk of a security with respect to the entire market. The market portfolio in the CAPM consists of all risky securities. To represent the performance of the market portfolio, investors use a broad index. For example, the Tokyo Price Index (TOPIX) and the S&P 500 often serve as proxies for the market portfolio in Japan and the United States, respectively, and are used for measuring and modeling systematic risk and market returns.

Security market indexes also serve as market proxies when measuring risk-adjusted performance. The beta of an actively managed portfolio allows investors to form a passive alternative with the same level of systematic risk. For example, if the beta of an actively managed portfolio of global stocks is 0.95 with respect to the MSCI World Index, investors can create a passive portfolio with the same systematic risk by investing 95 percent of their portfolio in a MSCI World Index fund and holding the remaining 5 percent in cash. Alpha, the difference between the return of the actively managed portfolio and the return of the passive portfolio, is a measure of risk-adjusted return or investment performance. Alpha can be the result of manager skill (or lack thereof), transaction costs, and fees.

5.3. Proxies for Asset Classes in Asset Allocation Models

Because indexes exhibit the risk and return profiles of select groups of securities, they play a critical role as proxies for asset classes in asset allocation models. They provide the historical data used to model the risks and returns of different asset classes.

5.4. Benchmarks for Actively Managed Portfolios

Investors often use indexes as benchmarks to evaluate the performance of active portfolio managers. The index selected as the benchmark should reflect the investment strategy used by the manager. For example, an active manager investing in global small-capitalization stocks should be evaluated using a benchmark index, such as the FTSE Global Small Cap Index, which includes approximately 4,400 liquid small-capitalization stocks across 47 countries as of August 2018.

The choice of an index to use as a benchmark is important because an inappropriate index could lead to incorrect conclusions regarding an active manager’s investment performance. Suppose that the small-cap manager underperformed the small-cap index but outperformed a broad equity market index. If investors use the broad market index as a benchmark, they might conclude that the small-cap manager is earning his or her fees and should be retained or given additional assets to invest. Using the small-cap index as a benchmark might lead to a very different conclusion.

5.5. Model Portfolios for Investment Products

Indexes also serve as the basis for the development of new investment products. Using indexes as benchmarks for actively managed portfolios has led some investors to conclude that they should invest in the benchmarks instead. Based on the CAPM’s conclusion that investors should hold the market portfolio, broad market index funds have been developed to function as proxies for the market portfolio.

Investment management firms initially developed and managed index portfolios for institutional investors. Eventually, mutual fund companies introduced index funds for individual investors. Subsequently, investment management firms introduced exchange-traded funds, which are managed the same way as index mutual funds but trade like stocks.

The first ETFs were based on existing indexes. As the popularity of ETFs increased, index providers created new indexes for the specific purpose of forming ETFs, leading to the creation of numerous narrowly defined indexes with corresponding ETFs. The VanEck Vectors Vietnam ETF, for example, allows investors to invest in the equity market of Vietnam.

The choice of indexes to meet the needs of investors is extensive. Index providers are constantly looking for opportunities to develop indexes to meet the needs of investors.

6. EQUITY INDEXES

LEARNING OUTCOMES

The candidate should be able to:

- ☐ describe types of equity indexes
- ☐ compare types of security market indexes

A wide variety of equity indexes exist, including broad market, multi-market, sector, and style indexes.

6.1. Broad Market Indexes

A broad equity market index, as its name suggests, represents an entire given equity market and typically includes securities representing more than 90 percent of the selected market. For example, the Shanghai Stock Exchange Composite Index (SSE) is a market-capitalization-weighted index of all shares that trade on the Shanghai Stock Exchange. In the United States, the Wilshire 5000 Total Market Index is a market-capitalization-weighted index that includes all US equities with readily available prices and is designed to represent the entire US equity market.³ The Russell 3000, consisting of the largest 3,000 stocks by market capitalization, represents approximately 98 percent of the US equity market.

6.2. Multi-Market Indexes

Multi-market indexes usually comprise indexes from different countries and regions and are designed to represent multiple security markets. Multi-market indexes may represent multiple national markets, geographic regions, economic development groups, and, in some cases, the entire world. World indexes are of importance to investors who take a global approach to equity investing without any particular bias toward a particular country or region. A number of index providers publish families of multi-market equity indexes.

MSCI offers a number of multi-market indexes. As shown in [Exhibit 7](#), MSCI classifies countries and regions along two dimensions: level of economic development and geographic region. Developmental groups, which MSCI refers to as market classifications, include developed markets, emerging markets, and frontier markets. The geographic regions are largely divided by longitudinal lines of the globe: the Americas, Europe with Africa, and Asia with the Pacific. MSCI provides country- and region-specific indexes for each of the developed and emerging markets within its multi-market indexes. MSCI periodically reviews the classifications of markets in its indexes for movement from frontier markets to emerging markets and from emerging markets to developed markets and reconstitutes the indexes accordingly.

Exhibit 7: MSCI Global Investable Market Indexes (as of October 2018)

Developed Markets					
Americas	Europe and Middle East			Pacific	
Canada, United States	Austria, Belgium, Denmark, Finland, France, Germany, Ireland, Israel, Italy, Netherlands, Norway, Portugal, Spain, Sweden, Switzerland, United Kingdom			Australia, Hong Kong SAR, Japan, New Zealand, Singapore	
Emerging Markets					
Americas	Europe, Middle East, Africa		Asia		
Brazil, Chile, Colombia, Mexico, Peru	Czech Republic, Egypt, Greece, Hungary, Poland, Qatar, Russia, South Africa, Turkey, United Arab Emirates		Chinese mainland, India, Indonesia, South Korea, Malaysia, Pakistan, Philippines, Taiwan region, Thailand		
Frontier Markets					
Americas	Europe & CIS	Africa	Middle East	Asia	
Argentina	Croatia, Estonia, Lithuania, Kazakhstan, Romania, Serbia, Slovenia	Kenya, Mauritius, Morocco, Nigeria, Tunisia, WAEMU ¹	Bahrain, Jordan, Kuwait, Lebanon, Oman	Bangladesh, Sri Lanka, Vietnam	
MSCI Standalone Market Indexes ²					
Europe, Middle East, and Africa		Americas	Europe and CIS	Africa	Middle East
Saudi Arabia		Jamaica, Panama, ³ Trinidad & Tobago	Bosnia Herzegovina, Bulgaria, Ukraine	Botswana, Ghana, Zimbabwe	Palestine

¹ The West African Economic and Monetary Union (WAEMU) consists of the following countries: Benin, Burkina Faso, Ivory Coast, Guinea-Bissau, Mali, Niger, Senegal, and Togo. Currently the MSCI WAEMU Indexes include securities classified in Senegal, Ivory Coast, and Burkina Faso.

² The MSCI Standalone Market Indexes are not included in the MSCI Emerging Markets Index or MSCI Frontier Markets Index. However, these indexes use either the Emerging Markets or the Frontier Markets methodological criteria concerning size and liquidity.

³ MSCI Panama Index has been launched as a Standalone Market Index.

Source: adapted from MSCI (<https://www.msci.com/en/market-cap-weighted-indexes>), October 2018.

6.2.1. Fundamental Weighting in Multi-Market Indexes

Some index providers weight the securities within each country/region by market capitalization and then weight each country/region in the overall index in proportion to its relative GDP, effectively creating fundamental weighting in multi-market indexes. GDP-weighted indexes were some of the first fundamentally weighted indexes created. Introduced in 1987 by MSCI to address the 60 percent weight of Japanese equities in the market-capitalization-weighted MSCI EAFE Index at the time, GDP-weighted indexes reduced the allocation to Japanese equities by half.⁴

6.3. Sector Indexes

Sector indexes represent and track different economic sectors—such as consumer goods, energy, finance, health care, and technology—on either a national, regional, or global basis. Because different sectors of the economy behave differently over the course of the business cycle, some investors may seek to overweight or underweight their exposure to particular sectors.

Sector indexes are organized as families; each index within the family represents an economic sector. Typically, the aggregation of a sector index family is equivalent to a broad market index. Economic sector classification can be applied on a global, regional, or country-specific basis, but no universally agreed upon sector classification method exists.

Sector indexes play an important role in performance analysis because they provide a means to determine whether a portfolio manager is more successful at stock selection or sector allocation. Sector indexes also serve as model portfolios for sector-specific ETFs and other investment products.

6.4. Style Indexes

Style indexes represent groups of securities classified according to market capitalization, value, growth, or a combination of these characteristics. They are intended to reflect the investing styles of certain investors, such as the growth investor, value investor, and small-cap investor.

6.4.1. Market Capitalization

Market-capitalization indexes represent securities categorized according to the major capitalization categories: large cap, midcap, and small cap. With no universal definition of these categories, the indexes differ on the distinctions between large cap and midcap and between midcap and small cap, as well as the minimum market-capitalization size required to be included in a small-cap index. Classification into categories can be based on absolute market capitalization (e.g., below €100 million) or relative market capitalization (e.g., the smallest 2,500 stocks).

6.4.2. Value/Growth Classification

Some indexes represent categories of stocks based on their classifications as either value or growth stocks. Different index providers use different factors and valuation ratios (low price-to-book ratios, low price-to-earnings ratios, high dividend yields, etc.) to distinguish between value and growth equities.

6.4.3. Market Capitalization and Value/Growth Classification

Combining the three market-capitalization groups with value and growth classifications results in six basic style index categories:

- Large-Cap Value ▪ Large-Cap Growth
- Mid-Cap Value ▪ Mid-Cap Growth
- Small-Cap Value ▪ Small-Cap Growth

Because indexes use different size and valuation classifications, the constituents of indexes designed to represent a given style, such as small-cap value, may differ—sometimes substantially.

Because valuation ratios and market capitalizations change over time, stocks frequently migrate from one style index category to another on reconstitution dates. As a result, style indexes generally have much higher turnover than do broad market indexes.

7. FIXED-INCOME INDEXES

LEARNING OUTCOMES

The candidate should be able to:

- ☐ describe types of fixed-income indexes
- ☐ compare types of security market indexes

A wide variety of fixed-income indexes exists, but the nature of the fixed-income markets and fixed-income securities leads to some very important challenges to fixed-income index construction and replication. These challenges are the number of securities in the fixed-income universe, the availability of pricing data, and the liquidity of the securities.

7.1. Construction

The fixed-income universe includes securities issued by governments, government agencies, and corporations. Each of these entities may issue a variety of fixed-income securities with different characteristics. As a result, the number of fixed-income securities is many times larger than the number of equity securities. To represent a specific fixed-income market or segment, indexes may include thousands of different securities. Over time, these fixed-income securities mature, and issuers offer new securities to meet their financing needs, leading to turnover in fixed-income indexes.

Another challenge in index construction is that fixed-income markets are predominantly dealer markets. This means that firms (dealers) are assigned to specific securities and are responsible for creating liquid markets for those securities by purchasing and selling them from their inventory. In addition, many securities do not trade frequently and, as a result, are relatively illiquid. As a result, index providers must contact dealers to obtain current prices on constituent securities to update the index or they must estimate the prices of constituent securities using the prices of traded fixed-income securities with similar characteristics.

These challenges can result in indexes with dissimilar numbers of bonds representing the same markets. The large number of fixed-income securities—combined with the lack of liquidity of some securities—has made it more costly and difficult, compared with equity indexes, for investors to replicate fixed-income indexes and duplicate their performance.

7.2. Types of Fixed-Income Indexes

The wide variety of fixed-income securities, ranging from zero-coupon bonds to bonds with embedded options (i.e., callable or putable bonds), results in a number of different types of fixed-income indexes. Similar to equities, fixed-income securities can be categorized according to the issuer's economic sector, the issuer's geographic region, or the economic development of the issuer's geographic region. Fixed-income securities can also be classified along the following dimensions:

- type of issuer (government, government agency, corporation);
- type of financing (general obligation, collateralized);
- currency of payments;
- maturity;
- credit quality (investment grade, high yield, credit agency ratings); and
- absence or presence of inflation protection.

Fixed-income indexes are based on these various dimensions and can be categorized as follows:

- aggregate or broad market indexes;
- market sector indexes;
- style indexes;
- economic sector indexes; and
- specialized indexes such as high-yield, inflation-linked, and emerging market indexes.

The first fixed-income index created, the Bloomberg Barclays US Aggregate Bond Index (formerly the Barclays Capital Aggregate Bond Index), is an example of a single-country aggregate index. Designed to represent the broad market of US fixed-income securities, it comprises approximately 8,000 securities, including US Treasury, government-related, corporate, mortgage-backed, asset-backed, and commercial mortgage-backed securities.

Aggregate indexes can be subdivided by market sector (government, government agency, collateralized, corporate); style (maturity, credit quality); economic sector, or some other characteristic to create more narrowly defined indexes. A common distinction reflected in indexes is between investment grade (e.g., those with a Standard & Poor’s credit rating of BBB– or better) and high-yield securities. Investment-grade indexes are typically further subdivided by maturity (i.e., short, intermediate, or long) and by credit rating (e.g., AAA, BBB, etc.).⁵ The wide variety of fixed-income indexes reflects the partitioning of fixed-income securities on the basis of a variety of dimensions.

Exhibit 8 illustrates how the major types of fixed-income indexes can be organized on the basis of various dimensions.

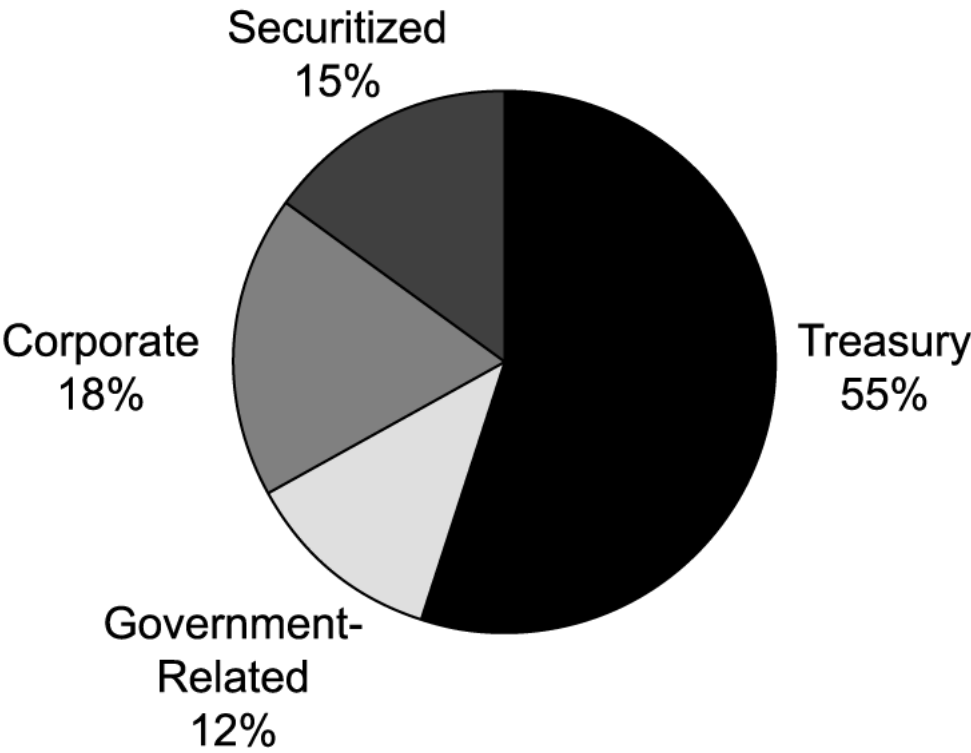
Exhibit 8: Dimensions of Fixed-Income Indexes

Market	Global			
	Regional			
	Country or currency zone			
Type	Corporate	Collateralized <i>Securitized</i> <i>Mortgage-backed</i>	Government agency	Government
Maturity	For example, 1–3, 3–5, 5–7, 7–10, 10+ years; short-term, medium-term, or long-term			
Credit quality	For example, AAA, AA, A, BBB, etc.; Aaa, Aa, A, Baa, etc.; investment grade, high yield			

All aggregate indexes include a variety of market sectors and credit ratings. The breakdown of the Bloomberg Barclays Global Aggregate Bond Index by market sectors and by credit rating is shown in Exhibit 9 and Exhibit 10, respectively.

Exhibit 9: Market Sector Breakdown of the Bloomberg Barclays Global Aggregate Bond Index

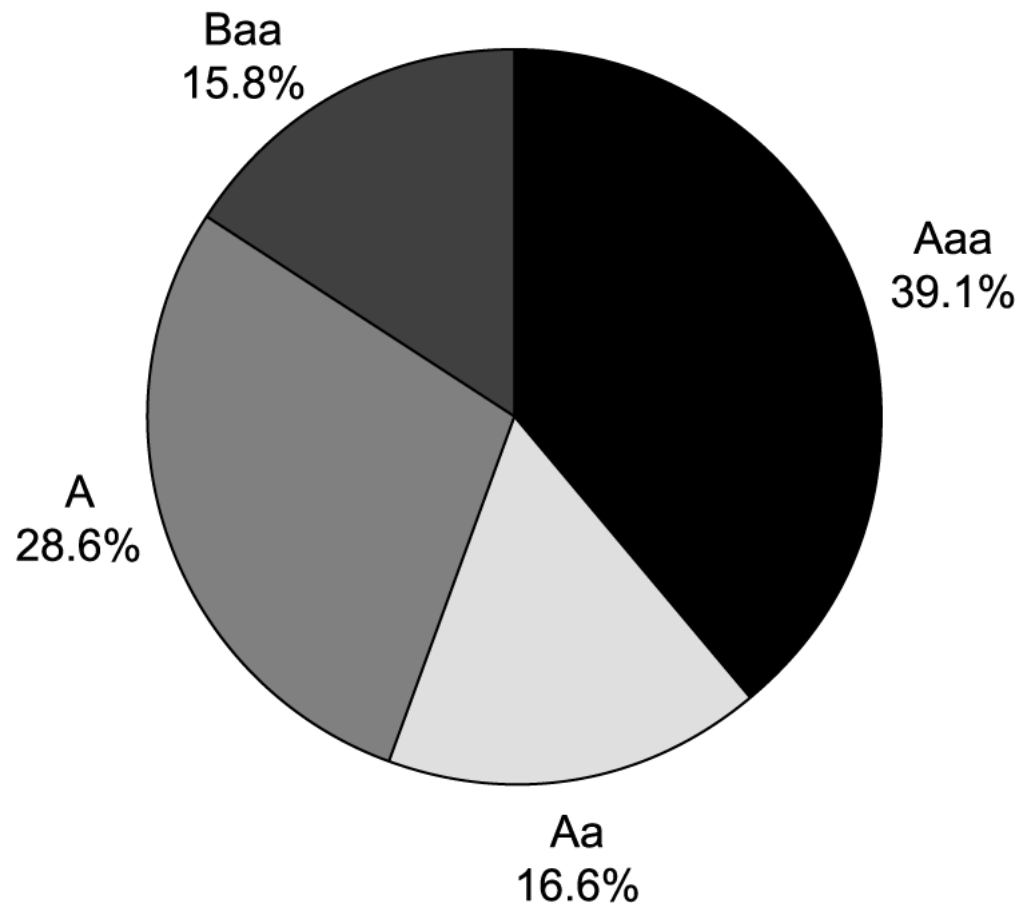
Sector Breakdown as of June 30, 2016



Source: Bloomberg Barclays Indices, Global Aggregate Index Factsheet, August 24, 2016. <https://data.bloomberglp.com/indices/sites/2/2016/08/Factsheet-Global-Aggregate.pdf>.

Exhibit 10: Credit Breakdown of the Bloomberg Barclays Global Aggregate Bond Index

Quality Breakdown as of June 30, 2016



Source: Bloomberg Barclays Indices, Global Aggregate Index Factsheet, August 24, 2016. <https://data.bloomberglp.com/indices/sites/2/2016/08/Factsheet-Global-Aggregate.pdf>.

8. INDEXES FOR ALTERNATIVE INVESTMENTS

LEARNING OUTCOMES

The candidate should be able to:

- ☐ describe indexes representing alternative investments
- ☐ compare types of security market indexes

Many investors seek to lower the risk or enhance the performance of their portfolios by investing in assets classes other than equities and fixed income. Interest in alternative assets and investment strategies has led to the creation of indexes designed to represent broad classes of alternative investments. Three of the most widely followed alternative investment classes are commodities, real estate, and hedge funds.

8.1. Commodity Indexes

Commodity indexes consist of futures contracts on one or more commodities, such as agricultural products (rice, wheat, sugar), livestock (cattle, hogs), precious and common metals (gold, silver, copper), and energy commodities (crude oil, natural gas).

Although some commodity indexes may include the same commodities, the returns of these indexes may differ because each index may use a different weighting method. Because commodity indexes do not have an obvious weighting mechanism, such as market capitalization, commodity index providers create their own weighting methods. Some indexes, such as the Thomson Reuters/Core Commodity CRB Index (TR/CC CRB Index), formerly known as the Commodity Research Bureau (CRB) Index, contain a fixed number of commodities that are weighted equally. The S&P GSCI uses a combination of liquidity measures and world production values in its weighting scheme and allocates more weight to commodities that have risen in price. Other indexes have fixed weights that are determined by a committee.

The different weighting methods can also lead to large differences in exposure to specific commodities. The S&P GSCI in 2018, for example, weights the energy-sector approximately 50% higher and the agriculture sector 40% lower than the CRB Index. These differences result in indexes with very different risk and return profiles. Unlike commodity indexes, broad equity and fixed-income indexes that target the same markets share similar risk and return profiles.

The performance of commodity indexes can also be quite different from their underlying commodities because the indexes consist of futures contracts on the commodities rather than the actual commodities. Index returns are affected by factors other than changes in the prices of the underlying commodities because futures contracts must be continually “rolled over” (i.e., replacing a contract nearing expiration with a new contract). Commodity index returns reflect the risk-free interest rate, the changes in future prices, and the roll yield. Therefore, a commodity index return can be quite different from the return based on changes in the prices of the underlying commodities.

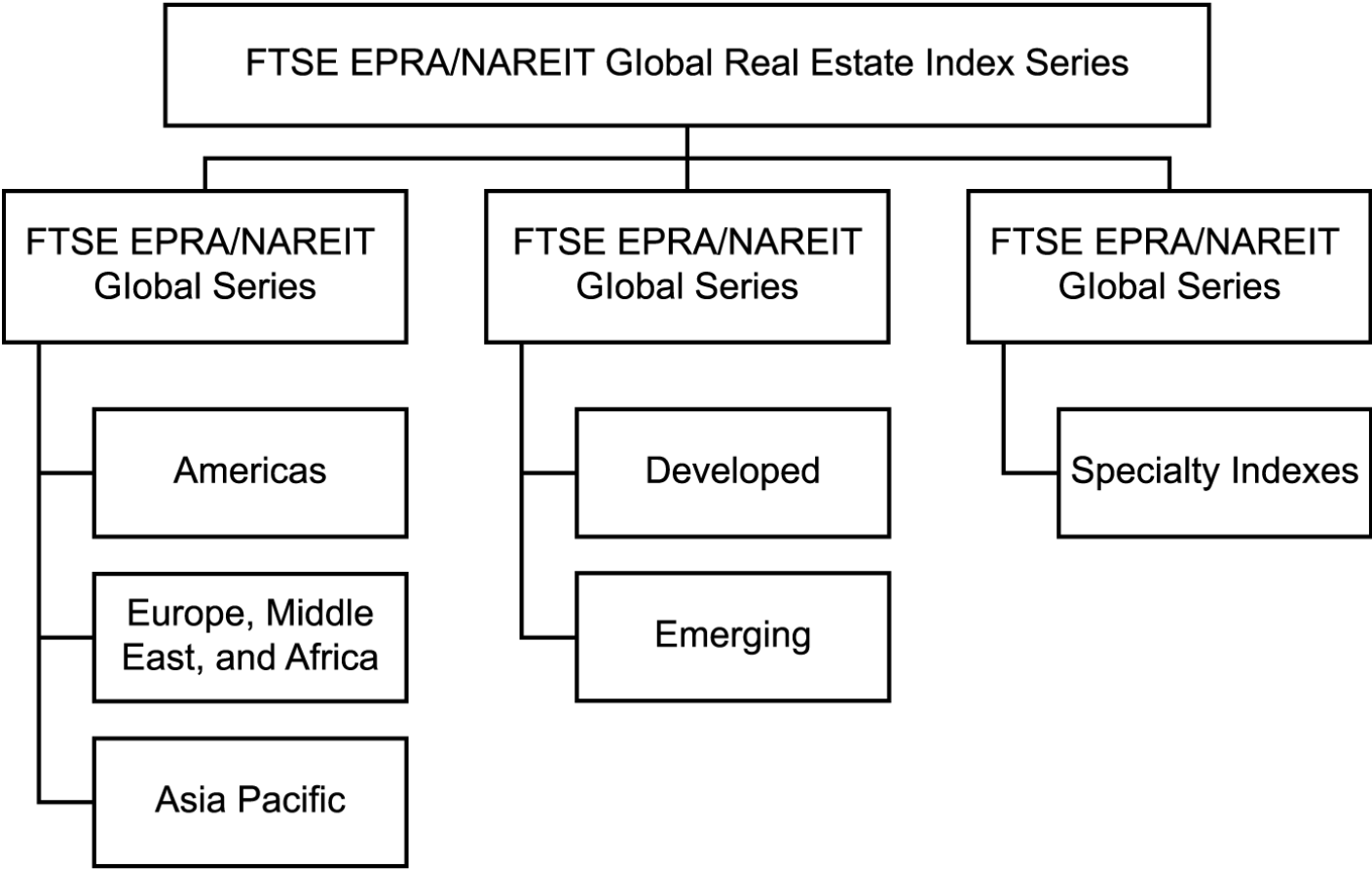
8.2. Real Estate Investment Trust Indexes

Real estate indexes represent not only the market for real estate securities but also the market for real estate—a highly illiquid market and asset class with infrequent transactions and pricing information. Real estate indexes can be categorized as appraisal indexes, repeat sales indexes, and real estate investment trust (REIT) indexes.

REIT indexes consist of shares of publicly traded REITs. REITs are public or private corporations organized specifically to invest in real estate, either through ownership of properties or investment in mortgages. Shares of public REITs are traded on the world's various stock exchanges and are a popular choice for investing in commercial real estate properties. Because REIT indexes are based on publicly traded REITs with continuous market pricing, the value of REIT indexes is calculated continuously.

The FTSE EPRA/NAREIT global family of REIT indexes shown in [Exhibit 11](#) seeks to represent trends in real estate stocks worldwide and includes representation from the European Public Real Estate Association (EPRA) and the National Association of Real Estate Investment Trusts (NAREIT).

Exhibit 11: The FTSE EPRA/NAREIT Global REIT Index Family



Source: FTSE International, "FTSE EPRA/NAREIT Global & Global Ex US Indices" Factsheet 2009). "FTSE®" is a trade mark of the London Stock Exchange Plc, "NAREIT®" is a trade mark of the National Association of Real Estate Investment Trusts ("NAREIT") and "EPRA®" is a trade mark of the European Public Real Estate Association ("EPRA") and all are used by FTSE International Limited ("FTSE") under license.

8.3. Hedge Fund Indexes

Hedge fund indexes reflect the returns on hedge funds. **Hedge funds** are private investment vehicles that typically use leverage and long and short investment strategies.

A number of research organizations maintain databases of hedge fund returns and summarize these returns into indexes. These database indexes are designed to represent the performance of the hedge funds on a very broad global level (hedge funds in general) or the strategy level. Most of these indexes are equal weighted and represent the performance of the hedge funds within a particular database.

Most research organizations rely on the voluntary cooperation of hedge funds to compile performance data. As unregulated entities, however, hedge funds are not required to report their performance to any party other than their investors. Therefore, each hedge fund decides to which database(s) it will report its performance. As a result, rather than index providers determining the constituents, the constituents determine the index.

Frequently, a hedge fund reports its performance to only one database. The result is little overlap of funds covered by the different indexes. With little overlap between their constituents, different global hedge fund indexes may reflect very different performance for the hedge fund industry over the same period of time.

Another consequence of the voluntary performance reporting is the potential for survivorship bias and, therefore, inaccurate performance representation. This means that hedge funds with poor performance may be less likely to report their performance to the database or may stop reporting to the database, so their returns may be excluded when measuring the return of the index. As a result, the index may not accurately reflect actual hedge fund performance so much as the performance of hedge funds that are performing well.

REPRESENTATIVE INDEXES WORLDWIDE

As indicated in this reading, the choice of indexes to meet the needs of investors is extensive. Investors using security market indexes must be careful in their selection of the index or indexes most appropriate for their needs. The following table illustrates the variety of indexes reflecting different asset classes, markets, and weighting methods.

Index	Representing	Number of Securities	Weighting Method	Comments
Dow Jones Industrial Average	US blue chip companies	30	Price	The oldest and most widely known US equity index. <i>Wall Street Journal</i> editors choose 30 stocks from among large, mature blue-chip companies.
Nikkei Stock Average	Japanese blue chip companies	225	Modified price	Known as the Nikkei 225 and originally formulated by Dow Jones & Company. Because of extreme variation in price levels of component securities, some high-priced shares are weighted as a fraction of share price.
TOPIX	All companies listed on the Tokyo Stock Exchange First Section	Varies	Float-adjusted market cap	Represents about 93 percent of the market value of all Japanese equities. Contains a large number of very small, illiquid stocks, making exact replication difficult.

Index	Representing	Number of Securities	Weighting Method	Comments
MSCI All Country World Index	Stocks of 23 developed and 24 emerging markets	Varies	Free-float-adjusted market cap	Composed of companies representative of the market structure of developed and emerging market countries in the Americas, Europe/Middle East, and Asia/Pacific regions. Price return and total return versions available in both USD and local currencies.
S&P Developed Ex-US BMI Energy Sector Index	Energy sector of developed global markets outside the United States	Varies	Float-adjusted market cap	Serves as a model portfolio for the SPDR® S&P Energy Sector Exchange-Traded Fund (ETF).
Bloomberg Barclays Global Aggregate Bond Index	Investment-grade bonds in the North American, European, and Asian markets	Varies	Market cap	Formerly known as Lehman Brothers Global Aggregate Bond Index.
Markit iBoxx Euro High-Yield Bond Indexes	Sub-investment-grade euro-denominated corporate bonds	Varies	Market cap and variations	Rebalanced monthly. Represents tradable part of market. Price and total return versions available with such analytical values as yield, duration, modified duration, and convexity. Provides platform for research and structured products.
FTSE EPRA/NAREIT Global Real Estate Index	Real estate securities in the North American, European, and Asian markets	Varies	Float-adjusted market cap	The stocks of REITs that constitute the index trade on public stock exchanges and may be constituents of equity market indexes.
HFRX Global Hedge Fund Index	Overall composition of the HFR database	Varies	Asset weighted	Comprises all eligible hedge fund strategies. Examples include convertible arbitrage, distressed securities, market neutral, event driven, macro, and relative value arbitrage. Constituent strategies are asset weighted on the basis of asset distribution within the hedge fund industry.
HFRX Equal Weighted Strategies EUR Index	Overall composition of the HFR database	Varies	Equal weighted	Denominated in euros and is constructed from the same strategies as the HFRX Global Hedge Fund Index.
Morningstar Style Indexes	US stocks classified by market cap and value/growth orientation	Varies	Float-adjusted market cap	The nine indexes defined by combinations of market cap (large, mid, and small) and value/growth orientation (value, core, growth) have mutually exclusive constituents and are exhaustive with respect to the Morningstar US Market Index. Each is a model portfolio for one of the iShares Morningstar ETFs.

SUMMARY

This reading explains and illustrates the construction, management, and uses of security market indexes. It also discusses various types of indexes. Security market indexes are invaluable tools for investors, who can select from among thousands of indexes representing a variety of security markets, market segments, and asset classes. These indexes range from those representing the global market for major asset classes to those representing alternative investments in specific geographic markets. To benefit from the use of security market indexes, investors must understand their construction and determine whether the selected index is appropriate for their purposes. Frequently, an index that is well suited for one purpose may not be well suited for other purposes. Users of indexes must be familiar with how various indexes are constructed in order to select the index or indexes most appropriate for their needs.

Among the key points made in this reading are the following:

- Security market indexes are intended to measure the values of different target markets (security markets, market segments, or asset classes).
- The constituent securities selected for inclusion in the security market index are intended to represent the target market.
- A price return index reflects only the prices of the constituent securities.
- A total return index reflects not only the prices of the constituent securities but also the reinvestment of all income received since the inception of the index.
- Methods used to weight the constituents of an index range from the very simple, such as price and equal weightings, to the more complex, such as market-capitalization and fundamental weightings.
- Choices in index construction—in particular, the choice of weighting method—affect index valuation and returns.
- Index management includes 1) periodic rebalancing to ensure that the index maintains appropriate weightings and 2) reconstitution to ensure the index represents the desired target market.
- Rebalancing and reconstitution create turnover in an index. Reconstitution can dramatically affect prices of current and prospective constituents.
- Indexes serve a variety of purposes. They gauge market sentiment and serve as benchmarks for actively managed portfolios. They act as proxies for measuring systematic risk and risk-adjusted performance. They also serve as proxies for asset classes in asset allocation models and as model portfolios for investment products.
- Investors can choose from security market indexes representing various asset classes, including equity, fixed-income, commodity, real estate, and hedge fund indexes.
- Within most asset classes, index providers offer a wide variety of indexes, ranging from broad market indexes to highly specialized indexes based on the issuer’s geographic region, economic development group, or economic sector or other factors.
- Proper use of security market indexes depends on understanding their construction and management.

PRACTICE PROBLEMS

1. A security market index represents the:
 - A. risk of a security market.
 - B. security market as a whole.
 - C. security market, market segment, or asset class.
2. One month after inception, the price return version and total return version of a single index (consisting of identical securities and weights) will be equal if:
 - A. market prices have not changed.
 - B. capital gains are offset by capital losses.
 - C. the securities do not pay dividends or interest.
3. The values of a price return index and a total return index consisting of identical equal-weighted dividend-paying equities will be equal:
 - A. only at inception.
 - B. at inception and on rebalancing dates.
 - C. at inception and on reconstitution dates.
4. Security market indexes are:
 - A. constructed and managed like a portfolio of securities.
 - B. simple interchangeable tools for measuring the returns of different asset classes.
 - C. valued on a regular basis using the actual market prices of the constituent securities.
5. When creating a security market index, an index provider must first determine the:
 - A. target market.
 - B. appropriate weighting method.
 - C. number of constituent securities.
6. An analyst gathers the following information for an equal-weighted index comprised of assets Able, Baker, and Charlie:

Security	Beginning of Period Price (€)	End of Period Price (€)	Total Dividends (€)
Able	10.00	12.00	0.75
Baker	20.00	19.00	1.00
Charlie	30.00	30.00	2.00

The price return of the index is:

- A. 1.7%.
- B. 5.0%.

C. 11.4%.

7. An analyst gathers the following information for an equal-weighted index comprised of assets Able, Baker, and Charlie:

Security	Beginning of Period Price (€)	End of Period Price (€)	Total Dividends (€)
Able	10.00	12.00	0.75
Baker	20.00	19.00	1.00
Charlie	30.00	30.00	2.00

The total return of the index is:

- A. 5.0%.
B. 7.9%.
C. 11.4%.

8. An analyst gathers the following information for a price-weighted index comprised of securities ABC, DEF, and GHI:

Security	Beginning of Period Price (£)	End of Period Price (£)	Total Dividends (£)
ABC	25.00	27.00	1.00
DEF	35.00	25.00	1.50
GHI	15.00	16.00	1.00

The price return of the index is:

- A. -4.6%.
B. -9.3%.
C. -13.9%.

9. An analyst gathers the following information for a market-capitalization-weighted index comprised of securities MNO, QRS, and XYZ:

Security	Beginning of Period Price (¥)	End of Period Price (¥)	Dividends per Share (¥)	Shares Outstanding
MNO	2,500	2,700	100	5,000
QRS	3,500	2,500	150	7,500
XYZ	1,500	1,600	100	10,000

The price return of the index is:

- A. -9.33%.
B. -10.23%.
C. -13.90%.

10. An analyst gathers the following information for a market-capitalization-weighted index comprised of securities MNO, QRS, and XYZ:

Security	Beginning of Period Price (¥)	End of Period Price (¥)	Dividends Per Share (¥)	Shares Outstanding
MNO	2,500	2,700	100	5,000

Security	Beginning of Period Price (¥)	End of Period Price (¥)	Dividends Per Share (¥)	Shares Outstanding
QRS	3,500	2,500	150	7,500
XYZ	1,500	1,600	100	10,000

The total return of the index is:

- A. 1.04%.
- B. -5.35%.
- C. -10.23%.

11. When creating a security market index, the target market:

- A. determines the investment universe.
- B. is usually a broadly defined asset class.
- C. determines the number of securities to be included in the index.

12. An analyst gathers the following data for a price-weighted index:

Security	Beginning of Period		End of Period	
	Price (€)	Shares Outstanding	Price (€)	Shares Outstanding
A	20.00	300	22.00	300
B	50.00	300	48.00	300
C	26.00	2,000	30.00	2,000

The price return of the index over the period is:

- A. 4.2%.
- B. 7.1%.
- C. 21.4%.

13. An analyst gathers the following data for a value-weighted index:

Security	Beginning of Period		End of Period	
	Price (£)	Shares Outstanding	Price (£)	Shares Outstanding
A	20.00	300	22.00	300
B	50.00	300	48.00	300
C	26.00	2,000	30.00	2,000

The return on the value-weighted index over the period is:

- A. 7.1%.
- B. 11.0%.
- C. 21.4%.

14. An analyst gathers the following data for an equally-weighted index:

Security	Beginning of Period		End of Period	
	Price (¥)	Shares Outstanding	Price (¥)	Shares Outstanding
A	20.00	300	22.00	300
B	50.00	300	48.00	300

Security	Beginning of Period		End of Period	
	Price (¥)	Shares Outstanding	Price (¥)	Shares Outstanding
C	26.00	2,000	30.00	2,000

The return on the index over the period is:

- A. 4.2%.
 - B. 6.8%.
 - C. 7.1%.
15. Which of the following index weighting methods requires an adjustment to the divisor after a stock split?
- A. Price weighting.
 - B. Fundamental weighting.
 - C. Market-capitalization weighting.
16. If the price return of an equal-weighted index exceeds that of a market-capitalization-weighted index comprised of the same securities, the *most likely* explanation is:
- A. stock splits.
 - B. dividend distributions.
 - C. outperformance of small-market-capitalization stocks.
17. A float-adjusted market-capitalization-weighted index weights each of its constituent securities by its price and:
- A. its trading volume.
 - B. the number of its shares outstanding.
 - C. the number of its shares available to the investing public.
18. Which of the following index weighting methods is most likely subject to a value tilt?
- A. Equal weighting.
 - B. Fundamental weighting.
 - C. Market-capitalization weighting.
19. Rebalancing an index is the process of periodically adjusting the constituent:
- A. securities' weights to optimize investment performance.
 - B. securities to maintain consistency with the target market.
 - C. securities' weights to maintain consistency with the index's weighting method.
20. Which of the following index weighting methods requires the most frequent rebalancing?
- A. Price weighting.
 - B. Equal weighting.
 - C. Market-capitalization weighting.
21. Reconstitution of a security market index reduces:

- A. portfolio turnover.
- B. the need for rebalancing.
- C. the likelihood that the index includes securities that are not representative of the target market.

22. Security market indexes are used as:

- A. measures of investment returns.
- B. proxies to measure unsystematic risk.
- C. proxies for specific asset classes in asset allocation models.

23. Uses of market indexes do not include serving as a:

- A. measure of systemic risk.
- B. basis for new investment products.
- C. benchmark for evaluating portfolio performance.

24. Which of the following statements regarding sector indexes is *most* accurate? Sector indexes:

- A. track different economic sectors and cannot be aggregated to represent the equivalent of a broad market index.
- B. provide a means to determine whether an active investment manager is more successful at stock selection or sector allocation.
- C. apply a universally agreed upon sector classification system to identify the constituent securities of specific economic sectors, such as consumer goods, energy, finance, health care.

25. Which of the following is an example of a style index? An index based on:

- A. geography.
- B. economic sector.
- C. market capitalization.

26. Which of the following statements regarding fixed-income indexes is *most* accurate?

- A. Liquidity issues make it difficult for investors to easily replicate fixed-income indexes.
- B. Rebalancing and reconstitution are the only sources of turnover in fixed-income indexes.
- C. Fixed-income indexes representing the same target market hold similar numbers of bonds.

27. An aggregate fixed-income index:

- A. comprises corporate and asset-backed securities.
- B. represents the market of government-issued securities.
- C. can be subdivided by market or economic sector to create more narrowly defined indexes.

28. Fixed-income indexes are *least likely* constructed on the basis of:

- A. maturity.
- B. type of issuer.
- C. coupon frequency.

29. In comparison to equity indexes, the constituent securities of fixed-income indexes are:

- A.** more liquid.
- B.** easier to price.
- C.** drawn from a larger investment universe.

30. Commodity index values are based on:

- A.** futures contract prices.
- B.** the market price of the specific commodity.
- C.** the average market price of a basket of similar commodities.

31. Which of the following statements is *most* accurate?

- A.** Commodity indexes all share similar weighting methods.
- B.** Commodity indexes containing the same underlying commodities offer similar returns.
- C.** The performance of commodity indexes can be quite different from that of the underlying commodities.

32. Which of the following is *not* a real estate index category?

- A.** Appraisal index.
- B.** Initial sales index.
- C.** Repeat sales index.

33. A unique feature of hedge fund indexes is that they:

- A.** are frequently equal weighted.
- B.** are determined by the constituents of the index.
- C.** reflect the value of private rather than public investments.

34. The returns of hedge fund indexes are *most likely*:

- A.** biased upward.
- B.** biased downward.
- C.** similar across different index providers.

SOLUTIONS

1. C is correct. A security market index represents the value of a given security market, market segment, or asset class.
2. C is correct. The difference between a price return index and a total return index consisting of identical securities and weights is the income generated over time by the underlying securities. If the securities in the index do not generate income, both indexes will be identical in value.
3. A is correct. At inception, the values of the price return and total return versions of an index are equal.
4. A is correct. Security market indexes are constructed and managed like a portfolio of securities.
5. A is correct. The first decision is identifying the target market that the index is intended to represent because the target market determines the investment universe and the securities available for inclusion in the index.
6. B is correct. The price return is the sum of the weighted returns of each security. The return of Able is 20 percent $[(12 - 10)/10]$; of Baker is -5 percent $[(19 - 20)/20]$; and of Charlie is 0 percent $[(30 - 30)/30]$. The price return index assigns a weight of 1/3 to each asset; therefore, the price return is $1/3 \times [20\% + (-5\%) + 0\%] = 5\%$.
7. C is correct. The total return of an index is calculated on the basis of the change in price of the underlying securities plus the sum of income received or the sum of the weighted total returns of each security. The total return of Able is 27.5 percent; of Baker is 0 percent; and of Charlie is 6.7 percent:

Able: $(12 - 10 + 0.75)/10 = 27.5\%$

Baker: $(19 - 20 + 1)/20 = 0\%$

Charlie: $(30 - 30 + 2)/30 = 6.7\%$

An equal-weighted index applies the same weight (1/3) to each security's return; therefore, the total return = $1/3 \times (27.5\% + 0\% + 6.7\%) = 11.4\%$.

8. B is correct. The price return of the price-weighted index is the percentage change in price of the index: $(68 - 75)/75 = -9.33\%$.

Security	Beginning of Period Price (£)	End of Period Price (£)
ABC	25.00	27.00
DEF	35.00	25.00
GHI	15.00	16.00
TOTAL	75.00	68.00

9. B is correct. The price return of the index is $(48,250,000 - 53,750,000)/53,750,000 = -10.23\%$.

Security	Beginning of Period Price (¥)	Shares Outstanding	Beginning of Period Value (¥)	End of Period Price (¥)	End of Period Value (¥)
MNO	2,500	5,000	12,500,000	2,700	13,500,000
QRS	3,500	7,500	26,250,000	2,500	18,750,000
XYZ	1,500	10,000	15,000,000	1,600	16,000,000

Security	Beginning of Period Price (¥)	Shares Outstanding	Beginning of Period Value (¥)	End of Period Price (¥)	End of Period Value (¥)
Total			53,750,000		48,250,000

10. B is correct. The total return of the market-capitalization-weighted index is calculated below:

Security	Beginning of Period Value (¥)	End of Period Value (¥)	Total Dividends (¥)	Total Return (%)
MNO	12,500,000	13,500,000	500,000	12.00
QRS	26,250,000	18,750,000	1,125,000	-24.29
XYZ	15,000,000	16,000,000	1,000,000	13.33
Total	53,750,000	48,250,000	2,625,000	-5.35

11. A is correct. The target market determines the investment universe and the securities available for inclusion in the index.

12. A is correct. The sum of prices at the beginning of the period is 96; the sum at the end of the period is 100. Regardless of the divisor, the price return is $100/96 - 1 = 0.042$ or 4.2 percent.

13. B is correct. It is the percentage change in the market value over the period:

Market value at beginning of period: $(20 \times 300) + (50 \times 300) + (26 \times 2,000) = 73,000$

Market value at end of period: $(22 \times 300) + (48 \times 300) + (30 \times 2,000) = 81,000$

Percentage change is $81,000/73,000 - 1 = 0.1096$ or 11.0 percent with rounding.

14. C is correct. With an equal-weighted index, the same amount is invested in each security. Assuming \$1,000 is invested in each of the three stocks, the index value is \$3,000 at the beginning of the period and the following number of shares is purchased for each stock:

Security A: 50 shares

Security B: 20 shares

Security C: 38.46 shares.

Using the prices at the beginning of the period for each security, the index value at the end of the period is \$3,213.8: $(\$22 \times 50) + (\$48 \times 20) + (\$30 \times 38.46)$. The price return is $\$3,213.8/\$3,000 - 1 = 7.1\%$.

15. A is correct. In the price weighting method, the divisor must be adjusted so the index value immediately after the split is the same as the index value immediately prior to the split.

16. C is correct. The main source of return differences arises from outperformance of small-cap securities or underperformance of large-cap securities. In an equal-weighted index, securities that constitute the largest fraction of the market are underrepresented and securities that constitute only a small fraction of the market are overrepresented. Thus, higher equal-weighted index returns will occur if the smaller-cap equities outperform the larger-cap equities.

17. C is correct. "Float" is the number of shares available for public trading.

18. B is correct. Fundamental weighting leads to indexes that have a value tilt.

19. C is correct. Rebalancing refers to adjusting the weights of constituent securities in an index to maintain consistency with the index's weighting method.

20. B is correct. Changing market prices will cause weights that were initially equal to become unequal, thus requiring rebalancing.

21. C is correct. Reconstitution is the process by which index providers review the constituent securities, re-apply the initial criteria for inclusion in the index, and select which securities to retain, remove, or add. Constituent securities that no longer meet the criteria are replaced with securities that do. Thus, reconstitution reduces the likelihood that the index includes securities that are not representative of the target market.
22. C is correct. Security market indexes play a critical role as proxies for asset classes in asset allocation models.
23. A is correct. Security market indexes are used as proxies for measuring market or systematic risk, not as measures of systemic risk.
24. B is correct. Sector indexes provide a means to determine whether a portfolio manager is more successful at stock selection or sector allocation.
25. C is correct. Style indexes represent groups of securities classified according to market capitalization, value, growth, or a combination of these characteristics.
26. A is correct. The large number of fixed-income securities—combined with the lack of liquidity of some securities—makes it costly and difficult for investors to replicate fixed-income indexes.
27. C is correct. An aggregate fixed-income index can be subdivided by market sector (government, government agency, collateralized, corporate), style (maturity, credit quality), economic sector, or some other characteristic to create more narrowly defined indexes.
28. C is correct. Coupon frequency is not a dimension on which fixed-income indexes are based.
29. C is correct. The fixed-income market has more issuers and securities than the equity market.
30. A is correct. Commodity indexes consist of futures contracts on one or more commodities.
31. C is correct. The performance of commodity indexes can be quite different from that of the underlying commodities because the indexes consist of futures contracts on the commodities rather than the actual commodities.
32. B is correct. It is not a real estate index category.
33. B is correct. Hedge funds are not required to report their performance to any party other than their investors. Therefore, each hedge fund decides to which database(s) it will report its performance. Thus, for a hedge fund index, constituents determine the index rather than index providers determining the constituents.
34. A is correct. Voluntary performance reporting may lead to survivorship bias, and poorer performing hedge funds will be less likely to report their performance.

NOTES

¹A stock split is an increase in the number of shares outstanding and a proportionate decrease in the price per share such that the total market value of equity, as well as investors' proportionate ownership in the company, does not change.

²According to the press release, final membership in the index would be published after market close on Friday, 26 June.

³Despite its name, the Wilshire 5000 has no constraint on the number of securities that can be included. It included approximately 5,000 securities at inception.

⁴Steven A. Schoenfeld, *Active Index Investing* (Hoboken, NJ: John Wiley & Sons, 2004):220.

⁵Credit ratings are discussed in depth in the Level I CFA Program reading "Fundamentals of Credit Analysis."

Market Efficiency

by Sean Cleary PhD, CFA, Howard J. Atkinson CIMA, ICD.D, CFA and Pamela Peterson Drake PhD, CFA

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LEARNING OUTCOMES

The candidate should be able to:

- ☐ describe market efficiency and related concepts, including their importance to investment practitioners
- ☐ contrast market value and intrinsic value
- ☐ explain factors that affect a market's efficiency
- ☐ contrast weak-form, semi-strong-form, and strong-form market efficiency
- ☐ explain the implications of each form of market efficiency for fundamental analysis, technical analysis, and the choice between active and passive portfolio management
- ☐ describe market anomalies
- ☐ describe behavioral finance and its potential relevance to understanding market anomalies

1. INTRODUCTION

Market efficiency concerns the extent to which market prices incorporate available information. If market prices do not fully incorporate information, then opportunities may exist to make a profit from the gathering and processing of information. The subject of market efficiency is, therefore, of great interest to investment managers, as illustrated in [Example 1](#).

EXAMPLE 1

Market Efficiency and Active Manager Selection

The chief investment officer (CIO) of a major university endowment fund has listed eight steps in the active manager selection process that can be applied both to traditional investments (e.g., common equity and fixed-income securities) and to alternative investments (e.g., private equity, hedge funds, and real assets). The first step specified is the evaluation of market opportunity:

What is the opportunity and why is it there? To answer this question, we start by studying capital markets and the types of managers operating within those markets. We identify market inefficiencies and try to understand their causes, such as regulatory structures or behavioral biases. We can rule out many broad groups of managers and strategies by simply determining that the degree of market inefficiency necessary to support a strategy is implausible. Importantly, we consider the past history of active returns meaningless unless we understand why markets will allow those active returns to continue into the future.¹

The CIO's description underscores the importance of not assuming that past active returns that might be found in a historical dataset will repeat themselves in the future. **Active returns** refer to returns earned by strategies that do *not* assume that all information is fully reflected in market prices.

Governments and market regulators also care about the extent to which market prices incorporate information. Efficient markets imply informative prices—prices that accurately reflect available information about fundamental values. In market-based economies, market prices help determine which companies (and which projects) obtain capital. If these prices do not efficiently incorporate information about a company's prospects, then it is possible that funds will be misdirected. By contrast, prices that are informative help direct scarce resources and funds available for investment to their highest-valued uses.² Informative prices thus promote economic growth. The efficiency of a country's capital markets (in which businesses raise financing) is an important characteristic of a well-functioning financial system.

The remainder of this reading is organized as follows. Section 2 provides specifics on how the efficiency of an asset market is described and discusses the factors affecting (i.e., contributing to and impeding) market efficiency. Section 3 presents an influential three-way classification of the efficiency of security markets and discusses its implications for fundamental analysis, technical analysis, and portfolio management. Section 4 presents several market anomalies (apparent market inefficiencies that have received enough attention to be individually identified and named) and describes how these anomalies relate to investment strategies. Section 5 introduces behavioral finance and how that field of study relates to market efficiency. A summary concludes the reading.

2. THE CONCEPT OF MARKET EFFICIENCY

LEARNING OUTCOMES

The candidate should be able to:

- ☐ describe market efficiency and related concepts, including their importance to investment practitioners
- ☐ contrast market value and intrinsic value

2.1. The Description of Efficient Markets

An **informationally efficient market** (an **efficient market**) is a market in which asset prices reflect new information quickly and rationally. An efficient market is thus a market in which asset prices reflect all past and present information.³

In this section we expand on this definition by clarifying the time frame required for an asset's price to incorporate information as well as describing the elements of information releases assumed under market efficiency. We discuss the difference between market value and intrinsic value and illustrate how inefficiencies or discrepancies between these values can provide profitable opportunities for active investment. As financial markets are generally not considered being either completely efficient or inefficient, but rather falling within a range between the two extremes, we describe a number of factors that contribute to and impede the degree of efficiency of a financial market. Finally, we conclude our overview of market efficiency by illustrating how the costs incurred by traders in identifying and exploiting possible market inefficiencies affect how we interpret market efficiency.

Investment managers and analysts, as noted, are interested in market efficiency because the extent to which a market is efficient affects how many profitable trading opportunities (market inefficiencies) exist. Consistent, superior, risk-adjusted returns (net of all expenses) are not achievable in an efficient market.⁴ In an efficient market, a **passive investment** strategy (i.e., buying and holding a broad market portfolio) that does not seek superior risk-adjusted returns can be preferred to an **active investment** strategy because of lower costs (for example, transaction and information-seeking costs). By contrast, in a very inefficient market, opportunities may exist for an active investment strategy to achieve superior risk-adjusted returns (net of all expenses in executing the strategy) as compared with a passive investment strategy. In inefficient markets, an active investment strategy may outperform a passive investment strategy on a risk-adjusted basis. Understanding the characteristics of an efficient market and being able to evaluate the efficiency of a particular market are important topics for investment analysts and portfolio managers.

An efficient market is a market in which asset prices reflect information quickly. But what is the time frame of “quickly”? Trades are the mechanism by which information can be incorporated into asset transaction prices. The time needed to execute trades to exploit an inefficiency may provide a baseline for judging speed of adjustment.⁵ The time frame for an asset's price to incorporate information must be at least as long as the shortest time a trader needs to execute a transaction in the asset. In certain markets, such as foreign exchange and developed equity markets, market efficiency relative to certain types of information has been studied using time frames as short as one minute or less. If the time frame of price adjustment allows many traders to earn profits with little risk, then the market is relatively inefficient. These considerations lead to the observation that market efficiency can be viewed as falling on a continuum.

Finally, an important point is that in an efficient market, prices should be expected to react only to the elements of information releases that are not anticipated fully by investors—that is, to the “unexpected” or “surprise” element of such releases. Investors process the unexpected information and revise expectations (for example, about an asset's future cash flows, risk, or required rate of return) accordingly. The revised expectations enter or get incorporated in the asset price through trades in the asset. Market participants who process the news and believe that at the current market price an asset does not offer sufficient compensation for its perceived risk will tend to sell it or even sell it short. Market participants with opposite views should be buyers. In this way the market establishes the price that balances the various opinions after expectations are revised.

EXAMPLE 2

Price Reaction to the Default on a Bond Issue

Suppose that a speculative-grade bond issuer announces, just before bond markets open, that it will default on an upcoming interest payment. In the announcement, the issuer confirms various reports made in the financial media in the period leading up to the announcement. Prior to the issuer's announcement, the financial news media reported the following: 1) suppliers of the company were making deliveries only for cash payment, reducing the company's liquidity; 2) the issuer's financial condition had probably deteriorated to the point that it lacked the cash to meet an upcoming interest payment; and 3) although public capital markets were closed to the company, it was negotiating with a bank for a private loan that would permit it to meet its interest payment and continue operations for at least nine months. If the issuer defaults on the bond, the consensus opinion of analysts is that bondholders will recover approximately \$0.36 to \$0.38 per dollar face value.

1. If the market for the bond is efficient, the bond's market price is *most likely* to fully reflect the bond's value after default:

- A. in the period leading up to the announcement.
- B. in the first trade prices after the market opens on the announcement day.
- C. when the issuer actually misses the payment on the interest payment date.

Solution to 1:

B is correct. The announcement removed any uncertainty about default. In the period leading up to the announcement, the bond's market price incorporated a probability of default, but the price would not have fully reflected the bond's value after default. The possibility that a bank loan might permit the company to avoid default was not eliminated until the announcement.

2. If the market for the bond is efficient, the piece of information that bond investors *most likely* focus on in the issuer's announcement is that the issuer had:

- A. failed in its negotiations for a bank loan.
- B. lacked the cash to meet the upcoming interest payment.
- C. been required to make cash payments for supplier deliveries.

Solution to 2:

A is correct. The failure of the loan negotiations first becomes known in this announcement. The failure implies default.

2.2. Market Value versus Intrinsic Value

Market value is the price at which an asset can currently be bought or sold. **Intrinsic value** (sometimes called **fundamental value**) is, broadly speaking, the value that would be placed on it by investors if they had a complete understanding of the asset's investment characteristics.⁶ For a bond, for example, such information would include its interest (coupon) rate, principal value, the timing of its interest and principal payments, the other terms of the bond contract (indenture), a precise understanding of its default risk, the liquidity of its market, and other issue-specific items. In addition, market variables such as the term structure of interest rates and the size of various market premiums applying to the issue (for default risk, etc.) would enter into a discounted cash flow estimate of the bond's intrinsic value (discounted cash flow models are often used for such estimates). The word *estimate* is used because in practice, intrinsic value can be estimated but is not known for certain.

If investors believe a market is highly *efficient*, they will usually accept market prices as accurately reflecting intrinsic values. Discrepancies between market price and intrinsic value are the basis for

profitable active investment. Active investors seek to own assets selling below perceived intrinsic value in the marketplace and to sell or sell short assets selling above perceived intrinsic value.

If investors believe an asset market is relatively *inefficient*, they may try to develop an independent estimate of intrinsic value. The challenge for investors and analysts is estimating an asset's intrinsic value. Numerous theories and models, including the dividend discount model, can be used to estimate an asset's intrinsic value, but they all require some form of judgment regarding the size, timing, and riskiness of the future cash flows associated with the asset. The more complex an asset's future cash flows, the more difficult it is to estimate its intrinsic value. These complexities and the estimates of an asset's market value are reflected in the market through the buying and selling of assets. The market value of an asset represents the intersection of supply and demand—the point that is low enough to induce at least one investor to buy while being high enough to induce at least one investor to sell. Because information relevant to valuation flows continually to investors, estimates of intrinsic value change, and hence, market values change.

EXAMPLE 3

Intrinsic Value

1. An analyst estimates that a security's intrinsic value is lower than its market value. The security appears to be:

- A. undervalued.
- B. fairly valued.
- C. overvalued.

Solution to 1:

C is correct. The market is valuing the asset at more than its true worth.

2. A market in which assets' market values are, on average, equal to or nearly equal to intrinsic values is *best described* as a market that is attractive for:

- A. active investment.
- B. passive investment.
- C. both active and passive investment.

Solution to 2:

B is correct because an active investment is not expected to earn superior risk-adjusted returns if the market is efficient. The additional costs of active investment are not justified in such a market.

3. Suppose that the future cash flows of an asset are accurately estimated. The asset trades in a market that you believe is efficient based on most evidence, but your estimate of the asset's intrinsic value exceeds the asset's market value by a moderate amount. The *most likely* conclusion is that you have:

- A. overestimated the asset's risk.
- B. underestimated the asset's risk.
- C. identified a market inefficiency.

Solution to 3:

B is correct. If risk is underestimated, the discount rate being applied to find the present value of the expected cash flows (estimated intrinsic value) will be too low and the intrinsic value estimate will be too high.

3. FACTORS AFFECTING MARKET EFFICIENCY INCLUDING TRADING COSTS

LEARNING OUTCOMES

The candidate should be able to:

- ☐ describe market efficiency and related concepts, including their importance to investment practitioners
- ☐ explain factors that affect a market's efficiency

For markets to be efficient, prices should adjust quickly and rationally to the release of new information. In other words, prices of assets in an efficient market should “fully reflect” all information. Financial markets, however, are generally not classified at the two extremes as either completely inefficient or completely efficient but, rather, as exhibiting various degrees of efficiency. In other words, market efficiency should be viewed as falling on a continuum between extremes of completely efficient, at one end, and completely inefficient, at the other. Asset prices in a highly efficient market, by definition, reflect information more quickly and more accurately than in a less-efficient market. These degrees of efficiency also vary through time, across geographical markets, and by type of market. A number of factors contribute to and impede the degree of efficiency in a financial market.

3.1. Market Participants

One of the most critical factors contributing to the degree of efficiency in a market is the number of market participants. Consider the following example that illustrates the relationship between the number of market participants and market efficiency.

EXAMPLE 4

Illustration of Market Efficiency

Assume that the shares of a small market capitalization (cap) company trade on a public stock exchange. Because of its size, it is not considered “blue-chip” and not many professional investors follow the activities of the company.⁷ A small-cap fund analyst reports that the most recent annual operating performance of the company has been surprisingly good, considering the recent slump in its industry. The company's share price, however, has been slow to react to the positive financial results because the company is not being recommended by the majority of research analysts. This mispricing implies that the market for this company's shares is less than fully efficient. The small-cap fund analyst recognizes the opportunity and immediately recommends the purchase of the company's shares. The share price gradually increases as more investors purchase the shares once the news of the mispricing spreads through the market. As a result, it takes a few days for the share price to fully reflect the information.

Six months later, the company reports another solid set of interim financial results. But because the previous mispricing and subsequent profit opportunities have become known in the market, the number of analysts following the company's shares has increased substantially. As a result, as soon

as unexpected information about the positive interim results are released to the public, a large number of buy orders quickly drive up the stock price, thereby making the market for these shares more efficient than before.

A large number of investors (individual and institutional) follow the major financial markets closely on a daily basis, and if mispricings exist in these markets, as illustrated by the example, investors will act so that these mispricings disappear quickly. Besides the number of investors, the number of financial analysts who follow or analyze a security or asset should be positively related to market efficiency. The number of market participants and resulting trading activity can vary significantly through time. A lack of trading activity can cause or accentuate other market imperfections that impede market efficiency. In fact, in many of these markets, trading in many of the listed stocks is restricted for foreigners. By nature, this limitation reduces the number of market participants, restricts the potential for trading activity, and hence reduces market efficiency.

EXAMPLE 5

Factors Affecting Market Efficiency

The expected effect on market efficiency of opening a securities market to trading by foreigners would *most likely* be to:

- A. decrease market efficiency.
- B. leave market efficiency unchanged.
- C. increase market efficiency.

Solution:

C is correct. The opening of markets as described should increase market efficiency by increasing the number of market participants.

3.2. Information Availability and Financial Disclosure

Information availability (e.g., an active financial news media) and financial disclosure should promote market efficiency. Information regarding trading activity and traded companies in such markets as the New York Stock Exchange, the London Stock Exchange, and the Tokyo Stock Exchange is readily available. Many investors and analysts participate in these markets, and analyst coverage of listed companies is typically substantial. As a result, these markets are quite efficient. In contrast, trading activity and material information availability may be lacking in smaller securities markets, such as those operating in some emerging markets.

Similarly, significant differences may exist in the efficiency of different types of markets. For example, many securities trade primarily or exclusively in dealer or over-the-counter (OTC) markets, including bonds, money market instruments, currencies, mortgage-backed securities, swaps, and forward contracts. The information provided by the dealers that serve as market makers for these markets can vary significantly in quality and quantity, both through time and across different product markets.

Treating all market participants fairly is critical for the integrity of the market and explains why regulators place such an emphasis on “fair, orderly, and efficient markets.”⁸ A key element of this fairness is that all investors have access to the information necessary to value securities that trade in the market. Rules and regulations that promote fairness and efficiency in a market include those pertaining to the disclosure of information and illegal insider trading.

For example, US Securities and Exchange Commission's (SEC's) Regulation FD (Fair Disclosure) requires that if security issuers provide nonpublic information to some market professionals or investors, they must also disclose this information to the public.⁹ This requirement helps provide equal and fair opportunities, which is important in encouraging participation in the market. A related issue deals with illegal insider trading. The SEC's rules, along with court cases, define illegal insider trading as trading in securities by market participants who are considered insiders "while in possession of material, nonpublic information about the security."¹⁰ Although these rules cannot guarantee that some participants will not have an advantage over others and that insiders will not trade on the basis of inside information, the civil and criminal penalties associated with breaking these rules are intended to discourage illegal insider trading and promote fairness. In the European Union, insider trading laws are generally enshrined in legislation and enforced by regulatory and judicial authorities.¹¹

3.3. Limits to Trading

Arbitrage is a set of transactions that produces riskless profits. Arbitrageurs are traders who engage in such trades to benefit from pricing discrepancies (inefficiencies) in markets. Such trading activity contributes to market efficiency. For example, if an asset is traded in two markets but at different prices, the actions of buying the asset in the market in which it is underpriced and selling the asset in the market in which it is overpriced will eventually bring these two prices together. The presence of these arbitrageurs helps pricing discrepancies disappear quickly. Obviously, market efficiency is impeded by any limitation on arbitrage resulting from operating inefficiencies, such as difficulties in executing trades in a timely manner, prohibitively high trading costs, and a lack of transparency in market prices.

Some market experts argue that restrictions on short selling limit arbitrage trading, which impedes market efficiency. **Short selling** is the transaction whereby an investor sells shares that he or she does not own by borrowing them from a broker and agreeing to replace them at a future date. Short selling allows investors to sell securities they believe to be overvalued, much in the same way they can buy those they believe to be undervalued. In theory, such activities promote more efficient pricing. Regulators and others, however, have argued that short selling may exaggerate downward market movements, leading to crashes in affected securities. In contrast, some researchers report evidence indicating that when investors are unable to borrow securities, that is to short the security, or when costs to borrow shares are high, market prices may deviate from intrinsic values.¹² Furthermore, research suggests that short selling is helpful in price discovery (that is, it facilitates supply and demand in determining prices).¹³

3.4. Transaction Costs and Information-Acquisition Costs

The costs incurred by traders in identifying and exploiting possible market inefficiencies affect the interpretation of market efficiency. The two types of costs to consider are transaction costs and information-acquisition costs.

- *Transaction costs:* Practically, transaction costs are incurred in trading to exploit any perceived market inefficiency. Thus, "efficient" should be viewed as efficient within the bounds of transaction costs. For example, consider a violation of the principle that two identical assets should sell for the same price in different markets. Such a violation can be considered to be a rather simple possible exception to market efficiency because prices appear to be inconsistently processing information. To exploit the violation, a trader could arbitrage by simultaneously shorting the asset in the higher-price market and buying the asset in the lower-price market. If the price discrepancy between the two markets is smaller than the transaction costs involved in the arbitrage for the lowest cost traders, the arbitrage will not occur, and both prices are in effect efficient within the bounds of arbitrage. These bounds of arbitrage are relatively narrow in highly liquid markets, such as the market for US Treasury bills, but could be wide in illiquid markets.
- *Information-acquisition costs:* Practically, expenses are always associated with gathering and analyzing information. New information is incorporated in transaction prices by traders placing trades based on their analysis of information. Active investors who place trades based on information they have gathered and analyzed play a key role in market prices adjusting to reflect new information. The classic view of market efficiency is that active investors incur information

acquisition costs but that money is wasted because prices already reflect all relevant information. This view of efficiency is very strict in the sense of viewing a market as inefficient if active investing can recapture any part of the costs, such as research costs and active asset selection. [Grossman and Stiglitz \(1980\)](#) argue that prices must offer a return to information acquisition; in equilibrium, if markets are efficient, returns net of such expenses are just fair returns for the risk incurred. The modern perspective views a market as inefficient if, after deducting such costs, active investing can earn superior returns. Gross of expenses, a return should accrue to information acquisition in an efficient market.

In summary, a modern perspective calls for the investor to consider transaction costs and information-acquisition costs when evaluating the efficiency of a market. A price discrepancy must be sufficiently large to leave the investor with a profit (adjusted for risk) after taking account of the transaction costs and information-acquisition costs to reach the conclusion that the discrepancy may represent a market inefficiency. Prices may somewhat less than fully reflect available information without there being a true market opportunity for active investors.

4. FORMS OF MARKET EFFICIENCY

LEARNING OUTCOME

The candidate should be able to:

- ☐ contrast weak-form, semi-strong-form, and strong-form market efficiency

Eugene Fama developed a framework for describing the degree to which markets are efficient.¹⁴ In his efficient market hypothesis, markets are efficient when prices reflect *all* relevant information at any point in time. This means that the market prices observed for securities, for example, reflect the information available at the time.

In his framework, Fama defines three forms of efficiency: weak, semi-strong, and strong. Each form is defined with respect to the available information that is reflected in prices.

Forms of Market Efficiency	Market Prices Reflect:		
	Past Market Data	Public Information	Private Information
Weak form of market efficiency	✓		
Semi-strong form of market efficiency	✓	✓	
Strong form of market efficiency	✓	✓	✓

A finding that investors can consistently earn **abnormal returns** by trading on the basis of information is evidence contrary to market efficiency. In general, abnormal returns are returns in excess of those expected given a security’s risk and the market’s return. In other words, abnormal return equals actual return less expected return.

4.1. Weak Form

In the **weak-form efficient market hypothesis**, security prices fully reflect *all past market data*, which refers to all historical price and trading volume information. If markets are weak-form efficient, past trading data are already reflected in current prices and investors cannot predict future price changes by extrapolating prices or patterns of prices from the past.¹⁵

Tests of whether securities markets are weak-form efficient require looking at patterns of prices. One approach is to see whether there is any serial correlation in security returns, which would imply a predictable pattern.¹⁶ Although there is some weak correlation in daily security returns, there is not enough correlation to make this a profitable trading rule after considering transaction costs.

An alternative approach to test weak-form efficiency is to examine specific trading rules that attempt to exploit historical trading data. If any such trading rule consistently generates abnormal risk-adjusted returns after trading costs, this evidence will contradict weak-form efficiency. This approach is commonly associated with **technical analysis**, which involves the analysis of historical trading information (primarily pricing and volume data) in an attempt to identify recurring patterns in the trading data that can be used to guide investment decisions. Many technical analysts, also referred to as “technicians,” argue that many movements in stock prices are based, in large part, on psychology. Many technicians attempt to predict how market participants will behave, based on analyses of past behavior, and then trade on those predictions. Technicians often argue that simple statistical tests of trading rules are not conclusive because they are not applied to the more sophisticated trading strategies that can be used and that the research excludes the technician’s subjective judgment. Thus, it is difficult to definitively refute this assertion because there are an unlimited number of possible technical trading rules.

Can technical analysts profit from trading on past trends? Overall, the evidence indicates that investors cannot consistently earn abnormal profits using past prices or other technical analysis strategies in developed markets.¹⁷ Some evidence suggests, however, that there are opportunities to profit on technical analysis in countries with developing markets, including Hungary, Bangladesh, and Turkey, among others.¹⁸

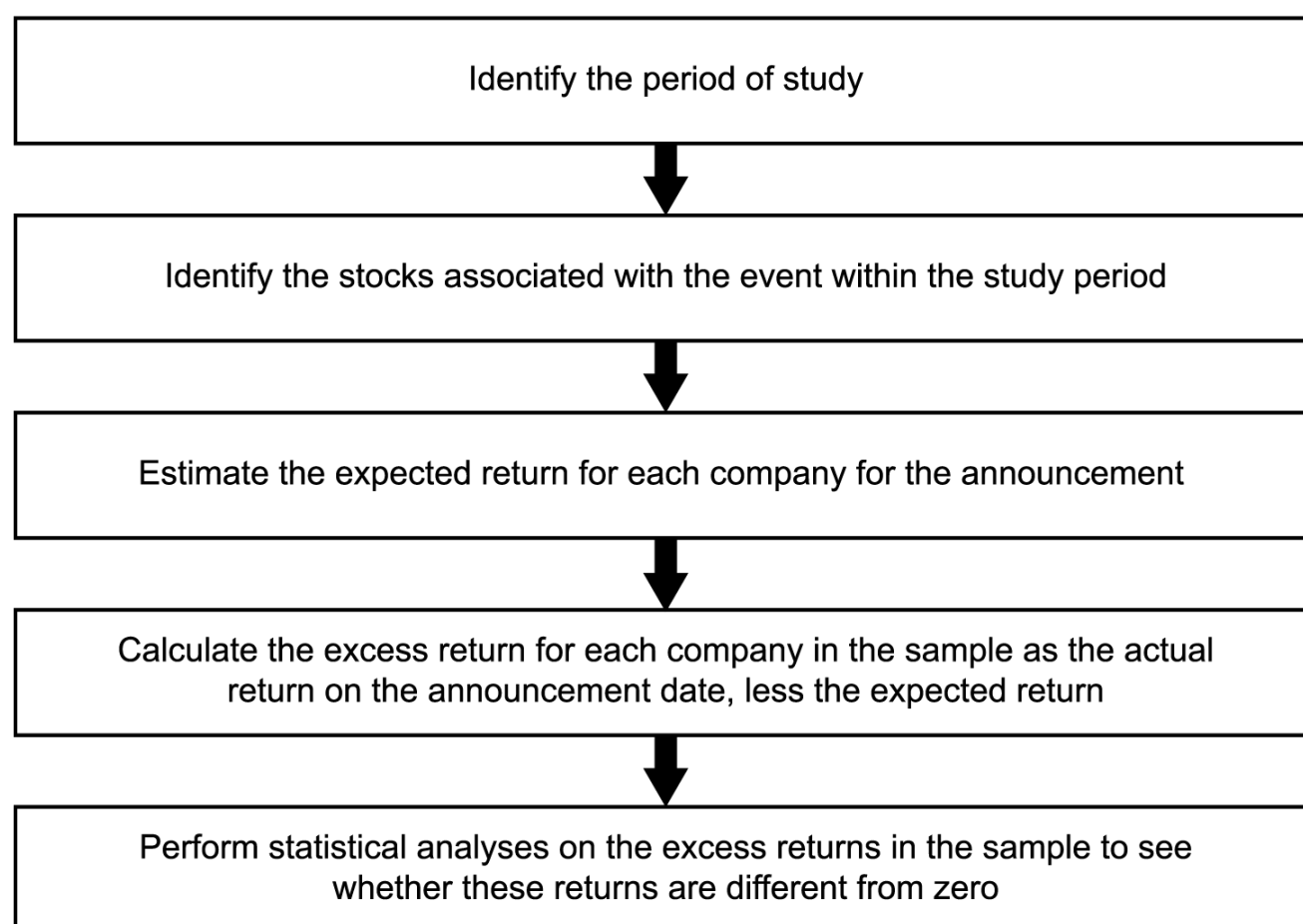
4.2. Semi-Strong Form

In a **semi-strong-form efficient market**, prices reflect all publicly known and available information. Publicly available information includes financial statement data (such as earnings, dividends, corporate investments, changes in management, etc.) and financial market data (such as closing prices, shares traded, etc.). Therefore, the semi-strong form of market efficiency encompasses the weak form. In other words, if a market is semi-strong efficient, then it must also be weak-form efficient. A market that quickly incorporates all publicly available information into its prices is semi-strong efficient.

In a semi-strong market, efforts to analyze publicly available information are futile. That is, analyzing earnings announcements of companies to identify underpriced or overpriced securities is pointless because the prices of these securities already reflect all publicly available information. If markets are semi-strong efficient, no single investor has access to information that is not already available to other market participants, and as a consequence, no single investor can gain an advantage in predicting future security prices. In a semi-strong efficient market, prices adjust quickly and accurately to new information. Suppose a company announces earnings that are higher than expected. In a semi-strong efficient market, investors would not be able to act on this announcement and earn abnormal returns.

A common empirical test of investors’ reaction to information releases is the event study. Suppose a researcher wants to test whether investors react to the announcement that the company is paying a special dividend. The researcher identifies a sample period and then those companies that paid a special dividend in the period and the date of the announcement. Then, for each company’s stock, the researcher calculates the expected return on the share for the event date. This expected return may be based on many different models, including the capital asset pricing model, a simple market model, or a market index return. The researcher calculates the excess return as the difference between the actual return and the expected return. Once the researcher has calculated the event’s excess return for each share, statistical tests are conducted to see whether the abnormal returns are statistically different from zero. The process of an event study is outlined in [Exhibit 1](#).

Exhibit 1: The Event Study Process



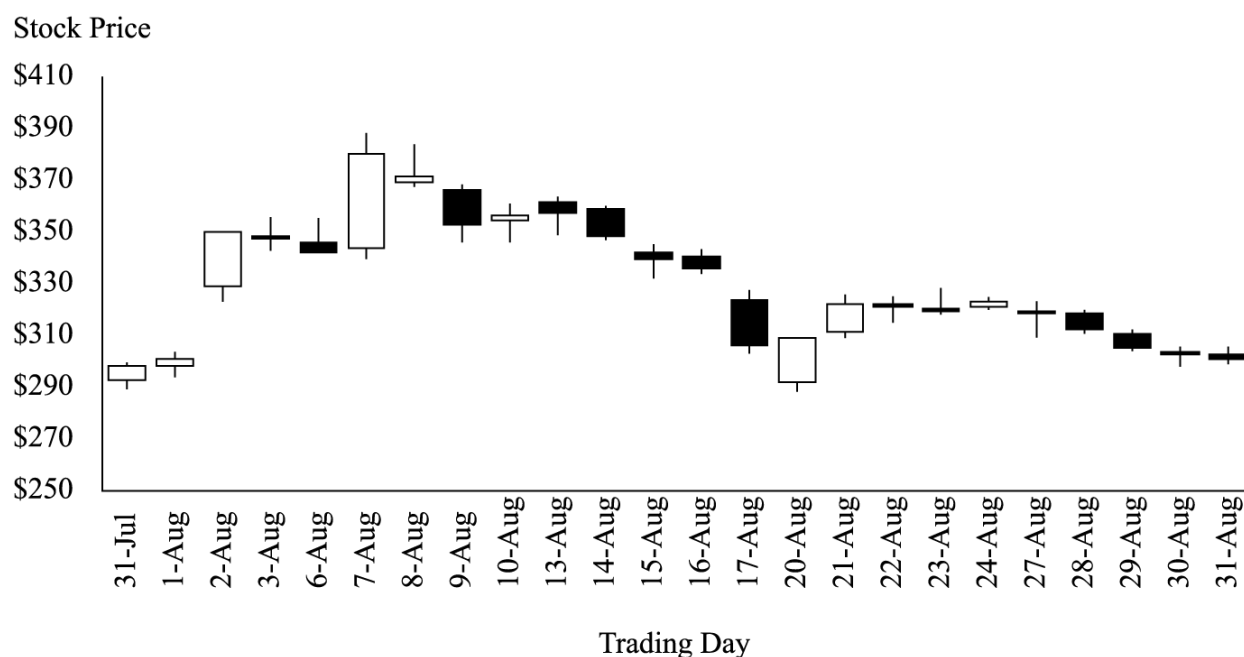
How do event studies relate to efficient markets? In a semi-strong efficient market, share prices react quickly and accurately to public information. Therefore, if the information is good news, such as better-than-expected earnings, one would expect the company's shares to increase immediately at the time of the announcement; if it is bad news, one would expect a swift, negative reaction. If actual returns exceed what is expected in absence of the announcement and these returns are confined to the announcement period, then they are consistent with the idea that market prices react quickly to new information. In other words, the finding of excess returns at the time of the announcement does not necessarily indicate market inefficiency. In contrast, the finding of consistent excess returns following the announcement would suggest a trading opportunity. Trading on the basis of the announcement—that is, once the announcement is made—would not, on average, yield abnormal returns.

EXAMPLE 6

Information Arrival and Market Reaction

Consider an example of a news item and its effect on a share's price. The following events related to Tesla, Inc. in August of 2018:

- | | |
|----------------|--|
| 1 August 2018 | After the market closes, Tesla, Inc., publicly reports that there was a smaller-than expected cash burn for the most recent quarter. |
| 2 August 2018 | Elon Musk, Chairman and CEO of Tesla, Inc., notifies Tesla's board of directors that he wants to take the company private. This is not public information at this point. |
| 7 August 2018 | Before the market opens, the <i>Financial Times</i> reports that a Saudi fund has a \$2 billion investment in Tesla.
During market trading, Musk announces on Twitter "Am considering taking Tesla private at \$420. Funding secured." [Twitter, Elon Musk @elonmusk, 9:48 a.m., 7 August 2018] |
| 24 August 2018 | After the market closed, Musk announces that he no longer intends on taking Tesla private. |



Note: Open-High-Low-Close graph of Tesla's stock price, with white rectangles indicating upward movement in the day and black rectangles indicating downward movement during the day.

Source of data: Yahoo! Finance.

1. Is the fact that the price of Tesla moves up immediately on the day after the Q2 earnings (the first day of trading with this information) indicative of efficiency regarding information?

Most likely.

2. Does the fact that the price of Tesla moves up but does not reach \$420 on the day the going-private Twitter announcement is made mean that investors underreacted?

Not necessarily. There was confusion and uncertainty about the going-private transaction at the time, so the price did not close in on the proposed \$420 per share for going private.

3. Does the fact that the market price of the stock declined well before the issue of going-private was laid to rest by Musk mean that the market is inefficient?

Not necessarily. There were numerous analyses, discussions, and other news regarding the likelihood of the transaction, all of which was incorporated in the price of the stock before the going-private transaction was dismissed by Musk.

Researchers have examined many different company-specific information events, including stock splits, dividend changes, and merger announcements, as well as economy-wide events, such as regulation changes and tax rate changes. The results of most research are consistent with the view that developed securities markets might be semi-strong efficient. But some evidence suggests that the markets in developing countries may not be semi-strong efficient.¹⁹

4.3. Strong Form

In a **strong-form efficient market**, security prices fully reflect both public and private information. A market that is strong-form efficient is, by definition, also semi-strong- and weak-form efficient. In the case of a strong-form efficient market, insiders would not be able to earn abnormal returns from trading on the basis of private information. A strong-form efficient market also means that prices reflect all private information, which means that prices reflect everything that the management of a company knows about the financial condition of the company that has not been publicly released. However, this is not likely because of the strong prohibitions against insider trading that are found in most countries. If a market is strong-form efficient, those with insider information cannot earn abnormal returns.

Researchers test whether a market is strong-form efficient by testing whether investors can earn abnormal profits by trading on nonpublic information. The results of these tests are consistent with the view that securities markets are not strong-form efficient; many studies have found that abnormal profits can be earned when nonpublic information is used.²⁰

5. IMPLICATIONS OF THE EFFICIENT MARKET HYPOTHESIS

LEARNING OUTCOME

The candidate should be able to:

- ☐ **explain the implications of each form of market efficiency for fundamental analysis, technical analysis, and the choice between active and passive portfolio management**

The implications of efficient markets to investment managers and analysts are important because they affect the value of securities and how these securities are managed. Several implications can be drawn from the evidence on efficient markets for developed markets:

- Securities markets are weak-form efficient, and therefore, investors cannot earn abnormal returns by trading on the basis of past trends in price.
- Securities markets are semi-strong efficient, and therefore, analysts who collect and analyze information must consider whether that information is already reflected in security prices and how any new information affects a security's value.
- Securities markets are not strong-form efficient because securities laws are intended to prevent exploitation of private information.

5.1. Fundamental Analysis

Fundamental analysis is the examination of publicly available information and the formulation of forecasts to estimate the intrinsic value of assets. Fundamental analysis involves the estimation of an asset's value using company data, such as earnings and sales forecasts, and risk estimates as well as industry and economic data, such as economic growth, inflation, and interest rates. Buy and sell decisions depend on whether the current market price is less than or greater than the estimated intrinsic value.

The semi-strong form of market efficiency says that all available public information is reflected in current prices. So, what good is fundamental analysis? Fundamental analysis is necessary in a well-functioning market because this analysis helps the market participants understand the value implications of information. In other words, fundamental analysis facilitates a semi-strong efficient market by disseminating value-relevant information. And, although fundamental analysis requires costly information, this analysis can be profitable in terms of generating abnormal returns if the analyst creates a comparative advantage with respect to this information.²¹

5.2. Technical Analysis

Investors using **technical analysis** attempt to profit by looking at patterns of prices and trading volume. Although some price patterns persist, exploiting these patterns may be too costly and, hence, would not produce abnormal returns.

Consider a situation in which a pattern of prices exists. With so many investors examining prices, this pattern will be detected. If profitable, exploiting this pattern will eventually affect prices such that this

pattern will no longer exist; it will be arbitrated away. In other words, by detecting and exploiting patterns in prices, technical analysts assist markets in maintaining weak-form efficiency. Does this mean that technical analysts cannot earn abnormal profits? Not necessarily, because there may be a possibility of earning abnormal profits from a pricing inefficiency. But would it be possible to earn abnormal returns on a consistent basis from exploiting such a pattern? No, because the actions of market participants will arbitrage this opportunity quickly, and the inefficiency will no longer exist.

5.3. Portfolio Management

If securities markets are weak-form and semi-strong-form efficient, the implication is that active trading, whether attempting to exploit price patterns or public information, is not likely to generate abnormal returns. In other words, portfolio managers cannot beat the market on a consistent basis, so therefore, passive portfolio management should outperform active portfolio management. Researchers have observed that mutual funds do not, on average, outperform the market on a risk-adjusted basis.²² Mutual funds perform, on average, similar to the market before considering fees and expenses and perform worse than the market, on average, once fees and expenses are considered. Even if a mutual fund is not actively managed, there are costs to managing these funds, which reduces net returns.

So, what good are portfolio managers? The role of a portfolio manager is not necessarily to beat the market but, rather, to establish and manage a portfolio consistent with the portfolio's objectives, with appropriate diversification and asset allocation, while taking into consideration the risk preferences and tax situation of the investor.

6. MARKET PRICING ANOMALIES - TIME SERIES AND CROSS-SECTIONAL

LEARNING OUTCOME

The candidate should be able to:

☐ **describe market anomalies**

Although considerable evidence shows that markets are efficient, researchers have identified a number of apparent market inefficiencies or anomalies. These market anomalies, if persistent, are exceptions to the notion of market efficiency. Researchers conclude that a **market anomaly** may be present if a change in the price of an asset or security cannot directly be linked to current relevant information known in the market or to the release of new information into the market.

The validity of any evidence supporting the potential existence of a market inefficiency or anomaly must be *consistent* over reasonably long periods. Otherwise, a detected market anomaly may largely be an artifact of the sample period chosen. In the widespread search for discovering profitable anomalies, many findings could simply be the product of a process called **data mining**, also known as **data snooping**. In generally accepted research practice, an initial hypothesis is developed which is based on economic rationale. Tests are then conducted on objectively selected data to either confirm or reject the original hypothesis. However, with data mining the process is typically reversed: data are examined with the intent to develop a hypothesis, instead of developing a hypothesis first. This is done by analyzing data in various manners, and even utilizing different empirical approaches until you find support for a desired result, in this case a profitable anomaly.

Can researchers look back on data and find a trading strategy that would have yielded abnormal returns? Absolutely. Enough data snooping often can detect a trading strategy that would have worked in the past by chance alone. But in an efficient market, such a strategy is unlikely to generate abnormal returns on a consistent basis in the future. Also, although identified anomalies may appear to produce excess returns,

it is generally difficult to profitably exploit the anomalies after accounting for risk, trading costs, and so on.

Several well-known anomalies are listed in [Exhibit 3](#). This list is by no means exhaustive, but it provides information on the breadth of the anomalies. A few of these anomalies are discussed in more detail in the following sections. The anomalies are placed into categories based on the research method that identified the anomaly. Time-series anomalies were identified using time series of data. Cross-sectional anomalies were identified based on analyzing a cross section of companies that differ on some key characteristics. Other anomalies were identified by a variety of means, including event studies.

Exhibit 3: Sampling of Observed Pricing Anomalies

Time Series	Cross-Sectional	Other
January effect	Size effect	Closed-end fund discount
Day-of-the-week effect	Value effect	Earnings surprise
Weekend effect	Book-to-market ratios	Initial public offerings
Turn-of-the-month effect	P/E ratio effect	Distressed securities effect
Holiday effect	Value Line enigma	Stock splits
Time-of-day effect		Super Bowl
Momentum		
Overreaction		

6.1. Time-Series Anomalies

Two of the major categories of time-series anomalies that have been documented are 1) calendar anomalies and 2) momentum and overreaction anomalies.

6.1.1. Calendar Anomalies

In the 1980s, a number of researchers reported that stock market returns in January were significantly higher compared to the rest of the months of the year, with most of the abnormal returns reported during the first five trading days in January. Since its first documentation in the 1980s, this pattern, known as the **January effect**, has been observed in most equity markets around the world. This anomaly is also known as the **turn-of-the-year effect**, or even often referred to as the “small firm in January effect” because it is most frequently observed for the returns of small market capitalization stocks.²³

The January effect contradicts the efficient market hypothesis because excess returns in January are not attributed to any new and relevant information or news. A number of reasons have been suggested for this anomaly, including tax-loss selling. Researchers have speculated that, in order to reduce their tax liabilities, investors sell their “loser” securities in December for the purpose of creating capital losses, which can then be used to offset any capital gains. A related explanation is that these losers tend to be small-cap stocks with high volatility.²⁴ This increased supply of equities in December depresses their prices, and then these shares are bought in early January at relatively attractive prices. This demand then drives their prices up again. Overall, the evidence indicates that tax-loss selling may account for a portion of January abnormal returns, but it does not explain all of it.

Another possible explanation for the anomaly is so-called “window dressing”, a practice in which portfolio managers sell their riskier securities prior to 31 December. The explanation is as follows: many portfolio managers prepare the annual reports of their portfolio holdings as of 31 December. Selling riskier securities is an attempt to make their portfolios appear less risky. After 31 December, a portfolio manager would then simply purchase riskier securities in an attempt to earn higher returns. However, similar to the tax-loss selling hypothesis, the research evidence in support of the window dressing hypothesis explains some, but not all, of the anomaly.

Recent evidence for both stock and bond returns suggests that the January effect is not persistent and, therefore, is not a pricing anomaly. Once an appropriate adjustment for risk is made, the January “effect” does not produce abnormal returns.²⁵

Several other calendar effects, including the day-of-the-week and the weekend effects,²⁶ have been found. These anomalies are summarized in [Exhibit 4](#).²⁷ But like the size effect, which will be described later, most of these anomalies have been eliminated over time. One view is that the anomalies have been exploited such that the effect has been arbitrated away. Another view, however, is that increasingly sophisticated statistical methodologies fail to detect pricing inefficiencies.

Exhibit 4: Calendar-Based Anomalies

Anomaly	Observation
Turn-of-the-month effect	Returns tend to be higher on the last trading day of the month and the first three trading days of the next month.
Day-of-the-week effect	The average Monday return is negative and lower than the average returns for the other four days, which are all positive.
Weekend effect	Returns on weekends tend to be lower than returns on weekdays.
Holiday effect	Returns on stocks in the day prior to market holidays tend to be higher than other days.

6.1.2. Momentum and Overreaction Anomalies

Momentum anomalies relate to short-term share price patterns. One of the earliest studies to identify this type of anomaly was conducted by Werner DeBondt and Richard Thaler, who argued that investors overreact to the release of unexpected public information.²⁸ Therefore, stock prices will be inflated (depressed) for those companies releasing good (bad) information. This anomaly has become known as the overreaction effect. Using the overreaction effect, they proposed a strategy that involved buying “loser” portfolios and selling “winner” portfolios. They defined stocks as winners or losers based on their total returns over the previous three- to five-year period. They found that in a subsequent period, the loser portfolios outperformed the market, while the winner portfolios underperformed the market. Similar patterns have been documented in many, but not all, global stock markets as well as in bond markets. One criticism is that the observed anomaly may be the result of statistical problems in the analysis.

A contradiction to weak-form efficiency occurs when securities that have experienced high returns in the short term tend to continue to generate higher returns in subsequent periods.²⁹ Empirical support for the existence of momentum in stock returns in most stock markets around the world is well documented. If investors can trade on the basis of momentum and earn abnormal profits, then this anomaly contradicts the weak form of the efficient market hypothesis because it represents a pattern in prices that can be exploited by simply using historical price information.³⁰

Researchers have argued that the existence of momentum is rational and not contrary to market efficiency because it is plausible that there are shocks to the expected growth rates of cash flows to shareholders and that these shocks induce a serial correlation that is rational and short lived.³¹ In other words, having stocks with some degree of momentum in their security returns may not imply irrationality but, rather, may reflect prices adjusting to a shock in growth rates.

6.2. Cross-Sectional Anomalies

Two of the most researched cross-sectional anomalies in financial markets are the size effect and the value effect.

6.2.1. *Size Effect*

The size effect results from the observation that equities of small-cap companies tend to outperform equities of large-cap companies on a risk-adjusted basis. Many researchers documented a small-company effect soon after the initial research was published in 1981. This effect, however, was not apparent in subsequent studies.³² Part of the reason that the size effect was not confirmed by subsequent studies may be because of the fact that if it were truly an anomaly, investors acting on this effect would reduce any potential returns. But some of the explanation may simply be that the effect as originally observed was a chance outcome and, therefore, not actually an inefficiency.

6.2.2. *Value Effect*

A number of global empirical studies have shown that value stocks, which are generally referred to as stocks that have below-average price-to-earnings (P/E) and market-to-book (M/B) ratios, and above-average dividend yields, have consistently outperformed growth stocks over long periods of time.³³ If the effect persists, the value stock anomaly contradicts semi-strong market efficiency because all the information used to categorize stocks in this manner is publicly available.

Fama and French developed a three-factor model to predict stock returns.³⁴ In addition to the use of market returns as specified by the capital asset pricing model (CAPM), the Fama and French model also includes the size of the company as measured by the market value of its equity and the company's book value of equity divided by its market value of equity, which is a value measure. The Fama and French model captures risk dimensions related to stock returns that the CAPM model does not consider. Fama and French find that when they apply the three-factor model instead of the CAPM, the value stock anomaly disappears.

7. OTHER ANOMALIES, IMPLICATIONS OF MARKET PRICING ANOMALIES

LEARNING OUTCOME

The candidate should be able to:

- ☐ **describe market anomalies**

A number of additional anomalies has been documented in the financial markets, including the existence of closed-end investment fund discounts, price reactions to the release of earnings information, returns of initial public offerings, and the predictability of returns based on prior information.

7.1. Closed-End Investment Fund Discounts

A closed-end investment fund issues a fixed number of shares at inception and does not sell any additional shares after the initial offering. Therefore, the fund capitalization is fixed unless a secondary public offering is made. The shares of closed-end funds trade on stock markets like any other shares in the equity market (i.e., their prices are determined by supply and demand).

Theoretically, these shares should trade at a price approximately equal to their net asset value (NAV) per share, which is simply the total market value of the fund's security holdings less any liabilities divided by the number of shares outstanding. An abundance of research, however, has documented that, on average, closed-end funds trade at a discount from NAV. Most studies have documented average discounts in the 4–10 percent range, although individual funds have traded at discounts exceeding 50 percent and others have traded at large premiums.³⁵

The closed-end fund discount presents a puzzle because conceptually, an investor could purchase all the shares in the fund, liquidate the fund, and end up making a profit. Some researchers have suggested that these discounts are attributed to management fees or expectations of the managers' performance, but these explanations are not supported by the evidence.³⁶ An alternative explanation for the discount is that tax liabilities are associated with unrealized capital gains and losses that exist prior to when the investor bought the shares, and hence, the investor does not have complete control over the timing of the realization of gains and losses.³⁷ Although the evidence supports this hypothesis to a certain extent, the tax effect is not large enough to explain the entire discount. Finally, it has often been argued that the discounts exist because of liquidity problems and errors in calculating NAV. The illiquidity explanation is plausible if shares are recorded at the same price as more liquid, publicly traded stocks; some evidence supports this assertion. But as with tax reasons, liquidity issues explain only a portion of the discount effect.

Can these discounts be exploited to earn abnormal returns if transaction costs are taken into account? No. First, the transaction costs involved in exploiting the discount—buying all the shares and liquidating the fund—would eliminate any profit.³⁸ Second, these discounts tend to revert to zero over time. Hence, a strategy to trade on the basis of these discounts would not likely be profitable.³⁹

7.2. Earnings Surprise

Although most event studies have supported semi-strong market efficiency, some researchers have provided evidence that questions semi-strong market efficiency. One of these studies relates to the extensively examined adjustment of stock prices to earnings announcements.⁴⁰ The unexpected part of the earnings announcement, or **earnings surprise**, is the portion of earnings that is unanticipated by investors and, according to the efficient market hypothesis, merits a price adjustment. Positive (negative) surprises should cause appropriate and rapid price increases (decreases). Several studies have been conducted using data from numerous markets around the world. Most of the results indicate that earnings surprises are reflected quickly in stock prices, but the adjustment process is not always efficient. In particular, although a substantial adjustment occurs prior to and at the announcement date, an adjustment also occurs after the announcement.⁴¹

As a result of these slow price adjustments, companies that display the largest positive earnings surprises subsequently display superior stock return performance, whereas poor subsequent performance is displayed by companies with low or negative earnings surprises.⁴² This finding implies that investors could earn abnormal returns using publicly available information by buying stocks of companies that had positive earnings surprises and selling those with negative surprises.

Although there is support for abnormal returns associated with earnings surprises, and some support for such returns beyond the announcement period, there is also evidence indicating that these observed abnormal returns are an artifact of studies that do not sufficiently control for transaction costs and risk.⁴³

7.3. Initial Public Offerings (IPOs)

When a company offers shares of its stock to the public for the first time, it does so through an initial public offering (or IPO). This offering involves working with an investment bank that helps price and market the newly issued shares. After the offering is complete, the new shares trade on a stock market for the first time. Given the risk that investment bankers face in trying to sell a new issue for which the true price is unknown, it is perhaps not surprising to find that, on average, the initial selling price is set too low and that the price increases dramatically on the first trading day. The percentage difference between the issue price and the closing price at the end of the first day of trading is often referred to as the degree of underpricing.

The evidence suggests that, on average, investors who are able to buy the shares of an IPO at their offering price may be able to earn abnormal profits. For example, during the internet bubble of 1995–2000, many IPOs ended their first day of trading up by more than 100 percent. Such performance, however, is not always the case. Sometimes the issues are priced too high, which means that share prices drop on their first day of trading. In addition, the evidence also suggests that investors buying after the

initial offering are not able to earn abnormal profits because prices adjust quickly to the “true” values, which supports semi-strong market efficiency. In fact, the subsequent long-term performance of IPOs is generally found to be below average. Taken together, the IPO underpricing and the subsequent poor performance suggests that the markets are overly optimistic initially (i.e., investors overreact).

Some researchers have examined closely why IPOs may appear to have anomalous returns. Because of the small size of the IPO companies and the method of equally weighting the samples, what appears to be an anomaly may simply be an artifact of the methodology.⁴⁴

7.4. Predictability of Returns Based on Prior Information

A number of researchers have documented that equity returns are related to prior information on such factors as interest rates, inflation rates, stock volatility, and dividend yields.⁴⁵ But finding that equity returns are affected by changes in economic fundamentals is not evidence of market inefficiency and would not result in abnormal trading returns.⁴⁶

Furthermore, the relationship between stock returns and the prior information is not consistent over time. For example, in one study, the relationship between stock prices and dividend yields changed from positive to negative in different periods.⁴⁷ Hence, a trading strategy based on dividend yields would not yield consistent abnormal returns.

7.5. Implications for Investment Strategies

Although it is interesting to consider the anomalies just described, attempting to benefit from them in practice is not easy. In fact, most researchers conclude that observed anomalies are not violations of market efficiency but, rather, are the result of statistical methodologies used to detect the anomalies. As a result, if the methodologies are corrected, most of these anomalies disappear.⁴⁸ Another point to consider is that in an efficient market, overreactions may occur, but then so do under-reactions.⁴⁹ Therefore, on average, the markets are efficient. In other words, investors face challenges when they attempt to translate statistical anomalies into economic profits. Consider the following quote regarding anomalies from the *Economist* (“Frontiers of Finance Survey,” 9 October 1993):

Many can be explained away. When transactions costs are taken into account, the fact that stock prices tend to over-react to news, falling back the day after good news and bouncing up the day after bad news, proves unexploitable: price reversals are always within the bid-ask spread. Others such as the small-firm effect, work for a few years and then fail for a few years. Others prove to be merely proxies for the reward for risk taking. Many have disappeared since (and because) attention has been drawn to them.

It is difficult to envision entrusting your retirement savings to a manager whose strategy is based on buying securities on Mondays, which tends to have negative returns on average, and selling them on Fridays. For one thing, the negative Monday returns are merely an average, so on any given week, they could be positive. In addition, such a strategy would generate large trading costs. Even more importantly, investors would likely be uncomfortable investing their funds in a strategy that has no compelling underlying economic rationale.

8. BEHAVIORAL FINANCE

LEARNING OUTCOME

The candidate should be able to:

□ describe behavioral finance and its potential relevance to understanding market anomalies

Behavioral finance examines investor behavior to understand how people make decisions, individually and collectively. Behavioral finance does not assume that people consider all available information in decision-making and act rationally by maximizing utility within budget constraints and updating expectations consistent with Bayes' formula. The resulting behaviors may affect what is observed in the financial markets.

In a broader sense, behavioral finance attempts to explain why individuals make the decisions that they do, whether these decisions are rational or irrational. The focus of much of the work in this area is on the behavioral biases that affect investment decisions. The behavior of individuals, in particular their behavioral biases, has been offered as a possible explanation for a number of pricing anomalies.

Most asset-pricing models assume that markets are rational and that the intrinsic value of a security reflects this rationality. But market efficiency and asset-pricing models do not require that each individual is rational—rather, only that the market is rational. If individuals deviate from rationality, other individuals are assumed to observe this deviation and respond accordingly. These responses move the market toward efficiency. If this does not occur in practice, it may be possible to explain some market anomalies referencing observed behaviors and behavioral biases.

8.1. Loss Aversion

In most financial models, the assumption is that investors are risk averse. **Risk aversion** refers to the tendency of people to dislike risk and to require higher expected returns to compensate for exposure to additional risk. Behavioral finance allows for the possibility that the dissatisfaction resulting from a loss exceeds the satisfaction resulting from a gain of the same magnitude. **Loss aversion** refers to the tendency of people to dislike losses more than they like comparable gains. This results in a strong preference for avoiding losses as opposed to achieving gains.⁵⁰ Some argue that behavioral theories of loss aversion can explain observed overreaction in markets. If loss aversion is more important than risk aversion, researchers should observe that investors overreact.⁵¹ Although loss aversion can explain the overreaction anomaly, evidence also suggests that under reaction is just as prevalent as overreaction, which counters these arguments.

8.2. Herding

Herding behavior has been advanced as a possible explanation of under reaction and overreaction in financial markets. **Herding** occurs when investors trade on the same side of the market in the same securities, or when investors ignore their own private information and/or analysis and act as other investors do. Herding is clustered trading that may or may not be based on information.⁵² Herding may result in under- or over-reaction to information depending upon the direction of the herd.

8.3. Overconfidence

A behavioral bias offered to explain pricing anomalies is overconfidence. If investors are overconfident, they overestimate their ability to process and interpret information about a security. Overconfident investors may not process information appropriately, and if there is a sufficient number of these investors, stocks will be mispriced.⁵³ But most researchers argue that this mispricing is temporary, with prices correcting eventually. If it takes a sufficiently long time for prices to become correctly priced and the mispricing is predictable, it may be possible for investors to earn abnormal profits.

Evidence has suggested that overconfidence results in mispricing for US, UK, German, French, and Japanese markets.⁵⁴ This overconfidence, however, is predominantly in higher-growth companies, whose prices react slowly to new information.⁵⁵

8.4. Information Cascades

An application of behavioral theories to markets and pricing focuses on the role of personal learning in markets. Personal learning is what investors learn by observing outcomes of trades and what they learn from “conversations”—ideas shared among investors about specific assets and the markets.⁵⁶ Social interaction and the resultant contagion is important in pricing and may explain such phenomena as price changes without accompanying news and mistakes in valuation.

Biases that investors possess can lead to herding behavior or information cascades. Herding and information cascades are related but not identical concepts. An **information cascade** is the transmission of information from those participants who act first and whose decisions influence the decisions of others. Those who are acting on the choices of others may be ignoring their own preferences in favor of imitating the choices of others. In particular, information cascades may occur with respect to the release of accounting information because accounting information may be difficult to interpret and may be noisy. For example, the release of earnings is difficult to interpret because it is necessary to understand how the number was arrived at and noisy because it is uncertain what the current earnings imply about future earnings.

Information cascades may result in serial correlation of stock returns, which is consistent with overreaction anomalies. Do information cascades result in correct pricing? Some argue that if a cascade is leading toward an incorrect value, this cascade is “fragile” and will be corrected because investors will ultimately give more weight to public information or the trading of a recognized informed trader.⁵⁷ Information cascades, although documented in markets, do not necessarily mean that investors can exploit knowledge of them as profitable trading opportunities.

Are information cascades rational? If the informed traders act first and uninformed traders imitate the informed traders, this behavior is consistent with rationality. The imitation trading by the uninformed traders may help the market incorporate relevant information and improve market efficiency.⁵⁸ However, the imitation trading may lead to an overreaction to information. The empirical evidence indicates that information cascades are greater for a stock when the information quality regarding the company is poor.⁵⁹ Information cascades may enhance the information available to investors.

8.5. Other Behavioral Biases

Other behavioral biases that have been put forth to explain observed investor behavior include the following:

- **representativeness**—investors assess new information and probabilities of outcomes based on similarity to the current state or to a familiar classification;
- **mental accounting**—investors keep track of the gains and losses for different investments in separate mental accounts and treat those accounts differently;
- **conservatism**—investors tend to be slow to react to new information and continue to maintain their prior views or forecasts; and
- **narrow framing**—investors focus on issues in isolation and respond to the issues based on how the issues are posed.⁶⁰

The basic idea behind behavioral finance is that investors are humans and, therefore, imperfect. These observed less than rational behaviors may help explain observed pricing anomalies. The beliefs investors have about a given asset’s value may not be homogeneous. But an issue, which is controversial, is whether these insights can help someone identify and exploit any mispricing. In other words, can investors use knowledge of behavioral biases to predict how asset prices will be affected and act based on the predictions to earn abnormal profits?

8.6. Behavioral Finance and Investors

Behavior biases can affect all market participants, from the novice investor to the most experienced investment manager. An understanding of behavioral finance can help market participants recognize their own and others' behavioral biases. As a result of this recognition, they may be able to respond and make improved decisions, individually and collectively.

8.7. Behavioral Finance and Efficient Markets

The use of behavioral finance to explain observed pricing is an important part of the understanding of how markets function and how prices are determined. Whether there is a behavioral explanation for market anomalies remains a debate. Pricing anomalies are continually being uncovered, and then statistical and behavioral explanations are offered to explain these anomalies.

On the one hand, if investors must be rational for efficient markets to exist, then all the imperfections of human investors suggest that markets cannot be efficient. On the other hand, if all that is required for markets to be efficient is that investors cannot consistently beat the market on a risk-adjusted basis, then the evidence does support market efficiency.

SUMMARY

This reading has provided an overview of the theory and evidence regarding market efficiency and has discussed the different forms of market efficiency as well as the implications for fundamental analysis, technical analysis, and portfolio management. The general conclusion drawn from the efficient market hypothesis is that it is not possible to beat the market on a consistent basis by generating returns in excess of those expected for the level of risk of the investment.

Additional key points include the following:

- The efficiency of a market is affected by the number of market participants and depth of analyst coverage, information availability, and limits to trading.
- There are three forms of efficient markets, each based on what is considered to be the information used in determining asset prices. In the weak form, asset prices fully reflect all market data, which refers to all past price and trading volume information. In the semi-strong form, asset prices reflect all publicly known and available information. In the strong form, asset prices fully reflect all information, which includes both public and private information.
- Intrinsic value refers to the true value of an asset, whereas market value refers to the price at which an asset can be bought or sold. When markets are efficient, the two should be the same or very close. But when markets are not efficient, the two can diverge significantly.
- Most empirical evidence supports the idea that securities markets in developed countries are semi-strong-form efficient; however, empirical evidence does not support the strong form of the efficient market hypothesis.
- A number of anomalies have been documented that contradict the notion of market efficiency, including the size anomaly, the January anomaly, and the winners–losers anomalies. In most cases, however, contradictory evidence both supports and refutes the anomaly.
- Behavioral finance uses human psychology, such as behavioral biases, in an attempt to explain investment decisions. Whereas behavioral finance is helpful in understanding observed decisions, a market can still be considered efficient even if market participants exhibit seemingly irrational behaviors, such as herding.

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PRACTICE PROBLEMS

1. If markets are efficient, the difference between the intrinsic value and market value of a company's security is:
 - A. negative.
 - B. zero.
 - C. positive.
2. The intrinsic value of an undervalued asset is:
 - A. less than the asset's market value.
 - B. greater than the asset's market value.
 - C. the value at which the asset can currently be bought or sold.
3. The market value of an undervalued asset is:
 - A. greater than the asset's intrinsic value.
 - B. the value at which the asset can currently be bought or sold.
 - C. equal to the present value of all the asset's expected cash flows.
4. In an efficient market, the change in a company's share price is *most likely* the result of:
 - A. insiders' private information.
 - B. the previous day's change in stock price.
 - C. new information coming into the market.
5. Regulation that restricts some investors from participating in a market will *most likely*:
 - A. impede market efficiency.
 - B. not affect market efficiency.
 - C. contribute to market efficiency.
6. With respect to efficient market theory, when a market allows short selling, the efficiency of the market is *most likely* to:
 - A. increase.
 - B. decrease.
 - C. remain the same.
7. Which of the following regulations will *most likely* contribute to market efficiency? Regulatory restrictions on:
 - A. short selling.
 - B. foreign traders.
 - C. insiders trading with nonpublic information.

8. Which of the following market regulations will *most likely* impede market efficiency?
- A. Restricting traders' ability to short sell.
 - B. Allowing unrestricted foreign investor trading.
 - C. Penalizing investors who trade with nonpublic information.
9. An increase in the time between when an order to trade a security is placed and when the order is executed *most likely* indicates that market efficiency has:
- A. decreased.
 - B. remained the same.
 - C. increased.
10. With respect to the efficient market hypothesis, if security prices reflect *only* past prices and trading volume information, then the market is:
- A. weak-form efficient.
 - B. strong-form efficient.
 - C. semi-strong-form efficient.
11. Which one of the following statements *best* describes the semi-strong form of market efficiency?
- A. Empirical tests examine the historical patterns in security prices.
 - B. Security prices reflect all publicly known and available information.
 - C. Semi-strong-form efficient markets are not necessarily weak-form efficient.
12. If prices reflect all public and private information, the market is *best* described as:
- A. weak-form efficient.
 - B. strong-form efficient.
 - C. semi-strong-form efficient.
13. If markets are semi-strong efficient, standard fundamental analysis will yield abnormal trading profits that are:
- A. negative.
 - B. equal to zero.
 - C. positive.
14. If markets are semi-strong-form efficient, then passive portfolio management strategies are *most likely* to:
- A. earn abnormal returns.
 - B. outperform active trading strategies.
 - C. underperform active trading strategies.
15. If a market is semi-strong-form efficient, the risk-adjusted returns of a passively managed portfolio relative to an actively managed portfolio are *most likely*:
- A. lower.
 - B. higher.

- C. the same.
16. Technical analysts assume that markets are:
- A. weak-form efficient.
 - B. weak-form inefficient.
 - C. semi-strong-form efficient.
17. Fundamental analysts assume that markets are:
- A. weak-form inefficient.
 - B. semi-strong-form efficient.
 - C. semi-strong-form inefficient.
18. If a market is weak-form efficient but semi-strong-form inefficient, then which of the following types of portfolio management is *most likely* to produce abnormal returns?
- A. Passive portfolio management.
 - B. Active portfolio management based on technical analysis.
 - C. Active portfolio management based on fundamental analysis.
19. Which of the following is *least likely* to explain the January effect anomaly?
- A. Tax-loss selling.
 - B. Release of new information in January.
 - C. Window dressing of portfolio holdings.
20. If a researcher conducting empirical tests of a trading strategy using time series of returns finds statistically significant abnormal returns, then the researcher has *most likely* found:
- A. a market anomaly.
 - B. evidence of market inefficiency.
 - C. a strategy to produce future abnormal returns.
21. Researchers have found that value stocks have consistently outperformed growth stocks. An investor wishing to exploit the value effect should purchase the stock of companies with above-average:
- A. dividend yields.
 - B. market-to-book ratios.
 - C. price-to-earnings ratios.
22. Which of the following market anomalies is inconsistent with weak-form market efficiency?
- A. Earnings surprise.
 - B. Momentum pattern.
 - C. Closed-end fund discount.
23. With respect to efficient markets, a company whose share price changes gradually after the public release of its annual report *most likely* indicates that the market where the company trades is:
- A. semi-strong-form efficient.

B. subject to behavioral biases.

C. receiving additional information about the company.

24. With respect to rational and irrational investment decisions, the efficient market hypothesis requires:

A. only that the market is rational.

B. that all investors make rational decisions.

C. that some investors make irrational decisions.

25. Observed overreactions in markets can be explained by an investor's degree of:

A. risk aversion.

B. loss aversion.

C. confidence in the market.

26. Like traditional finance models, the behavioral theory of loss aversion assumes that investors dislike risk; however, the dislike of risk in behavioral theory is assumed to be:

A. leptokurtic.

B. symmetrical.

C. asymmetrical.

SOLUTIONS

1. B is correct. A security's intrinsic value and market value should be equal when markets are efficient.
2. B is correct. The intrinsic value of an undervalued asset is greater than the market value of the asset, where the market value is the transaction price at which an asset can be currently bought or sold.
3. B is correct. The market value is the transaction price at which an asset can be currently bought or sold.
4. C is correct. Today's price change is independent of the one from yesterday, and in an efficient market, investors will react to new, independent information as it is made public.
5. A is correct. Reducing the number of market participants can accentuate market imperfections and impede market efficiency (e.g., restrictions on foreign investor trading).
6. A is correct. According to theory, reducing the restrictions on trading will allow for more arbitrage trading, thereby promoting more efficient pricing. Although regulators argue that short selling exaggerates downward price movements, empirical research indicates that short selling is helpful in price discovery.
7. C is correct. Regulation to restrict unfair use of nonpublic information encourages greater participation in the market, which increases market efficiency. Regulators (e.g., US SEC) discourage illegal insider trading by issuing penalties to violators of their insider trading rules.
8. A is correct. Restricting short selling will reduce arbitrage trading, which promotes market efficiency. Permitting foreign investor trading increases market participation, which makes markets more efficient. Penalizing insider trading encourages greater market participation, which increases market efficiency.
9. A is correct. Operating inefficiencies reduce market efficiency.
10. A is correct. The weak-form efficient market hypothesis is defined as a market where security prices fully reflect all market data, which refers to all past price and trading volume information.
11. B is correct. In semi-strong-form efficient markets, security prices reflect all publicly available information.
12. B is correct. The strong-form efficient market hypothesis assumes all information, public or private, has already been reflected in the prices.
13. B is correct. If all public information should already be reflected in the market price, then the abnormal trading profit will be equal to zero when fundamental analysis is used.
14. B is correct. Costs associated with active trading strategies would be difficult to recover; thus, such active trading strategies would have difficulty outperforming passive strategies on a consistent after-cost basis.
15. B is correct. In a semi-strong-form efficient market, passive portfolio strategies should outperform active portfolio strategies on a risk-adjusted basis.
16. B is correct. Technical analysts use past prices and volume to predict future prices, which is inconsistent with the weakest form of market efficiency (i.e., weak-form market efficiency). Weak-form market efficiency states that investors cannot earn abnormal returns by trading on the basis of past trends in price and volume.

17. C is correct. Fundamental analysts use publicly available information to estimate a security's intrinsic value to determine if the security is mispriced, which is inconsistent with the semi-strong form of market efficiency. Semi-strong-form market efficiency states that investors cannot earn abnormal returns by trading based on publicly available information.
18. C is correct. If markets are not semi-strong-form efficient, then fundamental analysts are able to use publicly available information to estimate a security's intrinsic value and identify misvalued securities. Technical analysis is not able to earn abnormal returns if markets are weak-form efficient. Passive portfolio managers outperform fundamental analysis if markets are semi-strong-form efficient.
19. B is correct. The excess returns in January are not attributed to any new information or news; however, research has found that part of the seasonal pattern can be explained by tax-loss selling and portfolio window dressing.
20. A is correct. Finding significant abnormal returns does not necessarily indicate that markets are inefficient or that abnormal returns can be realized by applying the strategy to future time periods. Abnormal returns are considered market anomalies because they may be the result of the model used to estimate the expected returns or may be the result of underestimating transaction costs or other expenses associated with implementing the strategy, rather than because of market inefficiency.
21. A is correct. Higher than average dividend yield is a characteristic of a value stock, along with low price-to-earnings and low market-to-book ratios. Growth stocks are characterized by low dividend yields and high price-to-earnings and high market-to-book ratios.
22. B is correct. Trading based on historical momentum indicates that price patterns exist and can be exploited by using historical price information. A momentum trading strategy that produces abnormal returns contradicts the weak form of the efficient market hypothesis, which states that investors cannot earn abnormal returns on the basis of past trends in prices.
23. C is correct. If markets are efficient, the information from the annual report is reflected in the stock prices; therefore, the gradual changes must be from the release of additional information.
24. A is correct. The efficient market hypothesis and asset-pricing models only require that the market is rational. Behavioral finance is used to explain *some* of the market anomalies as irrational decisions.
25. B is correct. Behavioral theories of loss aversion can explain observed overreaction in markets, such that investors dislike losses more than comparable gains (i.e., risk is not symmetrical).
26. C is correct. Behavioral theories of loss aversion allow for the possibility that the dislike for risk is not symmetrical, which allows for loss aversion to explain observed overreaction in markets such that investors dislike losses more than they like comparable gains.

NOTES

¹The CIO is Christopher J. Brightman, CFA, of the University of Virginia Investment Management Company, as reported in [Yau, Schneeweis, Robinson, and Weiss \(2007\)](#), pp. 481–482).

²This concept is known as allocative efficiency.

³This definition is convenient for making several instructional points. The definition that most simply explains the sense of the word *efficient* in this context can be found in [Fama \(1976\)](#): “An efficient capital market is a market that is efficient in processing information” (p. 134).

⁴The technical term for *superior* in this context is *positive abnormal* in the sense of higher than expected given the asset’s risk (as measured, according to capital market theory, by the asset’s contribution to the risk of a well-diversified portfolio).

⁵Although the original theory of market efficiency does not quantify this speed, the basic idea is that it is sufficiently swift to make it impossible to consistently earn abnormal profits. [Chordia, Roll, and Subrahmanyam \(2005\)](#) suggest that the adjustment to information on the New York Stock Exchange (NYSE) is between 5 and 60 minutes.

⁶Intrinsic value is often defined as the present value of all expected future cash flows of the asset.

⁷A “blue-chip” share is one from a well-recognized company that is considered to be high quality but low risk. This term generally refers to a company that has a long history of earnings and paying dividends.

⁸“The Investor’s Advocate: How the SEC Protects Investors, Maintains Market Integrity, and Facilitates Capital Formation,” US Securities and Exchange Commission (www.sec.gov/about/whatwedo.shtml).

⁹Regulation FD, “Selective Disclosure and Insider Trading,” 17 CFR Parts 240, 243, and 249, effective 23 October 2000.

¹⁰Although not the focus of this particular reading, it is important to note that a party is considered an insider not only when the individual is a corporate insider, such as an officer or director, but also when the individual is aware that the information is nonpublic information [Securities and Exchange Commission, Rules 10b5-1 (“Trading on the Basis of Material Nonpublic Information in Insider Trading Cases”) and Rule 10b5-2 “Duties of Trust or Confidence in Misappropriation Insider Trading Cases”]].

¹¹See the European Union’s Market Abuse Regulation (Regulation (EU) no. 596/2014 of the European Parliament and of the Council of 16 April 2014 on market abuse) and Directive for Criminal Sanctions for Market Abuse (Directive 2014/57/EU of the European Parliament and of the Council of 16 April 2014 on criminal sanctions for market abuse).

¹²See [Deng, Mortal, and Gupta \(2017\)](#) and references therein.”

¹³See [Bris, Goetzmann, and Zhu \(2009\)](#).

¹⁴ [Fama \(1970\)](#).

¹⁵Market efficiency should not be confused with the random walk hypothesis, in which price changes over time are independent of one another. A random walk model is one of many alternative expected return generating models. Market efficiency does not require that returns follow a random walk.

¹⁶Serial correlation is a statistical measure of the degree to which the returns in one period are related to the returns in another period.

¹⁷ [Bessembinder and Chan \(1998\)](#) and [Fifield, Power, and Sinclair \(2005\)](#).

¹⁸[Fifield, Power, and Sinclair \(2005\)](#), [Chen and Li \(2006\)](#), and [Mobarek, Mollah, and Bhuyan \(2008\)](#).

¹⁹See [Gan, Lee, Hwa, and Zhang \(2005\)](#) and [Raja, Sudhahar, and Selvam \(2009\)](#).

²⁰Evidence that finds that markets are not strong-form efficient include [Jaffe \(1974\)](#) and [Rozeff and Zaman \(1988\)](#).

²¹ [Brealey \(1983\)](#).

²²See [Malkiel \(1995\)](#). One of the challenges to evaluating mutual fund performance is that the researcher must control for survivorship bias.

²³There is also evidence of a January effect in bond returns that is more prevalent in high-yield corporate bonds, similar to the small-company effect for stocks.

²⁴See [Roll \(1983\)](#).

²⁵See, for example, [Kim \(2006\)](#).

²⁶For a discussion of several of these anomalous patterns, see [Jacobs and Levy \(1988\)](#).

²⁷The weekend effect consists of a pattern of returns around the weekend: abnormal positive returns on Fridays followed by abnormally negative returns on Mondays. This is a day-of-the-week effect that specifically links Friday and Monday returns. It is interesting to note that in 2009, the weekend effect in the United States was inverted, with 80 percent of the gains from March 2009 onward coming from the first trading day of the week.

²⁸[DeBondt and Thaler \(1985\)](#).

²⁹Notice that this pattern lies in sharp contrast to DeBondt and Thaler's reversal pattern that is displayed over longer periods of time. In theory, the two patterns could be related. In other words, it is feasible that prices are bid up extremely high, perhaps too high, in the short term for companies that are doing well. In the longer term (three-to-five years), the prices of these short-term winners correct themselves and they do poorly.

³⁰[Jegadeesh and Titman \(2001\)](#).

³¹[Johnson \(2002\)](#).

³²Although a large number of studies documents a small-company effect, these studies are concentrated in a period similar to that of the original research and, therefore, use a similar data set. The key to whether something is a true anomaly is persistence in out-of-sample tests. [Fama and French \(2008\)](#) document that the size effect is apparent only in microcap stocks but not in small- and large-cap stocks and these microcap stocks may have a significant influence in studies that document a size effect.

³³For example, see [Capaul, Rowley, and Sharpe \(1993\)](#) and [Fama and French \(1998\)](#).

³⁴[Fama and French \(1995\)](#).

³⁵See [Dimson and Minio-Kozerski \(1999\)](#) for a review of this literature.

³⁶See [Lee, Sheifer, and Thaler \(1990\)](#).

³⁷The return to owners of closed-end fund shares has three parts: 1) the price appreciation or depreciation of the shares themselves, 2) the dividends earned and distributed to owners by the fund, and 3) the capital gains and losses earned by the fund that are distributed by the fund. The explanation of the anomalous pricing has to do with the timing of the distribution of capital gains.

³⁸See, for example, the study by [Pontiff \(1996\)](#), which shows how the cost of arbitraging these discounts eliminates the profit.

³⁹See [Pontiff \(1995\)](#).

⁴⁰See [Jones, Rendleman, and Latané \(1984\)](#).

⁴¹Not surprisingly, it is often argued that this slow reaction contributes to a momentum pattern.

⁴²A similar pattern has been documented in the corporate bond market, where bond prices react too slowly to new company earnings announcements as well as to changes in company debt ratings.

⁴³See [Brown \(1997\)](#) for a summary of evidence supporting the existence of this anomaly. See [Zarowin \(1989\)](#) for evidence regarding the role of size in explaining abnormal returns to surprises; [Alexander, Goff, and Peterson \(1989\)](#) for evidence regarding transaction costs and unexpected earnings strategies; and [Kim and Kim \(2003\)](#) for evidence indicating that the anomalous returns can be explained by risk factors.

⁴⁴See [Brav, Geczy, and Gompers \(1995\)](#).

⁴⁵See, for example, [Fama and Schwert \(1977\)](#) and [Fama and French \(1988\)](#).

⁴⁶See [Fama and French \(2008\)](#).

⁴⁷[Schwert \(2003, Chapter 15\)](#).

⁴⁸[Fama \(1998\)](#).

⁴⁹This point is made by [Fama \(1998\)](#).

⁵⁰See [DeBondt and Thaler \(1985\)](#) and [Tversky and Kahneman \(1981\)](#).

⁵¹See [Fama \(1998\)](#).

⁵²The term used when there is herding without information is “spurious herding.”

⁵³Another aspect to overconfidence is that investors who are overconfident in their ability to select investments and manage a portfolio tend to use less diversification, investing in what is most familiar. Therefore, investor behavior may affect investment results—returns and risk—without implications for the efficiency of markets.

⁵⁴ [Scott, Stumpp, and Xu \(2003\)](#) and [Boujelbene Abbes, Boujelbene, and Bouri \(2009\)](#).

⁵⁵ [Scott, Stumpp, and Xu \(2003\)](#).

⁵⁶ [Hirshleifer and Teoh \(2009\)](#).

⁵⁷ [Avery and Zemsky \(1999\)](#).

⁵⁸Another alternative is that the uninformed traders are the majority of the market participants and the imitators are imitating not because they agree with the actions of the majority but because they are looking to act on the actions of the uninformed traders.

⁵⁹ [Avery and Zemsky \(1999\)](#) and [Bikhchandani, Hirshleifer, and Welch \(1992\)](#).

⁶⁰For a review of these behavioral issues, see [Hirshleifer \(2001\)](#).

Overview of Equity Securities

by Ryan C. Fuhrmann CFA and Asjeet S. Lamba PhD, CFA

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LEARNING OUTCOMES

The candidate should be able to:

- ☐ describe characteristics of types of equity securities
- ☐ describe differences in voting rights and other ownership characteristics among different equity classes
- ☐ compare and contrast public and private equity securities
- ☐ describe methods for investing in non-domestic equity securities
- ☐ compare the risk and return characteristics of different types of equity securities
- ☐ explain the role of equity securities in the financing of a company's assets
- ☐ contrast the market value and book value of equity securities
- ☐ compare a company's cost of equity, its (accounting) return on equity, and investors' required rates of return

1. IMPORTANCE OF EQUITY SECURITIES

Equity securities represent ownership claims on a company's net assets. As an asset class, equity plays a fundamental role in investment analysis and portfolio management because it represents a significant portion of many individual and institutional investment portfolios.

The study of equity securities is important for many reasons. First, the decision on how much of a client's portfolio to allocate to equities affects the risk and return characteristics of the entire portfolio. Second, different types of equity securities have different ownership claims on a company's net assets, which affect their risk and return characteristics in different ways. Finally, variations in the features of equity securities are reflected in their market prices, so it is important to understand the valuation implications of these features.

This reading provides an overview of equity securities and their different features and establishes the background required to analyze and value equity securities in a global context. It addresses the following questions:

- What distinguishes common shares from preference shares, and what purposes do these securities serve in financing a company's operations?
- What are convertible preference shares, and why are they often used to raise equity for unseasoned or highly risky companies?
- What are private equity securities, and how do they differ from public equity securities?
- What are depository receipts and their various types, and what is the rationale for investing in them?
- What are the risk factors involved in investing in equity securities?
- How do equity securities create company value?
- What is the relationship between a company's cost of equity, its return on equity, and investors' required rate of return?

The remainder of this reading is organized as follows. Section 2 provides an overview of global equity markets and their historical performance. Section 3 examines the different types and characteristics of equity securities, and Section 4 outlines the differences between public and private equity securities. Section 5 provides an overview of the various types of equity securities listed and traded in global markets. Section 6 discusses the risk and return characteristics of equity securities. Section 7 examines the role of equity securities in creating company value and the relationship between a company's cost of equity, its return on equity, and investors' required rate of return. The final section summarizes the reading.

1.1. Equity Securities in Global Financial Markets

This section highlights the relative importance and performance of equity securities as an asset class. We examine the total market capitalization and trading volume of global equity markets and the prevalence of equity ownership across various geographic regions. We also examine historical returns on equities and compare them to the returns on government bonds and bills.

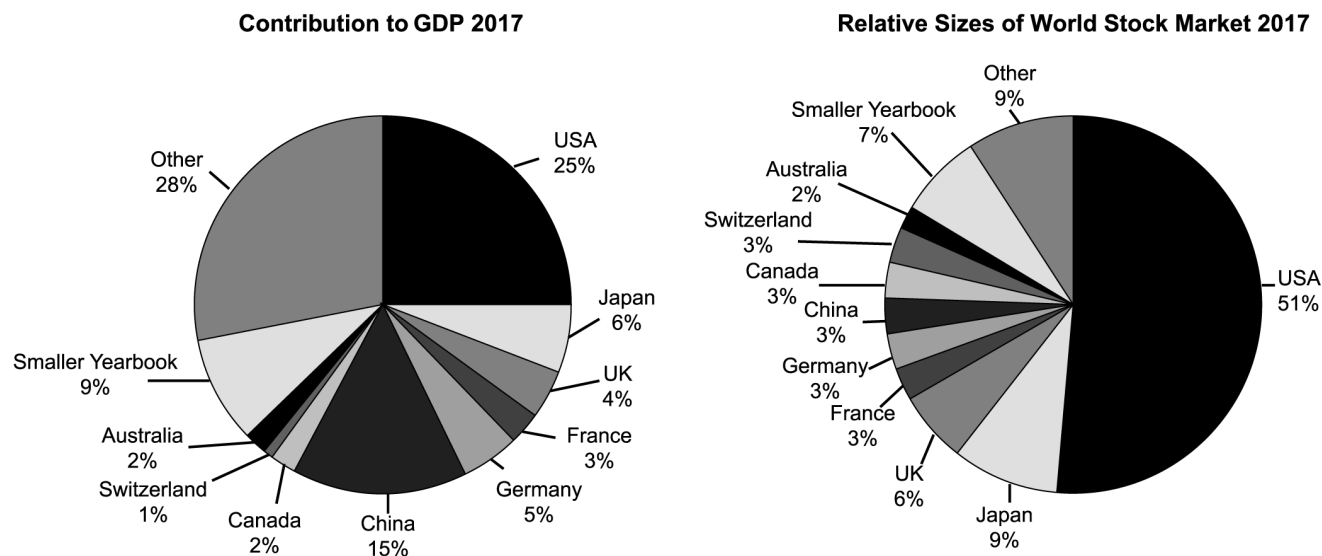
[Exhibit 1](#) summarizes the contributions of selected countries and geographic regions to global gross domestic product (GDP) and global equity market capitalization. Analysts may examine the relationship between equity market capitalization and GDP as a rough indicator of whether the global equity market (or a specific country's or region's equity market) is under, over, or fairly valued, particularly compared to its long-run average.

[Exhibit 1](#) illustrates the significant value that investors attach to publicly traded equities relative to the sum of goods and services produced globally every year. It shows the continued significance, and the

potential over-representation, of US equity markets relative to their contribution to global GDP. That is, while US equity markets contribute around 51 percent to the total capitalization of global equity markets, their contribution to the global GDP is only around 25 percent. Following the stock market turmoil in 2008, however, the market capitalization to GDP ratio of the United States fell to 59 percent, which is significantly lower than its long-run average of 79 percent.

As equity markets outside the United States develop and become increasingly global, their total capitalization levels are expected to grow closer to their respective world GDP contributions. Therefore, it is important to understand and analyze equity securities from a global perspective.

Exhibit 1: Country and Regional Contributions to Global GDP and Equity Market Capitalization (2017)



Source: The WorldBank Databank 2017, and [Dimson, Marsh, and Staunton \(2018\)](#).

[Exhibit 2](#) lists the top 10 equity markets at the end of 2017 based on total market capitalization (in billions of US dollars), trading volume, and the number of listed companies.¹ Note that the rankings differ based on the criteria used. For example, the top three markets based on total market capitalization are the NYSE Euronext (US), NASDAQ OMX, and the Japan Exchange Group; however, the top three markets based on total US dollar trading volume are the Nasdaq OMX, NYSE Euronext (US), and the Shenzhen Stock Exchange, respectively.²

Exhibit 2: Equity Markets Ranked by Total Market Capitalization at the End of 2017 (Billions of US Dollars)

Rank	Name of Market	Total US Dollar Market Capitalization	Total US Dollar Trading Volume	Number of Listed Companies
1	NYSE Euronext (US)	\$22,081.4	\$16,140.1	2,286
2	NASDAQ OMX	\$10,039.4	\$33,407.1	2,949
3	Japan Exchange Group ^a	\$6,220.0	\$6,612.1	3,604
4	Shanghai Stock Exchange	\$5,084.4	\$7,589.3	1,396
5	Euronext ^b	\$4,393.0	\$1,981.6	1,255
6	Hong Kong Exchanges	\$4,350.5	\$1,958.8	2,118
7	Shenzhen Stock Exchanges	\$3,617.9	\$9,219.7	2,089

Rank	Name of Market	Total US Dollar Market Capitalization	Total US Dollar Trading Volume	Number of Listed Companies
8	National Stock Exchange of India	\$2,351.5	\$1,013.3	1,897
9	BSE Limited ^c	\$2,331.6	\$183.0	5,616
10	Deutsche Börse	\$2,262.2	\$1,497.9	499

Notes:

^a Japan Exchange Group is the merged entity containing the Tokyo Stock Exchange and Osaka Securities Exchange.

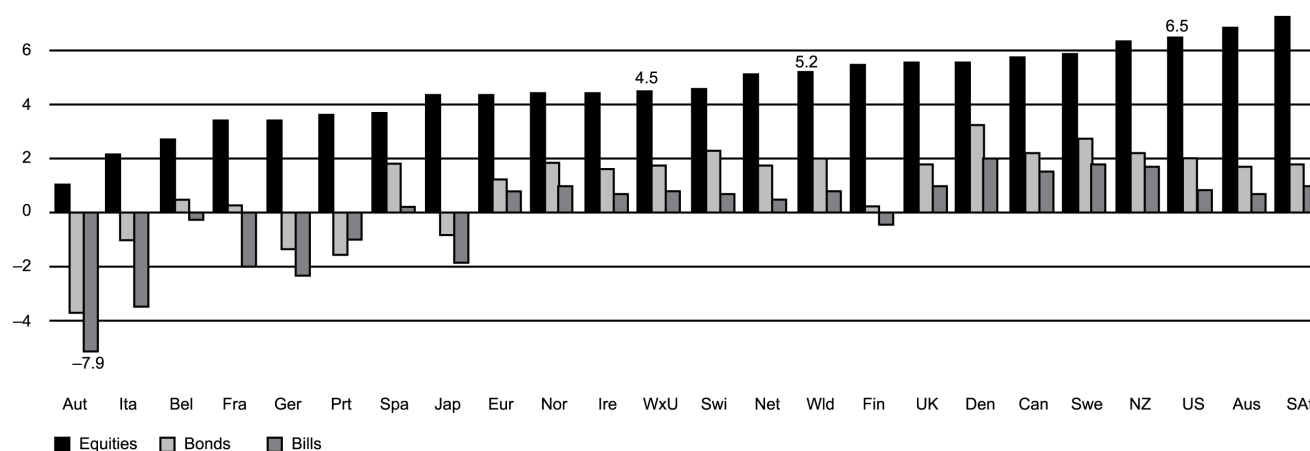
^b From 2001, includes Netherlands, France, England, Belgium, and Portugal.

^c Bombay Stock Exchange.

Source: Adapted from the *World Federation of Exchanges 2017 Report* (see <http://www.world-exchanges.org>). Note that market capitalization by company is calculated by multiplying its stock price by the number of shares outstanding. The market's overall capitalization is the aggregate of the market capitalizations of all companies traded on that market. The number of listed companies includes both domestic and foreign companies whose shares trade on these markets.

Exhibit 3 compares the *real* (or inflation-adjusted) compounded returns on government bonds, government bills, and equity securities in 21 countries plus the world index (“Wld”), the world ex-US (“WxU”), and Europe (“Eur”) during the 118 years 1900–2017.³ In real terms, government bonds and bills have essentially kept pace with the inflation rate, earning annualized real returns of less than 2 percent in most countries.⁴ By comparison, real returns in equity markets have generally been around 3.5 percent per year in most markets—with a world average return of around 5.2 percent and a world average return excluding the United States just under 5 percent. During this period, South Africa and Australia were the best performing markets followed by the United States, New Zealand, and Sweden.

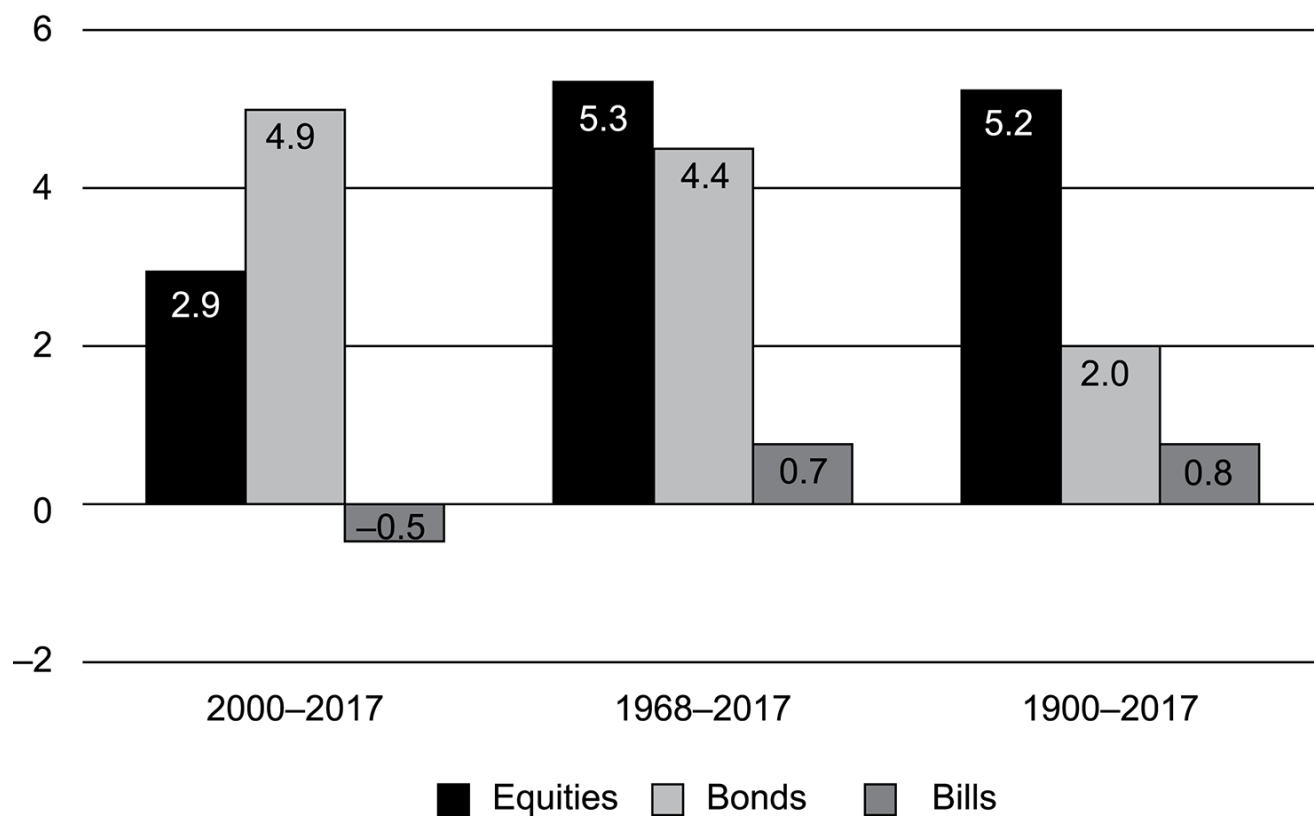
Exhibit 3: Real Returns on Global Equity Securities, Bonds, and Bills During 1900–2017



Source: Dimson, Marsh, and Staunton (2018).

Exhibit 4 shows the annualized real returns on major asset classes for the world index over 1900–2017.

Exhibit 4: Annualized Real Returns on Asset Classes for the World Index, 1900–2017

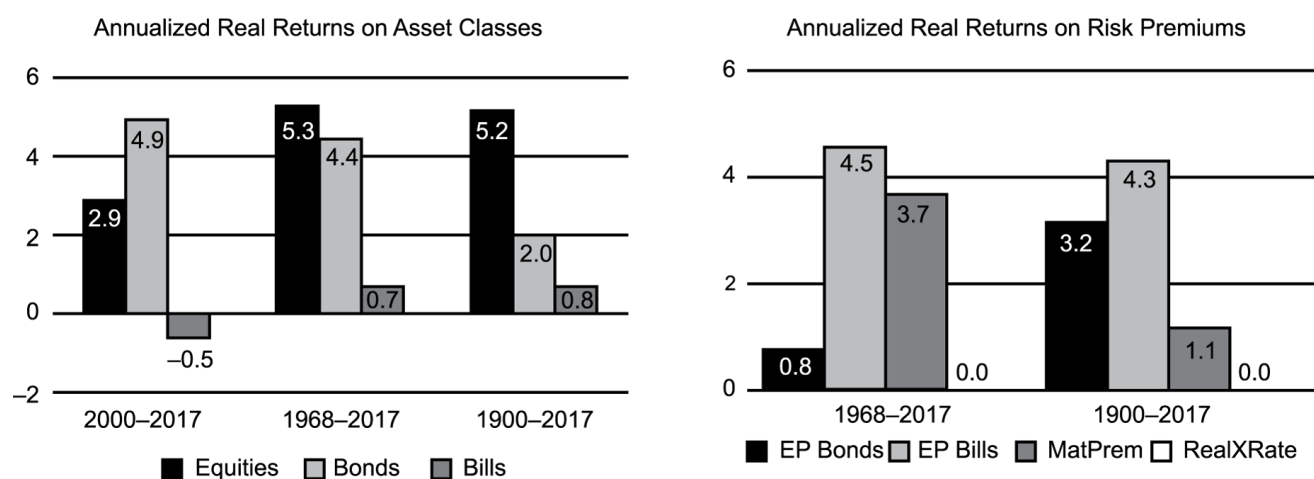


Source: [Dimson, Marsh, and Staunton \(2018\)](#).

The volatility in asset market returns is further highlighted in [Exhibit 5](#), which shows the annualized risk premia for equity relative to bonds (EP Bonds), and equity relative to treasury bills (EP Bills). Maturity premium for government bond returns relative to treasury bill returns (Mat Prem) is also shown.

These observations and historical data are consistent with the concept that the return on securities is directly related to risk level. That is, equity securities have higher risk levels when compared with government bonds and bills, they earn higher rates of return to compensate investors for these higher risk levels, and they also tend to be more volatile over time.

Exhibit 5: Annualized Real Returns on Asset Classes and Risk Premiums for the World Index since 1900-2017



Notes: Equities are total returns, including reinvested dividend income. Bonds are total return, including reinvested coupons, on long-term government bonds. Bills denotes the total return, including any income, from Treasury bills. All returns are adjusted for inflation and are expressed as geometric mean returns. EP bonds denotes the equity risk premium relative to long-term government bonds. EP Bills denotes the equity premium relative to Treasury bills. MatPrem denotes the maturity premium for government bond returns relative to bill returns. RealXRate denotes the real (inflation-adjusted) change in the exchange rate against the US dollar.

Source: [Dimson, Marsh, and Staunton \(2018\)](#).

Given the high risk levels associated with equity securities, it is reasonable to expect that investors' tolerance for risk will tend to differ across equity markets. This is illustrated in [Exhibit 6](#), which shows the results of a series of studies conducted by the Australian Securities Exchange on international differences in equity ownership. During the 2004–2014 period, equity ownership as a percentage of the population was lowest in South Korea (averaging 9.0 percent), followed by Germany (14.5 percent) and Sweden (17.7 percent). In contrast, Australia and New Zealand had the highest equity ownership as a percentage of the population (averaging more than 20 percent). In addition, there has been a relative decline in share ownership in several countries over recent years, which is not surprising given the recent overall uncertainty in global economies and the volatility in equity markets that this uncertainty has created.

Exhibit 6: International Comparisons of Stock Ownership: 2004–2014⁵

	2004	2006	2008	2010	2012	2014
Australia – Direct/Indirect	55%	46%	41%	43%	38%	36%
South Korea – Shares	8	7	10	10	10	N/A
Germany – Shares/Funds	16	16	14	13	15	13
Sweden – Shares	22	20	18	17	15	14
United Kingdom – Shares/Funds	22	20	18	N/A	17	N/A
New Zealand – Direct	23	26	N/A	22	23	26

Source: Adapted from the 2014 *Australian Share Ownership Study* conducted by the Australian Securities Exchange (see <http://www.asx.com.au>). For Australia and the United States, the data pertain to direct and indirect ownership in equity markets; for other countries, the data pertain to direct ownership in shares and share funds. Data not available in specific years are shown as “N/A.”

An important implication from the above discussion is that equity securities represent a key asset class for global investors because of their unique return and risk characteristics. We next examine the various types of equity securities traded on global markets and their salient characteristics.

2. CHARACTERISTICS OF EQUITY SECURITIES

LEARNING OUTCOMES

The candidate should be able to:

- ☐ describe characteristics of types of equity securities
- ☐ describe differences in voting rights and other ownership characteristics among different equity classes

Companies finance their operations by issuing either debt or equity securities. A key difference between these securities is that debt is a liability of the issuing company, whereas equity is not. This means that when a company issues debt, it is contractually obligated to repay the amount it borrows (i.e., the principal or face value of the debt) at a specified future date. The cost of using these funds is called interest, which the company is contractually obligated to pay until the debt matures or is retired.

When the company issues equity securities, it is not contractually obligated to repay the amount it receives from shareholders, nor is it contractually obligated to make periodic payments to shareholders for the use of their funds. Instead, shareholders have a claim on the company's assets after all liabilities have been paid. Because of this residual claim, equity shareholders are considered to be owners of the company. Investors who purchase equity securities are seeking total return (i.e., capital or price

appreciation and dividend income), whereas investors who purchase debt securities (and hold until maturity) are seeking interest income. As a result, equity investors expect the company's management to act in their best interest by making operating decisions that will maximize the market price of their shares (i.e., shareholder wealth).

In addition to common shares (also known as ordinary shares or common stock), companies may also issue preference shares (also known as preferred stock), the other type of equity security. The following sections discuss the different types and characteristics of common and preference securities.

2.1. Common Shares

Common shares represent an ownership interest in a company and are the predominant type of equity security. As a result, investors share in the operating performance of the company, participate in the governance process through voting rights, and have a claim on the company's net assets in the case of liquidation. Companies may choose to pay out some, or all, of their net income in the form of cash dividends to common shareholders, but they are not contractually obligated to do so.⁶

Voting rights provide shareholders with the opportunity to participate in major corporate governance decisions, including the election of its board of directors, the decision to merge with or take over another company, and the selection of outside auditors. Shareholder voting generally takes place during a company's annual meeting. As a result of geographic limitations and the large number of shareholders, it is often not feasible for shareholders to attend the annual meeting in person. For this reason, shareholders may **vote by proxy**, which allows a designated party—such as another shareholder, a shareholder representative, or management—to vote on the shareholders' behalf.

Regular shareholder voting, where each share represents one vote, is referred to as **statutory voting**. Although it is the common method of voting, it is not always the most appropriate one to use to elect a board of directors. To better serve shareholders who own a small number of shares, **cumulative voting** is often used. Cumulative voting allows shareholders to direct their total voting rights to specific candidates, as opposed to having to allocate their voting rights evenly among all candidates. Total voting rights are based on the number of shares owned multiplied by the number of board directors being elected. For example, under cumulative voting, if four board directors are to be elected, a shareholder who owns 100 shares is entitled to 400 votes and can either cast all 400 votes in favor of a single candidate or spread them across the candidates in any proportion. In contrast, under statutory voting, a shareholder would be able to cast only a maximum of 100 votes for each candidate.

The key benefit to cumulative voting is that it allows shareholders with a small number of shares to apply all of their votes to one candidate, thus providing the opportunity for a higher level of representation on the board than would be allowed under statutory voting.

Exhibit 7 describes the rights of Viacom Corporation's shareholders. In this case, a dual-share arrangement allows the founding chairman and his family to control more than 70 percent of the voting rights through the ownership of Class A shares. This arrangement gives them the ability to exert control over the board of director election process, corporate decision making, and other important aspects of managing the company. A cumulative voting arrangement for any minority shareholders of Class A shares would improve their board representation.

Exhibit 7: Share Class Arrangements at Viacom Corporation⁷

Viacom has two classes of common stock: Class A, which is the voting stock, and Class B, which is the non-voting stock. There is no difference between the two classes except for voting rights; they generally trade within a close price range of each other. There are, however, far more shares of Class B outstanding, so most of the trading occurs in that class.

- **Voting Rights**—Holders of Class A common stock are entitled to one vote per share. Holders of Class B common stock do not have any voting rights, except as required by Delaware law. Generally, all matters to be voted on by Viacom stockholders must be approved by a majority of the aggregate voting power of the shares of Class A common stock present in person or represented by proxy, except as required by Delaware law.

- **Dividends**—Stockholders of Class A common stock and Class B common stock will share ratably in any cash dividend declared by the Board of Directors, subject to any preferential rights of any outstanding preferred stock. Viacom does not currently pay a cash dividend, and any decision to pay a cash dividend in the future will be at the discretion of the Board of Directors and will depend on many factors.
- **Conversion**—So long as there are 5,000 shares of Class A common stock outstanding, each share of Class A common stock will be convertible at the option of the holder of such share into one share of Class B common stock.
- **Liquidation Rights**—In the event of liquidation, dissolution, or winding-up of Viacom, all stockholders of common stock, regardless of class, will be entitled to share ratably in any assets available for distributions to stockholders of shares of Viacom common stock subject to the preferential rights of any outstanding preferred stock.
- **Split, Subdivision, or Combination**—In the event of a split, subdivision, or combination of the outstanding shares of Class A common stock or Class B common stock, the outstanding shares of the other class of common stock will be divided proportionally.
- **Preemptive Rights**—Shares of Class A common stock and Class B common stock do not entitle a stockholder to any preemptive rights enabling a stockholder to subscribe for or receive shares of stock of any class or any other securities convertible into shares of stock of any class of Viacom.

As seen in [Exhibit 7](#), companies can issue different classes of common shares (Class A and Class B shares), with each class offering different ownership rights.⁸ For example, as shown in [Exhibit 8](#), the Ford Motor Company has Class A shares (“Common Stock”), which are owned by the investing public. It also has Class B shares, which are owned only by the Ford family. The exhibit contains an excerpt from Ford’s *2017 Annual Report* (p. 144). Class A shareholders have 60 percent voting rights, whereas Class B shareholders have 40 percent. In the case of liquidation, however, Class B shareholders will not only receive the first US\$0.50 per share that is available for distribution (as will Class A shareholders), but they will also receive the next US\$1.00 per share that is available for distribution before Class A shareholders receive anything else. Thus, Class B shareholders have an opportunity to receive a larger proportion of distributions upon liquidation than do Class A shareholders.⁹

Exhibit 8: Share Class Arrangements at Ford Motor Company¹⁰

NOTE 21. CAPITAL STOCK AND AMOUNTS PER SHARE

All general voting power is vested in the holders of Common Stock and Class B Stock. Holders of our Common Stock have 60% of the general voting power and holders of our Class B Stock are entitled to such number of votes per share as will give them the remaining 40%. Shares of Common Stock and Class B Stock share equally in dividends when and as paid, with stock dividends payable in shares of stock of the class held.

If liquidated, each share of Common Stock is entitled to the first \$0.50 available for distribution to holders of Common Stock and Class B Stock, each share of Class B Stock is entitled to the next \$1.00 so available, each share of Common Stock is entitled to the next \$0.50 so available, and each share of Common and Class B Stock is entitled to an equal amount thereafter.

2.2. Preference Shares

Preference shares (or preferred stock) rank above common shares with respect to the payment of dividends and the distribution of the company’s net assets upon liquidation.¹¹ However, preference

shareholders generally do not share in the operating performance of the company and do not have any voting rights, unless explicitly allowed for at issuance. Preference shares have characteristics of both debt securities and common shares. Similar to the interest payments on debt securities, the dividends on preference shares are fixed and are generally higher than the dividends on common shares. However, unlike interest payments, preference dividends are not contractual obligations of the company. Similar to common shares, preference shares can be perpetual (i.e., no fixed maturity date), can pay dividends indefinitely, and can be callable or puttable.

Exhibit 9 provides an example of callable preference shares issued by the GDL Fund to raise capital to redeem the remaining outstanding Series B Preferred shares. In this case, the purchaser of the shares will receive an ongoing dividend from the GDL Fund. If the GDL Fund chooses to buy back the shares, it must do so at the \$50 a share liquidation preference price. The purchasers of the shares also have the right to put back the shares to GDL at the \$50 a share price.

Exhibit 9: Callable Stock offering by the GDL Fund¹²

RYE, NY—March 26, 2018—The GDL Fund (NYSE:GDL) (the “Fund”) is pleased to announce the completion of a rights offering (the “Offering”) in which the Fund issued 2,624,025 Series C Cumulative Puttable and Callable Preferred Shares (the “Series C Preferred”), totaling \$131,201,250. Pursuant to the Offering, the Fund issued one non-transferable right (a “Right”) for each outstanding Series B Cumulative Puttable and Callable Preferred Share (the “Series B Preferred”) of the Fund to Series B Preferred shareholders of record as of February 14, 2018. Holders of Rights were entitled to purchase the Series C Preferred with any combination of cash or surrender of the Series B Preferred at liquidation preference. Therefore, one Right plus \$50.00, or one Right plus one share of Series B Preferred with a liquidation value of \$50.00 per share, was required to purchase each share of the Series C Preferred. The Offering expired at 5:00 PM Eastern Time on March 20, 2018.

Dividends on preference shares can be cumulative, non-cumulative, participating, non-participating, or some combination thereof (i.e., cumulative participating, cumulative non-participating, non-cumulative participating, non-cumulative non-participating).

Dividends on **cumulative preference shares** accrue so that if the company decides not to pay a dividend in one or more periods, the unpaid dividends accrue and must be paid in full before dividends on common shares can be paid. In contrast, **non-cumulative preference shares** have no such provision. This means that any dividends that are not paid in the current or subsequent periods are forfeited permanently and are not accrued over time to be paid at a later date. However, the company is still not permitted to pay any dividends to common shareholders in the current period unless preferred dividends have been paid first.

Participating preference shares entitle the shareholders to receive the standard preferred dividend plus the opportunity to receive an additional dividend if the company’s profits exceed a pre-specified level. In addition, participating preference shares can also contain provisions that entitle shareholders to an additional distribution of the company’s assets upon liquidation, above the par (or face) value of the preference shares. **Non-participating preference shares** do not allow shareholders to share in the profits of the company. Instead, shareholders are entitled to receive only a fixed dividend payment and the par value of the shares in the event of liquidation. The use of participating preference shares is much more common for smaller, riskier companies where the possibility of future liquidation is more of a concern to investors.

Preference shares can also be convertible. **Convertible preference shares** entitle shareholders to convert their shares into a specified number of common shares. This conversion ratio is determined at issuance. Convertible preference shares have the following advantages:

- They allow investors to earn a higher dividend than if they invested in the company’s common shares.
- They allow investors the opportunity to share in the profits of the company.

- They allow investors to benefit from a rise in the price of the common shares through the conversion option.
- Their price is less volatile than the underlying common shares because the dividend payments are known and more stable.

As a result, the use of convertible preference shares is a popular financing option in venture capital and private equity transactions in which the issuing companies are considered to be of higher risk and when it may be years before the issuing company “goes public” (i.e., issues common shares to the public).

[Exhibit 10](#) provides examples of the types and characteristics of preference shares as issued by Tsakos Energy Navigation Ltd (TNP.PRE).

Exhibit 10: Examples of Preference Shares Issued by TEN Ltd¹³

Athens, Greece, June 21, 2018—TEN Ltd. (“TEN”) (NYSE: TNP), a leading diversified crude, product and LNG tanker operator, today announced the pricing of its public offering of its Series F Fixed-to-Floating Rate Cumulative Redeemable Perpetual Preferred Shares, par value \$1.00 per share, liquidation preference \$25.00 per share (“Series F Preferred Shares”). TEN will issue 5,400,000 Series F Preferred Shares at a price to the public of \$25.00 per share. Dividends will be payable on the Series F Preferred Shares to July 30, 2028 at a fixed rate equal to 9.50% per annum and from July 30, 2028, if not redeemed, at a floating rate. In connection with the offering, TEN has granted the underwriters a 30-day option to purchase 810,000 additional Series F Preferred Shares, which, if exercised in full, would result in total gross proceeds of \$155,250,000. TEN intends to use the net proceeds from the offering for general corporate purposes, which may include making vessel acquisitions and/or strategic investments and preferred share redemptions. Following the offering, TEN intends to file an application to list the Series F Preferred Shares on the New York Stock Exchange. The offering is expected to close on or about June 28, 2018.

3. PRIVATE VERSUS PUBLIC EQUITY SECURITIES

LEARNING OUTCOME

The candidate should be able to:

- **compare and contrast public and private equity securities**

Our discussion so far has focused on equity securities that are issued and traded in public markets and on exchanges. Equity securities can also be issued and traded in private equity markets. **Private equity securities** are issued primarily to institutional investors via non-public offerings, such as private placements. Because they are not listed on public exchanges, there is no active secondary market for these securities. As a result, private equity securities do not have “market determined” quoted prices, are highly illiquid, and require negotiations between investors in order to be traded. In addition, financial statements and other important information needed to determine the fair value of private equity securities may be difficult to obtain because the issuing companies are typically not required by regulatory authorities to publish this information.

There are three primary types of private equity investments: venture capital, leveraged buyouts, and private investment in public equity (or PIPE). **Venture capital** investments provide “seed” or start-up capital, early-stage financing, or mezzanine financing to companies that are in the early stages of development and require additional capital for expansion. These funds are then used to finance the

company's product development and growth. Venture capitalists range from family and friends to wealthy individuals and private equity funds. Because the equity securities issued to venture capitalists are not publicly traded, they generally require a commitment of funds for a relatively long period of time; the opportunity to "exit" the investment is typically within 3 to 10 years from the initial start-up. The exit return earned by these private equity investors is based on the price that the securities can be sold for if and when the start-up company first goes public, either via an **initial public offering** (IPO) on the stock market or by being sold to other investors.

A **leveraged buyout** (LBO) occurs when a group of investors (such as the company's management or a private equity partnership) uses a large amount of debt to purchase all of the outstanding common shares of a publicly traded company. In cases where the group of investors acquiring the company is primarily comprised of the company's existing management, the transaction is referred to as a **management buyout** (MBO). After the shares are purchased, they cease to trade on an exchange and the investor group takes full control of the company. In other words, the company is taken "private" or has been privatized. Companies that are candidates for these types of transactions generally have large amounts of undervalued assets (which can be sold to reduce debt) and generate high levels of cash flows (which are used to make interest and principal payments on the debt). The ultimate objective of a buyout (LBO or MBO) is to restructure the acquired company and later take it "public" again by issuing new shares to the public in the primary market.

The third type of private investment is a **private investment in public equity**, or PIPE.¹⁴ This type of investment is generally sought by a public company that is in need of additional capital quickly and is willing to sell a sizeable ownership position to a private investor or investor group. For example, a company may require a large investment of new equity funds in a short period of time because it has significant expansion opportunities, is facing high levels of indebtedness, or is experiencing a rapid deterioration in its operations. Depending on how urgent the need is and the size of the capital requirement, the private investor may be able to purchase shares in the company at a significant discount to the publicly-quoted market price. **Exhibit 11** contains a recent PIPE transaction for the health care company TapImmune, which also included the proposed merger with Marker Therapeutics.

Exhibit 11: Example of a PIPE Transaction¹⁵

JACKSONVILLE, Florida, June 8, 2018—TapImmune Inc. (NASDAQ: TPIV), a clinical-stage immuno-oncology company, today announced that it has entered into security purchase agreements with certain institutional and accredited investors in connection with a private placement of its equity securities. The private placement will be led by New Enterprise Associates (NEA) with participation from Aisling Capital and Perceptive Advisors, among other new and existing investors. The private placement is expected to be completed concurrently with the closing of the proposed merger between TapImmune Inc. and Marker Therapeutics, Inc., which was previously announced on May 15, 2018.

Upon closing the private placement, TapImmune will issue 17,500,000 shares of its common stock at a price of \$4.00 per share. The aggregate offering size, before deducting the placement agent fees and other offering expenses, is expected to be \$70 million. Additionally, TapImmune will issue warrants to purchase 13,125,000 shares of TapImmune common stock at an exercise price of \$5.00 per share that will be exercisable for a period of five years from the date of issuance. The closing of the transaction, which is subject to the closing of the merger with Marker, the approval by TapImmune's stockholders as required by NASDAQ Stock Market Rules, and other customary closing conditions, is anticipated to occur by the end of the third quarter of 2018.

While the global private equity market is relatively small in comparison to the global public equity market, it has experienced considerable growth over the past three decades. According to a study of the private equity market sponsored by the *World Economic Forum* and spanning the period 1970–2007, approximately US\$3.6 trillion in debt and equity were acquired in leveraged buyouts. Of this amount, approximately 75 percent or US\$2.7 trillion worth of transactions occurred during 2001–2007.¹⁶ This pace continued with a further US\$2.9 trillion in transactions occurring during 2008–2017.¹⁷ While the US and the UK markets were the focus of most private equity investments during the 1980s and 1990s,

private equity investments outside of these markets have grown substantially in recent years. In addition, the number of companies operating under private equity ownership has also grown. For example, during the mid-1990s, fewer than 2,000 companies were under LBO ownership compared to more than 20,000 companies that were under LBO ownership globally at the beginning of 2017. The holding period for private equity investments has also increased during this time period from 3 to 5 years (1980s and 1990s) to approximately 10 years.¹⁸

The move to longer holding periods has given private equity investors the opportunity to more effectively and patiently address any underlying operational issues facing the company and to better manage it for long-term value creation. Because of the longer holding periods, more private equity firms are issuing convertible preference shares because they provide investors with greater total return potential through their dividend payments and the ability to convert their shares into common shares during an IPO.

In operating a publicly traded company, management often feels pressured to focus on short-term results¹⁹ (e.g., meeting quarterly sales and earnings targets from analysts biased toward near-term price performance) instead of operating the company to obtain long-term sustainable revenue and earnings growth. By “going private,” management can adopt a more long-term focus and can eliminate certain costs that are necessary to operate a publicly traded company—such as the cost of meeting regulatory and stock exchange filing requirements, the cost of maintaining investor relations departments to communicate with shareholders and the media, and the cost of holding quarterly analyst conference calls.

As described above, public equity markets are much larger than private equity networks and allow companies more opportunities to raise capital that is subsequently actively traded in secondary markets. By operating under public scrutiny, companies are incentivized to be more open in terms of corporate governance and executive compensation to ensure that they are acting for the benefit of shareholders. In fact, some studies have shown that private equity firms score lower in terms of corporate governance effectiveness, which may be attributed to the fact that shareholders, analysts, and other stakeholders are able to influence management when corporate governance and other policies are public.

4. NON-DOMESTIC EQUITY SECURITIES

LEARNING OUTCOME

The candidate should be able to:

☐ **describe methods for investing in non-domestic equity securities**

Technological innovations and the growth of electronic information exchanges (electronic trading networks, the internet, etc.) have accelerated the integration and growth of global financial markets. As detailed previously, global capital markets have expanded at a much more rapid rate than global GDP in recent years; both primary and secondary international markets have benefited from the enhanced ability to rapidly and openly exchange information. Increased integration of equity markets has made it easier and less expensive for companies to raise capital and to expand their shareholder base beyond their local market. Integration has also made it easier for investors to invest in companies that are located outside of their domestic markets. This has enabled investors to further diversify and improve the risk and return characteristics of their portfolios by adding a class of assets with lower correlations to local country assets.

One barrier to investing globally is that many countries still impose “foreign restrictions” on individuals and companies from other countries that want to invest in their domestic companies. There are three primary reasons for these restrictions. The first is to limit the amount of control that foreign investors can exert on domestic companies. For example, some countries prevent foreign investors from acquiring a majority interest in domestic companies. The second is to give domestic investors the opportunity to own shares in the foreign companies that are conducting business in their country. For example, the Swedish home furnishings retailer IKEA abandoned efforts to invest in parts of the Asia/Pacific region because local governments did not want IKEA to maintain complete ownership of its stores. The third

reason is to reduce the volatility of capital flows into and out of domestic equity markets. For example, one of the main consequences of the Asian Financial Crisis in 1997–98 was the large outflow of capital from such emerging market countries as Thailand, Indonesia, and South Korea. These outflows led to dramatic declines in the equity markets of these countries and significant currency devaluations and resulted in many governments placing restrictions on capital flows. Today, many of these same markets have built up currency reserves to better withstand capital outflows inherent in economic contractions and periods of financial market turmoil.

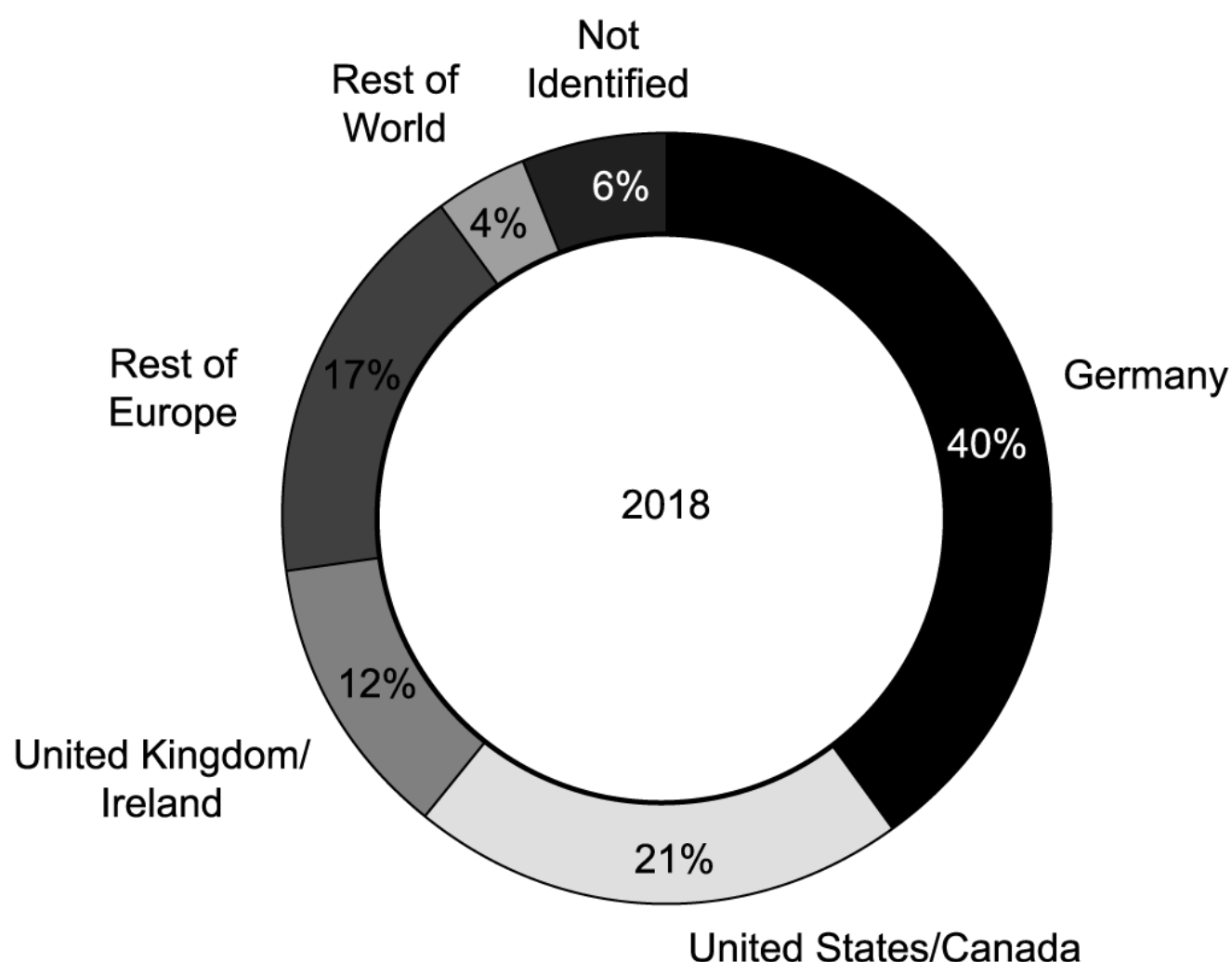
Studies have shown that reducing restrictions on foreign ownership has led to improved equity market performance over the long term.²⁰ Although restrictions vary widely, more countries are allowing increasing levels of foreign ownership. For example, Australia has sought tax reforms as a means to encourage international demand for its managed funds in order to increase its role as an international financial center.

Over the past two decades, three trends have emerged: a) an increasing number of companies have issued shares in markets outside of their home country; b) the number of companies whose shares are traded in markets outside of their home has increased; and c) an increasing number of companies are dual listed, which means that their shares are simultaneously issued and traded in two or more markets. Companies located in emerging markets have particularly benefited from these trends because they no longer have to be concerned with capital constraints or lack of liquidity in their domestic markets. These companies have found it easier to raise capital in the markets of developed countries because these markets generally have higher levels of liquidity and more stringent financial reporting requirements and accounting standards. Being listed on an international exchange has a number of benefits. It can increase investor awareness about the company's products and services, enhance the liquidity of the company's shares, and increase corporate transparency because of the additional market exposure and the need to meet a greater number of filing requirements.

Technological advancements have made it easier for investors to trade shares in foreign markets. The German insurance company Allianz SE recently delisted its shares from the NYSE and certain European markets because international investors increasingly traded its shares on the Frankfurt Stock Exchange. [Exhibit 12](#) illustrates the extent to which the institutional shareholder base at BASF, a large German chemical corporation, has become increasingly global in nature.

Exhibit 12: Example of Increased Globalization of Share Ownership²¹

BASF is one of the largest publicly owned companies with over 500,000 shareholders and a high free float. An analysis of the shareholder structure carried out in March 2018 showed that, at 21% of share capital, the United States and Canada made up the largest regional group of institutional investors. Institutional investors from Germany made up 12%. Shareholders from United Kingdom and Ireland held 12% of BASF shares, while a further 17% are held by institutional investors from the rest of Europe. Around 28% of the company's share capital is held by private investors, most of whom are resident in Germany.



4.1. Direct Investing

Investors can use a variety of methods to invest in the equity of companies outside of their local market. The most obvious is to buy and sell securities directly in foreign markets. However, this means that all transactions—including the purchase and sale of shares, dividend payments, and capital gains—are in the company's, not the investor's, domestic currency. In addition, investors must be familiar with the trading, clearing, and settlement regulations and procedures of that market. Investing directly often results in less transparency and more volatility because audited financial information may not be provided on a regular basis and the market may be less liquid. Alternatively, investors can use such securities as depository receipts and global registered shares, which represent the equity of international companies and are traded on local exchanges and in the local currencies. With these securities, investors have to worry less about currency conversions (price quotations and dividend payments are in the investor's local currency), unfamiliar market practices, and differences in accounting standards. The sections that follow discuss various securities that investors can invest in outside of their home market.

4.2. Depository Receipts

A **depository receipt**²² (DR) is a security that trades like an ordinary share on a local exchange and represents an economic interest in a foreign company. It allows the publicly listed shares of a foreign company to be traded on an exchange outside its domestic market. A depository receipt is created when the equity shares of a foreign company are deposited in a bank (i.e., the depository) in the country on whose exchange the shares will trade. The depository then issues receipts that represent the shares that were deposited. The number of receipts issued and the price of each DR is based on a ratio, which specifies the number of depository receipts to the underlying shares. Consequently, a DR may represent one share of the underlying stock, many shares of the underlying stock, or a fractional share of the underlying stock. The price of each DR will be affected by factors that affect the price of the underlying shares, such as company fundamentals, market conditions, analysts' recommendations, and exchange rate movements. In addition, any short-term valuation discrepancies between shares traded on multiple exchanges represent a quick arbitrage profit opportunity for astute traders to exploit. The responsibilities

of the **depository bank** that issues the receipts include acting as custodian and as a registrar. This entails handling dividend payments, other taxable events, stock splits, and serving as the transfer agent for the foreign company whose securities the DR represents. The Bank of New York Mellon is the largest depository bank; however, Deutsche Bank, JPMorgan, and Citibank also offer depository services.²³

A DR can be **sponsored** or **unsponsored**. A sponsored DR is when the foreign company whose shares are held by the depository has a direct involvement in the issuance of the receipts. Investors in sponsored DRs have the same rights as the direct owners of the common shares (e.g., the right to vote and the right to receive dividends). In contrast, with an unsponsored DR, the underlying foreign company has no involvement with the issuance of the receipts. Instead, the depository purchases the foreign company's shares in its domestic market and then issues the receipts through brokerage firms in the depository's local market. In this case, the depository bank, not the investors in the DR, retains the voting rights. Sponsored DRs are generally subject to greater reporting requirements than unsponsored DRs. In the United States, for example, sponsored DRs must be registered (meet the reporting requirements) with the US Securities and Exchange Commission (SEC). **Exhibit 13** contains an example of a sponsored DR issued by Alibaba in September 2014.

Exhibit 13: Sponsored Depository Receipts²⁴

NEW YORK—(BUSINESS WIRE)—Citi today announced that Alibaba Group Holding Limited ("Alibaba Group") has appointed Citi's Issuer Services business, acting through Citibank, N.A., as the depository bank for its American Depositary Receipt ("ADR") program. Alibaba Group's ADRs, which began trading on September 19, 2014, represent the largest Depositary Receipt program in initial public offering market history.

Alibaba Group's ADR program was established through a \$25.03 billion initial public offering of 368,122,000 American Depositary Shares ("ADSs"), representing ordinary shares of Alibaba Group, which was priced at \$68 per ADS on September 18, 2014. The IPO ranks as the largest in history. The ADRs are listed on the New York Stock Exchange (the "NYSE") under the trading symbol BABA. Each ADS represents one ordinary share of the Company. In its role as depository bank, Citibank will hold the underlying ordinary shares through its local custodian and issue ADSs representing such shares. Alibaba Group's ADSs trade on the NYSE in ADR form.

There are two types of depository receipts: Global depository receipts (GDRs) and American depository receipts (ADRs), which are described below.

4.2.1. *Global Depository Receipts*

A **global depository receipt** (GDR) is issued outside of the company's home country and outside of the United States. The depository bank that issues GDRs is generally located (or has branches) in the countries on whose exchanges the shares are traded. A key advantage of GDRs is that they are not subject to the foreign ownership and capital flow restrictions that may be imposed by the issuing company's home country because they are sold outside of that country. The issuing company selects the exchange where the GDR is to be traded based on such factors as investors' familiarity with the company or the existence of a large international investor base. The London and Luxembourg exchanges were the first ones to trade GDRs. Some other stock exchanges trading GDRs are the Dubai International Financial Exchange and the Singapore Stock Exchange. Currently, the London and Luxembourg exchanges are where most GDRs are traded because they can be issued in a more timely manner and at a lower cost. Regardless of the exchange they are traded on, the majority of GDRs are denominated in US dollars, although the number of GDRs denominated in pound sterling and euros is increasing. Note that although GDRs cannot be listed on US exchanges, they can be privately placed with institutional investors based in the United States.

4.2.2. *American Depository Receipts*

An **American depository receipt** (ADR) is a US dollar-denominated security that trades like a common share on US exchanges. First created in 1927, ADRs are the oldest type of depository receipts and are currently the most commonly traded depository receipts. They enable foreign companies to raise capital from US investors. Note that an ADR is one form of a GDR; however, not all GDRs are ADRs because GDRs cannot be publicly traded in the United States. The term **American depository share** (ADS) is often used in tandem with the term ADR. A depository share is a security that is actually traded in the issuing company’s domestic market. That is, while American depository receipts are the certificates that are traded on US markets, American depository shares are the underlying shares on which these receipts are based.

There are four primary types of ADRs, with each type having different levels of corporate governance and filing requirements. Level I Sponsored ADRs trade in the over-the-counter (OTC) market and do not require full registration with the Securities and Exchange Commission (SEC). Level II and Level III Sponsored ADRs can trade on the New York Stock Exchange (NYSE), NASDAQ, and American Stock Exchange (AMEX). Level II and III ADRs allow companies to raise capital and make acquisitions using these securities. However, the issuing companies must fulfill all SEC requirements.

The fourth type of ADR, an SEC Rule 144A or a Regulation S depository receipt, does not require SEC registration. Instead, foreign companies are able to raise capital by privately placing these depository receipts with qualified institutional investors or to offshore non-US investors. [Exhibit 14](#) summarizes the main features of ADRs.

Exhibit 14: Summary of the Main Features of American Depository Receipts

	Level I (Unlisted)	Level II (Listed)	Level III (Listed)	Rule 144A (Unlisted)
Objectives	Develop and broaden US investor base with existing shares	Develop and broaden US investor base with existing shares	Develop and broaden US investor base with existing/new shares	Access qualified institutional buyers (QIBs)
Raising capital on US markets?	No	No	Yes, through public offerings	Yes, through private placements to QIBs
SEC registration	Form F-6	Form F-6	Forms F-1 and F-6	None
Trading	Over the counter (OTC)	NYSE, NASDAQ, or AMEX	NYSE, NASDAQ, or AMEX	Private offerings, resales, and trading through automated linkages such as PORTAL
Listing fees	Low	High	High	Low
Size and earnings requirements	None	Yes	Yes	None

Source: Adapted from [Boubakri, Cosset, and Samet \(2010\)](#): Table 1.

More than 2,000 DRs, from over 80 countries, currently trade on US exchanges. Based on current statistics, the total market value of DRs issued and traded is estimated at approximately US\$2 trillion, or 15 percent of the total dollar value of equities traded in US markets.²⁵

4.2.3. Global Registered Share

A **global registered share** (GRS) is a common share that is traded on different stock exchanges around the world in different currencies. Currency conversions are not needed to purchase or sell them, because identical shares are quoted and traded in different currencies. Thus, the same share purchased on the Swiss exchange in Swiss francs can be sold on the Tokyo exchange for Japanese yen. As a result, GRSs offer more flexibility than depository receipts because the shares represent an actual ownership interest in the company that can be traded anywhere and currency conversions are not needed to purchase or sell them. GRSs were created and issued by Daimler Chrysler in 1998 and by UBS AG in 2011.

4.2.4. *Basket of Listed Depository Receipts*

Another type of global security is a **basket of listed depository receipts** (BLDR), which is an exchange-traded fund (ETF) that represents a portfolio of depository receipts. An ETF is a security that tracks an index but trades like an individual share on an exchange. An equity-ETF is a security that contains a portfolio of equities that tracks an index. It trades throughout the day and can be bought, sold, or sold short, just like an individual share. Like ordinary shares, ETFs can also be purchased on margin and used in hedging or arbitrage strategies. The BLDR is a specific class of ETF security that consists of an underlying portfolio of DRs and is designed to track the price performance of an underlying DR index. For example, the Invesco BLDRS Asia 50 ADR Index Fund is a capitalization-weighted ETF designed to track the performance of 50 Asian market-based ADRs.

5. RISK AND RETURN CHARACTERISTICS

LEARNING OUTCOME

The candidate should be able to:

- **compare the risk and return characteristics of different types of equity securities**

Different types of equity securities have different ownership claims on a company's net assets. The type of equity security and its features affect its risk and return characteristics. The following sections discuss the different return and risk characteristics of equity securities.

5.1. Return Characteristics of Equity Securities

There are two main sources of equity securities' total return: price change (or capital gain) and dividend income. The price change represents the difference between the purchase price (P_{t-1}) and the sale price (P_t) of a share at the end of time $t - 1$ and t , respectively. Cash or stock dividends (D_t) represent distributions that the company makes to its shareholders during period t . Therefore, an equity security's total return is calculated as:

Equation 1

$$\text{Total return, } R_t = (P_t - P_{t-1} + D_t) / P_{t-1}$$

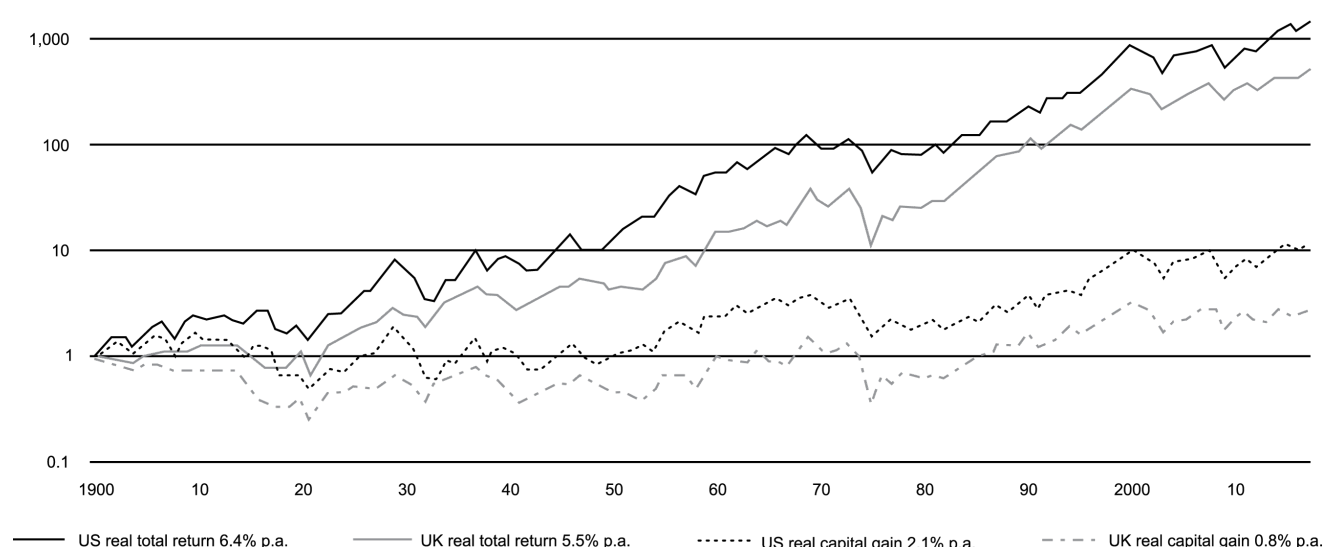
For non-dividend-paying stocks, the total return consists of price appreciation only. Companies that are in the early stages of their life cycle generally do not pay dividends because earnings and cash flows are reinvested to finance the company's growth. In contrast, companies that are in the mature phase of their life cycle may not have as many profitable growth opportunities; therefore, excess cash flows are often returned to investors via the payment of regular dividends or through share repurchases.

For investors who purchase depository receipts or foreign shares directly, there is a third source of return: **foreign exchange gains (or losses)**. Foreign exchange gains arise because of the change in the exchange rate between the investor's currency and the currency that the foreign shares are

denominated in. For example, US investors who purchase the ADRs of a Japanese company will earn an additional return if the yen appreciates relative to the US dollar. Conversely, these investors will earn a lower total return if the yen depreciates relative to the US dollar. For example, if the total return for a Japanese company was 10 percent in Japan and the yen depreciated by 10 percent against the US dollar, the total return of the ADR would be (approximately) 0 percent. If the yen had instead appreciated by 10 percent against the US dollar, the total return of the ADR would be (approximately) 20 percent.

Investors that only consider price appreciation overlook an important source of return: the compounding that results from reinvested dividends. Reinvested dividends are cash dividends that the investor receives and uses to purchase additional shares. As [Exhibit 15](#) shows, in the long run total returns on equity securities are dramatically influenced by the compounding effect of reinvested dividends. Between 1900 and 2016, US\$1 invested in US equities in 1900 would have grown in *real* terms to US\$1,402 with dividends reinvested, but to just US\$11.9 when taking only the price appreciation or capital gain into account. This corresponds to a real compounded return of 6.4 percent per year with dividends reinvested, versus only 2.1 percent per year without dividends reinvested. The comparable ending real wealth for bonds and bills are US\$9.8 and US\$2.60, respectively. These ending real wealth figures correspond to annualized real compounded returns of 2.0 percent on bonds and 0.8 percent on bills.

Exhibit 15: Impact of Reinvested Dividends on Cumulative Real Returns in the US and UK Equity Market: 1900–2016



Source: Dimson, Marsh, and Staunton (2017). This chart is updated annually and can be found at <http://publications.credit-suisse.com/index.cfm/publikationen-shop/research-institute/>.

5.2. Risk of Equity Securities

The risk of any security is based on the uncertainty of its future cash flows. The greater the uncertainty of its future cash flows, the greater the risk and the more variable or volatile the security's price. As discussed above, an equity security's total return is determined by its price change and dividends. Therefore, the risk of an equity security can be defined as the uncertainty of its expected (or future) total return. Risk is most often measured by calculating the standard deviation of the equity's expected total return.

A variety of different methods can be used to estimate an equity's expected total return and risk. One method uses the equity's average historical return and the standard deviation of this return as proxies for its expected future return and risk. Another method involves estimating a range of future returns over a specified period of time, assigning probabilities to those returns, and then calculating an expected return and a standard deviation of return based on this information.

The type of equity security, as well as its characteristics, affects the uncertainty of its future cash flows and therefore its risk. In general, preference shares are less risky than common shares for three main reasons:

1. Dividends on preference shares are known and fixed, and they account for a large portion of the preference shares' total return. Therefore, there is less uncertainty about future cash flows.
2. Preference shareholders receive dividends and other distributions before common shareholders.
3. The amount preference shareholders will receive if the company is liquidated is known and fixed as the par (or face) value of their shares. However, there is no guarantee that investors will receive that amount if the company experiences financial difficulty.

With common shares, however, a larger portion of shareholders' total return (or all of their total return for non-dividend shares) is based on future price appreciation and future dividends are unknown. If the company is liquidated, common shareholders will receive whatever amount (if any) is remaining after the company's creditors and preference shareholders have been paid. In summary, because the uncertainty surrounding the total return of preference shares is less than common shares, preference shares have lower risk and lower expected return than common shares.

It is important to note that some preference shares and common shares can be riskier than others because of their associated characteristics. For example, from an investor's point of view, putable common or preference shares are less risky than their callable or non-callable counterparts because they give the investor the option to sell the shares to the issuer at a pre-determined price. This pre-determined price establishes a minimum price that investors will receive and reduces the uncertainty associated with the security's future cash flow. As a result, putable shares generally pay a lower dividend than non-putable shares.

Because the major source of total return for preference shares is dividend income, the primary risk affecting all preference shares is the uncertainty of future dividend payments. Regardless of the preference shares' features (callable, putable, cumulative, etc.), the greater the uncertainty surrounding the issuer's ability to pay dividends, the greater risk. Because the ability of a company to pay dividends is based on its future cash flows and net income, investors try to estimate these amounts by examining past trends or forecasting future amounts. The more earnings and the greater amount of cash flow that the company has had, or is expected to have, the lower the uncertainty and risk associated with its ability to pay future dividends.

Callable common or preference shares are riskier than their non-callable counterparts because the issuer has the option to redeem the shares at a pre-determined price. Because the call price limits investors' potential future total return, callable shares generally pay a higher dividend to compensate investors for the risk that the shares could be called in the future. Similarly, putable preference shares have lower risk than non-putable preference shares. Cumulative preference shares have lower risk than non-cumulative preference shares because the cumulative feature gives investors the right to receive any unpaid dividends before any dividends can be paid to common shareholders.

6. EQUITY AND COMPANY VALUE

LEARNING OUTCOMES

The candidate should be able to:

- ☐ **explain the role of equity securities in the financing of a company's assets**
- ☐ **contrast the market value and book value of equity securities**
- ☐ **compare a company's cost of equity, its (accounting) return on equity, and investors' required rates of return**

Companies issue equity securities on primary markets to raise capital and increase liquidity. This additional liquidity also provides the corporation an additional "currency" (its equity), which it can use to make acquisitions and provide stock option-based incentives to employees. The primary goal of raising capital is to finance the company's revenue-generating activities in order to increase its net income and maximize the wealth of its shareholders. In most cases, the capital that is raised is used to

finance the purchase of long-lived assets, capital expansion projects, research and development, the entry into new product or geographic regions, and the acquisition of other companies. Alternatively, a company may be forced to raise capital to ensure that it continues to operate as a going concern. In these cases, capital is raised to fulfill regulatory requirements, improve capital adequacy ratios, or to ensure that debt covenants are met.

The ultimate goal of management is to increase the book value (shareholders' equity on a company's balance sheet) of the company and maximize the market value of its equity. Although management actions can directly affect the book value of the company (by increasing net income or by selling or purchasing its own shares), they can only indirectly affect the market value of its equity. The book value of a company's equity—the difference between its total assets and total liabilities—increases when the company retains its net income. The more net income that is earned and retained, the greater the company's book value of equity. Because management's decisions directly influence a company's net income, they also directly influence its book value of equity.

The market value of the company's equity, however, reflects the collective and differing expectations of investors concerning the amount, timing, and uncertainty of the company's future cash flows. Rarely will book value and market value be equal. Although management may be accomplishing its objective of increasing the company's book value, this increase may not be reflected in the market value of the company's equity because it does not affect investors' expectations about the company's future cash flows. A key measure that investors use to evaluate the effectiveness of management in increasing the company's book value is the accounting return on equity.

6.1. Accounting Return on Equity

Return on equity (ROE) is the primary measure that equity investors use to determine whether the management of a company is effectively and efficiently using the capital they have provided to generate profits. It measures the total amount of net income available to common shareholders generated by the total equity capital invested in the company. It is computed as net income available to ordinary shareholders (i.e., after preferred dividends have been deducted) divided by the average total book value of equity (BVE). That is:

Equation 2

$$\text{ROE}_t = \frac{\text{NI}_t}{\text{Average BVE}_t} = \frac{\text{NI}_t}{(\text{BVE}_t + \text{BVE}_{t-1})/2}$$

where NI_t is the net income in year t and the average book value of equity is computed as the book values at the beginning and end of year t divided by 2. Return on equity assumes that the net income produced in the current year is generated by the equity existing at the beginning of the year and any new equity that was invested during the year. Note that some formulas only use shareholders' equity at the beginning of year t (that is, the end of year $t - 1$) in the denominator. This assumes that only the equity existing at the beginning of the year was used to generate the company's net income during the year. That is:

Equation 3

$$\text{ROE}_t = \frac{\text{NI}_t}{\text{BVE}_{t-1}}$$

Both formulas are appropriate to use as long as they are applied consistently. For example, using beginning of the year book value is appropriate when book values are relatively stable over time or when computing ROE for a company annually over a period of time. Average book value is more appropriate if a company experiences more volatile year-end book values or if the industry convention is to use average book values in calculating ROE.

One caveat to be aware of when computing and analyzing ROE is that net income and the book value of equity are directly affected by management's choice of accounting methods, such as those relating to depreciation (straight line versus accelerated methods) or inventories (first in, first out versus weighted average cost). Different accounting methods can make it difficult to compare the return on equity of companies even if they operate in the same industry. It may also be difficult to compare the ROE of the same company over time if its accounting methods have changed during that time.

Exhibit 16 contains information on the net income and total book value of shareholders' equity for three **blue chip** (widely held large market capitalization companies that are considered financially sound and are leaders in their respective industry or local stock market) pharmaceutical companies: Pfizer, Novartis AG, and GlaxoSmithKline. The data are for their financial years ending December 2015 through December 2017.²⁶

Exhibit 16: Net Income and Book Value of Equity for Pfizer, Novartis AG, and GlaxoSmithKline (in Thousands of US Dollars)

	Financial Year Ending		
	31 Dec 2015	31 Dec 2016	31 Dec 2017
Pfizer			
Net income	\$6,960,000	\$7,215,000	\$21,308,000
Total stockholders' equity	\$64,998,000	\$59,840,000	\$71,287,000
Novartis AG			
Net income	\$17,783,000	\$6,712,000	\$7,703,000
Total stockholders' equity	\$77,122,000	\$74,891,000	\$74,227,000
GlaxoSmithKline			
Net income	\$12,420,000	\$1,126,000	\$2,070,700
Total stockholders' equity	\$11,309,250	\$6,127,800	\$4,715,800

Using the average book value of equity, the return on equity for Pfizer for the years ending December 2016 and 2017 can be calculated as:

Return on equity for the year ending December 2016

$$ROE_{2016} = \frac{NI_{2016}}{(BVE_{2015} + BVE_{2016})/2} = \frac{7,215,000}{(64,998,000 + 59,840,000)/2} = 11.6\%$$

Return on equity for the year ending December 2017

$$ROE_{2017} = \frac{NI_{2017}}{(BVE_{2016} + BVE_{2017})/2} = \frac{21,308,000}{(59,840,000 + 71,287,000)/2} = 32.5\%$$

Exhibit 17 summarizes the return on equity for Novartis and GlaxoSmithKline in addition to Pfizer for 2016 and 2017.

Exhibit 17: Return on Equity for Pfizer, Novartis AG, and GlaxoSmithKline

	31 Dec 2016 (%)	31 Dec 2017 (%)
Pfizer	11.6	32.5
Novartis AG	8.8	10.3
GlaxoSmithKline	12.9	38.2

In the case of Pfizer, the ROE of 32.5 percent in 2017 indicates that the company was able to generate a return (profit) of US\$0.325 on every US\$1.00 of capital invested by shareholders. GlaxoSmithKline almost tripled its return on equity over this period, from 12.9 percent to 38.2 percent. Novartis's ROE remained relatively unchanged.

ROE can increase if net income increases at a faster rate than shareholders' equity or if net income decreases at a slower rate than shareholders' equity. In the case of GlaxoSmithKline, ROE almost tripled between 2016 and 2017 due to its net income almost doubling during the period and due to its average shareholder's fund decreasing by almost 45 percent during the period. Stated differently, in 2017 compared to 2016, GlaxoSmithKline was significantly more effective in using its equity capital to generate profits. In the case of Pfizer, its ROE increased dramatically from 11.6 percent to 32.5 percent in 2017 versus 2016 even though its average shareholder equity increased by around 5 percent due to a nearly tripling of net income during the period.

An important question to ask is whether an increasing ROE is always good. The short answer is, "it depends." One reason ROE can increase is if net income decreases at a slower rate than shareholders' equity, which is not a positive sign. In addition, ROE can increase if the company issues debt and then uses the proceeds to repurchase some of its outstanding shares. This action will increase the company's leverage and make its equity riskier. Therefore, it is important to examine the source of changes in the company's net income *and* shareholders' equity over time. The DuPont formula, which is discussed in a separate reading, can be used to analyze the sources of changes in a company's ROE.

The book value of a company's equity reflects the historical operating and financing decisions of its management. The market value of the company's equity reflects these decisions as well as investors' collective assessment and expectations about the company's future cash flows generated by its positive net present value investment opportunities. If investors believe that the company has a large number of these future cash flow-generating investment opportunities, the market value of the company's equity will exceed its book value. [Exhibit 18](#) shows the market price per share, the total number of shares outstanding, and the total book value of shareholders' equity for Pfizer, Novartis AG, and GlaxoSmithKline at the end of December 2017. This exhibit also shows the total market value of equity (or market capitalization) computed as the number of shares outstanding multiplied by the market price per share.

Exhibit 18: Market Information for Pfizer, Novartis AG, and GlaxoSmithKline (in Thousands of US Dollars except market price)

	Pfizer	Novartis AG	GlaxoSmithKline
Market price	\$35.74	\$90.99	\$18.39
Total shares outstanding	5,952,900	2,317,500	4,892,200
Total shareholders' equity	\$71,287,000	\$74,227,000	\$4,715,800
Total market value of equity	\$212,756,646	\$210,869,325	\$89,967,558

Note that in [Exhibit 18](#), the total market value of equity for Pfizer is computed as:

Market value of equity = Market price per share × Shares outstanding

Market value of equity = US\$35.74 × 5,952,900 = US\$212,756,646.

The book value of equity per share for Pfizer can be computed as:

Book value of equity per share = Total shareholders' equity/Shares outstanding

Book value of equity per share = US\$71,287,000/5,952,900 = US\$11.98.

A useful ratio to compute is a company's price-to-book ratio, which is also referred to as the market-to-book ratio. This ratio provides an indication of investors' expectations about a company's future investment and cash flow-generating opportunities. The larger the price-to-book ratio (i.e., the greater the divergence between market value per share and book value per share), the more favorably investors will view the company's future investment opportunities. For Pfizer the price-to-book ratio is:

Price-to-book ratio = Market price per share/Book value of equity per share

Price-to-book ratio = US\$35.74/US\$11.98= 2.98

Exhibit 19 contains the market price per share, book value of equity per share, and price-to-book ratios for Novartis and GlaxoSmithKline in addition to Pfizer.

Exhibit 19: Pfizer, Novartis AG, and GlaxoSmithKline

	Pfizer	Novartis AG	GlaxoSmithKline
Market price per share	\$35.74	\$90.99	\$18.39
Book value of equity per share	\$11.98	\$32.03	\$0.96
Price-to-book ratio	2.98	2.84	19.16

The market price per share of all three companies exceeds their respective book values, so their price-to-book ratios are all greater than 1.00. However, there are significant differences in the sizes of their price-to-book ratios. GlaxoSmithKline has the largest price-to-book ratio, while the price-to-book ratios of Pfizer and Novartis are similar to each other. This suggests that investors believe that GlaxoSmithKline has substantially higher future growth opportunities than either Pfizer or Novartis.

It is not appropriate to compare the price-to-book ratios of companies in different industries because their price-to-book ratios also reflect investors' outlook for the industry. Companies in high growth industries, such as technology, will generally have higher price-to-book ratios than companies in slower growth (i.e., mature) industries, such as heavy equipment. Therefore, it is more appropriate to compare the price-to-book ratios of companies in the same industry. A company with relatively high growth opportunities compared to its industry peers would likely have a higher price-to-book ratio than the average price-to-book ratio of the industry.

Book value and return on equity are useful in helping analysts determine value but can be limited as a primary means to estimate a company's true or intrinsic value, which is the present value of its future projected cash flows. In Exhibit 20, Warren Buffett, one of the most successful investors in the world and CEO of Berkshire Hathaway, provides an explanation of the differences between the book value of a company and its intrinsic value in a letter to shareholders. As discussed above, market value reflects the collective and differing expectations of investors concerning the amount, timing, and uncertainty of a company's future cash flows. A company's intrinsic value can only be estimated because it is impossible to predict the amount and timing of its future cash flows. However, astute investors—such as Buffett—have been able to profit from discrepancies between their estimates of a company's intrinsic value and the market value of its equity.

Exhibit 20: Book Value versus Intrinsic Value²⁷

We regularly report our per-share book value, an easily calculable number, though one of limited use. Just as regularly, we tell you that what counts is intrinsic value, a number that is impossible to pinpoint but essential to estimate.

For example, in 1964, we could state with certitude that Berkshire's per-share book value was \$19.46. However, that figure considerably overstated the stock's intrinsic value since all of the company's resources were tied up in a sub-profitable textile business. Our textile assets had neither going-concern nor liquidation values equal to their carrying values. In 1964, then, anyone inquiring into the soundness of Berkshire's balance sheet might well have deserved the answer once offered up by a Hollywood mogul of dubious reputation: "Don't worry, the liabilities are solid."

Today, Berkshire's situation has reversed: Many of the businesses we control are worth far more than their carrying value. (Those we don't control, such as Coca-Cola or Gillette, are carried at current market values.) We continue to give you book value figures, however, because they serve as a rough, understated, tracking measure for Berkshire's intrinsic value.

We define intrinsic value as the discounted value of the cash that can be taken out of a business during its remaining life. Anyone calculating intrinsic value necessarily comes up with a highly

subjective figure that will change both as estimates of future cash flows are revised and as interest rates move. Despite its fuzziness, however, intrinsic value is all-important and is the only logical way to evaluate the relative attractiveness of investments and businesses.

To see how historical input (book value) and future output (intrinsic value) can diverge, let's look at another form of investment, a college education. Think of the education's cost as its "book value." If it is to be accurate, the cost should include the earnings that were foregone by the student because he chose college rather than a job.

For this exercise, we will ignore the important non-economic benefits of an education and focus strictly on its economic value. First, we must estimate the earnings that the graduate will receive over his lifetime and subtract from that figure an estimate of what he would have earned had he lacked his education. That gives us an excess earnings figure, which must then be discounted, at an appropriate interest rate, back to graduation day. The dollar result equals the intrinsic economic value of the education.

6.2. The Cost of Equity and Investors' Required Rates of Return

When companies issue debt (or borrow from a bank) or equity securities, there is a cost associated with the capital that is raised. In order to maximize profitability and shareholder wealth, companies attempt to raise capital efficiently so as to minimize these costs.

When a company issues debt, the cost it incurs for the use of these funds is called the cost of debt. The cost of debt is relatively easy to estimate because it reflects the periodic interest (or coupon) rate that the company is contractually obligated to pay to its bondholders (lenders). When a company raises capital by issuing equity, the cost it incurs is called the cost of equity. Unlike debt, however, the company is not contractually obligated to make any payments to its shareholders for the use of their funds. As a result, the cost of equity is more difficult to estimate.

Investors require a return on the funds they provide to the company. This return is called the investor's minimum required rate of return. When investors purchase the company's debt securities, their minimum required rate of return is the periodic rate of interest they charge the company for the use of their funds. Because all of the bondholders receive the same periodic rate of interest, their required rate of return is the same. Therefore, the company's cost of debt and the investors' minimum required rate of return on the debt are the same.

When investors purchase the company's equity securities, their minimum required rate of return is based on the future cash flows they expect to receive. Because these future cash flows are both uncertain and unknown, the investors' minimum required rate of return must be estimated. In addition, the minimum required return may differ across investors based on their expectations about the company's future cash flows. As a result, the company's cost of equity may be different from the investors' minimum required rate of return on equity.²⁸ Because companies try to raise capital at the lowest possible cost, the company's cost of equity is often used as a proxy for the investors' *minimum* required rate of return.

In other words, the cost of equity can be thought of as the minimum expected rate of return that a company must offer its investors to purchase its shares in the primary market and to maintain its share price in the secondary market. If this expected rate of return is not maintained in the secondary market, then the share price will adjust so that it meets the minimum required rate of return demanded by investors. For example, if investors require a higher rate of return on equity than the company's cost of equity, they would sell their shares and invest their funds elsewhere resulting in a decline in the company's share price. As the share price declined, the cost of equity would increase to reach the higher rate of return that investors require.

Two models commonly used to estimate a company's cost of equity (or investors' minimum required rate of return) are the dividend discount model (DDM) and the capital asset pricing model (CAPM). These models are discussed in detail in other curriculum readings.

The cost of debt (after tax) and the cost of equity (i.e., the minimum required rates of return on debt and equity) are integral components of the capital budgeting process because they are used to estimate a company's weighted average cost of capital (WACC). Capital budgeting is the decision-making process

that companies use to evaluate potential long-term investments. The WACC represents the minimum required rate of return that the company must earn on its long-term investments to satisfy all providers of capital. The company then chooses among those long-term investments with expected returns that are greater than its WACC.

SUMMARY

Equity securities play a fundamental role in investment analysis and portfolio management. The importance of this asset class continues to grow on a global scale because of the need for equity capital in developed and emerging markets, technological innovation, and the growing sophistication of electronic information exchange. Given their absolute return potential and ability to impact the risk and return characteristics of portfolios, equity securities are of importance to both individual and institutional investors.

This reading introduces equity securities and provides an overview of global equity markets. A detailed analysis of their historical performance shows that equity securities have offered average real annual returns superior to government bills and bonds, which have provided average real annual returns that have only kept pace with inflation. The different types and characteristics of common and preference equity securities are examined, and the primary differences between public and private equity securities are outlined. An overview of the various types of equity securities listed and traded in global markets is provided, including a discussion of their risk and return characteristics. Finally, the role of equity securities in creating company value is examined as well as the relationship between a company's cost of equity, its accounting return on equity, investors' required rate of return, and the company's intrinsic value.

We conclude with a summary of the key components of this reading:

- Common shares represent an ownership interest in a company and give investors a claim on its operating performance, the opportunity to participate in the corporate decision-making process, and a claim on the company's net assets in the case of liquidation.
- Callable common shares give the issuer the right to buy back the shares from shareholders at a price determined when the shares are originally issued.
- Putable common shares give shareholders the right to sell the shares back to the issuer at a price specified when the shares are originally issued.
- Preference shares are a form of equity in which payments made to preference shareholders take precedence over any payments made to common stockholders.
- Cumulative preference shares are preference shares on which dividend payments are accrued so that any payments omitted by the company must be paid before another dividend can be paid to common shareholders. Non-cumulative preference shares have no such provisions, implying that the dividend payments are at the company's discretion and are thus similar to payments made to common shareholders.
- Participating preference shares allow investors to receive the standard preferred dividend plus the opportunity to receive a share of corporate profits above a pre-specified amount. Non-participating preference shares allow investors to simply receive the initial investment plus any accrued dividends in the event of liquidation.
- Callable and putable preference shares provide issuers and investors with the same rights and obligations as their common share counterparts.
- Private equity securities are issued primarily to institutional investors in private placements and do not trade in secondary equity markets. There are three types of private equity investments: venture capital, leveraged buyouts, and private investments in public equity (PIPE).
- The objective of private equity investing is to increase the ability of the company's management to focus on its operating activities for long-term value creation. The strategy is to take the "private" company "public" after certain profit and other benchmarks have been met.

- Depository receipts are securities that trade like ordinary shares on a local exchange but which represent an economic interest in a foreign company. They allow the publicly listed shares of foreign companies to be traded on an exchange outside their domestic market.
- American depository receipts are US dollar-denominated securities trading much like standard US securities on US markets. Global depository receipts are similar to ADRs but contain certain restrictions in terms of their ability to be resold among investors.
- Underlying characteristics of equity securities can greatly affect their risk and return.
- A company's accounting return on equity is the total return that it earns on shareholders' book equity.
- A company's cost of equity is the minimum rate of return that stockholders require the company to pay them for investing in its equity.

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PRACTICE PROBLEMS

1. Which of the following is *not* a characteristic of common equity?
 - A. It represents an ownership interest in the company.
 - B. Shareholders participate in the decision-making process.
 - C. The company is obligated to make periodic dividend payments.
2. The type of equity voting right that grants one vote for each share of equity owned is referred to as:
 - A. proxy voting.
 - B. statutory voting.
 - C. cumulative voting.
3. All of the following are characteristics of preference shares *except*:
 - A. They are either callable or puttable.
 - B. They generally do not have voting rights.
 - C. They do not share in the operating performance of the company.
4. Participating preference shares entitle shareholders to:
 - A. participate in the decision-making process of the company.
 - B. convert their shares into a specified number of common shares.
 - C. receive an additional dividend if the company's profits exceed a pre-determined level.
5. Which of the following statements about private equity securities is *incorrect*?
 - A. They cannot be sold on secondary markets.
 - B. They have market-determined quoted prices.
 - C. They are primarily issued to institutional investors.
6. Venture capital investments:
 - A. can be publicly traded.
 - B. do not require a long-term commitment of funds.
 - C. provide mezzanine financing to early-stage companies.
7. Which of the following statements *most accurately* describes one difference between private and public equity firms?
 - A. Private equity firms are focused more on short-term results than public firms.
 - B. Private equity firms' regulatory and investor relations operations are less costly than those of public firms.
 - C. Private equity firms are incentivized to be more open with investors about governance and compensation than public firms.

8. Emerging markets have benefited from recent trends in international markets. Which of the following has *not* been a benefit of these trends?
- A. Emerging market companies do not have to worry about a lack of liquidity in their home equity markets.
 - B. Emerging market companies have found it easier to raise capital in the markets of developed countries.
 - C. Emerging market companies have benefited from the stability of foreign exchange markets.
9. When investing in unsponsored depository receipts, the voting rights to the shares in the trust belong to:
- A. the depository bank.
 - B. the investors in the depository receipts.
 - C. the issuer of the shares held in the trust.
10. With respect to Level III sponsored ADRs, which of the following is *least likely* to be accurate? They:
- A. have low listing fees.
 - B. are traded on the NYSE, NASDAQ, and AMEX.
 - C. are used to raise equity capital in US markets.
11. A basket of listed depository receipts, or an exchange-traded fund, would *most likely* be used for:
- A. gaining exposure to a single equity.
 - B. hedging exposure to a single equity.
 - C. gaining exposure to multiple equities.
12. Calculate the total return on a share of equity using the following data:
- Purchase price: \$50
- Sale price: \$42
- Dividend paid during holding period: \$2
- A. -12.0%
 - B. -14.3%
 - C. -16.0%
13. If a US-based investor purchases a euro-denominated ETF and the euro subsequently depreciates in value relative to the dollar, the investor will have a total return that is:
- A. lower than the ETF's total return.
 - B. higher than the ETF's total return.
 - C. the same as the ETF's total return.
14. Which of the following is *incorrect* about the risk of an equity security? The risk of an equity security is:
- A. based on the uncertainty of its cash flows.
 - B. based on the uncertainty of its future price.

- C. measured using the standard deviation of its dividends.
15. From an investor's point of view, which of the following equity securities is the *least* risky?
- A. Putable preference shares.
 - B. Callable preference shares.
 - C. Non-callable preference shares.
16. Which of the following is *least likely* to be a reason for a company to issue equity securities on the primary market?
- A. To raise capital.
 - B. To increase liquidity.
 - C. To increase return on equity.
17. Which of the following is *not* a primary goal of raising equity capital?
- A. To finance the purchase of long-lived assets.
 - B. To finance the company's revenue-generating activities.
 - C. To ensure that the company continues as a going concern.
18. Which of the following statements is *most accurate* in describing a company's book value?
- A. Book value increases when a company retains its net income.
 - B. Book value is usually equal to the company's market value.
 - C. The ultimate goal of management is to maximize book value.
19. Calculate the book value of a company using the following information:

Number of shares outstanding	100,000
Price per share	€52
Total assets	€12,000,000
Total liabilities	€7,500,000
Net Income	€2,000,000

- A. €4,500,000.
 - B. €5,200,000.
 - C. €6,500,000.
20. Which of the following statements is *least accurate* in describing a company's market value?
- A. Management's decisions do not influence the company's market value.
 - B. Increases in book value may not be reflected in the company's market value.
 - C. Market value reflects the collective and differing expectations of investors.
21. Calculate the return on equity (ROE) of a stable company using the following data:

Total sales	£2,500,000
Net income	£2,000,000
Beginning of year total assets	£50,000,000
Beginning of year total liabilities	£35,000,000

Number of shares outstanding at the end of the year	1,000,000
Price per share at the end of the year	£20

- A. 10.0%.
- B. 13.3%.
- C. 16.7%.

22. Holding all other factors constant, which of the following situations will *most likely* lead to an increase in a company's return on equity?

- A. The market price of the company's shares increases.
- B. Net income increases at a slower rate than shareholders' equity.
- C. The company issues debt to repurchase outstanding shares of equity.

23. Which of the following measures is the *most difficult* to estimate?

- A. The cost of debt.
- B. The cost of equity.
- C. Investors' required rate of return on debt.

24. A company's cost of equity is often used as a proxy for investors':

- A. average required rate of return.
- B. minimum required rate of return.
- C. maximum required rate of return.

SOLUTIONS

1. C is correct. The company is not obligated to make dividend payments. It is at the discretion of the company whether or not it chooses to pay dividends.
2. B is correct. Statutory voting is the type of equity voting right that grants one vote per share owned.
3. A is correct. Preference shares do not have to be either callable or puttable.
4. C is correct. Participating preference shares entitle shareholders to receive an additional dividend if the company's profits exceed a pre-determined level.
5. B is correct. Private equity securities do not have market-determined quoted prices.
6. C is correct. Venture capital investments can be used to provide mezzanine financing to companies in their early stage of development.
7. B is correct. Regulatory and investor relations costs are lower for private equity firms than for public firms. There are no stock exchange, regulatory, or shareholder involvements with private equity, whereas for public firms these costs can be high.
8. C is correct. The trends in emerging markets have not led to the stability of foreign exchange markets.
9. A is correct. In an unsponsored DR, the depository bank owns the voting rights to the shares. The bank purchases the shares, places them into a trust, and then sells shares in the trust—not the underlying shares—in other markets.
10. A is correct. The listing fees on Level III sponsored ADRs are high.
11. C is correct. An ETF is used to gain exposure to a basket of securities (equity, fixed income, commodity futures, etc.).
12. A is correct. The formula states $R_t = (P_t - P_{t-1} + D_t) / P_{t-1}$. Therefore, total return = $(42 - 50 + 2) / 50 = -12.0\%$.
13. A is correct. The depreciated value of the euro will create an additional loss in the form of currency return that is lower than the ETF's return.
14. C is correct. Some equity securities do not pay dividends, and therefore the standard deviation of dividends cannot be used to measure the risk of all equity securities.
15. A is correct. Puttable shares, whether common or preference, give the investor the option to sell the shares back to the issuer at a pre-determined price. This pre-determined price creates a floor for the share's price that reduces the uncertainty of future cash flows for the investor (i.e., lowers risk relative to the other two types of shares listed).
16. C is correct. Issuing shares in the primary (and secondary) market *reduces* a company's return on equity because it increases the total amount of equity capital invested in the company (i.e., the denominator in the ROE formula).
17. C is correct. Capital is raised to ensure the company's existence only when it is required. It is not a typical goal of raising capital.
18. A is correct. A company's book value increases when a company retains its net income.
19. A is correct. The book value of the company is equal to total assets minus total liabilities, which is $\text{€}12,000,000 - \text{€}7,500,000 = \text{€}4,500,000$.

- 20.** A is correct. A company's market value is affected by management's decisions. Management's decisions can directly affect the company's *book* value, which can then affect its market value.
- 21.** B is correct. A company's ROE is calculated as (NI_t / BVE_{t-1}) . The BVE_{t-1} is equal to the beginning total assets minus the beginning total liabilities, which equals $£50,000,000 - £35,000,000 = £15,000,000$. Therefore, $ROE = £2,000,000 / £15,000,000 = 13.3\%$.
- 22.** C is correct. A company's ROE will increase if it issues debt to repurchase outstanding shares of equity.
- 23.** B is correct. The cost of equity is not easily determined. It is dependent on investors' required rate of return on equity, which reflects the different risk levels of investors and their expectations about the company's future cash flows.
- 24.** B is correct. Companies try to raise funds at the lowest possible cost. Therefore, cost of equity is used as a proxy for the minimum required rate of return.

NOTES

¹The market capitalization of an individual stock is computed as the share price multiplied by the number of shares outstanding. The total market capitalization of an equity market is the sum of the market capitalizations of each individual stock listed on that market. Similarly, the total trading volume of an equity market is computed by value weighting the total trading volume of each individual stock listed on that market. Total dollar trading volume is computed as the average share price multiplied by the number of shares traded.

²NASDAQ is the acronym for the National Association of Securities Dealers Automated Quotations.

³The real return for a security is approximated by taking the nominal return and subtracting the observed inflation rate in that country.

⁴The exceptions are Austria, Belgium, Finland, France, Germany, Portugal, and Italy—where the average real returns on government bonds and/or bills have been negative. In general, that performance reflects the very high inflation rates in these countries during the World War years.

The percentages reported in the exhibit are based on samples of the adult population in each country who own equity securities either directly or indirectly through investment or retirement funds. For example, 36 percent of the adult population of Australia in 2014 (approximately 6.5 million people) owned equity securities either directly or indirectly. As noted in the study, it is not appropriate to make absolute comparisons across countries given the differences in methodology, sampling, timing, and definitions that have been used in different countries. However, trends across different countries can be identified.

⁶It is also possible for companies to pay more than the current period's net income as dividends. Such payout policies are, however, generally not sustainable in the long run.

⁷This information has been adapted from Viacom's investor relations website and its 10-K filing with the US Securities and Exchange Commission; see www.viacom.com.

⁸In some countries, including the United States, companies can issue different classes of shares, with Class A shares being the most common. The role and function of different classes of shares is described in more detail in Exhibit 8.

⁹For example, if US\$2.00 per share is available for distribution, the Common Stock (Class A) shareholders will receive US\$0.50 per share, while the Class B shareholders will receive US\$1.50 per share. However, if there is US\$3.50 per share available for distribution, the Common Stock shareholders will receive a total of US\$1.50 per share and the Class B shareholders will receive a total of US\$2.00 per share.

¹⁰Extracted from Ford Motor Company's 2017 Annual Report (http://s22.q4cdn.com/857684434/files/doc_financials/2017/annual/Final-Annual-Report-2017.pdf).

¹¹Preference shares have a lower priority than debt in the case of liquidation. That is, debt holders have a higher claim on a firm's assets in the event of liquidation and will receive what is owed to them first, followed by preference shareholders and then common shareholders.

¹² <https://www.businesswire.com/news/home/20180326005609/en/GDL-Fund-Successfully-Completes-Offering-Issues-131>

¹³ <https://www.tenn.gr/wp-content/uploads/2018/06/tenn062118.pdf>

¹⁴The term PIPE is widely used in the United States and is also used internationally, including in emerging markets.

¹⁵ <https://tapimmune.com/2018/06/tapimmune-announces-pricing-of-70-million-private-placement/>

¹⁶ Strömberg (2008).

¹⁷ <https://www.statista.com/statistics/270195/global-private-equity-deal-value/>

¹⁸See, for example, Bailey, Wirth, and Zapol (2005).

¹⁹See, for example, Graham, Harvey, and Rajgopal (2005).

²⁰See, for example, Henry and Chari (2004).

²¹Adapted from BASF's investor relations website (www.basf.com). **Free float** refers to the extent that shares are readily and freely tradable in the secondary market.

²²Note that the spellings *depository* and *depository* are used interchangeably in financial markets. In this reading, we use the spelling *depository* throughout.

²³ Boubakri, Cosset, and Samet (2010).

²⁴ <https://www.businesswire.com/news/home/20140924005984/en/Citi-Appointed-Depositary-Bank-Alibaba-Group-Holding>

²⁵ *JPMorgan Depositary Receipt Guide* (2005):4.

²⁶ Pfizer uses US GAAP to prepare its financial statements; Novartis and GlaxoSmithKline use International Financial Reporting Standards. Therefore, it would be inappropriate to compare the ROE of Pfizer to that of Novartis or GlaxoSmithKline.

²⁷ Extracts from Berkshire Hathaway's *2008 Annual Report* (www.berkshirehathaway.com).

²⁸ Another important factor that can cause a firm's cost of equity to differ from investors' required rate of return on equity is the flotation cost associated with equity

Company Analysis: Past and Present

LEARNING OUTCOMES

The candidate should be able to:

- ☐ describe the elements that should be covered in a thorough company research report
- ☐ determine a company's business model
- ☐ evaluate a company's revenue and revenue drivers, including pricing power
- ☐ evaluate a company's operating profitability and working capital using key measures
- ☐ evaluate a company's capital investments and capital structure

1. INTRODUCTION

An insightful and well-written company research report helps investors understand a company and make better investment decisions about the company's securities. This module is the first of three that will provide a framework to prepare a company research report by applying methods covered in previous modules to assess a company's business model, financial performance, and financial position.

Learning Module Overview

- Company research reports analyze a company's past and present financials, its industry and competitors, and forecast its future financial statements. Reports end with a valuation, an investment recommendation, and investment risks.
- The first step of company analysis requires an understanding of the issuer's business model, for which analysts rely on both issuer and third-party information sources.
- An understanding of the issuer's business model and analysis of historical financial statements will allow the analyst to identify key drivers of revenues, profitability, cash flows, and financial position.
- Revenue analysis can be done using a bottom-up or top-down approach. A bottom-up approach breaks down revenues into drivers such as sales volumes and prices, by product line or segment. A top-down approach expresses revenue as a function of drivers such as market share, market size, and GDP growth.
- While a company can change its prices at will, its ability to do so successfully (i.e., not causing a loss of volume) and relative to costs is driven by the company's pricing power. Pricing power is primarily a function of industry structure and competitive strategy.
- Cost analysis assesses a company's profitability and working capital management. Analysts calculate and interpret several cost and profitability measures, including gross, operating, and net margins.
- While a fixed versus variable cost analysis is useful, the analysis is often limited by issuer disclosures and accounting standards that emphasize cost reporting by function or nature.
- The degree of operating leverage measures the sensitivity of operating profit to changes in sales. Operating leverage is primarily driven by the fixed and variable cost composition of the issuer's operating expenses.
- The degree of financial leverage measures the sensitivity of net income to changes in operating income. Financial leverage is primarily driven by the issuer's capital structure.

Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of this learning module.

1. Identify two elements that are common to *all* company research reports and two elements that are common to *initial* research reports.

Solution:

Common to all research reports:

- Analyst's investment recommendation and target buy or sell prices
- Risks such as evaluation of material downside and upside risks

Common to initial research reports:

- Discussion of issuer's business model and strategy, and explanatory charts and figures that disaggregate revenues and profits by product or geography
- Detailed financial analysis and models
- Industry overview and competitive positioning such as industry size, growth rate, market share trends, and industry profitability. Evaluation of the company's competitive position and strategy in each product line or segment is also common.

2. Identify three information sources that analysts commonly use to determine and analyze a company's business model.

Solution:

- Regulatory filings if the issuer is public, especially the annual and quarterly reports
- Investor events and presentations by the issuer
- Visiting the company's properties and/or websites

3. Last year, a distributor of dental care products earned EUR800 million in revenues. Management estimated that its market share was 10%. Based on management's estimate, the market size for distribution of dental care products in this geography last year was *closest* to (in millions of EUR):

- A. 80
- B. 7,200
- C. 8,000

Solution:

C is correct.

$$\text{Market share} = \frac{\text{Revenue}}{\text{Market Size}}$$

Which can be rearranged to:

$$\text{Market size} = \frac{\text{Revenue}}{\text{Market Share}}$$

$$\text{Market size} = \frac{\text{EUR 800 million}}{10\%}$$

$$\text{Market size} = \text{EUR 8,000 million (8 billion)}$$

4. A beverage manufacturer recently introduced a new line of healthy tea-based drinks. The average selling price was USD20 a case last year and the company sold 2 million cases. This year, an analyst forecasts sales of 3 million cases and an increase in the average selling price—from a reduction in discounts and promotions—of 5%. The analyst's forecast for net sales growth this year over the prior year is *closest* to:

- A. 50%

B. 55%

C. 58%

Solution:

C is correct.

Net sales = average selling price × cases sold

Net sales last year = USD20 × 2 million = USD40 million

Net sales forecast this year = (USD20 × 1.05) × 3 million = USD63 million

Forecasted net sales growth rate = $63/40 - 1 = 0.575$ or 58%

5. Impression Ltd. is a fictional company that designs, manufactures, and sells skin care and beauty products. Identify whether each expense line below from Impression Ltd.'s last annual income statement is *most likely* a fixed or variable cost.

I. Amortization of acquired intangible assets	Fixed	Variable
II. Interest expense	Fixed	Variable
III. Costs of goods sold	Fixed	Variable
IV. General and administrative expenses	Fixed	Variable
V. Sales commissions	Fixed	Variable

Solution:

I. Fixed. Amortization expense generally does not change with levels of sales volume, as definite-lived intangible assets are amortized on a straight-line basis over the assets' useful lives.

II. Fixed. Interest expense is a function of the quantity of debt and its interest rate, not sales volume.

III. Variable. Costs of goods sold are incurred when sales are made.

IV. Fixed. General and administrative expenses are usually not related to sales volume each year.

V. Variable. Sales commissions, a form of performance-based compensation for salespeople, are a function of sales and are therefore variable.

6. Based on the data below, the degree of operating leverage for Company XYZ for the year ended 31 December 20X1 is *closest* to:

Company XYZ: Statement of Income for the Year Ended (in millions of EUR)		
	31 December 20X1	31 December 20X0
Revenue	9,707	9,256
Costs of goods sold	4,850	4,637
Selling, general, and administrative expenses	993	1,090
Research and development expenses	1,700	1,554
Other operating expenses	491	448
Interest expense	309	325
Other (income) expense	24	(71)
Income before income taxes	1,340	1,273

Company XYZ: Statement of Income for the Year Ended (in millions of EUR)

	31 December 20X1	31 December 20X0
Provision for income taxes	295	255
Net income	1,045	1,018

A. 0.51

B. 1.08

C. 1.96

C is correct.

$$20X1 \text{ Operating Income} = 9,707 - 4,850 - 993 - 1,700 - 491 = 1,673$$

$$20X0 \text{ Operating Income} = 9,256 - 4,637 - 1,090 - 1,554 - 448 = 1,527$$

$$DOL = \% \Delta \text{ Operating Income} / \% \Delta \text{ Sales}$$

$$DOL = (1,673/1,527 - 1) / (9,707/9,256 - 1)$$

$$DOL = 1.96$$

A is incorrect, as it inverts the equation; it is the ratio of the change in sales to the change in operating income.

B is incorrect; it is the ratio of the percentage change in pre-tax income, rather than operating income, to the percentage change in sales.

7. If a company reported trailing 12 months' operating cash flow of USD150 million, current assets excluding cash of USD40 million, and current liabilities of USD60 million, the company *most likely*:

A. was inefficient in managing its financing.

B. used trade credit to manage working capital.

C. over-extended itself and is at risk of bankruptcy.

Solution:

B is correct. Negative net working capital typically indicates that the company uses suppliers' trade credit to lengthen days payable, while also efficiently managing inventory days and days sales outstanding.

A is incorrect. Negative working capital is a source rather than a use of financing and gives an issuer more financial flexibility, as capital is not tied up in working capital.

C is incorrect. There is not enough information here to determine bankruptcy risk, but positive operating cash flows and negative net working capital are not bankruptcy risk indicators.

8. A company's management may choose *not* to use financial leverage because the company has:

A. an investment-grade credit rating.

B. a high degree of operating leverage.

C. a lower net debt to EBITDA ratio than its peers.

Solution:

B is correct. The degree of financial leverage and the degree of operating leverage together equal the degree of total leverage in the business. If the company already has a high degree of operating leverage, using financial leverage may increase total leverage and risk to too high a level.

A is incorrect. An investment-grade credit rating generally indicates that the company could borrow economically, which would be a reason to use financial leverage.

C is incorrect. A lower net debt to EBITDA ratio than its peers is an indicator of borrowing capacity, which would be a reason to use financial leverage.

2. COMPANY RESEARCH REPORTS

LEARNING OUTCOME

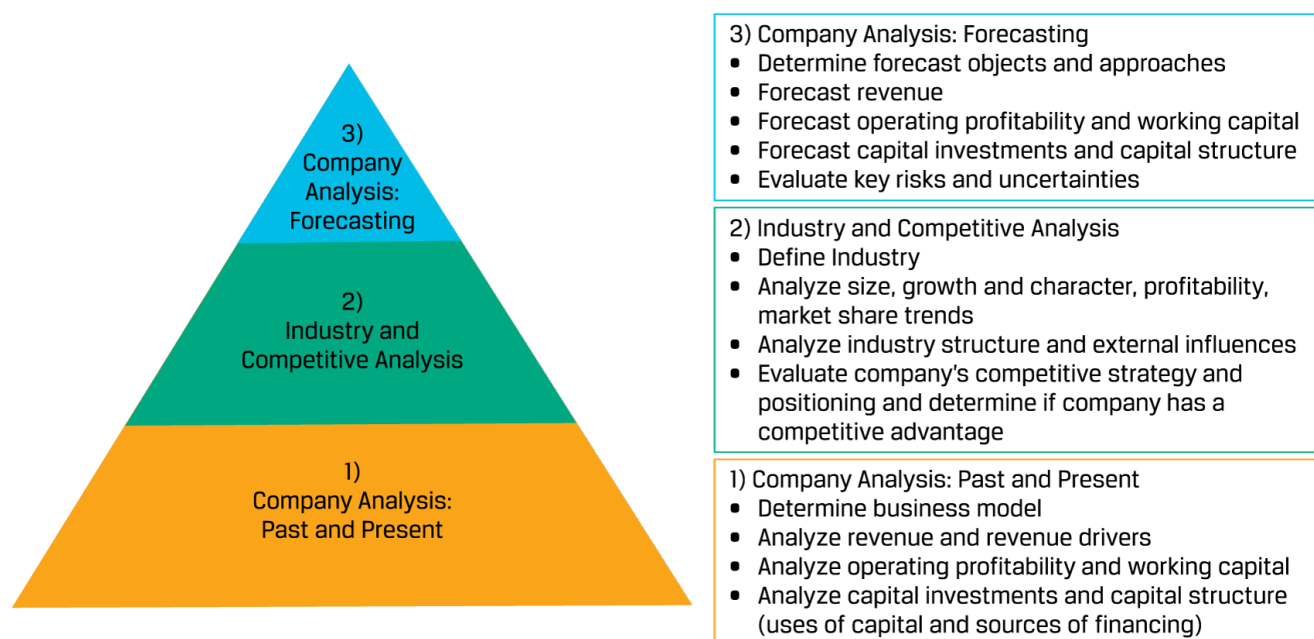
The candidate should be able to:

- **describe the elements that should be covered in a thorough company research report**

Analysts value and make investment recommendations on issuers' equity securities using scenarios of future earnings, cash flows, and financial position. These future scenarios are structured in the form of financial statements and are known as financial statement models. While a financial statement model is quantitative, it is not a mathematics problem to solve with a correct answer, but rather a quantitative expression of an analyst's forward-looking views. These views should be based on supporting evidence and justified by analysis.

The process of forming and justifying a view of an issuer's future financial results and position involves company and industry analysis and is the subject of this module and the next two. [Exhibit 1](#) introduces a framework for company and industry analysis. This module focuses on the foundational block Company Analysis: Past and Present.

Exhibit 1: Company and Industry Analysis Framework



Analysts present their company and industry analysis, as well as their valuation and investment recommendation, in a **company research report**. The structure, content, and tone of the report

depend on the analyst’s setting. Reports on a public issuer’s equity securities for distribution to external clients (a “sell-side report”), for example, often consist of an extensive initial report (an “initiating coverage” report, or “initiation”) when the analyst begins covering the security, followed by shorter reports on specific topics or that make updates to a recommendation after the analyst receives new information or conducts additional analyses, particularly after the issuer reports financial results. Analysts’ reports solely for internal distribution to an audience already familiar with the issuer might be far shorter and delivered verbally or in a handful of presentation slides.

An example of the structure of an *initial* company research report for external distribution is shown in [Exhibit 2](#). The primary audience is those who are not already knowledgeable about the issuer or security.

Exhibit 2: Initial Company Research Report Elements

Front Matter	▪ Issuer name and security identifiers (e.g., symbol, CUSIP) ▪ Analysts’ recommendation (buy, hold, sell) and target buy or sell prices ▪ Disclosures, disclaimers, and other legally required information
Recommendation	▪ Analysts’ recommendation ▪ Summary of key reasons supporting the recommendation
Company Description	▪ Discussion of issuer’s business model and strategy ▪ Explanatory charts and figures—for example, disaggregation of revenues and profits by product, geography, etc.
Industry Overview & Competitive Positioning	▪ Industry size, growth rate and key drivers, market share trends, profitability—historical and outlook ▪ Competitive analysis such as Porter’s Five Forces ▪ Analysis of external industry influences such as political, economic, social, technological, legal, and environmental (“PESTLE” analysis) ▪ Evaluation of company’s position and strategy in the industry
Financial Analysis and Model	▪ Evaluation of key drivers of revenue, costs, profitability, cash flows, and the issuer’s uses and sources of capital ▪ Forecasts of key drivers and supporting discussion ▪ Historical and forecasted financial statements
Valuation	▪ Estimates of company and security values with target price ▪ Typically includes relative and present value approaches ▪ Discussion of key inputs and scenario and sensitivity analyses

Environmental, Social, and Governance (ESG) Considerations	▪ Evaluation of ESG indicators and risks ▪ Ownership structure and management composition ▪ Executive compensation
Risks	▪ Evaluation of material downside and upside risks and discussion of how they are considered in the financial analysis and valuation

THE CFA INSTITUTE RESEARCH CHALLENGE

For examples of initial company research reports, we encourage you to visit the Past Research Challenge Champions section of [cfainstitute.org](https://www.cfainstitute.org), which has winning reports and presentations from the research competition that CFA Institute and CFA Societies around the world host each year. Teams of university students are assigned a public company to research by a local CFA Society and compete on the quality of their research and modeling, valuation, and communication skills.

Subsequent company research reports are often shorter than initial reports. Their primary audience is those who are already familiar with the issuer or security and require an update based on new information and analyses or a change in the analyst's recommendation. Such a report for a company that recently reported quarterly financial results may be structured in the form shown in [Exhibit 3](#). The structure of a subsequent report depends on the analyst's setting and the nature of the report.

Exhibit 3: Subsequent Company Research Report Elements

Front Matter	▪ Analysts' names ▪ Issuer name ▪ Security and exchange identifiers (e.g., symbol, CUSIP) ▪ Analysts' recommendation: buy, hold, sell ▪ Current security price and analysts' target price ▪ Disclosures, disclaimers, and other legally required information
Recommendation	▪ Analysts' recommendation ▪ Summary of changes to the recommendation and support explanations
Analysis of New Information	▪ Comparison of quarterly actual results with expected results ▪ Analysts' interpretation of new information and changes to forecasts ▪ Historical and updated forecasted financial statements
Valuation	▪ Estimates of company and security values ▪ Discussion of changes from valuation in prior report

Question Set

1. Which of the following statements about initiation reports is true?

- A. The structure, content, and tone of the report are independent of the analyst's setting.
- B. The report makes updates to the recommendation after the issuer reports financial results.
- C. The primary audience is those who are not already knowledgeable about the issuer or security.

Solution:

C is correct. The primary audience is those who are not already knowledgeable about the issuer or security.

A is incorrect, because the structure, content, and tone of the report are dependent on the analyst's setting. Examples include whether the report is meant for internal or external distribution and whether it is an in-depth initiation report or a subsequent report updating the reader after a change in recommendation or the release of the issuer's quarterly results.

B is incorrect, because the initial report is to help those who are not already knowledgeable to understand the issuer or security. A subsequent research report is used to update the audience after the release of financial results.

2. If you read a research report that includes both a Porter's Five Forces analysis and a PESTLE analysis, then *most likely*:

- A. the authoring analyst is initiating coverage.
- B. the report is updating a recommendation after an earnings release.
- C. the report is written by a firm's analyst who has covered the company for many years.

Solution:

A is correct. A Porter's Five Forces analysis and a PESTLE analysis are common elements in initiations and are less common in subsequent research reports, as they tend to evolve more slowly than other report elements such as financial analyses and models.

B and C are incorrect; these elements are common in subsequent rather than initial research reports.

3. The evaluation of a company's competitive strategy takes place before the:

- A. analysis of revenue and revenue drivers.
- B. analysis of operating profitability and working capital.
- C. forecasting of revenue, operating profitability, and working capital.

Solution:

C is correct. Forecasting takes place only after company analysis and industry and competitive analysis are complete. The evaluation of a company’s competitive strategy is part of the industry and competitive analysis.

A is incorrect, because the analysis of revenue and revenue drivers is part of the first stage of analysis, where the analyst studies the company’s past and present financial statements. As we shall see in Lesson 3, analysts start by analyzing the revenue line items first.

B is incorrect, because the analysis of operating profitability and working capital typically takes place before the industry and competitive analysis. The evaluation of the company’s competitive strategy is usually part of the industry and company analysis, which takes place after completion of the analysis of operating profitability and working capital.

3. DETERMINING THE BUSINESS MODEL

LEARNING OUTCOME

The candidate should be able to:

- ☐ **determine a company’s business model**

Determining the business model is the first step in our industry and company analysis framework because it summarizes important drivers of an issuer’s financial results and position, focuses the analyst on what requires further investigation, and should begin setting the analyst’s expectations for the issuer. A discussion of the business model is usually in the first part of a company research report (see the “Company Description” section in Exhibit 2).

Recall that a business model describes a company’s operations and includes the following elements, which analysts investigate by answering several key questions discussed in prior learning modules.

Business Model Element	Key Questions for Analysts
The product(s) or service(s) the company sells	What is the firm selling?
Customers and key customer groups	To whom does the company sell?
Sales channels, including customer acquisition and product/service delivery mechanisms	How does the company reach potential and current customers and how does it deliver products?
How the product(s) or service(s) are priced and paid for	How much does the company charge and how are prices and payment terms structured?
Resource, supplier, and partner relationships need to operate effectively	What does the company buy and rely on?

There are many corollary questions for each of the key questions, such as determining whether customers are few and concentrated, whether suppliers and resources are specialized or hard to replace, and so on. The answers to these questions are company specific, but the key questions to ask are common across industries and companies. Also, recall that many companies have a conventional business model, such as retailer or natural resource producer, which simplifies business model identification in practice. Analysts will often focus their analysis on the differences in a company’s business model, if any, from a conventional model or those of its competitors.

Information sources that analysts use to answer these questions include the following, grouped by origin: the issuer, public third party, proprietary third party, and proprietary primary research.

- Issuer sources (available freely if the company is public)
 - Regulatory filings, especially the annual and quarterly reports
 - Quarterly or semi-annual earnings conference calls
 - Presentations and events for investors
 - Press releases
 - Issuer management, investor relations, or other personnel
 - Company website or properties that the analyst may be able to visit as either a customer or an investor. It is often useful to experience an issuer's and its competitors' products firsthand, though it is not always possible (e.g., pharmaceuticals).
- Public third-party sources
 - Free industry white papers or analyst reports from a consultancy
 - Economic or industry indicators from governments and other organizations
 - General news outlets
 - Industry-specific news outlets
 - Social media
 - Miscellaneous sources available via search engines
- Proprietary third-party sources
 - Analyst reports and communications, including from "sell-side" or "Wall Street" analysts and credit rating agencies
 - Reports and data from platforms such as Bloomberg and FactSet
 - Reports and data from consultancies, often industry specific, such as Rystad in energy, IQVIA and Evaluate in biopharma, Gartner and IDC in information technology
- Proprietary primary research
 - Surveys, conversations, product comparisons, and other studies commissioned by the analyst or conducted directly

Analysts in an institutional investment setting (e.g., an asset management firm) are often able to conduct this initial research quickly, as they enjoy broad access to a variety of proprietary third-party sources, may have industry knowledge and experience themselves, can access prior analyses of the company or industry, and often have access to the issuer investor relations personnel and management.

We will demonstrate the determination of an issuer's business model with a fictional example adapted from a real company. We will use this case study throughout this module and the next two to demonstrate the use of the company and industry analysis framework. Another case study of a different company in the same industry is in the self-assessment section at the end of each lesson, so you can practice using the framework on your own.

Warehouse Club Inc.

Elaine Nguyen is an analyst at Fyleton Investments. To find investment candidates for one of Fyleton's funds, Nguyen ran a screen for companies in developed markets that exceeded a certain level of sales growth and return on invested capital over the last five years while also performing well in the last recession. One company that passed the screen is Warehouse Club Inc. ("Warehouse"). Neither Nguyen nor her colleagues have ever worked on Warehouse before. A portfolio manager suggested that Nguyen research the company and its industry for several days and then discuss initial findings with colleagues before writing a formal research report.

Warehouse is a public company, so Nguyen compiles several issuer information sources, including the company's latest annual report and a transcript of management's presentation at a recent investor event. The following paragraphs and figures are excerpts from these sources.

Warehouse Club Inc. ("we," "our," "us," "the company") is a leading discount retailer, consistently offering customers more than 20% savings on a broad range of food and other merchandise compared to supermarket, department store, convenience store, and e-commerce competitors. Warehouse operated 149, 145, and 141 stores as of 31 December 2X19, 2X18, and 2X17, respectively.

We sell a core assortment of packaged food and beverages, fresh foods, and non-food merchandise such as apparel, appliances, electronics and entertainment, housewares, and sporting goods. In addition to our core assortment, our stores feature a changing assortment of seasonal or discounted items based on the latest consumer trends and availability from suppliers, which vary by store. We also offer ancillary products and services such as petrol stations co-located with our stores, in-store takeaway prepared foods, pharmacy and optician centers, and hearing aids. In addition to carrying the leading branded goods, we also sell packaged food and household goods under our exclusive private-label brand that we source from contract manufacturers, which account for over 15% of our net sales and carry higher than average profitability.

We operate a member-based model that requires customers to show their membership card to enter and shop in our stores. The annual membership fee is \$60 per household. We estimate that members' annual savings versus shopping at competitors for the same basket of goods amounts to over five times their annual membership fee. Our membership fee revenues were \$601, \$576, and \$553 million in fiscal years 2X19, 2X18, and 2X17, respectively. Besides providing us with a source of recurring revenue, membership engenders shopper loyalty and drives our industry-low levels of inventory theft of less than 20 basis points as a percentage of net sales, which we believe is at least one-fifth that of competitors. Total memberships were 10.2, 9.8, and 9.4 million as of 31 December 2X19, 2X18, and 2X17, respectively.

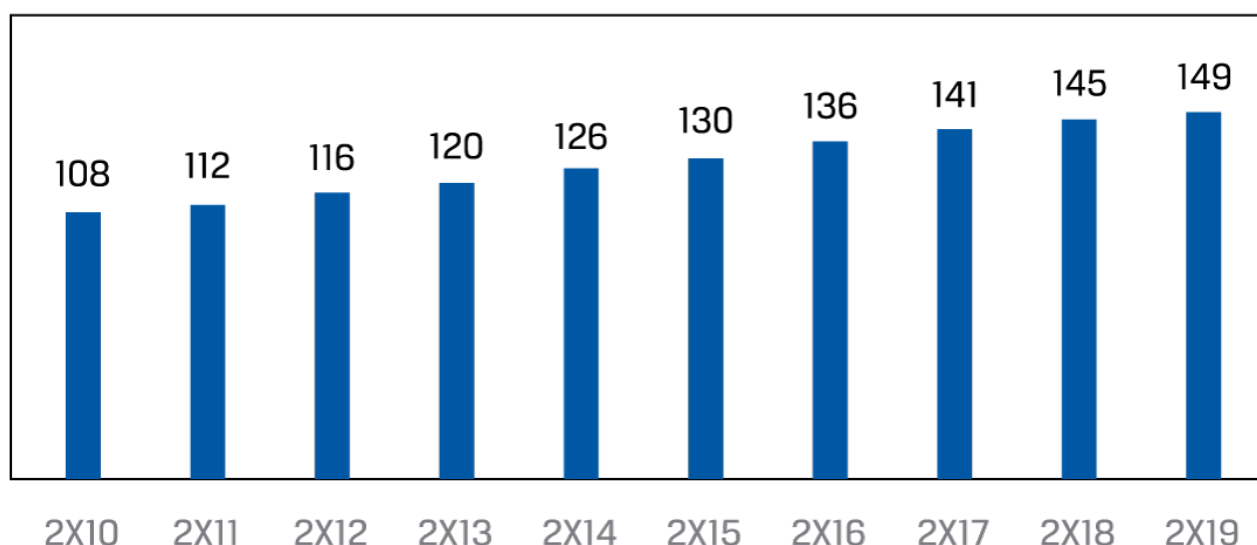
Our strategy is based on price and cost leadership. We stock a limited selection of high-quality branded and private-label products in a wide range of categories that produce high sales volumes and rapid inventory turnover. We limit items to fast-selling models, sizes, and colors and carry an average of approximately 4,000 unique products per store, less than a quarter of the assortment of our supermarket and supercenter competitors. Most items are sold only in bulk sizes. Wherever possible, we utilize direct-from-manufacturer distribution and move merchandise immediately to the sales floor, presenting on pallets in a no-frills warehouse atmosphere. Our high-volume purchases of a limited number of products, direct from manufacturer distribution, and reduced handling of merchandise enable us to hold a low-cost position in the retail industry in terms of merchandise and labor costs. Our low costs and membership fee revenues enable us to operate profitably while charging significantly lower prices than other retailers. Low prices for quality products drive greater membership and net sales over time, leveraging our selling, general, and administrative expenses, reducing them as a percentage of net sales.

Our stores operate on a seven-day, 70-hour week with predictable shifts for employees. Because the hours of operation are shorter than other retailers, and due to other efficiencies inherent in a warehouse-type operation, labor costs are lower relative to the volume of sales. We look to maintain a large base of full-time employees with above-average wages and benefits to establish long tenures, as we believe this maximizes

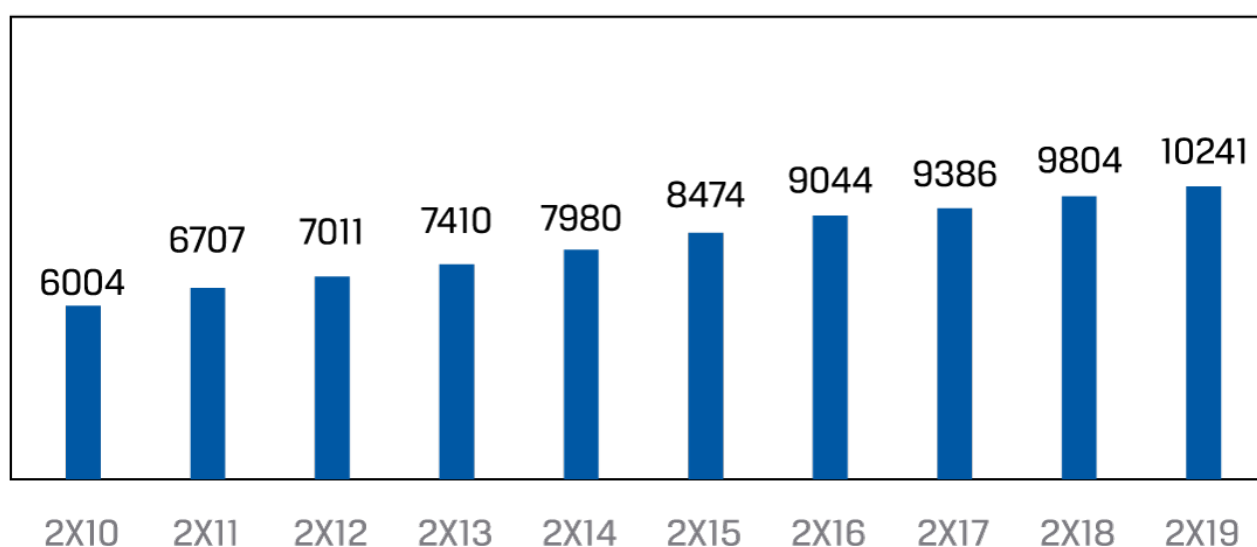
productivity and customer service at our stores. Our employee retention rate for employees who have been with us for at least a year is 90%, which we believe is substantially higher than the retail industry average.

Our retail operations, which represent substantially all our consolidated net sales, are our only reportable segment. We do not have significant sales outside our domestic geography: the United States. No single customer or supplier represents a material amount of revenues or merchandise costs, respectively.

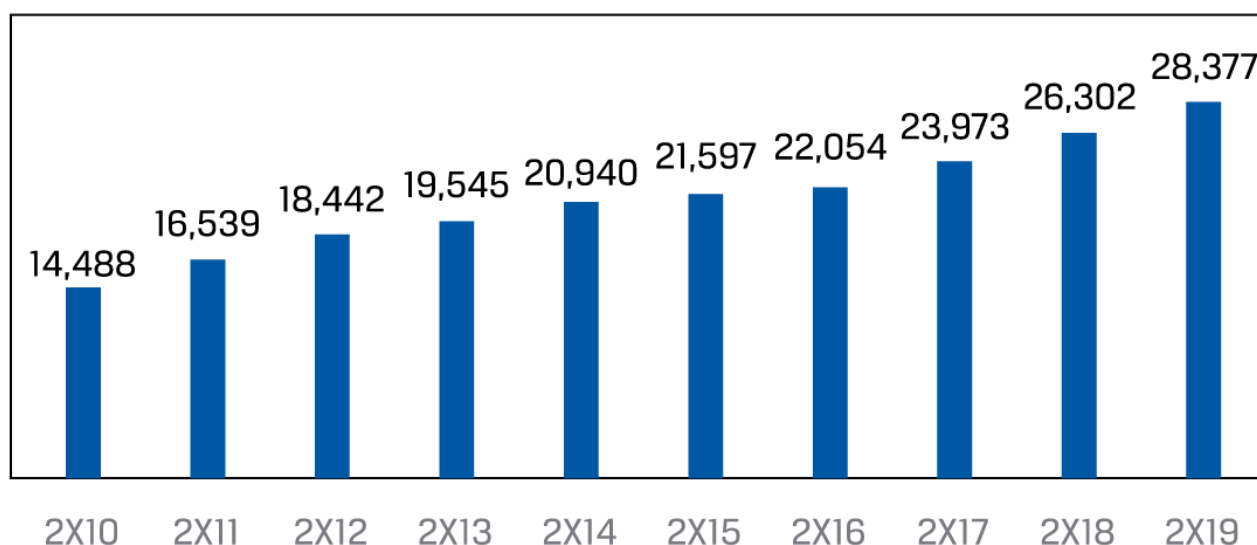
Warehouse Club Inc.
Number of Stores at Year-End, 2X10-2X19



Warehouse Club Inc.
Number of Members (thousands) at Year-End, 2X10-2X19



Warehouse Club Inc.
Annual Net Sales (USD millions), 2X10-2X19



Warehouse Club Inc. Income Statements for the Years Ended . . . (amounts in USD millions except per share amounts)

	31 December 2X17	31 December 2X18	31 December 2X19
Net sales	23,973	26,302	28,377
Membership fees	553	576	601
Total revenues	24,526	26,878	28,978
Merchandise costs	(21,258)	(23,399)	(25,248)
Selling, general, and administrative expenses	(2,201)	(2,363)	(2,566)
Depreciation and amortization	(260)	(273)	(283)
New-store opening expenses	(16)	(13)	(16)
Operating income	791	830	865
Interest expense	(25)	(30)	(29)
Interest income	10	14	24
Income before taxes	776	814	860
Income tax expense	(178)	(187)	(198)
Net income	598	627	662
Shares outstanding	83	83	84
Earnings per share	7.20	7.55	7.88

During its presentation at a recent investor event, Warehouse management made the following statements that were not in the annual report:

- Three years ago, the company launched “Buy Online, Pick-Up in Store,” which enables members to shop on Warehouse’s website or mobile app and pick up their order, assembled by Warehouse employees, at a store. The company does not plan to offer e-commerce or delivery itself or through third-party services.
- Membership is \$60 per year, paid upfront, and refundable on a prorated basis for days remaining in the year. Management has increased the price of membership every five years by \$5 and expects to continue to do so. The last price increase was three years ago.
- Management intends to open four new stores per year for the next five years.
- The company operates stores in urban and suburban areas in a single country. Over 90% of current members live within 10 kilometers of a store, and while membership has grown around existing stores over time, management believes the primary driver of membership growth is opening stores in new areas.
- Management does not plan to open stores in another country for the foreseeable future. An immaterial amount of merchandise is imported.

EXAMPLE 1

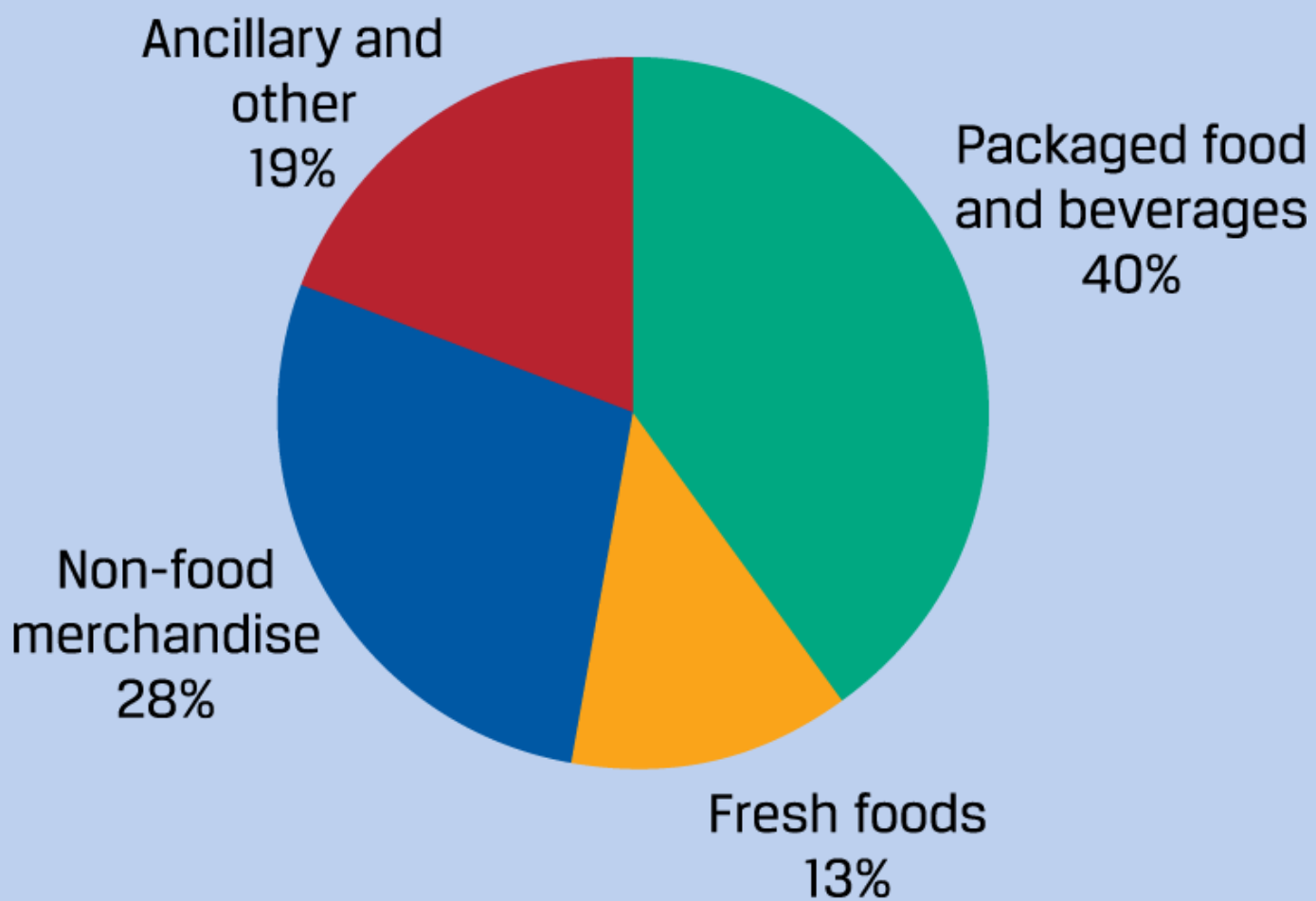
Determining Warehouse Club Inc.’s Business Model

Nguyen answers the following questions to determine Warehouse Club Inc.'s business model and to identify areas to conduct further research.

- What is the firm selling?

Warehouse sells a range of consumer goods, mostly consumables and perishables, approximately 4,000 unique items. For customers to shop at the stores, they must hold a membership, which Warehouse sells for \$60 per year, per household. Nguyen graphs the following net sales by category data with descriptions supplied by Warehouse, and notes that the composition of net sales by category has not changed materially in the last 10 years.

Warehouse Club Inc. Net Sales by Category, 2X19



“Non-food merchandise” includes a broad range of items such as appliances, electronics, health and personal care products, hardware, outdoor and sporting goods, jewelry, furniture, and housewares.

“Ancillary and other” are service businesses that operate in or next to Warehouse’s stores and include takeaway prepared foods, prescription pharmacies, petrol stations, optician centers, and hearing aids.

- To whom?

Warehouse’s members are primarily consumer households that live within 10 kilometers of a store, are value-conscious, are willing to forgo a more extensive selection for lower unit prices on bulk-sized goods, and are able to afford the annual membership fee.

- How?

Warehouse sells products and services both in and around its stores. Members can either shop at the store or shop online and pick up at the store (Nguyen notes that management did not disclose the size of this in terms of net sales). Management said that it does minimal advertising (besides promotional mail) in an area before it opens a store and also sends out membership renewal reminders; the company primarily relies on word of mouth and its stores’ physical presence to drive interest and foot traffic.

Memberships can be purchased online, by phone, or at any Warehouse store. Management said that over three-quarters of members purchase and renew their memberships online.

- For how much?

Warehouse has two revenue streams: goods and services (for which it earns net sales) and membership (for which it earns membership fees).

Management sets the prices of its merchandise by item. Gross margin and markup on aggregate net sales for the last three fiscal years were as follows:

	31 December 2X17	31 December 2X18	31 December 2X19
Net sales	23,973	26,302	28,377
Merchandise costs	(21,258)	(23,399)	(25,248)
Gross margin	2,715	2,903	3,129
Gross margin % of net sales	11.33%	11.04%	11.03%
Markup	12.77%	12.41%	12.39%

Therefore, the average good or service is sold at a price that is a little over 12% of the cost for Warehouse to purchase and ready the item for sale. While management claims that the company is a cost and price leader, Nguyen will have to corroborate that.

Membership is \$60 per year, per household. Membership is required to shop at Warehouse stores, thus effectively raising the price of all its goods and services.

- What does the company buy and rely on to operate effectively?
 - Merchandise from a broad range of manufacturers and private-label goods from contract manufacturers
 - Human capital in stores and in management
 - Fixed assets including land in attractive locations, buildings, fixtures, refrigerators and freezers, and a range of information technology systems such as point-of-sale payment systems
 - Credit and debit card payment networks and financial institutions to receive and make electronic payments

While these are all broadly available and the company has not had supply or service interruptions in the past, a key partner appears to be merchandise manufacturers, particularly those that manufacture Warehouse's private-label products. Since Warehouse is strict on cost and quality and carries a limited selection, the company does rely heavily on these contract manufacturers.

In summary, Warehouse's business model is aimed at creating a virtuous cycle whereby low prices drive memberships, which drive further sales, which drive negotiating leverage with suppliers to enable low prices at low costs.

Discussion Board Question

Based on the information provided, what would you change about or add to Elaine Nguyen's analysis of Warehouse Club Inc.'s business model? Why?

We encourage you to post your response, as well as read and reply to others' responses, on the Learning Ecosystem discussion board.

Question Set

Iliso Marketplace Ltd. (“we,” “our,” “us,” “the Company,” “Iliso”) is redefining e-commerce by delivering superior customer experiences at competitive prices through our Iliso Marketplace mobile app and website. While e-commerce customers have historically been forced to compromise on shipping times, product quality, and hassles with returns, we’ve built a vertically integrated system that delivers a vast selection of products with one-day delivery on most orders. We continue to make significant investments in both technology and our fulfillment infrastructure to improve the customer experience and expand into new product categories and geographies.

Customer Experience

We’re committed to delighting our customers every day. Hallmarks of our customer experience include:

- **Next-Day or Faster Delivery.** Customers are eligible for one-day delivery 365 days a year. In 2X19, over 93% of orders were delivered within 24 hours of receiving the order.
- **Free Shipping.** Customers are eligible for free shipping on orders greater than \$25, which we believe is the industry-leading minimum order threshold, and no membership fee is required.
- **Vast Selection of Items, Including Fresh Groceries.** Customers can order from a selection of items across almost every category of goods—from fresh produce to consumer electronics to home décor and furniture. To continue offering the highest-quality items, we modify our selection continuously based on customer reviews. We enforce strict standards for third-party merchants to protect customers from counterfeit goods.
- **Frictionless Returns.** Our customers simply tap a button on our app and leave the item outside their door for pickup. Refunds are initiated the moment the item is picked up at the door.

The number of active customer accounts as of the fourth quarter for the years ended 31 December 2X17, 2X18, and 2X19, were 29, 31, and 36 million, respectively. Active customer account refers to a unique email address and at least one purchase in the indicated period; we do not charge membership fees or require minimum purchases to maintain an account.

Our Fulfillment Network

To achieve our differentiated customer experience, we focus on urban markets and have built a vertically integrated system in which we control every step, from a customer’s purchase on our app to the photo confirmation of the delivery at the customer’s front door. We receive most of our products directly from manufacturers at our fulfillment centers and minimize the use of third-party carriers and contractors by directly employing over 65,000 people in fulfillment and delivery. We believe direct employment and control of fulfillment assets such as delivery trucks enable speed, efficiency, and waste reduction, which is especially important for product categories such as fresh foods.

We’ve made the strategic decision to focus on urban markets in the United States for the foreseeable future to achieve an economical route density for customer deliveries and returns and one-day shipping.

Merchant Services

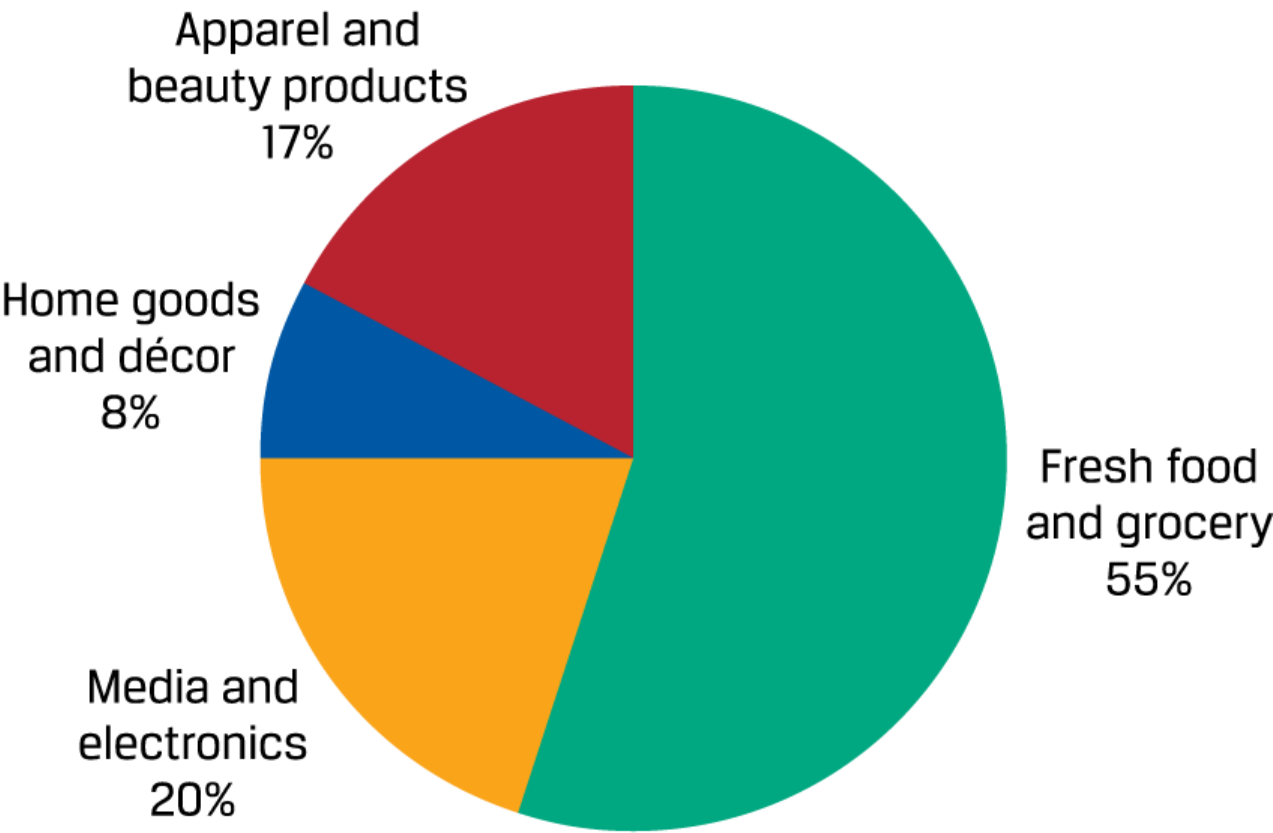
In 2X13, we launched Merchant Services, allowing third-party merchants to sell products on Iliso Marketplace and utilize our fulfillment network. When multiple merchants (including us) sell the same product, our algorithm shows the customer the lowest price, including shipping cost, though the customer can manually choose a different seller. Third-party

merchants must meet our rigorous quality standards but otherwise have discretion in product assortment and description, pricing, and shipping terms.

We receive sales commissions on third-party merchants’ sales, as well as service revenues if the merchant utilizes our fulfillment network for warehousing and delivery. We record revenue on a “net” basis for third-party merchant-related transactions because we act as an agent; the Company is not the seller of record, nor does it take control of the goods inventory. The number of active third-party merchants as of the fourth quarter for the years ended 31 December 2X17, 2X18, and 2X19 was 25,500, 31,500, and 40,500, respectively. Active third-party merchant refers to a unique email address and at least one sale in the indicated period; we do not charge merchant membership fees, nor do we require minimum sales levels.

Our annual gross merchandise value (GMV)—which includes the value of all products sold on our marketplace by us and third-party merchants, net of sales taxes and returns allowances but including any applicable shipping fees—was \$5,721, \$6,950, and \$8,605 million for the years ended 31 December 2X17, 2X18, and 2X19, respectively. GMV by product category for the year ended 31 December 2X19 was as follows:

***Iliso Marketplace Ltd.
GMV by Product Category, 2X19***



Iliso recently went public, so financial information is available only for the five years ended 31 December 2X19. Amounts are in millions of USD (except per share amounts).

Iliso Marketplace Ltd. GMV, Customer, and Merchant Counts for the Years Ended 31 December

	2X15	2X16	2X17	2X18	2X19
Iliso retail sales	2,589	3,424	5,100	6,052	7,336
Third-party merchant sales	279	432	621	898	1,269
Total GMV	2,868	3,856	5,721	6,950	8,605
Active customer accounts (millions)	16	23	29	31	36
Third-party merchants (#)	14,850	18,000	25,500	31,500	40,500

Iliso Marketplace Ltd. Income Statements for the Years Ended 31 December

	2X15	2X16	2X17	2X18	2X19
Net retail sales	2,589	3,424	5,100	6,052	7,336
Third-party merchant fees	42	65	94	134	190
Total revenues	2,631	3,489	5,194	6,186	7,526
Costs of revenues	(1,894)	(2,506)	(3,729)	(4,446)	(5,414)
Operating, general, and administrative expenses	(685)	(895)	(1,300)	(1,544)	(1,831)
Advertising and marketing	(53)	(70)	(104)	(124)	(151)
Operating income	(1)	18	61	72	130
Interest income	6	7	9	9	10
Interest expense	(5)	(13)	(15)	(12)	(8)
Pre-tax income	0	12	55	69	132
Provision for income taxes	0	(3)	(13)	(16)	(32)
Net income	0	9	42	52	100
Basic EPS	0.00	0.03	0.15	0.18	0.34
Diluted EPS	0.00	0.03	0.14	0.18	0.34
Basic share count	274	280	284	288	291
Diluted share count	274	280	291	288	297

The notes to Iliso's financial statements provide the following descriptions of accounts.

Net retail sales

Retail sales are earned from the Company's online product sales to consumers. Retail revenue is recognized when control of the goods is transferred to the customer, which occurs upon delivery to the customer.

Third-party merchant fees

Third-party merchant fees revenue includes commissions and service fees earned from merchants that sell their products through the Company's online business. The revenue is recognized when the order is completed and transmitted to the third-party merchant.

Costs of revenues

Costs of revenues primarily comprise the purchase price of products sold to customers where the Company records revenue gross, and include logistics center costs. Inbound shipping and handling costs to receive products from suppliers are included in inventory and recognized in costs of revenues as products are sold. Additionally, costs of revenues include outbound shipping and logistics-related expenses, delivery service costs from our Merchant Services business (where the Company is the delivery service provider), and depreciation expense.

Operating, general, and administrative expenses

Operating, general, and administrative expenses include costs incurred in operating and staffing the Company's fulfillment centers (including costs attributable to receiving, inspecting, picking, packaging, and preparing customer orders), customer service costs, payment processing fees, costs related to the design, execution, and maintenance of the Company's technology infrastructure and online offerings, and general corporate function costs.

Advertising and marketing expenses

Advertising and marketing expenses are expensed as incurred and include brand and performance advertising such as television and digital advertising, sponsored search, email marketing campaigns, and other similar initiatives.

1. What forms the customer base (the “who”) for Iliso’s merchant services segment?

- A. Third-party merchants
- B. E-commerce customers
- C. Fresh food and grocery consumers

Solution:

A is correct. In merchant services, Iliso acts as an agent for third-party merchants. Iliso is paid commissions and service fees by merchants that sell their products through the company’s online business. Third-party merchants are therefore the customer base for this business segment.

B is incorrect, because e-commerce customers are customers of Iliso’s retail sales segment. Iliso is paid by e-commerce customers for goods purchased through its online platform. The purchases of third-party merchant products by these e-commerce customers belong to the third-party merchant and are not recognized in Iliso’s retail sales.

C is incorrect, because fresh food and grocery consumers are a product line segmentation of both its retail sales and its third-party merchant GMV. In addition, fresh food and grocery forms only 55% of the overall GMV and does not encompass all of Iliso’s GMV sales.

2. Iliso’s business model(s) is(are) likely to benefit from _____ effects.

- A. network
- B. value chain
- C. supply chain

Solution:

A is correct. Iliso’s business model—in particular, merchant services—is an example of a two-sided network effect. Network effect refers to a network in which the value increases as more participants join the network. By providing an excellent service to merchants and to consumers, Iliso is able to attract both buyers and sellers to its online marketplace. The growth of users on Iliso’s platform attracts more merchants, which in turn attracts more users.

B is incorrect, because value chain refers to the systems and processes within a firm that create value for its customers—which answers the “how” aspect of a business model.

C is incorrect, because the supply chain is the network inside and outside the company involved in producing the product/service and delivering it to the end user.

3. Which of the following business features is *least likely* to increase consumer retention (“stickiness”) on Iliso’s platform?

- A. A focus on urban markets
- B. Next-day or faster delivery
- C. No membership fee is required.

Solution:

A is correct. A focus on urban markets does not change customer stickiness. The customer experience for those in urban markets would not necessarily change if the company began shipping to rural markets as well.

B is incorrect. The company states that next-day or faster delivery is a hallmark of its customer experience and may provide a competitive advantage.

C is incorrect, because the lack of a membership fee is likely to decrease the cost of a customer's staying on the platform, thus increasing retention for some customers.

4. The gross margin of Iliso's merchant services segment is *most likely* _____ the gross margin of Iliso's retail sales segment.

A. higher than

B. the same as

C. lower than

Solution:

A is correct. The cost of revenues for Iliso's retail sales includes the purchase price of products sold to customers, including shipping and handling costs, whereas the cost of revenues for merchant services includes only the outbound shipping and logistics expenses. The purchase price of products is a far higher proportion of total GMV sold to customers compared with shipping and logistics costs. Accordingly, the exclusion of the purchase price of products in both the numerator and the denominator will raise the reported gross margin of the merchant services segment.

B is incorrect, because the gross margin for merchant services cannot be the same as that of retail sales. The retail sales segment includes the purchase price of products sold and associated inbound logistics costs, which means the numerator and the denominator are significantly higher than in the merchant services segment.

C is incorrect, because the gross margin for merchant services cannot be lower than that of retail sales. Assuming a positive gross margin for both segments, the gross margin for merchant services must be higher since the denominator is smaller (being only the service commissions and service fees charged to merchants).

4. REVENUE ANALYSIS

LEARNING OUTCOME

The candidate should be able to:

- ☐ evaluate a company's revenue and revenue drivers, including pricing power

4.1. Revenue Drivers

After determining the business model, we turn to the analysis of historical financial results and position for the company. This analysis will be used as a basis for forecasting a financial statement model that supports security valuation. For most companies—perhaps with the exception of banks, for which the balance sheet comes first—analysts start by analyzing revenues. As with other financial statement lines, the analysis involves identifying **drivers**, which are causative factors that explain the level of and changes in an output variable (here, revenues), and understanding the evolution of the drivers over time.

Analysts can take a bottom-up or top-down approach to determining revenue drivers. A bottom-up approach decomposes revenues into drivers such as sales volume and price, or revenues by product line, segment, or geography, which may be further broken down into other drivers. A top-down approach expresses revenues as a function of drivers such as market share, the addressable market or market size, and GDP growth. The two approaches are often used together.

Bottom-Up Approach to Revenue Analysis: Warehouse Club Inc.

Warehouse's revenues are composed of net sales and membership fees.

Based on her business model work, Nguyen knows that net sales are sales of merchandise and services in Warehouse stores (the company does not operate an e-commerce business). A logical driver of net sales each year, therefore, is the number of stores open that year. Nguyen decomposes Warehouse's net sales into two drivers: the *average* number of stores open and net sales per average store. Nguyen also calculates the change in these drivers: the absolute change in stores each year (new-store openings; so far, Warehouse has never closed a store) and the annual percentage change in net sales per store.

	2X10	2X11	2X12	2X13	2X14	2X15	2X16	2X17	2X18	2X19
Stores, year-end	108	112	116	120	126	130	136	141	145	149
New-store openings	5	4	4	4	6	4	6	5	4	4
Average stores (#)	106	110	114	118	123	128	133	139	143	147
Net sales per average store	138	150	162	166	170	169	166	173	184	193
Growth rate		9%	8%	2%	3%	-1%	-2%	4%	6%	5%
Net sales	14,488	16,539	18,442	19,545	20,940	21,597	22,054	23,973	26,302	28,377

Net sales in 2X19 were 96% higher than in 2X10, driven by similar growth contributions from the two drivers: the average number of stores increased by 39% and sales per average store increased by 40%.

An encouraging sign for forecasting this company's results is that each driver appears relatively stable, indicating that they might be somewhat predictable. Sometimes, however, key revenue drivers are volatile, such as interest rates and oil prices for banks and oil producers, respectively. Volatility does not mean that the analyst has selected an incorrect driver; drivers are based on the business model.

Nguyen then decomposes Warehouse's membership fee revenues into the annual price of membership and the average number of members.

Members	2X10	2X11	2X12	2X13	2X14	2X15	2X16	2X17	2X18	2X19
Members, end of year	6,004	6,707	7,011	7,410	7,980	8,474	9,044	9,386	9,804	10,241
Members, annual average	5,900	6,360	6,855	7,218	7,691	8,218	8,764	9,217	9,600	10,017

Members	2X10	2X11	2X12	2X13	2X14	2X15	2X16	2X17	2X18	2X19
Avg. per member fee	50.00	50.00	55.00	55.00	55.00	55.00	55.00	60.00	60.00	60.00
Membership fee revenues	295	318	377	397	423	452	482	553	576	601

Membership growth has been a larger driver of membership fee revenue growth than price increases, as the number of members has grown by 70% from 2X10 to 2X19 while prices have increased by 20% over the same period.

Management indicated that over 90% of members live within 10 kilometers of a store and, from 2X10 to 2X19, the company opened 46 new stores. Nguyen adds to her membership analysis by calculating the average number of members per store; if this number has been falling over time, it could mean that existing stores are losing members and sales to new stores the company opens, an example of **cannibalization**.

	2X10	2X11	2X12	2X13	2X14	2X15	2X16	2X17	2X18	2X19
Avg. members per store	57,282	57,818	60,128	61,171	62,528	64,205	65,892	66,546	67,133	68,141
Growth rate		0.9%	4.0%	1.7%	2.2%	2.7%	2.6%	1.0%	0.9%	1.5%

Not only has the average number of members per store not declined, but it has also grown by 19%, indicating no cannibalization and that new-store openings are not the only driver of membership growth. This is a positive sign, though Nguyen notes that additional analysis is needed to estimate a ceiling on the number of members a store can reasonably support, and to determine whether stores that opened many years ago have already hit this ceiling.

4.2. Pricing Power

Prices are a driver of revenue for every firm, and it may appear that pricing is a unilateral decision by management (e.g., Warehouse management chooses its markups on merchandise net sales and has chosen to increase its annual membership fee by \$5 every five years). In fact, prices are constrained by a company's **pricing power**. Pricing power refers to a company's ability to set prices and other economic terms with customers without affecting its sales volumes. While Warehouse's management could choose to increase the price of membership from \$60 to \$600, such a decision would likely result in the loss of all its members. The membership price is constrained by the savings Warehouse can offer members on purchases, which itself is the subject of competition from other retailers that also use low prices to attract customers and of negotiations with suppliers that desire higher prices. Pricing power is primarily a function of both market structure (covered earlier in Economics) and a company's competitive positioning in its market.

In the most competitive markets, where firms are selling nearly identical products, firms are price *takers*—that is, price is dictated by the forces of supply and demand—and all firms generally sell at the same price. If a firm prices above the prevailing market price, it will sell a fraction or none of its output, as customers will buy elsewhere. It is also generally not rational for a firm to price below the market price, as doing so would not only reduce its profits but also cause competitors to do the same, leading to a price war. Returns on capital in highly competitive markets move in cycles based on supply and demand, but over the long run approximate firms' cost of capital as competition forces prices down to marginal cost. A **low-cost producer** (with costs below—in some cases, *well* below—those of a marginal producer) can, however, earn long-run returns above its cost of capital in a highly competitive industry, but it requires the low-cost position to be sustained against competition over time. A prominent example is state-owned oil producers that benefit from monopoly ownership of low-cost oil reserves in certain countries.

Other attributes of highly competitive markets include little to no product differentiation, low barriers to firm entry, available substitutes, a lack of customer loyalty, and low switching costs for customers. Many markets fit this description, including retail, oil and gas and other natural resources, bank loans and

deposits, insurance, fresh food, off-patent pharmaceuticals, transportation, and some types of technology hardware. Highly competitive markets often do not start as highly competitive but, rather, become more competitive as new firms enter, the pace of innovation slows, and imitation becomes widespread, a process known as **commoditization**.

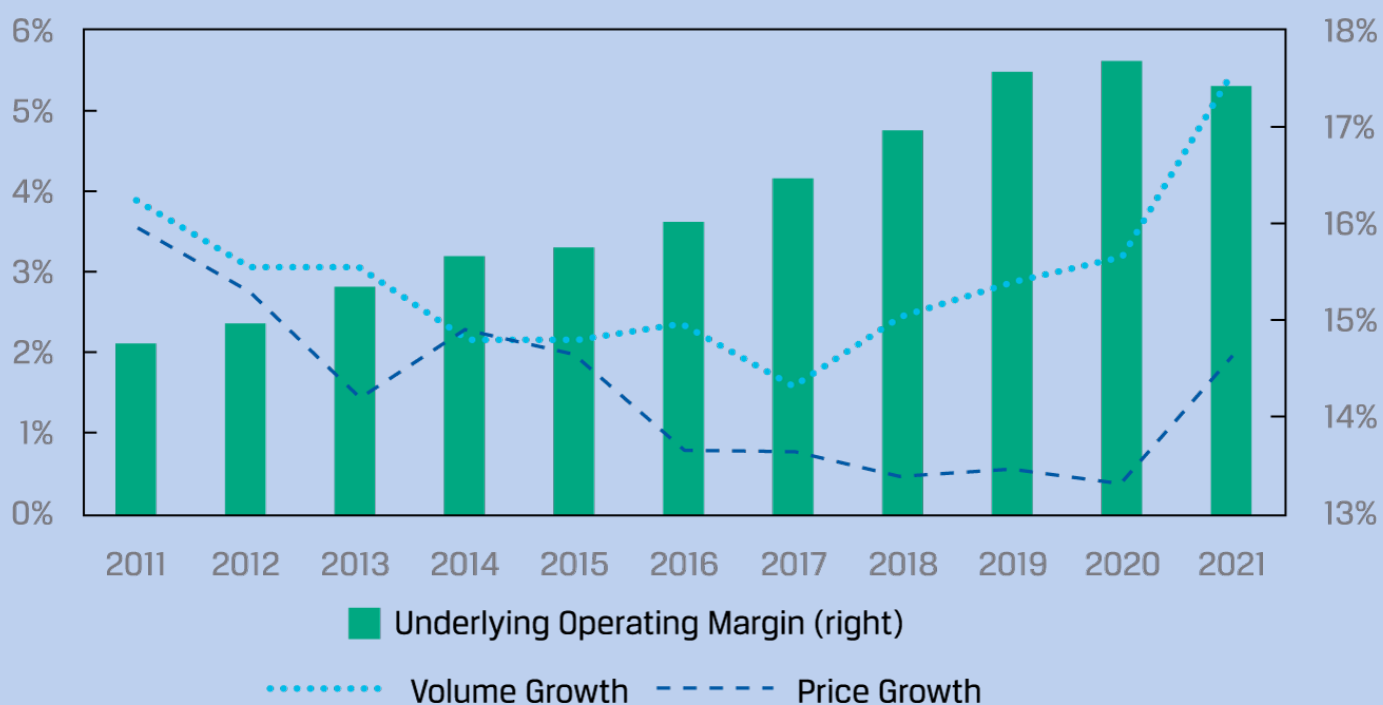
Firms operating in less competitive markets, such as monopolistic competition, oligopoly, or monopoly structures, tend to have some degree of pricing power. They can raise prices or change other economic terms (contract length, late fees, etc.) without materially affecting sales volumes. Attributes of these markets include product differentiation, barriers to entry, lack of economical substitutes, switching costs for customers, and a high degree of customer loyalty or retention. Markets and companies that fit this description include branded consumer goods, utilities, patented pharmaceuticals, software, restaurants, payment networks, medical devices, and some types of real estate. Firms with pricing power often apply value- or cost-based pricing as well as price discrimination strategies (outlined in the previous module). While product differentiation is most visible, differentiation can also include convenience, wrap-around services, and a secondary resale market.

Analysts evaluate pricing power not just by examining a firm's prices over time or relative to competitors, but also by comparing a firm's prices with its costs—in other words, its profit margins. While a company may be increasing prices by 3% per year, if its costs are rising by 5%, it is showing both an inability to pass along price increases to customers and declining profitability for investors. A company without pricing power may be unable to increase prices even as costs rise, because it is constrained by the availability of substitutes and customers can easily switch. For example, a movie theater might be unable to increase prices in line with costs, because streaming video subscriptions are cheaper and the company would likely lose customers if it raised prices significantly. Therefore, an important sign of pricing power is rising profitability over time, which demonstrates both an ability to retain economics for investors and that competitors, new entrants, or substitutes are not driving down prices faster than costs.

EXAMPLE 2

Nestlé SA is a global packaged food and beverage company headquartered in Switzerland. The company sells products with proprietary brand trademarks (such as Nespresso, Purina, Perrier, Maggi, and Stouffer's) that it acquires and develops with marketing and R&D investments, which totaled CHF19 billion in 2021, or 20% of the company's revenues. The distinctiveness of its products through branding can confer pricing power. From 2011 to 2021, the company demonstrated its pricing power through a record of increasing prices while increasing volumes and its operating profitability.

Nestle SA Pricing Power



4.3. Top-Down Revenue Analysis

Analysts can also take a top-down approach to analyzing revenue by expressing it as a function of drivers such as market size and a company's share of that market. This approach will be discussed in more detail in the next module (on industry and competitive analysis), but we will demonstrate this approach for Warehouse Club Inc.

Top-Down Approach to Revenue Analysis: Warehouse Club Inc.

A government agency reports total national retail and food services sales each month for the country that Warehouse operates in (the United States). The data capture consumer spending at hundreds of thousands of retailers and restaurants. Sales are categorized by the type of retailer or restaurant, which reflects the primary type of product or service it sells. An excerpt of a monthly release is shown below.

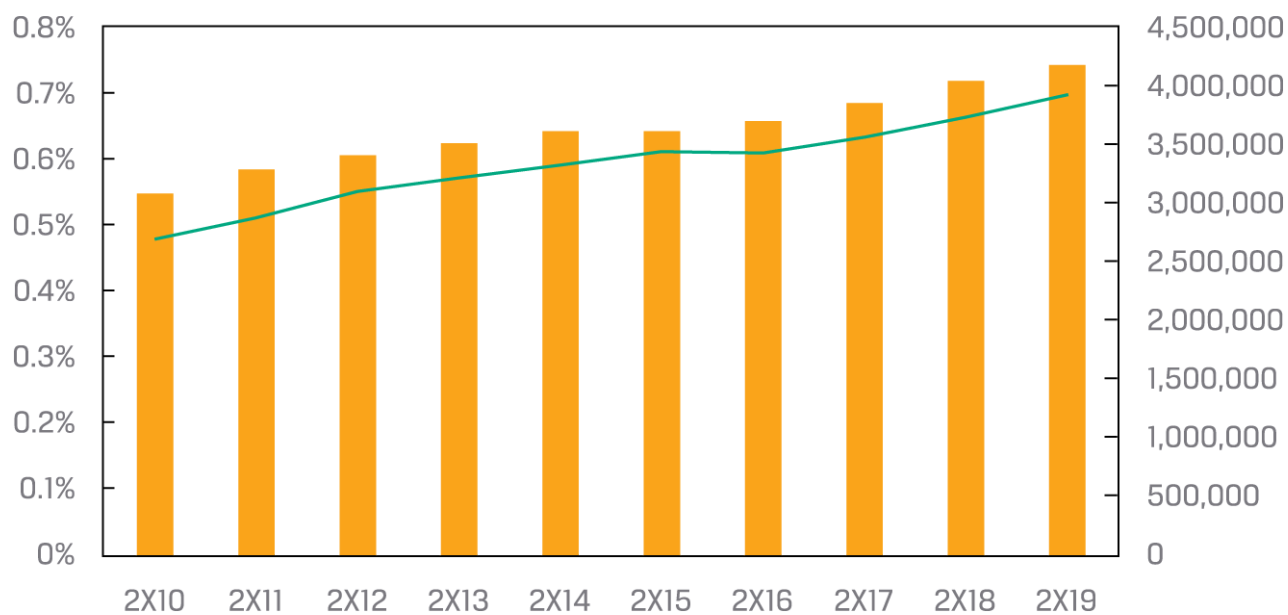
National Retail Sales, January 2X19 (billions of USD)	
Total retail and food services sales	456,012
Total retail sales	399,359
Automobile and parts dealers	88,152
Petrol stations	36,539
Pharmacies and drug stores	24,273
Grocery stores	57,392
Total food services sales	56,653
Full-service restaurants	24,730
Limited-service restaurants	25,273
Pubs, bars, and other	6,650

Warehouse sells a broad range of retail goods in most cities, so US national retail spending is generally representative of Warehouse's **market size**—the existing demand for goods or services offered by a company. By expressing Warehouse's revenues as a percentage of the market, or its **market share**, over time, we can assess whether the company has been gaining or losing relevance with its customers relative to competitors and substitutes.

A challenge for analysts is determining market size. The data might be difficult to gather (not the case here), and judgment is required for determining what to include and exclude: only sales of products identical to the company's, sales of similar products, all sales of competitors, and so on. The common practice is to include all sales of *similar* products and to consider *substitutes* separately (discussed in the next module on industry analysis). A substitute serves the same function as a product but differs in form (e.g., movie theater and streaming video, grocery stores and restaurants, print and digital advertising). This practice is subject to a wide range of opinions, however, and some analysts include sales of substitutes in market size estimates. The share of market that a firm does not have (i.e., $100\% - x\%$ market share) represents sales potential.

Elaine Nguyen decides to use total US retail sales as the market size for Warehouse, with one exclusion: automobiles and parts. Warehouse does not sell automobiles or automobile parts, nor does it sell substitute means of transportation or repair. Nguyen considers food service substitutes (restaurants and pubs) for Warehouse.

**Warehouse Club Inc. Market Share (% , left) of
Retail Sales excl. Autos and Parts (USD millions, right)**



Nguyen makes three observations:

- The market is large and growth has been positive, albeit at a slow (3.4%) rate. This finding aligns with Nguyen’s expectations, because this is a developed market that represents a broad range of consumer spending (a large portion of which is for necessities) and the country has been in an economic expansion.
- Warehouse Club Inc. has a small (70 basis points in 2019) but growing share of the market (up from 48 basis points in 2010), indicating that its value proposition relative to other retailers has been resonating with consumers.
- Warehouse’s market share increased in most years, with the exception of 2016, when its share was flat from the prior year. Nguyen will have to investigate further.

Discussion Board Question

Which of Nguyen’s analyses was more helpful to you in understanding Warehouse’s historical revenue growth: the bottom-up or the top-down approach? Why?

We encourage you to post your response, as well as read and reply to others’ responses, on the Learning Ecosystem discussion board.

Question Set

Questions 1–4 refer to the earlier vignette on Iliso Marketplace Ltd.

- In 2019, the percentage of GMV that Iliso collected from third-party merchants (the company’s “take rate”) was *closest* to:
 - 2.2%.
 - 14.7%.
 - 15.0%.

Solution:

C is correct. For e-commerce businesses, the take rate refers to the retention percentage of the value of transactions facilitated by the platform. In 2X19, Iliso third-party merchant fees were \$190 million against third-party gross merchandise value (GMV) of \$1,269 million on its platform. The take rate is therefore:

$$190/1,269 = 0.1497 \approx 15.0\%$$

A is incorrect, because it results from dividing third-party merchant fees by total GMV. B is incorrect, because 14.7% is the percentage of Iliso's GMV attributable to third-party merchant sales.

2. Which of the following measures would be used to best assess Iliso's pricing power?

A. Take rate

B. GMV year-on-year growth rate

C. GMV per active customer account

Solution:

A is correct. The percentage of transaction value retained by the platform reflects the price Iliso charges third-party merchants to use its platform. Both the stability and the level of its take rate reflect the state of competition that Iliso faces from other e-commerce platforms.

B is incorrect, because the GMV year-on-year growth rate can be affected by numerous factors, including customer volumes, product mix, and basket size. It does not carry information for analysts to estimate pricing power.

C is incorrect, because GMV per active customer account can be affected by product mix and the average basket size, which is driven by its \$25 minimum for free shipping. This measure does not carry information for analysts to estimate the pricing power of the platform.

3. Referring to the data table earlier in this lesson titled "National Retail Sales, January 2X19" and assuming 2X19 US monthly retail and food sales of \$456 billion and assuming Iliso's addressable geography is the United States, Iliso's market share of retail and food sales was *closest* to:

A. 0.134%.

B. 0.138%.

C. 0.157%.

Solution:

C is correct. In 2X19, annual US retail and food sales were: $456 \times 12 = \$5,472$ billion. Since GMV reflects the value of retail sales on Iliso's platform, the appropriate estimate of Iliso's market share is therefore:

$$2X19 \text{ GMV} / 2X19 \text{ US retail and food sales} = 8,605 / 5,472,000 = 0.157\%$$

A is incorrect, because this measure uses only Iliso's retail sales and ignores third-party GMV, even though the vignette states that merchants may sell the same products.

B is incorrect, because this measure uses Iliso's net revenues as a percentage of the market. This measure is inappropriate, since Iliso's net revenues do not reflect the gross value of third-party merchant sales on its platform.

4. Which of the following is an appropriate bottom-up driver of Iliso's revenues?

A. Iliso's market share of US retail and food sales

B. GMV per average active customer account

C. Iliso’s revenue as a function of GDP growth

Solution:

B is correct. In a bottom-up approach, analysts decompose the revenues into logical revenue drivers such as product line or segment. A driver of Iliso’s net sales must be the average number of active customer accounts during the period and the net retail sales per account. We use GMV, since this measure accurately reflects the total amount of retail sales (both Iliso’s first-party and third-party sales).

A and C are incorrect, because they are top-down, not bottom-up, drivers.

5. OPERATING PROFITABILITY AND WORKING CAPITAL ANALYSIS

LEARNING OUTCOME

The candidate should be able to:

- ☐ evaluate a company’s operating profitability and working capital using key measures

5.1. Operating Costs and Their Classification

Generally, operating costs are incurred in generating—or are otherwise related to—*current period* revenue: all costs related to the acquisition, production, sale, improvement, and delivery of goods and services; the management of business activities; and compliance with laws and regulations. Investing costs are related to the acquisition and production of long-term assets, including tangible assets like property and equipment and intangible assets such as software, trademarks, and patents. Financing costs include payments to debt and equity investors as a return on their investment.

MINDING THE GAAP: CLASSIFICATION OF COSTS

IFRS and US GAAP do not classify operating, investing, or financing costs in an intuitive manner, and there are differences in classification on the income statement and statement of cash flows. Several prominent examples include:

Cost	Intuition	IFRS and US GAAP Classification: Income Statement	IFRS and US GAAP Classification: Statement of Cash Flows
Research and development	Investing	Operating	Operating
Depreciation and amortization*	Investing	Operating	Non-cash add-back to Operating <i>Capital expenditures</i> in Investing
Interest expense	Financing	Other (not operating)	US GAAP: Operating IFRS: Operating or Financing
Income taxes	Separate	Other (not operating)	Operating

Recall that IFRS and US GAAP require the costs of acquiring and constructing long-term tangible and intangible assets to be capitalized and then, for those with a finite useful life, systematically expensed over time on the income statement as non-cash depreciation and amortization expense. The most common allocation method is straight line, where the cost is expensed evenly over the useful life of the asset.

Unless specified otherwise, we use the IFRS classifications of costs on the income statement in our discussion.

For most companies, operating costs account for the majority of costs and are primarily determined by the company's business model and size. As shown in [Exhibit 4](#), a company's operating costs can be categorized and analyzed in three ways: by their behavior with output, their nature, or their function.

Exhibit 4: Categorization of Operating Costs

Behavior with Output <i>Whether the cost varies or not with output in the short run</i>	Nature <i>What the cost is</i>	Function <i>The purpose of the cost</i>
• Variable costs vary with output • Fixed costs do not vary with output	• Compensation of employees • Raw materials • Merchandise • Office supplies	• Cost of goods sold • Sales and Marketing • Genral and administrative • Research and development

5.2. Behavior with Output: Fixed and Variable Costs

A useful classification of operating costs for analysts is fixed versus variable, which was introduced in an earlier module. The distinction is useful because the proportion of fixed to variable costs in operating costs affects the stability and predictability of operating profit. Using a fixed/variable operating cost classification, operating profit is defined in [Equation 1](#).

Equation 1

$$\text{Operating profit} = [Q \times (P - VC)] - FC$$

Q = units of outputs sold in a period.

P = price per unit of output.

VC = variable operating costs expressed per unit of output. Examples include merchandise costs for a retailer like Warehouse Club Inc. and materials and direct labor costs for a manufacturer.

FC = fixed operating costs, which do not change within a given range of output in the short run. Examples include compensation for salaried employees, depreciation and amortization, software and IT expenses, insurance, and certain utilities costs. FC is stated on a total dollar, not per unit, basis.

For a firm to be profitable, it's clear from [Equation 1](#) that $(P - VC)$, or the **contribution margin**, must be positive and that Q must be high enough such that FC is exceeded. Recall that the amount of fixed costs in the operating cost structure of an issuer is referred to as operating leverage and that it presents both benefits and risks. If operating costs are largely fixed and the contribution margin is positive, operating profit can increase rapidly with increases in Q . However, the reverse is true if Q declines, since fixed costs

do not change, so operating profit will fall. As shown in [Equation 2](#), operating leverage can be measured and compared across firms by using the **degree of operating leverage** (DOL):

Equation 2

$$DOL = \% \Delta \text{ Operating Profit} / \% \Delta \text{ Sales}$$

A firm can increase its degree of operating leverage by increasing the fixed costs and decreasing the variable costs in its cost base.

EXAMPLE 3

Variable and Fixed Cost Analysis

While analyzing operating costs using the variable and fixed framework is insightful, its use by independent investment analysts is limited because disclosure of fixed and variable costs is not required by IFRS or US GAAP, nor is the disclosure of output volume. Additionally, many companies have several lines of business that have different volume, price, and cost structures, the details of which are lost in the presentation of highly aggregated financial statements that investors receive. However, in some industries, it is common for issuers to disclose these figures, or it is relatively easy for an analyst to make assumptions. A prominent example is oil and gas producers, for which we provide an example here.

Ribbon Energy Ltd. (“Ribbon”) is a fictional oil and natural gas company focused on the exploration and development of oil and natural gas reserves. Selected operating and financial data for the last 12 months are as follows:

	Last 12 Months
Sales volume (thousands of barrels of oil equivalent)	154,812
Average price, per barrel of oil equivalent	\$49.66
Total revenues (millions)	\$7,688
<i>Production costs, per barrel of oil equivalent:</i>	
Lease operating expense	11.21
Production and ad valorem taxes	3.40
Gathering and transportation expense	2.63
Production costs	\$17.24
<i>Other operating expenses (millions of USD):</i>	
Depreciation, depletion, and accretion (DD&A)	1,275
General and administrative expenses (G&A)	150

Production costs are variable costs stated on a per barrel of oil equivalent basis, while other operating expenses (DD&A and G&A) are fixed and stated on an aggregate dollar basis.

Ribbon’s management provides the following guidance for the next 12 months. Management does not provide guidance for prices (and therefore revenue), since it sells in a commodity market with volatile prices that are difficult to predict. Ribbon does not hedge. At the time guidance was provided, the average oil price per barrel was \$62.50.

	Next 12 Months
Sales volume (thousands of barrels of oil equivalent)	156,360–167,197
Production costs, per barrel of oil equivalent	\$17.34
<i>Other operating expenses (millions of USD):</i>	
Depreciation, depletion, and accretion (DD&A)	1,415
General and administrative expenses (G&A)	150

1. Ribbon’s contribution margin per barrel for the last 12 months is *closest* to:

- A. \$23.2.
- B. \$32.4.
- C. \$45.3.

Solution:

B is correct. Recall that the contribution margin per barrel is $P - VC$. The average price per barrel, P , in the past 12 months was \$49.66, and the variable cost per barrel, VC , was \$17.24. Therefore, the contribution margin was $\$49.66 - \$17.24 = \$32.42$.

A is incorrect, because this amount is the company's EBIT, or operating profit, per barrel.

C is incorrect, because it results from using the current rather than the last 12 months' average oil price per barrel to calculate the contribution margin.

2. At the high end of management's guidance, Ribbon's operating profit for the next 12 months is *closest* to:
- A. \$5,496 million.
 - B. \$5,986 million.
 - C. \$6,002 million.

Solution:

B is correct.

$$\text{Operating profit} = [Q \times (P - VC)] - FC$$

$$\text{Operating profit} = [167.197 \text{ million} \times (\$62.50 - \$17.34)] - (\$1,415 + \$150)$$

$$\text{Operating profit} = \$5,986 \text{ million}$$

A is incorrect, because this amount is the operating profit using the low end of guided sales volume of 156.360 million barrels.

C is incorrect, because it relies on last year's production costs of \$17.24 per barrel.

3. Some analysts covering Ribbon believe that volume will be lower than what management expects. Assuming a 5% reduction over last year's volume, Ribbon's operating profit for the next 12 months is *closest* to:
- A. \$5,077 million.
 - B. \$5,091 million.
 - C. \$5,776 million.

Solution:

A is correct.

The last 12 months' volume was 154.812 million. A 5% reduction is $154.812 \times 0.95 = 147,071$ million barrels.

$$\text{Operating profit} = [Q \times (P - VC)] - FC$$

$$\text{Operating profit} = [147,071 \text{ million} \times (\$62.50 - \$17.34)] - (\$1,415 + \$150)$$

$$\text{Operating profit} = \$5,076.7 \text{ million}$$

B is incorrect, because it uses last year's production costs.

C is incorrect, because it assumes a 5% increase rather than decrease in volume from the prior year.

4. Ribbon's degree of operating leverage for the next 12 months at the low end of management's guidance is _____ at the high end of management's guidance.

A. lower than

B. the same as

C. higher than

Solution:

C is correct.

Last 12 months' sales: \$7,688

Last 12 months' operating profit: \$1,244

Low end of guidance

Next 12 months' sales: $156.360 \times \$62.50 = \$9,773$

Next 12 months' operating profit: $\$9,773 - (156.360 \times 17.34) - 1,565 = 5,496$

Degree of operating leverage: $(5,496/1,244 - 1)/(9,773/7,688 - 1) = 1.95$

High end of guidance

Next 12 months' sales: $167.197 \times \$62.50 = \$10,450$

Next 12 months' operating profit: $\$10,450 - (167.197 \times 17.34) - 1,565 = 5,986$

Degree of operating leverage: $(5,986/1,244 - 1)/(10,450/7,688 - 1) = 1.85$

5.3. Natural and Functional Operating Cost Classifications and Measures of Operating Profitability

Rather than fixed and variable, IFRS and US GAAP require issuers to disclose operating costs using either a natural or a functional cost classification based on "historical and industry factors and the nature of the entity." Most issuers, including Warehouse Club Inc., choose a functional classification. This approach leads to all issuers' income statements appearing similar in structure, with lines related to functional areas such as "cost of sales," "selling, general, and administrative expenses," and "research and development," even though their business models (and fixed versus variable cost composition) can differ significantly. The functional classification of operating costs is used to calculate and distinguish key profitability measures used in analysis and forecasting, as shown in [Exhibit 5](#).

Exhibit 5: Measures of Operating Profitability

Gross Profit:	EBITDA:	Operating Profit:
Revenue	Revenue	Revenue
Cost of Sales	Cost of Sales	Cost of Sales
Gross Profit	Operating Expenses i.e. SG&A, R&D, etc.	Operating Expenses i.e. SG&A, R&D, etc.
	EBITDA	Depreciation and Amortization
		Operating Profit (EBIT)

Gross Margin:	EBITDA Margin:	Operating/EBIT Margin:
Gross Profit	EBITDA	Operating Profit (EBIT)
Revenue	Revenue	Revenue

Gross profit, EBITDA, and EBIT are measures of operating profitability that capture different functional costs in their definitions. Generally, a significant amount of cost of sales is variable, so gross margin is an approximate measure of contribution margin. Other operating expenses such as sales and marketing, general and administrative, and research and development costs tend to be largely fixed, though they may have a variable component such as sales commissions. Depreciation and amortization expenses are fixed unless the issuer uses units of output as the depreciation method. An issuer's notes to its financial statements contain useful information about the composition and nature of operating costs.

For most companies, the major driver of operating costs over the long run is output, even for fixed costs, because output growth often requires growth in assets, human capital, and purchased goods and services. Warehouse Club Inc., for example, might be able to grow membership fee revenues to an extent without incurring costs, but growing net sales requires more stores and employees. This has been true even for so-called asset-light companies in the internet and software industries, which have shown large increases in operating costs along with large increases in revenue over time. Since output or revenue is a major cost driver, analysts often express operating costs as a percentage of revenue. Additional considerations include industry profitability, **economies of scale**, and **economies of scope**.

Industry profitability will be discussed in detail in the next module, but recall from the earlier discussion on pricing power that prices relative to costs (i.e., profitability) are largely dictated by market structure and a company's competitive positioning. Since industry participants are competing in the same product and factor markets, competitive forces within them determine industry profitability in the long run; analysts should not analyze profitability only on a company-by-company basis.

Economies of scale refer to a decline in costs per unit as output grows and generally result from having fixed costs in the cost structure that are spread over more units of output. A company with entirely variable costs can also exhibit some economies of scale over time if it increases its bargaining power over suppliers as it grows, driving down variable costs per unit. Economies of scale are usually evident from the business model and can be found empirically by comparing a company's size in revenues with its operating costs as percentages of sales or margins.

Economies of scope refer to a decline in costs per unit as the number of product or business lines increases and generally result from shared costs between the product lines. Economies of scope are usually evident from the business model and can be found empirically by comparing the profitability of

an integrated company with that of a standalone company. Examples of economies of scope can be found in financial services, where there are customer, client service, compliance, technology, and back-office similarities (and thus cost efficiencies) across commercial banking, consumer banking, brokerage, asset management, payment processing, and investment banking. Economies of scope have resulted in large global firms competing in many lines of business.

Recall that Warehouse Club Inc.’s management claimed that it has economies of scale, saying that “low prices for quality products drive greater membership and net sales over time, leveraging our selling, general, and administrative expenses, reducing them as a percentage of net sales.” While this makes intuitive sense as there is probably a fixed amount of SG&A costs each year to operate a store, Elaine Nguyen must corroborate it empirically.

Warehouse Club Inc. Operating Profitability Analysis

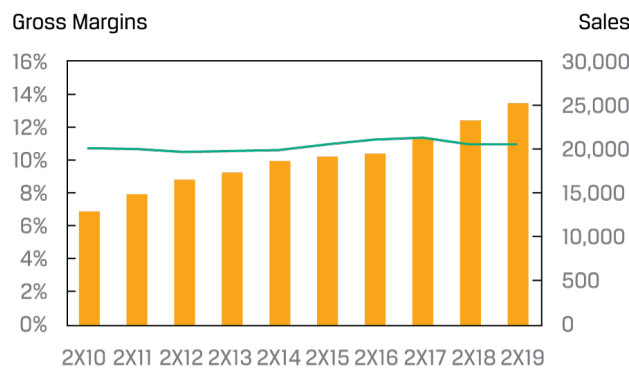
Warehouse Club Inc.’s operating costs are composed of merchandise costs (costs of sales), SG&A expenses, depreciation and amortization, and costs associated with opening new stores.

Based on her business model work, Elaine Nguyen believes the following about drivers for each of these operating costs.

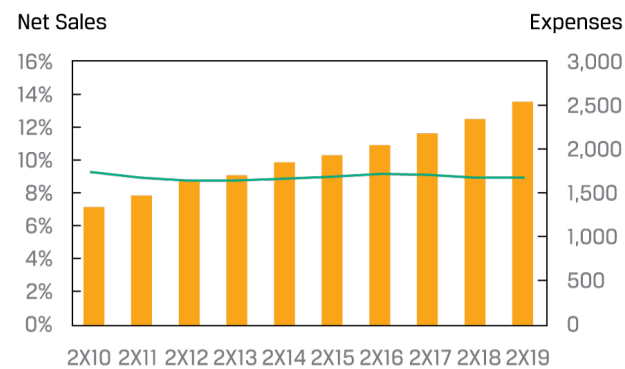
Operating Cost	Driver(s)
Merchandise costs (costs of sales)	Sales
SG&A expenses	Sales Number of employees and per-employee compensation
Depreciation and amortization	Long-term assets Useful lives
Store opening expenses	Store openings

Based on these drivers, Nguyen calculates Warehouse’s gross margin (1 – cost of sales as % of net sales), SG&A expenses as a percentage of net sales, gross fixed assets to depreciation and amortization (implied useful life), store opening expenses per new-store opening, and, finally, profit margins. Profit margins include membership fee revenues.

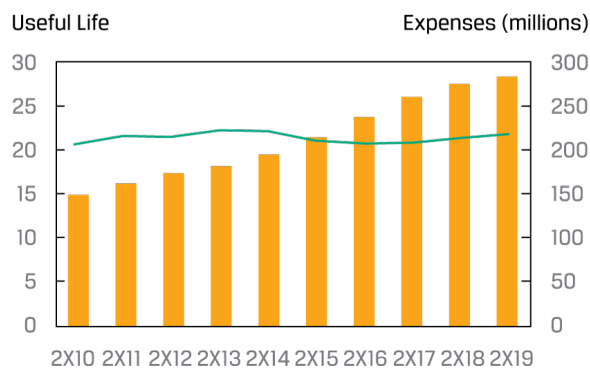
Warehouse Club Inc. Gross Margin (line, left) and Cost of Sales (bars, right, USD millions)



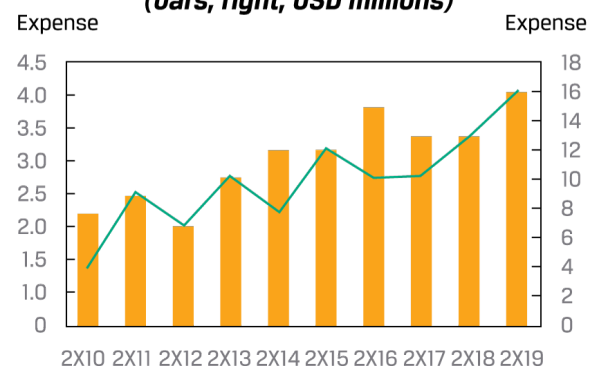
Warehouse Club Inc. SG&A % of Net Sales (line, left) & SG&A Expenses (bars, right, USD millions)



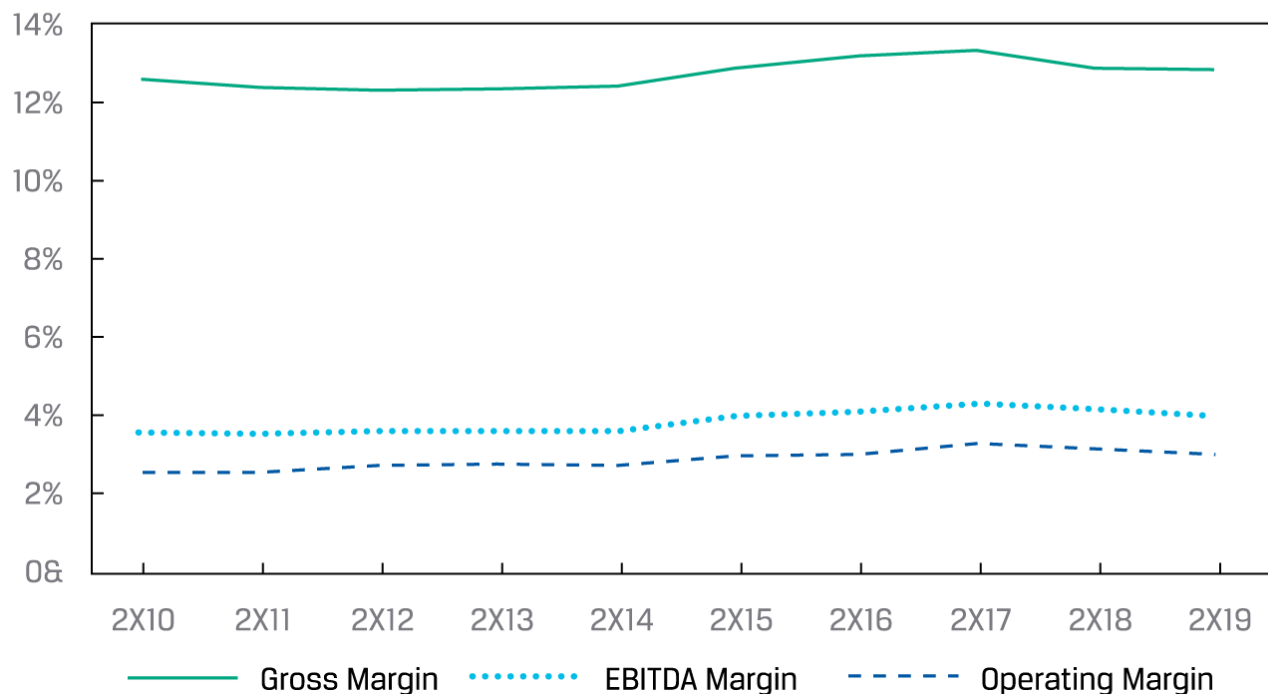
Warehouse Club Inc. Implied Useful Life (line, left) & D&A Expense (bars, right, USD millions)



Warehouse Club Inc. Store Opening Expense per Opening (line, left) & Store Opening Expense (bars, right, USD millions)



Warehouse Club Inc. Operating Profitability

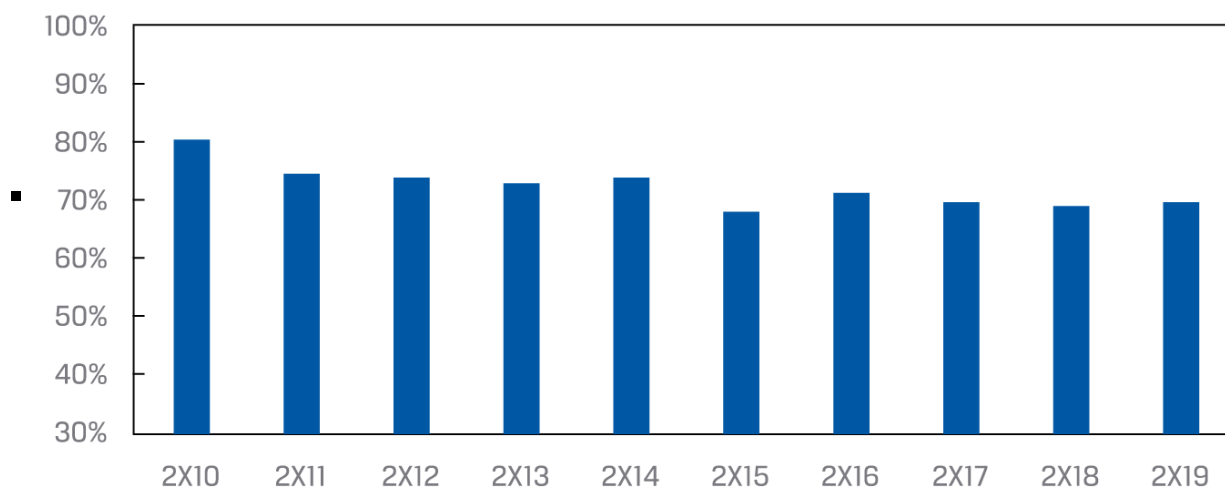


Nguyen makes the following observations:

- Operating costs have increased significantly on an absolute basis from 2X10 to 2X19, but on a relative-to-sales basis have remained stable over the same period, indicating that the driver of the increase has been net sales growth.
- One exception is store opening expenses per store opening, which have quadrupled from 2X10 to 2X19, though in aggregate remain a small cost (less than 1/100 of SG&A expenses). Note that these are operating costs related to store openings, such as employee training before opening, not the investing cost of land and fixed assets.
- SG&A expenses do not appear to be fixed relative to net sales, as they have remained roughly the same percentage of net sales over time.
- Margins have remained mostly stable, though they did increase from 2X15 to 2X17 before coming back down in 2X18 and 2X19. Margins are low for Warehouse, with EBITDA and operating margins of 4% and 3%, respectively, in 2X19.

- The amount of operating profit is similar to the amount of membership fee revenues each year. Nguyen creates the following chart:

Warehouse Club Inc. Membership Fee Revenue as % of Operating Profit



This chart implies that net sales from its stores—after deducting costs of sales, SG&A expenses, D&A, and store opening expenses—account for the remaining 30% of operating profit. This percentage is likely an understatement, since some amount of SG&A expenses is associated with membership fee revenues (marketing, customer service, payment processing, management), but it does align with Warehouse management’s assertion in its annual report that prices provide members with a significant return on membership: the company earns a thin margin on net sales, and it is membership fees that allow the company to run profitably.

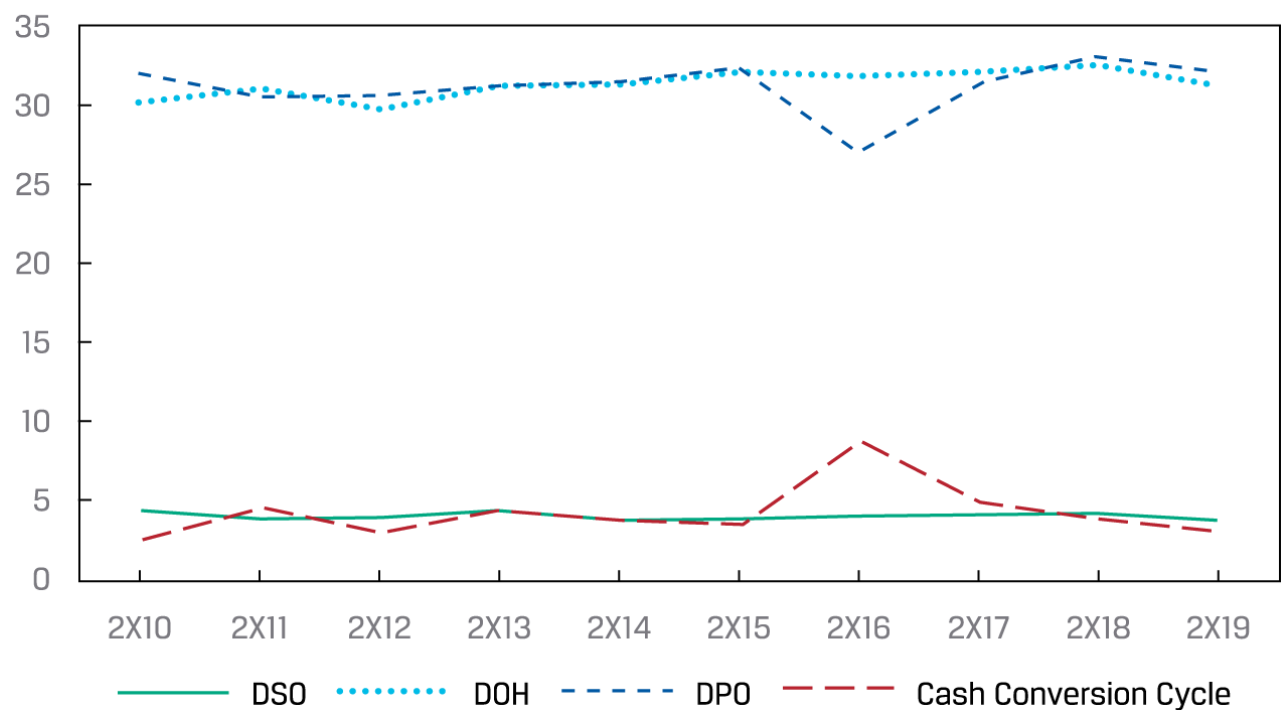
5.4. Working Capital

Recall from earlier modules on working capital that the primary measures of a company’s working capital management are activity ratios that determine its cash conversion cycle and the ratio of net working capital to sales. A short cash conversion cycle means that the company requires less external financing to fund operations, while net working capital requirements determine a minimum level of investment, in addition to capital investments, that cannot be distributed to investors. Negative net working capital means that suppliers are a source of financing.

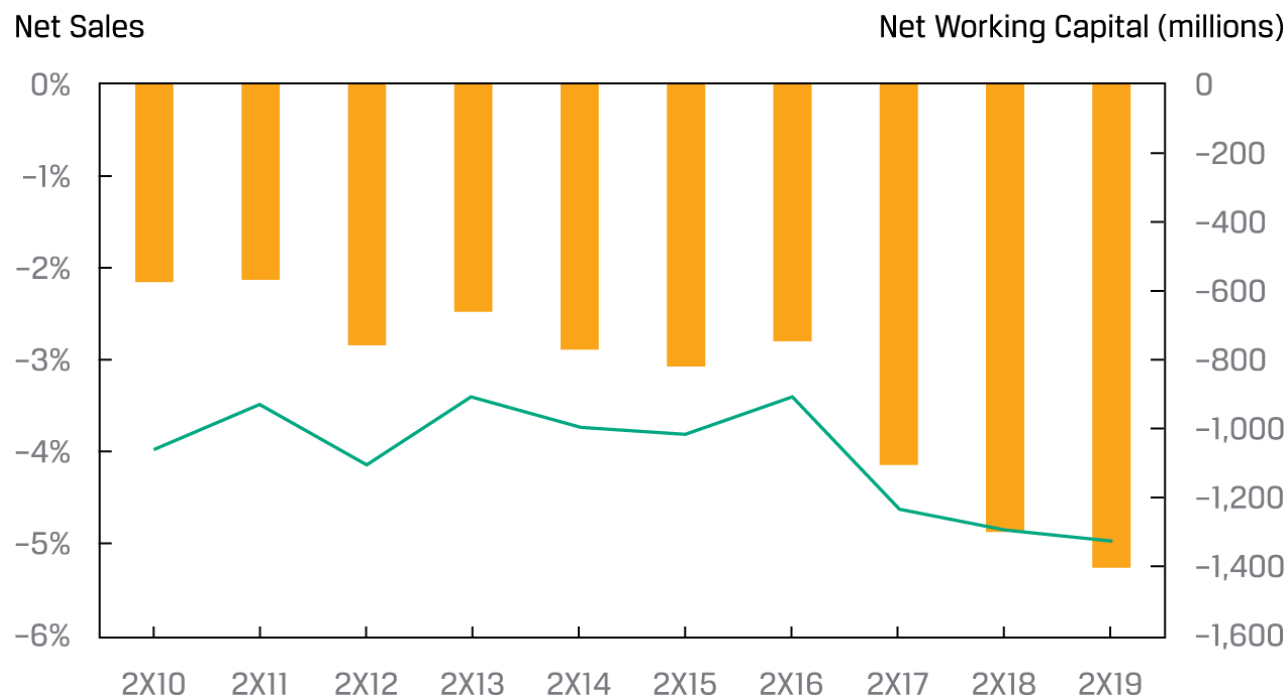
Warehouse Club Inc. Working Capital Management

Elaine Nguyen calculates the working capital measures for Warehouse.

Warehouse Club Inc. Activity Ratios and Cash Conversion Cycle (Days)



Warehouse Club Inc. Net Working Capital to Net Sales (line, left) and Net Working Capital (bars, right, USD millions)



Nguyen makes the following observations:

- The cash conversion cycle is short, less than a week, primarily as a result of a DSO of 4, because Warehouse’s customers pay in cash or with credit or debit cards that settle in less than five business days, and days of inventory on hand is nearly equal to days payable outstanding, indicating that inventory is financed by suppliers.
- Net working capital is negative, now down to -5.0% of net sales, indicating that it is a source of financing for the company.
- Generally, all the activity ratios have been stable since 2X10, except for a one-year increase in 2X16 from a decline in days payable outstanding. Nguyen looked into this increase in the notes to the financial statements in a historical annual report. The company indicated that it changed accounting information systems in early 2X17 and made accelerated payments to suppliers at the end of 2X16, in advance of the system change in case there were problems.

Question Set

Questions 1–4 relate to the working capital data below for Iliso Marketplace, as well as to Iliso's financial statements in the lesson "Determining the Business Model".

Working Capital	2X15	2X16	2X17	2X18	2X19
Accounts receivable, net	29	50	50	52	58
<i>DSO</i>	4	5	4	3	3
Inventories, net	190	250	325	384	439
<i>DOH</i>	37	36	32	32	30
Prepayments and other	23	32	35	56	59
<i>% of total revenue</i>	1%	1%	1%	1%	1%
Accounts payable	138	190	272	321	410
<i>DPO</i>	27	28	27	26	28
Taxes payable	2	8	10	12	29
<i>% of total revenue</i>	0%	0%	0%	0%	0%
Accrued expenses and other current	111	193	232	295	354
<i>% of total revenue</i>	4%	6%	4%	5%	5%
Net working capital	(8)	(59)	(104)	(136)	(237)
<i>NWC to sales</i>	0%	-2%	-2%	-2%	-3%
Cash conversion cycle	14	14	9	8	5

1. Compared with 2X18, Iliso's operating margin change in 2X19 is *closest* to:

- A. 0.07%.
- B. 0.57%.
- C. 80.56%.

Solution:

B is correct. Operating (EBIT) margin is Operating profit/Revenue:

$$2X18: 72/6,186 = 1.163\%$$

$$2X19: 130/7,526 = 1.727\%$$

Subtracting 2X18 EBIT margin from 2X19 EBIT margin gives:

$$1.727\% - 1.163\% = 0.564\%$$

A is incorrect, because it is the change in gross profit margin.

C is incorrect, because it is the year-over-year change in operating profit.

2. What is the *most likely* explanation for Iliso's improvement in inventory days on hand from 2X15 to 2X19?

- A. More customers paying in cash
- B. Strong growth in net retail sales
- C. Rising proportion of GMV from third-party merchant sales

Solution:

C is correct. With third-party merchant sales, Iliso acts as an agent for the third-party merchant and does not take control of inventory. Accordingly, as the proportion of third-party merchant sales rises in GMV and Iliso's revenues, the portion of revenues associated with inventories falls.

A is incorrect, because there is no indication that Iliso accepts cash on delivery (as is common in China and other geographies). It appears that, like other e-commerce platforms, Iliso accepts only online payments.

B is incorrect, because growth in retail sales is likely to grow accounts receivable as well; days sales outstanding should be unaffected.

3. Which of the following expenses are *most likely* to vary with output?

- A. depreciation
- B. costs of sales
- C. interest expense

Solution:

B is correct. Costs of sales are inventory costs that are recognized when a sale is made.

A is incorrect, because depreciation varies with fixed assets. This expense is a fixed cost in the short run. Recall that fixed operating costs do not change within a given range of output in the short run.

C is incorrect, because interest expense is a fixed cost component that arises from the company's capital structure, not its revenues.

4. Iliso doesn't disclose gross profit on third-party merchant sales. However, an analyst observes that a competing marketplace platform makes 85% gross profit margin on third-party merchant sales. Assuming that Iliso's gross profit margin is similar, what proportion of Iliso's gross profit *most likely* came from third-party merchant sales in 2X19?

- A. 7.6%
- B. 15.0%
- C. 28.1%

Solution:

A is correct.

Gross profit of third-party merchant sales = $85\% \times 190 = \$162$

Gross profit of Iliso = Total revenues – Costs of revenues = $\$7,526 - \$5,414 = \$2,112$

Proportion of gross profit arising from merchant sales = $\$162/\$2,112 = 7.6\%$

B is incorrect; it is the cost of revenues on third-party merchant sales.

C is incorrect; it is Iliso's gross profit margin in 2X19.

6. CAPITAL INVESTMENTS AND CAPITAL STRUCTURE

LEARNING OUTCOME

The candidate should be able to:

- ❑ evaluate a company’s capital investments and capital structure

6.1. Sources and Uses of Capital

Firms invest capital from debt and equity investors to earn returns above their investors’ required rates of return. An important consideration in company analysis is to evaluate whether required rates of return have been met or exceeded (i.e., whether economic value is created for investors) over the longer run and to assess risks and opportunities associated with the capital structure, such as the use of financial leverage. Before evaluation, a first step is to determine a company’s sources and uses of capital on a historical basis, as shown generically in [Exhibit 6](#).

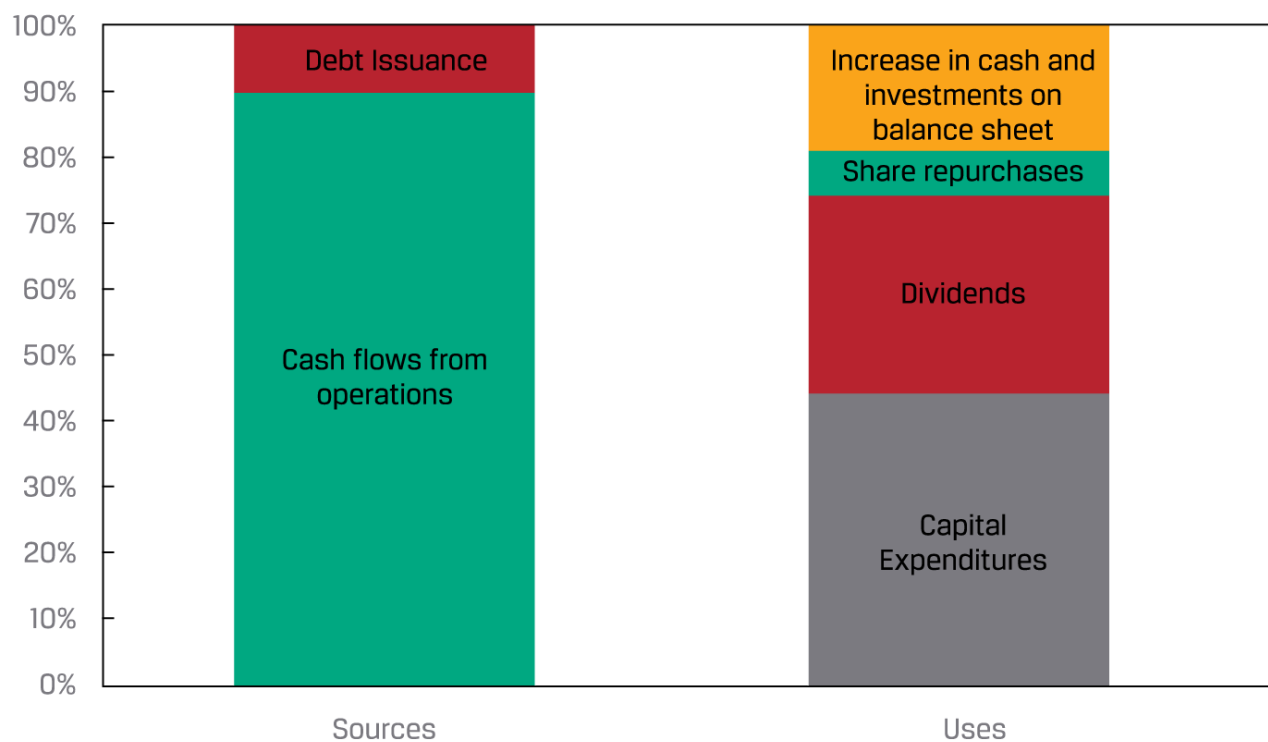
Exhibit 6: Sources and Uses of Capital

Sources of Capital	Uses of Capital
Cash flows from operations, including net working capital (if negative)	Cash and investments on hand
Debt issuance	Net working capital (if positive)
Equity issuance	Capital expenditures and additions to intangibles
Asset disposals	Acquisitions
	Debt paydown
	Dividends and share repurchases

Warehouse Club Inc.’s Sources and Uses of Capital

Elaine Nguyen calculates the following information on Warehouse over 2X10–2X19. Cash flows from operations include the effect of additional net working capital financing (i.e., Warehouse’s net working capital became more negative over time).

Warehouse Club Inc. Sources and Uses of Capital, 2X10–2X19



Nguyen makes the following observations:

- Capital expenditures and returns to shareholders (dividends and share repurchases) are almost equal, which is surprising considering that new-store openings performed well; Nguyen expected the company to devote a greater amount to capital expenditures for new stores.
- The company has made no acquisitions.
- Even after returns to shareholders, the company seems to prefer equity financing to debt financing. This preference is logical considering the company's thin operating margin, though the fact that it primarily sells a wide range of necessities reduces its sales risks (i.e., from changes in the business cycle and in technology or fashion) and membership fees are a source of predictable income.
- The fact that debt issuance is smaller than the increase in cash and investments on the balance sheet indicates that the company's net debt has decreased over time, implying a decline in financial leverage.

6.2. Evaluating Capital Investments and Capital Structure

Returns on invested capital over the longer run, compared with investors' required rates of return, can be used to evaluate whether management has used investors' capital wisely. Recall that independent investment analysts do not have access to the same information that management uses to assess individual projects with IRRs and NPVs, so analysts have to use more aggregated measures to discern whether there have been worsening or improving trends in value creation.

Risks related to the capital structure can be measured using leverage and coverage ratios, credit ratings by third-party rating agencies, and the sensitivity of net income to changes in operating income known as the **degree of financial leverage**, as shown in [Equation 3](#). The degree of financial leverage is increased by higher interest expenses that are fixed with respect to operating income. The degree of financial leverage is analogous to the degree of operating leverage but is determined by financing costs rather than fixed operating costs.

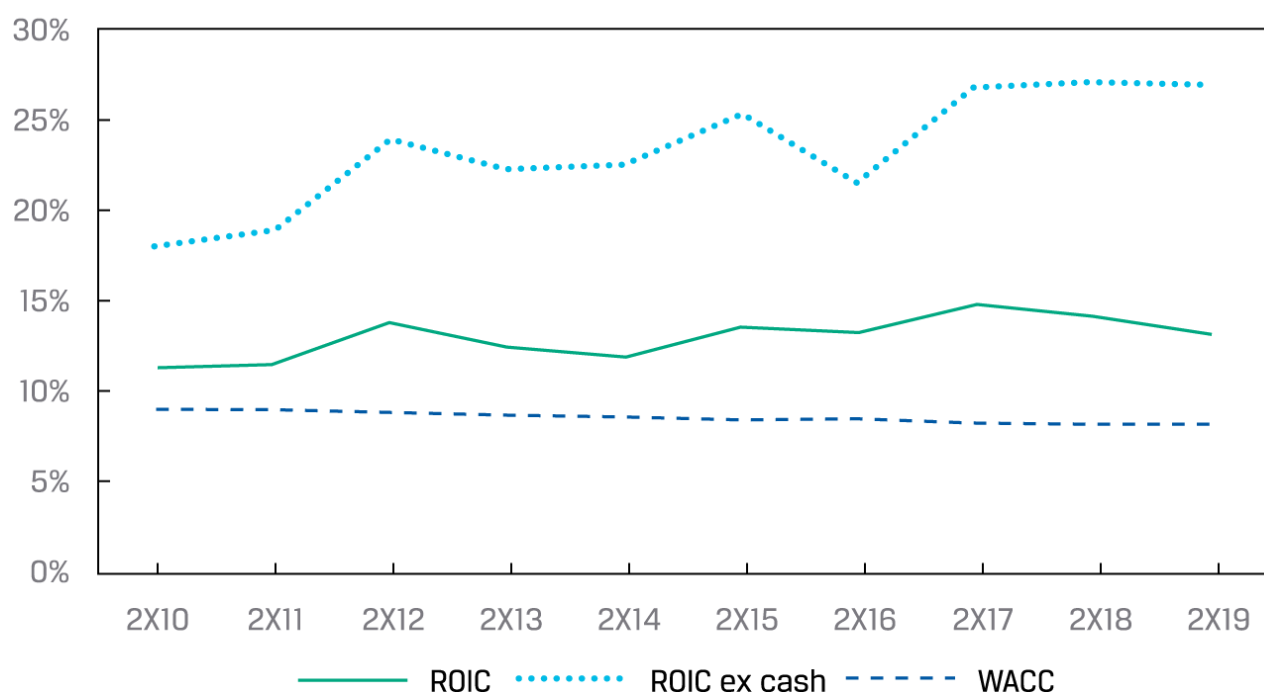
Equation 3

$$DFL = \% \Delta \text{ Net income} / \% \Delta \text{ Operating income}$$

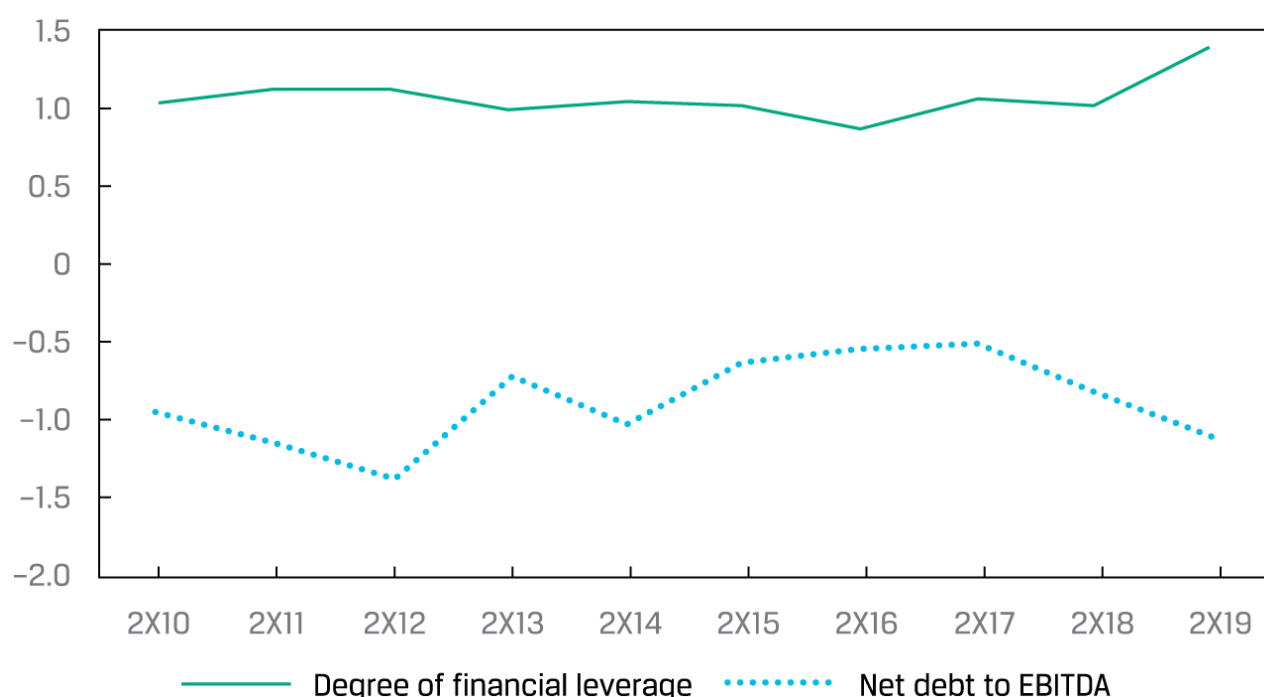
Evaluating Warehouse Club Inc.'s Capital Investments and Capital Structure

Elaine Nguyen calculates the following information on Warehouse over 2X10–2X19. Additionally, the company's interest coverage ratio (EBIT to interest expense) has been over 20 times and recently reached 30 times in 2X19, and the company's credit is rated investment grade by major credit rating agencies.

Warehouse Club Inc. Return on Invested Capital and Weighted Average Cost of Capital



Warehouse Club Inc. Financial Leverage Ratios



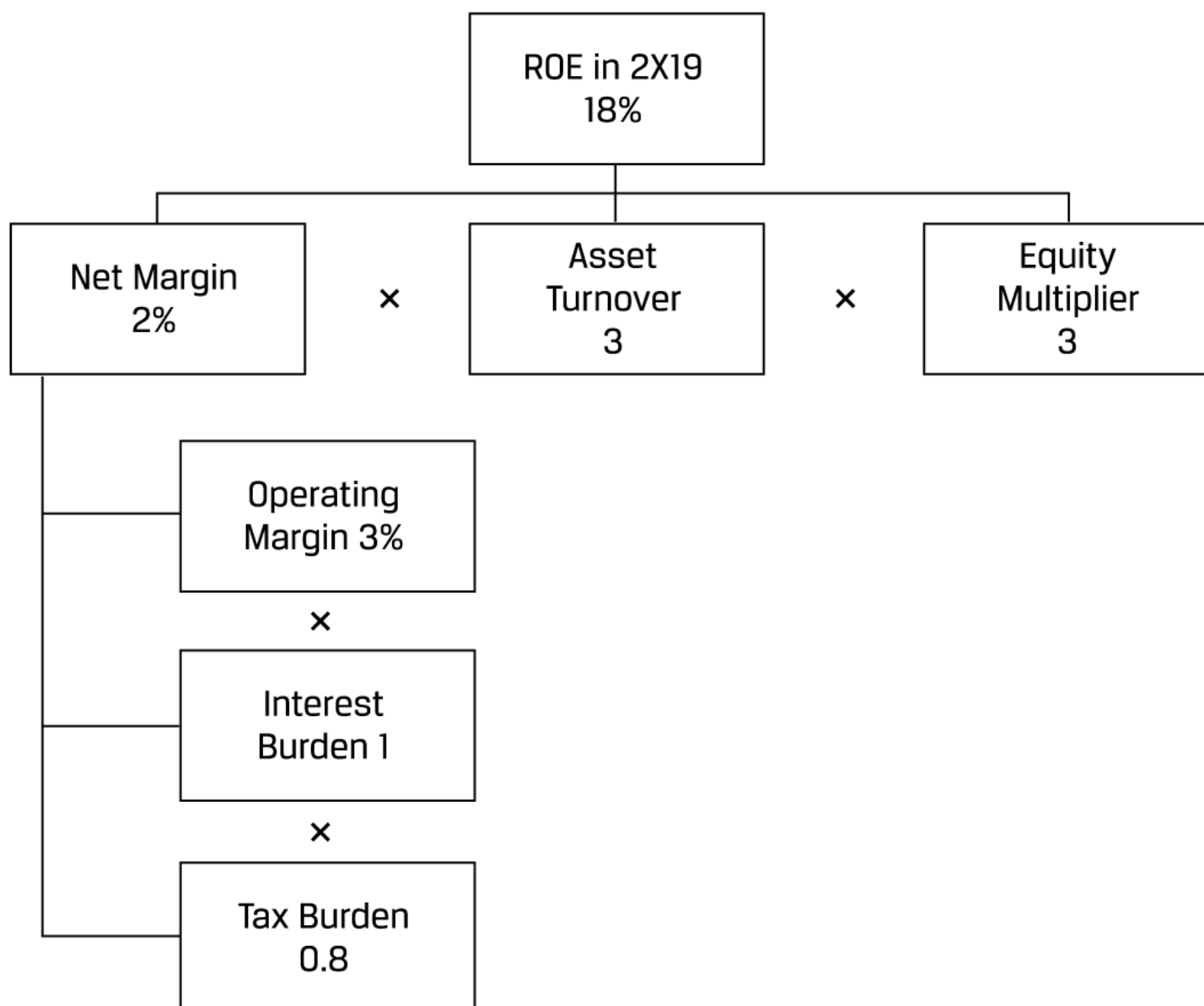
Nguyen makes the following observations:

- The company has created value for investors, with a spread of ROIC over WACC of 200–700 basis points since 2X10. WACC has declined slightly over time as interest rates have fallen; this calculation assumes a constant required rate of return on equity of 9%.
- The company's returns on capital are far higher than its low operating and net margins would suggest, owing to its high asset turnover and negative net working capital as a low-cost means of financing from suppliers, requiring less capital from investors.
- The company has a conservative capital structure, with negative net debt and a degree of financial leverage close to 1.0 because interest expense is largely covered by interest

income on cash holdings (i.e., net and operating income move together).

- The capital structure conservatism is evidenced in the counterfactual ROIC, which excludes cash and investments on the balance sheet. While an exaggeration (the company would need *some* cash on hand), ROIC could be almost 15 percentage points higher if the company reduced net cash by investing in value-creating projects or by returning capital.

Unlevered returns, as measured by ROIC or return on assets, are augmented by financial leverage to produce levered returns, or return on equity, which was introduced in an earlier module. We can use return on equity and its decomposition as a comprehensive measure of profitability for an issuer. The ROE decomposition for Warehouse Club Inc. illustrates the drivers of the company's levered returns, which are well in excess of its net margin.



Question Set

Questions 1–4 relate to Iliso's financial statements in the lesson "Determining the Business Model".

1. In 2X19, the market value of Iliso's equity was \$1,178 million, and the book value of debt, which was approximately equal to the market value, was \$144 million. Assuming a cost of debt of 5.5%, cost of equity of 12%, tax rate of 24%, and price-to-book ratio of 1.0, Iliso's WACC is *closest* to:

A. 11.15%.

B. 11.30%.

C. 12.00%.

Solution:

A is correct. Recall that:

$$WACC = [\text{weighting of debt} \times \text{gross cost of debt} \times (1 - \text{tax rate})] + (\text{weighting of equity} \times \text{cost of equity}).$$

Total capital is $\$1,178 + \$144 = \$1,322$ million. Weighting of debt is therefore $\$144/\$1,322 = 10.89\%$, and Weighting of equity is $1 - \text{Weight of debt} = 89.11\%$.

Therefore, $WACC = [10.89\% \times 5.5\% \times (1 - 24\%)] + (89.11\% \times 12\%) = 11.15\%$.

B is incorrect, because it incorrectly calculates WACC using a pre-tax rather than after-tax cost of debt.

C is incorrect; it is the cost of equity.

2. Why might Iliso's WACC be significantly higher than Warehouse Club's WACC, as calculated earlier in the module?

A. Warehouse Club's business model is less risky than e-commerce.

B. Warehouse Club's capital structure, with degree of financial leverage close to 1.0, is less conservative than Iliso's.

C. Iliso is an established retailer with an investment-grade rating.

Solution:

A is correct. The cost of equity is the compensation that equity investors demand for the business and financial risks of investing in an issuer's equity securities. Iliso's cost of equity is 12%, whereas Warehouse Club's is 9%, reflecting Warehouse Club's longer operating history, more proven business model, and competitive positioning.

B is incorrect, because Warehouse Club's capital structure, with degree of financial leverage of 1.0, would imply almost 100% weighting of equity, whereas Iliso has an 11% debt weighting and therefore Warehouse's capital structure is *more* conservative.

C is incorrect, because an investment-grade rating would reduce Iliso's cost of debt and WACC.

3. Iliso's degree of financial leverage in 2X19 is *closest* to:

A. 0.77.

B. 1.13.

C. 1.84.

Solution:

B is correct, calculated as $\text{Change in net income} / \text{Change in operating income} = (100/52 - 1) \text{ divided by } (130/72 - 1) = 1.146$.

A is incorrect, because it is incorrectly calculated as $2X19 \text{ Net income} / \text{Operating income} = 100/130 = 0.77$.

C is incorrect, because it is Iliso's equity multiplier: $\text{Total assets} / \text{Equity} = 2,168 / (592 + 586) = 1.84$.

4. Explain why Iliso's ROE differs from Warehouse Club's ROE.

Solution:

Iliso's ROE is lower than Warehouse Club's ROE.

$\text{ROE} = \text{Net margin} \times \text{Asset turnover} \times \text{Equity multiplier}$

$\text{ROE} = 1.3\% \times 3.5 \times 1.84 = 8.37\%$, which is less than half of Warehouse Club's 18% ROE.

This primarily results from Iliso's lower net margin (1% versus 2%) and an equity multiplier that is 40% lower (1.84 versus 3), which is somewhat offset by Iliso's higher asset turnover (3.5 versus 3).

PRACTICE PROBLEMS

The following information relates to questions 1-7

Joshua Hu, a research analyst, is initiating coverage on several companies in the ocean freight shipping industry. OldShips is a mature company with high fixed costs and a high capital expenditure to sales ratio because it owns and operates its own fleet of ships. CleanYards is a technologically advanced, sustainable shipyard with a focus on specialized repairs and ship construction. Hu has compiled the following data for the two companies:

	OldShips	CleanYards
Degree of Financial Leverage (DFL)	2.0	2.0
Degree of Operating Leverage (DOL)	1.0	2.0
Asset turnover ratio	0.84×	0.42×
DSO	27	98
DOH	12	46
DPO	55	40

NewShips, a third company, is a web-based shipping technology platform that connects ship operators such as OldShips with customers in a wide variety of industries who need ocean freight shipping. NewShips' customers place orders online and pay for freight to be placed on a container to any destination in the world. NewShips' partners, like OldShips, provide vessels on both long- and short-term charters.

In 2X19, NewShips' platform brokered orders for 900,000 twenty-foot equivalent unit (TEU) containers in aggregate, with an average gross freight rate of USD3,848 per TEU. On average, NewShips' commission, which it receives as a broker from the customer, was 6% of the freight rate.

- Which of the following is *most likely* to be an element of Hu's subsequent reports?
 - Industry size, growth rate, and key drivers
 - Summary of changes to the recommendation
 - Ownership structure and management composition
- Shipping is a highly cyclical industry. Which of the following companies *most likely* has the most ability to ramp up capacity to meet increased demand for shipping services?
 - OldShips
 - NewShips
 - CleanYards
- In 2X19, NewShips' revenue was *closest* to:
 - \$173 million.
 - \$1.46 billion.
 - \$3.63 billion.
- Assuming both companies have similar equity multipliers, OldShips is *most likely* to have a:

- A. much lower ROE than CleanYards.
 - B. similar ROE to CleanYards'.
 - C. much higher ROE than CleanYards.
5. When conducting company analysis, which of the following is *least* relevant to determining a company's business model?
- A. Company annual report describing the firm's product lines
 - B. Comments by management about competing products and substitutes
 - C. Industry white papers regarding the company's product pricing strategy
6. OldShips' cash conversion cycle (CCC) is:
- A. lower than CleanYards'.
 - B. similar to CleanYards'.
 - C. higher than CleanYards'.
7. The first step in reviewing a company's capital investments is to:
- A. evaluate management's investments.
 - B. identify operating cash flow drivers.
 - C. determine the major sources and uses of cash.
-

SOLUTIONS

1. B is correct. A subsequent company research report focuses on updating investors on new information or changes in an analyst's price target and recommendation.

A and C are incorrect, because in-depth information regarding industry size, growth rate, and key drivers or ownership structure and management composition is more suited to an initiation report.

2. B is correct. Only NewShips has a capital-light business model. Both OldShips and CleanYards require substantial capital investment in order to increase supply capacity. New ship construction (for OldShips) and acquiring new yards (for CleanYards) take time and cannot be met immediately.

3. A is correct. NewShips acts as a broker and earns 5% of the freight rate in commissions. In 2X19, NewShips brokered 900,000 TEUs at an average freight rate of 3,848 per TEU, earning a 5% commission: $900,000 \times 3,848 \times 0.05 = \173 million.

4. B is correct; $ROE = DTL \times \text{Asset turnover} \times \text{Equity multiplier}$.

OldShips has half the degree of CleanYards' total leverage. However, its asset turnover is double CleanYards' (0.84x versus 0.42x). Since the equity multiplier is the same, their ROEs must be similar.

Degree of total leverage (DTL) = $DFL \times DOL = (\% \Delta \text{ Net income} / \% \Delta \text{ Operating income}) \times (\% \Delta \text{ Operating income} / \% \Delta Q)$.

Thus, OldShips' $DTL = 2 \times 1 = 2$, whereas CleanYards' $DTL = 2 \times 2 = 4$.

A and C are incorrect, because both companies must have similar ROEs.

5. B is correct. Competing products and substitutes are analyzed in Porter's Five Forces. This analysis is usually performed to better understand a company's industry and its competitive positioning. Management comments are unlikely to guide the analyst to better understand the company's business model.

A and C are incorrect, because both are relevant sources for an analyst in determining the company's business model.

6. A is correct.

$CCC = DOH + DSO - DPO$, so:

OldShips' $CCC = 12 + 27 - 55 = -16$ days

CleanYards' $CCC = 46 + 98 - 40 = 104$ days

OldShips, in fact, has a negative cash conversion cycle. OldShips takes advantage of the favorable credit terms granted by its suppliers.

B and C are incorrect, because OldShips' CCC is lower than CleanYards'.

7. C is correct. Before a review of a company's capital investments, the first step is to determine the company's sources and uses of capital on a historical basis.

A and B are incorrect, because both take place after the analyst has identified the company's sources and uses of cash and determined what to focus on.

Industry and Competitive Analysis

LEARNING OUTCOMES

The candidate should be able to:

- ☐ describe the purposes of, and steps involved in, industry and competitive analysis
- ☐ describe industry classification methods and compare methods by which companies can be grouped
- ☐ determine an industry's size, growth characteristics, profitability, and market share trends
- ☐ analyze an industry's structure and external influences using Porter's Five Forces and PESTLE frameworks
- ☐ evaluate the competitive strategy and position of a company

1. INTRODUCTION

It is essential for analysts to understand the industry context for a company's financial performance and its evaluation. For example, is a company's revenue growth driven by a macroeconomic or industry-wide factor (such as an economic expansion) or a company-specific factor (such as increasing market share)? The answer has important implications for forecasting and valuation.

In this learning module, we discuss how an industry is defined and address the challenges associated with grouping companies that operate in multiple industries. Next, we introduce methods to survey an industry in terms of size, profitability, and market share trends. Further analysis includes frameworks to interpret a competitive environment using Porter's Five Forces and external trends using PESTLE analysis. Finally, we combine these approaches to understand a company's competitive strategy and its position relative to its industry peers.

Learning Module Overview

- Analysts use industry and competitive analysis to understand an industry's structural factors, relative competitive strengths and weaknesses, and their contribution to a company's economic profits.
- An analyst's first step in an industry and competitive analysis is to define the boundaries of the industry in question and its constituents. Industries are commonly defined as companies that sell similar products or services. However, this definition can be challenging to apply in practice, as companies can sell diverse products or services across many industries and product similarity is subjective.
- To define an industry, analysts can use third-party classification systems such as GICS but need to be aware of the methodologies and limitations of these systems.
- After defining the industry, the analyst surveys the industry by estimating its total annual sales, historical growth rate, and profitability metrics and then determining market shares and trends of key players.
- Industry metrics can be compared against broader economic trends during recessions or expansions to determine the industry's level of maturity, sensitivity to business cycles, and competitive rivalry.
- Analysts use Porter's Five Forces to understand an industry's structure, as defined by the level of competitive rivalry, through an assessment of the threat of new entrants, the threat of substitutes, customer bargaining power, and the bargaining power of suppliers.
- To supplement this research, analysts use PESTLE analysis to understand the external influences on and potential changes to an industry's growth rate and market share dynamics.
- To evaluate a company's competitive strategy, an analyst should determine whether the strategy creates a defense against industry forces and is aligned with the external forces acting on the industry, and whether the company has the resources and capabilities necessary to execute it.

Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of this learning module.

1. When assigning a company to an industry, which of the following is a limitation of using third-party industry classification schemes?
 - A. Currency differences
 - B. Single-product companies
 - C. Strictly hierarchical taxonomies

Solution:

C is correct. Commercial classification schemes such as GICS, IBC, and TRBC are examples of strictly hierarchical taxonomies that classify a company to a single group, regardless of whether the company sells multiple types of products or services.

2. A factor that determines sensitivity to the business cycle is *most likely*:
 - A. customer migration to substitute products.
 - B. interest rate exposure of the business model.
 - C. growth rates in line with broader economic activity.

Solution:

B is correct. Factors that determine sensitivity to the business cycle include the degree to which sales are discretionary or necessary for consumers, pricing power, the interest rate exposure of the business model, and whether the product is a durable or capital good versus a recurring purchase such as consumables and subscription services.

A and C are incorrect, as they reflect features of a mature industry but are not necessarily more cyclical.

3. A common measure of industry concentration is:
 - A. Porter's Five Forces.
 - B. the PESTLE framework.
 - C. the Herfindahl-Hirschman Index.

Solution:

C is correct. The Herfindahl-Hirschman Index (HHI) is a common measure of industry concentration that is calculated as the sum of the squares of competitor market shares. Porter's Five Forces model evaluates an industry's level of competitive rivalry and profitability. The PESTLE framework is more concerned with an industry's growth rate and market share dynamics.

4. Identify the following statement as true or false. Justify your answer.

Some of the forces included in Porter's Five Forces framework are the threat of new entrants, sensitivity to the business cycle, and the bargaining power of customers.

Solution:

False. Porter's Five Forces model uses the threat of substitutes, the threat of new entrants, the bargaining power of customers, and the bargaining power of suppliers to determine the rivalry among existing competitors. Sensitivity to the business cycle is *not* one of the forces.

5. The price competition historically demonstrated by automakers and aircraft manufacturers is *best* described as an example of:

- A. the threat of substitutes.
- B. the bargaining power of suppliers.
- C. the rivalry among existing competitors.

Solution:

C is correct. Despite relatively low risks from the other Five Forces, automakers and aircraft manufacturers compete fiercely on price, offering promotions and generous financing and warranty terms in an attempt to capture every sale.

6. Standardization of a product will _____ (increase/decrease/have no effect on) the bargaining power of customers.

Solution:

Standardization of a product will *increase* the bargaining power of customers. For example, oil refiners pay for crude oil based on its grade, not on its specific oil producer; crude oil within a grade is interchangeable.

7. Impression Ltd. is a fictional company that designs, manufactures, and sells skin care and beauty products. A PESTLE analysis for Impression Ltd. would *most likely* identify which of the following?

- A. A customer of Impression Ltd., a large e-commerce retailer, is acquiring a private-label manufacturer of skin care products.
- B. Increased discussion by legislatures around instituting an excise tax on disposable plastic packaging as part of a package of carbon taxes
- C. An online, direct sales competitor has launched a suite of photo filters that integrate with leading social media apps based on its beauty products.

Solution:

B is correct. PESTLE analysis is concerned with identifying and evaluating *external* forces on an industry, which include political forces such as a packaging tax. A and C are incorrect, as they are competitor moves within the beauty industry.

8. As a venture capital investor, you are on the board of Ridge Inc., a fictional company that is entering the auto industry, which is characterized by high capital intensity, minimal switching costs for customers because regulations require standardization of many features, and price consciousness of customers except among a relatively small percentage of affluent consumers.

Based on these observations, recommend and justify a competitive strategy to the rest of Ridge Inc.'s board.

Solution:

A cost leadership strategy is one competitive strategy that might be successful for Ridge Inc. Since most customers are price conscious and face minimal switching costs between automakers or models, a low selling price can be an effective way to gain market share. Combined with high capital intensity, such a strategy might enable Ridge to operate at an economic profit if it produces and sells enough volume and maintains fixed-cost discipline to keep unit costs low. Other competitive strategies that might be successful for Ridge include a differentiation strategy aimed at luxury or performance customers.

2. USES OF INDUSTRY ANALYSIS

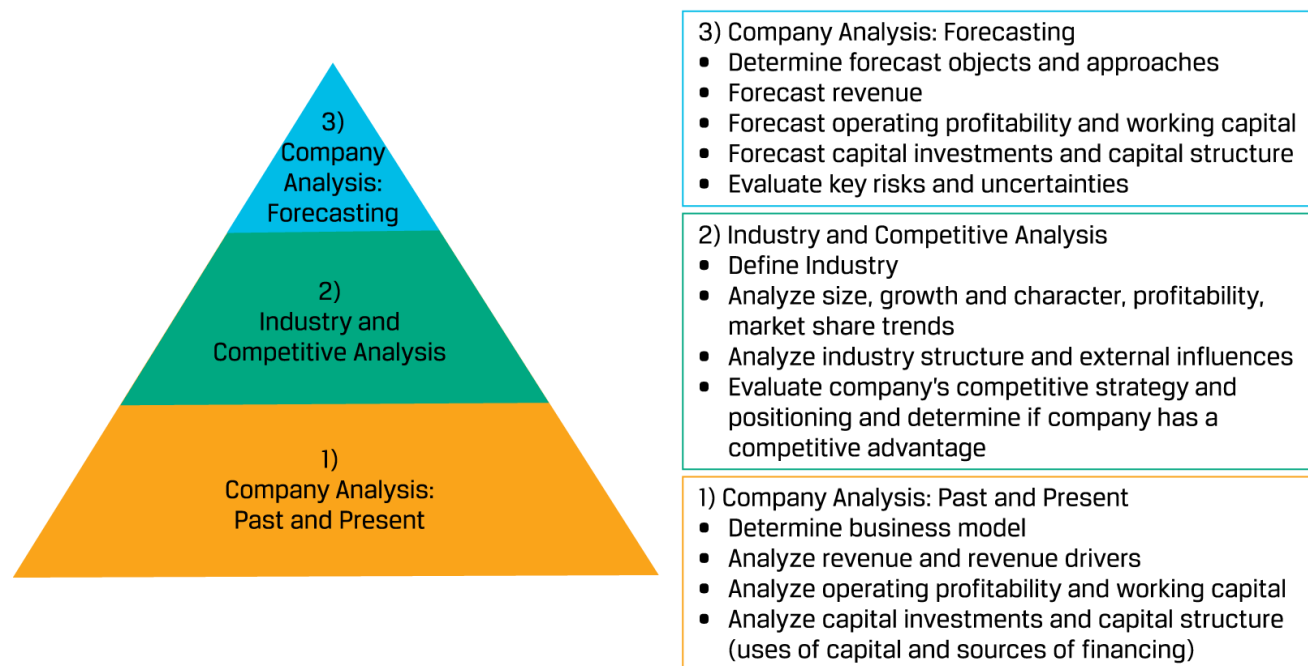
LEARNING OUTCOME

The candidate should be able to:

- describe the purposes of, and steps involved in, industry and competitive analysis

The next step in our company and industry analysis framework ([Exhibit 1](#)), introduced in the last module, is industry and competitive analysis, which involves the study of the drivers of an industry's size, profitability, and market shares and the evaluation of a company's competitive positioning in its industry.

Exhibit 1: Company and Industry Analysis Framework



2.1. Why Analyze an Industry?

Many industry participants have similar business models. They often compete in the same or similar product markets as sellers and in factor markets as buyers, so they tend to be exposed to the same demand and supply opportunities and risk factors. As will be discussed later with Porter's Five Forces model, these industry structural factors result in profitability differences by industry in the long run, with company-specific factors like business model variation, competitive strategy, size, and execution creating variance around the industry median.

[McGahan and Porter \(1997\)](#) estimated the relative importance of industry versus company-specific effects on both the emergence of economic profits (ROIC in excess of WACC) and their sustainability over time, finding that industry was the most important factor in the sustainability of economic profits. Importantly, they found that company-specific effects were much larger for low performers than for high performers in an industry, suggesting that industry and competitive forces can act as a ceiling on a company's returns but not as a floor.

A useful way to think about this dynamic is that competition "pulls" company profitability to an industry base rate over time. Industry and competitive analysis is used to estimate that base rate and its determinants, to form a forward-looking perspective on potential structural changes, and, finally, to analyze firms' relative strengths and weaknesses to determine their position against the base rate. It is important to emphasize, though, that these measures and outcomes are dynamic and that some

companies have outperformed or underperformed their industry in economic profitability for many years.

2.2. Improve Forecasts

Competitive forces from industry incumbents, substitutes, suppliers, and customers discipline companies' prices and costs, market share, and thus profitability. By taking an industry perspective and better understanding these drivers—as well as building a database of past competitive actions and strategies by firms in an industry and their track records of success—analysts can sharpen their forecasts and alertness to what is most important. [Guan, Wong, and Zhang \(2014\)](#) found that sell-side investment analysts who covered both companies and their suppliers made more accurate earnings forecasts than those who did not cover the suppliers. Without a broader perspective, an analyst may underrate competitive forces and overrate the degree to which a company controls its destiny.

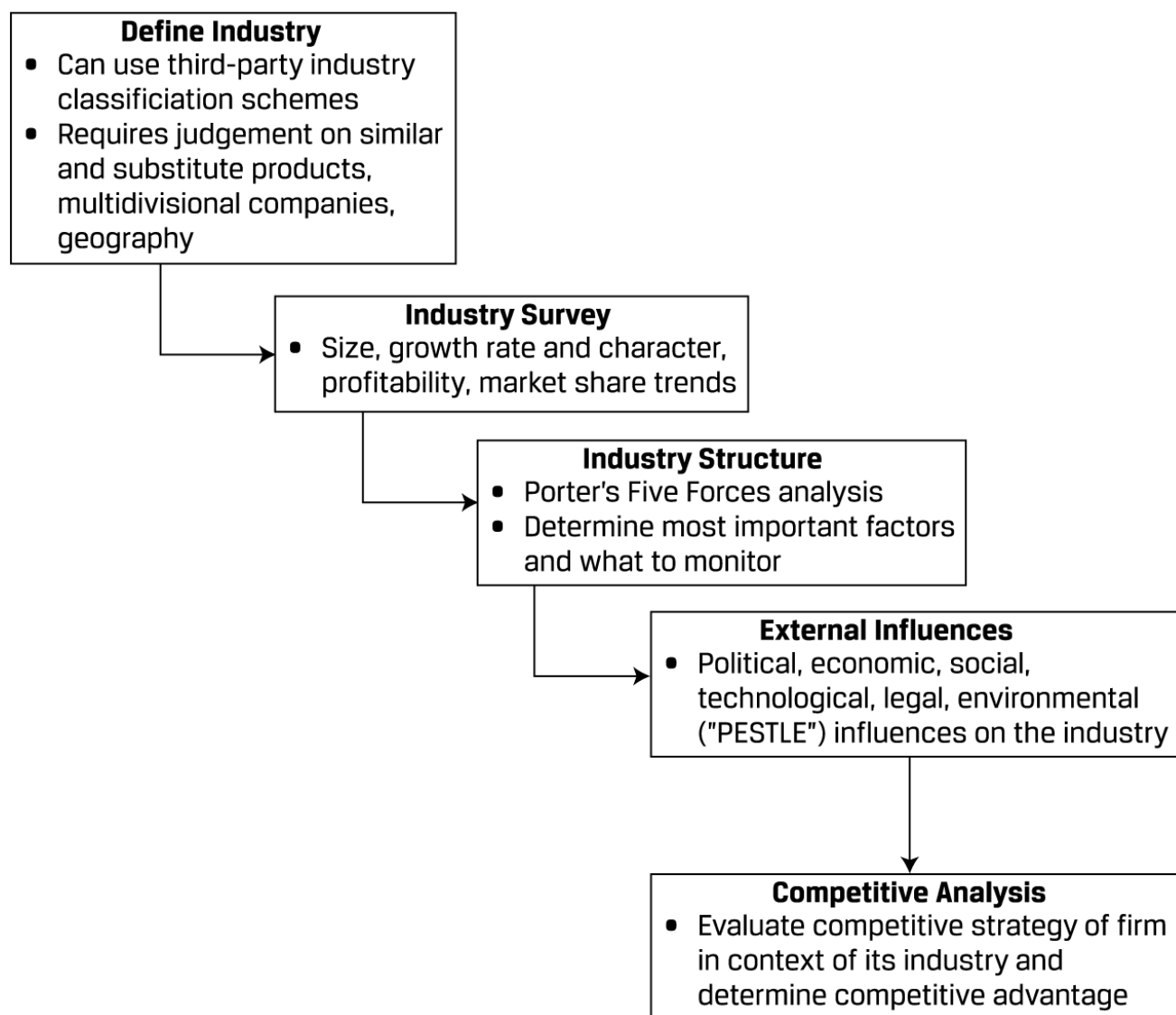
2.3. Identify Investment Opportunities

Industry analysis may uncover an attractive investment candidate that an analyst was previously unaware of or had not fully appreciated until assessing its strengths and weaknesses relative to industry peers and competitors. In addition, some investors may conclude that it is the industry, not any specific company, that they want exposure to, and in fact would like to diversify away the company-specific risks. These investors might take a basket approach by investing in several companies with position sizes scaled by size, liquidity, or relative attractiveness.

2.4. Industry and Competitive Analysis Steps

We approach industry and competitive analysis through the process shown in [Exhibit 2](#), which begins with defining an industry and ends with evaluating a company's competitive positioning in the context of that industry.

Exhibit 2: Industry and Competitive Analysis Steps



Question Set

1. Over the long run:

- A. business cycles result in profitability differences by industry.
- B. company-specific factors such as competitive strategy result in profitability differences by industry.
- C. structural factors result in profitability differences by industry.

Solution:

C is correct. Structural factors, such as Porter's Five Forces, drive profitability and profitability differences by industry in the long run. Company-specific factors result in company profitability levels around the industry median, while changes in the business cycle drive *short-run* profitability for both industries and companies.

2. Which of the following is an industry, *not* a company-specific, attribute?

- A. Competitive strategy
- B. Business model variation
- C. Sensitivity to the business cycle

Solution:

C is correct. Sensitivity to the business cycle is an example of a factor that influences an entire industry.

A and B are examples of company-specific attributes, which also include company size and strategy execution.

3. Explain why industry has a significant effect on company growth and profitability.

Solution:

Companies in the same industry compete in similar product markets as sellers and in factor markets as buyers, so they tend to be exposed to the same demand and supply opportunities and risk factors. For example, while a restaurant company that sells chicken has a different product than a restaurant that sells pastries, both are competing for a similar customer need (prepared food), employ similarly skilled workers, and have capital equipment and real estate.

3. INDUSTRY CLASSIFICATION

LEARNING OUTCOME

The candidate should be able to:

- ☐ **describe industry classification methods and compare methods by which companies can be grouped**

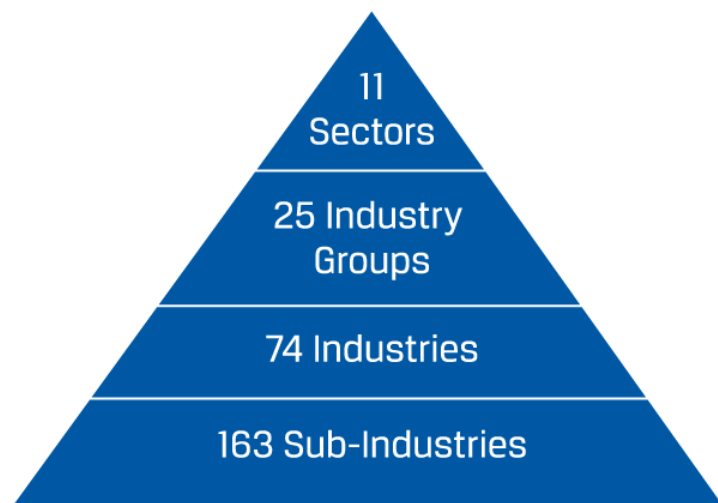
An industry is commonly defined as companies that sell similar products or services, from the perspective of a customer. Challenges in defining an industry include whether to include substitutes, classifying companies that operate in multiple industries, geographical considerations, and making updates since business models are always evolving. To help analysts contend with these difficulties, third-party organizations maintain industry classification schemes that are widely used in investment management.

3.1. Third-Party Industry Classification Schemes

Early third-party industry classification schemes such as SIC, NACE, and ISIC were devised by government agencies, tended to be country-specific, and grouped companies by their production characteristics into industries such as agriculture, manufacturing, distribution, retail, and services. The schemes were not updated frequently and became less useful as new technologies and business models emerged and both issuers and investors became increasingly global. These legacy schemes used by investment analysts were largely replaced with commercial schemes such as the Global Industry Classification Standard (GICS) by MSCI and S&P Dow Jones Indices, Industry Classification Benchmark (ICB) by FTSE Russell, and The Refinitiv Business Classification (TRBC) by Refinitiv. These schemes are global, reviewed and updated at least annually, with new companies added more frequently, and group companies by the similarity of the products or services they sell—a “demand” approach rather than the “supply” approach of the legacy schemes. GICS and ICB cover public companies, while TRBC also covers private companies, non-profits, and government entities. Data aggregators (such as Bloomberg, FactSet, and CapIQ) and stock exchange operators (such as the London Stock Exchange) either use these industry classification schemes or have their own that are substantially similar.

GICS, ICB, and TRBC are strictly hierarchical taxonomies, analogous to the species, genus, family, etc., taxonomy used in biology. In each scheme, a company is classified to a *single* group in the lowest tier, with the groups themselves making up the higher tiers. In other words, continuing with the biology metaphor, a company is classified to a single species, which automatically assigns it to a genus, family, and so on. GICS, ICB, and TRBC have slightly different groupings, tiers, and names. [Exhibit 3](#) shows the names and hierarchy of each scheme’s tiers, the number of groups in each tier, and the composition of the highest tier.

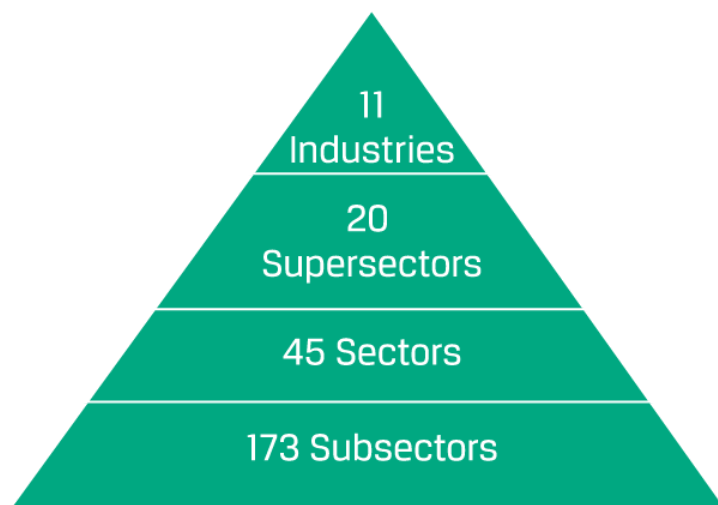
GICS by S&P DJI and MSCI



GICS Sectors

1. Energy
2. Financials
3. Materials
4. Information Technology
5. Industrials
6. Communication Services
7. Consumer Discretionary
8. Utilities
9. Consumer Staples
10. Real Estate
11. Health care

ICB by FTSE Russell



ICB Industries

1. Energy
2. Financials
3. Basic Materials
4. Information Technology
5. Industrials
6. Telecommunications
7. Consumer Discretionary
8. Utilities
9. Consumer Staples
10. Real Estate
11. Health care

TRBC by Refinitiv

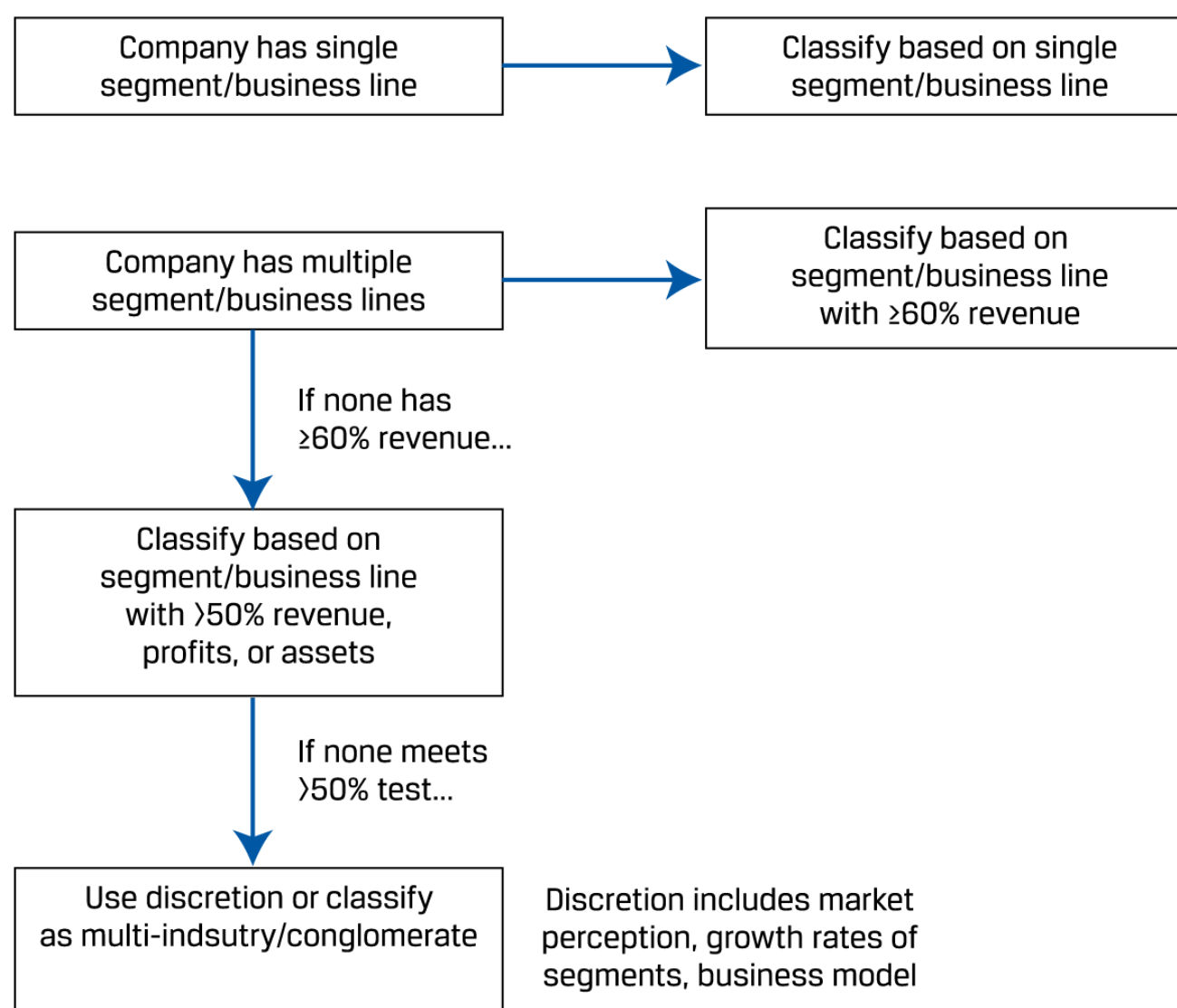


TRBC Economic Sectors

1. Energy
2. Financials
3. Basic Materials
4. Technology
5. Industrials
6. Communication Services
7. Consumer Cyclicals
8. Utilities
9. Consumer Non-Cyclical
10. Real Estate
11. Healthcare
12. Institutions, Associations, and Organizations
13. Government Activity
14. Academic and Educational Services

GICS, ICB, and TRBC also have slightly different rules for companies operating in multiple industries, all of which use some discretion. While each scheme differs slightly, the general process is shown in [Exhibit 4](#). The goal is to assign a company to a single grouping that describes the majority of its business.

Exhibit 4: Classifying Companies to Industries



3.2. Limitations of Third-Party Industry Classification Schemes

While these classification schemes are a useful starting place for research (and they have other uses like index construction and investment performance attribution, described elsewhere in the curriculum), they have four important limitations that analysts must contend with in doing industry research:

1. Groupings of companies with business model variations or that sell substitute products
2. The classification of multi-product companies
3. Geographical considerations
4. Changes in groupings over time that affect prior-period comparability of industry statistics

Analysts may find that industry groupings in the third-party schemes are either too narrow or too broad, as the scheme providers use discretion in constructing groups based on product similarity and business model. Therefore, analysts must either use classification tiers at a higher or lower level or modify the groupings based on their own discretion and purposes of analysis. On the one hand, many companies classified as “application software” sell products with completely different functionality and do not compete with one another—for example, Shopify, the Canadian company that sells front-end and payment processing software to small and mid-sized e-commerce companies, is grouped with Check Point Software, an Israeli security software company. On the other hand, as will be shown with Warehouse Club Inc., retailers are classified into numerous sub-industries and even across sectors, although many of their product offerings overlap and are thus competing for the same consumer dollar.

Because GICS, ICB, and TRBC are strict taxonomies, multi-product companies are assigned to a single grouping. This treatment can be a problem for an analyst identifying competitors if a competitor is a division of a company classified in a different industry or even sector. The most prominent examples are

in technology. A large software market is cloud infrastructure and platform services (sometimes called “public cloud”), which are computing, storage, and other IT-related capabilities sold on a metered basis to clients over the internet. Large players include Amazon Web Services, Microsoft Azure, Google Cloud Platform, and Alibaba Cloud, which are all divisions of larger parent companies with other segments that differ materially by business model. While Amazon and Alibaba are classified in the same industry, Alphabet (parent of Google Cloud Platform) and Microsoft are each classified in different sectors. Amazon’s classification as a consumer discretionary retailer has long inspired consternation, as most of its operating profits come from its Amazon Web Services segment, not e-commerce.

With their global nature, GICS, ICB, and TRBC are useful for some sectors and industries that are in fact global (primarily those that manufacture physical goods and digital products), but they can be less useful for more service-oriented companies that compete nationally or even locally because of customer behavior or regulations that restrict competition. For example, only a small minority of people travel a significant distance from their home to receive healthcare services, and insurers (whether private or government) typically restrict the choice of provider geographically. Governments also regulate the practice of medicine through the licensing of facilities, physicians, and other practitioners. Therefore, the healthcare services industry is best examined on a national or regional basis; the economics of hospitals in the UAE have little relevance to those in France. However, this approach should not be taken for granted, as industries can globalize or localize based on business model changes. While media (pay TV) companies were once regional, the shift to internet-based streaming video has resulted in a global industry with new, larger players.

Finally, industry classification scheme providers change groupings and classifications, which can affect industry statistics. One of the largest changes, for example, was the creation of the Real Estate sector, composed primarily of Real Estate Investment Trusts carved out of the Financials sector. Additionally, public offerings and de-listings from acquisitions and bankruptcies change the composition of the groupings over time (introducing survivorship bias in historical returns statistics). Analysts should be aware of these issues, make sure that historical statistics are restated to conform to the latest groupings, and construct survivorship bias-free datasets if possible.

Defining Warehouse Club Inc.’s Industry

Warehouse Club Inc. has a single line of business and would be assigned to the “Consumer Staples Merchandise Retail” sub-industry in GICS.

Sector	Industry Group	Industry	Sub-Industry	Other Sub-Industry Constituents
Consumer Staples	Consumer Staples Distribution & Retail	Consumer Staples Distribution & Retail	Consumer Staples Merchandise Retail	Costco, Walmart, BJ’s, Carrefour, Sun Art Retail Group, Coles Group, PriceSmart

While this classification is logical, the sub-industry includes companies outside Warehouse’s domestic geography that are less relevant to its customers and does not include many companies in its domestic geography that sell similar products to the same types of customers. For example, grocers and drugstores are included in a different sub-industry (Food Retail and Drug Retail, respectively), and e-commerce retailers that it competes with are in a different sector entirely (Consumer Discretionary).

The challenge in placing Warehouse in an industry is that it sells several categories of consumer goods, both staples and discretionary goods like apparel, as do many other companies. Retailers also have varying business models—including membership-based warehouses, e-commerce, and co-locations with other services like pharmacies—and have varying location, assortment, and pricing strategies.

Elaine Nguyen chooses to define Warehouse’s industry more broadly than its GICS sub-industry by taking a customer perspective: retailers of all kinds operating in Warehouse’s domestic geography, except for automobile and auto parts retailers as Warehouse does not

sell those products or even substitutes for them (and it is a large category of domestic retail sales). Nguyen also does not include restaurants and food services, instead considering them substitutes.

3.3. Alternative Methods of Grouping Companies

An industry or product approach to grouping companies is not the only grouping method and may be less useful in contexts outside industry analysis. Other approaches that are used in contexts such as index construction and investment performance evaluation include the following:

- Geography, in which companies are classified by country and then countries are aggregated into categories such as developed, emerging, and frontier markets. Classification by country is typically by the country where the issuer is incorporated, the country of the primary listing of its equity securities, the location of its headquarters, or market perception. Note that classification by the geographic composition of revenue is generally *not* the approach taken, though this aspect may be the foremost concern for an analyst. For example, Toyota is universally classified as a Japanese company. However, its largest market by revenues is North America, which is also the location of most of its assets. The classification of developed, emerging, and frontier markets is more controversial, and third parties like index providers use greater discretion. It is not a quantitative determination but considers variables such as the size and liquidity of equity securities markets in the country, income per capita, and legal restrictions on foreign investment.
- Sensitivity to the business cycle, with groupings such as “defensive” and “cyclical.” Defensive companies are those whose sales growth, profitability, and valuations are less affected by changes in broad macroeconomic activity (e.g., GDP growth), while the opposite is true of cyclical. Generally, this classification is done by grouping entire sectors from the industry classification schemes discussed earlier. Sectors such as consumer staples, healthcare, and utilities are considered defensive, while financials, basic materials, consumer discretionary, and industrials are considered cyclical. This grouping method can be combined with other methods—such as a geographic approach (e.g., Japanese cyclical), because countries may vary with respect to their current position in the business cycle, or credit ratings (e.g., Japanese cyclical with high-yield credit)—to further isolate exposure to certain risk factors.
- Statistical similarities, or the use of clustering analysis to group companies based on similarities of financial ratios and market data or co-movements of their securities’ investment returns. This approach includes grouping by size according to market capitalization or by other characteristics such as the following:
 - Valuation ratios
 - Growth rates of sales or earnings
 - Profitability ratios
 - Statistics based on price performance such as volatility and momentum
- ESG characteristics, such as the ratio of carbon emissions to revenues, measures of board and executive personnel diversity, and exposure to certain businesses such as tobacco and gambling. These metrics can be aggregated into composite ESG ratings or scores that enable cross-issuer comparability.

These groupings are usually relative (e.g., companies with dividend yields at or above the 75th percentile as of the measurement date) and tend to show far more turnover in their constituents than groupings based on industries and countries, because these statistics are less stable by company and companies’ rankings change.

1. Contrast the Global Industry Classification Standard (GICS) commercial scheme with the Standard Industrial Classification (SIC) third-party classification system in terms of geographic coverage, update frequency, and the addition of new companies.

Solution:

The GICS commercial scheme is determined on a global basis, updated at least annually, and adds new companies frequently, while the SIC is US-only, infrequently updated, and does not regularly add new companies.

2. The GICS, ICB, and TRBC commercial industry classification schemes suffer from which common problem when used by analysts?
 - A. They quickly become out of date.
 - B. They cover a limited number of countries.
 - C. They often classify multi-industry companies inaccurately.

Solution:

C is correct. Commercial industry classification schemes are strictly hierarchical and assign each company to a single industry. Multi-industry companies are classified using evaluations of segment revenues, profits, or assets or the discretion of the scheme creator. This approach often results in classifications that are different from analyst expectations or in “multi-industry” classifications that are challenging to use in practice.

3. Identify whether each of the following sectors is generally considered “cyclical” or “defensive.”

I. Consumer Staples	Cyclical	Defensive
II. Consumer Discretionary	Cyclical	Defensive
III. Energy	Cyclical	Defensive
IV. Healthcare	Cyclical	Defensive
V. Utilities	Cyclical	Defensive
VI. Financials	Cyclical	Defensive
VII. Industrials	Cyclical	Defensive
VIII. Technology	Cyclical	Defensive
IX. Materials	Cyclical	Defensive
X. Real Estate	Cyclical	Defensive

Solution:

I. Defensive

II. Cyclical

III. Cyclical

IV. Defensive

V. Defensive

VI. Cyclical

VII. Cyclical

VIII. Defensive

IX. Cyclical

4. INDUSTRY SURVEY

LEARNING OUTCOME

The candidate should be able to:

- ☐ **determine an industry's size, growth characteristics, profitability, and market share trends**

After defining an industry, the next step in industry and competitive analysis is to survey the industry by estimating its size, calculating its historical growth rate, evaluating the character of that growth rate, measuring its profitability, and identifying major industry players and market share trends over time. This industry survey provides a basis for the industry evaluation in the next step and makes an analyst aware of the main issues and opportunities.

4.1. Industry Size and Historical Growth Rate

Industry size is typically measured by total annual sales from the product or customer perspective, which is not necessarily all sales of each industry constituent. For example, Amazon's retail segment sales would be included in sizing the retail industry, while sales for its Amazon Web Services segment would not. The growth rate is calculated either as year-over-year rates each year or as a compounded annual growth rate over a multi-year period. If possible, industry growth should be broken out by contributions from volume and price/mix drivers.

Except for some industries that are dominated by large, publicly traded companies (e.g., autos, smartphones, airlines, pharmaceuticals), industry size will often include a potentially large amount in sales from private companies, sometimes small businesses, for which data are unavailable or impractical to tabulate. It's common practice to estimate industry size for these industries using economic indicators published by government agencies, data from third-party consultancies that use surveys, or industry data from issuers' investor presentations based on proprietary sources. These all need to be corroborated for reasonableness by the analyst. For Warehouse Club Inc., for example, we use retail sales published by a government agency for industry size, because the retail industry includes hundreds of thousands of small retailers.

4.2. Characterizing Industry Growth

The pattern of historical growth of an industry can be used to characterize it based on the magnitude of its growth rate and sensitivity to the business cycle. One approach to characterizing an industry's growth rate is a style box, as shown in [Exhibit 5](#). Essential to this approach is comparing the industry's historical growth rate with a broader measure of economic growth during recessions and expansions.

Defensive	-Utilities -Beverages -Pharmaceuticals	-Biotechnology -Software -Gaming
Cyclical	-Crude oil -Natural gas -Freight transportation	-Semiconductors -Fintech -Digital advertising
	Mature	Growth

Source: Authors' analysis.

Growth industries include those that have not yet reached full saturation or penetration of their total addressable market, and thus have idiosyncratic growth drivers separate from broader economic growth. Growth industries are often based on or benefit from an emerging technology. Key questions for analysts related to growth industries include whether historically high growth rates will persist and what the peak penetration rate is—which are more challenging to answer the “newer” the industry is. Conversely, mature industries have fully penetrated their addressable market and have growth rates either in line with broader economic activity (approximately zero) or declining as customer demand migrates to a substitute. Key issues for investors in a mature industry include monitoring for disruptive threats, changes in competitive intensity, and the speed of decline.

Sensitivity to the business cycle is driven by the business models of companies in an industry and correlates with industry maturity. Factors that determine sensitivity include the degree to which sales are discretionary or necessary for consumers, prices, the interest rate exposure of the business model, and whether the product is a durable or capital good versus a recurring purchase such as consumables and subscription services. Although most investors expect greater variability in returns from firms in cyclical industries, differing investor views as to the length and magnitude of such cycles lead to wide variations in the valuations of issuers over time.

Although commonly used by market participants, these distinctions can sometimes be of limited usefulness. For example, a severe economic downturn is likely to have a negative impact on all firms, so differences will be a matter of degree but not categorical. Second, companies in different stages of the life cycle within an industry can differ materially in terms of growth, defensiveness, and cyclicity; for example, growth firms based on an emerging technology may exist within a cyclical industry that is far less affected by an economic downturn.

4.3. Industry Profitability Measures

The best measure of industry profitability is a time series of the distribution of returns on invested capital (e.g., 25th, 50th, 75th percentiles), which captures after-tax operating profits for each dollar of invested capital and is agnostic about capital structure. However, unless the industry is dominated by public companies, obtaining these data is seldom feasible.

Two common methods to overcome this challenge in practice are to measure the profitability of the publicly traded players and assume that the profitability of private competitors is similar—either using various sources to estimate the profitability of non-public companies (e.g., for state-owned energy companies, there are estimates of their production costs per barrel, production, and capital expenditures, and prices are generally uniform throughout the industry) or obtaining data from a government agency or third-party consultancy. For Warehouse Club Inc., the same government agency that publishes retail sales data each month also does an annual survey of retailers’ gross margins and operating expenses, which can be used to calculate industry operating margins.

Clearly, a profitable industry is preferable to an unprofitable one, but often more important than a point estimate of profitability is the time series: is industry profitability rising or falling? If there is a discernible trend, it will be an important point of investigation for the analyst.

4.4. Market Share Trends and Major Players

Finally, market shares are measured by expressing industry participants' annual revenues as percentages of the industry size each year. Given the aforementioned measurement problems with estimating industry size, market shares should be interpreted as a range rather than a definitive point estimate. Again, more important than each company's market share is the trend over time, which can reveal whether customers have judged the company's products superior to competitors'. An important consideration is acquisitions, particularly large ones; a company buying a competitor is a market share gain, but more important is whether the company is increasing or decreasing its share on an organic (non-acquired) basis.

An additional consideration in this analysis is the degree of industry concentration. Lower concentration—that is, many small competitors in the market—is usually associated with a high degree of competitive intensity unless the industry is very service-oriented, is local in nature, or has high product differentiation. Rising concentration (consolidation) is associated with falling competitive intensity and higher profitability. A common measure of industry concentration is the **Herfindahl-Hirschman Index (HHI)**, calculated as the sum of the squares of competitor market shares:

$$HHI = \sum_{i=1}^{\infty} s_i^2,$$

where s is the market share of market participant i stated as a whole number (i.e., 50% share = 50, not 0.50).

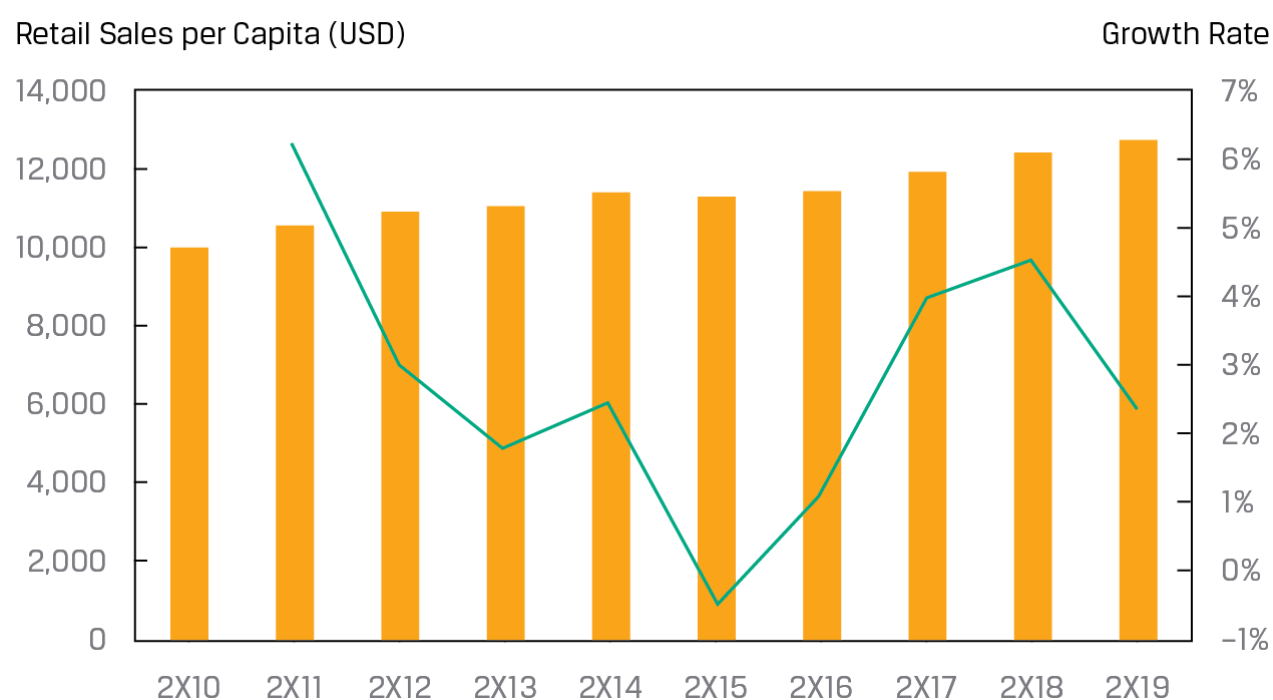
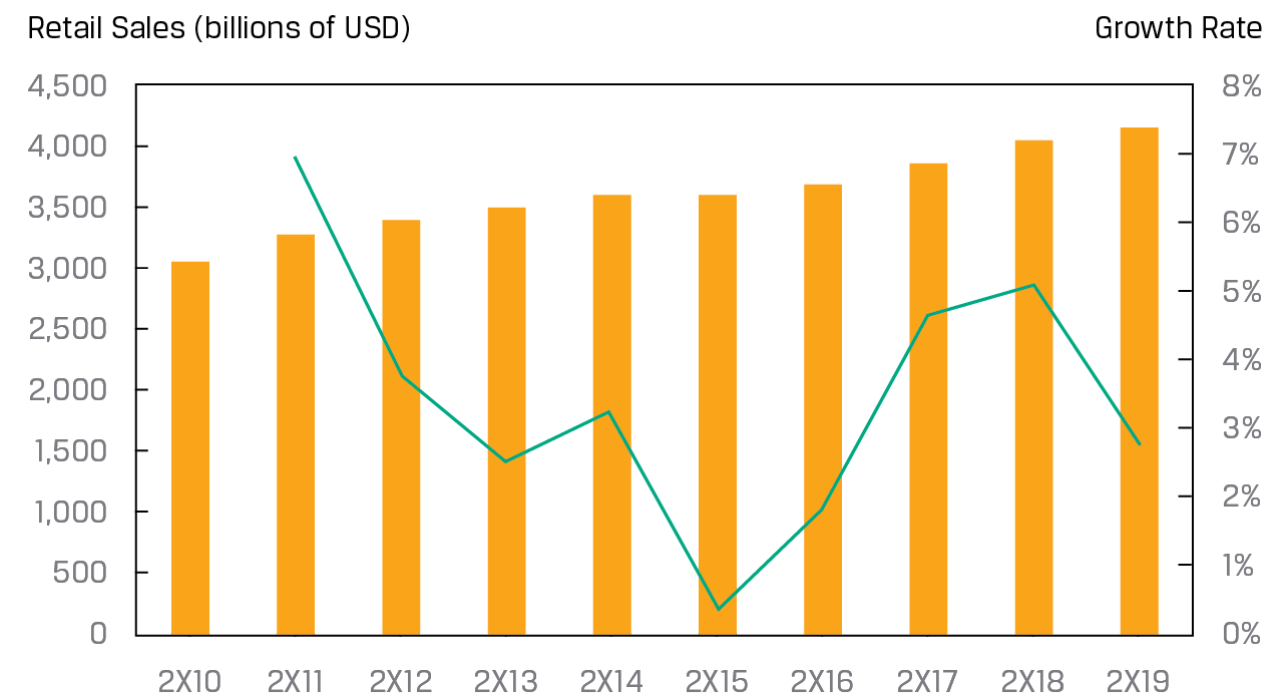
A market consisting of four firms with shares of 30, 30, 20, and 20 has an HHI of $30^2 + 30^2 + 20^2 + 20^2 = 2,600$. The maximum HHI is for a monopoly, with a value of $100^2 = 10,000$. Antitrust regulators in some countries consider markets with an HHI between 1,500 and 2,500 moderately concentrated and consider markets with an HHI over 2,500 highly concentrated. A rule of thumb is that acquisitions in highly concentrated markets that increase the HHI by more than 200 points are often subject to regulatory challenges.

Industry Survey for Warehouse Club Inc. and Iliso Marketplace Ltd.

Industry Size and Growth Rate

Elaine Nguyen sizes the US retail (excluding autos) industry using three measures: total annual revenues, annual revenues on a per capita basis, and the number of retailers. For all these measures, Nguyen uses data from the US Census Bureau, since the industry consists mostly of small businesses with no public filings and the data are high quality; the Bureau has been collecting such data since 1952.

US Annual Retail Sales ex Autos (Bars, left) and Growth Rate (Line, right)



At the end of 2019, there were 556,531 firms in the industry, but over 92% of them employed fewer than 20 people. Just 470 large firms with over 5,000 employees accounted for 62% of total industry employment. Of these large firms, 144 were publicly traded.

US Retailers (ex autos) as of 2019

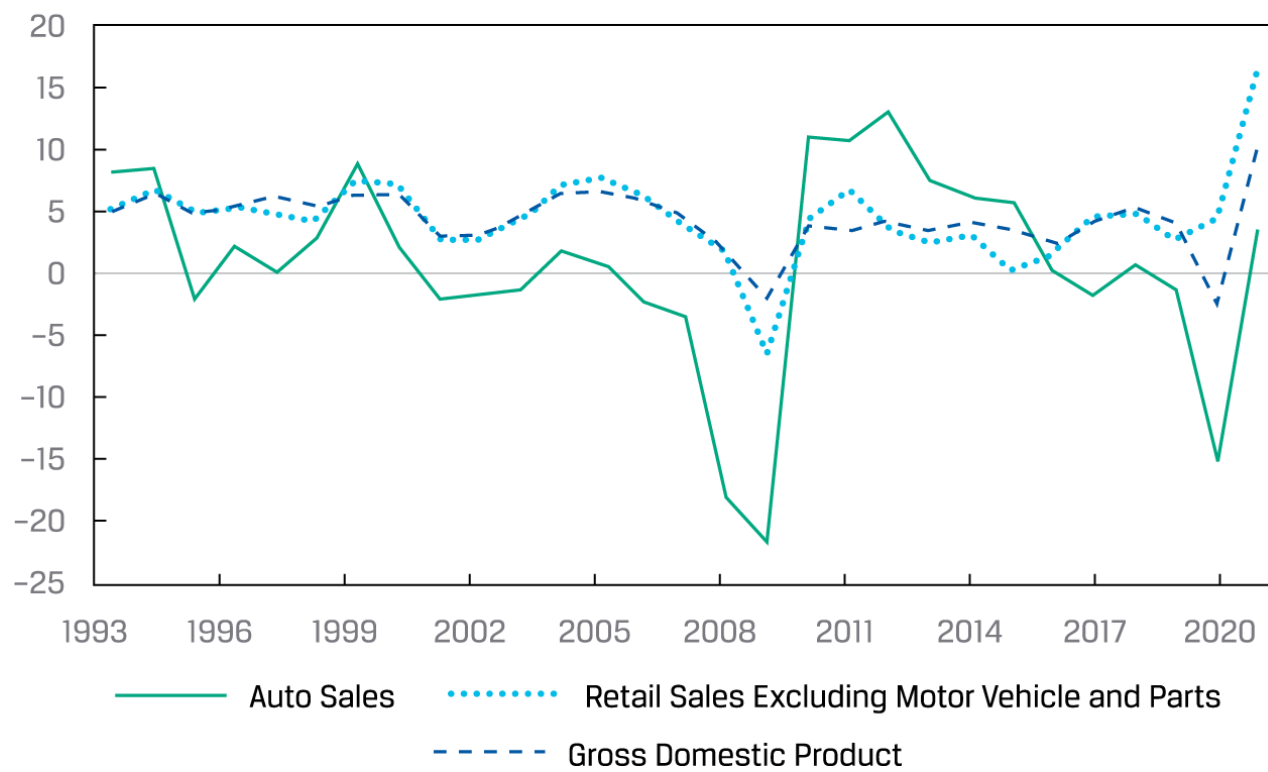
< 20 employees	516,766
Between 20 and 5,000 employees	39,295
> 5,000 employees	470
Total	556,531
<i>Note: public companies</i>	144

Character of Industry Growth

The retail industry has existed for hundreds of years in the USA, though there have been many changes in both form (e.g., department stores, discounters, e-commerce) and the products sold. Comparing industry sales growth with GDP growth since the early 1990s shows that the two are tightly related. However, the industry does not show extreme cyclicity, unlike the auto industry.

Elaine Nguyen classifies the industry as mature and moderately cyclical.

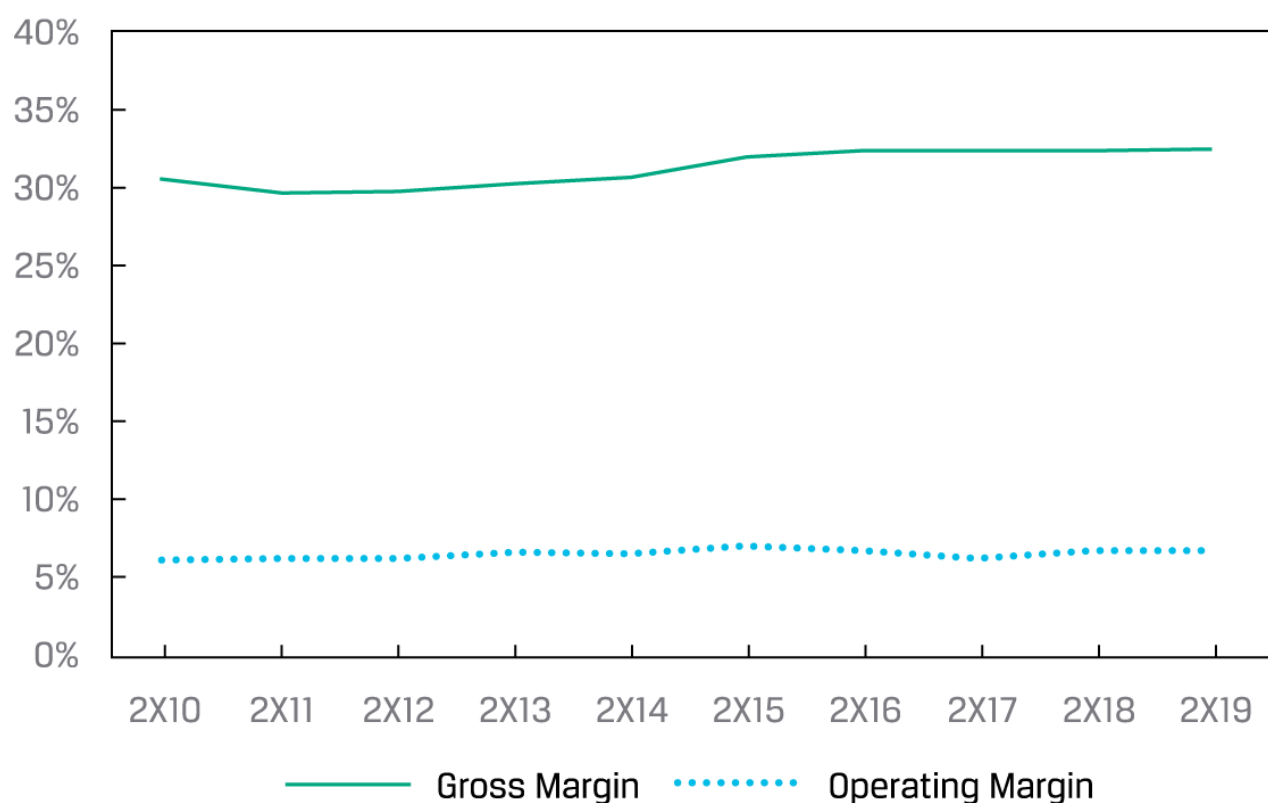
Annual Growth Rates: US Retail Sales ex Autos, Autos, and GDP



Industry Profitability

Nguyen analyzes industry profitability in two ways: (1) the gross and operating margins of all retailers (except autos) in the USA using Census Bureau data and (2) the profitability of the 10 largest publicly traded retailers in the USA by 2X19 sales. Because of segment disclosure changes, M&A, and other corporate transactions, Nguyen examines the last 5 rather than 10 years of data.

Gross and Operating Margins of US Retailers ex Autos



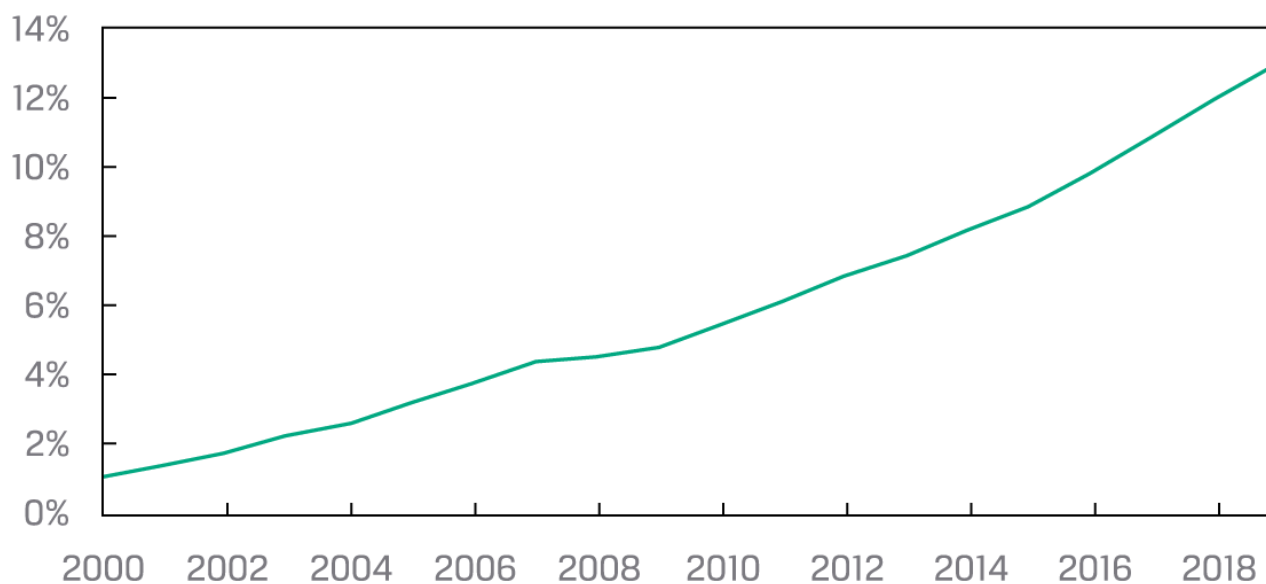
Operating Margins	2X15	2X16	2X17	2X18	2X19
Walmart (US segment)	6.4%	5.8%	5.3%	5.2%	5.1%
Walmart (Sam's Club segment)	3.2%	2.9%	1.5%	2.6%	2.8%
Amazon.com (North America segment)	2.2%	3.0%	2.7%	5.1%	4.1%
Costco Wholesale (US segment)	2.7%	2.7%	2.8%	2.7%	2.7%
The Home Depot, Inc.	13.3%	14.2%	14.5%	14.4%	14.4%

Operating Margins	2X15	2X16	2X17	2X18	2X19
Kroger Co.	3.3%	3.0%	2.1%	2.1%	1.8%
Target Corp.	6.9%	7.1%	5.8%	5.5%	6.0%
Lowe's Companies, Inc.	8.4%	9.0%	9.6%	7.9%	8.8%
Albertsons Companies, Inc.	0.7%	1.1%	0.3%	1.0%	1.5%
Ahold Delhaize (US segment)	4.0%	3.9%	4.1%	4.5%	4.3%
The TJX Companies, Inc. (US segments)	13.0%	12.8%	11.8%	11.8%	11.6%

Market Share Trends

The most significant market share trend in US retail is the shift toward e-commerce, which has been ongoing since the late 1990s. This shift has accrued largely to Amazon.com, the pioneer in US e-commerce, with its share of e-commerce at roughly 40% (Amazon has a large third-party merchant business but does not disclose GMV, so its exact share is unknown).

US E-Commerce % of Retail Sales ex Autos and Petrol



The industry is fragmented, with the top 10 largest firms holding less than 30% market share, the 10th largest holding less than 1%, with a large amount of share held by over 500,000 small retail businesses. However, the top 10 largest firms have gained share (~2%, from 26.3% in 2X15 to 28.6% in 2X19).

US Net Sales	2X15	2X16	2X17	2X18	2X19
Walmart US and Sam's Club	355,206	365,198	377,693	389,505	399,796
Amazon.com*	63,708	79,785	106,110	141,366	170,773
Costco Wholesale (US segment)	84,351	86,579	93,889	102,286	111,751
The Home Depot, Inc.	80,550	86,615	92,413	99,386	101,333
Kroger Co.	109,830	115,337	123,280	121,852	122,286
Target Corp.	73,717	69,414	71,786	74,433	77,130
Lowe's Companies, Inc.	59,074	61,311	63,263	65,872	67,147
Albertsons Companies, Inc.	58,734	59,678	56,925	60,535	62,455
Ahold Delhaize (US segment)	42,482	42,946	43,194	44,174	44,841
The TJX Companies, Inc. (US segments)	23,899	25,651	27,365	29,845	32,021
Total top 10	951,551	992,514	1,055,918	1,129,254	1,189,533
Share of US retail sales ex autos	26.3%	26.9%	27.4%	27.9%	28.6%

*Amazon.com's "North America" segment only.

Question Set

1. Identify the following statement as true or false. Justify your answer.

Industry size is typically measured by the total annual sales of all industry constituents.

Solution:

False. Industry size is typically measured by the total annual sales from the product or customer perspective, which is not necessarily the sum of total sales of each industry constituent as some constituents may have segments in other industries.

2. A section of the fast-food market consists of 10 firms operating 1,840 restaurant locations across North America and currently has a Herfindahl-Hirschman Index (HHI) of 1,516. The market leader announces plans to acquire its closest competitor and commits to not closing any locations at either company. As a result, the HHI will *most likely*:

A. decrease.

B. not change.

C. increase.

Solution:

C is correct. The Herfindahl-Hirschman Index (HHI) is calculated as the sum of the squares of competitor market shares. In this example, the market share of the leading company would increase and the number of competitors would decrease, which would increase the HHI. The number of store locations does not affect the HHI.

3. Identify the following statement as true or false. Justify your answer.

Companies in different stages of the life cycle within the same industry can differ materially in terms of cyclicalities.

Solution:

True. An industry is defined as companies with similar products or services from the perspective of a customer. A new entrant in an industry may have low but increasing market share, which can serve as a growth driver in addition to an economic expansion. The new entrant may therefore be less affected than a mature company by a recession.

Please refer to the Industry Survey for Warehouse Club Inc. and Iliso Marketplace Ltd. earlier in the lesson to answer the following questions.

4. Amazon.com's change in US retail market share from 2X15 to 2X19 is *best* described as an example of:
- A. a disruptive threat.
- B. achieving peak penetration rate.
- C. saturation of its addressable market.

Solution:

A is correct. As the pioneer in US e-commerce, Amazon achieved roughly 40% market share by using Amazon.com as an online shopping platform. Therefore, Amazon.com is an example of a disruptive threat to the established retail industry. Based on US net sales from 2X15 to 2X19, it is uncertain whether Amazon achieved its peak penetration rate or saturated its addressable market, as it continued to increase its market share component of the top 10 retailers' share from 7% to 14%.

5. Current economic forecasts call for an increase of 3% in US GDP, an inflation rate of 1.5%, and decreasing consumer sentiment. Therefore, over the next fiscal year, Nguyen would *most likely* expect Warehouse Club Inc.'s total sales to:

- A. decrease.
- B. increase 2%–5%.
- C. increase 5%–8%.

Solution:

B is correct. Nguyen classifies the industry as mature and moderately cyclical, and Warehouse Club Inc. operates a single line of business with no new products or services defined. As a result, Warehouse Club Inc.'s total sales would move in line with broader US GDP growth as a stable company in a mature industry.

5. INDUSTRY STRUCTURE AND EXTERNAL INFLUENCES

LEARNING OUTCOME

The candidate should be able to:

- ☐ **analyze an industry's structure and external influences using Porter's Five Forces and PESTLE frameworks**

Recall from earlier lessons in economics that Porter's Five Forces is a framework for assessing industry structure that determines an industry's long-run profitability measure by its returns on invested capital.



If some or all of the five industry forces are intense, almost no company earns attractive returns on capital, while benign forces lead to highly profitable industries. From the industry survey work in the previous lesson, we may have already measured the industry's profitability on a historical basis. The analysis of structural forces is aimed at not simply measuring profitability but rather characterizing its drivers in a qualitative fashion and improving alertness to key issues to monitor that can affect industry profitability. For new industries for which data are unavailable or which are still in an early, unprofitable stage, the analysis of industry structure can help point to potential profitability levels.

5.1. Assessing the Five Forces: A Checklist Approach

The assessment of the five forces is similar to the determination of a company's business model, from the prior module, in that it is a qualitative research investigation that is difficult to generalize. We provide illustrative questions to answer as an analyst and offer examples of real industry observations for each question.

5.1.1. Assessing the Threat of New Entrants

- Have there been significant numbers of new industry entrants in recent history?
 - In the USA, more than 50,000 new restaurants are opened per year. Conversely, from 2010 to 2021, fewer than three new commercial bank charters were issued per year in the USA.

- Are there network effects with the industry's product or service, such that its value to users is greater as the number of users increases and a new entrant would have a low-value product that has to achieve scale?
 - Payment networks such as Visa, Mastercard, JCB, and China UnionPay enable payments between hundreds of millions of cardholders and millions of merchants by connecting thousands of financial institutions on their network.
- Do incumbents benefit from economies of scale from significant fixed costs in the cost structure, such that a new entrant would require time and unprofitable economics to reach competitiveness?
 - Cloud computing services require enormous fixed assets in terms of servers and networking equipment. These services have virtually no variable cost, however, and large incumbents have decreased unit prices substantially over time while gaining scale.
- Do incumbents benefit from economies of scope across multiple business lines—from increased convenience in terms of customers and market share and/or from joint costs that can be leveraged across the businesses?
 - Warehouse Club Inc. operates pharmacies and gas stations, which leverages real estate costs and drives increased foot traffic. Many commercial banks operate consumer banks, asset and investment management, broker-dealers, and investment bank businesses, as there are common customers and joint operating costs.
- Are customers loyal to brands?
 - A common threat to branded consumer merchandise sold to retailers is merchandise manufactured by contract manufacturers at the direction of the retailer and sold under retailer-owned "private label" or "generic" brands. In some categories, such as dairy milk and packaged baked goods, private label has over 50% volume share, which has risen over time. Consumers do not appear to be driven by brands in these categories. However, energy drinks, alcoholic beverages, and tobacco products have a very low private label market share over time, with consumers making repeat purchases of the same brand. Note that an important consideration for the value of brands is the legal environment that protects trademarks, trade dress, copyright, and other intangible assets in a jurisdiction. Without these legal protections, brands and their likeness can be imitated by competitors.
- Are there significant switching costs for customers in terms of time, training, or replacement?
 - So-called infrastructure software such as SAP for businesses requires lengthy implementation processes led by third-party consulting firms, and connections to other mission-critical software systems need to be established. Switching from SAP to another provider is a costly and lengthy endeavor. Conversely, customers can and do switch between food delivery apps, as minimal useful information is needed (or stored) to switch apps (payment information).
- Do incumbents have exclusive or preferential access to hard-to-obtain inputs or end-customer interfaces?
 - In the beverages business, incumbents have shelf space arrangements with retailers, often own distribution assets, and in some countries have the same arrangements with restaurants and pubs. This advantage is a potent deterrent for the beverage companies' customers to start carrying new entrants' products.
- Are there government policies that restrict or slow industry entry?
 - These policies can take many forms, such as regulations that only larger companies can afford to comply with, subsidies for domestic companies, patents, and licenses.

Note that experience and expertise among incumbents are not sustainable barriers to industry entry. Entrants can imitate, poach employees, purchase inputs and services from the incumbents' suppliers, and engage with industry consultants to obtain industry expertise.

5.1.2. Assessing the Threat of Substitutes

- Is there another product or service that fulfills the same or similar customer need? Is it lower priced?
 - Video conference calls and collaboration software are a low-price substitute for business travel and office real estate.
- Can customers go without the industry's product?
 - Aesthetic and elective medical procedures can be canceled or deferred without loss of function for a patient.
- Is the substitute product improving in performance relative to the industry's product?
 - While digital advertising was rudimentary in the early 2000s, with pop-ups and display ads, and was not a material threat to television advertising, digital advertising improved to include highly targeted videos and advertising disguised as content from third-party influencers and has taken market share from television advertising.
- Is it difficult for customers to switch to a substitute?
 - In most countries, prescriptions for a branded pharmaceutical are automatically substituted with a generic at a pharmacy if there is one approved.

5.1.3. Assessing the Bargaining Power of Customers

- Are there few, concentrated customers for the industry?
 - For many electronic component companies, mobile device makers such as Apple, Samsung, and Xiaomi account for a significant percentage of their revenues.
- Are the industry's products standardized or undifferentiated?
 - Crude oil is graded (e.g., Brent, West Texas Intermediate) by its density and sulfur content, which determine its cost and the effort to refine it into products such as petroleum. Oil refiners pay for crude oil based on its grade, not by specific oil producer; crude oil within a grade is interchangeable.
- Are the industry's products critical for customers?
 - Pesticides and seeds, especially those made with biotechnology, are sold at high prices but are vital to sustaining high agricultural yields and the high prices are outweighed by the cost of crops lost to disease and insects.
- Are the industry's products a large part of customers' budget?
 - Customers may be more motivated to seek alternatives and negotiate for products that account for a large part of their budget and for which quality is not a paramount concern. For livestock farmers, the cost of feed accounts for over 60% of annual operating costs, while pharmaceuticals and vaccines for livestock account for a small percentage but come with a high cost of failure, which has given animal-pharmaceutical companies a strong bargaining position.
- Can customers backward-integrate—that is, “build” rather than “buy” what the industry is selling?
 - Streaming video services first licensed content from existing studios and media companies, but over time they vertically integrated by producing their own video content.

5.1.4. Assessing the Bargaining Power of Suppliers

- Are there few, concentrated suppliers for the industry?
 - Apple's "A" and "M" series processors for its iPhones and MacBooks are manufactured exclusively by TSMC. While Apple has more supplier choice for its other components, it is beholden to a single company for this crucial component.
- Are there significant costs for switching suppliers?
 - Asset managers need performance data and analytics on equity, fixed income, and other indexes for investment processes and client reporting of benchmarks. Indexes are owned by a small number of companies and charge licensing fees for data permissions and commercial use. An, Benetton, Song, and Index Providers (2021) found that managers of indexed exchange-traded funds paid index licensing fees equal to 35% of their management fee revenue. While asset managers can switch because each provider offers similar catalogues of indexes, doing so would require changing investment processes, client reporting, and conversations with clients—and can look bad to prospects.
- Are suppliers differentiated or specialized?
 - Healthcare providers, airlines, investment banks, technology companies, and professional services firms all require skilled, credentialed, or licensed labor, for which there is limited supply.
- Are there substitutes for the products from suppliers?
 - Retailers can substitute branded merchandise for private label merchandise from contract manufacturers.

5.1.5. Assessing Rivalry among Existing Competitors

- Is there a history of price competition among competitors?
 - Despite a low risk from every other industry force, automakers and aircraft manufacturers compete fiercely on price, offering promotions and generous financing and warranty terms to try to capture every sale.
- Are there numerous, roughly equal-sized competitors?
 - There are approximately a dozen automakers with an annual global market share over 1%. However, the largest company (Toyota) has had less than 15% market share for many years.
- Are products differentiated?
 - Pharmaceuticals without a patent (generics) are legally required to be chemically equivalent to their originator drug in order to be approved by regulators in an abbreviated process that doesn't involve a clinical trial. Pharmacies are indifferent as to whether an approved generic is sold by one company or another and manufacturers' costs are largely fixed, so the companies compete fiercely on price for volume. Price competition increases with the number of approved generics per originator drug; [Dave, Hartzema, and Kesselheim \(2017\)](#) found that a drug with one generic competitor had prices that were, on average, six times higher than the prices of one for which there were nine generic manufacturers.
- Are there high barriers to exiting?
 - If exiting an industry is difficult or costly, such as shutting down natural resource-producing assets, companies with low profitability may remain in the industry for much longer than in other industries, which can result in lingering competition.
- Is industry growth slow?
 - If penetration of the addressable market is low, companies may not feel a need to compete for customers using price and other economic terms. This was the case for mobile wireless plans

until the mid-2010s, when penetration plateaued and companies began engaging in aggressive promotional activity in competition for one another's customers.

After completing the evaluation of industry structure, an analyst must judge which are the most and least important factors. It is often clear that one or two forces are strongest, but there isn't a quantitative test or threshold by which to judge definitively. Posing scenarios to experienced industry insiders is often useful, such as asking, "What do you think would happen if company X increased its prices by 10% tomorrow?" If their answer is that a large percentage of customers are willing and able to switch to a competitor or substitute—perhaps it has happened before—it's likely that customer bargaining power and the threat of substitutes warrant further investigation.

Porter's Five Forces Analysis for US Retail (ex autos)

Threat of New Entrants: Very High

Opening a retailer, especially an e-commerce retailer that utilizes third-party merchant services, is relatively easy and common. In the USA, retailers are the most common type of business formed—by a wide margin, with over 40,000 new firms filing formation papers each month (the next most common is transportation and warehousing, with around half the number of retailer formations). Customers can easily switch retailers, as most do not have an ongoing relationship such as a subscription fee or contract, and there are minimal regulatory barriers such as licenses and patents.

Threat of Substitutes: Low

Broadly speaking, the substitute for retail is consumer services (e.g., restaurants, travel, healthcare), which also includes digital services (such as streaming video subscriptions and gaming). Most categories of goods, however, are not easily replaceable with services (e.g., apparel and home décor), or they enjoy a cost advantage because of lower labor intensity (e.g., fresh food). There have been periods when services grew faster than goods, but as shown in an earlier exhibit, retail sales have grown essentially in line with US nominal output/income. Retail is arguably one of the oldest industries, and while it has evolved over time, it has yet to be replaced.

Bargaining Power of Customers: Moderate

Retail customers are highly fragmented, with each consumer representing a distinct decision-maker, as there are generally no group purchasing organizations. However, many products are sold by many retailers, and the internet has enabled easy comparison shopping at retailers. Customers are price sensitive with respect to identical products sold by different retailers.

Bargaining Power of Suppliers: Low to Moderate

Key suppliers for retailers include manufacturers of goods, employees, and lessors of retail or fulfillment space. In most cases, numerous options are available. However, branded goods are sold exclusively by a sole manufacturer that may impose high prices and other economic terms, like shelf space and visibility, and may want to sell only to certain retailers (e.g., makers of luxury goods may not sell to discount retailers in order to maintain exclusivity).

Rivalry among Existing Competitors: High

Given the sheer number of similarly sized firms selling similar or identical products, retailers compete fiercely, often with price promotions and discounts as price is one of the few ways they can lure customers away from competitors.

5.2. External Influences on Industry Growth

Analyzing the structure of an industry primarily involves looking *inside* the industry and at close adjacencies such as customers, suppliers, and substitutes. It is important for analysts to look *outside* the industry for factors that influence the industry's economic outcomes. One framework for this purpose is a **PESTLE analysis** of the *political, economic, social, technological, legal, and environmental* influences on an industry. While Porter's Five Forces is concerned with the determinants of industry profitability, a PESTLE analysis is more concerned with an industry's growth rate and market share dynamics. For example, athletic apparel companies such as lululemon athletica inc. have long benefited from increasing consumer interest in fitness and wellness, a social influence on the apparel industry. This trend is not apparent from Porter's Five Forces but is obviously important for growth in athletic apparel. When the trend eventually plateaus or declines, it's likely to result in declining sales growth or sales for lululemon, so analysts carefully monitor the trend using a variety of metrics. PESTLE analysis is a framework for identifying "themes" or "narratives" that investors may take a perspective on and desire exposure to.

While analysts must look broadly for influences, several factors make this search easier in practice. First, external influences often do not change very quickly (though their magnitude and direction can), so this analysis does not need to be done every quarter. Second, analysts develop industry expertise, and those in an institutional setting can access industry specialists who offer their perspectives and surveillance of industry trends as a paid service. Finally, it is usually the case that some influences are far more important than others for a given industry.

5.2.1. Political Influences

Political influences on an industry include changing fiscal and monetary policies, governments' direct selling and purchasing activities, regulatory changes, and geopolitical conditions and actions. We highlight three sectors and their constituent industries for which political influences are often significant: energy, healthcare, and defense.

The energy sector is subject to three important political influences: the political appeal of low and stable energy prices in the short run, climate agreements and legislation calling for emissions reductions, and the actions of OPEC (Organization of the Petroleum Exporting Countries). Most governments desire low energy prices in the short run to maintain their appeal to consumers and businesses that have inelastic demand with respect to price. However, in the long run, higher energy prices (perhaps because of a carbon tax) might be preferable in order to curb demand and meet emissions targets, for which the consumption of fossil fuels is the main driver. Investors and producers may worry about the longer-term outlook, since governments are intending to shrink fossil fuels as a percentage of the energy mix, but reductions in fossil fuel production are at odds with low energy prices in the short run. Finally, energy companies owned by member states of OPEC produced 35% of crude oil globally in 2021 (*BP Statistical Review of World Energy*) and set production quotas as a bloc. Actions taken by OPEC significantly affect global and regional energy prices. OPEC members may choose to keep prices low enough to slow the transition to renewables, exploiting governments' need for low energy prices in the short run.

Governments are the largest buyers of healthcare products and services and, in most cases, are the exclusive buyers of defense goods and services globally. For political appeal and to shore up fiscal deficits, governments have at times pursued reforms such as expansions of healthcare coverage and subsidies, price controls or reductions, and restrictions on certain healthcare services or products.

Defense spending is determined by countries' geopolitical objectives, perceived threat environment, and alliance commitments. From the late 1990s to 2010, global military spending rose markedly, increasing as a share of global GDP, which benefited defense companies. Future defense spending depends on geopolitical events and other competing budget priorities. Finally, governments also extensively regulate healthcare and defense, the stringency of which changes over time. From 2000-2007, several high-profile pharmaceutical scandals led to a period of heightened regulatory scrutiny and a declining number of drug approvals. This scrutiny was relaxed by the mid-2010s. Countries are selective about which countries they permit domestic defense companies to do business with, and the contracting process is highly complex, involving third-party costs and performance audits.

5.2.2. Economic Influences

Economic influences on an industry include changes in GDP or personal income, inflation, interest rates, and exchange rates. These influences can be related to the business cycle (cyclical) or structural, such as how longer-run population and productivity growth rates have differed across countries, which affects the sales and costs of companies operating in different countries. Economic influences are most important for cyclical sectors such as financials, basic materials, and consumer discretionary, and exchange rate changes are most important for industries and companies operating in multiple currencies—especially if there are large currency compositional mismatches in revenues and costs and if the company does business in an illiquid currency with more volatile exchange rates.

The auto industry is among the most affected by economic influences, because it sells high-priced durable goods, the purchase of which can be delayed (consumer keeps current car longer) or substituted (used car) based on consumer confidence. Structural economic influences are important for many multinational companies that have sought to increase exposure to faster-growing emerging markets over time. As of 2022, over 40% of Nestlé SA's sales were in emerging markets, which have long shown faster volume growth than its developed-market sales.

5.2.3. Social Influences

Social influences include cultural and consumer trends, demographic changes such as the relative growth rates among different population cohorts, and changes in lifestyles. These influences are not entirely external to an industry, as companies influence culture through political advocacy, advertising, and a variety of other efforts such as product placements in media. There is growing reputational risk and/or pressure on companies via social media and the press to source sustainable inputs, as well as increased scrutiny of supply chains for violations of human rights.

Social influences are often more significant for industries that sell to consumers rather than to other businesses. In concert with technological and economic influences, the global beauty industry (comprising skin care, color cosmetics, hair care, fragrances, and select personal care product categories) has grown briskly since 2010, particularly premium products. High-quality cameras on mobile devices and social media have increased the salience of people's physical appearance and created efficient marketing channels for beauty companies that utilize third-party influencers and video tutorials.

5.2.4. Technological Influences

Technological changes can create new or improved products or render existing products obsolete, radically changing industries and companies. Based on the pioneering work of Clayton Christensen, we distinguish between two kinds of technological changes: sustaining and disruptive innovations.

Sustaining innovations are improvements in product or service performance and the addition of marginal features, but without a fundamental change in functionality or operation. These innovations are often led by industry incumbents making improvements to better serve existing or adjacent customers' needs. Successive improvements to cable television—including digital transmission, improved video quality, premium programming, DVRs, and additional channels—represent sustaining innovations. Disruptive innovation is a change that creates a new market or enters an existing one but with a very different value proposition that may not immediately appeal to existing customers who are happy with the current product. Cable television was disrupted by streaming video delivered over the internet. At first a rudimentary offering that appealed to people who did not have cable subscriptions, it eventually grew to overtake cable in the number of subscribers.

Disruptive innovation is most likely to come from new entrants, because they do not have an existing profitable business to protect. The difficulty faced by incumbents with disruptive innovation is termed the "innovator's dilemma": they can either invest in the disruptive innovation themselves, speeding the decline of their existing business (and potentially harming employees) but not losing market share, or ignore the innovation and lose market share while continuing to generate strong profits in the near to medium term.

5.2.5. Legal Influences

Legal influences on an industry include changes in laws and regulations from courts and policymakers that alter an industry's business practices or economic outcomes. These influences, which are in addition to the fiscal and monetary policy changes discussed earlier under political influences, are often more subtle changes to the "rules of the road" governing industry affairs. Companies look to shape these influences through policy advocacy and in legal proceedings of their own.

Examples of industries strongly affected by legal influences include tobacco and cannabis. Jurisdictions have wide-ranging laws and regulations on the manufacturing, distribution, marketing, sale, and use of tobacco products to discourage consumption and the associated adverse health consequences, including secondhand smoke. Laws cover advertising prohibitions, graphic images on packaging, minimum purchase age, bans on smoking in public places, and limitations on nicotine content and flavorings. These laws are in addition to the generally high taxes levied on tobacco, which encourage non-taxable counterfeit activity that governments must police. Adoption of or changes in these laws in various jurisdictions significantly affect tobacco industry sales. In response to these influences, as well as broader social influences that have resulted in declining cigarette sale volumes, the tobacco industry is primarily focused on developing so-called reduced risk products that may pose fewer health risks and therefore create a more favorable legal environment.

The cannabis industry is in an evolving legal environment, where certain jurisdictions have legalized its production, sale, and use (Canada), while some have varying laws at the sub-sovereign level (the USA) and most others still completely prohibit cannabis (most of Asia).

5.2.6. Environmental Influences

Finally, environmental influences are often closely associated with legal influences and include risks and opportunities related to the transition to a lower-carbon economy, laws and practices concerning waste and land use, and environmental protections.

Transition risks and opportunities are important influences for sectors such as energy, airlines, utilities, and autos, as they are involved in the production or consumption of fossil fuels, which are responsible for carbon emissions. Due to changes in regulations, taxes, consumer preferences, or voluntary actions, companies in these sectors must evolve their business operations or face share losses to competing solutions with lower-carbon emissions.

Question Set

1. In 2009, Amazon.com started Amazon Publishing, extending its business into publishing books in addition to book retailing. From the perspective of a competing publisher, this is an example of:
 - A. decreasing the threat of substitutes.
 - B. decreasing the bargaining power of suppliers.
 - C. increasing the bargaining power of customers.

Solution:

C is correct. Amazon Publishing is an example of backward-integration, as it could now produce and sell its own published content as well as other publishers' content. The creation of Amazon Publishing increased the bargaining power of Amazon as a customer and increased the level of competitive rivalry in the publishing industry. Amazon Publishing would also increase the threat of substitutes and the bargaining power of suppliers, as authors and end users could work directly with Amazon.

2. Proposed government regulations for the restaurant industry would introduce new compliance requirements for all new and existing facilities to reduce the environmental impact of take-out and delivery options. These regulations would apply only to restaurants with a physical address and would not apply to mobile restaurants (i.e., food trucks). For an existing restaurant with several physical locations, these regulations would:

- A. reduce competitive rivalry by increasing the barriers to entry.
- B. increase competitive rivalry by increasing the threat of substitutes.
- C. both reduce and increase competitive rivalry.

Solution:

C is correct. The proposed regulatory changes would have a mixed impact on the level of competitive rivalry in the restaurant industry. These environmental regulations would increase the entry barriers by increasing the costs of setting up a new restaurant, while also increasing the threat of substitutes by making it easier to open mobile restaurants, which do not have to comply with these regulations.

Please refer to the Industry Survey for Warehouse Club Inc. and Iliso Marketplace Ltd. in the previous lesson to answer the following questions.

3. To compete with online retailers, Warehouse Club Inc. is considering installing a sophisticated image recognition system in its stores to track and analyze consumer behavior in order to identify clothing items that were tried on but not purchased. For Warehouse, this change would be a _____ (*sustaining* or *disruptive*) innovation.

Solution:

For Warehouse, this change would be a *sustaining* innovation. Sustaining improvements focus on adding marginal features without a fundamental change in functionality or operation. For example, the proposed system would provide incremental customer and product knowledge but would not represent a fundamental change in Warehouse's go-to-market strategy. Disruptive innovation is a change that creates a new market or enters an existing one with a different value proposition.

4. Identify the following statement as true or false. Justify your answer.

The CEO of Iliso Marketplace Ltd. expects revenue growth to be in line with US GDP growth for the next two fiscal years. As a result, the CEO approves an extension to an existing aggressive sales promotion and discount strategy that has been in place for the past 18 months. This extension will increase the competitive rivalry in the industry.

Solution:

False. The sales promotion and discount strategy is already in place within this mature industry, thus demonstrating a history of price competition among the competitors.

5. A PESTLE analysis for Iliso Marketplace Ltd. would *most likely* identify which of the following items?

- A. A competitor is acquiring a fleet of delivery vehicles instead of using third-party parcel carriers.
- B. The increasing political discussion around instituting a tax on parcel deliveries of less than 2 kilograms
- C. A competitor is launching an augmented reality application to allow tech-savvy consumers to view home décor and furniture products in real time.

Solution:

B is correct. The political discussion around potential applications of a tax is an example of a political trend changing over time. This theme could potentially affect Iliso Marketplace Ltd. A (fleet of delivery vehicles) and C (augmented reality) are examples of competitive actions that could affect the level of competitive rivalry in the industry.

6. COMPETITIVE POSITIONING

LEARNING OUTCOME

The candidate should be able to:

☐ **evaluate the competitive strategy and position of a company**

All companies have a competitive strategy, whether intentional or not. An intentional strategy results from company-wide planning, performance measurement, and feedback loops to sharpen the strategy. An unintentional strategy results from various teams within a firm pursuing their own incentives, doing whatever they did in prior years, or following industry or professional norms. An unintentional strategy is rarely the best and often exacerbates communication and coordination problems present at every company, although the approach might perform well in areas such as discovery-oriented research. [Munos \(2009\)](#) found evidence of diseconomies of scale in pharmaceutical R&D, with most new drugs discovered by small companies operating in a relatively “unmanaged” way.

Ultimately, an effective competitive strategy is evidenced by a company’s track record of value added for its stakeholders, such as economic profits. Its effectiveness can be judged only in hindsight. To evaluate a competitive strategy on a forward-looking basis, an analyst should assess the strategy along three dimensions:

- Does the strategy create a defense against the five industry forces?
- Does the strategy benefit from, or is at least not at odds with, the expected external industry influences identified in the PESTLE analysis?
- Does the company have the resources and capabilities to execute the strategy?

Clearly, the analysis and the answers to these questions are company and industry specific (CFA Institute has published a helpful industry-by-industry reference titled *Sector Analysis: A Framework for Investors* with examples). We can, however, identify three well-known competitive strategies that have worked in a variety of industries, as shown in [Exhibit 6](#): cost leadership, differentiation, and focus. These three generic strategies are based on Michael Porter’s research on competition. While companies do tend to have specific variations, generic strategies explain a large percentage of strategies in practice—similar to how many companies’ business models are well described using conventional models. A poor situation is for a company to be “stuck in the middle”—not a cost leader, not differentiated, not focused.

Exhibit 6: Generic Competitive Strategies

	Cost Leadership	Differentiation	Focus
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	Cost Leadership	Differentiation	Focus
Means of executing strategy	▪ Economies of scale from fixed costs ▪ Favorable access to raw materials ▪ Culture of strict cost control ▪ Aggressive pricing to gain high volume ▪ Low-cost distribution ▪ Economies of scope	▪ Investments in advertising, brand, customer service, proprietary distribution channels ▪ Protection using trademarks, copyright, patents ▪ Superior quality, unique features ▪ Culture of strong customer experience ▪ Premium pricing ▪ Integration of services, software, and hardware	▪ Proximity to customers and strong understanding of their needs ▪ May incorporate elements of strategy from both cost leadership and differentiation, but focused on particular group
Which of the Five Forces it defends against (why it works)	▪ Threat of new entrants: Capital requirements and scale advantages deter entrants ▪ Bargaining power of customers: Customers can only bring prices down to the costs of the marginal producer, leaving margin for the cost leaders ▪ Industry rivalry: Rivals may not be able to compete on price with cost leaders	▪ Threat of new entrants and of substitutions: Customer loyalty to unique product can deter switching, protect market share ▪ Bargaining power of customers: Customers may be unable to unwilling to comparison shop or switch ▪ Bargaining power of suppliers: The company may have the ability to pass along price increases to customers and/or margin to absorb cost increases	▪ Threat of new entrants and of substitutes: Customer loyalty to unique product can deter switching, protect market share ▪ Bargaining power of customers: Customers may be unable or unwilling to comparison shop or switch

	Cost Leadership	Differentiation	Focus
Industry appropriateness	▪ Capital intensive ▪ Price-conscious customers ▪ Customers do not value or notice product differences ▪ Minimal innovation in industry	▪ Price is not foremost concern for customers ▪ Customers value distinctiveness ▪ Innovation in industry, with products varying in features and forms	▪ Difficult (or uneconomical) to serve customer group, product, or geography for other players
Risks to the strategy	▪ Cost inflation, loss of discipline ▪ Technological change that results in loss of cost leadership or market share ▪ Desire for premiumization among customers	▪ Imitation by competitors ▪ Buyers become sophisticated, no longer demand level of service ▪ Pricing premium becomes too high for customers to bear ▪ May preclude high market share, as customers value exclusivity	▪ Larger competitors outcompete on price ▪ The differences in demand between the narrow group and industry as a whole narrow ▪ Buyers become sophisticated, no longer demand level of service

Question Set

- Manitou Resorts is a 25-room boutique hotel located in a rural mountain area approximately three hours from a major urban center. It provides accommodations, high-end dining options, and an on-site health and wellness spa. Manitou's primary customers are young professional couples in the middle-to-upper-income brackets. The CEO is contemplating several options for expansion, as the resort is frequently at full capacity. Which of the following initiatives would be consistent with Manitou's current competitive strategy?
 - The development of 10 stand-alone lakefront cottages to accommodate families of up to six people, providing access to family-related activities such as pools, water sports, and evening entertainment
 - The addition of conference room facilities to accommodate corporate events that could be priced at a 10%–15% discount to a 150-room franchise hotel approximately 30 minutes from Manitou

- C. Developing a management and reservation system to identify recommendations to referral partners such as golf courses, taxi/shuttle services, or other guest experiences to increase fee revenue

Solution:

C is correct. Manitou Resorts' current competitive strategy is a focus strategy targeting a specific customer segment: young professionals in middle-to-upper-income brackets located in a nearby city and focused on wellness and relaxation. The other options represent an expansion into new customer segments (family vacations and corporate events), which could be inconsistent with its current customer base. A represents a cost leadership competitive strategy, while B represents a differentiation strategy.

2. Warehouse Club Inc. is considering launching a "private label" brand of non-perishable food products. Based on its current competitive strategy, which of the following is the *most appropriate* rationale to support?

- A. Customers do not notice product differences.
- B. Customers value new product features and forms.
- C. Products can be segmented into regular and premium brands.

Solution:

A is correct. Warehouse Club Inc.'s current competitive strategy is cost leadership. In a cost leadership strategy, industry appropriateness is determined by capital intensity, price-conscious customers, customers not valuing or noticing product differences, and minimal innovation in the industry. B represents an example of a differentiation strategy. C represents an example of a focus strategy.

3. To defend against the bargaining power of suppliers, customers may be unable or unwilling to comparison shop or switch in which of the following competitive strategies (select all that apply):

- ☐ Cost leadership
- ☐ Differentiation
- ☐ Focus

Solution:

Both differentiation and focus. That "customers may be unable or unwilling to comparison shop or switch" is a common feature of both differentiation and focus strategies. In a cost leadership strategy, "customers can only bring prices down to the costs of the marginal producer, leaving margin for the cost leaders."

4. A risk to a differentiation competitive strategy is:

- A. limited market share due to exclusivity.
- B. larger competitors outcompete on price.
- C. technological change that results in loss of market share.

Solution:

A is correct. Since a differentiation strategy pursues a unique product or service offering, it creates a risk that the strategy may preclude high market share, as customers value exclusivity. B is a risk to a focus strategy where larger competitors outcompete on price. C is a risk to a cost leadership strategy where the technological change results in cost leadership or market share loss.

5. A large online retailer in the USA has begun to offer a “price-matching” guarantee on its website to match any competitor’s online price for an identical product, including Iliso Marketplace. Recommend and justify whether Iliso should launch a similar guarantee to protect its market share.

Solution:

No, Iliso should not launch a similar guarantee. A price-matching guarantee is a strategy pursued by companies with a cost leadership approach by using aggressive pricing to gain volume. Iliso Marketplace Ltd. operates with a differentiation strategy that permits premium pricing if aligned with a strong customer experience. If Iliso maintained its customer experience offering and began to reduce its revenue to match cost leaders, the company would likely decrease its overall profitability.

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PRACTICE PROBLEMS

1. An alternative method of grouping companies by geography is *least likely* to be completed using:
 - A. location of head office.
 - B. geographic composition of revenue.
 - C. primary listing of its equity securities.
2. A market consists of three firms with market shares of 50%, 30%, and 20%. The Herfindahl-Hirschman Index (HHI) is *closest* to:
 - A. 0.38.
 - B. 2,500.
 - C. 3,800.
3. Lower industry concentration is usually associated with a high degree of competitive intensity unless the industry is *most likely*:
 - A. global.
 - B. service-oriented.
 - C. one with low product differentiation.
4. Increased environmental regulations on automotive manufacturers will *most likely* have which of the following effects on the threat of new entrants?
 - A. An increase
 - B. A decrease
 - C. No change
5. The CEO of a large law firm is concerned about a new mobile application that uses algorithms to auto-complete legal forms and questions in discovery and provides recommendations for small-claims matters at a much lower cost than a traditional law firm charges. As a result, the CEO is contemplating whether the firm should develop and launch its own branded application, which would affect its short-term profitability but maintain its existing customer base, or give up market share to focus on more complex claims with higher profitability. This scenario is an example of:
 - A. defensiveness.
 - B. innovator's dilemma.
 - C. sustaining innovation.
6. PESTLE analysis is a framework for identifying:
 - A. industry themes.
 - B. the level of industry concentration.
 - C. determinants of industry profitability.
7. The means of executing cost leadership competitive strategies *do not include* which of the following?

- A.** proximity to customers
- B.** favorable access to raw materials
- C.** economies of scale from fixed costs

8. Which of the following is *most likely* a risk of executing a differentiation competitive strategy?

- A.** pricing premiums become too high
- B.** larger competitors outcompete on price
- C.** a desire for premium-ization among customers

SOLUTIONS

1. B is correct. Classification by country is typically by the country where the issuer is incorporated, the country of the primary listing of its equity securities, the location of its headquarters, or market perception. Note that classification by the geographic composition of revenue is generally *not* the approach taken, though this aspect may be the foremost concern for an analyst.
2. C is correct. The Herfindahl-Hirschman Index (HHI) is calculated as the sum of the squares of competitor market shares. In this example, the HHI is calculated as $(50^2 + 30^2 + 20^2 = 3,800)$. A incorrectly calculates the HHI using the percentage market shares $(0.5^2 + 0.3^2 + 0.2^2 = 0.38)$. B incorrectly calculates the HHI using only the market leader concentration $(50^2 = 2,500)$.
3. B is correct. Lower industry concentration, defined as many small competitors in the market, is usually associated with a high degree of competitive intensity unless the industry is service-oriented, is local in nature, or has high product differentiation.
4. B is correct. Increased environmental regulations on automotive manufacturers represent an additional cost for compliance when building a new manufacturing facility and increase the barriers to entry into the industry.
5. B is correct. The described mobile application represents a fundamental change in the business model of a law firm. The application is a disruptive, not sustaining, innovation, as it brings new entrants into the market but with a very different value proposition. The CEO's decision represents the "innovator's dilemma": the firm can either invest in the disruptive innovation, speeding the decline of its existing business but not losing market share, or ignore the innovation and lose market share while continuing to generate strong profits in the near term.
6. A is correct. PESTLE analysis is a framework for identifying "themes" or "narratives" that investors may take a perspective on and desire exposure to. B refers to the Herfindahl-Hirschman Index. C refers to Porter's Five Forces.
7. A is correct. Proximity to customers is a means of executing a focus strategy. The means of executing a cost leadership strategy include economies of scale from fixed costs, favorable access to raw materials, a culture of strict cost control, aggressive pricing to gain high volume, low-cost distribution, and economies of scope.
8. A is correct. The risks of a differentiation strategy include imitation by competitors, buyers becoming sophisticated and no longer demanding level of service, and pricing premiums becoming too high for customers to beat; such a strategy may also preclude high market share, as customers value exclusivity. B relates to a focus strategy, while C relates to a cost leadership strategy.

Company Analysis: Forecasting

LEARNING OUTCOMES

The candidate should be able to:

- ☐ explain principles and approaches to forecasting a company's financial results and position
- ☐ explain approaches to forecasting a company's revenues
- ☐ explain approaches to forecasting a company's operating expenses and working capital
- ☐ explain approaches to forecasting a company's capital investments and capital structure
- ☐ describe the use of scenario analysis in forecasting

1. INTRODUCTION

Forecasts of companies' financial statements are used by analysts in valuation and to make investment recommendations. Developing the forecasts or projections is an important aspect of an analyst's job and is the focus of this module. In the first lesson, what to forecast, approaches to forecasting, and selecting a forecast horizon are discussed. The next three lessons focus on particular forecasts: revenues, operating expenses and working capital, and capital investments and capital structure. The final lesson discusses the use of scenario analysis in considering multiple outcomes.

Learning Module Overview

- Four common types of forecast objects are drivers of financial statement lines, individual financial statement lines, summary measures, and ad hoc objects. An analyst's choice of forecast object depends on available information, efficiency, accuracy, explanatory value, and verifiability.
 - Forecast approaches generally are based on historical results, historical base rates and convergence, management guidance, or analyst discretion. An analyst's choice of forecast approach depends on the company's industry structure, sensitivity to the business cycle, and business model, as well as the reliability and availability of information.
 - The choice of the forecast time horizon is determined by the investment strategy for which the security is being considered, the cyclicity of the industry, company-specific factors, and the analyst's employer's preferences.
 - Revenue forecasts may be based on top-down or bottom-up forecast objects, using any of the four forecast approaches. Using different forecast objects and approaches to project revenue can be useful in uncovering implicit assumptions or errors in any single approach.
 - Top-down revenue drivers include growth relative to GDP growth, and market growth and market share. Bottom-up revenue drivers include volumes and average selling prices; revenue by product line, geographic area, or reporting segment; capacity-based measures; and return-based measures.
 - Analysts often use aggregated forecast objects or summary measures to forecast operating expenses because of a lack of disaggregated information. However, forecasts for operating expenses should be coherent with revenue forecasts. The choice of forecast object can vary depending on the forecast horizon.
 - Working capital forecasts typically use efficiency ratios combined with revenue and operating expense forecasts to project accounts receivable, inventory, accounts payable, and other current assets and liabilities.
 - Forecasts for capital expenditures may differentiate between maintenance and growth capital expenditures. Maintenance capital expenditure forecasts are often based on depreciation and amortization expenses. Growth capital expenditure forecasts are tied to a company's strategy, expansion plans, and revenue growth.
 - Forecasts about a company's capital structure consider historical leverage ratios and capital structure, the company's financial strategy, and capital expenditure forecasts.
 - Based on a company's risk factors, an analyst may develop several forecast scenarios rather than develop a single forecast. The analyst will judge the likelihood of each scenario occurring.
-

Learning Module Self-Assessment

These initial questions are intended to help you gauge your current level of understanding of this learning module.

1. A benefit of using summary measures as forecast objects is *most likely*:

- A. efficiency.
- B. transparency.
- C. explanatory value.

Solution:

A is correct. The benefit of using summary measures as a forecast object is efficiency, as fewer variables need to be forecast, but a disadvantage is less transparency. Forecasting financial statement lines or their drivers has the benefit of improved explanatory value.

2. Which of the following approaches is *most* appropriate for forecasting annual revenues for a company in a cyclical industry?

- A. Historical results
- B. Management guidance
- C. Analyst's discretionary forecast

Solution:

C is correct. An analyst's discretionary forecast is most appropriate, because it allows the analyst to consider both the current phase and the expected future phase of the business cycle. Historical results are a less appropriate forecast approach for companies in cyclical industries, because a future period is likely to be at a different point in the business cycle than the current or past period, and so results will differ. The use of guidance for companies that are highly sensitive to the business cycle is less appropriate, as management does *not* have an informational advantage over investors in forecasting macroeconomic variables like GDP or the prices of commodities.

3. An example of a top-down driver for revenue is a company's:

- A. market share.
- B. sales growth rate.
- C. products' average selling price.

Solution:

A is correct. Top-down drivers of revenue include the company's market share and the industry's market growth. The company's sales growth rate and products' average selling price are bottom-up drivers of revenue.

4. In developing a revenue forecast, non-recurring revenue *most likely*:

- A. has the same drivers as recurring revenue.
- B. will be disclosed by a company's management.
- C. is considered separately from recurring revenue.

Solution:

C is correct. Non-recurring items and effects should be considered separately. The non-recurring objects, such as changes in exchange rates or provisions for legal costs, generally do not have the same drivers as the recurring objects. While management may disclose some non-recurring items and effects, there are also non-recurring items and effects that management does not disclose, which require analyst judgment.

5. An analyst should consider the effects of changing commodity prices when forecasting:

- A. revenue only.
- B. cost of sales only.
- C. both revenue and cost of sales.

Solution:

C is correct. Both a company's cost of sales through input prices and its revenue may be affected by commodity prices. The effect on revenue depends on the company's competitive positioning and the price elasticity of demand for the company's products.

6. Working capital forecasts for a mature company with a unique business model are *most likely* made by:

- A. increasing the current assets and liabilities by the sales growth rate.
- B. using the company's efficiency ratios combined with sales and operating forecasts.
- C. using a historical base rate and convergence approach to develop forecasts of revenue, operating expenses, and efficiency ratios.

Solution:

B is correct. For a company with a unique business model, using a historical results approach is appropriate. Working capital forecasts are typically made by using efficiency ratios, which are combined with sales and operating cost forecasts to project accounts receivable, inventories, accounts payable, and other current assets and liabilities. The historical base rate and convergence approach is appropriate if the company is converging toward the norm for the industry. Adjusting current assets and liabilities by the sales growth rate is not appropriate, because changes in sales will affect each item differently.

7. Depreciation and amortization expenses are often used as the basis for:

- A. growth capital expenditures only.
- B. maintenance capital expenditures only.
- C. both growth and maintenance capital expenditures.

Solution:

B is correct. Maintenance capital expenditure forecasts are often based on depreciation and amortization expenses. Growth capital expenditure forecasts are more discretionary and are tied to management's expansion plans and revenue growth.

8. Which of the following generic risk factors is *most likely* to affect an established company in the consumer staples sector?

- A. Inflation or deflation
- B. Technological developments
- C. Changes in the business cycle

Solution:

A is correct. While all of these choices are generic risk factors, inflation or deflation is the one most likely to affect an established company in the consumer staples sector. The consumer staples sector is not likely to experience significant technological developments and is generally not sensitive to changes in the business cycle.

2. FORECAST OBJECTS, PRINCIPLES, AND APPROACHES

LEARNING OUTCOME

The candidate should be able to:

- ☐ **explain principles and approaches to forecasting a company's financial results and position**

Approaches to forecasting vary by analyst and setting. Analysts who publish company research for external distribution tend to maintain quarterly forecasts and invest a great deal of effort to achieve forecast accuracy of revenue and earnings per share estimates. An investor with a controlling position in a private company has access to far more information, which may be used to build highly detailed models with a multi-year or even multi-decade perspective. Additionally, there are forecasting norms that vary by business model of the target company, such that models for banks appear markedly different from those for oil and gas producers.

In this module, we take the perspective of a longer-term-oriented analyst forecasting relatively straightforward public issuers—Warehouse Club Inc. and Iliso Marketplace Ltd., introduced in the prior two modules—but the principles discussed are broadly applicable. The more technical details of financial statement modeling using spreadsheets are covered in lessons in Financial Statement Analysis.

2.1. What to Forecast?

Analysts may focus on different **forecast objects** related to issuers' financial statements. Below are four common forecast objects.

- 1. Drivers of financial statement lines*, which were illustrated in the earlier module on company analysis. Recall that for Warehouse Club Inc., net sales each year was analyzed using two bottom-up drivers: the average number of stores open and the average net sales per store, both of which were also analyzed using year-over-year percentage changes over the last 10 years. Forecasts of net sales (at the top of Warehouse's income statement) can be made by forecasting these drivers individually and multiplying them. Gross margin and SG&A expenses as percentages of net sales were identified as key drivers of operating expenses, which can be forecast and then multiplied by the net sales forecast to obtain the forecast cost of sales and SG&A expenses. Forecasting drivers rather than financial statement lines outright has the benefit of improved explanatory value and may improve accuracy; a financial statement line item can have multiple drivers moving in different directions, which might be difficult to forecast on an aggregated basis.
- 2. Individual financial statement lines*. Rather than drivers, analysts can also directly forecast individual financial statement lines. This approach is often used for lines without clear drivers, for less-material items, and for items that the analyst does not have a perspective on. In some cases, the analyst may look for an estimate by management or simply assume the quantity will remain the same in future periods as in the past. Examples include lines such as amortization expense on the income statement (a forecast for which is often provided by management in the annual report),

“other non-current assets” on the balance sheet (when small), and various lines on statements of cash flows for which minimal disclosures are provided.

3. *Summary measures* such as free cash flow, earnings per share, and total assets. The benefit of using these as forecast objects is efficiency. The disadvantage is less transparency, making it difficult to audit the forecast. This approach is most appropriate if the summary measure is stable and predictable, or if issuer disclosures are severely limited.
4. *Ad hoc objects*, which may not yet be reported on historical financial statements. In some cases, an analyst may need to forecast a loss or gain and its timing to make an investment decision with respect to the company’s equity or debt securities before the issuer recognizes an accrual on its financial statements. Examples of such cases include an issuer’s announcing a material legal proceeding, government regulatory action, a tax dispute, or a natural disaster.

2.2. Focus on Objects That Are Regularly Disclosed

We recommend using forecast objects that either are disclosed regularly or can be directly calculated using what is disclosed regularly (e.g., for Warehouse Club: net sales, number of stores, net sales per store, national retail sales published monthly by a government agency). Information that is *not* disclosed regularly (such as the size of a market from a third-party consultancy’s report) is suitable for *informing* forecasts but can be problematic for direct use because forecasts cannot be confirmed in a timely manner. While it may be intuitive to forecast sales and gross margin by individual product line for a company, if gross margin is disclosed only on a consolidated basis, it will be difficult for the analyst to verify product-line gross margin estimates. If the company’s reported gross margin results differ from gross margin estimates, is the forecast off on product A, B, or C? Similarly, if a quantity is disclosed only on an annual basis, a forecast of it can be confirmed only once a year.

We also caution against overly complex models, because they require more forecasts and take more time to update, often without any improvement in accuracy. For example, the Swiss pharmaceutical company Novartis AG discloses revenues on a quarterly basis for over 35 individual drugs by US and Rest of World regions. Most of the drugs are approved for more than one type of illness. The company also has over 50 drugs in late-stage clinical development that may drive future revenue. While an analyst could forecast Novartis AG’s revenues by forecasting sales for each of its individual drugs by illness and geography, it would take months to create such a model and weeks to update it every quarter, and the forecasts by illness are not verifiable over time based on the company’s disclosures. Instead, the analyst should use judgment to focus on the most important drivers and use aggregations and shortcuts where the value-added of discrete forecasting is low. For example, it would not be a good use of an analyst’s time to forecast sales for drugs with small addressable markets or those that have reached maturity and will exhibit slow growth until they lose patent protection.

2.3. Forecast Approaches

For any object, there are four general forecast approaches, as shown in [Exhibit 1](#). In practice, they are often combined.

Exhibit 1: Forecast Approaches for Financial Statement Objects



2.3.1. *Historical Results: Assume Past is Precedent*

This approach uses past observed or calculated values (such as a historical median) as a forecast. This is the easiest approach and is, perhaps, the default. Because the past occurred at least once before, it seems plausible to expect that it will again. However, past results were produced in past conditions, which may differ from current and future conditions.

USING HISTORICAL RESULTS AS FORECASTS

An analyst is forecasting the market size for energy drinks, a type of non-alcoholic ready-to-drink beverage. The analyst uses two drivers: a forecast of the market for non-alcoholic ready-to-drink beverages and energy drinks' share of that market. Sales of non-alcoholic ready-to-drink beverages have grown, on average, by 3% per year for the last 10 years. Since the analyst does not expect any material changes in conditions, she forecasts that the market will continue to grow by 3% per year.

The analyst believes that energy drinks bear some similarities to sports drinks, another type of non-alcoholic ready-to-drink beverage, which has achieved a market share of 20%. The analyst thus forecasts that energy drinks will also reach a market of 20% by 2035. These two forecasts are multiplied to arrive at a forecast market size for energy drinks.

This approach may be appropriate for companies operating in industries where the analyst does not expect the industry structure (e.g., Porter's Five Forces, PESTLE influences) to change, as well as for

companies that have a low sensitivity to changes in the business cycle. This approach is also commonly used for forecast objects that are not material or that the analyst does not hold an opinion on.

Historical results are a less appropriate forecast approach for companies in cyclical industries, because a future period is likely to be at a different point in the business cycle than the current or past period, and so results will differ. An “over the cycle” average or median may be appropriate for a multi-year forecast for a cyclical company, but less appropriate for a specific year as such an average or median disguises the year-to-year volatility. This forecast approach is also less appropriate for companies that are changing their competitive strategy or restructuring in some way, such as making a large acquisition or divestiture that renders historical results non-comparable.

2.3.2. Historical Base Rates and Convergence

This approach uses an industry or peer group average or median, computed over a long period of time, as a “base rate” for forecasting that an object will converge to over some time frame. The base rate could also be a macroeconomic variable such as GDP growth. This approach is not a rote historical exercise, as analyst discretion is required in selecting the object, the sample to calculate the base rate, and a time frame for convergence to the base rate.

HISTORICAL BASE RATES AND CONVERGENCE

Tesla is a relatively new entrant in the automotive industry, with an innovative suite of electric vehicles. An analyst may forecast Tesla’s income statement by forecasting that its sales growth, gross and operating margins, capital intensity, and capital structure will converge to historical automotive industry averages (using a peer group of publicly traded global automakers) over the next 10 years.

This forecast may be contested by other analysts who believe that the shift from internal combustion to electric vehicles, Tesla’s innovative battery technology, and its direct-to-consumer sales approach make historical auto industry averages irrelevant for Tesla.

Historical base rates and convergence may be appropriate for forecasting companies in well-established industries with many publicly traded peers, such as banks, airlines, restaurants, automakers, and retailers, especially when the analyst’s industry analysis does not call for material changes in industry structure or external industry influences. This approach is also logical for smaller companies that are “maturing into” a financial profile similar to that of larger peers with scale. For example, a biotechnology company that recently launched its first drug likely spends more than half its revenues on R&D, a ratio that is likely to decline toward the industry norm as the company scales.

This approach is less appropriate for companies in changing or new industries for which calculating a base rate is difficult or less relevant, for companies in highly cyclical industries for which a longer-term base rate and smooth convergence to it would obscure year-to-year volatility, and for companies that are industry leaders by a wide margin and account for a substantial proportion of any industry base rate. For example, “digital advertising industry” averages for sales growth and operating margins would be dominated by Alphabet and Meta, such that using them to forecast Alphabet’s and Meta’s financial results would simply be using their own historical results as the forecast, which may ignore any benefit from economies of scale (as the companies were smaller in the past).

2.3.3. Management guidance

Management of public companies may publicly provide targets for earnings, revenues, and other measures (e.g., capital expenditures) for the next quarter, year, or longer term, known as **management guidance** (or simply as guidance). Guidance can be detailed or rather directional and is often updated throughout the year. Initial guidance for next fiscal year might be provided during the fourth-quarter earnings call and updated for completed quarters, and new information provided at the first-, second-, and third-quarter earnings calls. [Beyer et al. \(2010\)](#) found that guidance accounts for a majority of the

quarterly financial reporting information used by investors. Its importance stems from its forward-looking rather than backward-looking nature, and because management has an abundance of industry and company information that investors do not.

Guidance is often provided as a range (e.g., “sales growth of 2%–4%”) and embeds many sub-forecasts and assumptions by management, including macroeconomic growth, cost inflation, market share changes, pricing actions, and changes in exchange rates. If management’s estimates for these change, so will guidance, and it is not uncommon for companies to suspend guidance altogether in periods of high uncertainty, such as during the COVID-19 pandemic or in recessions. A key focus of investors is understanding management’s assumptions embedded in guidance and scrutinizing their plausibility.

While the midpoint of a guidance range might be intuitively interpreted as management’s “true” expectations, [Ciconte et al. \(2013\)](#) found that the upper bound often better represents management’s expectations, which are “padded” by associating it with a pessimistic lower bound to create a hurdle that is more easily cleared, for which management can be rewarded.

SHORT- AND LONG-TERM MANAGEMENT GUIDANCE

Novo Nordisk A/S is a large biopharmaceutical company with shares and American Depositary Receipts (ADRs) listed on the NASDAQ Copenhagen and NYSE, respectively. Its management provides annual guidance that it updates each quarter, as well as less detailed but longer-term “strategic aspirations” during its annual Capital Markets Day presentations. Guidance includes management’s point and range estimates for the following (for each fiscal year):

- Sales growth
- Operating profit growth
- Net interest income/expense
- Effective tax rate
- Capital expenditures
- Depreciation and amortization
- Free cash flow

Using guidance for forecasts is appropriate when it is provided and when management has demonstrated a track record of reliable estimates (analysts should analyze past guidance versus actuals). We caution against the use of guidance for companies that are highly sensitive to the business cycle, as management does *not* have an informational advantage over investors at forecasting macroeconomic variables like GDP or the prices of commodities. [Hutton et al. \(2012\)](#) found that investors have an informational advantage at the macroeconomic level, while management has an advantage at the firm level and tends to make more accurate forecasts for objects under greater management control, such as operating expenses and capital expenditures.

2.3.4. *Analyst’s discretionary forecast*

We group all other forecast approaches as analysts’ discretionary forecasts, which include those based on surveys, quantitative models, probability distributions, analogies to historical precedents that differ from comparable companies or industry averages, and other unobservable inputs. This approach is most common for companies in cyclical industries, companies that have no or few comparables, those that do not provide management guidance, and/or those undergoing a fundamental change like a shift in the competitive or regulatory environment. For example, while shifts in energy production have happened before (e.g., coal to natural gas), climate change and the potential shift to renewable energy and adoption of technologies like electric vehicles are unprecedented, with no historical observations to use as forecasts. Instead, an energy sector analyst must make discretionary forecasts based on many sources,

such as government emission reduction commitments, proposed legislation, capital expenditure constraints, and so on.

DISCRETIONARY FORECASTS

An analyst is building a financial statement model for a medical technology company developing a new device to treat a type of stroke for which no other effective treatments exist. The analyst uses information from the device's clinical trials and epidemiology data from various medical journals to estimate the total addressable market of patients that are treatable with the device. The analyst also conducts a survey of specialist physicians to gauge their interest in using the device and to estimate the number of patients they would use it for (depending on the results of the clinical trials), and uses recent launches of other medical devices for neuro- and cardiovascular diseases to estimate market share for this device for the next 10 years. The analyst develops several market share scenarios based on the results of the clinical trials (i.e., the safer and more effective the device, the higher sales growth will be). While the analyst uses comparable, historical results as inputs, significant judgment is also required because this situation is unique.

2.4. Selecting a Forecast Horizon

The choice of the forecast time horizon is determined by the investment strategy for which the security is being considered, the cyclicity of the industry, company-specific factors, and the analyst's employer's preferences. Most professionally managed investment strategies describe the investment time frame and average holding period in the investment objectives of the strategy. For example, long-term-oriented fund managers may focus their forecasting primarily on a time period of three to five years, while shorter-term-oriented managers may focus more on the next one or two quarters.

The cyclicity of the industry could also influence the analyst's choice of time frame, because the forecast period should be long enough to allow the business to reach an expected mid-cycle level of sales and profitability. Similar to cyclicity, various company-specific factors, including recent acquisition or restructuring activity, can influence the selection of the forecast period to allow enough time for the realization of the expected benefits from such activity to be reflected in the financial statements. In other cases, there might be no individual analyst choice in the sense that the analyst's employer has specified fixed parameters.

Question Set

1. A company discloses financial results by geographic area, and the economic exposure of each geographic area is similar. Explain why an analyst should or should not develop forecasts by geographic area and aggregate the results.

Solution:

An analyst should *not* develop forecasts by geographic area, because each area has similar economic exposure. Developing forecasts for each geographic area would require more time to develop and update, without a significant improvement in accuracy over an aggregate forecast.

2. Which approach is *most* appropriate for an analyst to use to forecast revenue for an industry leader in a mature, non-cyclical industry?
 - A. Historical results
 - B. Analyst's discretionary forecast
 - C. Historical base rates and convergence

Solution:

A is correct. Historical results is appropriate for companies operating in mature industries where the analyst does not expect the industry structure to change and for companies with low sensitivity to the business cycle. Analyst's discretionary forecast is most common for companies that have no or few comparables, do not provide management guidance, and/or are undergoing a fundamental change. Historical base rates and convergence is not appropriate, because the company is an industry leader and would account for a substantial proportion of any industry base rate.

3. Which of the following statements about management guidance is *most* accurate?

- A. The availability of guidance increases during periods of high uncertainty.
- B. The midpoint of guidance best represents management's "true" expectations.
- C. Understanding the assumptions behind the guidance is critical to assessing the guidance.

Solution:

C is correct. A key focus of an analyst is to understand management's assumptions embedded in guidance and to scrutinize their plausibility.

It is not uncommon for companies to suspend guidance altogether in periods of high uncertainty. While the midpoint of a guidance range might be intuitively interpreted as management's "true" expectations, management may provide a range specifically intended to create a hurdle that is more easily cleared, for which management can be rewarded.

4. Which of the following is an example of an ad hoc forecast object?

- A. Amortization expense
- B. Average sales per store
- C. Pending legal proceedings

Solution:

C is correct. Ad hoc objects are those that may not yet be reflected in historical financial statements. Amortization expense is an individual financial statement line. Average sales per store is a bottom-up driver of a financial statement line (revenue).

3. FORECASTING REVENUES

LEARNING OUTCOME

The candidate should be able to:

- ☐ explain approaches to forecasting a company's revenues

After the forecast time horizon is determined, a revenue forecast requires the selection of the forecast objects and approach.

3.1. Forecast Objects for Revenues

Forecast objects for revenues are typically either top-down or bottom-up drivers, as discussed in the earlier module on company analysis. Common top-down forecast objects include “growth relative to GDP growth” and “market growth and market share.”

Growth relative to GDP growth. The analyst first forecasts the growth rate of nominal GDP. The analyst then considers how the growth rate of the specific company being examined will compare with nominal GDP growth. The analyst can use a forecast for real GDP growth to project volumes and a forecast for inflation to project prices. Analysts often think in terms of percentage point premiums or discounts derived from a company’s position in its life cycle (e.g., growth, mature, or decline) or business cycle sensitivity. Thus, an analyst’s conclusion might be that a healthcare company’s revenue will grow at a rate of 200 bps above the nominal GDP growth rate. The forecast could also be in relative terms. Thus, if GDP is forecast to grow at 4% and the company’s revenue is forecast to grow at a 50% faster rate, the forecast percentage change in revenue would be $4\% \times (1 + 0.50) = 6.0\%$, or 200 bps higher than the GDP growth rate forecast in absolute terms.

Market growth and market share. The analyst first forecasts a growth rate for a company’s product market, and then considers the company’s current market share and how that share is likely to change over time. For example, if a company is expected to maintain an 8% market share of a given product market and the product market is forecast to grow from CNY144 billion to CNY154 billion in annual revenue, the forecast growth in company revenue goes from a level of $8\% \times \text{CNY144 billion} = \text{CNY11.5 billion}$ to a level of $8\% \times \text{CNY154 billion} = \text{CNY12.3 billion}$ (considering this product market alone). If the product market revenue has a predictable relationship with GDP, regression analysis might be used to estimate the relationship.

In contrast, an analyst could select bottom-up drivers as the forecast objects, which are then aggregated to arrive at a forecast of total revenue for the company. Examples of bottom-up drivers for revenue forecasts include the following:

- *Volumes and average selling prices.* Forecasts for volumes and prices of the company’s products are prepared individually and multiplied to arrive at a revenue forecast. This approach is commonly used for companies that disclose these quantities, such as media and internet companies (users and revenues per user), airlines, asset managers (i.e., AUM and management fee rates), and commodity producers. Many companies disclose price and volume contributions to overall revenue growth, which can be used as the forecast object as well.
- *Product-line or segment revenues.* Forecasts for individual products, product or business lines, geographic areas, or reporting segments are made and then aggregated into a total revenue forecast. This approach is commonly used if a company makes such disclosures and if the objects have different economic exposures.
- *Capacity-based measures.* Forecasts, for example, in retailing, based on the number of stores and sales per store, or same-store sales growth (for stores that have been open for at least 12 months) and sales related to new-store openings.
- *Return- or yield-based measures.* Forecasts based on account balances and revenue yields on them. For example, net interest income for a bank can be calculated as loans multiplied by the average interest rate minus the product of deposits and liabilities and their average interest rate.

Using elements of both top-down and bottom-up objects can be useful for uncovering implicit assumptions or errors that could arise from using a single approach (e.g., forecasting revenues using bottom-up drivers but checking against what this approach would imply for the company’s market size and share).

3.1.1. Separating Recurring and Non-Recurring Revenue or Revenue Growth

An important principle is that non-recurring items and effects should be excluded from a forecast object and considered separately. The non-recurring objects generally do not have the same drivers as the

recurring object and may inflate or deflate its size. There are two types of non-recurring items and effects: those that are disclosed by a company’s management and those that are not, which require analyst judgment to identify.

For revenue, the first type includes the effect of changes in exchange rates, extra selling days and weeks in the period, acquisitions/divestitures, and other “one-time” revenues or gains. These are separated from a revenue forecast object so that “underlying” revenue or growth can be forecast apart from the non-recurring items (for which a discrete forecast may not be made at all). This is generally a straightforward task unless the credibility of an issuer’s financial reporting is in doubt, in which case the analyst may judge certain “non-recurring” items as, in fact, recurring. Some analysts may also use proprietary exchange rate forecasts in revenue projections.

The second type of non-recurring items is those *not* quantified by management and that require analyst judgment to estimate. For example, the COVID-19 pandemic had numerous economic effects, including a rapid increase in e-commerce sales as a percentage of retail sales in most geographies. Many believed that this effect was a “new normal,” that the increase in e-commerce sales would recur and subsequent periods would show growth from the newly achieved level. By the second half of 2021, this belief was called into question, because e-commerce sales as a percentage of retail sales began to decline, returning to a trend established prior to the pandemic. Many e-commerce companies saw revenue decelerate or decline in 2022, confirming that some amount of e-commerce’s COVID-19-related growth was, in fact, non-recurring. As a second example, graphic processing units (GPUs) are types of semiconductors used in a variety of applications, including gaming PCs. GPU manufacturers saw a large increase in sales in the late 2010s, which many believed was partly attributable to the use of GPUs in cryptocurrency mining, but the companies either could not or would not quantify the precise amount. Although that amount is unknown, an analyst may judge such sales as non-recurring (e.g., due to government restrictions on cryptocurrency mining) and forecast these GPU sales separately from those attributable to gaming.

3.2. Forecast Approaches for Revenues

Any of the four forecast approaches introduced in Exhibit 1 in the prior lesson (historical results, historical base rates and convergence, management guidance, discretionary forecasts) can be used for forecasting revenue objects, with the selection based on the analyst’s judgment, the information environment, and the issuer’s business model. We illustrate how Elaine Nguyen might use the four approaches to forecast revenues for Warehouse Club Inc. in [Exhibit 2](#).

Exhibit 2: Forecast Approaches for Warehouse Club Inc.’s Revenues

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Top-down drivers (e.g., market size, market share)	Historical results	US retail sales excluding autos has grown on average by 3.4% each year for the last 10 years and the company’s share of that (70 basis points in 2X19) has increased by 2 basis points per year. Nguyen can forecast revenues by assuming 3.4% market growth and 2 basis points of market share gain each year going forward.
	Historical base rates and convergence	Rather than assuming continued share gains, Nguyen may forecast that Warehouse’s sales growth rate will converge to the market growth rate over the next five years, and forecast revenues by assuming a smooth decline in annual market share gains to 0 basis points by 2X25.
	Management guidance	If Warehouse provides annual or longer-term expectations for market size, market share, or sales, Nguyen can use these in her revenue forecast.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
	Analyst's discretionary forecast	Warehouse has operated stores in some regional geographies for a long time. Nguyen may commission a study or survey to determine Warehouse's market share in its most established markets (which could be multiples of its all-in market share) and develop market size and share estimates by established/growing/new regions for Warehouse and aggregate them into a sales forecast.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Bottom-up drivers (e.g., stores, sales per store, membership, membership annual price)	Historical results	Warehouse has opened 4–6 new stores, grown net sales per store by 3.8%, and added ~450,000 members per year for the last 10 years. Nguyen could assume these same quantities going forward as the revenue forecast.
	Historical base rates and convergence	There are several mature retailers in the USA that ceased opening new stores several years ago. Nguyen may slow the rate of new-store openings gradually to zero in the USA.
	Management guidance	Management has provided guidance on new-store openings per year and the annual price of membership, which are bottom-up drivers for the revenue forecast.
	Analyst's discretionary forecast	Nguyen may develop a discretionary forecast for stores and members by determining stores per capita and the ratio of members to stores in Warehouse's more mature regional geographies and then applying these quantities, or fractions of them, to different regions where Warehouse is opening stores today. This approach would require a geographic analysis of stores and commissioning surveys to determine regional membership counts or the use of third-party or government data, if available.

Analysts must incorporate their view (implicitly or explicitly) on key risk factors in their revenue forecasts. Four risk factors to consider for all companies are competition, changes in the business cycle, inflation and deflation, and technological developments. For Warehouse Club Inc., two key risk factors are competition and technological developments, embodied in e-commerce, which has been taking share in US retail for many years. Views on these risk factors are also embedded in forecasts of profitability, working capital, capital investments, and capital structure. Given the potential wide range of outcomes, analysts often construct several forecasts for an issuer's financial statements called *scenarios*, each based on different views of key risk factors. We discuss scenario analysis in the last lesson of this module, after forecasting other key financial statement drivers.

Question Set

An analyst gathers the following information for Iliso Marketplace Ltd. (Iliso), the e-commerce retailer introduced in earlier modules. Recall that sales by third-party merchants are recognized on a net basis, because Iliso does not take control of the inventory. Iliso receives 15% of the gross merchandise value (GMV) of third-party sales as commission.

2X19
(USD millions)

	2X19 (USD millions)
National retail sales excluding autos	4,165,000
Iliso's retail sales	7,336
Iliso's third-party merchant sales	1,269
Total GMV	8,605
Iliso's retail sales	7,336
Third-party merchant fees	190
Total Revenues	7,526

Gross merchandise value has increased by an average of 3.6% per year from 2X15 to 2X19 and by 2.7% from 2X18 to 2X19. Iliso's GMV as a percentage of national retail sales has increased steadily, from 0.08% in 2X15 to 0.21% in 2X19.

	2X19	Average Growth per Year, 2X15 to 2X19	Growth from 2X18 to 2X19
Average sales/customer account (USD)	204	5.9%	4.6%
# accounts (millions)	36	22.5%	16.1%
Iliso's retail sales (USD millions)	7,336	29.7%	21.4%
Average sales/third-party merchant (USD)	31,300	13.6%	9.8%
# third-party merchants	40,500	28.5%	28.6%
Third-party merchant sales (USD millions)	1,269	46.0%	41.2%

1. Explain how an analyst would use top-down drivers to forecast Iliso's 2X20 GMV.

Solution:

An analyst would first forecast national retail sales excluding autos and then forecast Iliso's GMV as a percentage of that market, which represents Iliso's market share. National retail sales excluding autos have been increasing each year, and the analyst is likely to use a growth rate between 2.7% and 3.6%, depending on economic conditions. Iliso's market share in 2X19 was 0.21%, and given past increases, an increase of 2 to 3 bps seems reasonable.

2. Explain why an analyst using top-down drivers needs to allocate Iliso's forecast GMV sales between its own retail sales and sales by third-party merchants. Describe two alternative approaches to doing so.

Solution:

It is important that an analyst allocate the forecast total GMV between Iliso and third-party merchants, because Iliso's revenue and cost structure differs for each. Iliso recognizes revenue on a gross basis for its own retail sales and on a net basis for sales made by third-party merchants. Iliso's take rate is 15% on third-party merchant sales, but its gross margin percentage is far greater.

Currently, Iliso's retail sales account for 85% of its GMV, while third-party merchant sales account for the remaining 15%. An analyst could allocate the forecast increase in GMV sales based on those percentages. Alternatively, an analyst could allocate an

increasing proportion to third-party merchant sales, because they have been growing at a faster rate than Iliso's retail sales.

3. Given the following assumptions, the analyst's forecast of Iliso's 2X20 total revenues is *closest* to:

- National retail sales excluding autos grows by 3.4%.
- Iliso's market share (i.e., its GMV as a percentage of national retail sales excluding autos) increases by 2 bps.
- GMV is split 75%/25% between retail sales and third-party merchant sales.
- The take rate on third-party merchant sales remains the same as in 2X19.

A. 7,800.

B. 8,441.

C. 9,906.

Solution:

A is correct.

Forecast of national retail sales excluding autos = $1.034 \times 4,165,000 = 4,306,610$

Forecast of Iliso's 2X20 GMV = $4,307,000 \times (0.21\% + 2 \text{ bps}) = 9,906$

Forecast of Iliso's 2X20 retail sales = $9,905 \times 75\% = 7,430$

Forecast of 2X20 third-party merchant sales = $9,905 \times 25\% = 2,477$

Forecast of 2X20 third-party merchant fees = $2,476 \times 15\% = 371$

Forecast of Iliso's 2X20 total revenues = $7,429 + 371 = 7,801$

4. An analyst who uses bottom-up drivers forecasts the number of customer accounts to increase by 15%, the average sales per customer to increase by 5%, the number of third-party merchants to increase by 28%, and the average sales per third-party merchant to increase by 40%. The analyst's forecast for Iliso's 2X20 GMV (in USD millions) is *closest* to:

A. 8,860.

B. 9,210.

C. 11,140.

Solution:

C is correct.

Forecast of Iliso's 2X20 retail sales = $204(1.05) \times 36(1.15) = 8,868$ or $= 7,336(1.05)(1.15) = 8,858$

Forecast of 2X20 third-party merchant sales = $[40,500(1.28) \times 31,300(1.40)] / 1,000,000 = 2,272$ or $= 1,269(1.28)(1.40) = 2,274$

Forecast of Iliso's 2X20 GMV = $8,868 + 2,272 = 11,140$

4. FORECASTING OPERATING EXPENSES AND WORKING CAPITAL

LEARNING OUTCOME

The candidate should be able to:

- explain approaches to forecasting a company’s operating expenses and working capital

Issuers’ disclosures about operating costs are typically less detailed than revenue disclosures. Rather than modeling costs separately for different geographic regions, business segments, or product lines, analysts are often forced to use more aggregated forecast objects such as consolidated financial statement lines (e.g., cost of sales, SG&A) or summary measures like EBITDA margins on a consolidated or segment basis. The forecasts for revenues and costs should still be coherent. If sales of a relatively low-margin product, segment, or geography are forecast to grow faster than other revenues, some level of overall profit margin deterioration should be forecast, even if the analyst is uncertain about the precise margins earned on each object. For example, food and grocery typically has low gross profit margins, so an analyst of a grocery chain that also sells higher-margin items, such as alcoholic products or pharmaceutical products, would want to forecast any change in product mix sold.

4.1. Cost of Sales and Gross Margins

Cost of sales (cost of goods sold, or COGS) is typically the single largest cost for companies that make and/or sell products. Because it has a direct link with sales, forecasting this item as a percentage of sales (or as a gross margin) is usually a good approach. If a company is losing market share to new substitute products with lower prices, gross margins are likely to decline. But if the company is gaining market share because it has introduced differentiated products—especially if it has done so in combination with achieving cost advantages—gross margins are likely to rise.

Cost of sales is typically a large cost, so even a single basis point in a gross margin forecast can have a material impact on forecasts of operating profit and free cash flow. Analysts should consider whether a more detailed analysis of these costs (e.g., by segment, input, or product line, or by volume and price components) is possible to better justify the forecast. For example, some companies face fluctuating input costs that can be passed on to customers only with a time lag. Particularly for companies with low gross margins, sudden shocks in input costs can affect operating profit significantly. A good example is the sensitivity of airlines’ profits to unhedged changes in jet fuel costs. In these cases, a breakdown of both costs and sales into volume and price components is essential for developing short-term forecasts, even if analysts use the overall relationship between sales and input costs for developing longer-term forecasts, as shown in [Exhibit 3](#).

Exhibit 3: The Effect of Prices and Costs on Gross Profit and Margin

Assume that a company’s COGS as a percentage of sales equals 25% and that the quantity sold is the same in Period 2 as in Period 1. If input costs double in Period 2 and the company can pass the entire increase on to its customers through a 25% price increase, COGS as a percentage of sales will increase (to 40%) because an equal absolute amount has been added to the numerator and to the denominator.

	Period 1	Period 2
Sales	100.0	125.0
COGS	25.0	50.0
Gross profit	75.0	75.0

	Period 1	Period 2
COGS as % of sales	25%	40%
Gross margin %	75%	60%

Thus, although the absolute amount of gross profit will remain constant, the gross margin will decrease (from 75% to 60%).

Analysts should incorporate a company's hedging strategy into a forecast. For example, commodity-driven companies' gross margins almost automatically decline if input prices of needed commodities increase significantly, because the variable costs increase at a faster rate than output prices. Through hedging strategies, a company can mitigate the impact on profitability. For example, brewers often hedge the cost of barley, a key raw material needed for brewing beer, one year in advance. Although companies usually do not disclose specific hedging positions, their hedging strategy is often disclosed in the notes to the financial statements. Further, the negative impact of increasing sales prices on sales volume (if product demand is price elastic) can be mitigated by a policy of gradual sales price increases. For example, if the brewer expects higher barley prices because of a poor harvest, the brewer can slowly increase prices to avoid a strong price jump next year.

While competitors' gross margins can provide a useful cross-check for forecasting gross margin, the margins may not be comparable given differences in business models. For example, some retailers own and operate their own stores, whereas other companies operate as wholesalers with franchised retail operations. In the franchise model, most of the operating costs are incurred by the franchisee; the wholesaler sells products with only a small markup to these franchisees. Compared with a retailer with its own stores, a wholesaler will have a much lower gross margin but also much lower operating costs.

4.2. SG&A Expenses

SG&A expenses are the other main type of operating costs. In contrast to cost of sales, SG&A expenses often have a less direct relationship with revenues. As an illustration of the profit impact of COGS and SG&A, consider the case of a Thai cement and building materials company, Siam Cement Group, from 2017 to 2018. A summary of the company's key income statement items is shown in [Exhibit 4](#).

Exhibit 4: Siam Cement Group 2017–18 Financials

	2017 (THB billions)	2018 (THB billions)	YoY%	Percentage of Sales	
				2017	2018
Net sales	450.92	478.44	6.1	100.0	100.0
COGS	349.31	383.46	9.8	77.5	80.1
Gross profit	101.61	94.98	−6.5	22.5	19.9
SG&A	52.58	55.09	4.8	11.7	11.5
<i>Selected SG&A items:</i>					
Salary and personnel expenses	24.24	23.98	−1.1	5.4	5.0
Freight costs	11.63	11.55	−0.7	2.6	2.4
Research and development	4.18	4.67	11.7	0.9	1.0
Promotion and advertising	2.62	2.58	−1.5	0.6	0.5
Operating income	49.03	39.89	−18.6	10.8	8.4

Note: "YoY%" means year-over-year percentage change.

Source: Based on information in Siam Cement Group's annual reports.

As shown in [Exhibit 4](#), Siam Cement was affected in 2018 by higher input costs that could not be fully passed on to customers. Consequently, despite sales growth of 6.1%, gross profit fell by 6.5% and gross margin declined. The company was able to limit its other operating costs; SG&A expenses grew 4.8%, declining slightly as a percentage of revenue. Operating income fell 18.6%. This experience contrasts with the company's experience in 2016 (not shown), when lower input costs resulted in a widening of the gross margin by 240 basis points versus the prior year.

Although SG&A expenses overall are generally less closely linked to revenue than is cost of sales, certain SG&A expenses could be more variable than others. Specifically, selling and distribution expenses often have a large variable component and can be modeled as a percentage of sales, while general corporate costs are more fixed and might be better modeled using a fixed growth rate based on expected wage inflation.

Segment disclosures often include operating and EBITDA margin profitability by segment, but generally do not include cost items such as cost of sales, SG&A, etc., by segment. If an analyst is constructing a model based on segment forecasts, the analyst might instead use summary measures on a segment basis, as shown in [Example 1](#).

EXAMPLE 1

Analysis of a Consumer Goods Company with Multiple Segments

A consumer goods company reported an overall underlying operating margin of 19.1% in 2X19. As shown in [Exhibit 5](#), the operating margin is lowest in its fastest-growing segment, Home Care. The other two segments enjoy higher margins but are growing more slowly.

Exhibit 5: Segment Revenue and Profits (EUR millions, unless noted)

Segment	2X19	2X18	'19/'18 YoY	Avg. Growth Rate, 2X17- 2X19
Beauty & Personal Care	21,868	20,624	6%	3%
Foods & Refreshments	19,287	20,227	-5%	-5%
Home Care	10,825	10,131	7%	3%
Total revenues	51,980	50,982	2%	0%
<i>Underlying operating profit</i>				
Beauty & Personal Care	4,960	4,543	9%	
Foods & Refreshments	3,382	3,576	-5%	
Home Care	1,605	1,344	19%	
Total	9,947	9,463	5%	
<i>Underlying operating profit margin</i>				
Beauty & Personal Care	22.7%	22.0%		
Foods & Refreshments	17.5%	17.7%		
Home Care	14.8%	13.3%		

Segment	2X19	2X18	'19/'18 YoY	Avg. Growth Rate, 2X17- 2X19
Total	19.1%	18.6%		

Note: Underlying operating profit is a non-IFRS operating profit measure, equal to IFRS operating profit adjusted for items such as restructuring costs.

Source: Example adapted from Unilever.

1. Determine the estimated sales, operating profit, and operating profit margin by using the following two approaches: (A) Assume total sales growth of 2.0% and overall underlying operating margin of 19.1% for the next five years, and (B) assume each individual segment's sales growth and underlying operating margin continue at the same rate reported in 2019. Which approach will result in a higher estimated underlying operating profit after five years?

Solution:

[Exhibit 6](#) shows that operating profit after five years will be EUR10,962 million under Approach A and EUR11,549 million under Approach B.

Exhibit 6: Sales and Operating Profit, 2X19–2X24E (EUR millions, unless noted)

Approach A	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Total revenues	51,980	53,020	54,080	55,162	56,265	57,390
Growth rate		2%	2%	2%	2%	2%
Underlying operating profit	9,947	10,127	10,329	10,536	10,747	10,962
Growth rate		2%	2%	2%	2%	2%
Underlying operating profit margin	19.1%	19.1%	19.1%	19.1%	19.1%	19.1%
Approach B	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Sales	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	21,868	23,187	24,586	26,069	27,641	29,308
Growth rate		6%	6%	6%	6%	6%
Foods & Refreshments	19,287	18,391	17,536	16,721	15,944	15,203
Growth rate		–5%	–5%	–5%	–5%	–5%
Home Care	10,825	11,567	12,359	13,205	14,110	15,077
Growth rate		7%	7%	7%	7%	7%
Total revenues	51,980	53,144	54,481	55,995	57,695	59,588
Margins	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	22.7%	22.7%	22.7%	22.7%	22.7%	22.7%
Foods & Refreshments	17.5%	17.5%	17.5%	17.5%	17.5%	17.5%
Home Care	14.8%	14.8%	14.8%	14.8%	14.8%	14.8%

Underlying operating profit	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	4,960	5,259	5,576	5,913	6,269	6,648
Foods & Refreshments	3,382	3,225	3,075	2,932	2,796	2,666
Home Care	1,605	1,715	1,832	1,958	2,092	2,235
Total	9,947	10,199	10,484	10,803	11,157	11,549

2. Compare and explain the results under the two alternative approaches (A and B) in Question 1.

Solution:

Approach A assumes a constant 2% total sales growth rate and a stable 19.1% underlying operating margin. Thus, the operating profit growth rate is in line with the revenue growth rate and constant at 2%, which therefore assumes no difference in growth rates and profitability of the segments. Approach B assumes growth rates of 6%, –5%, and 7% of sales for the Beauty & Personal Care, Foods & Refreshments, and Home Care segments. This approach results in a faster overall compounded growth rate than under Approach A (3% versus 2%) and an annual increase, on average, in the total underlying operating profit margin of 6 bps due to the mix effect of different segment margins. In 2024E, Approach A yields an underlying operating profit margin of 19.1%.

3. Assume the company can grow segment revenues over the next five years at the following rates: Beauty & Personal Care, 3.0%; Foods & Refreshments, 2.0%; Home Care, 4.0%. But underlying operating profit margins in Beauty & Personal Care will fall 20 bps annually for the next five years (because of high competition, limited growth, and costs resulting from the adoption of sustainable packaging), and operating profit margins in the Foods & Refreshments and Home Care segments will increase by 15 and 50 bps, respectively, each year for the next five years (helped by increasing demand for the company's products and better utilization of its factories). Calculate the overall underlying operating profit margin in each of the next five years.

Solution:

As shown in [Exhibit 7](#), the overall underlying operating profit margin improves from 19.1% in 2X19 to 19.5% in 2X24, because the margin decline in Beauty & Personal Care is more than offset by the margin increase in Foods & Refreshments and the faster-growing Home Care segment.

Exhibit 7: Sales and Operating Profit, 2X19–2X24E (EUR millions, unless noted)

Sales	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	21,868	22,524	23,200	23,896	24,613	25,351
Growth rate		3%	3%	3%	3%	3%
Foods & Refreshments	19,287	19,673	20,066	20,468	20,877	21,294
Growth rate		2%	2%	2%	2%	2%
Home Care	10,825	11,258	11,708	12,177	12,664	13,170
Growth rate		4%	4%	4%	4%	4%
Total revenues	51,980	53,455	54,974	56,540	58,153	59,816

Margins	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	22.7%	22.5%	22.3%	22.1%	21.9%	21.7%
Foods & Refreshments	17.5%	17.7%	17.8%	18.0%	18.1%	18.3%
Home Care	14.8%	15.3%	15.8%	16.3%	16.8%	17.3%

Underlying operating profit	2X19A	2X20E	2X21E	2X22E	2X23E	2X24E
Beauty & Personal Care	4,960	5,064	5,169	5,277	5,386	5,496
Foods & Refreshments	3,382	3,479	3,579	3,681	3,786	3,894
Home Care	1,605	1,725	1,853	1,988	2,131	2,282
Total	9,947	10,268	10,601	10,946	11,303	11,672
Margin	19.1%	19.2%	19.3%	19.4%	19.4%	19.5%

4.3. Working Capital Forecasts

Working capital forecasts are typically made by using efficiency ratios, discussed in earlier modules, as the forecast object, which are combined with sales and costs forecasts to project accounts receivable, inventories, accounts payable, and other current assets and liabilities, as demonstrated in [Example 2](#).

EXAMPLE 2

Working Capital Forecasts with Efficiency Ratios

[Exhibit 8](#) shows revenue, COGS, and year-end working capital account balances for YY Ltd., a fictional company, for years 1–3. Based on the data in the exhibit, answer Questions 1–3.

Exhibit 8: YY Ltd. Financial Data (CNY millions)

Year	1	2	3
Revenue	174,915	205,839	245,866
COGS	152,723	177,285	209,114
Accounts receivable	5,598	6,949	10,161
Inventory	29,481	32,585	41,671
Accounts payable	46,287	59,528	72,199

1. Calculate days sales outstanding (DSO), inventory days on hand (DOH), and days payable outstanding (DPO) for years 1, 2, and 3, using year-end balances and assuming a 365-day fiscal year.

Solution:

Recall that days sales outstanding is equal to accounts receivable/(revenues/365), inventory days on hand is equal to inventories/(COGS/365), and days payable outstanding is equal to accounts payable/(COGS/365). Using the data in [Exhibit 8](#), the three ratios for years 1–3 are as follows:

Year	1	2	3
Days sales outstanding	12	12	15
Inventory days on hand	70	67	73
Days payable outstanding	111	123	126

2. Your colleague Liang forecasts revenue growth of 18%, 16%, and 13% and gross margins of 17%, 17%, and 16% in years 4, 5, and 6, respectively, for YY Ltd. Using Liang's forecasts, calculate expected revenue and COGS in years 4, 5, and 6.

Solution:

Using Liang's forecasts for annual revenue growth and gross margins and the data in [Exhibit 8](#), expected revenue and COGS for years 4–6 are as follows:

Year (CNY millions)	1	2	3	4	5	6
Revenue	174,915	205,839	245,866	290,122	336,541	380,292
Growth rate		18%	19%	18%	16%	13%
COGS	152,723	177,285	209,114	240,801	279,329	319,445
Gross margin	13%	14%	15%	17%	17%	16%

3. Liang forecasts that days sales outstanding, inventory days on hand, and days payable outstanding in years 4, 5, and 6 will remain the same as the year 3 amounts. Using Liang's forecasts as well as forecast revenue and COGS, calculate expected accounts receivable, inventories, and accounts payable year-end balances for years 4, 5, and 6.

Solution:

Each of the efficiency ratios can be rearranged to yield working capital balances, because we have values for two of the three variables in them: the efficiency ratios are assumed to remain constant from year 3 levels, and revenue and COGS have already been forecast.

Days sales outstanding is equal to accounts receivable/(revenues/365); thus, accounts receivable is equal to days sales outstanding $\times$ (revenues/365). Similarly, inventories is equal to inventory days on hand $\times$ (COGS/365), and accounts payable is equal to days payable outstanding $\times$ (COGS/365).

Using the data in [Exhibit 8](#) and the revenue and COGS forecasts, year-end working capital account balances are as follows:

Year (CNY millions)	1	2	3	4	5	6
Revenue	174,915	205,839	245,866	290,122	336,541	380,292
COGS	152,723	177,285	209,114	240,801	279,329	319,445
Accounts receivable	5,598	6,949	10,161	11,990	13,908	15,716
Inventories	29,481	32,585	41,671	47,985	55,663	63,657
Accounts payable	46,287	59,528	72,199	83,139	96,442	110,292
Days sales outstanding	12	12	15	15	15	15
Inventory days on hand	70	67	73	73	73	73
Days payable outstanding	111	123	126	126	126	126

While a historical results approach is common for working capital efficiency ratios, analysts can also use the other forecast approaches. [Exhibit 9](#) discusses forecast objects and approaches that Elaine Nguyen can use for Warehouse Club Inc.'s operating costs and working capital.

Exhibit 9: Forecast Approaches for Warehouse Club Inc.'s Operating Costs and Working Capital

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Gross margin (on net sales)	Historical results	From 2X10 to 2X19, Warehouse's gross margin increased by an average of 2 bps per year. Such an increase may reflect increasing bargaining power over suppliers with scale, and Nguyen can use the same 2 bps per year increase in gross margin as a forecast.
	Historical base rates and convergence	Both the industry average gross margin and that of larger, more mature retailers such as Walmart are much greater than Warehouse's gross margin. But Warehouse's business model is to use a low gross margin on net sales to drive high-margin membership revenues. Therefore, Nguyen probably will not use this approach.
	Management guidance	If Warehouse management provides guidance on margins, Nguyen can use this approach as the forecast, after evaluating management's forecasting credibility.
	Analyst's discretionary forecast	Nguyen may formulate a discretionary forecast for gross margin by exploring the relationship between it, inflation/deflation in Warehouse's product categories, and management's desire for member savings vs. competitors to equal 5× the annual price of membership.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Selling, general, and administrative expenses (SG&A) as a percentage of sales	Historical results	SG&A, which captures the cost of operating stores and some merchandise distribution, is primarily composed of employee payroll. Relative to sales, it has remained mostly stable at or around 9%. Nguyen may forecast it to remain the same.
	Historical base rates and convergence	Warehouse's SG&A as a percentage of sales is already lower than that of larger retailer competitors (and substantially lower than the retail industry average), owing to its business model differences that emphasize store productivity and cost leadership. Nguyen will therefore probably not use this approach to forecast SG&A.
	Management guidance	If Warehouse management provides guidance on SG&A, Nguyen can use this approach as the forecast, after evaluating management's forecasting credibility.
	Analyst's discretionary forecast	Since SG&A is primarily composed of employee payroll, Nguyen may forecast it based on projected employee productivity (e.g., employees to stores) and projected wage rates, which may be based on macroeconomic variables like projected economy-wide wage growth.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
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Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Working capital ratios (e.g., DSO, DOH, DPO)	Historical results	As of 2X19, Warehouse's DSO, DOH, and DPO were 3, 30, and 28, respectively. DSO and DPO have been stable for the last 10 years, while DOH has decreased by 7 days since 2X10. Nguyen might forecast a stable DSO and DPO and a modestly declining DOH over the next several years.
	Historical base rates and convergence	Warehouse's working capital ratios and cash conversion cycle are materially better than the industry average and some larger peers and have been for many years, owing to its business model differences. Nguyen will therefore probably not use this forecast approach, unless Warehouse changes its business model (e.g., it launches an e-commerce business).
	Management guidance	If Warehouse management provides guidance on working capital (e.g., a projection for the net working capital to sales ratio), Nguyen can use this approach as the forecast, after evaluating management's forecasting credibility.
	Analyst's discretionary forecast	Nguyen may develop a discretionary forecast for working capital by, for example, estimating net working capital to sales ratios by product category (e.g., fresh foods, packaged foods, non-food merchandise) and applying them to a net sales forecast also broken down by category.

Question Set

Iliso Marketplace Ltd., the e-commerce retailer introduced in prior modules, reports its GMV by product category: in 2X19, the mix was 55% food and grocery, 20% media and electronics, 8% home goods and décor, and 17% apparel and beauty products. However, Iliso presents only aggregated financial results (no segment disclosures).

Iliso's cost of revenue is primarily composed of the inventory costs of products sold to customers where Iliso records revenue gross. Inventory costs include shipping and handling costs. Additionally, cost of revenue includes outbound shipping and logistics-related expenses, as well as delivery service costs from the third-party merchant business, where Iliso is the delivery service provider. Cost of revenue also includes depreciation expense related to inventory handling and shipping assets.

Iliso's operating, general, and administrative expenses include costs incurred in operating and staffing its fulfillment centers, customer service costs, payment processing fees, costs related to the design, execution, and maintenance of Iliso's technology infrastructure, and general corporate function costs.

Iliso has minimal accounts receivable, because customers pay with credit cards that settle in less than a week. Iliso does have significant inventory, accounts payable, and accrued expenses on its balance sheet.

1. How might an analyst use Iliso's disclosure of GMV by product category?

Solution:

An analyst could estimate cost of sales/sales and gross profit margin by category by looking at other retailers that primarily sell products in the specific category. Information on those companies could be used to check whether Iliso's gross profit margin is consistent with that implied by the product category mix. Also, if the mix is expected to change, an analyst can adjust the aggregate cost of sales/sales and gross profit margins accordingly. The information should probably not be used directly to

develop detailed forecasts by category, because those forecasts would not be verifiable and would not be an efficient use of the analyst's time.

2. It is *most* appropriate to forecast Iliso's cost of revenue as a percentage of its:

- A. retail sales.
- B. total revenues.
- C. total GMV sales.

Solution:

A is correct. Cost of revenue should be based on Iliso's retail sales. Cost of revenue is unaffected by third-party merchandise sales unless the merchant uses Iliso's fulfillment network for warehousing and delivery. The company does not provide information on such use by third parties, and it would be a small cost compared with cost of revenue for Iliso's retail sales. Iliso's cost of revenue is primarily composed of the inventory costs of products sold to customers where Iliso records revenue gross.

3. Discuss the use of a historical base rates and convergence approach, calculated using other retailers' financial results, for forecasting Iliso's operating expenses.

Solution:

Iliso's e-commerce business model and the breadth of the retail sales industry may make industry averages inappropriate to use for Iliso. However, if there are larger, more mature e-commerce retailers to calculate statistics for, this approach may be appropriate.

4. If food and grocery increases as a proportion of Iliso's retail sales, which of the following would *most likely* increase?

- A. Inventory turnover
- B. Gross profit margin
- C. Current assets/sales

Solution:

A is correct. Inventory turnover would increase because of the perishable nature of food and grocery. Inventory as a percentage of sales would decrease, resulting in a decrease in current assets to sales.

5. FORECASTING CAPITAL INVESTMENTS AND CAPITAL STRUCTURE

LEARNING OUTCOME

The candidate should be able to:

- ☐ explain approaches to forecasting a company's capital investments and capital structure

Projections for long-term assets are based on cash flow statement and income statement projections, because net PP&E and intangible assets on the balance sheet primarily increase due to capital

expenditures and decrease due to depreciation and amortization expenses. Capital expenditures can be broken down into maintenance capital expenditures necessary to sustain the current business and growth capital expenditures needed to expand the business. Maintenance capital expenditure forecasts are often based on historical depreciation and amortization expenses, usually with a small adjustment upward to account for inflation in capital goods. For businesses with low fixed asset turnover, maintenance capital expenditure requirements can be quite high. Growth capital expenditure forecasts are more discretionary and are tied to management’s expansion plans and revenue growth. Depreciation and amortization forecasts are based on net PP&E and intangibles on the balance sheet (which increase due to capital expenditures) and their useful lives as assumed by management’s accounting policies, which can be approximated by the ratio of gross fixed assets to depreciation and amortization expenses. Information may also be found in the notes to the financial statements.

Analysts must also make projections about a company’s future capital structure. Leverage ratios—such as debt to capital, debt to equity, and debt to EBITDA—are often used as the forecast object to project future debt and equity levels. Analysts should consider historical company practice, management’s financial strategy, and the capital requirements implied by the capital expenditure assumptions when projecting the future capital structure. Management may provide guidance on target capital structure, debt covenant ratios (e.g., net debt to EBITDA), and capital expenditures, sometimes broken down into maintenance, growth, and acquisitions. [Example 3](#) demonstrates the use of these forecast objects for YY Ltd., the same company in Example 2.

EXAMPLE 3

Balance Sheet Modeling

[Exhibit 10](#) shows financial data for YY Ltd. related to its PP&E and intangible assets. Based on the data in [Exhibit 10](#) and the data and analysis in Example 2, answer Questions 1 and 2.

Exhibit 10: YY Ltd. Long-Term Asset Data (CNY millions)

Year	1	2	3
PP&E, net	5,068	6,992	6,306
Goodwill	282	248	253
Intangible assets, net	1,779	1,424	4,013
Total fixed assets	7,129	8,664	10,572
Capital expenditures – PP&E	3,785	3,405	3,026
Capital expenditures – intangibles	333	142	3,310
Depreciation expense	220	324	518
Amortization expense	529	486	666

Note: PP&E and intangibles asset account balances were also affected each year by changes in exchange rates and by disposals, which are not shown in the exhibit. Assume that such effects are zero in years 4–6.

1. Calculate the following:

- A. Capital expenditures (for both PP&E and intangibles) as a percentage of revenue (for years 1–3)
- B. Depreciation expense as a percentage of beginning-of-year PP&E, net (for years 2 and 3)
- C. Amortization expense as a percentage of beginning-of-year intangible assets, net (for years 2 and 3).

Solution:

Using the data, the following percentages were calculated.

Year	1	2	3
Revenue	174,915	205,839	245,866
Capital expenditures – PP&E (% of revenue)	2.2%	1.7%	1.2%
Capital expenditures – intangibles (% of revenue)	0.2%	0.1%	1.3%
Depreciation (% of beginning PP&E)		6%	7%
Amortization (% of beginning intangibles)		27%	47%

2. Given the following assumptions and forecast revenue from Example 2, calculate expected total fixed assets for years 4–6.

- Capital expenditures for PP&E as a percentage of revenue to remain at the year 3 level
- Capital expenditures for intangibles as a percentage of revenue to remain at the year 1 level
- Goodwill to remain at the year 3 level
- Depreciation and amortization expenses as a percentage of beginning-of-year PP&E, net, and intangible assets, net, to remain at year 3 levels

Solution:

Using the data provided, expected total fixed assets for years 4–6 was calculated as CNY12,351 million; CNY15,179 million; and CNY18,662 million, respectively. PP&E, net, for each year was calculated as the prior period balance plus capital expenditures minus depreciation expense. Intangible assets, net, for each year was calculated in a similar fashion. Goodwill was held constant at CNY253 million.

Year	1	2	3	4	5	6
Revenue	174,915	205,839	245,866	290,122	336,541	380,292
PP&E, net	5,068	6,992	6,306	9,410	12,854	16,583
Goodwill	282	248	253	253	253	253
Other intangible assets, net	1,779	1,424	4,013	2,688	2,072	1,827
Total fixed assets	7,129	8,664	10,572	12,351	15,179	18,662
Capital expenditures – PP&E	3,785	3,405	3,026	3,571	4,142	4,680
Capital expenditures – PP&E (% of revenue)	2.2%	1.7%	1.2%	1.2%	1.2%	1.2%
Capital expenditures – intangibles	333	142	3,310	552	641	724
Capital expenditures – intangibles (% of revenue)	0.2%	0.1%	1.3%	0.2%	0.2%	0.2%
Depreciation expense	220	324	518	467	697	952
Depreciation (% of beginning PP&E)		6%	7%	7%	7%	7%
Amortization expense	529	486	666	1,877	1,257	969

Year	1	2	3	4	5	6
Amortization (% of beginning intangibles)		27%	47%	47%	47%	47%

3. Based on [Exhibit 11](#), which shows financial data for YY Ltd. related to its capital structure and profitability, answer Question 3.

Exhibit 11: YY Ltd. Debt and Profitability Data (CNY millions)

Year	1	2	3
Gross debt	10,931	17,624	17,597
Revenue	174,915	205,839	245,866
EBITDA	9,304	12,343	14,190
EBITDA margin	5.3%	6.0%	5.8%

YY Ltd. management has a year 6 gross debt to EBITDA ratio target of 2.0.

- A. Assuming an EBITDA margin of 6.0%, revenue forecasts from Example 2, and gross debt to EBITDA ratios of 1.25, 1.50, and 2.0 for years 4, 5, and 6, respectively, calculate expected gross debt for years 4–6.

- B. Given the results of part A, how much incremental borrowing does the forecast imply from year 3 to year 6?

Solution:

Gross debt of CNY21,579 million; CNY30,289 million; and CNY45,635 million are estimated for years 4–6, respectively, for YY Ltd. This forecast is made by first multiplying forecast revenue by the forecast EBITDA margin to calculate forecast EBITDA. Then, the expected gross debt to EBITDA ratio is multiplied by the forecast EBITDA to calculate forecast gross debt.

Year	1	2	3	4	5	6
Gross debt	10,931	17,624	17,597	21,759	30,289	45,635
Revenue	174,915	205,839	245,866	290,122	336,541	380,292
EBITDA	9,304	12,343	14,190	17,407	20,192	22,818
EBITDA margin	5.3%	6.0%	5.8%	6.0%	6.0%	6.0%
Gross debt to EBITDA	1.17	1.43	1.24	1.25	1.50	2.00

[Exhibit 12](#) discusses forecast objects and approaches that Elaine Nguyen can use for Warehouse Club Inc.'s capital expenditures and gross debt to EBITDA.

Exhibit 12: Forecast Approaches for Warehouse Club Inc.'s Capital Expenditures and Structure

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Capital expenditures as a percentage of net sales	Historical results	Warehouse has spent 2% of net sales on capital expenditures for the last 10 years, which Nguyen can use as the forecast going forward.
	Historical base rates and convergence	More mature retailers (e.g., Walmart in the USA) have all but stopped opening new stores, which has reduced their capital requirements to maintenance and new initiatives like e-commerce. If Warehouse does not intend to go to new geographies or launch an e-commerce business, Nguyen might forecast capital expenditures trending down to maintenance level, along with a decline in new-store openings per year, as Warehouse matures.
	Management guidance	Warehouse management provides guidance on new-store openings, which Nguyen can use to forecast capital expenditures (along with a maintenance capital expenditure forecast and a forecast for growth capital expenditures per new store).
	Analyst's discretionary forecast	Nguyen might instead use a discretionary forecast for capital expenditures by projecting a maintenance CAPEX amount (not disclosed by management) and growth CAPEX associated with new-store openings.

Forecast Object	Forecast Approach	Example of Object and Approach for Warehouse Club Inc.
Gross debt to EBITDA	Historical results	For the last five years, Warehouse's ratio of debt to trailing twelve months EBITDA has been 1.10, which Nguyen can project going forward. The product of this ratio and projected EBITDA (from earlier forecasts of sales and operating expenses) is Warehouse's gross debt.
	Historical base rates and convergence	Warehouse's more mature competitors and peers use a greater amount of debt in their capital structure, and return more capital to shareholders with buybacks and dividends. Nguyen might forecast the same for Warehouse over time, with an increasing ratio of debt to EBITDA and percentage of free cash flow returned to shareholders.
	Management guidance	Warehouse's management may provide a target capital structure through a target debt to EBITDA ratio or credit rating, which Nguyen can use as the forecast.
	Analyst's discretionary forecast	Nguyen might instead make a discretionary forecast linked to her free cash flow forecast, projecting a debt to EBITDA ratio and return of capital to shareholders. Depending on Fyleton Investments' strategy, Nguyen might engage with Warehouse's management to advocate for such a capital structure.

Question Set

An analyst gathers the following information related to Iliso's capital expenditures and capital structure (USD millions):

	2X15	2X16	2X17	2X18	2X19
Iliso's retail sales	2,589	3,424	5,100	6,052	7,336

	2X15	2X16	2X17	2X18	2X19
Operating income	-1	18	61	72	130
Depreciation expense	25	43	65	91	123
PP&E (net)	198	294	435	590	768
Short-term debt	56	255	198	66	0
Long-term debt	43	157	168	144	144
Equity	548	732	858	1,006	1,178
Capital expenditures	79	140	208	247	301

1. Which of the following would the analyst *most likely* use in estimating Iliso's maintenance capital expenditures?

- A. Operating income
- B. Depreciation expense
- C. Earnings before interest, taxes, depreciation, and amortization (EBITDA)

Solution:

B is correct. Depreciation expense can serve as the basis for maintenance capital expenditures, as it is management's estimate of the cost of fixed assets expensed on the income statement in a manner that tracks its use.

2. Which of the following approaches is *most* appropriate for forecasting Iliso's capital expenditures?

- A. Historical results
- B. Analyst's discretionary forecast
- C. Historical base rates and convergence

Solution:

A is correct. Capital expenditures have been 4.1% of retail sales for the past four years (2X16–2X19). The historical results approach—using the bottom-up driver retail sales—is most appropriate. An analyst's discretionary forecast is not required, because Iliso is not in a cyclical industry or undergoing a fundamental change. Iliso's unique model and the breadth of the industry make the approach using historical base rates and convergence to the industry inappropriate.

3. The analyst forecasts 2X20 total revenues of 8,411 million, operating income of 2% of total revenues, capital expenditures of 4% of total revenues, depreciation expense of 20% of opening PP&E (net), and debt/EBITDA of 0.50. The analyst's forecast for debt (in millions) is *closest* to:

- A. 84.
- B. 161.
- C. 195.

Solution:

B is correct.

$$2X20 \text{ operating income} = 0.02(8,411) = 168$$

$$2X20 \text{ depreciation expense} = 0.20(768) = 154$$

$$\text{EBITDA} = 168 + 154 = 322$$

$$\text{Debt} = 0.50(322) = 161$$

4. The analyst notes that Iliso's common stock and paid-in capital accounts increased, in total, by 75 million from 2X18 to 2X19, but only 42 million was reported as stock issuance on the statement of cash flows. Retained earnings increased by the amount of net income. Which of the following *most likely* explains the difference of 33 million?

- A. A stock split
- B. A stock dividend
- C. Stock-based compensation

Solution:

C is correct. Stock-based compensation is a non-cash expense in the income statement and is reported as an addition to net income in the indirect method of reporting cash flow from operations. The issuance also increases common stock and paid-in capital. A stock split typically simply increases the number of shares outstanding and decreases any stated par value per share. A stock dividend increases common stock and paid-in capital and reduces retained earnings by the same amount.

6. SCENARIO ANALYSIS

LEARNING OUTCOME

The candidate should be able to:

- ☐ describe the use of scenario analysis in forecasting

Recall from earlier lessons that industry and business risks may result in future outcomes that differ from expectations. The final step in forecasting involves incorporating the possibility of different outcomes based on key risk factors as well as judging their likelihood of occurrence. Generic risk factors that affect all companies—but to varying degrees—include changes in the business cycle, competition, inflation and deflation, and technological developments. Rather than develop single point estimate forecasts, analysts make several forecast scenarios that vary based on different outcomes with respect to key risk factors. Investors compare these scenarios with other analysts' (e.g., sell-side analysts) forecasts for a company, as well as forecasts implied by current valuations, to make investment decisions.

We demonstrate the use of scenario analysis for technological developments that threaten to cannibalize demand for an existing product in the following historical case study. Technological developments can affect demand for a product, the quantity supplied of a product, or both. When changes in technology lead to lower manufacturing costs, the supply curve will shift to the right as suppliers produce more of the product at the same price. Conversely, if changes in technology result in the development of attractive substitute products, the demand curve will shift to the left.

EXAMPLE 4

Quantifying the Tablet Market's Potential to Cannibalize Demand for Personal Computers

The worldwide tablet market experienced a major technological development with the introduction of the Apple iPad tablet in April 2010, which was expected to have (and indeed did have) important

implications for the manufacturers of desktop and laptop computers. A tablet promised to offer the capabilities of a portable personal computer (PC) with a touchscreen interface instead of a keyboard. Another distinguishing feature of the then-new tablets was that—unlike the majority of PCs, which run on the Microsoft Windows platform—they would run on a non-Microsoft operating system: Apple’s iOS or Google’s Android. Given the tablet’s ability to perform many of the most common tasks of a PC—including emailing, browsing the web, sharing photos, playing music, watching movies, playing games, keeping a calendar, and managing contacts—an analyst at that time might reasonably have wondered to what extent sales of tablets might cannibalize demand for PCs and what the potential impact might be on Microsoft’s sales and earnings. [Exhibit 13](#) presents one approach to answering these questions. It is set at the start of 2012, just over a year after the launch of the iPad. It is presented from the position of an analyst assessing the impact of the tablet on the PC market and Microsoft.

Exhibit 13: Unit and Revenue Projections (in thousands, unless noted)

Pre-Cannibalization PC Projections	FY2011	FY2012E	FY2013E	FY2014E	3-Year CAGR
Consumer PC shipments	170,022	174,430	184,120	193,811	4.5%
Non-consumer PC shipments	180,881	185,570	195,880	206,189	4.5%
Total global PC shipments	350,903	360,000	380,000	400,000	4.5%
% of which is consumer	48%	48%	48%	48%	
% of which is non-consumer	52%	52%	52%	52%	
Consumer tablet shipments	36,785	82,800	111,250	148,750	59.3%
Non-consumer tablet shipments	1,686	7,200	13,750	26,250	149.7%
Total global tablet shipments	38,471	90,000	125,000	175,000	65.7%
% of which is consumer	96%	92%	89%	85%	
% of which is non-consumer	4%	8%	11%	15%	
Cannibalization factor, consumer	30%	30%	30%	30%	
Cannibalization factor, non-consumer	10%	10%	10%	10%	
# of consumer PCs cannibalized by tablets	11,036	24,840	33,375	44,625	
# of non-consumer PCs cannibalized by tablets	169	720	1,375	2,625	
Total PCs cannibalized by tablets	11,205	25,560	34,750	47,250	
% of total PCs cannibalized by tablets	3.2%	7.1%	9.1%	11.8%	
Post-Cannibalization PC Projections					
Consumer PC shipments	158,987	149,590	150,745	149,186	–2.1%

Pre-Cannibalization PC Projections	FY2011	FY2012E	FY2013E	FY2014E	3-Year CAGR
Non-consumer PC shipments	180,712	184,850	194,505	203,564	4.0%
Total global PC shipments	339,699	334,440	345,250	352,750	1.3%
<i>Microsoft implied average selling price (USD)</i>					
Consumer	85	85	85	85	
Non-consumer	155	155	155	155	
<i>Revenue impact for Microsoft (USD millions)</i>					
Consumer	938	2,111	2,837	3,793	
Non-consumer	26	112	213	407	
Total revenue impact	964	2,223	3,050	4,200	

Note: CAGR is compound annual growth rate. Non-consumer includes enterprise, education, and government purchasers.

Sources: Based on data from Gartner, JPMorgan, Microsoft, and authors' analysis.

Worldwide market shipments of PCs in FY2011 totaled 350.9 million units, while worldwide shipments of tablets totaled 38.5 million units (source: Gartner Personal Computer Quarterly Statistics Worldwide Database). Shipments to consumers represented 96% of total tablet shipments during fiscal year 2011. We next estimate the magnitude of the potential substitution effect, or cannibalization factor, that tablets will have on the PC market. Because the cannibalization factor depends on many different variables—including user preferences, end-use application, and whether the purchaser already owns a PC—we use a range of potential estimates. Moreover, we also divide the worldwide PC market into consumer and non-consumer (enterprise, education, and government purchasers), because the degree of substitution is likely to differ between the two. For purposes of illustration, we assume a cannibalization factor of 30% for the consumer market and 10% for the non-consumer market in our base case scenario.

In addition, the base case scenario assumes that non-consumer adoption of tablets increases to 15% of the market from 4% in 2011. Moreover, although the composition of the global PC market is roughly evenly divided between consumers and non-consumers (48% and 52% in fiscal year 2011, respectively), the non-consumer segment is significantly more profitable for Microsoft, because approximately 80% of the company's Office products are sold to enterprise, education, and government institutions. The average selling price (ASP) estimates are derived by dividing Microsoft's estimated average revenue by customer type for the prior three years by Microsoft's estimated PC shipments for each type of customer. By multiplying the projected number of PCs cannibalized by tablets by the estimated ASP, we can derive an estimate of the revenue impact for Microsoft. For example, in FY2012, it is projected that 24.8 million consumer PCs will be cannibalized by sales of tablets. With a consumer ASP of USD85, this cannibalization implies a revenue loss for Microsoft of USD2.1 billion (24.8 million units × USD85 per unit = USD2.1 billion).

Once the revenue impact has been projected, the next step is to estimate the impact of lower PC unit volumes on operating costs and margins. We begin by analyzing the cost structure of Microsoft and, more specifically, the breakdown between fixed and variable costs. Most software companies have a cost structure with a relatively high proportion of fixed costs and a low proportion of variable costs, because costs related to product development and marketing (mostly fixed) are sunk and unrecoverable, whereas the cost of producing an additional copy of the software (mostly variable) is relatively low. Because very few, if any, companies provide an explicit breakdown of fixed versus variable costs, an estimate almost always needs to be made. One method is to use the formula

$$\% \Delta (\text{Cost of revenue} + \text{Operating expenses}) / \% \Delta \text{ revenue},$$

where %Δ is “percentage change in,” used as a proxy for variable cost percentage. Another approach is to assign an estimate of the percentage of fixed and variable costs to the various components of operating expenses. The two approaches are illustrated in [Exhibit 14](#) and [Exhibit 15](#).

Exhibit 14: Estimation of Variable Costs for Microsoft, Method 1 (USD millions)

Selected Operating Segments	FY2009	FY2010	FY2011	FY2011/FY2009 Percentage Change
Revenue:				
Windows and Windows Live	15,563	18,792	18,778	
Microsoft business division	19,211	19,345	21,986	
Total segment revenue	34,774	38,137	40,764	17%
Operating expenses:				
Windows and Windows Live	6,191	6,539	6,810	
Microsoft business division	8,058	7,703	8,159	
Total operating expenses	14,249	14,242	14,969	5%

%Variable cost estimate $\approx$ %Δ (Cost of revenue + Operating expenses)/%Δ revenue $\approx$ 5%/17% $\approx$ **29%**

%Fixed cost $\approx$ 1 – %Variable cost $\approx$ 1 – 29% $\approx$ **71%**

Exhibit 15: Estimation of Variable Costs for Microsoft, Method 2 (USD millions)

Operating Expenses	FY2009	FY2010	FY2011	FY2009–FY2011 Average	% of Total Op. Expenses	Estimated % of Fixed Costs	Fixed Cost Contribution
Cost of revenue (excl. depreciation)	10,455	10,595	13,577	11,542	29%	20%	6%
Depreciation expense	1,700	1,800	2,000	1,833	5%	100%	5%
Total cost of revenue	12,155	12,395	15,577	13,376	34%		10%
R&D	9,010	8,714	9,043	8,922	22%	100%	22%
Sales and marketing	12,879	13,214	13,940	13,344	34%	80%	27%
General and admin.	4,030	4,063	4,222	4,105	10%	100%	10%

Operating Expenses	FY2009	FY2010	FY2011	FY2009– FY2011 Average	% of Total Op. Expenses	Estimated % of Fixed Costs	Fixed Cost Contribution
Total operating expenses	38,074	38,386	42,782	39,747	100%		60%
Estimated percentage of Microsoft's total cost structure that is fixed:							70%

Note: Fiscal year ends in June.

Sources: Microsoft 2011 Form 10-K and authors' analysis.

As can be seen, Microsoft's cost structure appears to consist of approximately 70% fixed costs and 30% variable costs. Note, however, that a growing company like Microsoft will typically re-invest in PP&E to support future growth, so even those expenses that appear to be "fixed" will increase over time. To adjust for this expected growth in fixed costs, this example includes an assumption that the change in fixed costs will be half the rate of the change in sales. Variable costs are projected to change at the same rate as sales. As shown in [Exhibit 16](#), after incorporating these assumptions into the projections, an assumed 7.0% compound annual growth rate (CAGR) in revenue through FY2014 would translate into a 10.6% CAGR in operating income $[(36,757/27,161)^{1/3} - 1 = 0.106$, or 10.6%]. In addition, these assumptions would result in an operating margin expansion of 410 bps over the same period (42.9% – 38.8% = 4.1 pps, or 410 bps), because of the significant amount of operating leverage that exists as a result of a relatively large fixed cost base. With the further assumptions of no change in other income, a constant effective tax rate, and no change in shares outstanding, the pre-cannibalization model, shown in [Exhibit 16](#), results in projected revenue of USD85.7 billion, operating income of USD36.8 billion, an operating margin of 42.9%, and EPS that increases at a CAGR of 10.3% to USD3.62 in FY2014.

Exhibit 16: Microsoft Pre-Cannibalization EPS Projections (USD millions, except EPS)

	FY2011	FY2012E	FY2013E	FY2014E	3- Year CAGR
Revenue	69,943	74,839	80,078	85,683	7.0%
YoY%		7.0%	7.0%	7.0%	
<i>Operating Expenses</i>					
Fixed (70%)	29,947	30,996	32,080	33,203	3.5%
Variable (30%)	12,835	13,733	14,694	15,723	7.0%
Total operating expenses	42,782	44,729	46,774	48,926	4.6%
Operating income	27,161	30,110	33,303	36,757	10.6%
Operating margin	38.83%	40.23%	41.59%	42.90%	
Other income (expense)	910	910	910	910	
Pre-tax income	28,071	31,020	34,213	37,667	
Provision for income taxes	4,921	5,438	5,998	6,603	
Effective tax rate	17.53%	17.53%	17.53%	17.53%	

	FY2011	FY2012E	FY2013E	FY2014E	3- Year CAGR
Net income	23,150	25,582	28,215	31,064	
Weighted average shares outstanding, diluted	8,593	8,593	8,593	8,593	
Estimated EPS pre- cannibalization	2.69	2.98	3.28	3.62	10.3%

In the post-cannibalization scenario, as shown in [Exhibit 17](#), revenue is reduced each year to reflect the expected impact of cannibalization, which results in a decrease in the CAGR of revenue over the period to 5.2%, down from 7.0% in the pre-cannibalization scenario. Given the reduction in revenue growth and holding the cost structure constant at 70/30 fixed versus variable costs, operating income growth slows to a CAGR of 8.0%, down from 10.6% in the pre-cannibalization scenario. Operating margin at the end of the period is reduced by approximately 100 bps, from 42.9% to 41.9%, because the company is unable to leverage its fixed cost base to the same degree as a result of slower revenue growth. Overall, in the post-cannibalization scenario, Microsoft is expected to generate revenue of USD81.5 billion, operating income of USD34.2 billion, an operating margin of 41.9%, and EPS that increases at a CAGR of 7.7% to USD3.37 in FY2014. Thus, the cannibalization of PCs as a result of projected growth in the tablet market is expected to reduce the company's annual revenues in FY2014 by USD4.2 billion, operating income by USD2.6 billion, operating margins by 96 bps, and EPS by USD0.25.

**Exhibit 17: Microsoft Post-Cannibalization EPS Projections, Base Case Scenario
(USD millions, except EPS and unless noted)**

	FY2011	FY2012E	FY2013E	FY2014E	3- Year CAGR
Revenue	69,943	72,616	77,028	81,483	5.2%
YoY%		3.8%	6.1%	5.8%	
<i>Operating Expenses</i>					
Fixed (70%)	29,947	30,520	31,447	32,356	2.6%
Variable (30%)	12,835	13,325	14,135	14,952	5.2%
Total operating expenses	42,782	43,845	45,581	47,308	3.4%
Operating income	27,161	28,771	31,446	34,175	8.0%
Operating margin	38.83%	39.62%	40.82%	41.94%	
Other income (expense)	910	910	910	910	
Pre-tax income	28,071	29,681	32,356	35,085	
Provision for income taxes	4,921	5,203	5,672	6,151	
Effective tax rate	17.53%	17.53%	17.53%	17.53%	
Net income	23,150	24,478	26,684	28,934	
Weighted average shares outstanding, diluted	8,593	8,593	8,593	8,593	

	FY2011	FY2012E	FY2013E	FY2014E	3- Year CAGR
Estimated EPS post-cannibalization	2.69	2.85	3.11	3.37	7.7%
Compared with pre-cannibalization:					
Estimated impact on operating margin		-61 bps	-77 bps	-96 bps	
Estimated impact on EPS (USD)		-0.13	-0.17	-0.25	-2.6%

Knowledge Check

Answer the following questions using the data from the exhibits on Microsoft:

1. Estimate post-cannibalization global PC shipments in FY2012 assuming a cannibalization factor of 40% for consumers and 15% for non-consumers.

Solution:

The number of PCs cannibalized by tablets is equal to the product of the expected number of global tablet shipments, the percentage representation of each category, and the cannibalization factor for the category. Tablet shipments in FY2012 are projected to be 90 million units; (90 million tablets × 92% consumer representation × 40% consumer cannibalization factor = 33.12 million consumer PCs cannibalized by tablets) + (90 million tablets × 8% non-consumer representation × 15% cannibalization = 1.08 million non-consumer PCs cannibalized by tablets) = 34.2 million total PCs cannibalized by tablets. Post-cannibalization shipments are equal to pre-cannibalization shipments minus expected cannibalization, or 360 million – 34.2 million = 325.8 million.

2. Using the results derived in Question 1, estimate the post-cannibalization revenue in FY2012 for Microsoft.

Solution:

The estimated impact on revenue is equal to the product of the number of PCs cannibalized and the ASP. Using the results obtained in Question 1 and the ASP data, the expected revenue impact can be calculated as (33.12 million consumer PCs cannibalized by tablets × USD85 ASP = USD2.815 billion) + (1.08 million non-consumer PCs cannibalized by tablets × USD155 ASP = USD167.4 million) = USD2.983 billion total impact on revenue for Microsoft. Post-cannibalization revenue is equal to pre-cannibalization revenue minus the estimated impact of cannibalization on revenue, or USD74.839 billion – USD2.983 billion = USD71.856 billion.

3. Using the estimate for post-cannibalization revenue derived in Question 2 and the cost structure provided, estimate post-cannibalization operating income and operating margin in FY2012 for Microsoft. Assume that fixed costs change at half the rate of the change in sales.

Solution:

Exhibit 18: Solution to Question 3 (USD millions)

	FY2011	FY2012E	Notes
Revenue	69,943	71,856	Derived from Question 2
YoY%		2.74%	Rate of change in sales used to estimate operating expenses
<i>Operating Expenses</i>			
Fixed (70%)	29,947	30,357	Fixed costs change at half the rate of the change in sales, or $29,947 \times (1 + 2.74\%/2)$
Variable (30%)	12,835	13,186	Variable costs change at the same rate as the change in sales, or $12,835 \times (1 + 2.74\%)$
Total operating expenses	42,782	43,543	Although not shown, operating expenses include COGS
Operating income	27,161	28,313	Revenue minus total operating expenses, or $71,856 - 43,543 = 28,313$
Operating margin	38.8%	39.4%	Operating income divided by revenue, or $28,313/71,856 = 39.4\%$

Post-cannibalization operating income and operating margin in FY2012 for Microsoft are USD28,314 million and 39.4%, respectively.

4. Using the estimate for operating income derived in Question 3 and the data in the exhibits, calculate the expected post-cannibalization EPS in FY2012 for Microsoft. Assume that other income (expense), the effective tax rate, and the diluted weighted average shares outstanding provided for FY2011 remain constant in FY2012.

Solution:

Exhibit 19: Solution to Question 4 (USD millions, except EPS and unless noted)

	FY2011	FY2012E	Notes
Revenue	69,943	71,856	
YoY%		2.74%	
<i>Operating Expenses</i>			
Fixed (70%)	29,947	30,357	
Variable (30%)	12,835	13,186	
Total operating expenses	42,782	43,543	
Operating income	27,161	28,314	
Operating margin	38.8%	39.4%	

	FY2011	FY2012E	Notes
Other income (expense)	910	910	
Pre-tax income	28,071	29,224	Operating income + Other income (expense), or 28,314 + 910 = 29,224
Provision for income taxes	4,921	5,123	Pre-tax income × Effective tax rate, or 29,224 × 17.53% = 5,123
Effective tax rate	17.53%	17.53%	
Net income	23,150	24,101	Pre-tax income – Provision for income taxes, or 29,224 – 5,123 = 24,101
Weighted average shares outstanding, diluted	8,593	8,593	
Estimated EPS post-cannibalization (USD)	2.69	2.80	Net income/Wtd. avg. shs. out., or 24,101/8,593 = USD2.80

Whenever one is estimating something that depends on many different variables that are difficult to measure, we recommend altering some of the assumptions to generate a range of estimates based on various scenarios. Thus, having developed a forecast under a base case cannibalization scenario, we can analyze the sensitivity of the results by altering the cannibalization assumptions. The base case scenario corresponds to the assumptions reflected in the boxed center of the table in [Exhibit 20](#). [Exhibit 21](#) summarizes the results of both bull and bear case scenarios, showing the estimated FY2014 EPS under alternative estimated cannibalization factors.

Exhibit 20: Estimated FY2014 EPS Sensitivity to Changes in Cannibalization Rates (USD)

		Non-Consumer Cannibalization				
		0%	5%	10%	15%	20%
Consumer Cannibalization	15%	-0.11	-0.12	-0.14	-0.15	-0.16
	20%	-0.15	-0.16	-0.17	-0.19	-0.20
	25%	-0.19	-0.20	-0.21	-0.22	-0.23
	30%	-0.22	-0.24	-0.25	-0.26	-0.27
	35%	-0.26	-0.27	-0.28	-0.30	-0.31
	40%	-0.30	-0.31	-0.32	-0.33	-0.35
	45%	-0.34	-0.35	-0.36	-0.37	-0.38

Exhibit 21: Post-Cannibalization EPS Projections for Bull and Bear Case Scenarios (USD millions, except EPS and unless noted)

Bull Case Scenario (Cannibalization Factor: 15% Consumer/5% Non-Consumer)					
	FY2011	FY2012E	FY2013E	FY2014E	3-Year CAGR
Revenue	69,943	73,728	78,553	83,583	6.1%
YoY%		5.4%	6.5%	6.4%	
<i>Operating Expenses</i>					
Fixed (70%)	29,947	30,758	31,764	32,781	3.1%
Variable (30%)	12,835	13,529	14,414	15,338	6.1%
Total operating expenses	42,782	44,287	46,178	48,119	4.0%
Operating income	27,161	29,441	32,374	35,464	9.3%
Operating margin	38.83%	39.93%	41.21%	42.43%	
Other income (expense)	910	910	910	910	
Pre-tax income	28,071	30,351	33,284	36,374	
Provision for income taxes	4,921	5,321	5,835	6,377	
Effective tax rate	17.53%	17.53%	17.53%	17.53%	
Net income	23,150	25,030	27,449	29,998	
Weighted average shares outstanding, diluted	8,593	8,593	8,593	8,593	
Estimated EPS post-cannibalization (USD)	2.69	2.91	3.19	3.49	9.0%
Compared with pre-cannibalization:					
Estimated impact on operating margin		-30 bps	-38 bps	-47 bps	
Estimated impact on EPS (USD)		-0.07	-0.09	-0.13	-1.3%
Bear Case Scenario (cannibalization factor: 40% consumer, 20% non-consumer)					
	FY2011	FY2012E	FY2013E	FY2014E	3-Year CAGR
Revenue	69,943	71,801	75,869	79,812	4.5%
YoY%		2.7%	5.7%	5.2%	
<i>Operating Expenses</i>					
Fixed (70%)	29,947	30,345	31,205	32,016	2.3%
Variable (30%)	12,835	13,175	13,922	14,646	4.5%
Total operating expenses	42,782	43,520	45,127	46,662	2.9%
Operating income	27,161	28,280	30,742	33,151	6.9%
Operating margin	38.83%	39.39%	40.52%	41.54%	
Other income (expense)	910	910	910	910	
Pre-tax income	28,071	29,190	31,652	34,061	

Bear Case Scenario (cannibalization factor: 40% consumer, 20% non-consumer)

	FY2011	FY2012E	FY2013E	FY2014E	3-Year CAGR
Provision for income taxes	4,921	5,117	5,549	5,971	
Effective tax rate	17.53%	17.53%	17.53%	17.53%	
Net income	23,150	24,073	26,103	28,090	
Weighted average shares outstanding, diluted	8,593	8,593	8,593	8,593	
Estimated EPS post- cannibalization (USD)	2.69	2.80	3.04	3.27	6.7%
Compared with pre- cannibalization:					
Estimated impact on operating margin		-84 bps	-107 bps	-136 bps	
Estimated impact on EPS (USD)		-0.18	-0.24	-0.35	-3.6%

Question Set

1. For which of the following companies is it *most* crucial for an analyst to conduct scenario analysis to estimate a decline in earnings from a recession?

- A. A bank
- B. A chain of large grocery stores
- C. A provider of residential electricity

Solution:

A is correct, because both lending volume and interest rates typically fall in recessions, which results in declining earnings for banks. Demand for both food-at-home and residential electricity is generally not as sensitive to changes in the business cycle.

2. An analyst is preparing scenarios for a company's gross margin. Selected financial data from the prior period include the following:

Average selling price per product EUR6.50

Quantity sold (000s) 2,300

Total revenues (EUR 000s) 14,950

Cost of sales (EUR 000s) 7,475

Gross profit (EUR 000s) 7,475

Gross profit margin 50%

If input prices increase by 20%, the analyst expects the company to increase selling prices by 10%, resulting in a 5% decrease in volume sold. The forecast decrease in gross profit margin is *closest* to:

- A. 4.5%
- B. 7.4%.
- C. 15.0%

Solution:

A is correct.

$$\text{Average selling price} = \text{EUR}6.50(1.10) = \text{EUR}7.15$$

$$\text{Quantity sold in 000s} = 2,300(0.95) = 2,185$$

$$\text{Average input cost} = (7,475/2,300)(1.20) = \text{EUR}3.90$$

$$\text{Total revenues in 000s} = 2,185(\text{EUR}7.15) = 15,622.75$$

$$\text{Gross profit in 000s} = 2,185(\text{EUR}7.15 - \text{EUR}3.90) = 7,101.25$$

$$\text{Gross profit margin} = 7,101.25/15,622.75 \approx 45.5\%, \text{ which is } 4.5\% \text{ less than } 50\%$$

3. The invention of a new technology that leads to lower product manufacturing costs in a highly competitive industry is *most likely* to result in:

- A. lower gross profit margins.
- B. higher gross profit margins.
- C. no impact on gross profit margins.

Solution:

C is correct. In a highly competitive industry, savings in manufacturing costs are likely to be passed on to customers through competition in the form of lower prices.

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PRACTICE PROBLEMS

1. Forecast objects should be:
 - A. based on independent third-party reports.
 - B. based on individual discrete items, not aggregations.
 - C. disclosed regularly or based on items that are disclosed regularly.
2. Management guidance is *most* appropriately used by analysts in forecasting:
 - A. capital expenditures of a company.
 - B. pricing of commodities sold by a company.
 - C. revenue of a company sensitive to the business cycle.
3. Which of the following is an example of a top-down driver of revenues for a passenger airline?
 - A. Market share of routes
 - B. Number of planes in fleet
 - C. Average ticket price per mile flown
4. An analyst forecasts a company's sales using a historical results approach and top-down drivers (expected industry sales and expected market share). The analyst observes that the forecast is within the range of management guidance on sales but is toward the upper end and not closer to the midpoint. Which of the following is the *most likely* explanation for this observation?
 - A. The analyst has been overly optimistic in estimating the company's expected market share.
 - B. Management's true expectations are toward the upper end, but management has created a range using a pessimistic lower bound that is more easily cleared.
 - C. The midpoint represents management's true expectations, and management has a lower and better estimate of market growth based on macroeconomic variables.
5. A clothing company, which initially produced designer clothing, has increasingly entered into the mass market. Which of the following approaches to forecasting revenues is *least* appropriate?
 - A. Historical results
 - B. Analyst's discretionary forecast
 - C. Historical base rates and convergence
6. An analyst is forecasting operating costs for a company with relatively high fixed costs, sensitivity to economic conditions, and commodity inputs with volatile pricing. The company does not follow a hedging strategy for commodity purchases but tries to buy when prices are low. Which of the following is *most* appropriate to use in forecasting operating costs? The analyst uses:
 - A. analyst discretion to forecast all operating costs.
 - B. management guidance to forecast all operating costs.
 - C. management guidance to forecast fixed operating costs and analyst discretion to forecast variable operating costs.

7. An analyst notes that a company's capital expenditures do not follow a discernible pattern; the company seems to have periods of very low capital expenditures and periods of high capital expenditures. Management does not provide any guidance on capital expenditures. The analyst should develop a forecast of capital expenditures based on:
- A. the company's usage of PP&E capacity.
 - B. the industry's average capital expenditures.
 - C. the company's average capital expenditures.
8. An analyst predicts that if a company's technological developments are a success, the company's operating costs will be reduced by 15%. As a result of the reduction in costs, the company will reduce the average selling price of its products by 5% and the volume of sales will increase by 8%. The company's current gross profit margin is 40%. If technological developments occur, the company's gross profit margin will be *closest* to:
- A. 44.8%.
 - B. 46.3%.
 - C. 47.5%.

SOLUTIONS

1. C is correct. Forecast objects should either be disclosed regularly or be directly calculable using what is disclosed regularly. Information that is *not* disclosed regularly (such as a third-party consultancy's report) is suitable to *inform* forecasts but can be problematic for direct use, because forecasts cannot be confirmed in a timely manner. While it may be intuitive to forecast individual discrete items, such as sales and gross margin by individual product line, if gross margin is disclosed only on a consolidated basis, it will be difficult for analysts to verify their product-line gross margin estimates.
2. A is correct. Management has an advantage at the firm level and tends to make more accurate forecasts for objects that are subject to management actions, such as capital expenditures. Management does *not* have an informational advantage over investors in forecasting macroeconomic variables, such as GDP, the business cycle, or the prices of commodities. Hutton et al. (2012) found that investors have an informational advantage over management at the macroeconomic level.
3. A is correct. Market share of routes is a top-down driver of revenues and can be used to estimate an airline's market share. The number of planes in fleet and the average ticket price per mile flown are bottom-up drivers of revenues.
4. B is correct. Management has a tendency to pad its guidance to make it easier to achieve the forecast results for which management will be rewarded. Management does not have an informational advantage at the macroeconomic level.
5. A is correct. Historical results is least appropriate, because the company has changed its competitive strategy. Analyst's discretionary forecast may be appropriate, because the company is undergoing a fundamental change. The company is growing and may be transitioning to be more like the industry, in which case historical base rates and convergence may be an appropriate approach.
6. C is correct. Management has an advantage in forecasting for objects that are subject to its actions (such as capital expenditures and inventories, which affect fixed costs). The analyst is likely to have an informational advantage when it comes to forecasting economic conditions and commodity prices, which affect revenues and variable costs.
7. A is correct. Based on the company's spending pattern, it most likely makes capital expenditures based on capacity needs as it grows. If it is approaching full usage of existing capacity, it will expand.
8. B is correct.

$$\text{Sales} = 100(0.95)(1.08) = 102.6$$

$$\text{Cost of sales} = 60(0.85)(1.08) = 55.1$$

$$\text{Gross profit} = 102.2 - 55.1 = 47.5$$

$$\text{Gross profit margin} = 47.5/102.6 = 46.3\%$$

Equity Valuation: Concepts and Basic Tools

by John J. Nagorniak CFA and Stephen E. Wilcox PhD, CFA

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LEARNING OUTCOMES

The candidate should be able to:

- ☐ evaluate whether a security, given its current market price and a value estimate, is overvalued, fairly valued, or undervalued by the market
- ☐ describe major categories of equity valuation models
- ☐ describe regular cash dividends, extra dividends, stock dividends, stock splits, reverse stock splits, and share repurchases
- ☐ describe dividend payment chronology
- ☐ explain the rationale for using present value models to value equity and describe the dividend discount and free-cash-flow-to-equity models
- ☐ explain advantages and disadvantages of each category of valuation model
- ☐ calculate the intrinsic value of a non-callable, non-convertible preferred stock
- ☐ calculate and interpret the intrinsic value of an equity security based on the Gordon (constant) growth dividend discount model or a two-stage dividend discount model, as appropriate
- ☐ identify characteristics of companies for which the constant growth or a multistage dividend discount model is appropriate
- ☐ explain the rationale for using price multiples to value equity, how the price to earnings multiple relates to fundamentals, and the use of multiples based on comparables
- ☐ calculate and interpret the following multiples: price to earnings, price to an estimate of operating cash flow, price to sales, and price to book value
- ☐ describe enterprise value multiples and their use in estimating equity value
- ☐ describe asset-based valuation models and their use in estimating equity value

1. INTRODUCTION

Analysts gather and process information to make investment decisions, including buy and sell recommendations. What information is gathered and how it is processed depend on the analyst and the purpose of the analysis. Technical analysis uses such information as stock price and trading volume as the basis for investment decisions. Fundamental analysis uses information about the economy, industry, and company as the basis for investment decisions. Examples of fundamentals are unemployment rates, gross domestic product (GDP) growth, industry growth, and quality of and growth in company earnings. Whereas technical analysts use information to predict price movements and base investment decisions on the direction of predicted change in prices, fundamental analysts use information to estimate the value of a security and to compare the estimated value to the market price and then base investment decisions on that comparison.

This reading introduces equity valuation models used to estimate the **intrinsic value** (synonym: **fundamental value**) of a security; intrinsic value is based on an analysis of investment fundamentals and characteristics. The fundamentals to be considered depend on the analyst's approach to valuation. In a top-down approach, an analyst examines the economic environment, identifies sectors that are expected to prosper in that environment, and analyzes securities of companies from previously identified attractive sectors. In a bottom-up approach, an analyst typically follows an industry or industries and forecasts fundamentals for the companies in those industries in order to determine valuation. Whatever the approach, an analyst who estimates the intrinsic value of an equity security is implicitly questioning the accuracy of the market price as an estimate of value. Valuation is particularly important in active equity portfolio management, which aims to improve on the return-risk trade-off of a portfolio's benchmark by identifying mispriced securities.

This reading is organized as follows. Section 2 discusses the implications of differences between estimated value and market price. Section 3 introduces three major categories of valuation model. Section 4 presents an overview of present value models with a focus on the dividend discount model. Section 5 describes and examines the use of multiples in valuation. Section 6 explains asset-based valuation and demonstrates how these models can be used to estimate value. Section 7 states conclusions and summarizes the reading.

2. ESTIMATED VALUE AND MARKET PRICE

LEARNING OUTCOME

The candidate should be able to:

- ☐ **evaluate whether a security, given its current market price and a value estimate, is overvalued, fairly valued, or undervalued by the market**

By comparing estimates of value and market price, an analyst can arrive at one of three conclusions: The security is *undervalued*, *overvalued*, or *fairly valued* in the marketplace. For example, if the market price of an asset is \$10 and the analyst estimates intrinsic value at \$10, a logical conclusion is that the security is fairly valued. If the security is selling for \$20, the security would be considered overvalued. If the security is selling for \$5, the security would be considered undervalued. Basically, by estimating value, the analyst is assuming that the market price is not necessarily the best estimate of intrinsic value. If the estimated value exceeds the market price, the analyst infers the security is *undervalued*. If the estimated value equals the market price, the analyst infers the security is *fairly valued*. If the estimated value is less than the market price, the analyst infers the security is *overvalued*.

In practice, the conclusion is not so straightforward. Analysts must cope with uncertainties related to model appropriateness and the correct value of inputs. An analyst's final conclusion depends not only on the comparison of the estimated value and the market price but also on the analyst's confidence in the estimated value (i.e., in the model selected and the inputs used in it). One can envision a spectrum

running from relatively high confidence in the valuation model *and* the inputs to relatively low confidence in the valuation model *and/or* the inputs. When confidence is relatively low, the analyst might demand a substantial divergence between his or her own value estimate and the market price before acting on an apparent mispricing. For instance, if the estimate of intrinsic value is \$10 and the market price is \$10.05, the analyst might reasonably conclude that the security is fairly valued and that the 1/2 of 1 percent market price difference from the estimated value is within the analyst's confidence interval.

Confidence in the convergence of the market price to the intrinsic value over the investment time horizon relevant to the objectives of the portfolio must also be taken into account before an analyst acts on an apparent mispricing or makes a buy, sell, or hold recommendation: The ability to benefit from identifying a mispriced security depends on the market price converging to the estimated intrinsic value.

In seeking to identify mispricing and attractive investments, analysts are treating market prices with skepticism, but they are also treating market prices with respect. For example, an analyst who finds that many securities examined appear to be overvalued will typically recheck models and inputs before acting on a conclusion of overvaluation. Analysts also often recognize and factor into recommendations that different market segments—such as securities closely followed by analysts versus securities relatively neglected by analysts—may differ in how common or persistent mispricing is. Mispricing may be more likely in securities neglected by analysts.

EXAMPLE 1

Valuation and Analyst Response

1. An analyst finds that all the securities analyzed have estimated values higher than their market prices. The securities all appear to be:

- A. overvalued.
- B. undervalued.
- C. fairly valued.

Solution to 1:

B is correct. The estimated intrinsic value for each security is greater than the market price. The securities all appear to be undervalued in the market. Note, however, that the analyst may wish to reexamine the model and inputs to check that the conclusion is valid.

2. An analyst finds that nearly all companies in a market segment have common shares which are trading at market prices above the analyst's estimate of the shares' values. This market segment is widely followed by analysts. Which of the following statements describes the analyst's *most appropriate* first action?

- A. Issue a sell recommendation for each share issue.
- B. Issue a buy recommendation for each share issue.
- C. Reexamine the models and inputs used for the valuations.

Solution to 2:

C is correct. It seems improbable that all the share issues analyzed are overvalued, as indicated by market prices in excess of estimated value—particularly because the market segment is widely followed by analysts. Thus, the analyst will not issue a sell recommendation for each issue. The analyst will *most appropriately* reexamine the models and inputs prior to issuing any recommendations. A buy recommendation is not an appropriate response to an overvalued security.

3. An analyst, using a number of models and a range of inputs, estimates a security's value to be between ¥250 and ¥270. The security is trading at ¥265. The security appears to be:

- A. overvalued.
- B. undervalued.
- C. fairly valued.

Solution to 3:

C is correct. The security's market price of ¥265 is within the range estimated by the analyst. The security appears to be fairly valued.

Analysts often use a variety of models and inputs to achieve greater confidence in their estimates of intrinsic value. The use of more than one model and a range of inputs also helps the analyst understand the sensitivity of value estimates to different models and inputs.

3. CATEGORIES OF EQUITY VALUATION MODELS

LEARNING OUTCOME

The candidate should be able to:

- ☐ **describe major categories of equity valuation models**

Three major categories of equity valuation models are as follows:

- **Present value models** (synonym: **discounted cash flow models**). These models estimate the intrinsic value of a security as the present value of the future benefits expected to be received from the security. In present value models, benefits are often defined in terms of cash expected to be distributed to shareholders (**dividend discount models**) or in terms of cash flows available to be distributed to shareholders after meeting capital expenditure and working capital needs (**free-cash-flow-to-equity models**). Many models fall within this category, ranging from the relatively simple to the very complex. In Section 4, we discuss in detail two of the simpler models, the Gordon (constant) growth model and the two-stage dividend discount models.
- **Multiplier models** (synonym: **market multiple models**). These models are based chiefly on share price multiples or enterprise value multiples. The former model estimates intrinsic value of a common share from a price multiple for some fundamental variable, such as revenues, earnings, cash flows, or book value. Examples of the multiples include price to earnings (P/E, share price divided by earnings per share) and price to sales (P/S, share price divided by sales per share). The fundamental variable may be stated on a forward basis (e.g., forecasted EPS for the next year) or a trailing basis (e.g., EPS for the past year), as long as the usage is consistent across companies being examined. Price multiples are also used to compare relative values. The use of the ratio of share price to EPS—that is, the P/E multiple—to judge relative value is an example of this approach to equity valuation.

Enterprise value (EV) multiples have the form (Enterprise value)/(Value of a fundamental variable). Two possible choices for the denominator are earnings before interest, taxes, depreciation, and amortization (EBITDA) and total revenue. Enterprise value, the numerator, is a measure of a company's total market value from which cash and short-term investments have been subtracted (because an acquirer could use those assets to pay for acquiring the company). An estimate of common share value can be calculated indirectly from the EV multiple; the value of liabilities and preferred shares can be subtracted from the EV to arrive at the value of common equity.

- **Asset-based valuation models.** These models estimate intrinsic value of a common share from the estimated value of the assets of a corporation minus the estimated value of its liabilities and preferred shares. The estimated market value of the assets is often determined by making adjustments to the **book value** (synonym: **carrying value**) of assets and liabilities. The theory underlying the asset-based approach is that the value of a business is equal to the sum of the value of the business's assets.

As already mentioned, many analysts use more than one type of model to estimate value. Analysts recognize that each model is a simplification of the real world and that there are uncertainties related to model appropriateness and the inputs to the models. The choice of model(s) will depend on the availability of information to input into the model(s) and the analyst's confidence in the information and in the appropriateness of the model(s).

EXAMPLE 2

Categories of Equity Valuation Models

1. An analyst is estimating the intrinsic value of a new company. The analyst has one year of financial statements for the company and has calculated the average values of a variety of price multiples for the industry in which the company operates. The analyst plans to use at least one model from each of the three categories of valuation models. The analyst is *least likely* to rely on the estimate(s) from the:

- A. multiplier model(s).
- B. present value model(s).
- C. asset-based valuation model(s).

Solution to 1:

B is correct. Because the company has only one year of data available, the analyst is *least likely* to be confident in the inputs for a present value model. The values on the balance sheet, even before adjustment, are likely to be close to market values because the assets are all relatively new. The multiplier models are based on average multiples from the industry.

2. Based on a company's EPS of €1.35, an analyst estimates the intrinsic value of a security to be €16.60. Which type of model is the analyst *most likely* to be using to estimate intrinsic value?

- A. Multiplier model.
- B. Present value model.
- C. Asset-based valuation model.

Solution to 2:

A is correct. The analyst is using a multiplier model based on the P/E multiple. The P/E multiple used was $16.60/1.35 = 12.3$.

As you begin the study of specific equity valuation models in the next section, you must bear in mind that any model of value is, by necessity, a simplification of the real world. Never forget this simple fact! You may encounter models much more complicated than the ones discussed here, but even those models will be simplifications of reality.

4. BACKGROUND FOR THE DIVIDEND DISCOUNT MODEL

LEARNING OUTCOMES

The candidate should be able to:

- ☐ describe regular cash dividends, extra dividends, stock dividends, stock splits, reverse stock splits, and share repurchases
- ☐ describe dividend payment chronology

Present value models follow a fundamental tenet of economics which states that individuals defer consumption—that is, they invest—for the future benefits expected. Individuals and companies make an investment because they expect thereby to earn a rate of return over the investment period. Logically, the value of an investment should be equal to the present value of the expected future benefits. For common shares, an analyst can equate benefits to the cash flows to be generated by the investment. The simplest present value model of equity valuation is the dividend discount model (DDM), which specifies cash flows from a common stock investment to be dividends.

The next section describes aspects of dividends that users of dividend discount models should understand.

4.1. Dividends: Background for the Dividend Discount Model

Generally, there are two sources of return from investing in equities: (1) cash dividends received by an investor over his or her holding period and (2) the change in the market price of equities over that holding period.

A **dividend** is a distribution paid to shareholders based on the number of shares owned, and a cash dividend is a cash distribution made to a company's shareholders. Cash dividends are typically paid out regularly at known intervals; such dividends are known as regular cash dividends. By contrast, an **extra dividend** or **special dividend** is a dividend paid by a company that does not pay dividends on a regular schedule or a dividend that supplements regular cash dividends with an extra payment. Companies in cyclical industries and companies undergoing corporate and/or financial restructuring are among those observed to use extra dividends.¹

The payment of dividends is not a legal obligation: dividends must be declared (i.e., authorized) by a company's board of directors; in some jurisdictions, they must also be approved by shareholders. Regular cash dividends are customarily declared and paid out quarterly in the United States and Canada; semiannually in Europe and Japan; and annually in some other countries, including China.

Dividend discount models address discounting expected cash dividends. A **stock dividend** (also known as a **bonus issue of shares**) is a type of dividend in which a company distributes additional shares of its common stock (typically, 2%–10% of the shares then outstanding) to shareholders instead of cash. A stock dividend divides the “pie” (the market value of shareholders' equity) into smaller pieces without affecting the value of the pie or any shareholder's proportional ownership in the company. Thus, stock dividends are not relevant for valuation. Stock splits and reverse stock splits are similar to stock dividends in that they have no economic effect on the company or shareholders. A **stock split** involves an increase in the number of shares outstanding with a consequent decrease in share price. An example of a stock split is a two-for-one stock split in which each shareholder is issued an additional share for each share currently owned. A **reverse stock split** involves a reduction in the number of shares outstanding with a corresponding increase in share price. In a one-for-two reverse stock split, each shareholder would receive one new share for every two old shares held, thereby reducing the number of shares outstanding by half.

In contrast to stock dividends and stock splits, share repurchases are an alternative to cash dividend payments. A **share repurchase** (or **buyback**) is a transaction in which a company uses cash to buy back its own shares. Shares that have been repurchased are not considered for dividends, voting, or computing earnings per share. A share repurchase is viewed as equivalent to the payment of cash dividends of equal value in terms of the effect on shareholders' wealth, all other things being equal. Company managements have expressed several key reasons for engaging in share repurchases—namely,

(1) signaling a belief that their shares are undervalued (or, more generally, to support share prices), (2) flexibility in the amount and timing of distributing cash to shareholders, (3) tax efficiency in markets where tax rates on dividends exceed tax rates on capital gains, and (4) the ability to absorb increases in outstanding shares because of the exercise of employee stock options.

The payout of regular cash dividends to common shareholders follows a fairly standard chronology that is set in motion once the company's board of directors votes to pay the dividend. First is the **declaration date**, the day that the company issues a statement declaring a specific dividend. Next comes the **ex-dividend date** (or ex-date), the first date that a share trades without (i.e., "ex") the dividend. This is followed closely (one or two business days later) by the **holder-of-record date** (also called the owner-of-record date, shareholder-of-record date, record date, date of record, or date of book closure), the date that a shareholder listed on the company's books will be deemed to have ownership of the shares for purposes of receiving the upcoming dividend; the amount of time between the ex-date and the holder-of-record date is linked to the trade settlement cycle in force. The final milestone is the **payment date** (or **payable date**), which is the day that the company actually mails out (or electronically transfers) a dividend payment to shareholders.

EXAMPLE 3

Total S.A. Dividend Payment Time Line

On 26 May 2017, Total S.A., one of the world's largest integrated energy companies, declared an annual dividend of €2.48 per share, payable on a quarterly basis. The first quarterly dividend of $€2.48/4 = €0.62$ was payable on 12 October 2017. The holder-of-record date was 26 September, and the ex-dividend date was 25 September. A timeline for the upcoming Total S.A. quarterly dividend is shown in [Exhibit 1](#).

Exhibit 1: Timeline for Total S.A. Quarterly Dividend

Source: Total S.A. website: www.total.com.

Because buyers of a company's shares on the ex-dividend date are no longer eligible to receive the upcoming dividend, all else being equal, on that day the company's share price immediately decreases by the amount of the foregone dividend. [Exhibit 2](#) illustrates the decrease in share price that occurs for a hypothetical company that has declared a \$1.00 per share dividend as trading begins on its ex-dividend date.

Exhibit 2: Stock Price Change for Hypothetical Company on Ex-Dividend Date

Note: Assumes dividend declared is \$1 per share and convention for stock trade settlement is $T + 3$.

5. DIVIDEND DISCOUNT MODEL (DDM) AND FREE-CASH-FLOW-TO-EQUITY MODEL (FCFE)

LEARNING OUTCOMES

The candidate should be able to:

- explain the rationale for using present value models to value equity and describe the dividend discount and free-cash-flow-to-equity models
- explain advantages and disadvantages of each category of valuation model

If the issuing company is assumed to be a going concern, the intrinsic value of a share is the present value of expected future dividends. If a constant required rate of return is also assumed, then the DDM expression for the intrinsic value of a share is [Equation 1](#):

Equation 1

$$V_0 = \sum_{t=1}^{\infty} \frac{D_t}{(1+r)^t}$$

where

V_0 = value of a share of stock today, at $t = 0$

D_t = expected dividend in year t , assumed to be paid at the end of the year

r = required rate of return on the stock

At the shareholder level, cash received from a common stock investment includes any dividends received and the proceeds when shares are sold. If an investor intends to buy and hold a share for one year, the value of the share today is the present value of two cash flows—namely, the expected dividend *plus* the expected selling price in one year:

Equation 2

$$V_0 = \frac{D_1 + P_1}{(1+r)^1} = \frac{D_1}{(1+r)^1} + \frac{P_1}{(1+r)^1}$$

where P_1 = the expected price per share at $t = 1$.

To estimate the expected selling price, P_1 , the analyst could estimate the price another investor with a one-year holding period would pay for the share in one year. If V_0 is based on D_1 and P_1 , it follows that P_1 could be estimated from D_2 and P_2 :

$$P_1 = \frac{D_2 + P_2}{(1+r)}$$

Substituting the right side of this equation for P_1 in [Equation 2](#) results in V_0 estimated as

$$V_0 = \frac{D_1}{(1+r)} + \frac{D_2 + P_2}{(1+r)^2} = \frac{D_1}{(1+r)} + \frac{D_2}{(1+r)^2} + \frac{P_2}{(1+r)^2}$$

Repeating this process, we find the value for n holding periods is the present value of the expected dividends for the n periods plus the present value of the expected price in n periods:

$$V_0 = \frac{D_1}{(1+r)^1} + \cdots + \frac{D_n}{(1+r)^n} + \frac{P_n}{(1+r)^n}$$

Using summation notation to represent the present value of the n expected dividends, we arrive at the general expression for an n -period holding period or investment horizon:

Equation 3

$$V_0 = \sum_{t=1}^n \frac{D_t}{(1+r)^t} + \frac{P_n}{(1+r)^n}$$

The expected value of a share at the end of the investment horizon—in effect, the expected selling price—is often referred to as the **terminal stock value** (or **terminal value**).

EXAMPLE 4

Estimating Share Value for a Three-Year Investment Horizon

For the next three years, the annual dividends of a stock are expected to be €2.00, €2.10, and €2.20. The stock price is expected to be €20.00 at the end of three years. If the required rate of return on the shares is 10 percent, what is the estimated value of a share?

Solution:

The present values of the expected future cash flows can be written as follows:

$$V_0 = \frac{2.00}{(1.10)^1} + \frac{2.10}{(1.10)^2} + \frac{2.20}{(1.10)^3} + \frac{20.00}{(1.10)^3}$$

Calculating and summing these present values gives an estimated share value of $V_0 = 1.818 + 1.736 + 1.653 + 15.026 = €20.23$.

The three dividends have a total present value of €5.207, and the terminal stock value has a present value of €15.026, for a total estimated value of €20.23.

Extending the holding period into the indefinite future, we can say that a stock's estimated value is the present value of all expected future dividends as shown in [Equation 1](#).

Consideration of an indefinite future is valid because businesses established as corporations are generally set up to operate indefinitely. This general form of the DDM applies even in the case in which the investor has a finite investment horizon. For that investor, stock value today depends *directly* on the dividends the investor expects to receive before the stock is sold and depends *indirectly* on the expected dividends for periods subsequent to that sale, because those expected future dividends determine the expected selling price. Thus, the general expression given by [Equation 1](#) holds irrespective of the investor's holding period.

In practice, many analysts prefer to use a free-cash-flow-to-equity (FCFE) valuation model. These analysts assume that dividend-paying *capacity* should be reflected in the cash flow estimates rather than *expected dividends*. FCFE is a measure of dividend-paying capacity. Analysts may also use FCFE valuation models for a non-dividend-paying stock. To use a DDM, the analyst needs to predict the timing and amount of the first dividend and all the dividends or dividend growth thereafter. Making these predictions for non-dividend-paying stock accurately is typically difficult, so in such cases, analysts often resort to FCFE models.

The calculation of FCFE starts with the calculation of cash flow from operations (CFO). CFO is simply defined as net income plus non-cash expenses minus investment in working capital. FCFE is a measure of cash flow generated in a period that is available for distribution to common shareholders. What does "available for distribution" mean? The entire CFO is *not* available for distribution; the portion of the CFO needed for fixed capital investment (FCInv) during the period to maintain the value of the company as a going concern is *not* viewed as available for distribution to common shareholders. Net amounts borrowed (borrowings minus repayments) are considered to be available for distribution to common shareholders. Thus, FCFE can be expressed as

Equation 4

$$\text{FCFE} = \text{CFO} - \text{FCInv} + \text{Net borrowing}$$

The information needed to calculate historical FCFE is available from a company's statement of cash flows and financial disclosures. Frequently, under the assumption that management is acting in the interest of maintaining the value of the company as a going concern, reported capital expenditure is taken to represent FCInv. Analysts must make projections of financials to forecast future FCFE. Valuation obtained by using FCFE involves discounting expected future FCFE by the required rate of return on equity; the expression parallels [Equation 1](#):

$$V_0 = \sum_{t=1}^{\infty} \frac{\text{FCFE}_t}{(1+r)^t}$$

EXAMPLE 5

Present Value Models

1. An investor expects a share to pay dividends of \$3.00 and \$3.15 at the end of Years 1 and 2, respectively. At the end of the second year, the investor expects the shares to trade at \$40.00. The required rate of return on the shares is 8 percent. If the investor's forecasts are accurate and the market price of the shares is currently \$30, the *most likely* conclusion is that the shares are:
 - A. overvalued.
 - B. undervalued.
 - C. fairly valued.

Solution to 1:

B is correct.

$$V_0 = \frac{3.00}{(1.08)^1} + \frac{3.15}{(1.08)^2} + \frac{40.00}{(1.08)^2} = 39.77$$

The value estimate of \$39.77 exceeds the market price of \$30, so the conclusion is that the shares are undervalued.

2. Two investors with different holding periods but the same expectations and required rate of return for a company are estimating the intrinsic value of a common share of the company. The investor with the shorter holding period will *most likely* estimate a:
 - A. lower intrinsic value.
 - B. higher intrinsic value.
 - C. similar intrinsic value.

Solution to 2:

C is correct. The intrinsic value of a security is independent of the investor's holding period.

3. An equity valuation model that focuses on expected dividends rather than the capacity to pay dividends is the:
 - A. dividend discount model.
 - B. free cash flow to equity model.
 - C. cash flow return on investment model.

Solution to 3:

A is correct. Dividend discount models focus on expected dividends.

How is the required rate of return for use in present value models estimated? To estimate the required rate of return on a share, analysts frequently use the capital asset pricing model (CAPM):

Equation 5

Required rate of return on share i = Current expected risk
– free rate of return
+ Beta_i [Market (equity) risk premium]

Equation 5 states that the required rate of return on a share is the sum of the current expected risk-free rate plus a risk premium that equals the product of the stock's beta (a measure of non-diversifiable risk) and the market risk premium (the expected return of the market in excess of the risk-free return, where in practice, the "market" is often represented by a broad stock market index). However, even if analysts agree that the CAPM is an appropriate model, their inputs into the CAPM may differ. Thus, there is no uniquely correct answer to the question: What is the required rate of return?

Other common methods for estimating the required rate of return for the stock of a company include adding a risk premium that is based on economic judgments, rather than the CAPM, to an appropriate risk-free rate (usually a government bond) and adding a risk premium to the yield on the company's bonds. Good business and economic judgment is paramount in estimating the required rate of return. In many investment firms, required rates of return are determined by firm policy.

6. PREFERRED STOCK VALUATION

LEARNING OUTCOMES

The candidate should be able to:

- ☐ calculate the intrinsic value of a non-callable, non-convertible preferred stock
- ☐ explain advantages and disadvantages of each category of valuation model

General dividend discount models are relatively easy to apply to preferred shares. In its simplest form, **preferred stock** is a form of equity (generally, non-voting) that has priority over common stock in the receipt of dividends and on the issuer's assets in the event of a company's liquidation. It may have a stated maturity date at which time payment of the stock's par (face) value is made or it may be perpetual with no maturity date; additionally, it may be callable or convertible.

For a non-callable, non-convertible perpetual preferred share paying a level dividend D and assuming a constant required rate of return over time, Equation 1 reduces to the formula for the present value of a perpetuity. Its value is:

Equation 6

$$V_0 = \frac{D_0}{r}$$

For example, a \$100 par value non-callable perpetual preferred stock offers an annual dividend of \$5.50. If its required rate of return is 6 percent, the value estimate would be $\$5.50/0.06 = \91.67 .

For a non-callable, non-convertible preferred stock with maturity at time n , the estimated intrinsic value can be estimated by using Equation 3 but using the preferred stock's par value, F , instead of P_n :

Equation 7

$$V_0 = \sum_{t=1}^n \frac{D_t}{(1+r)^t} + \frac{F}{(1+r)^n}$$

When [Equation 7](#) is used, the most precise approach is to use values for n , r , and D that reflect the payment schedule of the dividends. This method is similar to the practice of fixed-income analysts in valuing a bond. For example, a non-convertible preferred stock with a par value of £20.00, maturity in six years, a nominal required rate of return of 8.20 percent, and semiannual dividends of £2.00 would be valued by using an n of 12, an r of 4.10 percent, a D of £2.00, and an F of £20.00. The result would be an estimated value of £31.01. Assuming payments are annual rather than semiannual (i.e., assuming that $n = 6$, $r = 8.20$ percent, and $D = £4.00$) would result in an estimated value of £30.84.

Preferred stock issues are frequently callable (redeemable) by the issuer at some point prior to maturity, often at par value or at prices in excess of par value that decline to par value as the maturity date approaches. Such call options tend to reduce the value of a preferred issue to an investor because the option to redeem will be exercised by the issuer when it is in the issuer's favor and ignored when it is not. For example, if an issuer can redeem shares at par value that would otherwise trade (on the basis of dividends, maturity, and required rate of return) above par value, the issuer has motivation to redeem the shares.

Preferred stock issues can also include a retraction option that enables the holder of the preferred stock to sell the shares back to the issuer prior to maturity on prespecified terms. Essentially, the holder of the shares has a put option. Such put options tend to increase the value of a preferred issue to an investor because the option to retract will be exercised by the investor when it is in the investor's favor and ignored when it is not. Although the precise valuation of issues with such embedded options is beyond the scope of this reading, [Example 6](#) includes a case in which [Equation 7](#) can be used to approximate the value of a callable, retractable preferred share.

EXAMPLE 6

Preferred Share Valuation: Two Cases

Case 1: Non-callable, Non-convertible, Perpetual Preferred Shares

The following facts concerning the Union Electric Company 4.75 percent perpetual preferred shares are as follows:

- Issuer: Union Electric Co. (owned by Ameren)
- Par value: US\$100
- Dividend: US\$4.75 per year
- Maturity: perpetual
- Embedded options: none
- Credit rating: Moody's Investors Service/Standard & Poor's Ba1/BB
- Required rate of return on Ba1/BB rated preferred shares as of valuation date: 7.5 percent.

1. Estimate the intrinsic value of this preferred share.

Solution

Basing the discount rate on the required rate of return on Ba1/BB rated preferred shares of 7.5 percent gives an intrinsic value estimate of $\text{US\$}4.75 / 0.075 = \text{US\$}63.33$.

2. Explain whether the intrinsic value of this issue would be higher or lower if the issue were callable (with all other facts remaining unchanged).

Solution

The intrinsic value would be lower if the issue were callable. The option to redeem or call the issue is valuable to the issuer because the call will be exercised when doing so is in the issuer's interest. The intrinsic value of the shares to the investor will typically be lower if the issue is callable. In this case, because the intrinsic value without the call is much less than the par value, the issuer would be unlikely to redeem the issue if it were callable; thus, callability would reduce intrinsic value, but only slightly.

Case 2: Retractable Term Preferred Shares

Retractable term preferred shares are a type of preferred share that has been previously issued by Canadian companies, and have now begun to be offered by companies in other jurisdictions, including Japan. This type of issue specifies a "retraction date" when the preferred shareholders have the option to sell back their shares to the issuer at par value (i.e., the shares are "retractable" or "putable" at that date).² At predetermined dates prior to the retraction date, the issuer has the option to redeem the preferred issue at predetermined prices (which are always at or above par value).

An example of a retractable term preferred share currently outstanding is TMC (Toyota Motor Corporation), First Series Model AA class shares, with a 0.5 percent dividend rate, increasing by 0.5 percent every year until 2020 and thereafter, becoming fixed at 2.5 percent. TMC is leading global automobile manufacturer, with headquarters in Japan and global operations. The issue is in Japanese Yen. The shares have a ¥10,598 par value and pay a semiannual dividend of ¥26.5 [= (0.5 percent × ¥10,598)/2] on 31 March 2016. The semiannual dividend is expected to increase to ¥132.5 [= (2.5 percent × ¥10,598)/2] on 31 March 2020 and beyond. As of 31 December 2017 the company carried ratings from Moody's and Standard & Poor's of Aa3 and AA-, respectively. Thus, the shares are viewed by Moody's and Standard & Poor's as having "adequate" credit quality, qualified by "Aa3 and AA-," which means relatively high quality within that group. Beginning from 2 April 2021, the shares are redeemable at the option of TMC at ¥10,598 (par value). The retraction date is the last day of March, June, September, and December of each year, starting from 1 September 2020, with the shares retractable at par value. The Series AA shares have voting rights and may exercise their voting rights and other rights held by holders of common shares of TMC in the same manner. The Series AA shares were issued at a 20% premium to the common shares price in 2015, and since then the share price has decreased to ¥7,243 as at 31 December 2017, and with a current required rate of 3.05 per year (1.525 percent semiannual). Because the issue's market price is so far below the prices at which TMC could redeem or call the issue, redemption is considered to be unlikely by TMC, whereas the retraction option for the Series AA holders appears to have a significant value since they will potentially be able to put back the shares to TMC at approximately 45 percent over the current market value (¥10,598 compared to ¥7,243).

3. Assume that the issue will be retracted in December 2020; the holders of the shares will put the shares to the company in December 2020. Based on the information given, estimate the intrinsic value of a share. Assume it is December 2017.

Solution

An intrinsic value estimate of a share of this preferred issue is ¥10,279.

Expected semiannual dividends:

Year ended March 31, 2018: ¥79.5 [= (1.5 percent × ¥10,598)/2]

Year ended March 31, 2019: ¥106 [= (2.0 percent × ¥10,598)/2]

Year ended March 31, 2020: ¥132.5 [= (2.5 percent × ¥10,598)/2]

$$V_0 = \left[\frac{¥79.5}{1.01525} + \frac{¥79.5}{1.01525^2} + \frac{¥106}{1.01525^3} + \frac{¥106}{1.01525^4} + \frac{¥132.5}{1.01525^5} + \frac{¥132.5}{1.01525^6} + \frac{¥10,598}{1.01525^6} \right] \\ \approx ¥10,279$$

The difference between the current market price of ¥7,243 and the intrinsic value of ¥10,279 is the implied value of retractable option given to the holders of the Series AA shares.

7. THE GORDON GROWTH MODEL

LEARNING OUTCOMES

The candidate should be able to:

- ☐ calculate and interpret the intrinsic value of an equity security based on the Gordon (constant) growth dividend discount model or a two-stage dividend discount model, as appropriate
- ☐ identify characteristics of companies for which the constant growth or a multistage dividend discount model is appropriate
- ☐ explain advantages and disadvantages of each category of valuation model

A rather obvious problem when one is trying to implement [Equation 1](#) for common equity is that it requires the analyst to estimate an infinite series of expected dividends. To simplify this process, analysts frequently make assumptions about how dividends will grow or change over time. The Gordon (constant) growth model (Gordon, 1962) is a simple and well-recognized DDM. The model assumes dividends grow indefinitely at a constant rate.

Because of its assumption of a constant growth rate, the Gordon growth model is particularly appropriate for valuing the equity of dividend-paying companies that are relatively insensitive to the business cycle and in a mature growth phase. Examples might include an electric utility serving a slowly growing area or a producer of a staple food product (e.g., bread). A history of increasing the dividend at a stable growth rate is another practical criterion if the analyst believes that pattern will hold in the future.

With a constant growth assumption, [Equation 1](#) can be written as [Equation 8](#), where g is the constant growth rate:

Equation 8

$$V_0 = \sum_{t=1}^{\infty} \frac{D_0(1+g)^t}{(1+r)^t} = D_0 \left[\frac{(1+g)}{(1+r)} + \frac{(1+g)^2}{(1+r)^2} + \dots + \frac{(1+g)^{\infty}}{(1+r)^{\infty}} \right]$$

If required return r is assumed to be strictly greater than growth rate g , then the square-bracketed term in [Equation 8](#) is an infinite geometric series and sums to $[(1+g)/(r-g)]$. Substituting into [Equation 8](#) produces the Gordon growth model as presented in [Equation 9](#):

Equation 9

$$V_0 = \frac{D_0(1+g)}{r-g} = \frac{D_1}{r-g}$$

For an illustration of the expression, suppose the current (most recent) annual dividend on a share is €5.00 and dividends are expected to grow at 4 percent per year. The required rate of return on equity is 8 percent. The Gordon growth model estimate of intrinsic value is, therefore, €5.00(1.04)/(0.08 – 0.04) = €5.20/0.04 = €130 per share. Note that the numerator is D_1 not D_0 . (Using the wrong numerator is a common error.)

The Gordon growth model estimates intrinsic value as the present value of a growing perpetuity. If the growth rate, g , is assumed to be zero, [Equation 8](#) reduces to the expression for the present value of a perpetuity, given earlier as [Equation 6](#).

In estimating a long-term growth rate, analysts use a variety of methods, including assessing the growth in dividends or earnings over time, using the industry median growth rate, and using the relationship shown in [Equation 10](#) to estimate the sustainable growth rate:

Equation 10

$$g = b \times \text{ROE}$$

where

g = dividend growth rate

b = earnings retention rate = $(1 - \text{Dividend payout ratio})$

ROE = return on equity

Example 7 illustrates the application of the Gordon growth model to the shares of a large industrial manufacturing company. The analyst believes it will continue to grow at a rate that it achieved in the previous three years and remain stable in the future. The example asks how much the dividend growth assumption adds to the intrinsic value estimate. The question is relevant to valuation because if the amount is high on a percentage basis, a large part of the value of the share depends on the realization of the growth estimate. One can answer the question by subtracting from the intrinsic value estimate determined by [Equation 9](#) the value determined by [Equation 6](#), which assumes no dividend growth.³

EXAMPLE 7

Applying the Gordon Growth Model

Siemens AG operates in the capital goods and technology space. It is involved in the engineering, manufacturing, automation, power, and transportation sectors. It operates globally and is one of the largest companies in the sectors in which it operates. It is a substantial employer in both its original, domestic German market, as well as dozens of countries around the world. Selected financial information for Siemens appears in [Exhibit 3](#).

Exhibit 3: Selected Financial Information for Siemens AG

Year	2017	2016	2015	2014	2013
EPS	€7.45	€6.74	€8.85	€6.37	€5.08
DPS	€3.7	€3.6	€3.5	€3.3	€3.0
Payout ratio	50%	53%	40%	52%	59%
ROE	15.6%	15.9%	22.3%	18.2%	14.6%
Share price (XETRA - Frankfurt)	€119.2	€104.2	€79.94	€94.37	€89.06

Note: DPS stands for “dividends per share.”

Source: Morningstar, www.siemens.com.

The analyst estimates the growth rate to be approximately 5.4 percent based on the dividend growth rate over the period 2013 to 2017 [$3(1 + g)^4 = 3.7$, so $g = 5.4\%$]. To verify that the estimated growth rate of 5.4 percent is feasible in the future, the analyst also uses the average of Siemens’s retention rate and ROE for the previous five years ($g \approx 0.49 \times 17.3\% \approx 8.5\%$) to estimate the sustainable growth rate.

Using a number of approaches, including adding a risk premium to a long-term German government bond and using the CAPM, the analyst estimates a required return of 7.5 percent. The most recent dividend of €3.70 is used for D_0 .

1. Use the Gordon growth model to estimate Siemens’s intrinsic value.

Solution to 1:

$$V_0 = \frac{\text{€}3.70(1+0.054)}{0.075-0.054} = \text{€}185.70$$

2. How much does the dividend growth assumption add to the intrinsic value estimate?

Solution to 2:

$$\text{€}185.70 - \frac{\text{€}3.70}{0.075} = \text{€}136.37$$

3. Based on the estimated intrinsic value, is a share of Siemens undervalued, overvalued, or fairly valued?

Solution to 3:

A share of Siemens appears to be undervalued. The analyst, before making a recommendation, might consider how realistic the estimated inputs are and check the sensitivity of the estimated value to changes in the inputs.

4. What is the intrinsic value if the growth rate estimate is lowered to 4.4 percent?

Solution to 4:

$$V_0 = \frac{\text{€}3.70(1+0.044)}{0.075-0.044} = \text{€}124.61$$

5. What is the intrinsic value if the growth rate estimate is lowered to 4.4 percent and the required rate of return estimate is increased to 8.5 percent?

Solution to 5:

$$V_0 = \frac{\text{€}3.70(1+0.044)}{0.085-0.044} = \text{€}94.21$$

The Gordon growth model estimate of intrinsic value is extremely sensitive to the choice of required rate of return r and growth rate g . It is possible that the growth rate assumption and the required return assumption used initially were too high. Worldwide economic growth is typically in the low single digits, which may mean that a large company such as Siemens may struggle to grow dividends at 5.4 percent into perpetuity. [Exhibit 4](#) presents a further sensitivity analysis of Siemens's intrinsic value to the required return and growth estimates.

Exhibit 4: Sensitivity Analysis of the Intrinsic-Value Estimate for Siemens AG

	$g = 2.5\%$	$g = 3.5\%$	$g = 4.5\%$	$g = 5.5\%$	$g = 6.5\%$
$r = 6\%$	€108.4	€153.2	€257.8	€780.7	—
$r = 7\%$	€84.3	€109.4	€154.7	€260.2	€788.1
$r = 8\%$	€69.0	€85.1	€110.5	€156.1	€262.7
$r = 9\%$	€58.3	€69.6	€85.9	€111.5	€157.6
$r = 10\%$	€50.6	€58.9	€70.3	€86.7	€112.6

Note that no value is shown when the growth rate exceeds the required rate of return. The Gordon growth model assumes that the growth rate cannot be greater than the required rate of return.

The assumptions of the Gordon model are as follows:

- Dividends are the correct metric to use for valuation purposes.
- The dividend growth rate is forever: It is perpetual and never changes.
- The required rate of return is also constant over time.
- The dividend growth rate is strictly less than the required rate of return.

An analyst might be dissatisfied with these assumptions for many reasons. The equities being examined might not currently pay a dividend. The Gordon assumptions might be too simplistic to reflect the characteristics of the companies being evaluated. Some alternatives to using the Gordon model are as follows:

- Use a more robust DDM that allows for varying patterns of growth.
- Use a cash flow measure other than dividends for valuation purposes.
- Use some other approach (such as a multiplier method) to valuation.

Applying a DDM is difficult if the company being analyzed is not currently paying a dividend. A company may not be paying a dividend if 1) the investment opportunities the company has are all so attractive that the retention and reinvestment of funds is preferable, from a return perspective, to the distribution of a dividend to shareholders or 2) the company is in such shaky financial condition that it cannot afford to pay a dividend. An analyst might still use a DDM to value such companies by assuming that dividends will begin at some future point in time. The analyst might further assume that constant growth occurs after that date and use the Gordon growth model for valuation. Extrapolating from no current dividend, however, generally yields highly uncertain forecasts. Analysts typically choose to use one or more of the alternatives instead of or as a supplement to the Gordon growth model.

EXAMPLE 8

Gordon Growth Model in the Case of No Current Dividend

A company does not currently pay a dividend but is expected to begin to do so in five years (at $t = 5$). The first dividend is expected to be \$4.00 and to be received five years from today. That dividend is expected to grow at 6 percent into perpetuity. The required return is 10 percent. What is the estimated current intrinsic value?

Solution:

The analyst can value the share in two pieces:

1. The analyst uses the Gordon growth model to estimate the value at $t = 5$; in the model, the year-ahead dividend is \$4(1.06). Then the analyst finds the present value of this value as of $t = 0$.
2. The analyst finds the present value of the \$4 dividend not “counted” in the estimate in Piece 1 (which values dividends from $t = 6$ onward). Note that the statement of the problem implies that D_0, D_1, D_2, D_3 , and D_4 are zero.

Piece 1: The value of this piece is \$65.818:

$$\begin{aligned} V_n &= \frac{D_n(1+g)}{r-g} = \frac{D_{n+1}}{r-g} \\ V_5 &= \frac{\$4(1+0.06)}{0.10-0.06} = \frac{\$4.24}{0.04} = \$106 \\ V_0 &= \frac{\$106}{(1+0.10)^5} = \$65.818 \end{aligned}$$

Piece 2: The value of this piece is \$2.484:

$$V_0 = \frac{\$4}{(1+0.10)^5} = \$2.484$$

The sum of the two pieces is $\$65.818 + \$2.484 = \$68.30$.

Alternatively, the analyst could value the share at $t = 4$, the point at which dividends are expected to be paid in the following year and from which point they are expected to grow at a constant rate.

$$V_4 = \frac{\$4.00}{0.10-0.06} = \frac{\$4.00}{0.04} = \$100$$

$$V_0 = \frac{\$100}{(1+0.10)^4} = \$68.30$$

The next section addresses the application of the DDM with more flexible assumptions as to the dividend growth rate.

8. MULTISTAGE DIVIDEND DISCOUNT MODELS

LEARNING OUTCOMES

The candidate should be able to:

- ☐ calculate and interpret the intrinsic value of an equity security based on the Gordon (constant) growth dividend discount model or a two-stage dividend discount model, as appropriate
- ☐ identify characteristics of companies for which the constant growth or a multistage dividend discount model is appropriate
- ☐ explain advantages and disadvantages of each category of valuation model

Multistage growth models are often used to model rapidly growing companies. The *two-stage DDM* assumes that at some point the company will begin to pay dividends that grow at a constant rate, but prior to that time the company will pay dividends that are growing at a higher rate than can be sustained in the long run. That is, the company is assumed to experience an initial, finite period of high growth, perhaps prior to the entry of competitors, followed by an infinite period of sustainable growth. The two-stage DDM thus makes use of two growth rates: a high growth rate for an initial, finite period followed by a lower, sustainable growth rate into perpetuity. The Gordon growth model is used to estimate a terminal value at time n that reflects the present value at time n of the dividends received during the sustainable growth period.

[Equation 11](#) will be used here as the starting point for a two-stage valuation model. The two-stage valuation model is similar to [Example 8](#) except that instead of assuming zero dividends for the initial period, the analyst assumes that dividends will exhibit a high rate of growth during the initial period. [Equation 11](#) values the dividends over the short-term period of high growth and the terminal value at the end of the period of high growth. The short-term growth rate, g_S , lasts for n years. The intrinsic value per share in year n , V_n , represents the year n value of the dividends received during the sustainable growth period or the terminal value at time n . V_n can be estimated by using the Gordon growth model as shown in [Equation 12](#), where g_L is the long-term or sustainable growth rate. The dividend in year $n + 1$, D_{n+1} , can be determined by using [Equation 13](#):

Equation 11

$$V_0 = \sum_{t=1}^n \frac{D_0(1 + g_S)^t}{(1 + r)^t} + \frac{V_n}{(1+r)^n}$$

Equation 12

$$V_n = \frac{D_{n+1}}{r - g_L}$$

Equation 13

$$D_{n+1} = D_0(1 + g_S)^n (1 + g_L)$$

EXAMPLE 9

Applying the Two-Stage Dividend Discount Model

The current dividend, D_0 , is \$5.00. Growth is expected to be 10 percent a year for three years and then 5 percent thereafter. The required rate of return is 15 percent. Estimate the intrinsic value.

Solution:

$$D_1 = \$5.00(1 + 0.10) = \$5.50$$

$$D_2 = \$5.00(1 + 0.10)^2 = \$6.05$$

$$D_3 = \$5.00(1 + 0.10)^3 = \$6.655$$

$$D_4 = \$5.00(1 + 0.10)^3(1 + 0.05) = \$6.98775$$

$$V_3 = \frac{\$6.98775}{0.15 - 0.05} = \$69.8775$$

$$V_0 = \frac{\$5.50}{(1+0.15)} + \frac{\$6.05}{(1+0.15)^2} + \frac{\$6.655}{(1+0.15)^3} + \frac{\$69.8775}{(1+0.15)^3} \approx \$59.68$$

The DDM can be extended to as many stages as deemed appropriate. For most publicly traded companies (that is, companies beyond the start-up stage), practitioners assume growth will ultimately fall into three stages: 1) growth, 2) transition, and 3) maturity. This assumption supports the use of a *three-stage DDM*, which makes use of three growth rates: a high growth rate for an initial finite period, followed by a lower growth rate for a finite second period, followed by a lower, sustainable growth rate into perpetuity.

One can make the case that a three-stage DDM would be most appropriate for a fairly young company, one that is just entering the growth phase. The two-stage DDM would be appropriate to estimate the value of an older company that has already moved through its growth phase and is currently in the transition phase (a period with a higher growth rate than the sustainable growth rate) prior to moving to the maturity phase (the period with a lower, sustainable growth rate).

However, the choice of a two-stage DDM need not rely solely on the age of a company. Long-established companies sometimes manage to restart above-average growth through, for example, innovation, expansion to new markets, or acquisitions. Or a company's long-run growth rate may be interrupted by a period of subnormal performance. If growth is expected to moderate (in the first case) or improve (in the second case) toward some long-term growth rate, a two-stage DDM may be appropriate. Thus, we chose a two-stage DDM to value International Business Machines Corporation in [Example 10](#).

EXAMPLE 10

Two-Stage Dividend Discount Model: International Business Machines Corporation

International Business Machines Corporation (IBM) is a US based leading technology company. IBM was founded in 1911, initially as a company that manufactured machinery for sale and lease, ranging from commercial scales and industrial time recorders, meat and cheese slicers, to punched cards. IBM introduced the personal computer in 1981; however, by the 1990s, it began to suffer losses in its core computer manufacturing business and by the 2000s, it had begun to diversify into business

consulting, which was finalized in 2005 when it sold its personal computer business to Chinese company Lenovo. IBM now operates through five segments: Cognitive Solutions, Global Business Services (GBS), Technology Services & Cloud Platforms, Systems, and Global Financing. The Cognitive Solutions segment delivers a spectrum of capabilities from descriptive, predictive, and prescriptive analytics to cognitive systems. Cognitive Solutions includes Watson, a cognitive computing platform that has the ability to interact in natural language, process big data, and learn from interactions with people and computers. The GBS segment provides clients with consulting, application management services, and global process services. The Technology Services & Cloud Platforms segment provides information technology infrastructure services. The Systems segment provides clients with infrastructure technologies. The Global Financing segment includes client financing, commercial financing, and remanufacturing and remarketing.

The 30 July 2018 *Value Line* report on IBM appears in [Exhibit 5](#). IBM has increased its dividends every year over the past 15 years. Information from *Value Line* shows that the dividend growth is around 17.0 percent for the past 10 years, 13.5 percent for the past 5 years, and estimated to be 4.5 percent for 2015 to 2023. After a period of growth through acquisition and merger, the pattern suggests that IBM may be transitioning to a mature growth phase.

The two-stage DDM is arguably a good choice for valuing IBM because the company appears to be transitioning from a high-growth phase (note the 13.5 percent dividend growth for the past 5 years) to a lower-growth phase (note the forecast of 4.5 percent dividend growth to 2015–2023).

The CAPM can be used to estimate the required return, r , for IBM. The *Value Line* report (in the top left corner) estimates beta to be 0.95. Using the yield of about 2.0 percent on 10-year US Treasury notes as a proxy for the risk-free rate and assuming an equity risk premium of 5.0 percent, we find the estimate for r would be 6.75 percent [$2.0\% + 0.95(5.0\%)$].

To estimate the intrinsic value at the end of 2018, we use the 2018 dividend of US\$6.21 from the *Value Line* report. The dividend is assumed to grow at a rate of 5.0 percent for one year and then 2.33 percent thereafter. The growth rate assumption for the first stage is consistent with the *Value Line* forecast for the 2018 and 2019 dividends. Our assumption of a 2.33 percent perpetual growth rate produces an 8-year growth rate assumption near 4.5%,⁴ which is consistent with the *Value Line* forecast of 4.5 percent growth from 2015–2022. Thus:

$$D_{2019} = \text{US\$}6.21(1 + 0.05) = \text{US\$}6.5205$$

$$D_{2020} = \text{US\$}6.21(1 + 0.05)(1 + 0.0233) = \text{US\$}6.6724$$

$$D_{2021} = \text{US\$}6.21(1 + 0.05)(1 + 0.0233)^2 = \text{US\$}6.8279$$

$$V_{2020} = \frac{\text{US\$}6.8279}{0.0675 - 0.0233} = \text{US\$}154.4774$$

$$V_{2018} = \frac{\text{US\$}6.5205}{(1+0.0675)} + \frac{\text{US\$}6.6724}{(1+0.0675)^2} + \frac{\text{US\$}154.4774}{(1+0.0675)^2} \approx \text{US\$}147.523$$

Given a recent price of US\$148.56, as noted at the top left corner of the *Value Line* report, the intrinsic-value estimate of US\$147.523 suggests that IBM is approximately fairly valued.

Exhibit 5: *Value Line* Report on IBM

Source: © 2018 Value Line, Inc. All Rights Reserved Worldwide. "Value Line" is a registered trademark of Value Line, Inc. Value Line Geometric and Arithmetic Indices calculated by Thomson Reuters. Information supplied by Thomson Reuters.

9. MULTIPLIER MODELS AND RELATIONSHIP AMONG PRICE MULTIPLES, PRESENT VALUE MODELS, AND FUNDAMENTALS

LEARNING OUTCOMES

The candidate should be able to:

- ☐ explain the rationale for using price multiples to value equity, how the price to earnings multiple relates to fundamentals, and the use of multiples based on comparables
- ☐ calculate and interpret the following multiples: price to earnings, price to an estimate of operating cash flow, price to sales, and price to book value
- ☐ explain advantages and disadvantages of each category of valuation model

The term **price multiple** refers to a ratio that compares the share price with some sort of monetary flow or value to allow evaluation of the relative worth of a company's stock. Some practitioners use price ratios as a screening mechanism. If the ratio falls below a specified value, the shares are identified as candidates for purchase, and if the ratio exceeds a specified value, the shares are identified as candidates for sale. Many practitioners use ratios when examining a group or sector of stocks and consider the shares for which the ratio is relatively low to be attractively valued securities.

Price multiples that are used by security analysts include the following:

- Price-to-earnings ratio (P/E). This measure is the ratio of the stock price to earnings per share. P/E is arguably the price multiple most frequently cited by the media and used by analysts and investors ([Block 1999](#)). The seminal works of [McWilliams \(1966\)](#), [Miller and Widmann \(1966\)](#), [Nicholson \(1968\)](#), [Dreman \(1977\)](#), and [Basu \(1977\)](#) presented evidence of a return advantage to low-P/E stocks.
- Price-to-book ratio (P/B). The ratio of the stock price to book value per share. Considerable evidence suggests that P/B multiples are inversely related to future rates of return ([Fama and French 1995](#)).
- Price-to-sales ratio (P/S). This measure is the ratio of stock price to sales per share. [O'Shaughnessy \(2005\)](#) provided evidence that a low P/S multiple is the most useful multiple for predicting future returns.
- Price-to-cash-flow ratio (P/CF). This measure is the ratio of stock price to some per-share measure of cash flow. The measures of cash flow include free cash flow (FCF) and operating cash flow (OCF).

A common criticism of all of these multiples is that they do not consider the future. This criticism is true if the multiple is calculated from trailing or current values of the divisor. Practitioners seek to counter this criticism by a variety of techniques, including forecasting fundamental values (the divisors) one or more years into the future. The resulting forward (leading or prospective) price multiples may differ markedly from the trailing price multiples. In the absence of an explicit forecast of fundamental values, the analyst is making an implicit forecast of the future when implementing such models. The choice of price multiple—trailing or forward—should be used consistently for companies being compared.

Besides the traditional price multiples used in valuation, just presented, analysts need to know how to calculate and interpret other ratios. Such ratios include those used to analyze business performance and financial condition based on data reported in financial statements. In addition, many industries have specialized measures of business performance that analysts covering those industries should be familiar with. In analyzing cable television companies, for example, the ratio of total market value of the company to the total number of subscribers is commonly used. Another common measure is revenue per subscriber. In the oil industry, a commonly cited ratio is proved reserves per common share. Industry-

specific or sector-specific ratios such as these can be used to understand the key business variables in an industry or sector as well as to highlight attractively valued securities.

9.1. Relationships among Price Multiples, Present Value Models, and Fundamentals

Price multiples are frequently used independently of present value models. One price multiple valuation approach, the method of comparables, does not involve cash flow forecasts or discounting to present value. A price multiple is often related to fundamentals through a discounted cash flow model, however, such as the Gordon growth model. Understanding such connections can deepen the analyst’s appreciation of the factors that affect the value of a multiple and often can help explain reasons for differences in multiples that do not involve mispricing. The expressions that are developed can be interpreted as the *justified value* of a multiple—that is, the value justified by (based on) fundamentals or a set of cash flow predictions. These expressions are an alternative way of presenting intrinsic-value estimates.

As an example, using the Gordon growth model identified previously in [Equation 9](#) and assuming that price equals intrinsic value ($P_0 = V_0$), we can restate [Equation 9](#) as follows:

Equation 14

$$P_0 = \frac{D_1}{r-g}$$

To arrive at the model for the justified forward P/E given in [Equation 15](#), we divide both sides of [Equation 14](#) by a forecast for next year’s earnings, E_1 . In [Equation 15](#), the dividend payout ratio, p , is the ratio of dividends to earnings:

Equation 15

$$\frac{P_0}{E_1} = \frac{D_1/E_1}{r-g} = \frac{p}{r-g}$$

[Equation 15](#) indicates that the P/E is inversely related to the required rate of return and positively related to the growth rate; that is, as the required rate of return increases, the P/E declines, and as the growth rate increases, the P/E increases. The P/E and the payout ratio appear to be positively related. This relationship may not be true, however, because a higher payout ratio may imply a slower growth rate as a result of the company retaining a lower proportion of earnings for reinvestment. This phenomenon is referred to as the dividend displacement of earnings.

EXAMPLE 11

Value Estimate Based on Fundamentals

Petroleo Brasileiro SA, commonly known as Petrobras, was once labeled “the most expensive oil company” by Bloomberg.com. Data for Petrobras and the oil industry, including the trailing twelve-month (TTM) P/E and payout ratios, appear below.

	Petrobras	Industry
P/E ratio (TTM)	39.61	13.0
Return on assets (TTM) (%)	3.0	3.2
EPS 3-year growth rate (%)	NM	66.00
EPS (MRQ) vs. Qtr. 1 yr. ago (% change)	138.96	−12.0

Note: NM stands for non-quantifiable. Petrobras EPS has decreased from a loss of BRL 1.14 per share to a profit of BRL 0.16 per share in the most recent period. MRQ stands for “most recent quarter.”

Source: Reuters.

Explain how the information shown supports a higher P/E for Petrobras than for the industry.

Solution:

The data support a higher P/E for Petrobras because its (MRQ) EPS growth rate exceed those of the industry. [Equation 15](#) implies a positive relationship between the payout ratio and the P/E multiple. Petrobras has had a negative EPS for the period 2014 to 2017, and has paid no dividend during that period. A higher payout ratio supports a higher P/E. Furthermore, to the extent that higher EPS growth implies a high growth rate in dividends, the high EPS growth rate supports a high P/E. The higher P/E ratio is due to an improvement in the underlying financial performance of the company and the expected higher growth potential of the stock compared to the median firms in the industry.

EXAMPLE 12

Determining Justified Forward P/E

Heinrich Gladisch, CFA, is estimating the justified forward P/E for Nestlé, one of the world's leading nutrition and health companies. Gladisch notes that sales for 2017 were SFr89.78 billion (US\$90.3 billion) and that net income was SFr7.18 billion (US\$7.25 billion). He organizes the data for EPS, dividends per share, and the dividend payout ratio for the years 2013–2017 in the following table:

	2013	2014	2015	2016	2017
Earnings per share	SFr3.24	SFr4.54	SFr2.90	SFr2.76	SFr2.32
Year over year % change		44.6%	–36.1%	–4.8%	–15.9%
Dividend per share	SFr2.15	SFr2.2	SFr2.25	SFr2.3	SFr2.35
Year over year % change		2.3%	2.3%	2.2%	2.2%
Dividend payout ratio	68.5%	48.5%	77.6%	83.3%	99.2%

Gladisch calculates that ROE averaged 15.5 percent in the period 2013–2017 but was below that level at 11.7 percent in 2017. In that year, however, Nestlé's reported net income included a large nonrecurring component. The company reported 2017 "underlying earnings," which it defined as net income "from continuing operations before impairments, restructuring costs, results on disposals and significant one-off items," to be SFr2.93, giving an adjusted 14.8% ROE. Predicting increasing improvement in Nestlé's profit margins from growth in its product markets, Gladisch estimates a long-run ROE of 21.5 percent.

Gladisch decides that the dividend payout ratios of the 2013–2016 period—averaging 67.7 percent—are more representative of Nestlé's future payout ratio than is the high 2017 dividend payout ratio (when based on reported earnings). The dividend payout ratio in 2017 was higher because management apparently based the 2017 dividend on the components of net income that were expected to continue into the future. But basing a dividend on net income including non-recurring items creates the potential need to increase dividends in the future. Rounding up the 2013–2017 average, Gladisch settles on an estimate of 68 percent for the dividend payout ratio for use in calculating a justified forward P/E using [Equation 15](#).

Gladisch's firm estimates that the required rate of return for Nestlé's shares is 9 percent per year. Gladisch also finds the following data at the opposite ends of the spectrum of external research analyst forecasts:

	2018E	2019E
Most optimistic analyst forecast:		
EPS	SFr3.99	SFr4.33
Year over year % change	71.9%	8.5%
P/E (based on a target price of SFr105)	26.3	24.2
Least optimistic analyst forecast:		

	2018E	2019E
EPS	SFr3.52	SFr3.59
Year over year % change	51.7%	2.0%
P/E (based on a target price of SFr68)	19.3	18.9

1. Based only on information and estimates developed by Gladisch and his firm, estimate Nestlé's justified forward P/E.

Solution to 1:

The estimate of the justified forward P/E is 32.38. The dividend growth rate can be estimated by using [Equation 10](#) as $(1 - \text{Dividend payout ratio}) \times \text{ROE} = (1 - 0.68) \times 0.215 = 0.069$, or 6.9 percent. Therefore,

$$\frac{P_0}{E_1} = \frac{\text{Payout}}{r-g} = \frac{0.68}{0.09-0.069} = 32.38$$

2. Compare and contrast the justified forward P/E estimate from Question 1 to the estimates from each end of the spectrum of external research analysts forecasts.

Solution to 2:

The estimated justified forward P/E of 32.38 is higher than the justified 2018 P/E estimates of 26.3 and 19.3 of the two analysts. Using a required rate of return of 9.5 percent rather than 9 percent results in a justified forward P/E estimate of $26.2 = 0.68 / (0.095 - 0.069)$. Using an ROE of 16.5 percent (the average ROE of the 2013–2016 period) rather than 21.5 percent results in a justified forward P/E estimate of $18.4 = 0.68 / [0.09 - (0.32)(0.165)] = 0.68 / (0.09 - 0.053)$. The justified forward P/E is very sensitive to changes in the inputs.

Justified forward P/E estimates can be sensitive to small changes in assumptions. Therefore, analysts can benefit from carrying out a sensitivity analysis, as shown in [Exhibit 6](#), which is based on [Example 12](#). [Exhibit 6](#) shows how the justified forward P/E varies with changes in the estimates for the dividend payout ratio (columns) and return on equity. The dividend growth rate (rows) changes because of changes in the retention rate $(1 - \text{Payout rate})$ and ROE. Recall $g = \text{ROE times retention rate}$.

Exhibit 6: Estimates for Nestlé's Justified Forward P/E (Required Rate of Return = 9 Percent)

Constant Dividend Growth Rate (%)	Dividend Payout Ratio				
	55%	60%	65%	70%	75%
4.0	11.0	12.0	13.0	14.0	15.0
4.5	12.2	13.3	14.4	15.6	16.7
5.0	13.8	15.0	16.3	17.5	18.8
5.5	15.7	17.1	18.6	20.0	21.4
6.0	18.3	20.0	21.7	23.3	25.0
6.5	22.0	24.0	26.0	28.0	30.0
7.0	27.5	30.0	32.5	35.0	37.5
7.5	36.7	40.0	43.3	46.7	50.0

10. METHOD OF COMPARABLES AND VALUATION BASED ON PRICE MULTIPLES

LEARNING OUTCOMES

The candidate should be able to:

- ☐ explain the rationale for using price multiples to value equity, how the price to earnings multiple relates to fundamentals, and the use of multiples based on comparables
- ☐ calculate and interpret the following multiples: price to earnings, price to an estimate of operating cash flow, price to sales, and price to book value
- ☐ explain advantages and disadvantages of each category of valuation model

The method of comparables is the most widely used approach for analysts reporting valuation judgments on the basis of price multiples. This method essentially compares relative values estimated using multiples or the relative values of multiples. The economic rationale underlying the method of comparables is the **law of one price**: Identical assets should sell for the same price. The methodology involves using a price multiple to evaluate whether an asset is fairly valued, undervalued, or overvalued in relation to a benchmark value of the multiple. Choices for the benchmark multiple include the multiple of a closely matched individual stock or the average or median value of the multiple for the stock's industry. Some analysts perform trend or time-series analyses and use past or average values of a price multiple as a benchmark.

Identifying individual companies or even an industry as the “comparable” may present a challenge. Many large corporations operate in several lines of business, so the scale and scope of their operations can vary significantly. When identifying comparables (sometimes referred to as “comps”), the analyst should be careful to identify companies that are most similar according to a number of dimensions. These dimensions include (but are not limited to) overall size, product lines, and growth rate. The type of analysis shown in Section 5.1 relating multiples to fundamentals is a productive way to identify the fundamental variables that should be taken into account in identifying comparables.

EXAMPLE 13

Method of Comparables (1)

As noted previously, P/E is a price multiple frequently used by analysts. Using P/E in the method of comparables can be problematic, however, as a result of business cycle effects on EPS. An alternative valuation tool that is useful during periods of economic slowdown or extraordinary growth is the P/S multiple. Although sales will decline during a recession and increase during a period of economic growth, the change in sales will be less than the change in earnings in percentage terms because earnings are heavily influenced by fixed operating and financing costs (operating and financial leverage).

The following data provide the P/S for most of the major automobile manufacturers as at December 2017:

Company	P/S
Peugeot	0.28
Ford Motor	0.33
General Motors	0.36
Nissan Motor	0.38
Honda Motor	0.46
Tata Motors	0.49

Company	P/S
Daimler	0.55
BMW	0.57
Toyota Motor	0.80

Sources: Morningstar and company websites.

Based on the data presented, which stock appears to be undervalued when compared with the others?

Solution:

The P/S analysis suggests that Peugeot shares offer the best value. An analyst must be alert for a range of potential explanations of apparently low or high multiples when performing comparables analysis, rather than just assuming a relative mispricing.

EXAMPLE 14

Method of Comparables (2)

Incorporated in the Netherlands, Airbus is active in the aerospace and defense industry. It is a dominant aerospace company in Europe. Its largest business, Airbus Commercial Aircraft, is a manufacturing company with bases in several European countries and accounts for the majority of Airbus SE profits. Airbus and its primary competitor, Boeing, control most of the global commercial airplane industry.

Comparisons are frequently made between Airbus and Boeing. As noted in [Exhibit 7](#), the companies are broadly similar in size as measured by total revenues. Converting total forecast revenues from euros to US dollars using the average exchange rate for 2017 of US\$1.13/€ results in a value of \$75.5 billion for Airbus's total revenues. Thus, total revenues for Boeing are expected to be about a fifth higher than those for Boeing.

The companies do differ, however, in several important areas. Airbus derives a greater share of its revenue from commercial aircraft production than does Boeing, and the order backlog for Airbus is much higher than that for Boeing. Converting the Airbus order backlog from euros to US dollars using the quarter-end rate for September 2017 of \$1.1813/€ results in a value of \$1.12 billion for Airbus's order backlog. Thus, the order backlog for Airbus is more than twice as high as the backlog for Boeing.⁵

Exhibit 7: Data for EADS and Boeing

	Airbus	Boeing
Total revenues (billions, 2017)	€66.8	\$92.2
Annual revenue growth (2015–2017 average)	1.8%	–2.1%
Percent of revenues from commercial aircraft	75%	69%
Order backlog (billions)	€945	\$474
Share price, 12/Dec/17	€86.96	\$283.73
EPS (basic)	€3.33	\$10.18
DPS	€1.48	\$5.7
Dividend payout ratio	44%	56%
P/E ratio	26.1	27.9

Note: 2017 forecast data and YTD average exchange rate as of 12 December 2017. Order backlog as of 30 September 2017.

Sources: Company websites: www.airbus.com and www.boeing.com, *Financial Times*.

What data shown in [Exhibit 7](#) support a higher P/E for Boeing than for Airbus?

Solution:

Recall from [Equation 14](#) and the discussion that followed it that P/E is directly related to the payout ratio and the dividend growth rate. The P/E is inversely related to the required rate of return. The only data presented in [Exhibit 7](#) that support a higher P/E for Boeing is the company's higher dividend payout ratio (expected at 56 percent versus 44 percent for Airbus).

The following implicitly supports a higher P/E for Airbus: Airbus has higher revenue growth (as reported for 2016 and expected for 2017) and a higher backlog of orders, suggesting that it may have a higher future growth rate.

EXAMPLE 15

Method of Comparables (3)

Canon Inc. is a leading worldwide manufacturer of business machines, cameras, and optical products. Canon was founded in 1937 as a camera manufacturer and is incorporated in Tokyo. The corporate philosophy of Canon is *kyosei* or "living and working together for the common good." The following data can be used to determine a P/E for Canon over the time period 2013–2017. Analyze the P/E of Canon over time and discuss the valuation of Canon.

Year	Price (a)	EPS (b)	P/E (a) ÷ (b)
2013	¥3,330	¥200.8	16.6
2014	¥3,840.5	¥229.0	16.8
2015	¥3,675	¥201.7	18.2
2016	¥3,295	¥138	23.9
2017	¥4,200	¥222.88	18.8

Sources: EPS, year-end prices, and P/E data are from Capital IQ and the *Financial Times*.

Solution:

Trend analysis of Canon's P/E reveals a peak of 23.9 at the end of 2016. The 2013 P/E of 16.6 is the lowest of the five years reported. This finding suggests that Canon's share price may be fairly price as of year-end 2017. A bearish case for Canon's stock can be made if an analyst believes that P/E will return to its historical low (16.6 over this five-year period) or be lower. Such a bearish prediction requires that a decrease in P/E not be offset by an increase in EPS. A bullish case can be made if the analyst believes the stock deserves re-rating and an even higher than trend P/E.

10.1. Illustration of a Valuation Based on Price Multiples

Telefónica S.A., a world leader in the telecommunication sector, provides communication, information, and entertainment products and services in Europe, Africa, and Latin America. It has operated in its home

country of Spain since 1924, but as of 2017, more than 75 percent of its business was outside its home market.

Deutsche Telekom AG provides network access, communication services, and value-added services via fixed and mobile networks. It generates more than half of its revenues outside its home country, Germany.

[Exhibit 8](#) provides comparable data for these two communication giants for 2015–2017.

Exhibit 8: Data for Telefónica and Deutsche Telekom

	Telefónica			Deutsche Telekom		
	2017	2016	2015	2017	2016	2015
(1) Total assets (€ billions)	115.0	123.6	120.3	141.3	148.5	143.9
<i>Asset growth</i>	−6.9%	2.7%	—	−4.9%	3.2%	—
(2) Net revenues (€ billions)	52.0	52.0	54.9	77.3	75.2	71.3
<i>Revenue growth</i>	0%	−5.2%	—	2.8%	5.5%	—
(3) Net cash flow from operating activities (€ billions)	13.8	13.3	13.6	17.2	15.5	15.0
<i>Cash flow growth</i>	3.4%	−2.0%	—	11.0%	3.3%	—
(4) Book value of common shareholders' equity (€ billions)	16.9	18.2	15.8	42.5	38.8	38.2
<i>Debt ratio:</i> $1 - [(4) \div (1)]$	85.3%	85.3%	86.9%	70.0%	73.9%	73.5%
(5) Net profit (€ billions)	2.9	2.1	0.4	3.5	2.7	3.3
<i>Earnings growth</i>	38.1%	425.0%	—	29.6%	−18.2%	—
(6) Weighted average number of shares outstanding (millions)	5,122.9	4,896.6	4,833.6	4,740.2	4,654.9	4,584.8
(7) Price per share (€)	7.93	9.52	9.13	13.42	16.21	15.44
<i>Price-to-revenue ratio (P/R):</i> $(7) \div [(2) \div (6)]$	0.8	0.9	0.8	0.8	1.0	1.0
<i>P/CF:</i> $(7) \div [(3) \div (6)]$	2.9	3.5	3.2	3.7	4.9	4.7
<i>P/B:</i> $(7) \div [(4) \div (6)]$	2.4	2.6	2.8	1.5	1.9	1.9
<i>P/E:</i> $(7) \div [(5) \div (6)]$	14.0	22.2	110.3	18.2	27.9	21.5

Sources: Company websites: www.telefonica.es and www.deutschetelekom.com.

Time-series analysis of all price multiples in [Exhibit 8](#) suggests that both companies are currently attractively valued. For example, the 2017 price-to-revenue ratio (P/R) of 0.78 for Telefónica is below the 2015–2017 average for this ratio of approximately 0.83. The 2017 P/CF of 3.7 for Deutsche Telekom is below the 2015–2017 average for this ratio of approximately 4.4.

A comparative analysis produces somewhat mixed results. The 2017 values for Deutsche Telekom for the P/R, P/CF, P/E multiples are higher than those for Telefónica. This result suggests that Telefónica is attractively valued when compared with Deutsche Telekom. The 2017 P/B for Telefónica, however, is higher than for Deutsche Telekom.

An analyst investigating these contradictory results would look for information not reported in [Exhibit 8](#). For example, the earnings before interest, taxes, depreciation, and amortization (EBITDA) for Telefónica was €16.4 billion in 2017. The EBITDA value for Deutsche Telekom was €20.7 billion in 2017. The 2017 price-to-EBITDA ratio for Telefónica is $[(7.93 \times 5,123)/16,400]$ or $[7.92/(16,400/5,123)] = 2.5$, whereas the 2017 price-to-EBITDA ratio for Deutsche Telekom is 3.1. Thus, the higher P/E for Deutsche Telekom can-not be explained by higher depreciation charges, higher interest costs, and/or a greater tax burden, but appears to be due to a better quality of earnings.

In summary, the major advantage of using price multiples is that they allow for relative comparisons, both cross-sectional (versus the market or another comparable) and in time series. The approach can be especially beneficial for analysts who are assigned to a particular industry or sector and need to identify the expected best performing stocks within that sector. Price multiples are popular with investors because the multiples can be calculated easily and many multiples are readily available from financial websites and newspapers.

Caution is necessary. A stock may be relatively undervalued when compared with its benchmarks but overvalued when compared with an estimate of intrinsic value as determined by one of the discounted cash flow methodologies. Furthermore, differences in reporting rules among different markets and in chosen accounting methods can result in revenues, earnings, book values, and cash flows that are not easily comparable. These differences can, in turn, result in multiples that are not easily comparable. Finally, the multiples for cyclical companies may be highly influenced by current economic conditions.

11. ENTERPRISE VALUE

LEARNING OUTCOMES

The candidate should be able to:

- ☐ describe enterprise value multiples and their use in estimating equity value
- ☐ explain advantages and disadvantages of each category of valuation model

An alternative to estimating the value of equity is to estimate the value of the enterprise. Enterprise value is most frequently determined as market capitalization plus market value of preferred stock plus market value of debt minus cash and investments (cash equivalents and short-term investments). Enterprise value is often viewed as the cost of a takeover: In the event of a buyout, the acquiring company assumes the acquired company's debt but also receives its cash. Enterprise value is most useful when comparing companies with significant differences in capital structure.

Enterprise value (EV) multiples are widely used in Europe, with EV/EBITDA arguably the most common. EBITDA is a proxy for operating cash flow because it excludes depreciation and amortization. EBITDA may include other non-cash expenses, however, and non-cash revenues. EBITDA can be viewed as a source of funds to pay interest, dividends, and taxes. Because EBITDA is calculated prior to payment to any of the company's financial stakeholders, using it to estimate enterprise value is logically appropriate.

Using enterprise value instead of market capitalization to determine a multiple can be useful to analysts. Even where the P/E is problematic because of negative earnings, the EV/EBITDA multiple can generally be computed because EBITDA is usually positive. An alternative to using EBITDA in EV multiples is to use operating income.

In practice, analysts may have difficulty accurately assessing enterprise value if they do not have access to market quotations for the company's debt. When current market quotations are not available, bond values may be estimated from current quotations for bonds with similar maturity, sector, and credit characteristics. Substituting the book value of debt for the market value of debt provides only a rough estimate of the debt's market value. This is because market interest rates change and investors' perception of the issuer's credit risk may have changed since the debt was issued.

Estimating the Market Value of Debt and Enterprise Value

Cameco Corporation is one of the world's largest uranium producers; it accounts for 16 percent of world production from its mines in Canada and the United States. Cameco estimates it has about 458 million kilograms of proven and probable reserves and holds premier land positions in the world's most promising areas for new uranium discoveries in Canada and Australia. Cameco is also a leading provider of processing services required to produce fuel for nuclear power plants. It generates 1,000 megawatts of electricity through a partnership in North America's largest nuclear generating station located in Ontario, Canada.

For simplicity of exposition in this example, we will present share counts in thousands and all dollar amounts in thousands of Canadian dollars. In 2017, Cameco had 395,793 shares outstanding. Its 2017 year-end share price was \$14.11. Therefore, Cameco's 2017 year-end market capitalization was \$5,584,640.

In its 2017 Annual Report (available at www.cameco.com), Cameco reported total debt and other liabilities of \$2,919,100. The company presented the following schedule for long-term debt payments:

Year	Payment
2018	\$69,000
2019 and 2020	610,000
2021 and 2022	482,000
Thereafter	744,000
Total	\$1,905,000

Cameco's longest maturity debt matures in 2042. We will assume that the amounts paid in 2019 and 2020, and in 2021 and 2022, will be paid equally during the two years. The "thereafter" period includes two debenture tranches, the first one maturing in 2024 for a total value of \$620,000 and the second tranche maturing in 2042 for the remaining \$124,000. A yield curve for zero-coupon Canadian government securities was available from the Bank of Canada. The yield-curve data and assumed risk premiums in [Exhibit 9](#) were used to estimate the market value of Cameco's long-term debt:

Exhibit 9: Estimated Market Value

Year	Yield on Zero-Coupon Government Security (%)	Assumed Risk Premium (%)	Discount Rate (%)	Book Value	Market Value
2018	0.89	0.50	1.39	\$69,000	\$68,054
2019	1.11	1.00	2.11	\$305,000	\$292,525
2020	1.39	1.50	2.89	\$305,000	\$280,014
2021	1.65	2.00	3.65	\$241,000	\$208,804
2022	1.88	2.50	4.38	\$241,000	\$194,505
2023	2.10	3.00	5.10	\$0	\$0
2024	2.30	3.50	5.80	\$620,000	\$417,823
2025	2.50	4.00	6.50	\$0	\$0
...	...	...	...	\$...	\$...
2042	2.92	5.00	7.92	\$124,000	\$18,445
				\$1,905,000	\$1,480,170

Note from [Exhibit 9](#) that the book value of long-term debt is \$1,905,000 and its estimated market value is \$1,480,170. The book value of total debt and liabilities of \$2,919,100 minus the book value of long-term debt of \$1,905,000 is \$1,014,100. If we assume that the market value of that remaining debt is equal to its book value of \$1,014,100 an estimate of the market value of total debt and liabilities is that amount plus the estimated market value of long-term debt of \$1,480,170 or \$2,494,270.

At the end of 2017, Cameco had cash and equivalents of \$591,600. Enterprise value can be estimated as the \$5,584,640 market value of stock plus the \$2,494,270 market value of debt minus the \$591,600 cash and equivalents, or \$7,487,310. Cameco's 2017 EBITDA was \$606,000; an estimate of EV/EBITDA is, therefore, \$7,487,310 divided by \$606,000, or 12.4.

EXAMPLE 17

EV/Operating Income

[Exhibit 10](#) presents data for twelve major mining companies. Based only on the information in [Exhibit 10](#), which two mining companies seem to be the *most* undervalued?

Exhibit 10: Data for Twelve Major Mining Companies

Company	EV (US\$ millions)	Operating Income (OI) (US\$ millions)	EV/OI
BHP Billiton	119,712.3	11,753	10.19
Rio Tinto	93,856.1	6,471	14.5
Vale	82,051.2	6,366	12.89
Glencore	80,772.0	-549	-147.13
Southern Copper	37,817.0	1,564	24.18
Freeport-McMoRan	33,452.0	-2,766	-12.09
Anglo American	32,870.3	2,562	12.83
Norilsk Nickel	22,483.0	3,377	6.66
Coal India	21,652.1	1,382	15.67
Barrick Gold	21,549.8	2,424	8.89
Newmont Mining	20,683.0	-65	-318.20
Goldcorp	12,986.7	369	35.19

Source: www.miningfeeds.com, Morningstar.

Solution:

Norilsk Nickel and Barrick Gold have the lowest EV/OI and thus appear to be the *most* undervalued or favorably priced on the basis of the EV/OI. Note the negative ratio for Glencore, Freeport-McMoRan, and Newmont Mining. Negative ratios are difficult to interpret, so other means are used to evaluate companies with negative ratios.

12. ASSET-BASED VALUATION

LEARNING OUTCOMES

The candidate should be able to:

- ☐ **describe asset-based valuation models and their use in estimating equity value**
- ☐ **explain advantages and disadvantages of each category of valuation model**

An asset-based valuation of a company uses estimates of the market or fair value of the company's assets and liabilities. Thus, asset-based valuations work well for companies that do not have a high proportion of intangible or "off the books" assets and that do have a high proportion of current assets and current liabilities. The analyst may be able to value these companies' assets and liabilities in a reasonable fashion by starting with balance sheet items. For most companies, however, balance sheet values are different from market (fair) values, and the market (fair) values can be difficult to determine.

Asset-based valuation models are frequently used together with multiplier models to value private companies. As public companies increase reporting or disclosure of fair values, asset-based valuation may be increasingly used to supplement present value and multiplier models of valuation. Important facts that the practitioner should realize are as follows:

- Companies with assets that do not have easily determinable market (fair) values—such as those with significant property, plant, and equipment—are very difficult to analyze using asset valuation methods.
- Asset and liability fair values can be very different from the values at which they are carried on the balance sheet of a company.
- Some assets that are "intangible" are shown on the books of the company. Other intangible assets, such as the value from synergies or the value of a good business reputation, may not be shown on the books. Because asset-based valuation may not consider some intangibles, it can give a "floor" value for a situation involving a significant amount of intangibles. When a company has significant intangibles, the analyst should prefer a forward-looking cash flow valuation.
- Asset values may be more difficult to estimate in a hyper-inflationary environment.

We begin by discussing asset-based valuation for hypothetical nonpublic companies and then move on to a public company example. Analysts should consider the difficulties and rewards of using asset-based valuation for companies that are suited to this measure. Owners of small privately held businesses are familiar with valuations arrived at by valuing the assets of the company and then subtracting any relevant liabilities.

EXAMPLE 18

An Asset-Based Valuation of a Family-Owned Laundry

A family owns a laundry and the real estate on which the laundry stands. The real estate is collateral for an outstanding loan of \$100,000. How can asset-based valuation be used to value this business?

Solution:

The analyst should get at least two market appraisals for the real estate (building and land) and estimate the cost to extinguish the \$100,000 loan. This information would provide estimated values for everything except the laundry as a going concern. That is, the analyst has market values for the building and land and the loan but needs to value the laundry business. The analyst can value the assets of the laundry: the equipment and inventory. The equipment can be valued at depreciated value, inflation-adjusted depreciated value, or replacement cost. Replacement cost in this case means the amount that would have to be spent to buy equivalent used machines. This amount is the market value of the used machines. The analyst will recognize that any intangible value of the laundry (prime location, clever marketing, etc.) is being excluded, which will result in an inaccurate asset-based valuation.

[Example 18](#) shows some of the subtleties present in applying asset-based valuation to determine company value. It also shows how asset-based valuation does not deal with intangibles. [Example 19](#) emphasizes this point.

EXAMPLE 19

An Asset-Based Valuation of a Restaurant

The business being valued is a restaurant that serves breakfast and lunch. The owner/proprietor wants to sell the business and retire. The restaurant space is rented, not owned. This particular restaurant is hugely popular because of the proprietor's cooking skills and secret recipes. How can the analyst value this business?

Solution:

Because of the intangibles, setting a value on this business is challenging. A multiple of income or revenue might be considered. But even those approaches overlook the fact that the proprietor may not be selling his secret recipes and, furthermore, does not intend to continue cooking. Some (or all) of the intangible assets may vanish when the business is sold. Asset-based valuation for this restaurant would begin with estimating the value of the restaurant equipment and inventory and subtracting the value of any liabilities. This approach will provide only a good baseline, however, for a minimum valuation.

For public companies, the assets will typically be so extensive that a piece-by-piece analysis will be impossible, and the transition from book value to market value is a nontrivial task. The asset-based valuation approach is most applicable when the market value of the corporate assets is readily determinable and the intangible assets, which are typically difficult to value, are a relatively small proportion of corporate assets. Asset-based valuation has also been applied to financial companies, natural resource companies, and formerly going-concerns that are being liquidated. Even for other types of companies, however, asset-based valuation of tangible assets may provide a baseline for a minimal valuation.

EXAMPLE 20

An Asset-Based Valuation of an Airline

Consider the value of an airline company that has few routes, high labor and other operating costs, has stopped paying dividends, and is losing millions of dollars each year. Using most valuation approaches, the company will have a negative value. Why might an asset-based valuation approach be appropriate for use by one of the company's competitors that is considering acquisition of this airline?

Solution:

The airline's routes, landing rights, leases of airport facilities, and ground equipment and airplanes may have substantial value to a competitor. An asset-based approach to valuing this company would value the company's assets separately and aside from the money-losing business in which they are presently being utilized.

Analysts recognizing the uncertainties related to model appropriateness and the inputs to the models frequently use more than one model or type of model in valuation to increase their confidence in their

estimates of intrinsic value. The choice of models will depend on the availability of information to put into the models. [Example 21](#) illustrates the use of three valuation methods.

EXAMPLE 21

A Simple Example of the Use of Three Major Equity Valuation Models

Company data for dividend per share (DPS), earnings per share (EPS), share price, and price-to-earnings ratio (P/E) for the most recent five years are presented in [Exhibit 11](#). In addition, estimates (indicated by an “E” after the amount) of DPS and EPS for the next five years are shown. The valuation date is at the end of Year 5. The company has 1,000 shares outstanding.

Exhibit 11: Company DPS, EPS, Share Price, and P/E Data

Year	DPS	EPS	Share Price	TTM P/E
10	\$3.10E	\$5.20E	—	—
9	\$2.91E	\$4.85E	—	—
8	\$2.79E	\$4.65E	—	—
7	\$2.65E	\$4.37E	—	—
6	\$2.55E	\$4.30E	—	—
5	\$2.43	\$4.00	\$50.80	12.7
4	\$2.32	\$3.90	\$51.48	13.2
3	\$2.19	\$3.65	\$59.86	16.4
2	\$2.14	\$3.60	\$54.72	15.2
1	\$2.00	\$3.30	\$46.20	14.0

The company’s balance sheet at the end of Year 5 is given in [Exhibit 12](#).

Exhibit 12: Balance Sheet as of End of Year 5

Cash	\$ 5,000
Accounts receivable	15,000
Inventories	30,000
Net fixed assets	50,000
Total assets	\$100,000
Accounts payable	\$ 3,000
Notes payable	17,000
Term loans	25,000
Common shareholders’ equity	55,000
Total liabilities and equity	\$100,000

1. Using a Gordon growth model, estimate intrinsic value. Use a discount rate of 10 percent and an estimate of growth based on growth in dividends over the next five years.

Solution to 1:

$$D_5 (1 + g)^5 = D_{10} 2.43(1 + g)^5 = 3.10$$

$$g \approx 5.0\%$$

$$\text{Estimate of value} = V_5 = 2.55 / (0.10 - 0.05) = \$51.00$$

2. Using a multiplier approach, estimate intrinsic value. Assume that a reasonable estimate of P/E is the average trailing twelve-month (TTM) P/E ratio over Years 1 through 4.

Solution to 2:

$$\text{Average P/E} = (14.0 + 15.2 + 16.4 + 13.2) / 4 = 14.7$$

$$\text{Estimate of value} = \$4.00 \times 14.7 = \$58.80$$

3. Using an asset-based valuation approach, estimate value per share from adjusted book values. Assume that the market values of accounts receivable and inventories are as reported, the market value of net fixed assets is 110 percent of reported book value, and the reported book values of liabilities reflect their market values.

Solution to 3:

$$\text{Market value of assets} = 5,000 + 15,000 + 30,000 + 1.1(50,000) = \$105,000$$

$$\text{Market value of liabilities} = \$3,000 + 17,000 + 25,000 = \$45,000$$

$$\text{Adjusted book value} = \$105,000 - 45,000 = \$60,000$$

$$\text{Estimated value (adjusted book value per share)} = \$60,000 \div 1,000 \text{ shares} = \$60.00$$

Given the current share price of \$50.80, the multiplier and the asset-based valuation approaches indicate that the stock is undervalued. Given the intrinsic value estimated using the Gordon growth model, the analyst is likely to conclude that the stock is fairly priced. The analyst might examine the assumptions in the multiplier and the asset-based valuation approaches to determine why their estimated values differ from the estimated value provided by the Gordon growth model and the market price.

SUMMARY

The equity valuation models used to estimate intrinsic value—present value models, multiplier models, and asset-based valuation—are widely used and serve an important purpose. The valuation models presented here are a foundation on which to base analysis and research but must be applied wisely. Valuation is not simply a numerical analysis. The choice of model and the derivation of inputs require skill and judgment.

When valuing a company or group of companies, the analyst wants to choose a valuation model that is appropriate for the information available to be used as inputs. The available data will, in most instances, restrict the choice of model and influence the way it is used. Complex models exist that may improve on the simple valuation models described in this reading; but before using those models and assuming that complexity increases accuracy, the analyst would do well to consider the “law of parsimony.” A model should be kept as simple as possible in light of the available inputs. Valuation is a fallible discipline, and any method will result in an inaccurate forecast at some time. The goal is to minimize the inaccuracy of the forecast.

Among the points made in this reading are the following:

- An analyst estimating intrinsic value is implicitly questioning the market’s estimate of value.

- If the estimated value exceeds the market price, the analyst infers the security is *undervalued*. If the estimated value equals the market price, the analyst infers the security is *fairly valued*. If the estimated value is less than the market price, the analyst infers the security is *overvalued*. Because of the uncertainties involved in valuation, an analyst may require that value estimates differ markedly from market price before concluding that a misvaluation exists.
- Analysts often use more than one valuation model because of concerns about the applicability of any particular model and the variability in estimates that result from changes in inputs.
- Three major categories of equity valuation models are present value, multiplier, and asset-based valuation models.
- Present value models estimate value as the present value of expected future benefits.
- Multiplier models estimate intrinsic value based on a multiple of some fundamental variable.
- Asset-based valuation models estimate value based on the estimated value of assets and liabilities.
- The choice of model will depend upon the availability of information to input into the model and the analyst's confidence in both the information and the appropriateness of the model.
- Companies distribute cash to shareholders using dividend payments and share repurchases.
- Regular cash dividends are a key input to dividend valuation models.
- Key dates in dividend chronology are the declaration date, ex-dividend date, holder-of-record date, and payment date.
- In the dividend discount model, value is estimated as the present value of expected future dividends.
- In the free cash flow to equity model, value is estimated as the present value of expected future free cash flow to equity.
- The Gordon growth model, a simple DDM, estimates value as $D_1 / (r - g)$.
- The two stage dividend discount model estimates value as the sum of the present values of dividends over a short-term period of high growth and the present value of the terminal value at the end of the period of high growth. The terminal value is estimated using the Gordon growth model.
- The choice of dividend model is based upon the patterns assumed with respect to future dividends.
- Multiplier models typically use multiples of the form: $P / \text{measure of fundamental variable}$ or $EV / \text{measure of fundamental variable}$.
- Multiples can be based upon fundamentals or comparables.
- Asset-based valuations models estimate value of equity as the value of the assets less the value of liabilities.

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PRACTICE PROBLEMS

1. An analyst estimates the intrinsic value of a stock to be in the range of €17.85 to €21.45. The current market price of the stock is €24.35. This stock is *most likely*:
 - A. overvalued.
 - B. undervalued.
 - C. fairly valued.
2. An analyst determines the intrinsic value of an equity security to be equal to \$55. If the current price is \$47, the equity is *most likely*:
 - A. undervalued.
 - B. fairly valued.
 - C. overvalued.
3. In asset-based valuation models, the intrinsic value of a common share of stock is based on the:
 - A. estimated market value of the company's assets.
 - B. estimated market value of the company's assets plus liabilities.
 - C. estimated market value of the company's assets minus liabilities.
4. Which of the following is *most likely* used in a present value model?
 - A. Enterprise value.
 - B. Price to free cash flow.
 - C. Free cash flow to equity.
5. Book value is *least likely* to be considered when using:
 - A. a multiplier model.
 - B. an asset-based valuation model.
 - C. a present value model.
6. An analyst is attempting to calculate the intrinsic value of a company and has gathered the following company data: EBITDA, total market value, and market value of cash and short-term investments, liabilities, and preferred shares. The analyst is *least likely* to use:
 - A. a multiplier model.
 - B. a discounted cash flow model.
 - C. an asset-based valuation model.
7. An analyst who bases the calculation of intrinsic value on dividend-paying capacity rather than expected dividends will *most likely* use the:
 - A. dividend discount model.
 - B. free cash flow to equity model.

- C. cash flow from operations model.
8. An investor expects to purchase shares of common stock today and sell them after two years. The investor has estimated dividends for the next two years, D_1 and D_2 , and the selling price of the stock two years from now, P_2 . According to the dividend discount model, the intrinsic value of the stock today is the present value of:
- A. next year's dividend, D_1 .
 - B. future expected dividends, D_1 and D_2 .
 - C. future expected dividends and price— D_1 , D_2 and P_2 .
9. In the free cash flow to equity (FCFE) model, the intrinsic value of a share of stock is calculated as:
- A. the present value of future expected FCFE.
 - B. the present value of future expected FCFE plus net borrowing.
 - C. the present value of future expected FCFE minus fixed capital investment.
10. With respect to present value models, which of the following statements is *most accurate*?
- A. Present value models can be used only if a stock pays a dividend.
 - B. Present value models can be used only if a stock pays a dividend or is expected to pay a dividend.
 - C. Present value models can be used for stocks that currently pay a dividend, are expected to pay a dividend, or are not expected to pay a dividend.
11. A Canadian life insurance company has an issue of 4.80 percent, \$25 par value, perpetual, non-convertible, non-callable preferred shares outstanding. The required rate of return on similar issues is 4.49 percent. The intrinsic value of a preferred share is *closest to*:
- A. \$25.00.
 - B. \$26.75.
 - C. \$28.50.
12. Two analysts estimating the value of a non-convertible, non-callable, perpetual preferred stock with a constant dividend arrive at different estimated values. The *most likely* reason for the difference is that the analysts used different:
- A. time horizons.
 - B. required rates of return.
 - C. estimated dividend growth rates.
13. The Beasley Corporation has just paid a dividend of \$1.75 per share. If the required rate of return is 12.3 percent per year and dividends are expected to grow indefinitely at a constant rate of 9.2 percent per year, the intrinsic value of Beasley Corporation stock is *closest to*:
- A. \$15.54.
 - B. \$56.45.
 - C. \$61.65.
14. An investor is considering the purchase of a common stock with a \$2.00 annual dividend. The dividend is expected to grow at a rate of 4 percent annually. If the investor's required rate of return is 7 percent, the intrinsic value of the stock is *closest to*:

- A. \$50.00.
- B. \$66.67.
- C. \$69.33.

15. The Gordon growth model can be used to value dividend-paying companies that are:

- A. expected to grow very fast.
- B. in a mature phase of growth.
- C. very sensitive to the business cycle.

16. Which of the following is *most likely* considered a weakness of present value models?

- A. Present value models cannot be used for companies that do not pay dividends.
- B. Small changes in model assumptions and inputs can result in large changes in the computed intrinsic value of the security.
- C. The value of the security depends on the investor's holding period; thus, comparing valuations of different companies for different investors is difficult.

17. An analyst gathers or estimates the following information about a stock:

Current price per share	€22.56
Current annual dividend per share	€1.60
Annual dividend growth rate for Years 1–4	9.00%
Annual dividend growth rate for Years 5+	4.00%
Required rate of return	12%

Based on a dividend discount model, the stock is *most likely*:

- A. undervalued.
- B. fairly valued.
- C. overvalued.

18. An analyst is attempting to value shares of the Dominion Company. The company has just paid a dividend of \$0.58 per share. Dividends are expected to grow by 20 percent next year and 15 percent the year after that. From the third year onward, dividends are expected to grow at 5.6 percent per year indefinitely. If the required rate of return is 8.3 percent, the intrinsic value of the stock is *closest* to:

- A. \$26.00.
- B. \$27.00.
- C. \$28.00.

19. Hideki Corporation has just paid a dividend of ¥450 per share. Annual dividends are expected to grow at the rate of 4 percent per year over the next four years. At the end of four years, shares of Hideki Corporation are expected to sell for ¥9000. If the required rate of return is 12 percent, the intrinsic value of a share of Hideki Corporation is *closest* to:

- A. ¥5,850.
- B. ¥7,220.
- C. ¥7,670.

20. The best model to use when valuing a young dividend-paying company that is just entering the growth phase is *most likely* the:
- A. Gordon growth model.
 - B. two-stage dividend discount model.
 - C. three-stage dividend discount model.
21. An equity analyst has been asked to estimate the intrinsic value of the common stock of Omega Corporation, a leading manufacturer of automobile seats. Omega is in a mature industry, and both its earnings and dividends are expected to grow at a rate of 3 percent annually. Which of the following is *most likely* to be the best model for determining the intrinsic value of an Omega share?
- A. Gordon growth model.
 - B. Free cash flow to equity model.
 - C. Multistage dividend discount model.
22. A price earnings ratio that is derived from the Gordon growth model is inversely related to the:
- A. growth rate.
 - B. dividend payout ratio.
 - C. required rate of return.
23. The primary difference between P/E multiples based on comparables and P/E multiples based on fundamentals is that fundamentals-based P/Es take into account:
- A. future expectations.
 - B. the law of one price.
 - C. historical information.
24. An analyst makes the following statement: "Use of P/E and other multiples for analysis is not effective because the multiples are based on historical data and because not all companies have positive accounting earnings." The analyst's statement is *most likely*:
- A. inaccurate with respect to both historical data and earnings.
 - B. accurate with respect to historical data and inaccurate with respect to earnings.
 - C. inaccurate with respect to historical data and accurate with respect to earnings.
25. An analyst has gathered the following information for the Oudin Corporation:
- Expected earnings per share = €5.70
 - Expected dividends per share = €2.70
 - Dividends are expected to grow at 2.75 percent per year indefinitely
 - The required rate of return is 8.35 percent
- Based on the information provided, the price/earnings multiple for Oudin is *closest* to:
- A. 5.7.
 - B. 8.5.
 - C. 9.4.

26. An analyst has prepared a table of the average trailing twelve-month price-to-earning (P/E), price-to-cash flow (P/CF), and price-to-sales (P/S) for the Tanaka Corporation for the years 2014 to 2017.

Year	P/E	P/CF	P/S
2014	4.9	5.4	1.2
2015	6.1	8.6	1.5
2016	8.3	7.3	1.9
2017	9.2	7.9	2.3

As of the date of the valuation in 2018, the trailing twelve-month P/E, P/CF, and P/S are, respectively, 9.2, 8.0, and 2.5. Based on the information provided, the analyst may reasonably conclude that Tanaka shares are *most likely*:

- A. overvalued.
- B. undervalued.
- C. fairly valued.

27. An analyst gathers the following information about two companies:

	Alpha Corp.	Delta Co.
Current price per share	\$57.32	\$18.93
Last year's EPS	\$3.82	\$1.35
Current year's estimated EPS	\$4.75	\$1.40

Which of the following statements is *most accurate*?

- A. Delta has the higher trailing P/E multiple and lower current estimated P/E multiple.
- B. Alpha has the higher trailing P/E multiple and lower current estimated P/E multiple.
- C. Alpha has the higher trailing P/E multiple and higher current estimated P/E multiple.

28. An analyst gathers the following information about similar companies in the banking sector:

	First Bank	Prime Bank	Pioneer Trust
P/B	1.10	0.60	0.60
P/E	8.40	11.10	8.30

Which of the companies is *most likely* to be undervalued?

- A. First Bank.
- B. Prime Bank.
- C. Pioneer Trust.

29. The market value of equity for a company can be calculated as enterprise value:

- A. minus market value of debt, preferred stock, and short-term investments.
- B. plus market value of debt and preferred stock minus short-term investments.
- C. minus market value of debt and preferred stock plus short-term investments.

30. Which of the following statements regarding the calculation of the enterprise value multiple is *most likely* correct?

- A. Operating income may be used instead of EBITDA.

B. EBITDA may not be used if company earnings are negative.

C. Book value of debt may be used instead of market value of debt.

31. An analyst has determined that the appropriate EV/EBITDA for Rainbow Company is 10.2. The analyst has also collected the following forecasted information for Rainbow Company:

EBITDA = \$22,000,000

Market value of debt = \$56,000,000

Cash = \$1,500,000

The value of equity for Rainbow Company is *closest* to:

A. \$169 million.

B. \$224 million.

C. \$281 million.

32. Enterprise value is most often determined as market capitalization of common equity and preferred stock minus the value of cash equivalents plus the:

A. book value of debt.

B. market value of debt.

C. market value of long-term debt.

33. A disadvantage of the EV method for valuing equity is that the following information may be difficult to obtain:

A. Operating income.

B. Market value of debt.

C. Market value of equity.

34. Asset-based valuation models are best suited to companies where the capital structure does not have a high proportion of:

A. debt.

B. intangible assets.

C. current assets and liabilities.

35. Which of the following is *most likely* a reason for using asset-based valuation?

A. The analyst is valuing a privately held company.

B. The company has a relatively high level of intangible assets.

C. The market values of assets and liabilities are different from the balance sheet values.

36. Which type of equity valuation model is *most likely* to be preferable when one is comparing similar companies?

A. A multiplier model.

B. A present value model.

C. An asset-based valuation model.

SOLUTIONS

1. A is correct. The current market price of the stock exceeds the upper bound of the analyst's estimate of the intrinsic value of the stock.
2. A is correct. The market price is less than the estimated intrinsic, or fundamental, value.
3. C is correct. Asset-based valuation models calculate the intrinsic value of equity by subtracting liabilities from the market value of assets.
4. C is correct. FCFE can be used in a form of present value, or discounted cash flow, model. Both EV and price to free cash flow are forms of multiplier models.
5. C is correct. Multiplier valuation models (in the form of P/B) and asset-based valuation models (in the form of adjustments to book value) use book value, whereas present value models typically discount future expected cash flows.
6. B is correct. To use a discounted cash flow model, the analyst will require FCFE or dividend data. In addition, the analyst will need data to calculate an appropriate discount rate.
7. B is correct. The FCFE model assumes that dividend-paying capacity is reflected in FCFE.
8. C is correct. According to the dividend discount model, the intrinsic value of a stock today is the present value of all future dividends. In this case, the intrinsic value is the present value of D_1 , D_2 , and P_2 . Note that P_2 is the present value at Period 2 of all future dividends from Period 3 to infinity.
9. A is correct. In the FCFE model, the intrinsic value of stock is calculated by discounting expected future FCFE to present value. No further adjustments are required.
10. C is correct. Dividend discount models can be used for a stock that pays a current dividend or a stock that is expected to pay a dividend. FCFE can be used for both of those stocks and for stocks that do not, or are not expected to, pay dividends in the near future. Both of these models are forms of present value models.
11. B is correct. The expected annual dividend is $4.80\% \times \$25 = \1.20 . The value of a preferred share is $\$1.20/0.0449 = \26.73 .
12. B is correct. The required rate of return, r , can vary widely depending on the inputs and is not unique. A preferred stock with a constant dividend would not have a growth rate to estimate, and the investor's time horizon would have no effect on the calculation of intrinsic value.
13. C is correct. $P_0 = D_1/(r - g) = 1.75(1.092)/(0.123 - 0.092) = \61.65 .
14. C is correct. According to the Gordon growth model, $V_0 = D_1/(r - g)$. In this case, $D_1 = \$2.00 \times 1.04 = \2.08 , so $V_0 = \$2.08/(0.07 - 0.04) = \$69.3333 = \$69.33$.
15. B is correct. The Gordon growth model (also known as the constant growth model) can be used to value dividend-paying companies in a mature phase of growth. A stable dividend growth rate is often a plausible assumption for such companies.
16. B is correct. Very small changes in inputs, such as required rate of return or dividend growth rate, can result in large changes to the valuation model output. Some present value models, such as FCFE models, can be used to value companies without dividends. Also, the intrinsic value of a security is independent of the investor's holding period.

- 17.** A is correct. The current price of €22.56 is less than the intrinsic value (V_0) of €24.64; therefore, the stock appears to be currently undervalued. According to the two-stage dividend discount model:

$$V_0 = \sum_{t=1}^n \frac{D_0(1+g_S)^t}{(1+r)^t} + \frac{V_n}{(1+r)^n} \text{ and } V_n = \frac{D_{n+1}}{r-g_L}$$

$$D_{n+1} = D_0(1+g_S)^n(1+g_L)$$

$$D_1 = €1.60 \times 1.09 = €1.744$$

$$D_2 = €1.60 \times (1.09)^2 = €1.901$$

$$D_3 = €1.60 \times (1.09)^3 = €2.072$$

$$D_4 = €1.60 \times (1.09)^4 = €2.259$$

$$D_5 = [€1.60 \times (1.09)^4](1.04) = €2.349$$

$$V_4 = €2.349 / (0.12 - 0.04) = €29.363$$

$$V_0 = \frac{1.744}{(1.12)^1} + \frac{1.901}{(1.12)^2} + \frac{2.072}{(1.12)^3} + \frac{2.259}{(1.12)^4} + \frac{29.363}{(1.12)^4}$$

$$= 1.557 + 1.515 + 1.475 + 1.436 + 18.661$$

$$= €24.64 \text{ (which is greater than the current price of €22.56)}$$

- 18.** C is correct.

$$\begin{aligned} V_0 &= \frac{D_1}{(1+r)} + \frac{D_2}{(1+r)^2} + \frac{P_2}{(1+r)^2} \\ &= \frac{0.70}{(1.083)} + \frac{0.80}{(1.083)^2} + \frac{31.29}{(1.083)^2} \\ &= \$28.01 \end{aligned}$$

Note that $D_1 = 0.58(1.20) = 0.70$, $D_2 = 0.58(1.20)(1.15) = 0.80$, and $P_2 = D_3 / (k - g) = 0.80(1.056) / (0.083 - 0.056) = 31.29$

- 19.** B is correct.

$$\begin{aligned} V_0 &= \frac{D_1}{(1+r)} + \frac{D_2}{(1+r)^2} + \frac{D_3}{(1+r)^3} + \frac{D_4}{(1+r)^4} + \frac{P_4}{(1+r)^4} \\ &= \frac{468}{(1.12)} + \frac{486.72}{(1.12)^2} + \frac{506.19}{(1.12)^3} + \frac{526.44}{(1.12)^4} + \frac{9000}{(1.12)^4} \\ &= 7,220 \end{aligned}$$

- 20.** C is correct. The Gordon growth model is best suited to valuing mature companies. The two-stage model is best for companies that are transitioning from a growth stage to a mature stage. The three-stage model is appropriate for young companies just entering the growth phase.

- 21.** A is correct. The company is a mature company with a steadily growing dividend rate. The two-stage (or multistage) model is unnecessary because the dividend growth rate is expected to remain stable. Although an FCFE model could be used, that model is more often chosen for companies that currently pay no dividends.

- 22.** C is correct. The justified forward P/E is calculated as follows:

$$\frac{P_0}{E_1} = \frac{\frac{D_1}{E_1}}{r-g}$$

P/E is inversely related to the required rate of return, r , and directly related to the growth rate, g , and the dividend payout ratio, D/E .

- 23.** A is correct. Multiples based on comparables are grounded in the law of one price and take into account historical multiple values. In contrast, P/E multiples based on fundamentals can be based on the Gordon growth model, which takes into account future expected dividends.
- 24.** A is correct. The statement is inaccurate in both respects. Although multiples can be calculated from historical data, forecasted values can be used as well. For companies without accounting earnings, several other multiples can be used. These multiples are often specific to a company's industry or sector and include price-to-sales and price-to-cash flow.
- 25.** B is correct.

$$\frac{P_0}{E_1} = \frac{\frac{D_1}{E_1}}{r-g} = \frac{\frac{2.7}{5.7}}{0.0835-0.0275} = 8.5$$

- 26.** A is correct. Tanaka shares are most likely overvalued. As the table below shows, all the 2018 multiples are currently above their 2014–2017 averages.

Year	P/E	P/CF	P/R
2014	4.9	5.4	1.2
2015	6.1	8.6	1.5
2016	8.3	7.3	1.9
2017	9.2	7.9	2.3
Average	7.1	7.3	1.7

- 27.** B is correct. P/E = Current price/EPS, and Estimated P/E = Current price/Estimated EPS.

$$\text{Alpha P/E} = \$57.32/\$3.82 = 15.01$$

$$\text{Alpha estimated P/E} = \$57.32/4.75 = 12.07$$

$$\text{Delta P/E} = \$18.93/\$1.35 = 14.02$$

$$\text{Delta estimated P/E} = \$18.93/\$1.40 = 13.52$$

- 28.** C is correct. Relative to the others, Pioneer Trust has the lowest P/E multiple and the P/B multiple is tied for the lowest with Prime Bank. Given the law of one price, similar companies should trade at similar P/B and P/E levels. Thus, based on the information presented, Pioneer is most likely to be undervalued.
- 29.** C is correct. Enterprise value is calculated as the market value of equity plus the market value of debt and preferred stock minus short-term investments. Therefore, the market value of equity is enterprise value minus the market value of debt and preferred stock plus short-term investments.
- 30.** A is correct. Operating income may be used in place of EBITDA when calculating the enterprise value multiple. EBITDA may be used when company earnings are negative because EBITDA is usually positive. The book value of debt cannot be used in place of market value of debt.
- 31.** A is correct.
- $$\begin{aligned} \text{EV} &= 10.2 \times 22,000,000 = \$224,400,000 \\ \text{Equity value} &= \text{EV} - \text{Debt} + \text{Cash} \\ &= 224,400,000 - 56,000,000 + 1,500,000 \\ &= \$169,900,000 \end{aligned}$$
- 32.** B is correct. The market value of debt must be calculated and taken out of the enterprise value. Enterprise value, sometimes known as the cost of a takeover, is the cost of the purchase of the company, which would include the assumption of the company's debts at market value.
- 33.** B is correct. According to the reading, analysts may have not have access to market quotations for company debt.

- 34.** B is correct. Intangible assets are hard to value. Therefore, asset-based valuation models work best for companies that do not have a high proportion of intangible assets.
- 35.** A is correct. Asset-based valuations are most often used when an analyst is valuing private enterprises. Both B and C are considerations in asset-based valuations but are more likely to be reasons to avoid that valuation model rather than reasons to use it.
- 36.** A is correct. Although all models can be used to compare various companies, multiplier models have the advantage of reducing varying fundamental data points into a format that allows direct comparisons. As long as the analyst applies the data in a consistent manner for all the companies, this approach provides useful comparative data.

NOTES

¹Another type of dividend is a liquidating dividend, which is a return of capital rather than a distribution from earnings or retained earnings. Liquidating dividends are used when a company goes out of business and distributes its net assets, sells a portion of its business for cash and distributes the sale's proceeds, or pays a dividend that exceeds its accumulated retained earnings.

²"Retraction" refers to this option, which is a put option. The terminology is not completely settled: The type of share being called "retractable term preferred" is also known as "hard retractable preferred," with "hard" referring to payment in cash rather than common shares at the retraction date.

³A related concept, the present value of growth opportunities (PVGO), is discussed in more advanced readings.

⁴The exact geometric average annual growth rate for 2015–2023 can be determined as $[(1 + 0.10)(1 + 0.0727)(1 + 0.0525)(1 + 0.499)(1 + 0.022)(1 + 0.022)(1 + 0.022)(1 + 0.022)]^{1/8} - 1 = 4.5\%$.

⁵Exchange rate data are available from FRED (Federal Reserve Economic Data) at <http://research.stlouisfed.org/fred2/>. Each company uses slightly different methodology for calculating order backlog.

GLOSSARY

Abandonment option

The option to terminate an investment at some future time if the financial results are disappointing.

Abnormal return

The amount by which a security's actual return differs from its expected return, given the security's risk and the market's return.

Absolute dispersion

The amount of variability present without comparison to any reference point or benchmark.

Accelerated book build

An offering of securities by an investment bank acting as principal that is accomplished in only one or two days.

Accounting profit

Income as reported on the income statement, in accordance with prevailing accounting standards, before the provisions for income tax expense. Also called *income before taxes* or *pretax income*.

Accredited investors

Investors that meet certain minimum regulatory net worth or other requirements in order to invest in certain types of alternative assets.

Accrued interest

The amount of interest in currency or par value terms of a fixed-income instrument that accumulates from the last coupon payment until the trade settlement date. The amount is paid by the buyer to the seller.

Action lag

Delay from policy decisions to implementation.

Active investment

An approach to investing in which the investor seeks to outperform a given benchmark.

Active return

The return on a portfolio minus the return on the portfolio's benchmark.

Activist

Short for "activist shareholder." Managers secure sufficient equity holdings to allow them to seek a position in a company's board and influence corporate policies or direction.

Activity ratios

Ratios that measure how well a company is managing key current assets and working capital over time.

Ad hoc committee

A small group of lenders or bondholders who negotiate with an issuer on debt restructuring and refinancing before the issuer submits a final proposal to the wider group of all lenders and bondholders.

Add-on pricing

A pricing approach based on high-margin optional features, customizations, and additional content.

Add-on rate

A yield or pricing convention for money market instrument quotations. It is the interest earned on an instrument, derived from the difference between the price and face value, expressed as a percentage of the price and multiplied by the periodicity of the annual rate.

Agency costs

Direct and indirect costs borne by the principal in a principal-agent relationship owing primarily to information asymmetries. Agency costs include the costs of monitoring and assessing the agent as well as missed opportunities.

Agency RMBS

Securities created by the pooling of residential mortgage-backed securities in the United States by either the Federal National Mortgage Association (Fannie Mae) or the Federal Home Loan Mortgage Corporation (Freddie Mac). These RMBS carry the full faith and credit of the government, essentially a guarantee with respect to timely payment of interest and repayment of principal.

All-or-nothing (AON) orders

An order that includes the instruction to trade only if the trade fills the entire quantity (size) specified.

Allocationally efficient

A characteristic of a market, a financial system, or an economy that promotes the allocation of resources to their highest value uses.

Altcoin

A cryptocurrency other than Bitcoin.

Alternative data

Data that are generated from non-traditional sources, such as social media and sensor networks.

Alternative hypothesis

The hypothesis that is accepted if the null hypothesis is rejected.

Alternative investment markets

Market for investments other than traditional securities investments (i.e., traditional common and preferred shares and traditional fixed income instruments). The term usually encompasses direct and indirect investment in real estate (including timberland and farmland) and commodities (including precious metals); hedge funds, private equity, and other investments requiring specialized due diligence.

Alternative trading systems

Trading venues that function like exchanges but that do not exercise regulatory authority over their subscribers except with respect to the conduct of the subscribers' trading in their trading systems. Also called *electronic communications networks* or *multilateral trading facilities*.

American depository receipt

A US dollar-denominated security that trades like a common share on US exchanges.

American depository share

The underlying shares on which American depository receipts are based. They trade in the issuing company's domestic market.

American options

Options that may be exercised at any time from contract inception until maturity.

American-style

Type of option contract that can be exercised at any time up to the option's expiration date.

Amortization

The process of allocating the cost of intangible long-term assets having a finite useful life to accounting periods; the allocation of the amount of a bond premium or discount to the periods remaining until bond maturity.

Amortizing debt

A loan or bond with a payment schedule that calls for periodic payments of interest and repayments of principal.

Analysis of variance (ANOVA)

A table that presents the sums of squares, degrees of freedom, mean squares, and F-statistic for a regression model.

Analytical duration

Estimates of duration using mathematical formulas. Estimates of the impact of yield changes on bond prices using analytical duration implicitly assume that benchmark yields and spreads are independent variables and are uncorrelated.

Anchoring and adjustment bias

An information-processing bias in which the use of a psychological heuristic influences the way people estimate probabilities.

Annual general meeting (AGM)

A yearly meeting of the corporate board of directors and shareholders, typically held in person and digitally, during which votes on directors, compensation plans, shareholder resolutions, and any other matters properly brought forward at the meeting are held. Issuer management may also make presentations and hold events.

Anomalies

Apparent deviations from market efficiency.

Antidilutive

With reference to a transaction or a security, one that would increase earnings per share (EPS) or result in EPS higher than the company's basic EPS—antidilutive securities are not included in the calculation of diluted EPS.

Arbitrage

1) The simultaneous purchase of an undervalued asset or portfolio and sale of an overvalued but equivalent asset or portfolio, in order to obtain a riskless profit on the price differential. Taking advantage of a market inefficiency in a risk-free manner. 2) The condition in a financial market in which equivalent assets or combinations of assets sell for two different prices, creating an

opportunity to profit at no risk with no commitment of money. In a well-functioning financial market, few arbitrage opportunities are possible. 3) A risk-free operation that earns an expected positive net profit but requires no net investment of money.

Arbitrageurs

Traders who engage in arbitrage. See *arbitrage*.

Arithmetic mean

The sum of the observations divided by the number of observations.

Artificial intelligence (AI)

Computer systems that are capable of performing tasks that previously required human intelligence. AI methods are sometimes better suited to identify complex, non-linear relationships than are traditional quantitative and statistical methods.

Ask

The price at which a dealer or trader is willing to sell an asset, typically qualified by a maximum quantity (ask size). See *offer*.

Ask size

The maximum quantity of an asset that pertains to a specific ask price from a trader. For example, if the ask for a share issue is \$30 for a size of 1,000 shares, the trader is offering to sell at \$30 up to 1,000 shares.

Asset allocation

The process of determining how investment funds should be distributed among asset classes.

Asset class

A group of assets that have similar characteristics, attributes, and risk–return relationships.

Asset utilization ratios

Ratios that measure how efficiently a company performs day-to-day tasks, such as the collection of receivables and management of inventory.

Asset-backed commercial paper

Secured form of commercial paper issuance. Loans or receivables are sold to a special purpose entity that issues the ABCP and makes interest and principal payments to investors from asset cash flows.

Asset-backed securities (ABS)

A type of bond issued by a legal entity called a special purpose entity created solely to own assets such as loans, receivables, and mortgages and to distribute cash flows to ABS investors. Generally, ABS backed by mortgages are known as mortgage-backed securities (MBS) while ABS refer to non-mortgage ABS.

Asset-backed token

A token that represents the ownership of a physical asset that does not exist on the blockchain and whose value is based on the underlying asset.

Asset-based valuation models

Valuation based on estimates of the market value of a company's assets.

Asymmetric information

Also known as *information asymmetry*; the differential of information between corporate insiders and outsiders regarding the company's performance and prospects. Managers typically have more information about the company's performance and prospects than owners and creditors.

At-the-money

Describes a unique situation in which the price of the underlying is equal to an option's exercise price. Like an out-of-the-money option, the intrinsic value is zero.

Auction/reverse auction models

Pricing models that establish prices through bidding (by sellers in the case of reverse auctions).

Autarky

Countries seeking political self-sufficiency with little or no external trade or finance. State-owned enterprises control strategic domestic industries.

Automatic stabilizer

A countercyclical factor that automatically comes into play as an economy slows and unemployment rises.

Availability bias

An information-processing bias in which people take a heuristic approach to estimating the probability of an outcome based on how easily the outcome comes to mind.

Available-for-sale

Under US GAAP, debt securities not classified as either held-to-maturity or held-for-trading securities. The investor is willing to sell but not actively planning to sell. In general, available-for-

sale debt securities are reported at fair value on the balance sheet, with unrealized gains included as a component of other comprehensive income.

Average revenue (AR)

Total revenue divided by quantity sold.

Backfill Bias

A problem whereby certain surviving hedge funds may be added to databases and various hedge fund indexes only after they are initially successful and start to report their returns. Also see *survivorship bias*.

Backup line of credit

A type of credit enhancement provided by a bank to an issuer of commercial paper to ensure that the issuer will have access to sufficient liquidity to repay maturing commercial paper if issuing new paper is not a viable option.

Backwardation

A downward-sloping, or inverted, forward curve in a futures market.

Balance sheet ratios

Financial ratios involving balance sheet items only.

Balanced

With respect to a government budget, one in which spending and revenues (taxes) are equal.

Balloon payment

A large payment required at maturity to retire a bond's outstanding principal amount.

Base rates

The reference rate on which a bank bases lending rates to all other customers.

Base-rate neglect

A type of representativeness bias in which the base rate or probability of the categorization is not adequately considered.

Basic EPS

Net earnings available to common shareholders (i.e., net income minus preferred dividends) divided by the weighted average number of common shares outstanding.

Basis risk

The possibility that the expected value of a derivative differs unexpectedly from that of the underlying.

Basket of listed depository receipts (BLDR)

An exchange-traded fund (ETF) that represents a portfolio of depository receipts.

Bayes' formula

The rule for updating the probability of an event of interest—given a set of prior probabilities for the event, information, and information given the event—if you receive new information.

Bearer bonds

Bonds for which ownership is not recorded; only the clearing system knows who the bond owner is.

Behavioral finance

A field of finance that examines the psychological variables that affect and often distort the investment decision making of investors, analysts, and portfolio managers.

Behind the market

Said of prices specified in orders that are worse than the best current price; e.g., for a limit buy order, a limit price below the best bid.

Benchmark

A bond used to compare against another bond to discern attributes, often a government bond with the same or similar time-to-maturity as the bond under analysis.

Benchmark spread

The difference in yield-to-maturity between a bond and that of a benchmark bond.

Best bid

The highest bid in the market.

Best effort offering

An offering of a security using an investment bank in which the investment bank, as agent for the issuer, promises to use its best efforts to sell the offering but does not guarantee that a specific amount will be sold.

Best offer

The lowest offer (ask price) in the market.

Best-in-class

An ESG implementation approach that seeks to identify the most favorable companies and sectors based on ESG considerations. Also called *positive screening*.

Beta

A measure of the sensitivity of a given investment or portfolio to movements in the overall market.

Bid

The price at which a dealer or trader is willing to buy an asset, typically qualified by a maximum quantity.

Bid size

The maximum quantity of an asset that pertains to a specific bid price from a trader.

Big data

The vast amount of information being generated by both traditional sources—for example, stock exchanges, companies, governments—and non-traditional sources—for example, electronic devices, social media, sensor networks, and company exhaust.

Bilateralism

The conduct of political, economic, financial, or cultural cooperation between two countries. Countries engaging in bilateralism may have relations with many different countries but in one-at-a-time agreements without multiple partners. Typically, countries exist on a spectrum between bilateralism and multilateralism.

Bimodal

A distribution that has two most frequently occurring values.

Bitcoin

A cryptocurrency using blockchain technology that was created in 2009.

Bivariate correlation

Also known as Pearson correlation. A parametric measure of the relationship between two variables.

Black swan risk

An event that is rare and difficult to predict but has an important impact.

Block brokers

A broker (agent) that provides brokerage services for large-size trades.

Blockchain

A type of digital ledger in which information is recorded sequentially and then linked together and secured using cryptographic methods.

Blue chip

Widely held large market capitalization companies that are considered financially sound and are leaders in their respective industry or local stock market.

Board of directors

A body or individual selected by a limited company's member(s) or shareholder(s), in a manner determined by the company's charter, that manages the company. Typically, for larger companies, boards of directors appoint and oversee executive management.

Bond equivalent yield

A money market interest rate quoted on a 365-day add-on rate basis.

Bond indenture

A legal document between a bond issuer and investors that governs each party's rights and responsibilities.

Bond market vigilantes

Bond market participants who might reduce their demand for long-term bonds, thus pushing up their yields.

Bondholders

Investors in an entity's securitized debt claims, such as commercial paper, notes, and bonds.

Common types of bondholders include investment funds and institutional investors.

Bonds

Contractual agreements between an issuer and bondholders.

Bonus issue of shares

A type of dividend in which a company distributes additional shares of its common stock to shareholders instead of cash.

Book building

Investment bankers' process of compiling a "book" or list of indications of interest to buy part of an offering.

Book value

The net amount shown for an asset or liability on the balance sheet; book value may also refer to the company's excess of total assets over total liabilities. Also called *carrying value*.

Boom

An expansionary phase characterized by economic growth "testing the limits" of the economy.

Bootstrap

A resampling method that repeatedly draws samples with replacement of the selected elements from the original observed sample. Bootstrap is usually conducted by using computer simulation and is often used to find standard error or construct confidence intervals of population parameters.

Bottom-up analysis

An investment selection approach that focuses on company-specific circumstances rather than emphasizing economic cycles or industry analysis.

Box and whisker plot

A graphic for visualizing the dispersion of data across quartiles. It consists of a box with “whiskers” connected to the box.

Breakeven point

Represents the price of the underlying in a derivative contract in which the profit to both counterparties would be zero.

Bridge financing

Interim financing that provides funds until permanent financing can be arranged.

Broker

An agent who executes orders to buy or sell securities on behalf of a client in exchange for a commission.

Brokered market

A market in which brokers arrange trades among their clients.

Broker-dealer

A financial intermediary (often a company) that may function as a principal (dealer) or as an agent (broker) depending on the type of trade.

Brownfield investments

The third stage of development of an infrastructure asset. Brownfield investments involve expanding existing facilities and may involve privatization of public assets or a sale leaseback of completed greenfield projects. They are characterized by a shorter investment period with immediate cash flows and an operating history.

Budget surplus/deficit

The difference between government revenue and expenditure for a stated fixed period of time.

Bullet bond

A bond whose principal repayment is made entirely at maturity.

Bundling

A pricing approach that refers to combining multiple products or services so that customers are incentivized or required to buy them together.

Business cycles

Are recurrent expansions and contractions in economic activity affecting broad segments of the economy.

Business model

A concise description of how a business works and makes revenues and profits, including its customers, products or services, channels for reaching customers, and pricing.

Businesses

Organization entities formed and managed for the purpose of providing a return or economic benefits to its investors and owners.

Buy-side firm

An investment management company or other investor that uses the services of brokers or dealers (i.e., the client of the sell side firms).

Buyback

A transaction in which a company buys back its own shares. Unlike stock dividends and stock splits, share repurchases use corporate cash.

Cabotage

The right to transport passengers or goods within a country by a foreign firm. Many countries—including those with multilateral trade agreements—impose restrictions on cabotage across transportation subsectors, meaning that shippers, airlines, and truck drivers are not allowed to transport goods and services within another country's borders.

Call market

A market in which trades occur only at a particular time and place (i.e., when the market is called).

Call money rate

The interest rate that buyers pay for their margin loan.

Call option

The right to buy an underlying.

Call period

The time during which the issuer of a callable bond can exercise the call option.

Call price

The price at which the issuer of a callable bond has the right to purchase the bond from investors.

Call protection period

The time during which the issuer of a callable bond is not allowed to exercise the call option.

Call risk

The uncertain maturity and limited price appreciation associated with callable bonds.

Callable bond

A bond containing an embedded call option that gives the issuer the right to buy the bond back from the investor at specified prices on predetermined dates.

Cannibalization

A transfer of sales or market share from one product to another product owned by the same company. It tends to occur when the two products are actual or perceived substitutes.

Capacity

The ability of the borrower to make its debt payments on time.

Capital

Other company resources available that reduce reliance on debt

Capital allocation

The process that companies use for decision making on capital investments—those projects with a life of one year or longer.

Capital allocation line (CAL)

A graph line that describes the combinations of expected return and standard deviation of return available to an investor from combining the optimal portfolio of risky assets with the risk-free asset.

Capital asset pricing model (CAPM)

An equation describing the expected return on any asset (or portfolio) as a linear function of its beta relative to the market portfolio.

Capital expenditure

Expenditure on physical capital (fixed assets).

Capital investments

An expenditure for an asset or resource with a useful life of more than one year.

Capital market expectations (CME)

Expectations concerning the risk and return prospects of asset classes.

Capital market line (CML)

The line with an intercept point equal to the risk-free rate that is tangent to the efficient frontier of risky assets; represents the efficient frontier when a risk-free asset is available for investment.

Capital market securities

Fixed-income securities with original maturities greater than one year.

Capital markets

Financial markets that trade securities of longer duration, such as bonds and equities.

Capital restrictions

Controls placed on foreigners' ability to own domestic assets and/or domestic residents' ability to own foreign assets.

Capital structure

The mix of debt and equity that a company uses to finance its business; a company's specific mix of long-term financing.

Capital-indexed bond

A type of index-linked bond for which changes in the index are captured with adjustments to the principal. A common example is Treasury Inflation Protected Securities (TIPS) issued by the United States government.

Capital-intensive businesses

Companies or business activities that are characterized by a relatively low fixed asset turnover, a high percentage of capital expenditures to sales, or a high net-working-capital-to-sales ratio.

Capital-light businesses

Also known as *asset light businesses*, companies or business activities characterized by relatively high fixed asset turnover, a low percentage of capital expenditures to sales, or a low net-working-capital-to-sales ratio.

Carried interest

A performance fee (also referred to as an incentive fee, or carry) that is applied based on excess returns above a hurdle rate.

Carrying

Investing and holding an asset for a period of time.

Carrying amount

The amount at which an asset or liability is valued according to accounting principles.

Carrying value

Of a fixed-income instrument is the purchase price plus (minus) the amortized amount of the discount (premium) if the bond is purchased at a price below (above) par value.

Cartel

Participants in collusive agreements that are made openly and formally.

Cash conversion cycle

The amount of time between an issuer paying its suppliers in cash and receiving cash from its customers.

Cash flow additivity principle

The principle that dollar amounts indexed at the same point in time are additive.

Cash flow from operations

A cash profit measure over a period for an issuer's primary business activities. It includes cash from customers as well as interest and dividends received from financial investments, less cash paid to employees and suppliers as well as taxes paid to governments and interest paid to lenders.

Cash flow hedge

Refers to a specific hedge accounting classification in which a derivative is designated as absorbing the variable cash flow of a floating-rate asset or liability, such as foreign exchange, interest rates, or commodities.

Cash markets

Markets in which specific assets are exchanged at current prices. Cash markets are often referred to as spot markets.

Cash prices

The current prices prevailing in cash markets.

Cash ratio

A measure of liquidity that is the ratio of cash and marketable securities to current liabilities.

Catch-up clause

A clause in an agreement that favors the GP. For a GP who earns a 20% performance fee, a catch-up clause allows the GP to receive 100% of the distributions above the hurdle rate *until* she receives 20% of the profits generated, and then every excess dollar is split 80/20 between the LPs and GP.

CDS credit spread

Reflects the credit spread of a credit default swap (CDS) derivative contract. As with cash bonds, CDS credit spreads depend on the probability of default (POD) and the loss given default (LGD).

Central bank digital currencies (CBDCs)

A tokenized version of the currency issued by the central bank, such as a digital bank note or coin, and a digital liability of the central bank.

Central bank funds market

The market in which deposit-taking banks that have an excess reserve with their national central bank can lend money to banks that need funds for maturities ranging from overnight to one year. Called the federal or fed funds market in the United States.

Central bank funds rate

The interest rate at which central bank funds are bought (borrowed) and sold (lent) for maturities ranging from overnight to one year. Called federal or fed funds rate in the United States.

Central clearing mandate

A requirement instituted by global regulatory authorities following the 2008 global financial crisis that most over-the-counter (OTC) derivatives be cleared by a central counterparty (CCP).

Central counterparty (CCP)

An economic entity that assumes the counterparty credit risk between derivative counterparties, one of which is typically a financial intermediary. CCPs provide clearing and settlement for most derivative contracts.

Central limit theorem

The theorem that states the sum (and the mean) of a set of independent, identically distributed random variables with finite variances is normally distributed, whatever distribution the random variables follow.

Certificate of deposit (CD)

An instrument that represents a specified amount of funds on deposit with a bank for a specified maturity and interest rate. CDs are issued in various denominations and can be negotiable or non-negotiable.

Channels

Venues where a company markets and/or delivers its products and services.

Character

The quality of a debt issuer's management.

Checking accounts

Bank deposits with no stated maturity available for transactional purposes that pay little or no interest. Also known as a *demand deposit*.

Circuit breaker

A pause in intraday trading for a brief period if a price limit is reached.

Classical cycle

Refers to fluctuations in the level of economic activity when measured by GDP in volume terms.

Clawback

A requirement that the general partner return any funds distributed as incentive fees until the limited partners have received their initial investment and a percentage of the total profit.

Clearing

An exchange's process of verifying the execution of a transaction, exchange of payments, and recording of participants.

Clearing instructions

Instructions that indicate how to arrange the final settlement ("clearing") of a trade.

Clearinghouse

An entity associated with a futures market that acts as middleman between the contracting parties and guarantees to each party the performance of the other.

Closed-end fund

A mutual fund in which no new investment money is accepted. New investors invest by buying existing shares, and investors in the fund liquidate by selling their shares to other investors.

Cluster sampling

A procedure that divides a population into subpopulation groups (clusters) representative of the population and then randomly draws certain clusters to form a sample.

Co-investing

In co-investing, the investor invests in assets *indirectly* through the fund but also possesses rights (known as co-investment rights) to invest *directly* in the same assets. Through co-investing, an investor is able to make an investment *alongside* a fund when the fund identifies deals.

Code of ethics

An established guide that communicates an organization's values and overall expectations regarding member behavior. A code of ethics serves as a general guide for how community members should act.

Coefficient of determination (R^2)

The percentage of the variation of the dependent variable that is explained by the independent variable. It is a measure of goodness of fit of a regression model.

Coefficient of variation

The ratio of a set of observations' standard deviation to the observations' mean value.

Cognitive cost

The effort involved in processing new information and updating beliefs.

Cognitive dissonance

The mental discomfort that occurs when new information conflicts with previously held beliefs or cognitions.

Cognitive errors

Behavioral biases resulting from faulty reasoning; cognitive errors stem from basic statistical, information-processing, or memory errors.

Coincident economic indicators

Turning points that are usually close to those of the overall economy; they are believed to have value for identifying the economy's present state.

Collateral

Assets or financial guarantees underlying a debt obligation that are above and beyond the issuer's promise to pay.

Collateral manager

Buys and sells debt obligations for and from the CDO's collateral pool to generate sufficient cash flows to meet the obligations to the CDO bondholders.

Collateralized bond obligations (CBOs)

CDOs backed by high-yield corporate and emerging market bonds.

Collateralized debt obligations (CDOs)

Securities backed by a diversified pool of one or more debt obligations. CDOs can be backed by a broad range of debt.

Collateralized loan obligations (CLOs)

CDOs backed by leveraged bank loans.

Collateralized mortgage obligations

Securitize mortgage pass-through securities or multiple pools of loans. CMOs are structured to redistribute the cash flows to different bond classes or tranches and create securities that have different exposures to prepayment risk.

Commercial paper (CP)

Short-term, negotiable, unsecured promissory note that represents a debt obligation of the issuer.

Committed (regular) lines of credit

Bank commitments to extend credit; the commitment is considered a short-term liability and is usually in effect for 364 days (one day short of a full year).

Committed capital

The amount that the limited partners have agreed to provide to the private equity fund.

Commodities

A product or service from a firm that is indistinguishable from products or services of competing firms, usually conforming to a common standard or grade imposed by convention or regulation.

Commoditization

A process by which competing products become less differentiated over time and become interchangeable “commodities” in the eyes of customers. This process is typically associated with declining profitability for the selling firms.

Commodity producers

A firm that makes and/or sells commodities.

Commodity swap

A type of swap involving the exchange of payments over multiple dates as determined by specified reference prices or indexes relating to commodities.

Common market

Level of economic integration that incorporates all aspects of the customs union and extends it by allowing free movement of factors of production among members.

Common shares

A type of security that represents an ownership interest in a company. Also called *common stock*.

Common stock

A type of security that represents an ownership interest in a company. Also called *common shares*.

Common-size analysis

The restatement of financial statement items using a common denominator or reference item that allows one to identify trends and major differences; an example is an income statement in which all items are expressed as a percent of revenue.

Companies

Organization entities formed and managed for the purpose of providing a return or economic benefits to its investors and owners.

Company research report

A document that presents an analyst's investment recommendation on an issuer and its securities, supported by financial modeling, industry overviews and competitive analyses, valuation scenarios, ESG considerations, and investment risks.

Complete markets

Informally, markets in which the variety of distinct securities traded is so broad that any desired payoff in a future state-of-the-world is achievable.

Concession agreement

A legal agreement in infrastructure investing that governs the investor's obligations to construct and maintain infrastructure as well as the exclusive right to operate and earn fees for a pre-determined period.

Conditional expected value

The expected value of a stated event given that another event has occurred.

Conditional pass-through covered bonds

Convert to pass-through securities after the original maturity date if all bond payments have not yet been made.

Conditional variances

The variance of one variable, given the outcome of another.

Conditions

The general economic, competitive, and business environment faced by all borrowers that may affect their ability to service or refinance debt.

Confidence level

The complement of the level of significance.

Confirmation bias

A belief perseverance bias in which people tend to look for and notice what confirms their beliefs, to ignore or undervalue what contradicts their beliefs, and to misinterpret information as support for their beliefs.

Consensus protocol

A set of rules governing how blocks can join the blockchain that is designed to resist attempts at malicious manipulation up to a certain level of security; it can be either a proof of work or a proof of stake.

Conservatism bias

A belief perseverance bias in which people maintain their prior views or forecasts by inadequately incorporating new information.

Constant yield-price trajectory

A graphical depiction of the relationship between time to maturity and a bond price, assuming no default, that shows that a bond price approaches par as time passes.

Constituent securities

With respect to an index, the individual securities within an index.

Contango

Refers to spot price below forward price in a futures market.

Contingency provision

Clause in a legal document that allows for some action if a specific event or circumstance occurs.

Contingency table

A table of the frequency distribution of observations classified on the basis of two discrete variables.

Contingent claim

A type of derivative in which one of the *counterparties* determines whether and when the trade will settle. An *option* is a common type of contingent claim.

Contingent convertible bonds

Bonds that automatically convert to equity if a specific event or circumstance occurs, such as the issuer's equity capital falling below the minimum requirement set by regulators.

Continuous trading market

A market in which trades can be arranged and executed any time the market is open.

Continuously compounded return

The natural logarithm of 1 plus the holding period return, or equivalently, the natural logarithm of the ending price over the beginning price.

Contract manufacturers

Companies that make products for other companies that meet specific terms and specifications.

Contract size

Amount(s) used for calculation to price and value the derivative. The contract size is often referred to as "notional amount or notional principal."

Contraction

The period of a business cycle after the peak and before the trough; often called a *recession* or, if exceptionally severe, called a *depression*.

Contraction risk

The risk that the borrower repays the principal or a portion of the principal in a shorter period of time than the contractually agreed scheduled payment, reducing the amount of future payments the investor receives.

Contractionary

Tending to cause the real economy to contract.

Contractionary fiscal policy

A fiscal policy that has the objective to make the real economy contract.

Contribution margin

A profitability measure using variable costs: unit price less unit variable cost. It can also be expressed as a percentage of price or sales.

Controlling shareholder

An individual or entity that owns a majority of the voting rights in a corporation.

Convenience sampling

A procedure of selecting an element from a population on the basis of whether or not it is accessible to a researcher or how easy it is for a researcher to access the element.

Convenience yield

A non-cash benefit of holding a physical commodity versus a derivative.

Conversion price

For a convertible bond, the price per share at which the bond can be converted into shares.

Conversion ratio

For a convertible bond, the number of common shares that each bond can be converted into.

Conversion value

For a convertible bond, the value of the bond if it is converted at the market price of the shares.

Also called *parity value*.

Convertible bond

A bond that gives the bondholder the right to exchange the bond for a specified number of common shares in the issuing company.

Convertible debt

A debt instrument that gives the holder the right to exchange the instrument for a specified number of common shares in the issuing company.

Convertible preference shares

A type of equity security that entitles shareholders to convert their shares into a specified number of common shares.

Convexity

Used to describe the curved, non-linear shape of the price/yield relationship of fixed-income instruments as well as a measure that is used to complement modified duration, sometimes called the convexity adjustment, to capture the second-order effect of yield changes on a bond's price.

Convexity adjustment

A measure that is used to complement modified duration to capture the second-order effect of yield changes on a bond's price. It is equal to the annual convexity statistic times one-half times the given change in the yield-to-maturity squared.

Convexity bias

Refers to the difference in price changes for a given change in yield between interest rate futures and interest rate forward contracts. That is, interest rate forwards exhibit a non-linear or convex relationship between price and yield, while the price–yield relationship is linear for interest rate futures.

Cooperation

The process by which countries work together toward some shared goal or purpose. These goals may, and often do, vary widely—from strategic or military concerns, to economic influence, to cultural preferences.

Cooperative country

A country that engages and reciprocates in rules standardization; harmonization of tariffs; international agreements on trade, immigration, or regulation; and allowing the free flow of information, including technology transfer.

Core real estate strategies

Strategies with exposure to well-leased, high-quality commercial and residential real estate in the best markets, generally offered by open-end funds. Investors expect core real estate to deliver stable returns, primarily from income from the property.

Core-plus real estate strategies

Value-add investments that require modest redevelopment or upgrades to lease any vacant space together with possible alternative use of the underlying properties. Compared to core real estate strategies, these may be appealing for investors seeking higher returns and willing to accept additional risks from development, redevelopment, repositioning, and leasing.

Corporate issuers

Limited companies or corporations that seek financing in financial markets by, for example, issuing debt or equity securities.

Corporations

Another term for limited companies, though often used to refer to public limited companies. See *limited company*, *private limited company*, and *public limited company*.

Correlation

A measure of the linear relationship between two random variables.

Correlation coefficient

A number between -1 and $+1$ that measures the consistency or tendency for two investments to act in a similar way. It is used to determine the effect on portfolio risk when two assets are combined.

Cost averaging

The periodic investment of a fixed amount of money.

Cost of capital

The cost of financing for a company; the rate of return that suppliers of capital require as compensation for their contribution of capital (also called *opportunity cost of funds*).

Cost of carry

The net of the costs and benefits related to owning an underlying asset for a specific period.

Cost of debt

The required return on debt financing for a company, such as when it issues a bond, takes out a bank loan, or leases an asset through a finance lease.

Cost of equity

The return required by equity investors to compensate for both the time value of money and the risk. Also referred to as the required rate of return on common stock or the required return on equity.

Counterparty

Legal entities entering a derivative contract.

Counterparty credit risk

The likelihood that a counterparty is unable to meet its financial obligations under the contract.

Counterparty risk

The risk that the other party to a contract will fail to honor the terms of the contract.

Country

The geopolitical environment as well as the legal and political system faced by all issuers in a jurisdiction that may affect debt payment.

Coupon

Periodic interest payments paid by a bond issuer to investors, typically expressed as a percentage of par on an annual basis.

Cournot assumption

Assumption in which each firm determines its profit-maximizing production level assuming that the other firms' output will not change.

Covariance

A measure of the co-movement (linear association) between two random variables.

Covenants

The terms and conditions of lending agreements that the issuer must comply with; they specify the actions that an issuer is obligated to perform (affirmative covenant) or prohibited from performing (negative covenant).

Credit default swap (CDS)

A type of credit derivative in which one party, the credit protection buyer who is seeking credit protection against a third party, makes a series of regularly scheduled payments to the other party, the credit protection seller. The seller makes no payments until a credit event occurs.

Credit enhancements

Provisions or methods that allow a borrower improve their creditworthiness in a structured transaction.

Credit event

An event that defines a payout in a credit derivative. Events are usually defined as bankruptcy, failure to pay an obligation, or an involuntary debt restructuring.

Credit facilities

Loan agreements with pre-specified terms and limits but with fluctuating balances based on borrower-specific needs at different points in time, analogous to a credit card.

Credit migration risk

The risk that a bond issuer's creditworthiness deteriorates, or migrates lower, leading investors to believe the risk of default is higher. Also called downgrade risk.

Credit rating

Letter-grade, qualitative measures of an issuer's ability to meet its debt obligations based on both the probability of default and the expected loss under a default scenario.

Credit rating agencies

Institutions that issue and maintain credit ratings. The three largest are Standard & Poor's, Moody's, and Fitch Ratings.

Credit risk

the expected economic loss under a potential borrower default over the life of the contract

Credit spread

The compensation for the risk of default in a debt security, typically measured by the yield-to-maturity difference between a bond and a comparable government benchmark security.

Credit spread risk

The risk of greater expected loss due to changes in credit conditions as a result of macroeconomic, market, and/or issuer-related factors.

Credit tranching

Internal credit enhancement where cash flows into a senior/subordinate structure.

Credit-linked notes

Bonds whose coupon changes when the bonds' credit rating changes.

Critical values

Values of the test statistic at which the decision changes from fail to reject the null hypothesis to reject the null hypothesis.

Cross-default clause

Covenant or contract clause that specifies borrowers are considered in default if they default on another debt obligation.

Cross-sectional analysis

Also called relative analysis. Analysis that involves comparisons across individuals in a group over a given time period or at a given point in time.

Crossing networks

Trading systems that match buyers and sellers who are willing to trade at prices obtained from other markets.

Crowdsourcing

A business model that enables users to contribute directly to a product, service, or online content.

Cryptocurrency

An electronic medium of exchange that lacks physical form.

Cryptocurrency wallet

A storage unit for public and/or private keys for cryptocurrency transactions. These wallets may be a physical device, program, or service.

Cryptography

An algorithmic process to encrypt data, making the data unusable if received by unauthorized parties.

Cumulative preference shares

Preference shares for which any dividends that are not paid accrue and must be paid in full before dividends on common shares can be paid.

Cumulative voting

A voting process whereby shareholders can accumulate and vote all their shares for a single candidate in an election, as opposed to having to allocate their voting rights evenly among all candidates.

Currency

Money issued by national monetary authorities.

Currency swap

A swap in which each party makes interest payments to the other in different currencies.

Current government spending

With respect to government expenditures, spending on goods and services that are provided on a regular, recurring basis including health, education, and defense.

Current ratio

A measure of liquidity that is the ratio of current assets to current liabilities.

Current yield (CY)

The sum of the coupon payments received over the year divided by the flat price. Also called the income, interest yield, or running yield.

Customs union

Extends the free trade area (FTA) by not only allowing free movement of goods and services among members, but also creating a common trade policy against nonmembers.

CVaR

Conditional VaR, a tail loss measure. The weighted average of all loss outcomes in the statistical distribution that exceed the VaR loss.

Daily settlement

A specific process of *mark-to-market* by a central clearing party in which the profits and losses of all counterparties to derivatives contracts are determined using settlement prices for each contract.

Dark pools

Alternative trading systems that do not display the orders that their clients send to them.

Data mining

The practice of determining a model by extensive searching through a dataset for statistically significant patterns.

Data science

An interdisciplinary field that harnesses advances in computer science, statistics, and other disciplines for the purpose of extracting information from big data (or data in general).

Data snooping

The practice of determining a model by extensive searching through a dataset for statistically significant patterns.

Day order

An order that is good for the day on which it is submitted. If it has not been filled by the close of business, the order expires unfilled.

Days of inventory on hand (DOH)

The average number of days it would take to sell the amount of inventory on hand. It is calculated as either the ending or average balance of inventories divided by (cost of goods sold/days in the period).

Days payable outstanding (DPO)

The average number of days it takes a company to pay its suppliers. It is calculated as either the ending or average balance of accounts payable divided by (cost of goods sold/days in the period).

Days sales outstanding (DSO)

The average number of days it takes for a company to receive payment from customers who purchase goods or services on credit. It is calculated as either the ending or average balance of accounts receivable divided by (revenues/days in the period).

Dealers

Financial intermediaries, such as commercial banks or investment banks, who transact as counterparties with derivative end users.

Debt

A claim against an entity to receive cash, stock, or other assets at a future date. From the perspective of the debtor or borrower, an obligation to pay cash, stock, or other assets at a future date. Generally, debt claims are unconditional and are senior to equity claims.

Debt service coverage ratio

The ratio of project revenue available to cover interest and principal payments to debt service used as a measure of creditworthiness for revenue bonds.

Debt tax shield

The tax benefit from interest paid on debt being tax deductible from income, equal to the marginal tax rate multiplied by the value of the debt.

Debt-to-assets ratio

A solvency ratio calculated as total debt divided by total assets.

Debt-to-capital ratio

A solvency ratio calculated as total debt divided by total debt plus total shareholders' equity.

Debt-to-equity ratio

A solvency ratio calculated as total debt divided by total shareholders' equity.

Debt-to-income ratio (DTI)

Residential lending metric that compares an individual's monthly debt payments to their monthly pre-tax, gross income.

Debut issuer

An issuer approaching the bond market for the first time.

Deciles

Quantiles that divide a distribution into 10 equal parts.

Declaration date

The day that the corporation issues a statement declaring a specific dividend.

Decreasing returns to scale

When a production process leads to increases in output that are proportionately smaller than the increase in inputs.

Deductible temporary differences

Temporary differences that result in a reduction of or deduction from taxable income in a future period when the balance sheet item is recovered or settled.

Deep learning

An area of artificial intelligence in which a system uses neural networks to perform multistage, non-linear data processing to identify patterns. Also called *deep learning nets*.

Deep learning nets

See *Deep learning*.

Deep-in-the-money option

An option that is highly likely to be exercised.

Deep-out-of-the-money option

An option that is highly unlikely to be exercised.

Default

When a borrower on a mortgage loan fails to meet the obligations of the loan.

Default risk premium

An extra return that compensates investors for the possibility that the borrower will fail to make a promised payment at the contracted time and in the contracted amount.

Defeasance

Mechanism that allows prepayment on mortgage, but the borrower must purchase a portfolio of government securities that fully replicates the cash flows of the remaining scheduled principal and interest payments, including the balloon loan balance, on the loan.

Defensive interval ratio

A liquidity ratio that estimates the number of days that an entity could meet cash needs from liquid assets; calculated as (cash + short-term marketable investments + receivables) divided by daily cash expenditures.

Deferred coupon bonds

Bonds that pay no coupons for their first few years but then pay a higher coupon than they otherwise normally would for the remainder of their life. Also called *split coupon bonds*.

Deferred tax asset

A balance sheet asset that arises when an excess amount is paid for income taxes relative to accounting profit. The taxable income is higher than accounting profit and income tax payable exceeds tax expense. The company expects to recover the difference during the course of future operations when tax expense exceeds income tax payable.

Deferred tax liabilities

A balance sheet liability that arises when a deficit amount is paid for income taxes relative to accounting profit. The taxable income is less than the accounting profit and income tax payable is less than tax expense. The company expects to eliminate the liability over the course of future operations when income tax payable exceeds tax expense.

Defined benefit pension plans (DB plans)

Plans in which the company promises to pay a certain annual amount (defined benefit) to the employee after retirement. The company bears the investment risk of the plan assets.

Defined contribution pension plans

Individual accounts to which an employee and typically the employer makes contributions during their working years and expect to draw on the accumulated funds at retirement. The employee bears the investment and inflation risk of the plan assets.

Deflation

Negative inflation.

Degree of financial leverage

The ratio of percentage change in net income to percentage change in operating income over a period. It is a measure of how sensitive net income is to changes in operating income, driven by the firm's use of debt in its capital structure.

Degree of operating leverage (DOL)

The ratio of percentage change in operating income to percentage change in sales over a period. It is a measure of how sensitive operating income is to changes in sales, driven by the fixed and variable cost composition of operating expenses.

Delta

The relationship between the option price and the underlying price, which reflects the sensitivity of the price of the option to changes in the price of the underlying. Delta is a good approximation of how an option price will change for a small change in the stock.

Demand shock

A typically unexpected disturbance to demand, such as an unexpected interruption in trade or transportation.

Dependent variable

The variable that is explained by a regression model.

Depository bank

A bank that raises funds from depositors and other investors and lends it to borrowers.

Depository institutions

Commercial banks, savings and loan banks, credit unions, and similar institutions that raise funds from depositors and other investors and lend it to borrowers.

Depository receipt

A security that trades like an ordinary share on a local exchange and represents an economic interest in a foreign company.

Depreciation

The process of systematically allocating the cost of long-lived (tangible) assets to the periods during which the assets are expected to provide economic benefits.

Derivative

A financial instrument whose value depends on the value of some underlying asset or factor (e.g., a stock price, an interest rate, or exchange rate).

Derivative contract

A legal agreement between counterparties with a specific maturity, or length of time, until the closing of the transaction, or settlement.

Derivative pricing rule

A pricing rule used by crossing networks in which a price is taken (derived) from the price that is current in the asset's primary market.

Differentiated products

A product or service from a firm that is distinguishable or distinct from those of competing firms. It is customers who determine and value whether a product is differentiated.

Diffuse prior

The assumption of equal prior probabilities.

Diffusion index

Reflects the proportion of the index's components that are moving in a pattern consistent with the overall index.

Digital assets

The umbrella term covering assets that can be created, stored, and transmitted electronically and have associated ownership or use rights. Digital assets include a variety of assets, such as cryptocurrencies, tokens (security and utility), and digital collectables.

Diluted EPS

The EPS that would result if all dilutive securities were converted into common shares.

Dilution

An increase in the number of shares outstanding from share issuance that decreases the percentage of shares owned by existing shareholders.

Direct investing

Occurs when an investor makes a direct investment in an asset without the use of an intermediary.

Direct lending

Providing capital directly from private debt investors.

Direct listing

Where the equity of a security is floated on the public markets directly, without underwriters, reducing the complexity and cost of the transaction.

Direct sales

Marketing and/or delivering products and services to customers without an intermediary or third party between the customer and seller.

Direct taxes

Taxes levied directly on income, wealth, and corporate profits.

Discount factor

The price equivalent of a zero rate. Also may be stated as the present value of a currency unit on a future date.

Discount rate

A yield or pricing convention for money market instrument quotations. It is the interest earned on an instrument, derived from the difference between the price and face value, expressed as a percentage of the face value and multiplied by the periodicity of the annual rate.

Discounted cash flow models

Valuation models that estimate the intrinsic value of a security as the present value of the future benefits expected to be received from the security.

Discriminatory pricing rule

A pricing rule used in continuous markets in which the limit price of the order or quote that first arrived determines the trade price.

Diseconomies of scale

Increase in cost per unit resulting from increased production.

Dispersion

The variability of a population or sample of observations around the central tendency.

Display size

The size of an order displayed to public view.

Disposition effect

As a result of loss aversion, an emotional bias whereby investors are reluctant to dispose of losers.

This results in an inefficient and gradual adjustment to deterioration in fundamental value.

Distressed debt

Debt of mature companies in financial difficulty, in bankruptcy, or likely to default on debt.

Distressed/restructuring

These strategies focus on securities of companies either in or perceived to be near bankruptcy. In

one approach, hedge funds simply purchase fixed-income securities trading at a significant discount to par but that are still senior enough to be backed by sufficient corporate assets.

Distributed ledger

A type of database that can be shared among entities in a network.

Distributed ledger technology (DLT)

Technology based on a distributed ledger.

Diversification ratio

The ratio of the standard deviation of an equally weighted portfolio to the standard deviation of a randomly selected security.

Dividend discount model (DDM)

A present value model of stock value that views the intrinsic value of a stock as present value of the stock's expected future dividends.

Dividend payout ratio

The ratio of cash dividends paid to earnings for a period.

Dividends

Distributions of profits and/or net assets from a corporation to its shareholders. While often in cash, dividends can be also be paid in stock or assets, such as property.

Divisor

A number (denominator) used to determine the value of a price return index. It is initially chosen at the inception of an index and subsequently adjusted by the index provider, as necessary, to avoid changes in the index value that are unrelated to changes in the prices of its constituent securities.

Domestic bonds

A type of bond for which the issuer's domicile and jurisdiction of issuance are the same.

Domestic content provisions

Stipulate that some percentage of the value added or components used in production should be of domestic origin.

Double taxation

The taxation of business income at both the entity and personal or owner levels. In most jurisdictions, this taxation scheme applies to public limited companies.

Downside risk

Risk of incurring returns below a specified value.

Drag on liquidity

An action or event that reduces available funds or delays cash inflows.

Drivers

Causative factors that explain the level of and changes in an output variable

DSC ratio

A property's annual net operating income (NOI) divided by the debt service.

Dual-class structure

A capital structure that includes at least two classes of equity shares with unequal voting rights.

Dupont analysis

An approach to decomposing return on investment, e.g., return on equity, as the product of other financial ratios.

Duration

A measure of the approximate sensitivity of a security to a change in interest rates (i.e., a measure of interest rate risk).

Duration gap

The difference between a bond's Macaulay duration and its investor's investment horizon.

Dynamic pricing

A pricing approach that charges different prices at different times. Specific examples include off-peak pricing, "surge" pricing, and "congestion" pricing.

Early repayment option

May entitle the borrower to prepay all or part of the outstanding mortgage principal prior to maturity. This creates a risk from the lender's or investor's viewpoint because the cash flow amounts and timing cannot be known with certainty.

Earnings surprise

The portion of a company's earnings that is unanticipated by investors and, according to the efficient market hypothesis, merits a price adjustment.

Economic indicators

Economic statistics provided by government and established private organizations that contain information on an economy's recent past activity or its current or future position in the business cycle.

Economic infrastructure investments

A category of infrastructure investments that support economic activity through transportation assets, information and communication technology assets, and utility and energy assets.

Economic stabilization

Reduction of the magnitude of economic fluctuations.

Economic union

Incorporates all aspects of a common market and in addition requires common economic institutions and coordination of economic policies among members.

Economies of scale

A decline in costs per unit as output grows, generally resulting from having fixed costs in the cost structure that are spread over more units of output.

Economies of scope

A decline in costs per unit as the number of product or business lines increases, generally resulting from having shared costs between the product lines.

Effective annual rate

An interest rate with a periodicity of one.

Effective convexity

An interest rate risk statistic that measures the non-linear/second-order effect of changes in the benchmark yield curve on a bond's price.

Effective duration

The sensitivity of the bond's price to an instantaneous parallel shift in a benchmark yield curve—for example, the government par curve.

Efficient market

A market in which asset prices reflect new information quickly and rationally. See also, *informationally efficient market*.

Either/or fee

A custom fee arrangement whereby major investors are offered a structure where managers agree to charge *either* a lower management fee *or* a higher incentive fee, whichever is greater.

Electronic communications networks (ECNs)

See *alternative trading systems* and *multilateral trading facilities*.

Embedded derivative

A derivative within an underlying, such as a callable, puttable, or convertible bond.

Embedded options

Contingency provisions found in a bond's indenture representing rights that enable their holders to take advantage of interest rate movements. They can be exercised by the issuer, by the bondholder, or automatically depending on the course of interest rates.

Emotional biases

Behavioral biases resulting from reasoning influenced by feelings; emotional biases stem from impulse or intuition.

Empirical duration

Estimates of duration calculated over time and in different interest rate environments. Unlike analytical duration, empirical duration estimates do not assume that benchmark yields and spreads are independent variables and are uncorrelated

Employee stock ownership plan (ESOP)

A type of employee benefit plan in which a company sets up a trust fund to receive contributions of newly issued shares or cash to buy existing shares. Contributions are tax deductible up to certain limits. Shares in the trust fund are allocated to individual employees based on relative pay or a formula.

Endowment bias

An emotional bias in which people value an asset more when they hold rights to it than when they do not.

Enterprise risk management

An overall assessment of a company's risk position. A centralized approach to risk management sometimes called firmwide risk management.

Enterprise value (EV)

Total company value (the market value of debt, common equity, and preferred equity) minus the value of cash and investments.

Equal weighting

An index weighting method in which an equal weight is assigned to each constituent security at inception.

Equity

Ownership interest in an entity. A residual claim on the assets of an entity after more senior claims, such as debt, have been satisfied. Also known as *net assets*.

Equity swap

A swap transaction in which at least one cash flow is tied to the return on an equity portfolio position, often an equity index.

Error term

Represents the difference between the observed value of the independent variable and that expected from the true underlying population relation between the dependent and independent variable.

Estimated parameters

In a simple linear regression, the estimated parameters are the intercept and slope of the fitted line.

Ether

A programmable cryptocurrency created on the Ethereum blockchain in 2015 that allows for the execution of smart contracts.

Ethical principles

Beliefs regarding what is good, acceptable, or obligatory behavior and what is bad, unacceptable, or forbidden behavior.

Ethics

The study of moral principles or of making good choices. Ethics encompasses a set of moral principles and rules of conduct that provide guidance for our behavior.

Eurobonds

A type of bond issued internationally, outside the jurisdiction of the country in whose currency the bond is denominated.

European options

Options that may be exercised only at contract maturity.

European-style

Said of an option contract that can only be exercised on the option's expiration date.

Event risk

Risk that evolves around set dates, such as elections, new legislation, or other date-driven milestones, such as holidays or political anniversaries, known in advance. Example: Brexit referendum.

Ex-dividend date

The first date that a share trades without (i.e., "ex") the right to receive the declared dividend for the period.

Excess kurtosis

Degree of kurtosis (fatness of tails) relative to the kurtosis of the normal distribution.

Excess spread

Surplus difference of yield remaining after payments to bondholders are made after expenses are made and losses are covered.

Exchange

A rules-based, open access market venue where financial instruments are traded, with price and volume transparency accessible by issuers, investors, and their intermediaries.

Exchange-traded derivative (ETD)

Futures, options, and other financial contracts available on exchanges.

Exchanges

Places where traders can meet to arrange their trades.

Execution instructions

Instructions that indicate how to fill an order.

Exercise

The decision to transact the underlying by an option holder.

Exercise date

The day that an option is exercised by its holder. For a call option, the day the strike price is paid and underlying is purchased. For a put option, when the strike price is received and the underlying is sold.

Exercise price

The pre-agreed execution price specified in an option contract. Sometimes, this price is referred to as the strike price.

Exogenous risk

A sudden or unanticipated risk that impacts either a country's cooperative stance, the ability of non-state actors to globalize, or both. Examples include sudden uprisings, invasions, or the aftermath of natural disasters.

Expansion

The period of a business cycle after its lowest point and before its highest point.

Expansionary

Tending to cause the real economy to grow.

Expansionary fiscal policy

Fiscal policy aimed at achieving real economic growth.

Expected exposure (EE)

The size of the investor's claim at the time of default

Expected loss (EL)

Default probability times loss severity given default.

Expected return on the portfolio

Denoted as $E(R_p)$. The weighted average of the expected returns (R_1 to R_n) on the component securities using their respective weights (w_1 to w_n).

Expected value of a random variable

The probability-weighted average of the possible outcomes of a random variable.

Expert system

A type of computer programming, often based on "if-then" rules, that attempts to simulate the knowledge base and analytical abilities of human experts in specific problem-solving contexts.

Export subsidy

Paid by the government to the firm when it exports a unit of a good that is being subsidized.

Exposure at default (EAD)

The size of the investor's claim at the time of default

Extension risk

The risk that the borrower repays the principal or a proportion of the principal in a longer time period than the contractually agreed scheduled payment.

External credit enhancements

Provisions or methods from a third party that allow a borrower improve their creditworthiness in a structured transaction..

External debt

Sovereign debt owed to foreign creditors.

Extra dividend

A dividend paid by a company that does not pay dividends on a regular schedule, or a dividend that supplements regular cash dividends with an extra payment.

Extraordinary general meetings (EGMs)

Meetings besides an AGM of the corporate board and shareholders, typically held to deliberate and vote on urgent matters. Corporate charters and bylaws specify who can call an EGM and under what conditions.

Extreme value theory

A branch of statistics that focuses primarily on extreme outcomes.

Face value

The amount of principal on a bond, also known as par value.

Factoring arrangement

When a company sells its accounts receivable to a lender (known as a factor) that assumes responsibility for the credit-granting and collection process.

Fair value

A market-based measure of an investment based on observable or derived assumptions to determine a price that market participants would use to exchange an asset or liability in an orderly transaction at a specific time.

Fair value hedge

Refers to a specific hedge accounting designation that applies when a derivative is deemed to offset the fluctuation in fair value of an asset or liability.

Fallen angels

Formerly investment-grade issuers whose credit quality has deteriorated since the time of issuance.

Fat-Tailed

Describes a distribution that has fatter tails than a normal distribution (also called leptokurtic).

Federal funds rate

The US interbank lending rate on overnight borrowings of reserves. Also known as *Fed Funds rate*.

Fiat money

Money that is not convertible into any other commodity.

Fiduciary call

A combination of a purchased call option and investment in a risk-free bond with face value of the option's exercise price.

Fill or kill

See *immediate or cancel order*.

Finance lease

A type of lease which is more akin to the purchase or sale of the underlying asset.

Financial leverage

The use of debt in the capital structure. Measured using ratios such as operating income to operating income less interest expense, total assets to total equity, or debt to equity.

Financial leverage ratio

A measure of financial leverage calculated as average total assets divided by average total equity.

Financial risk

The risk arising from a company's capital structure and, specifically, from the level of debt and debt-like obligations.

Fintech

Technological innovation in the financial services industry, specifically with the design and delivery of financial services and products. It may also refer more broadly to companies involved in developing the new technologies and their applications, as well as the business sector that includes such companies.

Firm commitment

A pre-determined amount (price and quantity) is agreed to be exchanged at settlement. Examples of firm commitments include forward contracts, futures contracts, and swaps.

First lien

Security interest in a property that gives the lender the right to seize the collateral if the borrower does not pay as agreed.

First lien debt

Debt secured by a pledge of certain assets that could include buildings, but it may also include property and equipment, licenses, patents, brands, etc.

First mortgage debt

Debt secured by a pledge of a specific property.

Fiscal multiplier

The ratio of a change in national income to a change in government spending.

Fiscal policy

The use of taxes and government spending to affect the level of aggregate expenditures.

Fixed charge coverage

A solvency ratio measuring the number of times interest and lease payments are covered by operating income, calculated as (EBIT + lease payments) divided by (interest payments + lease payments).

Fixed charge coverage ratio

A measure of how well a company's earnings covers its fixed expenses, which may include debt payments, interest expense, and lease costs.

Fixed-income instruments

Debt instruments such as loans or bonds.

Fixed-income securities

Fixed-income instruments designed to be more easily tradeable than a loan, such as a bond.

Fixed-price call

A contingency provision that grants an issuer the right to buy back a bond at a predetermined price in the future.

Fixed-rate payer

The counterparty paying fixed cash flows in a swap contract. May also be referred to as the floating-rate receiver.

Flat price

The full price of a bond minus accrued interest. Flat prices are usually quoted by bond dealers.

Float-adjusted market-capitalization weighting

An index weighting method in which the weight assigned to each constituent security is determined by adjusting its market capitalization for its market float.

Floating-rate notes (FRNs)

Notes on which interest payments are not fixed but instead vary from period to period depending on the current level of a reference interest rate. Also known as floaters.

Floating-rate payer

The counterparty paying the variable cash flows in a swap contract. May also be referred to as the fixed-rate receiver.

Forecast object

A variable on or related to an issuer's financial statements that an analyst makes a projection for. Examples include drivers of financial statements, financial statement lines, and summary measures like EBITDA.

Foreclosure

Allows a lender to take possession of the property and ultimately sell the property to recover funds toward satisfying the outstanding debt obligation.

Foreign bonds

A type of bond for which the issuer's domicile and jurisdiction of issuance are different.

Foreign currency reserves

Holding by the central bank of non-domestic currency deposits and non-domestic bonds.

Foreign direct investments (FDI)

Long-term investments in the productive capacity of a foreign country.

Foreign exchange gains (or losses)

Gains (or losses) that occur when the exchange rate changes between the investor's currency and the currency that foreign securities are denominated in.

Forward contract

A derivative contract for the future exchange of an underlying at a fixed price set at contract signing.

Forward price

Represents the price agreed upon in a forward contract to be exchanged at the contract's maturity date, T . This price is shown in equations as $F_0(T)$.

Forward price-to-earnings ratio

A P/E calculated on the basis of a forecast of EPS; a stock's current price divided by next year's expected earnings.

Forward rate agreement (FRA)

An OTC derivatives contract in which counterparties agree to apply a specific interest rate to a future time period.

Founders class shares

A way to entice early participation in startup funds whereby managers offer incentives that entitle investors to a lower fee structure and/or other favorable terms.

Framing bias

An information-processing bias in which a person answers a question differently based on the way in which it is asked (framed).

Franchising

A situation where an owner of an asset and associated intellectual property divests the asset and licenses intellectual property to a third-party operator (franchisee) in exchange for royalties. Franchisees operate under the constraints of a franchise agreement.

Free cash flow

The actual cash that would be available to the company's investors after making all investments necessary to maintain the company as an ongoing enterprise (also referred to as free cash flow to the firm); the internally generated funds that can be distributed to the company's investors (e.g., shareholders and bondholders) without impairing the value of the company.

Free cash flow hypothesis

The hypothesis that higher debt levels discipline managers by forcing them to make fixed debt service payments and by reducing the company's free cash flow.

Free float

The portion of a listed company's equity securities that are not held by insiders, strategic investors, sponsors, founders, and so on, that are more freely available for trading.

Free trade areas

One of the most prevalent forms of regional integration, in which all barriers to the flow of goods and services among members have been eliminated.

Free-cash-flow-to-equity models

Valuation models based on discounting expected future free cash flow to equity.

Freemium business model

A pricing approach that allows customers a certain level of usage or functionality at no charge. Those who wish to use more must pay.

Frequency table

A representation of the frequency of occurrence of two discrete variables.

Full price

The price of a bond including any accrued interest owed to the seller. It is the flat price plus accrued interest.

Fully amortizing loan

A loan or bond with a payment schedule that calls for the complete repayment of principal over the instrument's time to maturity.

Fund investing

In fund investing, the investor invests in assets indirectly by contributing capital to a fund as part of a group of investors. Fund investing is available for all major alternative investment types.

Fund of funds

Funds that hold a portfolio of hedge funds; also called *funds of hedge funds*.

Fundamental analysis

The examination of publicly available information and the formulation of forecasts to estimate the intrinsic value of assets.

Fundamental growth

These strategies use fundamental analysis to identify companies expected to exhibit high growth and capital appreciation.

Fundamental long/short

In this strategy, the hedge fund takes a long position in companies that are trading at inexpensive levels compared to their potential intrinsic value and shorts those that trade in the other direction, with the intention of reversing this trade to obtain alpha.

Fundamental value

These strategies use fundamental analysis to identify undervalued and unloved companies for which there is a possibility that a corporate turnaround, with future revenue and cash flow growth, will result in higher valuations.

Fundamental weighting

An index weighting method in which the weight assigned to each constituent security is based on its underlying company's size. It attempts to address the disadvantages of market-capitalization weighting by using measures that are independent of the constituent security's price.

Fungible

Freely exchangeable, interchangeable, or substitutable with other things of the same type. Money and commodities are the most common examples.

Futures contract

A variation of a forward contract that has essentially the same basic definition but with some additional features, such as a clearinghouse guarantee against credit losses, a daily settlement of gains and losses, and an organized electronic or floor trading facility.

Futures contract basis point value (BPV)

The change in price of a futures contract given a 1 basis point (0.01%) change in yield.

Futures contracts

Forward contracts with standardized sizes, dates, and underlyings that trade on futures exchanges.

Futures margin account

An account held by an exchange clearinghouse for each derivatives counterparty. The funds in such an account are used to ensure that counterparties do not default on their contract obligation.

Futures price

The pre-agreed price at which a futures contract buyer (seller) agrees to pay (receive) for the underlying at the maturity date of the futures contract.

FX swap

The combination of a spot and a forward FX transaction.

G-spread

Yield spread in basis points between a bond's yield-to-maturity and that of an actual or interpolated government bond. It represents the return for bearing risks relative to the government bond.

Game theory

The set of tools decision makers use to incorporate responses by rival decision makers into their strategies.

Gamma

A numerical measure of how sensitive an option's delta (the sensitivity of the derivative's price) is to a change in the value of the underlying.

Gate

A provision that when implemented limits or restricts redemptions for a period of time.

General collateral repo

Rather than involving a specific security, a repo that instead references a specific group of securities as eligible collateral (such as government bonds of a specific maturity).

General collateral repo rate

The interest rate on a general collateral repo.

General obligation (GO) bonds

Unsecured bonds issued by a non-sovereign government which are backed by the taxing authority of the issuer.

General partners (GPs)

Owners of a general partnership or limited partnership with unlimited liability and other attributes as specified in the partnership agreement.

General partnership

A business organizational form owned entirely by general partners.

Geophysical resource endowment

Includes such factors as livable geography and climate as well as access to food and water, which are necessary for sustainable growth. Geophysical resource endowment is highly unequal among countries.

Geopolitics

The study of how geography affects politics and international relations. These relations matter for investments because they contribute to important drivers of investment performance, including economic growth, business performance, market volatility, and transaction costs.

Gilts

Bonds issued by the UK government.

Global depository receipt (GDR)

A depository receipt that is issued outside of the company's home country and outside of the United States.

Global minimum-variance portfolio

The portfolio on the minimum-variance frontier with the smallest variance of return.

Global registered share (GRS)

A common share that is traded on different stock exchanges around the world in different currencies.

Globalization

The process of interaction and integration among people, companies, and governments worldwide. It is marked by the spread of products, information, jobs, and culture across borders.

Gold standard

With respect to a currency, if a currency is on the gold standard a given amount can be converted into a prespecified amount of gold.

Good-on-close

An execution instruction specifying that an order can only be filled at the close of trading. Also called *market-on-close*.

Good-on-open

An execution instruction specifying that an order can only be filled at the opening of trading.

Good-till-cancelled order

An order specifying that it is valid until the entity placing the order has cancelled it (or, commonly, until some specified amount of time such as 60 days has elapsed, whichever comes sooner).

Goodwill

An intangible asset that represents the excess of the purchase price of an acquired company over the value of the net identifiable assets acquired.

Governance tokens

In permissionless networks, governance tokens serve as votes to determine how the particular network is run.

Government debt management

Government policies that relate to the issuance of debt securities, typically handled by a treasurer or finance ministry.

Government equivalent yield

Measures quoted using actual/actual day counts.

Grant date

The day that terms of compensation are communicated by an issuer and accepted by an employee recipient.

Green bonds

Bonds used in green finance whereby the proceeds are earmarked toward environmental-related products.

Greenfield investments

The first stage of development of an infrastructure asset. Greenfield investments involve developing new assets and new infrastructure with the intention either to lease or sell the assets to the government after construction or to hold and operate the assets. Greenfield investors typically invest alongside strategic investors or developers that specialize in developing the underlying assets.

Gross profit margin

The ratio of gross profit to revenues.

Groupthink

The practice of thinking or making decisions as a group in a way that discourages creativity or individual responsibility. For scenario analysis to be useful in portfolio management, teams must work hard to build creative processes, identify scenarios, track these scenarios, and assess the need for action on a regular cadence.

Growth cycle

Refers to fluctuations in economic activity around the long-term potential trend growth level, focusing on how much actual economic activity is below or above trend growth in economic activity.

Growth option

The option to make additional investments in a project at some future time if the financial results are strong. Also called an *expansion option*.

Growth rate cycle

Refers to fluctuations in the growth rate of economic activity.

Haircut

The difference between the market value of the security used as collateral and the value of the loan. Also called *repo margin*.

Halo effect

An emotional bias that extends a favorable evaluation of some characteristics to other characteristics.

Hard commodities

Traded natural resources, such as crude oil and metals, with markets often involving the physical delivery of the underlying upon settlement.

Hard hurdle rate

Hurdle rate where the manager earns fees on annual returns in excess of the hurdle rate

Hard-bullet covered bonds

Type of security where if payments do not occur according to the original schedule of a covered bond, a bond default is triggered and bond payments are accelerated.

Harmonic mean

A type of weighted mean computed as the reciprocal of the arithmetic average of the reciprocals.

Hedge

The derivative contract used in hedging an exposure.

Hedge accounting

Accounting standard(s) that allow an issuer to offset a hedging instrument (usually a derivative) against a hedged transaction or balance sheet item to reduce financial statement volatility.

Hedge funds

Private investment vehicles that may invest in public equities or publicly traded fixed-income assets, private capital, and/or real assets, but they are distinguished by their investment *approach* rather than by the investments themselves.

Hedge ratio

The proportion of an underlying that will offset the risk associated with a derivative position.

Hedging

The use of a derivative contract to offset or neutralize existing or anticipated exposure to an underlying.

Hegemony

Countries that are regional or even global leaders and use their political or economic influence of others to control resources.

Held-to-maturity

Debt (fixed-income) securities that a company intends to hold to maturity; these are presented at their original cost, updated for any amortisation of discounts or premiums.

Herding

Clustered trading that may or may not be based on information.

Herfindahl-Hirschman Index (HHI)

A measure of market concentration, calculated as the sum of the squares of competitor market shares. Antitrust regulators in some countries consider markets with an HHI between 1,500 and 2,500 moderately concentrated and consider markets with an HHI over 2,500 highly concentrated.

Heteroskedasticity

Non-constant variance across all observations.

Hidden order

An order that is exposed not to the public but only to the brokers or exchanges that receive it.

Hidden revenue business model

Business models that provide services to users at no charge and generate revenues elsewhere.

High yield

Bond issuers and issues rated BB+ (Ba1 on Moody's scale) or lower. Also known as speculative grade and junk.

High-water mark

The highest value, net of fees, that a fund has reached in history. It reflects the highest cumulative return used to calculate an incentive fee.

Hindsight bias

A bias with selective perception and retention aspects in which people may see past events as having been predictable and reasonable to expect.

Holder-of-record date

The date that a shareholder listed on the corporation's books will be deemed to have ownership of the shares for purposes of receiving an upcoming dividend.

Holding period return

The return that an investor earns during a specified holding period; a synonym for total return.

Home bias

A preference for securities listed on the exchanges of one's home country.

Homogeneity of expectations

The assumption that all investors have the same economic expectations and thus have the same expectations of prices, cash flows, and other investment characteristics.

Homoskedasticity

Constant variance across all observations.

Horizon yield

An investor's total rate of return on a fixed income instrument over their holding period, including reinvested coupon payments. It is an internal rate of return expressed as an annualized rate.

Hostile takeover

When a potential acquirer seeks to acquire a company (the target) against the wishes of the target's board of directors. Typically, a tender offer is used to carry out the hostile takeover, against which a board might use a poison pill in its defense.

Household

A person or a group of people living in the same residence, taken as a basic unit in economic analysis.

Human capital

The accumulated knowledge and skill that workers acquire from education, training, or life experience and the corresponding present value of future earnings to be generated by said skilled individual.

Hurdle rate

Also called "preferred return." The minimum rate of return on investment that a fund must reach before a GP receives carried interest.

Hypothesis

A proposed explanation or theory that can be tested.

Hypothesis testing

The process of testing of hypotheses about one or more populations using statistical inference.

I-spread

Also known as interpolated spread, it is the yield spread for a bond over the standard swap rate in that currency of the same tenor.

Iceberg order

An order in which the display size is less than the order's full size.

If-converted method

A method for accounting for the effect of convertible securities on earnings per share (EPS) that specifies what EPS would have been if the convertible securities had been converted at the beginning of the period, taking account of the effects of conversion on net income and the weighted average number of shares outstanding.

Illusion of control bias

A bias in which people tend to believe that they can control or influence outcomes when, in fact, they cannot.

Immediate or cancel order

An order that is valid only upon receipt by the broker or exchange. If such an order cannot be filled in part or in whole upon receipt, it cancels immediately. Also called *fill or kill*.

Impact lag

The lag associated with the result of actions affecting the economy with delay.

Implied forward rate (IFR)

An interest rate or yield over a future period implied by the current term structure of interest rates.

Import license

Specifies the quantity of a good that can be imported into a country.

In-the-money

Describes an option with a positive intrinsic value.

Income tax paid

The actual amount paid for income taxes in the period; not a provision, but the actual cash outflow.

Income tax payable

The income tax owed by the company on the basis of taxable income.

Increasing returns to scale

When a production process leads to increases in output that are proportionately larger than the increase in inputs.

Incurrence test

A financial ratio or other measurement taken prior to an action such as debt issuance, usually on a pro forma basis taking the action into account. Satisfaction of the test (e.g., leverage ratio below a certain value) is linked to covenants between the issuer and investors.

Indenture

A written contract between a lender and borrower that specifies the terms of the loan, such as interest rate, interest payment schedule, or maturity.

Independent

With reference to events, the property that the occurrence of one event does not affect the probability of another event occurring. With reference to two random variables X and Y , they are independent if and only if $P(X, Y) = P(X)P(Y)$.

Independent directors

Members of a corporation's board of directors who do not have an employment or familial relationship with the company, nor do they have a relationship that would impair their independence such as an economic interest in a vendor or competitor of the company.

Independent variable

An explanatory variable in a regression model.

Independently and identically distributed

With respect to random variables, the property of random variables that are independent of each other but follow the identical probability distribution.

Index-linked bonds

A bond whose coupon payments or principal repayment is linked to a specified index.

Indexing

An investment strategy in which an investor constructs a portfolio to mirror the performance of a specified index.

Indicator variable

A variable that takes on only one of two values, 0 or 1, based on a condition. In simple linear regression, the slope is the difference in the dependent variable for the two conditions. Also referred to as a *dummy variable*.

Indifference curve

A curve representing all the combinations of two goods or attributes such that the consumer is entirely indifferent among them.

Indirect taxes

Taxes such as taxes on spending, as opposed to direct taxes.

Inflation premium

An extra return that compensates investors for expected inflation.

Inflation reports

A type of economic publication put out by many central banks.

Inflation-linked bonds

A type of index-linked bond that offer investors protection against inflation by linking the bonds' coupon payments and/or the principal repayment to an index of consumer prices. Also called *linkers*.

Information cascade

The transmission of information from those participants who act first and whose decisions influence the decisions of others.

Information-motivated traders

Traders that trade to profit from information that they believe allows them to predict future prices.

Informationally efficient market

A market in which asset prices reflect new information quickly and rationally.

Infrastructure

A type of real asset that is intended for public use and provides essential services. These assets are typically long-lived fixed assets, such as bridges and toll roads.

Initial coin offering (ICO)

An unregulated process whereby companies raise capital by selling crypto-tokens to investors in exchange for fiat money or another agreed-upon cryptocurrency.

Initial margin

The ratio of the price of collateral to the value of cash exchanged in a repo; a value over 1.0 or 100% indicates overcollateralization.

Initial margin requirement

The margin requirement on the first day of a transaction as well as on any day in which additional margin funds must be deposited.

Initial public offering (IPO)

The first issuance of common shares to the public by a formerly private corporation.

Inside directors

Members of a corporation's board of directors who are not independent. Typically, inside directors are employees or founders (and their family) of the company.

Insolvency

Refers to the condition in which firm value is below the face value of debt used to finance the firm's assets.

Institution

An established organization or practice in a society or culture. An institution can be a formal structure, such as a university, organization, or process backed by law; or it can be informal, such as a custom or behavioral pattern important to society. Institutions can, but need not be, formed by national governments. Examples of institutions include non-governmental organizations, charities, religious customs, family units, the media, political parties, and educational practice.

Intangible assets

Assets without a physical form, such as patents and trademarks.

Interbank market

The market of loans and deposits between banks for maturities ranging from overnight to one year.

Intercept

The estimated value of the dependent variable when the independent variable is zero.

Interest coverage

A solvency ratio calculated as EBIT divided by interest payments.

Interest coverage ratio

A measure of an issuer's ability to service its debt, typically the ratio of operating income or EBIT to interest expense.

Interest rate

A rate of return that reflects the relationship between differently dated cash flows; a discount rate.

Interest rate swap

A swap in which the underlying is an interest rate. Can be viewed as a currency swap in which both currencies are the same and can be created as a combination of currency swaps.

Interest-indexed bond

A type of index-linked bond for which changes in the index are captured with adjustments to interest payments.

Internal credit enhancements

Provisions or methods a borrower initiates to improve their creditworthiness in a structured transaction, such as overcollateralization or excess spread.

Internal rate of return (IRR)

The discount rate that makes net present value equal 0; the discount rate that makes the present value of an investment's costs (outflows) equal to the present value of the investment's benefits (inflows).

Internet of things

The vast array of physical devices, home appliances, smart buildings, vehicles, and other items that are embedded with electronics, sensors, software, and network connections that enable the objects in the system to interact and share information.

Interquartile range

The difference between the third and first quartiles of a dataset.

Intrinsic value

The amount gained (per unit) by an option buyer if an option is exercised at any given point in time. May be referred to as the exercise value of the option.

Investment banks

Financial intermediaries that provide advice to their mostly corporate clients and help them arrange transactions such as initial and seasoned securities offerings.

Investment grade

Bond issuers and issues rated BBB- (Baa3 on Moody's scale).

Investment policy statement

A written planning document that describes a client's investment objectives and risk tolerance over a relevant time horizon, along with the constraints that apply to the client's portfolio.

Issue rating

A rating which seeks to capture the probability of default or expected loss of the issuer's senior unsecured bonds

Issuer rating

A rating which seeks to capture the credit risk of a specific financial obligation of an issuer which takes such factors as seniority into account.

J-curve effect

Represents the initial negative return in the capital commitment phase followed by an acceleration of returns through the capital deployment phase.

Jackknife

A resampling method that repeatedly draws samples by taking the original observed data sample and leaving out one observation at a time (without replacement) from the set.

January effect

Calendar anomaly that stock market returns in January are significantly higher compared to the rest of the months of the year, with most of the abnormal returns reported during the first five trading days in January. Also called *turn-of-the-year effect*.

Joint probability function

A function giving the probability of joint occurrences of values of stated random variables.

Judgmental sampling

A procedure of selectively handpicking elements from the population based on a researcher's knowledge and professional judgment.

Junior debt

Debt obligation with lower priority of payment than senior debt obligations.

Key rate duration

A method of measuring interest rate sensitivities of a fixed-income instrument or portfolio to shifts in key points along the yield curve.

Keynesians

Economists who believe that fiscal policy can have powerful effects on aggregate demand, output, and employment when there is substantial spare capacity in an economy.

Kurtosis

The statistical measure that indicates the combined weight of the tails of a distribution relative to the rest of the distribution.

Lagging economic indicators

Turning points that take place later than those of the overall economy; they are believed to have value in identifying the economy's past condition.

Law of one price

A principle that states that if two investments have the same or equivalent future cash flows regardless of what will happen in the future, then these two investments should have the same current price.

Lead underwriter

The lead investment bank in a syndicate of investment banks and broker-dealers involved in a securities underwriting.

Leading economic indicators

Turning points that usually precede those of the overall economy; they are believed to have value for predicting the economy's future state, usually near-term.

Legal tender

Something that must be accepted when offered in exchange for goods and services.

Lender of last resort

An entity willing to lend money when no other entity is ready to do so.

Leptokurtic

Describes a distribution that has fatter tails than a normal distribution (also called fat-tailed).

Lessee

The party obtaining the use of an asset through a lease.

Lessor

The owner of an asset that grants the right to use the asset to another party.

Level of significance

The probability of a Type I error in testing a hypothesis.

Leverage

A measure for identifying a potentially influential high-leverage point.

Leveraged buyout

A transaction whereby the target company management team converts the target to a privately held company by using heavy borrowing to finance the purchase of the target company's outstanding shares.

Leveraged buyout (LBO)

An acquirer (typically an investment fund specializing in LBOs) uses a significant amount of debt to finance the acquisition of a target and then pursues restructuring actions, with the goal of exiting the target with a sale or public listing.

Leveraged buyouts

Transactions whereby the target company's management team converts the target to a privately held company by using heavy borrowing to finance the purchase of the target company's outstanding shares.

Leveraged loan

Where private debt investor firms borrow money to make a direct loan to a borrower.

Leveraged loans

Loans made to a borrower or issuer with relatively lower credit quality and/or higher leverage.

Liability-driven investing

An investment industry term that generally encompasses asset allocation that is focused on funding an investor's liabilities in institutional contexts.

Licensing arrangements

Rights to produce a product or have access to intangible assets using someone else's brand name in return for a royalty (often a percentage of revenues).

Lien

A legal right or claim to property by a creditor.

Likelihood

The probability of an observation, given a particular set of conditions.

Limit order

Instructions to a broker or exchange to obtain the best price immediately available when filling an order, but in no event accept a price higher than a specified (limit) price when buying or accept a price lower than a specified (limit) price when selling.

Limit order book

The book or list of limit orders to buy and sell that pertains to a security.

Limited company

A business organizational form owned by shareholders or members with limited liability who elect a board of directors to appoint management. Generally, limited companies have indefinite life and easier transfer of ownership interests than limited partnerships.

Limited liability partnership (LLP)

A business organizational form available in some jurisdictions owned entirely by limited partners with limited liability.

Limited partners (LPs)

Owners of a limited partnership with limited liability and other attributes as specified in the partnership agreement.

Limited partnership

A business organizational form owned by a general partner and limited partners.

Limited partnership agreement (LPA)

A legal document that outlines the rules of the partnership and establishes the framework that ultimately guides the fund's operations throughout its life.

Lin-log model

A functional form for transforming regression model data in which the dependent variable is linear but the independent variable is logarithmic.

Linear derivatives

Firm commitment derivative contracts in which the contract's payoff/profit function is linear with respect to the price of the underlying.

Liquid market

Said of a market in which traders can buy or sell with low total transaction costs when they want to trade.

Liquidity

The extent to which a company is able to meet its short-term obligations using cash flows and those assets that can be readily transformed into cash.

Liquidity premium

An extra return that compensates investors for the risk of loss relative to an investment's fair value if the investment needs to be converted to cash quickly.

Liquidity ratios

Financial ratios measuring the company's ability to meet its short-term obligations to creditors as they come due.

Liquidity risk

A divergence in the cash flow timing of a derivative versus that of an underlying transaction.

Liquidity trap

A condition in which the demand for money becomes infinitely elastic (horizontal demand curve) so that injections of money into the economy will not lower interest rates or affect real activity.

Load fund

A mutual fund in which, in addition to the annual fee, a percentage fee is charged to invest in the fund and/or for redemptions from the fund.

Loan-to-value ratio (LTV)

Ratio of the amount of the mortgage to the property's value. The lower the LTV, the higher the borrower's equity. From the lender's perspective, the higher the borrower's equity, the less likely the borrower is to default.

Loans

Debt instruments agreed to between a borrower and lender, typically a bank.

Lockout or revolving period

For an ABS with a non-amortizing collateral pool, such as credit card debt, is the period in which the cash proceeds from principal repayments are reinvested in additional loans with a principal equal to the principal repaid. During this period, there is no prepayment risk and potential default risk is generally limited. When the lockout period is over, principal repayments are used to pay off the outstanding principal on the ABS. Lockout period and revolving period are interchangeable.

Lockup period

The minimum holding period before investors are allowed to make withdrawals or redeem shares from a fund. Its purpose is to allow the hedge fund manager the required time to implement and potentially realize a strategy's expected results.

Log-lin model

A functional form for transforming regression model data in which the dependent variable is logarithmic but the independent variable is linear.

Log-log model

A functional form for transforming regression model data in which both the dependent and independent variables are in logarithmic form.

Long

A trading position in a derivative contract that gains value as the price of the underlying moves higher.

Long position

A position in an asset or contract in which one owns the asset or has an exercisable right under the contract.

Long-run average total cost

The curve describing average total cost when no costs are considered fixed.

Loss aversion

The tendency of people to dislike losses more than they like comparable gains.

Loss given default (LGD)

The investor's loss conditional on an issuer event of default.

Loss severity

Portion of a bond's value (including unpaid interest) an investor loses in the event of default.

Loss-aversion bias

A bias in which people tend to strongly prefer avoiding losses as opposed to achieving gains.

Low-cost producer

A firm with lower production costs than its industry competitors.

M²

A measure of what a portfolio would have returned if it had taken on the same total risk as the market index.

M² alpha

Difference between the risk-adjusted performance of the portfolio and the performance of the benchmark.

Macauley duration

The present-value weighted average time to receipt of cash flows for fixed-income instrument, also the holding period needed to balance coupon reinvestment risk and price risk for a one-time instantaneous "parallel" shift in the yield curve once the bond purchase is settled. It is named after Frederick Macauley, the Canadian economist who introduced the concept in 1938.

Machine learning (ML)

Involves computer-based techniques that seek to extract knowledge from large amounts of data without making any assumptions about the data's underlying probability distribution. The goal of ML algorithms is to automate decision-making processes by generalizing, or "learning," from known examples to determine an underlying structure in the data.

Maintenance capital expenditures

Investments in assets to keep them in operation or increase their efficiency without extending their useful lives.

Maintenance margin

Minimum balance set below the initial margin that each contract buyer and seller must hold in the futures margin account from trade initiation until final settlement at maturity.

Maintenance margin requirement

The margin requirement on any day other than the first day of a transaction.

Management buy-in

A type of leveraged buyout where the current management team is replaced with the acquiring team involved in managing the company.

Management buyout

A type of leveraged buyout where the current management team participates in the acquisition.

Management guidance

Management of public companies may publicly provide targets for earnings, revenues, and other measures (e.g., capital expenditures) for the next quarter, year, or longer term. Guidance can be detailed or rather directional and is often updated throughout the year. Initial guidance for next fiscal year might be provided during the fourth-quarter earnings call and updated for completed quarters, and new information provided at the first-, second-, and third-quarter earnings calls. Also known simply as *guidance*.

Margin call

Request to a derivatives contract counterparty to immediately deposit funds to return the futures margin account balance to the initial margin.

Margin financing

A financing arrangement whereby the prime broker lends shares, bonds, or derivatives and the hedge fund (or investment manager) deposits cash or other collateral into a margin account at the prime broker based on certain fractions of the investment positions.

Margin loan

Money borrowed from a broker to purchase securities.

Marginal propensity to consume

The proportion of an additional unit of disposable income that is consumed or spent; the change in consumption for a small change in income.

Marginal propensity to save

The proportion of an additional unit of disposable income that is saved (not spent).

Mark to market (MTM)

The practice in which a central clearing party assigns profits and losses to counterparties to derivative contracts. In exchange-traded markets, this practice takes place daily and is often referred to as daily settlement.

Market anomaly

Change in the price or return of a security that cannot directly be linked to current relevant information known in the market or to the release of new information into the market.

Market bid–ask spread

The difference between the best bid and the best offer.

Market discount rate

The rate of return required by investors given the risk of the bond investment, also known as the required yield or required rate of return.

Market float

The number of shares that are available to the investing public.

Market makers

Over-the-counter (OTC) dealers who typically enter into offsetting bilateral transactions with one another to transfer risk to other parties.

Market model

A regression model with the return on a stock as the dependent variable and the returns on a market index as the independent variable.

Market multiple models

Valuation models based on share price multiples or enterprise value multiples.

Market neutral

These strategies use quantitative, fundamental, and technical analysis to identify under- and overvalued equity securities. The hedge fund takes long positions in undervalued securities and short positions in overvalued securities, while seeking to maintain a market-neutral net position.

Market order

Instructions to a broker or exchange to obtain the best price immediately available when filling an order.

Market reference rate

A market-determined interest rate used as the underlying in financial instruments and contracts such as variable-rate debt and interest rate swaps. An example is the Secured Overnight Financing Rate (SOFR), which is an overnight cash borrowing rate collateralized by US Treasuries. Other MRRs include the euro short-term rate (€STR) and the Sterling Overnight Index Average (SONIA).

Market reference rate (MRR)

The interest rate underlying used in interest rate swaps. These rates typically match those of loans or other short-term obligations. Survey-based Libor rates used as reference rates in the past have been replaced by rates based on a daily average of observed market transaction rates. For example, the Secured Overnight Financing Rate (SOFR) is an overnight cash borrowing rate collateralized by US Treasuries. Other MRRs include the euro short-term rate (€STR) and the Sterling Overnight Index Average (SONIA).

Market risk

The risk that arises from movements in interest rates, stock prices, exchange rates, and commodity prices.

Market share

A company's or product's revenue expressed as a percentage of its market size.

Market size

Total sales for a good or service, which can be calculated on a global or more regional basis.

Market value

The price at which an asset or security can currently be bought or sold in an open market.

Market-capitalization weighting

An index weighting method in which the weight assigned to each constituent security is determined by dividing its market capitalization by the total market capitalization (sum of the market capitalization) of all securities in the index. Also called *value weighting*.

Market-on-close

An execution instruction specifying that an order can only be filled at the close of trading.

Marketable limit order

A buy limit order in which the limit price is placed above the best offer, or a sell limit order in which the limit price is placed below the best bid. Such orders generally will partially or completely fill right away.

Markowitz efficient frontier

The graph of the set of portfolios offering the maximum expected return for their level of risk (standard deviation of return).

Master limited partnership (MLP)

Has similar features to limited partnerships but is usually a more liquid investment that is often publicly traded.

Master repurchase agreement

A legal document governing all repo trades between two parties.

Match funding

Financing an asset with a source, such as a loan or bond, that is aligned with certain attributes of the asset, such as duration and the respective streams of income and financing costs.

Material

(materiality) Refers to information that is decision-useful for a reasonable investor.

Matrix pricing

An estimation process for financial instruments based on the prices of comparable instruments.

Maturity

The date of a fixed-income instrument's final payment to investors.

Maturity premium

An extra return that compensates investors for the increased sensitivity of the market value of debt to a change in market interest rates as maturity is extended.

Maturity structure of interest rates

Also known as the term structure of interest rates, refers to the difference in interest rates or benchmark yields by time-to-maturity.

Mean absolute deviation

With reference to a sample, the mean of the absolute values of deviations from the sample mean.

Mean square error (MSE)

Calculated as the sum of squares error (SSE) divided by the degrees of freedom, which are the number of observations minus the number of independent variables minus one. Since simple linear regression has just one independent variable, the degrees of freedom calculation is the number of observations minus 2.

Mean square regression (MSR)

Calculated as the sum of squares regression (SSR) divided by the number of independent variables in the regression model. In simple linear regression, there is only one independent variable, so MSR equals SSR.

Mean-variance analysis

An approach to portfolio analysis using expected means, variances, and covariances of asset returns.

Measure of central tendency

A quantitative measure that specifies where data are centered.

Measures of location

Quantitative measures that describe the location or distribution of data. They include not only measures of central tendency but also other measures, such as percentiles.

Median

The value of the middle item of a set of items that has been sorted into ascending or descending order (i.e., the 50th percentile).

Meme coin

A type of altcoin that is often inspired by a joke.

Mental accounting bias

An information-processing bias in which people treat one sum of money differently from another equal-sized sum based on which mental account the money is assigned to.

Merger arbitrage

Generally, these strategies involve going long (buying) the stock of the company being acquired at a discount to its announced takeover price and going short (selling) the stock of the acquiring

company when the merger or acquisition is announced.

Mesokurtic

Describes a distribution with kurtosis equal to that of the normal distribution, namely, kurtosis equal to three.

Mezzanine debt

Refers to private credit subordinated to senior secured debt but senior to equity in the borrower's capital structure.

Mezzanine-stage financing

Mezzanine venture capital that prepares a company to go public as it continues to expand capacity and enhance its growth trajectory. It represents the bridge financing needed to fund a private firm until it can execute an IPO or be sold.

Miner

A validator of transactions on the blockchain that locks blocks of transactions into the blockchain and receives compensation for this process in the form of a digital asset.

Minimum efficient scale

The smallest output that a firm can produce such that its long-run average total cost is minimized.

Minimum-variance portfolio

The portfolio with the minimum variance for each given level of expected return.

Minority shareholder

An individual or entity that owns less than a majority of the voting rights in a corporation.

Mode

The most frequently occurring value in a distribution.

Modern portfolio theory (MPT)

The analysis of rational portfolio choices based on the efficient use of risk.

Modified duration

The first derivative of a bond's price with respect to its yield, this statistic is a measure of interest rate risk used to estimate the percentage price change for a given change in yield-to-maturity.

Monetarists

Economists who believe that the rate of growth of the money supply is the primary determinant of the rate of inflation.

Monetary policy

Actions taken by a nation's central bank to affect aggregate output and prices through changes in bank reserves, reserve requirements, or its target interest rate.

Monetary transmission mechanism

The process whereby a central bank's interest rate gets transmitted through the economy and ultimately affects the rate of increase of prices.

Monetary union

An economic union in which the members adopt a common currency.

Money convexity

A measure that is used to complement modified duration to capture the second-order effect of yield changes on a bond's price, expressed in currency terms.

Money duration

A measure of the price change of a fixed-income instrument in currency units from a change in yield-to-maturity. The money duration can be stated per 100 of par value or in terms of the actual position size. In the United States, money duration is commonly called "dollar duration."

Money market

The market for short-term debt instruments (one-year maturity or less).

Money market securities

Fixed-income securities with original maturities of one year or less.

Money-weighted return

The internal rate of return on a portfolio, taking account of all cash flows.

Moneyneess

Expresses the relationship between an option's value and its exercise price across the full range of possible underlying prices.

Monopolistic competition

Highly competitive form of imperfect competition; the competitive characteristic is a notably large number of firms, while the monopoly aspect is the result of product differentiation.

Monopoly

In pure monopoly markets, there are no substitutes for the given product or service. There is a single seller, which exercises considerable power over pricing and output decisions.

Monte Carlo simulation

A technique that uses the inverse transformation method for converting a randomly generated uniformly distributed number into a simulated value of a random variable of a desired distribution. Each key decision variable in a Monte Carlo simulation requires an assumed statistical distribution; this assumption facilitates incorporating non-normality, fat tails, and tail dependence as well as solving high-dimensionality problems.

Moral principles

Beliefs regarding what is good, acceptable, or obligatory behavior and what is bad, unacceptable, or forbidden behavior.

Mortgage loan

Agreement to finance real estate by the collateral of a specified property that obliges the borrower to make a predetermined series of payments to the lender.

Mortgage pass-through security

Security created when mortgage lenders pool mortgages together and sell securities to investors. The cash flow from the mortgage pool—monthly payments of principal, interest, and prepayments—are “passed through” to the security holders.

Mortgage-backed securities

Debt obligations that represent claims to the cash flows from pools of mortgage loans, most commonly on residential property.

Mortgage-backed securities (MBS)

Bonds created from the securitization of mortgages.

Multi-factor model

A model that explains a variable in terms of the values of a set of factors.

Multi-market indexes

Comprised of indexes from different countries, designed to represent multiple security markets.

Multilateral trading facilities

See *alternative trading systems*.

Multilateralism

The conduct of countries who participate in mutually beneficial trade relationships and extensive rules harmonization. Private firms are fully integrated into global supply chains with multiple trade partners. Examples of multilateral countries include Germany and Singapore.

Multiple of invested capital (MOIC)

A simplified calculation that measures the total value of all distributions and residual asset values relative to an initial total investment; also known as a *money multiple*.

Multiple-price auction

A debt securities auction in which bidders receive distinct prices based on their bids.

Multiplier models

Valuation models based on share price multiples or enterprise value multiples.

Mutual fund

A comingled investment pool in which investors in the fund each have a pro-rata claim on the income and value of the fund.

Nash equilibrium

When two or more participants in a non-cooperative game have no incentive to deviate from their respective equilibrium strategies given their opponent's strategies.

Nationalism

The promotion of a country's own economic interests to the exclusion or detriment of the interests of other nations. Nationalism is marked by limited economic and financial cooperation. These actors may focus on national production and sales, limited cross-border investment and capital flows, and restricted currency exchange.

Natural language processing (NLP)

A field of research within the field of text analytics and at the intersection of computer science, AI, and linguistics that focuses on developing computer programs to analyze and interpret human language.

Natural resources

These include commodities (hard and soft), agricultural land (farmland), and timberland.

Negative externalities

A cost to a third party because of the production or consumption of a good or service.

Negative pledge clause

Limitations on investments, the disposal of assets, or issuance of debt senior to existing obligations. Negative covenants seek to ensure that an issuer maintains the ability to make interest and principal payments.

Net cash

An issuer's total debt less cash and marketable securities. When the balance is negative it is referred to as net cash.

Net debt

An issuer's total debt less cash and marketable securities. When the balance is positive it is referred to as net debt.

Net investment hedge

Refers to a specific hedge accounting designation that applies when either a foreign currency bond or a derivative, such as an FX swap or forward, is used to offset the exchange rate risk of the equity of a foreign operation.

Net present value (NPV)

The present value of an investment's cash inflows (benefits) minus the present value of its cash outflows (costs).

Net profit margin

An indicator of profitability, calculated as net income divided by revenue; indicates how much of each dollar of revenues is left after all costs and expenses. Also called *profit margin* or *return on sales*.

Net tax rate

The tax rate net of transfer payments.

Net working capital

Working capital excluding short-term items unrelated to business operations, such as cash, marketable securities, and short-term debt.

Network effects

A business model that enables users to contribute directly to a product, service, or online content.

Neural networks

A type of computer program design based on how the human brain learns and processes information.

Neutral rate of interest

The rate of interest that neither spurs on nor slows down the underlying economy.

No-load fund

A mutual fund in which there is no fee for investing in the fund or for redeeming fund shares, although there is an annual fee based on a percentage of the fund's net asset value.

Node

Each value on a binomial tree from which successive moves or outcomes branch.

Non-agency RMBS

MBS backed by residential mortgages that are issued by private entities and not guaranteed by a federal agency or a GSE.

Non-amortizing loans

Type of debt where there are no scheduled principal repayments.

Non-cooperative country

A country with inconsistent and even arbitrary rules; restricted movement of goods, services, people, and capital across borders; retaliation; and limited technology exchange.

Non-cumulative preference shares

Preference shares for which dividends that are not paid in the current or subsequent periods are forfeited permanently (instead of being accrued and paid at a later date).

Non-financial risks

Risks that arise from sources other than changes in the external financial markets, such as changes in accounting rules, legal environment, or tax rates.

Non-fungible token (NFT)

A unique cryptographic token on the blockchain that cannot be replicated and is used to represent ownership of physical assets, such as artwork, real estate, or other assets.

Non-linear derivatives

Derivatives, such as options or other contingent claims, with payoff/profit profiles that are non-linear (asymmetric) with respect to the price of the underlying.

Non-participating preference shares

Preference shares that do not entitle shareholders to share in the profits of the company. Instead, shareholders are only entitled to receive a fixed dividend payment and the par value of the shares in the event of liquidation.

Non-probability sampling

A sampling plan dependent on factors other than probability considerations, such as a sampler's judgment or the convenience to access data.

Non-recourse loan

Loan in which the lender does not have a claim against the borrower and thus can look only to the property to recover the outstanding mortgage balance.

Non-state actors

Those that participate in global political, economic, or financial affairs but do not directly control national security or country resources. Examples of non-state actors are non-governmental organizations (NGOs), multinational companies, charities, and even influential individuals, such as business leaders or cultural icons.

Nonparametric test

A test that is not concerned with a parameter or that makes minimal assumptions about the population from which a sample comes.

Nonsystematic risk

Unique risk that is local or limited to a particular asset or industry that need not affect assets outside of that asset class.

Normal distribution

A continuous, symmetric probability distribution that is completely described by its mean and its variance.

Normalized earnings

The expected level of mid-cycle earnings for a company in the absence of any unusual or temporary factors that affect profitability (either positively or negatively).

Notching

Ratings adjustment methodology where specific issues from the same borrower may be assigned different credit ratings.

Notice period

The length of time (typically 30–90 days) in advance that investors may be required to notify a fund of their intent to redeem some or all of their investment. This allows a fund manager to liquidate a position in an orderly fashion without magnifying losses.

Novation process

A process that substitutes the initial swap execution facility (SEF) contract with identical trades facing the central counterparty (CCP). The CCP serves as counterparty for both financial intermediaries, eliminating bilateral counterparty credit risk and providing clearing and settlement services.

Null hypothesis

The hypothesis that is tested.

Off-the-run

Seasoned government bonds that are often less liquid.

Off-the-run securities

Sovereign debt securities outstanding other than on-the-run securities. Off-the-run securities are less liquid than on-the-run securities.

Offer

The price at which a dealer or trader is willing to sell an asset, typically qualified by a maximum quantity (ask size).

Official interest rate

An interest rate that a central bank sets and announces publicly; normally the rate at which it is willing to lend money to the commercial banks. Also called *official policy rate* or *policy rate*.

Official policy rate

An interest rate that a central bank sets and announces publicly; normally the rate at which it is willing to lend money to the commercial banks.

Oligopoly

Market structure with a relatively small number of firms supplying the market.

Omnichannel

Refers to a company selling its products or services in multiple channels, such as in store and online.

On-the-run

Most recently issued, and liquid, government bonds.

On-the-run securities

The most recently issued and liquid sovereign debt securities.

Open interest

The number of outstanding contracts.

Open market operations

The purchase or sale of bonds by the national central bank to implement monetary policy. The bonds traded are usually sovereign bonds issued by the national government.

Open-end fund

A mutual fund that accepts new investment money and issues additional shares at a value equal to the net asset value of the fund at the time of investment.

Operating cycle

The length of time between a company's acquisition of goods or raw materials and the collection of cash from sales to customers.

Operating efficiency ratios

Ratios that measure how efficiently a company performs day-to-day tasks, such as the collection of receivables and management of inventory.

Operating leases

A type of lease which is more akin to the rental of the underlying asset.

Operating leverage

The sensitivity of a firm's operating profit to a change in revenues, determined by the composition of fixed and variable operating costs.

Operating profit margin

A profitability ratio calculated as operating income (i.e., income before interest and taxes) divided by revenue. Also called *operating margin*.

Operational deposits

Bank deposits generated by clearing, custody, and cash management activities.

Operational independence

A bank's ability to execute monetary policy and set interest rates in the way it thought would best meet the inflation target.

Operational risk

The risk that arises from inadequate or failed people, systems, and internal policies, procedures, and processes, as well as from external events that are beyond the control of the organization but that affect its operations.

Operationally efficient

Said of a market, a financial system, or an economy that has relatively low transaction costs.

Opportunistic real estate strategies

Include major redevelopment, repurposing of assets, taking on large vacancies, or speculating on significant improvement in market conditions. These may be appealing for investors seeking higher returns and willing to accept additional risks from development, redevelopment, repositioning, and leasing.

Opportunity cost

The value that investors forgo by choosing a particular course of action; the value of something in its best alternative use.

Optimal capital structure

The capital structure at which the value of the company is maximized.

Option

A primary example of a contingent claim. A derivative contract that provides the buyer the right, but not the obligation, to buy or sell an underlying.

Option contract

See *option*.

Option premium

An amount that is paid upfront from the option buyer to the option seller. Reflects the value of the option buyer's right to exercise in the future.

Option-adjusted price

The sum of a bond's flat price and value of an embedded option.

Option-adjusted spread

Or OAS for a bond is its Z-spread adjusted for the value of an embedded option.

Option-adjusted yield

A yield measure for a bond adjusted for embedded options.

Order

A specification of what instrument to trade, how much to trade, and whether to buy or sell.

Order precedence hierarchy

With respect to the execution of orders to trade, a set of rules that determines which orders execute before other orders.

Order-driven markets

A market (generally an auction market) that uses rules to arrange trades based on the orders that traders submit; in their pure form, such markets do not make use of dealers.

Ordinary shares

Equity shares that are subordinate to all other types of equity (e.g., preferred equity). Also called *common stock* or *common shares*.

Organizational form

A legal and tax classification of a business, specific to a jurisdiction, that determines the organization's legal identity, owner–manager relationship, owner liability, taxation, and access to financing.

Out-of-the-money

Describes an option with zero intrinsic value because the option buyer would not rationally exercise the option. An example of such would be the case in which the price of the underlying is less than the option's exercise price for a call option.

Over-the-counter (OTC)

Refers to derivative markets in which derivative contracts are created and traded between derivatives end users and dealers, or financial intermediaries, such as commercial banks or investment banks.

Overcollateralization

Credit enhancement technique where collateral underlying the transaction exceeds the face value of the issued bonds.

Overconfidence bias

A bias in which people demonstrate unwarranted faith in their own intuitive reasoning, judgments, and/or cognitive abilities.

Overfitting

When a machine learning model learns the input and target dataset too precisely, making the system more likely to discover false relationships or unsubstantiated patterns that will lead to prediction errors.

p-Value

The smallest level of significance at which the null is rejected.

Par rate

A yield-to-maturity that makes the present value of a bond's cash flows equal to par.

Par swap rate

The fixed swap rate that equates the present value of all future expected floating cash flows to the present value of fixed cash flows.

Par value

The amount of principal on a bond, also known as face value.

Parallel shift

When all maturities along a yield curve increase or decrease in yield in the same direction by the same magnitude. A parallel shift in the yield curve is implicitly assumed in analytical duration and convexity.

Parameter

A descriptive measure computed from or used to describe a population of data, conventionally represented by Greek letters.

Parametric test

Any test (or procedure) concerned with parameters or whose validity depends on assumptions concerning the population generating the sample.

Pari passu clause

A covenant or contract clause that ensures a debt obligation is treated the same as the borrower's other senior debt instruments and is not subordinated to similar obligations.

Partially amortizing bond

A loan or bond with a payment schedule that calls for the complete repayment of principal over the instrument's time to maturity.

Participating preference shares

Preference shares that entitle shareholders to receive the standard preferred dividend plus the opportunity to receive an additional dividend if the company's profits exceed a pre-specified level.

Pass-through businesses

Businesses that, by virtue of their organizational form and/or other legal and regulatory attributes, do not pay entity-level taxes on income or loss; income or loss is passed through to owners, who pay personal taxes.

Pass-through rate

The coupon rate of a mortgage pass-through security that is received by the investor after administrative charges. It is lower than the weighted average mortgage rate earned on the underlying pool of mortgages because of administrative charges. The pass-through rate that the investor receives is said to be “net interest” or “net coupon.”

Passive investment

A buy and hold approach in which an investor does not make portfolio changes based on short-term expectations of changing market or security performance.

Payable date

The day that the company actually mails out (or electronically transfers) a dividend payment.

Payment date

The day that the company actually mails out (or electronically transfers) a dividend payment.

Payment-in-kind

A bond feature whereby coupon payments can be fully or partially paid in the form of additional issuance or added to the principal amount.

Payments system

The system for the transfer of money.

Pearson correlation

A parametric measure of the relationship between two variables.

Pecking order theory

The theory that managers consider how their actions might be interpreted by outsiders and thereby order their preferences for various forms of corporate financing. Forms of financing that are least visible to outsiders (e.g., internally generated funds) are most preferable to managers, and those that are most visible (e.g., equity issuance) are least preferable.

Penetration pricing

A discount pricing approach used when a firm willingly sacrifices margins in order to build scale and market share.

Percentiles

Quantiles that divide a distribution into 100 equal parts that sum to 100.

Perfect competition

A market structure in which the individual firm has virtually no impact on market price, because it is assumed to be a very small seller among a very large number of firms selling essentially identical products.

Performance evaluation

The measurement and assessment of the outcomes of investment management decisions.

Performance fee

Fee paid to the general partner from the limited partner(s) based on realized net profits.

Period costs

Costs (e.g., executives' salaries) that cannot be directly matched with the timing of revenues and which are thus expensed immediately.

Periodicity

Number of periods in a year, used for compound interest. The periodicity of a fixed-income instrument usually matches the frequency of its coupon payments.

Permanent differences

Differences between tax and financial reporting of revenue (expenses) that will not be reversed at some future date. These result in a difference between the company's effective tax rate and statutory tax rate and do not result in a deferred tax item.

Permissioned networks

Networks that are fully open only to select participants on a DLT network.

Permissionless networks

Networks that are fully open to any user on a DLT network.

Perpetual bonds

Bonds with no stated maturity date.

Perpetuity

A perpetual annuity, or a set of never-ending level sequential cash flows, with the first cash flow occurring one period from now.

PESTLE analysis

A framework for analyzing factors that influence an industry's economic outcomes.

Pet projects

A capital investment that is pursued by management but is not economically justifiable by a disinterested party. Motivations for pet projects include self-dealing and vanity.

Physical risks

Economic and financial losses from the increase in the severity and frequency of extreme weather due to climate change—for example, the loss of coastal real estate from a storm.

PIPE (private investment in public equity)

A private offering to select investors with fewer disclosures and lower transaction costs that allows the issuer to raise capital more quickly and cost effectively.

Platykurtic

Describes a distribution that has relatively less weight in the tails than the normal distribution (also called thin-tailed).

Pledge

A legal right or claim to property by a creditor. Also called a lien.

Poison pill

Officially known as a shareholder rights plan, a poison pill is a hostile-takeover defense adopted by boards of directors according to rules specified in the corporate charter. There are several types of poison pills. Generally, they allow shareholders, *excluding* the shareholder making the hostile bid and their affiliates, to buy newly issued shares at a discounted price. The share issuance would dilute the bidder's ownership percentage, rendering it impossible for the bidder to attain control.

Policy rate

An interest rate that a central bank sets and announces publicly; normally the rate at which it is willing to lend money to the commercial banks.

Portfolio companies

The individual companies owned by a private equity firm.

Portfolio investment flows

Short-term investments in foreign assets, such as stocks or bonds.

Portfolio planning

The process of creating a plan for building a portfolio that is expected to satisfy a client's investment objectives.

Position

The quantity of an asset that an entity owns or owes.

Posterior probability

An updated probability that reflects or comes after new information.

Power of a test

The probability of correctly rejecting the null—that is, rejecting the null hypothesis when it is false.

Pre-funding period

Allows the trust to acquire during a certain period of time after the close of the transaction.

Preference shares

A type of equity interest which ranks above common shares with respect to the payment of dividends and the distribution of the company's net assets upon liquidation. They have characteristics of both debt and equity securities. Also called *preferred stock*.

Preferred stock

See *preference shares*.

Premium

In the case of bonds, premium refers to the amount by which a bond is priced above its face (par) value. In the case of an option, the amount paid for the option contract.

Prepayment option

May entitle the borrower to prepay all or part of the outstanding mortgage principal prior to maturity. This creates a risk from the lender's or investor's viewpoint because the cash flow amounts and timing cannot be known with certainty.

Prepayment risk

The risk that the principal or a proportion of the principal is paid back at a different pace from the contractually agreed scheduled payment plan by the borrower.

Present value models

Valuation models that estimate the intrinsic value of a security as the present value of the future benefits expected to be received from the security. Also called *discounted cash flow models*.

Pretax margin

A profitability ratio calculated as earnings before taxes divided by revenue.

Price discrimination

A pricing approach that charges different prices to different customers based on their willingness to pay.

Price index

Represents the average prices of a basket of goods and services.

Price limits

Establish a band relative to the previous day's settlement price within which all trades must occur.

Price multiple

A ratio that compares the share price with some sort of monetary flow or value to allow evaluation of the relative worth of a company's stock.

Price priority

The principle that the highest priced buy orders and the lowest priced sell orders execute first.

Price return

Measures *only* the price appreciation or percentage change in price of the securities in an index or portfolio.

Price return index

An index that reflects *only* the price appreciation or percentage change in price of the constituent securities. Also called *price index*.

Price stability

In economics, refers to an inflation rate that is low on average and not subject to wide fluctuation.

Price takers

Producers that must accept whatever price the market dictates.

Price value of a basis point (PVBP)

An estimate of the change in the full price of a bond given a 1 bp change in its yield-to-maturity. The PVBP is also called the "PV01," standing for the "price value of an 01" or "present value of an 01," where "01" means 1 bp. In the United States, it is commonly called the "DV01" for the "dollar value" of 1 bp.

Price weighting

An index weighting method in which the weight assigned to each constituent security is determined by dividing its price by the sum of all the prices of the constituent securities.

Price-setting option

The option to adjust prices when demand or supply varies from what is forecast.

Price-to-earnings ratio (P/E)

The ratio of share price to earnings per share.

Pricing power

The ability of a firm to set prices for its products or services without materially losing volume share. Generally this means that the product or service is not a commodity because the firm is not a price taker.

Primary bond markets

Fixed-income markets comprised of issuers issuing bonds to investors to raise capital, often intermediated by a third-party such as an investment bank.

Primary capital markets (primary markets)

The market where securities are first sold and the issuers receive the proceeds.

Primary dealer

Financial institution that is authorized to deal in new issues of sovereign bonds and that serves primarily as a trading counterparty of the office responsible for issuing sovereign bonds.

Primary market

The market where securities are first sold and the issuers receive the proceeds.

Prime broker

A broker that provides services that commonly include custody, administration, lending, short borrowing, and trading.

Prime loans

Lending made to borrowers of high credit quality with strong employment and credit histories, a low DTI, substantial equity in the underlying property, and a first lien on the mortgaged property serving as the collateral for the loan.

Principal

The amount that an issuer agrees to repay the debtholders on the maturity date.

Principal-agent relationship

An arrangement in which one party (the agent) has authority to act for or on behalf of another party (the principal). Such an arrangement imposes a duty on the agent to act in the principal's best interest.

Prior probabilities

Probabilities reflecting beliefs prior to the arrival of new information.

Priority of claims

Priority of payment, with the most senior or highest ranking debt having the first claim on the cash flows and assets of the issuer.

Private capital

Funding provided to companies that is not sourced from the public markets.

Private company

A company, typically a limited company, that does not list its equity securities on an exchange.

Private debt

Capital extended to companies through a loan or other form of debt.

Private debtholders

Investors in an entity's non-securitized debt claims, such as a loan or lease. The most common type of private debtholder is a bank.

Private equity

Equity investment capital raised from sources other than public markets and traditional institutions.

Private equity fund

A hedge fund that seeks to buy, optimize, and ultimately sell portfolio companies to generate profits. See *venture capital fund*.

Private equity securities

Securities that are not listed on public exchanges and have no active secondary market. They are issued primarily to institutional investors via non-public offerings, such as private placements.

Private investment in public equity (PIPE)

An investment in the equity of a publicly traded firm that is made at a discount to the market value of the firm's shares.

Private limited company

A type of limited company in many jurisdictions with pass-through taxation but restrictions on the number of shareholders or members and on the transfer of ownership interest.

Private placement

A sale of debt or equity securities to a small group of investors on an unregulated basis. The terms of the offering are negotiated by the issuer and investors.

Probability of default (POD)

The likelihood that an issuer fails to make full and timely payments of principal and interest; typically an annualized measure.

Probability sampling

A sampling plan that allows every member of the population to have an equal chance of being selected.

Probability tree diagram

A diagram with branches emanating from nodes representing either mutually exclusive chance events or mutually exclusive decisions.

Production flexibility option

The option to alter production when demand varies from what is forecast.

Profession

An occupational group that has specific education, expert knowledge, and a framework of practice and behavior that underpins community trust, respect, and recognition.

Profit margin

An indicator of profitability, calculated as net income divided by revenue; indicates how much of each dollar of revenues is left after all costs and expenses.

Profitability ratios

Ratios that measure a company's ability to generate profitable sales from its resources (assets).

Prospectus

Legal document in securitization that describes the structure of the transaction, including the priority and amount of payments to be made to the servicer, administrators, and the ABS holders, as well as the credit enhancements used in the securitization.

Protective put

A strategy of purchasing an underlying asset and purchasing a put on the same asset.

Proxy contest

When a shareholder or group of shareholders campaigns for certain matters they have submitted to a shareholder vote, often a slate of directors who oppose the incumbent board and management. The incumbent board and management simultaneously campaign for their side.

Proxy voting

A form of casting a ballot in an election in which a voter authorizes a representative to vote on their behalf according to instructions. In corporate elections, proxy ballots are cast by shareholders that direct a representative, typically the corporate secretary, to enter their votes as instructed.

Public (listed) company

A company with its equity securities traded on an exchange.

Public limited companies

A type of limited company in many jurisdictions with entity-level taxation but no restrictions on the number of shareholders or transferability of ownership interest; the most suitable organizational form for a company that seeks to go public.

Public-private partnership (PPP)

An agreement between the public sector and the private sector to finance, build, and operate public infrastructure, such as hospitals and toll roads.

Pull on liquidity

An action or event that accelerates cash outflows.

Purchase agreement

Legal document in a securitization transaction that outlines the representations and warranties that the seller makes about the assets sold.

Pure discount bonds

Bonds that do not pay interest during their life. They are issued at a discount to par value and redeemed at par. Also called zero-coupon bonds.

Put

An option that gives the holder the right to sell an underlying asset to another party at a fixed price over a specific period of time.

Put option

The right to sell an underlying.

Puttable bonds

Bonds that give the bondholder the right to sell the bond back to the issuer at a predetermined price on specified dates.

Put-call forward parity

Describes the no-arbitrage condition in which at $t = 0$ the present value of the price of a long forward commitment plus the price of the long put must equal the price of the long call plus the price of the risk-free asset (with face value of the exercise price of both the call and the put).

Put-call parity

Describes the no-arbitrage condition in which at $t = 0$ the price of the long underlying asset plus the price of the long put must equal the price of the long call plus the price of the risk-free asset (with face value of the exercise price of both the call and the put).

Quantile

A value at or below which a stated fraction of the data lies. Also referred to as a fractile.

Quantitative easing

An expansionary monetary policy based on aggressive open market purchase operations.

Quartiles

Quantiles that divide a distribution into four equal parts.

Quick ratio

A measure of liquidity that is the ratio of cash, marketable securities, and receivables to current liabilities.

Quintiles

Quantiles that divide a distribution into five equal parts.

Quota rents

Profits that foreign producers can earn by raising the price of their goods higher than they would without a quota.

Quotas

Government policies that restrict the quantity of a good that can be imported into a country, generally for a specified period of time.

Quote-driven market

A market in which dealers acting as principals facilitate trading.

Quoted margin

Specified spread of a floating rate instrument over a market reference rate or benchmark.

Range

The difference between the maximum and minimum values in a dataset.

Rapid amortization provisions

Provisions in receivable ABS that may require early principal amortization if specific events occur. Such provisions are referred to as early amortization and are included to safeguard the credit quality of the issue, particularly during the revolving period.

Razor, razorblade pricing

A pricing approach that combines a low price on a piece of equipment and high-margin pricing on repeat-purchase consumables.

Real assets

Generally, these are tangible physical assets, such as real estate, infrastructure, and natural resources, but they also include such intangibles as patents, intellectual property, and goodwill. Real assets generate current or expected future cash flows and/or are considered a store of value.

Real estate

Includes borrowed or ownership capital in buildings or land. Developed land includes commercial and industrial real estate, residential real estate, and infrastructure.

Real option

A right, but not an obligation, for management to make a decision with respect to a capital investment that alters future cash flows from the original forecasted scenario.

Real risk-free interest rate

The single-period interest rate for a completely risk-free security if no inflation were expected.

Rebalancing

In the context of asset allocation, a discipline for adjusting the portfolio to align with the strategic asset allocation.

Rebalancing policy

The set of rules that guide the process of restoring a portfolio's asset class weights to those specified in the strategic asset allocation.

Recapitalization

Recapitalization via private equity describes the steps a firm takes to increase or introduce leverage to its portfolio company and pay itself a dividend out of the new capital structure.

Recognition lag

The lag in government response to an economic problem resulting from the delay in confirming a change in the state of the economy.

Recourse loan

Loan in which the lender has a claim against the borrower for the shortfall (deficiency) between the amount of the outstanding mortgage balance and the proceeds received from the sale of the property.

Recovery rate (RR)

The percentage of an outstanding debt claim recovered when an issuer defaults

Redemption fee

A fee charged to discourage redemptions and to offset the transaction costs for remaining investors in the fund.

Refinancing rate

A type of central bank policy rate.

Regionalism

In between the two extremes of bilateralism and multilateralism. In regionalism, a group of countries cooperate with one another. Both bilateralism and regionalism can be conducted at the exclusion of other groups. For example, regional blocs may agree to provide trade benefits to one another and increase barriers for those outside of that group.

Registered bonds

Bonds for which ownership is recorded by either name or serial number.

Regression analysis

Allows us to test hypotheses about the relationship between two variables, by quantifying the strength of the relationship between the two variables, and to use one variable to make predictions about the other variable.

Regression coefficients

The collective term for the intercept and slope coefficients in the regression model.

Regret

The feeling that an opportunity has been missed; typically, an expression of *hindsight bias*.

Regret-aversion bias

An emotional bias in which people tend to avoid making decisions that will result in action out of fear that the decision will turn out poorly.

Relative dispersion

The amount of dispersion relative to a reference value or benchmark.

Reopening

Issuing bonds by increasing the size of an existing bond issue with a price significantly different from par.

Replication

A strategy in which a derivative's cash flow stream may be recreated using a combination of long or short positions in an underlying asset and borrowing or lending cash.

Repo rate

The interest rate on a repurchase agreement.

Representativeness bias

A belief perseverance bias in which people tend to classify new information based on past experiences and classifications.

Repurchase agreement (Repo)

A form of collateralized loan involving the sale of a security with a simultaneous agreement by the seller to buy back the same security from the purchaser at an agreed-on price and future date. The party who sells the security at the inception of the repurchase agreement and buys it back at maturity is borrowing money from the other party, and the security sold and subsequently repurchased represents the collateral.

Repurchase date

The date when the party who sold the security at the inception of a repurchase agreement buys back the security from the cash lending counterparty.

Repurchase price

The price at which the party who sold the security at the inception of the repurchase agreement buys back the security from the cash lending counterparty.

Required margin

Yield spread of a floating rate instrument such that the instrument is priced at par value on a rate reset date.

Required rate of return

The rate of return required by investors given the risk of the bond investment, also known as the market discount rate or required yield.

Required yield

The rate of return required by investors given the risk of the bond investment, also known as the market discount rate or required rate of return.

Required yield spread

The difference in yield-to-maturity between a bond and that of a government benchmark bond with the same or similar time-to-maturity.

Resampling

A statistical method that repeatedly draws samples from the original observed data sample for the statistical inference of population parameters.

Reserve currency

A currency held by global central banks in significant quantities and widely used to conduct international trade and financial transactions.

Reserve requirement

The requirement for banks to hold reserves in proportion to the size of deposits.

Residual

The amount of deviation of an observed value of the dependent variable from its estimated value based on the fitted regression line.

Restricted domestic currency

A currency with limited convertibility into other currencies due to illiquidity.

Return on assets (ROA)

A profitability ratio calculated as net income divided by average total assets; indicates a company's net profit generated per dollar invested in total assets.

Return on equity (ROE)

A profitability ratio calculated as net income divided by average shareholders' equity.

Return on invested capital (ROIC)

A measure of the profitability of a company relative to the amount of capital invested by the equityholders and debtholders.

Return on sales

An indicator of profitability, calculated as net income divided by revenue; indicates how much of each dollar of revenues is left after all costs and expenses. Also referred to as *net profit margin*.

Return-generating model

A model that can provide an estimate of the expected return of a security given certain parameters and estimates of the values of the independent variables in the model.

Revenue bonds

Bonds issued by non-sovereign governments related to a government sponsored project expected to generate future cash flow as a primary source of repayment.

Reverse repurchase agreement

A repurchase agreement viewed from the perspective of the cash lending counterparty.

Reverse stock split

A reduction in the number of shares outstanding with a corresponding increase in share price, but no change to the company's underlying fundamentals.

Revolving credit agreements

The most reliable form of short-term bank borrowing facilities; they are in effect for multiple years (e.g., three to five years) and can have optional medium-term loan features. Also known as *revolvers*.

Rho

The change in a given derivative instrument for a given small change in the risk-free interest rate, holding everything else constant. Rho measures the sensitivity of the option to the risk-free interest rate.

Ricardian equivalence

An economic theory that implies that it makes no difference whether a government finances a deficit by increasing taxes or issuing debt.

Risk

Exposure to uncertainty. The chance of a loss or adverse outcome as a result of an action, inaction, or external event.

Risk averse

The assumption that an investor will choose the least risky alternative.

Risk aversion

The degree of an investor's unwillingness to take risk; the inverse of risk tolerance.

Risk budgeting

The establishment of objectives for individuals, groups, or divisions of an organization that takes into account the allocation of an acceptable level of risk.

Risk exposure

The state of being exposed or vulnerable to a risk. The extent to which an organization is sensitive to underlying risks.

Risk governance

The top-down process and guidance that directs risk management activities to align with and support the overall enterprise.

Risk management

The process of identifying the level of risk an organization wants, measuring the level of risk the organization currently has, taking actions that bring the actual level of risk to the desired level of risk, and monitoring the new actual level of risk so that it continues to be aligned with the desired level of risk.

Risk management framework

The infrastructure, process, and analytics needed to support effective risk management in an organization.

Risk premium

An extra return expected by investors for bearing some specified risk.

Risk shifting

Actions to change the distribution of risk outcomes.

Risk tolerance

The amount of risk an investor is willing and able to bear to achieve an investment goal.

Risk transfer

Actions to pass on a risk to another party, often, but not always, in the form of an insurance policy.

Risk-neutral pricing

A no-arbitrage derivative value established separately from investor views on risk that uses underlying asset volatility and the risk-free rate to calculate the present value of future cash flows.

Risk-neutral probability

The computed probability used in binomial option pricing by which the discounted weighted sum of expected values of the underlying equal the current option price. Specifically, this probability is computed using the risk-free rate and assumed up gross return and down gross return of the underlying.

Rollover risk

The possibility that an issuer will be unable to refinance maturing (typically short-term) debt at economical terms or at all.

Safety-first rules

Rules for portfolio selection that focus on the risk that portfolio value or portfolio return will fall below some minimum acceptable level over some time horizon.

Sample correlation coefficient

A standardized measure of how two variables in a sample move together. It is the ratio of the sample covariance to the product of the two variables' standard deviations.

Sample covariance

A measure of how two variables in a sample move together.

Sample excess kurtosis

A sample measure of the degree of a distribution's kurtosis in excess of the normal distribution's kurtosis.

Sample mean

The sum of the sample observations divided by the sample size.

Sample skewness

A sample measure of the degree of asymmetry of a distribution.

Sample standard deviation

The positive square root of the sample variance.

Sample variance

The sum of squared deviations around the mean divided by the degrees of freedom.

Sample-size neglect

A type of representativeness bias in which financial market participants incorrectly assume that small sample sizes are representative of populations (or “real” data).

Sampling distribution

The distribution of all distinct possible values that a statistic can assume when computed from samples of the same size randomly drawn from the same population.

Sampling error

The difference between the observed value of a statistic and the estimate resulting from using subsets of the population.

Sampling plan

The set of rules used to select a sample.

Saving deposits

Bank deposits typically held for non-transactional purposes that often have a stated term.

Scatter plot

A two-dimensional graphical plot of paired observations of values for the independent and dependent variables in a simple linear regression.

Scenario analysis

A technique for exploring the performance and risk of investment strategies in different structural regimes.

Scraping

An automated, large-scale, algorithm-driven approach that retrieves otherwise unstructured data available on websites and creates data in a more structured format.

Seasoned offering

An offering in which an issuer sells additional units of a previously issued security.

Secondary bond markets

Fixed-income markets comprised of investors trading existing bonds amongst themselves.

Secondary market

The market where securities are traded among investors.

Secondary precedence rules

Rules that determine how to rank orders placed at the same time.

Secondary sale

Sale of a private company stake to another private equity firm or group of financial buyers.

Secondary-stage investments

The second stage of development of an infrastructure asset. Secondary-stage investments involve existing infrastructure facilities or fully operational assets that do not require further investment or development over the investment horizon. These assets generate immediate cash flow and returns expected over the investment period.

Sector indexes

Indexes that represent and track different economic sectors—such as consumer goods, energy, finance, health care, and technology—on either a national, regional, or global basis.

Secured

With collateral; secured debt is backed by the cash flows of the issuer and the collateral as a secondary source of repayment.

Secured loans

Loans collateralized by an asset of the borrower.

Security

Evidence of equity or debt interest or in an entity or a related right, such as a derivative. Often standardized to conform to security exchange requirements.

Security characteristic line

A plot of the excess return of a security on the excess return of the market.

Security market index

A portfolio of securities representing a given security market, market segment, or asset class.

Security market line (SML)

The graph of the capital asset pricing model.

Security selection

The process of selecting individual securities; typically, security selection has the objective of generating superior risk-adjusted returns relative to a portfolio's benchmark.

Security tokens

Digitizes the ownership rights associated with publicly traded securities.

Segmenting

A process of identifying and grouping customers by decision-useful attributes.

Self-attribution bias

A bias in which people take too much credit for successes (*self-enhancing*) and assign responsibility to others for failures (*self-protecting*).

Self-control bias

A bias in which people fail to act in pursuit of their long-term, overarching goals because of a lack of self-discipline.

Self-investment limits

With respect to investment limitations applying to pension plans, restrictions on the percentage of assets that can be invested in securities issued by the pension plan sponsor.

Sell-side firm

A broker/dealer that sells securities and provides independent investment research and recommendations to their clients (i.e., buy-side firms).

Semi-strong-form efficient market

A market in which security prices reflect all publicly known and available information.

Semiannual bond basis yield

Also known as a semiannual bond equivalent yield, it is an annualized interest rate with a periodicity of two.

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Also known as a semiannual bond basis yield, it is an annualized interest rate with a periodicity of two.

Senior debt

A debt obligation with higher priority of payment than junior debt obligations.

Senior unsecured debt

The highest-ranked debt in an issuer's capital structure which is a general obligation of the borrower.

Seniority

Priority of payment of various debt obligations.

Sensitivity analysis

Analysis that shows the range of possible outcomes as specific assumptions are changed.

Separately managed account (SMA)

An investment portfolio managed exclusively for the benefit of an individual or institution.

Separately managed accounts

Accounts that are managed in accordance with an investor's specific investment preferences and risk tolerance.

Service period

The time between the grant and vesting dates for an employee share-based award, usually measured in years.

Settlement

The closing date at which the counterparties of a derivative contract exchange payment for the underlying as required by the contract.

Settlement price

The price determined by an exchange's clearinghouse in the daily settlement of the mark-to-market process. The price reflects an average of the final futures trades of the day.

Share class

Types of equity securities that have different voting rights—for example, an issuer may issue Class

A shares that carry one vote per share and Class B shares that carry ten votes per share.

Share repurchase

A transaction in which a company buys back its own shares. Unlike stock dividends and stock splits, share repurchases use corporate cash.

Shareholder activism

A range of actions by a corporation's shareholders that are intended to result in some change in the corporation, typically a change in the board of directors, management, or business strategy.

Shareholder derivative lawsuit

A legal action by a shareholder on behalf of a company, not the shareholder personally, against a third party. Often, the third party is a director or manager who the shareholder believes has harmed the company.

Shareholder engagement

Shareholder engagement reflects active ownership by investors in which the investor seeks to influence a corporation's decisions on ESG matters, either through dialogue with corporate officers or votes at a shareholder assembly (in the case of equity).

Shareholder theory of corporate governance

Espoused by Milton Friedman in his famous 1970 essay, the shareholder theory holds that the objective of a business is to increase profits and shareholder value.

Shareholders

Individuals or entities that own a share.

Shares

Units of ownership interest in a limited company.

Sharpe ratio

The average return in excess of the risk-free rate divided by the standard deviation of return; a measure of the average excess return earned per unit of standard deviation of return.

Shelf registration

A type of public offering that allows the issuer to file a single, all-encompassing offering circular that covers a series of bond issues.

Short

A trading position in a derivative contract that gains value as the price of the underlying moves lower.

Short biased

These strategies use quantitative, technical, and fundamental analysis to short overvalued equity securities with limited or no long-side exposures.

Short position

A position in an asset or contract in which one has sold an asset one does not own, or in which a right under a contract can be exercised against oneself.

Short selling

A transaction in which borrowed securities are sold with the intention to repurchase them at a lower price at a later date and return them to the lender.

short-run average total cost

The curve describing average total cost when some costs are considered fixed.

Shortfall risk

The risk that portfolio value or portfolio return will fall below some minimum acceptable level over some time horizon.

Shutdown point

The point at which average revenue is equal to the firm's average variable cost.

Side letter

A side agreement created between the GP and specific LPs. These agreements exist *outside* the LPA. These agreements provide additional terms and conditions related to the investment agreement.

Signpost

An indicator, market level, data piece, or event that signals a risk is becoming more or less likely. An analyst can think of signposts like a traffic light.

Simple linear regression (SLR)

An approach for estimating the linear relationship between a dependent variable and a single independent variable by minimizing the sum of the squared deviations between the fitted line and the observed values.

Simple random sample

A subset of a larger population created in such a way that each element of the population has an equal probability of being selected to the subset.

Simple random sampling

The procedure of drawing a sample to satisfy the definition of a simple random sample.

Simple yield

The sum of the coupon payments plus the straight-line amortized share of the gain or loss divided by the bond's flat price. Simple yields are used mostly to quote JGBs.

Simulation

A technique for exploring how a target variable (e.g. portfolio returns) would perform in a hypothetical environment specified by the user, rather than a historical setting.

Simulation trial

A complete pass through the steps of a simulation.

Single-price auction

A debt securities auction in which all bidders pay the same price.

Sinking fund

Provisions that reduce the credit risk of a bond issue by requiring the issuer to retire a portion of the bond's principal outstanding each year.

Situational influences

External factors, such as environmental or cultural elements, that shape our behavior.

Skewed

Not symmetrical.

Skewness

A quantitative measure of skew (lack of symmetry); a synonym of skew. It is computed as the average cubed deviation from the mean standardized by dividing by the standard deviation cubed.

Slope coefficient

The change in the estimated value of the dependent variable for a one-unit change in the value of the independent variable.

Small country

A country that is a price taker in the world market for a product and cannot influence the world market price.

Smart beta

Involves the use of simple, transparent, rules-based strategies as a basis for investment decisions.

Smart contracts

Computer programs that are designed to self-execute on the basis of pre-specified terms and conditions agreed to by parties to a contract.

Social infrastructure investments

A category of infrastructure investments that are directed toward human activities and include such assets as educational, health care, social housing, and correctional facilities, with the focus on providing, operating, and maintaining the asset infrastructure.

Soft commodities

Standardized agricultural products, such as cattle and corn, with markets often involving the physical delivery of the underlying upon settlement.

Soft hurdle rate

Hurdle rate where the fee is calculated on the entire return when the hurdle is exceeded. With a soft hurdle, GPs are able to catch up performance fees once the hurdle threshold is exceeded.

Soft power

A means of influencing another country's decisions without force or coercion. Soft power can be built over time through actions, such as cultural programs, advertisement, travel grants, and university exchange.

Soft-bullet covered bonds

Delay the bond default and payment acceleration of bond cash flows until a new final maturity date, which is usually up to a year after the original maturity date.

Solvency

Refers to the condition in which firm value exceeds the face value of debt used to finance the firm's assets.

Solvency ratios

Ratios that measure a company's ability to meet its long-term obligations.

Solvency risk

The risk that an organization does not survive or succeed because it runs out of cash, even though it might otherwise be solvent.

Sophisticated investors

Individuals or entities that are permitted in a jurisdiction to trade unregistered or, generally, less regulated securities, including shares of privately held companies; also called *accredited investors*.

Sovereign immunity

A principle limiting the legal recourse of bondholders holding national government debt from forcing the issuer to declare bankruptcy or liquidate assets to settle debt claims.

Spearman rank correlation coefficient

A measure of correlation applied to ranked data.

Special dividend

A dividend paid by a company that does not pay dividends on a regular schedule, or a dividend that supplements regular cash dividends with an extra payment.

Special purpose acquisition company (SPAC)

A company that has gone public, raising money to acquire a private company using stock and, often, private investments from institutional investors. In connection with the acquisition (known as a de-SPAC), the private company becomes a public company.

Special purpose entity (SPE)

A legal entity created for a temporary or narrow legal purpose to isolate a parent entity from risk. Sometimes also called a special purpose vehicle (SPV) or a special purpose company.

Special purpose vehicle

See *special purpose entity*.

Special situations

These strategies focus on opportunities to buy the equity of companies engaged in security issuance or repurchase, special capital distributions, rescue finance, asset sales/spin-offs, or other catalyst-oriented situations.

Split ratings

Complex risks viewed very differently by rating agencies

Sponsored

A type of depository receipt in which the foreign company whose shares are held by the depository has a direct involvement in the issuance of the receipts.

Spot curve

Yields-to-maturity on a series of default-risk-free zero-coupon bonds.

Spot markets

Markets in which specific assets are exchanged at current prices. Spot markets are often referred to as cash markets.

Spot prices

The current prices prevailing in spot markets.

Spot rates

Yields-to-maturity on default-risk-free zero-coupon bonds.

Spread

The difference in yield-to-maturity between a bond and that of another bond.

Spread risk

Bond price risk arising from changes in the yield spread on credit-risky bonds; reflects changes in the market's assessment and/or pricing of credit migration (or downgrade) risk and market liquidity risk.

Spurious correlation

Refers to: 1) correlation between two variables that reflects chance relationships in a particular dataset; 2) correlation induced by a calculation that mixes each of two variables with a third variable; and 3) correlation between two variables arising not from a direct relation between them but from their relation to a third variable.

Stablecoin

A cryptocurrency that aims to maintain a stable value relative to a specified asset or to a pool or basket of assets.

Stackelberg model

A prominent model of strategic decision making in which firms are assumed to make their decisions sequentially.

Staggered board

A structure of board elections in which only part of the board is elected simultaneously—for example, only one-third of the board may be up for election each year, so the board can be replaced over three years, not in one year if all seats were elected annually. This structure fosters greater continuity of board members but is an obstacle for shareholders seeking to effect change.

Stakeholder theory of corporate governance

An expansion of the shareholder theory of corporate governance under which the objective of a business is to maximize value for, and balance the interests of, a broad group of stakeholders, including shareholders, employees, society, and the non-human environment.

Stakeholders

Any party with an interest, financial or non-financial, in an entity or its actions.

Standard deviation

The positive square root of the variance; a measure of dispersion in the same units as the original data.

Standard error of the estimate

A measure of the distance between the observed values of the dependent variable and those predicted from the estimated regression. The smaller this value, the better the fit of the model. Also known as the standard error of the regression and the root mean square error.

Standard error of the forecast

Used to provide an interval estimate around the estimated regression line. It is necessary because the regression line does not describe the relationship between the dependent and independent variables perfectly.

Standard error of the slope coefficient

Calculated for simple linear regression by dividing the standard error of the estimate by the square root of the variation of the independent variable.

Standardization

The process of creating protocols for the production, sale, transport, or use of a product or service. Standardization occurs when relevant parties agree to follow these protocols together. It helps support expanded economic and financial activities, such as trade and capital flows that support higher economic growth and standards of living, across borders.

Standards of conduct

Behaviors required by a group; established benchmarks that clarify or enhance a group's code of ethics.

Standing limit orders

A limit order at a price below market and which therefore is waiting to trade.

State actors

Typically national governments, political organizations, or country leaders that exert authority over a country's national security and resources. The South African President, Sultan of Brunei, Malaysia's Parliament, and the British Prime Minister are all examples of state actors.

Statement of cash flows

A financial statement that details the movement of cash over a period. The statement is classified into operating, investing, and financing activities.

Static trade-off theory of capital structure

A theory pertaining to a company's optimal capital structure; the optimal level of debt is found at the point where additional debt would cause the costs of financial distress to increase by a greater amount than the benefit of the additional tax shield.

Statistically significant

A result indicating that the null hypothesis can be rejected; with reference to an estimated regression coefficient, frequently understood to mean a result indicating that the corresponding population regression coefficient is different from zero.

Status quo bias

An emotional bias in which people do nothing (i.e., maintain the status quo) instead of making a change.

Statutory voting

A common method of voting where each share represents one vote.

Step-up bonds

Bonds for which the coupon, be it fixed or floating, increases by specified margins at specified dates.

Stock dividend

A type of dividend in which a company distributes additional shares of its common stock to shareholders instead of cash.

Stock exchange

An exchange in which equity securities are traded. See *exchanges*.

Stock split

An increase in the number of shares outstanding with a consequent decrease in share price, but no change to the company's underlying fundamentals.

Stockholder overhang

The downward pressure on the share price of stock as large blocks of shares are being sold on the open market.

Stop order

An order in which a trader has specified a stop price condition. Also called *stop-loss order*.

Stop-loss order

See *stop order*.

Stranded assets

A resource that is no longer economically valuable owing to changes in demand, regulations, or availability of substitutes—for example, a newly discovered oil well that will not be brought into production.

Strategic asset allocation

Strategic asset allocation is an investment strategy premised on long-term asset allocation. This strategy only rebalances its portfolio when the asset mix represents significant deviation from the original or targeted allocation mix.

Stratified random sampling

A procedure that first divides a population into subpopulations (strata) based on classification criteria and then randomly draws samples from each stratum in sizes proportional to that of each stratum in the population.

Street convention

For yield measures on fixed-income instruments that assume payments are made on scheduled dates and ignore weekends and holidays.

Stress testing

A specific type of scenario analysis that estimates losses in rare and extremely unfavorable combinations of events or scenarios.

Strong-form efficient market

A market in which security prices reflect all public and private information.

Structural budget deficit

Also known as the cyclically adjusted budget deficit. The deficit that would exist if the economy was at full employment (or full potential output).

Structural subordination

Arises in a holding company structure when the debt of operating subsidiaries is serviced by the cash flow and assets of the subsidiaries before funds can be passed to the holding company to service debt at the parent level.

Structured notes

A broad category of securities that incorporate the features of debt instruments and one or more embedded derivatives designed to achieve a particular issuer or investor objective.

Subordinated debt

A class of unsecured debt that ranks below a firm's senior unsecured obligations.

Subordination

A form of internal credit enhancement that relies on creating more than one bond tranche and ordering the claim priorities for ownership or interest in an asset between the tranches. The ordering of the claim priorities is called a senior/subordinated structure, where the tranches of highest seniority are called senior, followed by subordinated or junior tranches. Also called credit tranching.

Subprime loans

Lending to borrowers with lower credit quality, high DTI, and/or are loans with higher LTV, and include loans that are secured by second liens otherwise subordinated to other loans.

Sum of squares error (SSE)

A measure of the total deviation between observed and estimated values of the dependent variable. It is calculated by subtracting each estimated value $\hat{Y}_i$ from its corresponding observed value Y_i , squaring each of these differences, and then summing all of these squared differences.

Sum of squares regression (SSR)

A measure of the explained variation in the dependent variable, calculated as the sum of the squared differences between the predicted value of the dependent variable, $\hat{Y}_i$, based on the estimated regression line, and the mean of the dependent variable, $\bar{Y}$.

Sum of squares total (SST)

A measure of the total variation in the dependent variable in a simple linear regression. It is calculated by subtracting the mean of the observed values $\bar{Y}$ from each of the observed values Y_i , squaring each of these differences, and then summing all of these squared differences.

Sunk costs

A cost that has already been incurred.

Supervised learning

A type of machine learning in which the system attempts to learn to model relationships based on labeled training data.

Supervisory board

In some jurisdictions, a corporation's board of directors is formally composed of a supervisory board and a management board. The supervisory board appoints and oversees the management board and often includes representatives of employees and other non-shareholder stakeholders.

Supply chain

The sequence of processes involved in the creation and delivery of a physical product to the end customer, both within and external to a firm, regardless of whether those steps are performed by a single firm.

Supply shock

A typically unexpected disturbance to supply.

Survivorship bias

Relates to the inclusion of only current investment funds in a database. As such, the returns of funds that are no longer available in the marketplace (have been liquidated) are excluded from the database. Also see *backfill bias*.

Swap

A firm commitment involving a periodic exchange of cash flows.

Swap contract

An agreement between two parties to exchange a series of future cash flows.

Swap execution facility (SEF)

A swap trading platform accessed by multiple dealers.

Swap rate

The fixed rate to be paid by the fixed-rate payer specified in a swap contract.

Syndicate

A group of lenders, typically made up of banks.

Synthetic protective put

The combination of a synthetic long underlying position (i.e., a long forward and risk-free borrowing) and a purchased put on the underlying.

Systematic risk

Risk that affects the entire market or economy; it cannot be avoided and is inherent in the overall market. Systematic risk is also known as non-diversifiable or market risk.

Systematic sampling

A procedure of selecting every k th member until reaching a sample of the desired size. The sample that results from this procedure should be approximately random.

Systemic risk

Refers to risks supervisory authorities believe are likely to have broad impact across the financial market infrastructure and affect a wide swath of market participants.

Tactical asset allocation

Asset allocation that involves making short-term adjustments to asset class weights based on short-term predictions of relative performance among asset classes.

Target capital structure

Management's desired proportions of debt and equity financing, usually stated on a book value basis or indirectly using a financial leverage metric, such as net or gross debt to EBITDA or credit rating.

Target independent

A bank's ability to determine the definition of inflation that they target, the rate of inflation that they target, and the horizon over which the target is to be achieved.

Target semideviation

A measure of downside risk, calculated as the square root of the average of the squared deviations of observations below the target (also called target downside deviation).

Tariffs

Taxes that a government levies on imported goods.

Tax base

The amount at which an asset or liability is valued for tax purposes.

Tax expense

An aggregate of an entity's income tax payable (or recoverable in the case of a tax benefit) and any changes in deferred tax assets and liabilities. It is essentially the income tax payable or recoverable if these had been determined based on accounting profit rather than taxable income.

Taxable income

The portion of an entity's income that is subject to income taxes under the tax laws of its jurisdiction.

Taxable temporary differences

Temporary differences that result in a taxable amount in a future period when determining the taxable profit as the balance sheet item is recovered or settled.

Technical analysis

A form of security analysis that uses price and volume data, often displayed graphically, in decision making.

Tender offer

A solicitation by a current or prospective shareholder to other shareholders to acquire a substantial percentage, including 100%, of shares at a specified price. This action is usually undertaken by a potential acquirer whose bid was rejected by the issuer's board of directors, prompting the potential acquirer to appeal directly to shareholders.

Tenor

The remaining time to maturity for a bond or derivative contract. Also called term to maturity.

Term repos

Repos with a maturity longer than one day.

Term structure of interest rates

Also known as the maturity structure of interest rates, refers to the difference in interest rates or benchmark yields by time-to-maturity.

Terminal stock value

The expected value of a share at the end of the investment horizon—in effect, the expected selling price. Also called *terminal value*.

Terminal value

The expected value of a share at the end of the investment horizon—in effect, the expected selling price.

Test of the mean of the differences

A statistical test for differences based on paired observations drawn from samples that are dependent on each other.

Text analytics

Involves the use of computer programs to analyze and derive meaning typically from large, unstructured text- or voice-based datasets, such as company filings, written reports, quarterly earnings calls, social media, email, internet postings, and surveys.

Thematic risks

Known risks that evolve and expand over a period of time. Climate change, pattern migration, the rise of populist forces, and the ongoing threat of terrorism fall into this category.

Thin-tailed

Describes a distribution that has relatively less weight in the tails than the normal distribution (also called platykurtic).

Tiered pricing

A pricing approach that charges different prices to different buyers, commonly based on volume purchased.

Timberland investment management organizations

Entities that support institutional investors by managing their investments in timberland by analyzing and acquiring suitable timberland holdings.

Time tranching

Structure of a securitization that allows for the redistribution of “prepayment risk” among bond classes by creating bond classes of different expected maturities.

Time value

The difference between an option's premium and its intrinsic value.

Time value decay

The process by which the time value of an option declines toward zero as the option's expiration date is approached.

Time-weighted rate of return

The compound rate of growth of one unit of currency invested in a portfolio during a stated measurement period; a measure of investment performance that is not sensitive to the timing and amount of withdrawals or additions to the portfolio.

Tokenization

The process of representing ownership rights to physical assets on a blockchain or distributed ledger.

Top-down analysis

An investment selection approach that begins with consideration of macroeconomic conditions and then evaluates markets and industries based upon such conditions.

Total probability rule for expected value

A rule explaining the expected value of a random variable in terms of expected values of the random variable conditional on mutually exclusive and exhaustive scenarios.

Total return

Measures the price appreciation, or percentage change in price of the securities in an index or portfolio, plus any income received over the period.

Total return index

An index that reflects the price appreciation or percentage change in price of the constituent securities plus any income received since inception.

Total working capital

The difference between current assets and current liabilities.

Tracking error

The standard deviation of the differences between a portfolio's returns and its benchmark's returns; a synonym of *active risk*. Also called *tracking risk*.

Tracking risk

The standard deviation of the differences between a portfolio's returns and its benchmarks returns. Also called *tracking error*.

Trade creation

When regional integration results in the replacement of higher cost domestic production by lower cost imports from other members.

Trade diversion

When regional integration results in lower-cost imports from non-member countries being replaced with higher-cost imports from members.

Trade sale

A portion or division of a private company sold via either direct sale or auction to a strategic buyer interested in increasing the scale and scope of an existing business.

Trade settlement date

The date when the buyer and seller transfer consideration and securities.

Traditional investment markets

Markets for traditional investments, which include all publicly traded debts and equities and shares in pooled investment vehicles that hold publicly traded debts and/or equities.

Tranches

A grouping of securities within an issue with characteristics that vary from other tranches, such as different credit quality and seniority.

Transfer payments

Welfare payments made through the social security system that exist to provide a basic minimum level of income for low-income households.

Transition risks

Economic and financial losses from the transition to a lower-carbon economy in response to climate change—for example, the abandonment of an oil well that is no longer economical.

Treasury Inflation-Protected Securities (TIPS)

US Treasury bonds with a principal that is adjusted for changes in the Consumer Price Index. TIPS are issued in 5-, 10-, and 30-year maturities.

Treynor ratio

A measure of risk-adjusted performance that relates a portfolio's excess returns to the portfolio's beta.

Trimmed mean

A mean computed after excluding a stated small percentage of the lowest and highest observations.

Triparty repo

A repurchase agreement in which the transacting parties agree to use a third-party agent that provides access to a larger collateral pool and multiple counterparties, as well as valuation and safekeeping of assets.

True yield

Measures on fixed-income instruments use actual payment dates, accounting for weekends and holidays. The true yield on an instrument is always lower than the street convention yield.

Turn-of-the-year effect

Calendar anomaly that stock market returns in January are significantly higher compared to the rest of the months of the year, with most of the abnormal returns reported during the first five trading days in January.

Two-fund separation theorem

The theory that all investors regardless of taste, risk preferences, and initial wealth will hold a combination of two portfolios or funds: a risk-free asset and an optimal portfolio of risky assets.

Two-way table

A table of the frequency distribution of observations classified on the basis of two discrete variables. Also known as *Contingency table*.

Two-week repo rate

The interest rate on a two-week repurchase agreement; may be used as a policy rate by a central bank.

Type I error

The error of rejecting a true null hypothesis; a false positive.

Type II error

The error of not rejecting a false null hypothesis; false negative.

Uncommitted lines of credit

Sources of bank credit that a bank can refuse to honor. Uncommitted credit lines are made up to a certain principal amount for a pre-determined maximum maturity, charging a market reference rate plus an issuer-specific spread on only the principal outstanding for the period of use.

Underfitted

When a machine learning model treats true parameters as if they are noise and is unable to recognize relationships in the training data, making the model more likely to fail to fully discover patterns that underlie the data.

Underlying

The asset referred to in a derivative contract.

Underwritten offering

A type of securities issue mechanism in which the investment bank guarantees the sale of the securities at an offering price that is negotiated with the issuer. Also known as *firm commitment offering*.

Unearned revenue

A liability account for money that has been collected for goods or services that have not yet been delivered; payment received in advance of providing a good or service. Also called *deferred revenue* or *deferred income*.

Unimodal

A distribution with a single value that is most frequently occurring.

Unit economics

The expression of revenues and costs on a per-unit basis.

Unitranche debt

A hybrid or blended loan structure combining different tranches of secured and unsecured debt into a single loan with a single, blended interest rate.

Unsecured

Without collateral; unsecured debt is backed only by cash flows of the issuer.

Un-sponsored

A type of depository receipt in which the foreign company whose shares are held by the depository has no involvement in the issuance of the receipts.

Unsupervised learning

A type of machine learning in which the system tries to learn the structure of unlabeled data.

Utility tokens

Tokens that provide services within a network, such as paying for services and network fees.

Validity instructions

Instructions which indicate when the order may be filled.

Value added resellers

Businesses that distribute a product and also handle more complex aspects of product installation, customization, service, or support.

Value at risk

A money measure of the minimum value of losses expected during a specified time period at a given level of probability.

Value chain

The systems and processes in a firm that create value for its customers.

Value proposition

The product or service attributes valued by a firm's target customer that lead those customers to prefer that firm's offering.

Value-add real estate strategies

Strategies that involve larger-scale redevelopment and repositioning of existing assets and that may allow the investor to earn a higher return compared with core-plus real estate strategies.

Value-based pricing

Pricing set primarily by reference to the value of the product or service to customers.

VaR

See *value at risk*.

Variance

The expected value (the probability-weighted average) of squared deviations from a random variable's expected value.

Variance of a random variable

The expected value (the probability-weighted average) of squared deviations from a random variable's expected value.

Variation margin

The difference between current margin required and the current collateral price in a repurchase agreement.

Vega

The change in a given derivative instrument for a given small change in volatility, holding everything else constant. A sensitivity measure for options that reflects the effect of volatility.

Velocity

The pace at which geopolitical risk impacts an investor portfolio.

Venture capital

Investments or financing to private companies with high growth potential.

Venture capital fund

A hedge fund that seeks to buy, optimize, and ultimately sell portfolio companies to generate profits. See *private equity fund*.

Venture debt

Private debt funding that provides venture capital backing to start-up or early-stage companies that may be generating little or negative cash flow.

Vest

To become unconditionally entitled to.

Vesting date

The day that an employee becomes unconditionally entitled to compensation.

Vintage year

The year in which a private capital fund makes its first investment.

Volatility

The standard deviation of the continuously compounded returns on the underlying asset.

Vote by proxy

A mechanism that allows a designated party—such as another shareholder, a shareholder representative, or management—to vote on the shareholder's behalf.

Voting rights

The power of shareholders to cast votes in corporate elections for directors and other matters submitted to a shareholder vote.

Warrant

An attached option that gives its holder the right to buy the underlying stock of the issuing company at a fixed exercise price until the expiration date.

Waterfall structures

These represent the distribution order for cash flows and risk to different tranches in a financing structure.

Weak-form efficient market hypothesis

The belief that security prices fully reflect all past market data, which refers to all historical price and volume trading information.

Weighted average coupon rate (WAC)

Rate calculated for a mortgage pass-through security by weighting the mortgage rate of each mortgage in the pool by the percentage of the outstanding mortgage balance relative to the outstanding amount of all the mortgages in the pool.

Weighted average maturity (WAM)

Calculated for a mortgage pass-through security by weighting the remaining number of months to maturity of each mortgage in the pool by the outstanding mortgage balance relative to the outstanding amount of all the mortgages in the pool.

Weighted-average cost of capital (WACC)

A weighted average of the after-tax required rates of return on a company's equity, debt, preferred stock, and other long-term financing sources, where the weights are the fraction of each source of financing in the company's current market value or management's target capital structure.

Winsorized mean

A mean computed after assigning a stated percentage of the lowest values equal to one specified low value and a stated percentage of the highest values equal to one specified high value.

Write-off/liquidation

Refers to a transaction that has not gone well, and the investment is likely to lose value. The private equity firm revises the value of its investment downward or liquidates the portfolio company.

Yield curve

A graphical depiction of yields-to-maturity of bonds from the same issuer across maturities.

Yield spread

The difference in yield-to-maturity between a bond and that of another bond.

Yield-to-call

An internal rate of return on a fixed-income instrument's cash flows assuming cash flows are received on scheduled dates and the bond is called at a certain call price and date.

Yield-to-maturity

Internal rate of return on a fixed-income instrument's cash flows assuming all cash flows are received on their scheduled dates.

Yield-to-worst

The lowest among a fixed-income instrument's yields-to-call and yield-to-maturity. A commonly cited yield measure for fixed-rate callable bonds.

Z-spread

Z-spread (or zero-volatility spread) is a constant yield spread for a bond over a government or swap curve.

Zero-coupon bond

A bond that does not pay a coupon but is priced at a discount and pays its full face value at maturity.

Zero-coupon bonds

Bonds that do not pay interest during their life. They are issued at a discount to par value and redeemed at par. Also called pure discount bond.